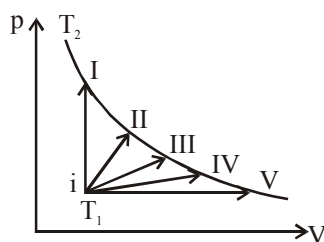


14. An ideal gas is taken reversibly from state  $i$ , at temperature  $T_1$ , to any of the other states labeled, I, II, III, IV and V on the  $p$ - $V$  diagram below. All are at the same temperature  $T_2$ . Rank the five processes according to the change in entropy of the gas, least to greatest :



- (A) I, II, III, IV, V  
(B) V, IV, III, II, I  
(C) I, then II, III, IV, and V tied  
(D) I, II, III and IV tied, then V  
(E) I and V tied, then II, III, IV
15. According to the second law of thermodynamics:  
(A) Heat energy cannot be completely converted to work  
(B) Work cannot be completely converted to heat energy  
(C) For all cyclic processes we have  $dQ/T < 0$   
(D) The reason all heat engine efficiencies are less than 100% is friction, which is unavoidable  
(E) All of the above are true
16. A reversible refrigerator operates between a low temperature reservoir at  $T_C$  and a high temperature reservoir at  $T_H$ . Its coefficient of performance is given by :  
(A)  $(T_H - T_C)/T_C$  (B)  $T_C/(T_H - T_C)$  (C)  $(T_H - T_C)/T_H$  (D)  $T_H/(T_H - T_C)$   
(E)  $T_H(T_H + T_C)$
17. An inventor claims to have a heat engine that has an efficiency of 40% when it operates between a high temperature reservoir of  $150^\circ\text{C}$  and a low temperature reservoir of  $30^\circ\text{C}$ . This engine:  
(A) Must violate the zeroth law of thermodynamics  
(B) Must violate the first law of thermodynamics  
(C) Must violate the second law of thermodynamics  
(D) Must violate the third law of thermodynamics  
(E) Does not necessarily violate any of the laws of thermodynamics
18. A Carnot heat engine runs between a cold reservoir at temperature  $T_C$  and a hot reservoir at temperature  $T_H$ . You want to increase its efficiency. Of the following, which change results in the greatest increase in efficiency? The value of  $\Delta T$  is the same for all changes.  
(A) Raise the temperature of the hot reservoir by  $\Delta T$   
(B) Raise the temperature of the cold reservoir by  $\Delta T$   
(C) Lower the temperature of the hot reservoir by  $\Delta T$   
(D) Lower the temperature of the cold reservoir by  $\Delta T$   
(E) Lower the temperature of the hot reservoir by  $1/2\Delta T$  and raise the temperature of the cold reservoir by  $1/2\Delta T$ .
19. A Carnot engine takes  $3 \times 10^6$  cal of heat from reservoir at  $627^\circ$  and gives it to a sink at  $27^\circ\text{C}$ . Then work done by the engine is :-  
(A)  $4.2 \times 10^6$  J (B)  $8.4 \times 10^6$  J (C)  $16.8 \times 10^6$  J (D) zero

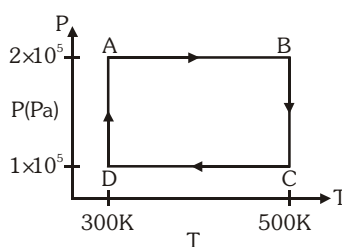
## EXERCISE (J-M)

1. One kg of a diatomic gas is at a pressure of  $8 \times 10^4 \text{ N/m}^2$ . The density of the gas is  $4 \text{ kg/m}^3$ . What is the energy of the gas due to its thermal motion ? [AIEEE-2009]  
 (1)  $6 \times 10^4 \text{ J}$  (2)  $7 \times 10^4 \text{ J}$  (3)  $3 \times 10^4 \text{ J}$  (4)  $5 \times 10^4 \text{ J}$

KT0097

**Directions :** Question number 2, 3 and 4 are based on the following paragraph.

Two moles of helium gas are taken over the cycle ABCDA, as shown in the P-T diagram.



2. Assuming the gas to be ideal the work done on the gas in taking it from A to B is :- [AIEEE-2009]  
 (1) 400 R (2) 500 R (3) 200 R (4) 300

KT0098

3. The work done on the gas in taking it from D to A is :- [AIEEE-2009]  
 (1) -690 R (2) +690 R (3) -414 R (4) +414 R

KT0099

4. The net work done on the gas in the cycle ABCDA is :- [AIEEE-2009]  
 (1) 1076 R (2) 1904 R (3) Zero (4) 276 R

KT0100

5. A diatomic ideal gas is used in a carnot engine as the working substance. If during the adiabatic expansion part of the cycle the volume of the gas increases from  $V$  to  $32V$ , the efficiency of the engine is :- [AIEEE-2010]  
 (1) 0.25 (2) 0.5 (3) 0.75 (4) 0.99

KT0101

6. A thermally insulated vessel contains an ideal gas of molecular mass  $M$  and ratio of specific heats  $\gamma$ . It is moving with speed  $v$  and is suddenly brought to rest. Assuming no heat is lost to the surroundings, its temperature increases by :- [AIEEE-2011]

(1)  $\frac{\gamma M v^2}{2R}$  (2)  $\frac{(\gamma - 1)}{2R} M v^2$  (3)  $\frac{(\gamma - 1)}{2(\gamma + 1)R} M v^2$  (4)  $\frac{(\gamma - 1)}{2\gamma R} M v^2$

KT0102

7. A Carnot engine operating between temperatures  $T_1$  and  $T_2$  has efficiency  $\frac{1}{6}$ . When  $T_2$  is lowered by 62 K, its efficiency increases to  $\frac{1}{3}$ . Then  $T_1$  and  $T_2$  are, respectively:- [AIEEE-2011]

(1) 330 K and 268 K (2) 310 K and 248 K (3) 372 K and 310 K (4) 372 K and 330 K

KT0103

8. Three perfect gases at absolute temperatures  $T_1$ ,  $T_2$  and  $T_3$  are mixed. The masses of molecules are  $m_1$ ,  $m_2$ , and  $m_3$  and the number of molecules are  $n_1$ ,  $n_2$  and  $n_3$  respectively. Assuming no loss of energy, then final temperature of the mixture is :- [AIEEE-2011]

(1)  $\frac{n_1 T_1^2 + n_2 T_2^2 + n_3 T_3^2}{n_1 T_1 + n_2 T_2 + n_3 T_3}$  (2)  $\frac{n_1^2 T_1^2 + n_2^2 T_2^2 + n_3^2 T_3^2}{n_1 T_1 + n_2 T_2 + n_3 T_3}$   
 (3)  $\frac{T_1 + T_2 + T_3}{3}$  (4)  $\frac{n_1 T_1 + n_2 T_2 + n_3 T_3}{n_1 + n_2 + n_3}$

KT0104

9. The specific heat capacity of a metal at low temperature ( $T$ ) is given as  $C_p(\text{kJ kg}^{-1} \text{K}^{-1}) = 32 \left( \frac{T}{400} \right)^3$

A 100 gram vessel of this metal is to be cooled from  $20^\circ\text{K}$  to  $4^\circ\text{K}$  by a special refrigerator operating at room temperature ( $27^\circ\text{C}$ ). The amount of work required to cool the vessel is:- [AIEEE-2011]

- (1) equal to 0.002 kJ (2) greater than 0.148 kJ  
 (3) between 0.148 kJ and 0.028 kJ (4) less than 0.028 kJ

KT0105

10. A container with insulating walls is divided into two equal parts by a partition fitted with a valve. One part is filled with an ideal gas at a pressure  $P$  and temperature  $T$ , whereas the other part is completely evacuated. If the valve is suddenly opened, the pressure and temperature of the gas will be :-

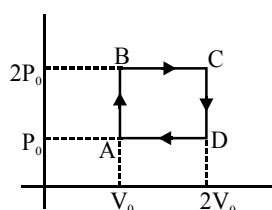
[AIEEE-2011]

- (1)  $\frac{P}{2}, T$  (2)  $\frac{P}{2}, \frac{T}{2}$  (3)  $P, T$  (4)  $P, \frac{T}{2}$

KT0106

11. Helium gas goes through a cycle ABCDA (consisting of two isochoric and two isobaric lines) as shown in figure. Efficiency of this cycle is nearly (Assume the gas to be close to ideal gas) :-

[AIEEE-2012]



- (1) 12.5% (2) 15.4% (3) 9.1% (4) 10.5%

KT0107

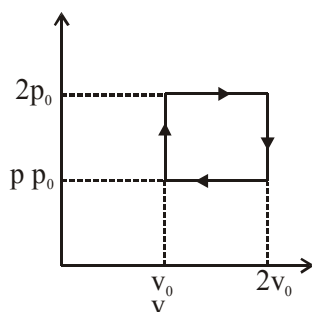
12. A Carnot engine, whose efficiency is 40% takes in heat from a source maintained at a temperature of 500 K. It is desired to have an engine of efficiency 60%. Then, the intake temperature for the same exhaust (sink) temperature must be :- [AIEEE-2012]

- (1) 600 K  
 (2) efficiency of Carnot engine cannot be made larger than 50%  
 (3) 1200 K  
 (4) 750 K

KT0108

13. The above p-v diagram represents the thermodynamic cycle of an engine, operating with an ideal monoatomic gas. The amount of heat, extracted from the source in a single cycle is :

[JEE-Mains-2013]



- (1)  $p_0 v_0$                       (2)  $\left(\frac{13}{2}\right)p_0 v_0$                       (3)  $\left(\frac{11}{2}\right)p_0 v_0$                       (4)  $4p_0 v_0$

KT0109

14. An open glass tube is immersed in mercury in such a way that a length of 8 cm extends above the mercury level. The open of the tube is then closed and sealed and the tube is raised vertically up by addition 46 cm. What will be length of the air column above mercury in the tube now ?

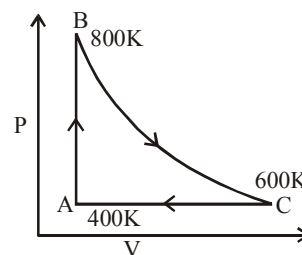
(Atmospheric pressure = 76 cm of Hg)

[JEE-Mains-2014]

- (1) 38 cm                      (2) 6 cm                      (3) 16 cm                      (4) 22 cm

KT0110

15. One mole of diatomic ideal gas undergoes a cyclic process ABC as shown in figure. The process BC is adiabatic. The temperatures at A, B and C are 400 K, 800 K and 600 K respectively. Choose the correct statement : [JEE-Mains-2014]



- (1) The change in internal energy in the process AB is  $-350 R$ .  
 (2) The change in internal energy in the process BC is  $-500R$   
 (3) The change in internal energy in whole cyclic process is  $250 R$ .  
 (4) The change in internal energy in the process CA is  $700 R$ .

KT0111

16. A solid body of constant heat capacity  $1 \text{ J}^\circ/\text{C}$  is being heated by keeping it in contact with reservoirs in two ways - [JEE-Mains-2015]

- (i) Sequentially keeping in contact with 2 reservoirs such that each reservoir supplies same amount of heat.  
 (ii) Sequentially keeping in contact with 8 reservoirs such that each reservoir supplies same amount of heat.

In both the cases body is brought from initial temperature  $100^\circ\text{C}$  to final temperature  $200^\circ\text{C}$ . Entropy change of the body in the two cases respectively is -

- (1)  $\ln 2, 2\ln 2$                       (2)  $2\ln 2, 8\ln 2$                       (3)  $\ln 2, 4\ln 2$                       (4)  $\ln 2, \ln 2$

KT0112



17. Consider a spherical shell of radius  $R$  at temperature  $T$ . The black body radiation inside it can be considered as an ideal gas of photons with internal energy per unit volume  $u = \frac{U}{V} \propto T^4$  and pressure

$p = \frac{1}{3} \left( \frac{U}{V} \right)$ . If the shell now undergoes an adiabatic expansion the relation between  $T$  and  $R$  is -

[JEE-Mains-2015]

- (1)  $T \propto \frac{1}{R}$       (2)  $T \propto \frac{1}{R^3}$       (3)  $T \propto e^{-R}$       (4)  $T \propto e^{-3R}$

KT0113

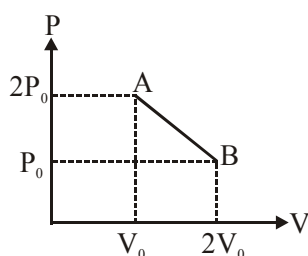
18. Consider an ideal gas confined in an isolated closed chamber. As the gas undergoes an adiabatic expansion, the average time of collision between molecules increases as  $V^q$ , where  $V$  is the volume

of the gas. The value of  $q$  is :-  $\left( \gamma = \frac{C_p}{C_v} \right)$  [JEE-Mains-2015]

- (1)  $\frac{\gamma+1}{2}$       (2)  $\frac{\gamma-1}{2}$       (3)  $\frac{3\gamma+5}{6}$       (4)  $\frac{3\gamma-5}{6}$

KT0114

19. 'n' moles of an ideal gas undergoes a process  $A \rightarrow B$  as shown in the figure. The maximum temperature of the gas during the process will be : [JEE-Mains-2016]



- (1)  $\frac{9P_0V_0}{nR}$       (2)  $\frac{9P_0V_0}{4nR}$       (3)  $\frac{3P_0V_0}{2nR}$       (4)  $\frac{9P_0V_0}{2nR}$

KT0115

20. An ideal gas undergoes a quasi static, reversible process in which its molar heat capacity  $C$  remains constant. If during this process the relation of pressure  $P$  and volume  $V$  is given by  $PV^n = \text{constant}$ , then  $n$  is given by (Here  $C_p$  and  $C_v$  are molar specific heat at constant pressure and constant volume, respectively) :- [JEE-Mains-2016]

- (1)  $n = \frac{C-C_v}{C-C_p}$       (2)  $n = \frac{C_p}{C_v}$       (3)  $n = \frac{C-C_p}{C-C_v}$       (4)  $n = \frac{C_p-C}{C-C_v}$

KT0116

21. The temperature of an open room of volume  $30 \text{ m}^3$  increases from  $17^\circ\text{C}$  to  $27^\circ\text{C}$  due to sunshine. The atmospheric pressure in the room remains  $1 \times 10^5 \text{ Pa}$ . If  $n_i$  and  $n_f$  are the number of molecules in the room before and after heating, then  $n_f - n_i$  will be :- [JEE-Main 2017]

(1)  $2.5 \times 10^{25}$  (2)  $-2.5 \times 10^{25}$  (3)  $-1.61 \times 10^{23}$  (4)  $1.38 \times 10^{23}$

KT0117

22.  $C_p$  and  $C_v$  are specific heats at constant pressure and constant volume respectively. It is observed that  $C_p - C_v = a$  for hydrogen gas  
 $C_p - C_v = b$  for nitrogen gas  
The correct relation between  $a$  and  $b$  is : [JEE-Main 2017]

(1)  $a = 14b$  (2)  $a = 28b$  (3)  $a = \frac{1}{14}b$  (4)  $a = b$

KT0118

23. Two moles of an ideal monoatomic gas occupies a volume  $V$  at  $27^\circ\text{C}$ . The gas expands adiabatically to a volume  $2V$ . Calculate (a) the final temperature of the gas and (b) change in its internal energy.

[JEE-Main 2018]

(1) (a) 195 K (b)  $-2.7 \text{ kJ}$  (2) (a) 189 K (b)  $-2.7 \text{ kJ}$   
(3) (a) 195 K (b)  $2.7 \text{ kJ}$  (4) (a) 189 K (b)  $2.7 \text{ kJ}$

KT0119

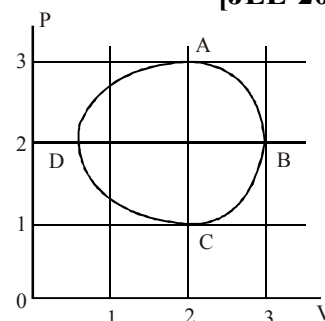
# EXERCISE (JA)

1.  $C_v$  and  $C_p$  denote the molar specific heat capacities of a gas at constant volume and constant pressure, respectively. Then [JEE-2009]
  - (A)  $C_p - C_v$  is larger for a diatomic ideal gas than for a monoatomic ideal gas
  - (B)  $C_p + C_v$  is larger for a diatomic ideal gas than for a monoatomic ideal gas
  - (C)  $C_p / C_v$  is larger for a diatomic ideal gas than for a monoatomic ideal gas
  - (D)  $C_p \cdot C_v$  is larger for a diatomic ideal gas than for a monoatomic ideal gas

KT0120

2. The figure shows the P-V plot of an ideal gas taken through a cycle ABCDA. The part ABC is a semicircle and CDA is half of an ellipse. Then, [JEE-2009]

- (A) the process during the path  $A \rightarrow B$  is isothermal
- (B) heat flows out of the gas during the path  $B \rightarrow C \rightarrow D$
- (C) work done during the path  $A \rightarrow B \rightarrow C$  is zero
- (D) positive work is done by the gas in the cycle ABCDA



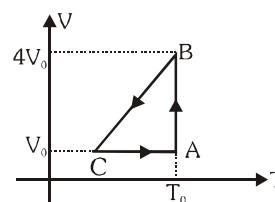
KT0121

3. A real gas behaves like an ideal gas if its
  - (A) pressure and temperature are both high
  - (B) pressure and temperature are both low
  - (C) pressure is high and temperature is low
  - (D) pressure is low and temperature is high

KT0122

4. One mole of an ideal gas in initial state A undergoes a cyclic process ABCA, as shown in the figure. its pressure at A is  $P_0$ . Choose the correct option(s) from the following. [JEE-2010]

- (A) Internal energies at A and B are the same
- (B) Work done by the gas in process AB is  $P_0 V_0 \ln 4$
- (C) Pressure at C is  $\frac{P_0}{4}$
- (D) Temperature at C is  $\frac{T_0}{4}$



KT0123

5. A diatomic ideal gas is compressed adiabatically to  $1/32$  of its initial volume. In the initial temperature of the gas is  $T_i$  (in Kelvin) and the final temperature is  $aT_i$ , the value of  $a$  is [JEE-2010]

KT0124

6. 5.6 liter of helium gas at STP is adiabatically compressed to 0.7 liter. Taking the initial temperature to be  $T_1$ , the work done in the process is [JEE-2011]

(A)  $\frac{9}{8}RT_1$  (B)  $\frac{3}{2}RT_1$  (C)  $\frac{15}{8}RT_1$  (D)  $\frac{9}{2}RT_1$

KT0125

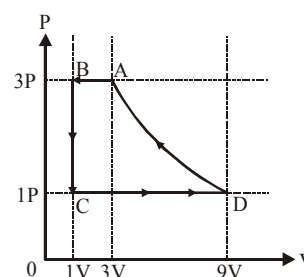
7. One mole of a monatomic ideal gas is taken through a cycle ABCDA as shown in P-V diagram. **Column II** gives the characteristics involved in the cycle. Match them with each of the processes given in **Column I**. [JEE-2011]

**Column I**

- (A) Process A→B  
(B) Process B→C  
(C) Process C→D  
(D) Process D→A

**Column II**

- (P) Internal energy decreases.  
(Q) Internal energy increases.  
(R) Heat is lost.  
(S) Heat is gained.  
(T) Work is done on the gas.



KT0126

8. A mixture of 2 moles of helium gas (atomic mass = 4 amu) and 1 mole of argon gas (atomic mass = 40 amu) is kept at 300 K in a container. The ratio of the rms speeds  $\left( \frac{v_{rms}(\text{helium})}{v_{rms}(\text{argon})} \right)$  is [JEE-2012]

(A) 0.32 (B) 0.45 (C) 2.24 (D) 3.16

KT0127

9. Two moles of ideal helium gas are in a rubber balloon at  $30^\circ\text{C}$ . The balloon is fully expandable and can be assumed to require no energy in its expansion. The temperature of the gas in the balloon is slowly changed to  $35^\circ\text{C}$ . The amount of heat required in raising the temperature is nearly [JEE-2012]

(Take :  $R = 8.31 \text{ J/mol.K}$ )

(A) 62 J (B) 104 J (C) 124 J (D) 208 J

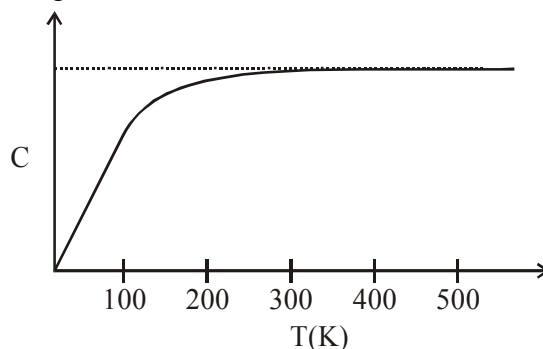
KT0128

10. Two non-reactive monoatomic ideal gases have their atomic masses in the ratio 2 : 3. The ratio of their partial pressures, when enclosed in a vessel kept at a constant temperature, is 4 : 3. The ratio of their densities is :- [JEE-2013]

(A) 1 : 4 (B) 1 : 2 (C) 6 : 9 (D) 8 : 9

KT0129

11. The figure below shows the variation of specific heat capacity ( $C$ ) of a solid as a function of temperature ( $T$ ). The temperature is increased continuously from 0 to 500 K at a constant rate. Ignoring any volume change, the following statement(s) is (are) correct to a reasonable approximation :



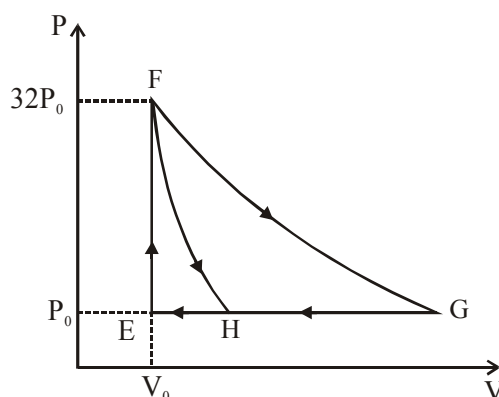
- (A) the rate at which heat is absorbed in the range 0–100 K varies linearly with temperature  $T$ .  
 (B) heat absorbed in increasing the temperature from 0–100 K is less than the heat required for increasing the temperature from 400–500 K.  
 (C) there is no change in the rate of heat absorption in the range 400–500 K  
 (D) the rate of heat absorption increases in the range 200–300 K

[JEE-2013]

KT0130

12. One mole of a monatomic ideal gas is taken along two cyclic processes  $E \rightarrow F \rightarrow G \rightarrow E$  and  $E \rightarrow F \rightarrow H \rightarrow E$  as shown in the  $PV$  diagram. The processes involved are purely isochoric, isobaric, isothermal or adiabatic.

[JEE-2013]



Match the paths in List I with the magnitudes of the work done in the List II and select the correct answer using the codes given below the lists.

- List I**  
 P.  $G \rightarrow E$   
 Q.  $G \rightarrow H$   
 R.  $F \rightarrow H$   
 S.  $F \rightarrow G$

**Codes :**

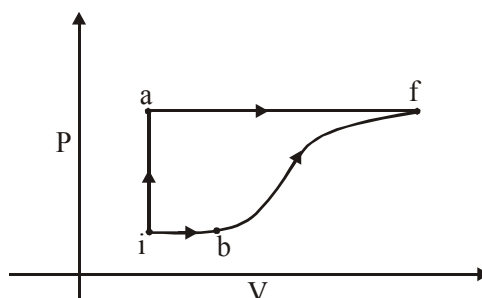
|     | P | Q | R | S |
|-----|---|---|---|---|
| (A) | 4 | 3 | 2 | 1 |
| (B) | 4 | 3 | 1 | 2 |
| (C) | 3 | 1 | 2 | 4 |
| (D) | 1 | 3 | 2 | 4 |

- List II**  
 1.  $160 P_0 V_0 \ln 2$   
 2.  $36 P_0 V_0$   
 3.  $24 P_0 V_0$   
 4.  $31 P_0 V_0$

KT0131

13. A thermodynamic system is taken from an initial state  $i$  with internal energy  $U_i = 100$  J to the final state  $f$  along two different paths  $iaf$  and  $ibf$ , as schematically shown in the figure. The work done by the system along the paths  $af$ ,  $ib$  and  $bf$  are  $W_{af} = 200$  J,  $W_{ib} = 50$  J and  $W_{bf} = 100$  J respectively. The heat supplied to the system along the path  $iaf$ ,  $ib$  and  $bf$  are  $Q_{iaf}$ ,  $Q_{ib}$  and  $Q_{bf}$  respectively. If the internal energy of the system in the state  $b$  is  $U_b = 200$  J and  $Q_{iaf} = 500$  J, the ratio  $Q_{bf}/Q_{ib}$  is.

[JEE-Advance-2014]

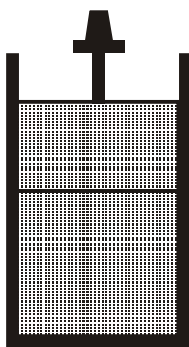


### Paragraph for Questions 14 and 15

In the figure a container is shown to have a movable (without friction) piston on top. The container and the piston are all made of perfectly insulating material allowing no heat transfer between outside and inside the container. The container is divided into two compartments by a rigid partition made of a thermally conducting material that allows slow transfer of heat. The lower compartment of the container is filled with 2 moles of an ideal monatomic gas at 700 K and the upper compartment is filled with 2 moles of an ideal diatomic gas at 400 K. The heat capacities per mole of an ideal monatomic

gas are  $C_V = \frac{3}{2}R$ ,  $C_P = \frac{5}{2}R$ , and those for an ideal diatomic gas are  $C_V = \frac{5}{2}R$ ,  $C_P = \frac{7}{2}R$ .

[JEE-Advance-2014]



KT0132

14. Consider the partition to be rigidly fixed so that it does not move. When equilibrium is achieved, the final temperature of the gases will be

(A) 550 K (B) 525 K (C) 513 K (D) 490 K

KT0133

15. Now consider the partition to be free to move without friction so that the pressure of gases in both compartments is the same. Then total work done by the gases till the time they achieve equilibrium will be

(A) 250 R (B) 200 R (C) 100 R (D) -100 R

KT0133

16. A container of fixed volume has a mixture of one mole of hydrogen and one mole of helium in equilibrium at temperature  $T$ . Assuming the gases are ideal, the correct statement(s) is (are) :-

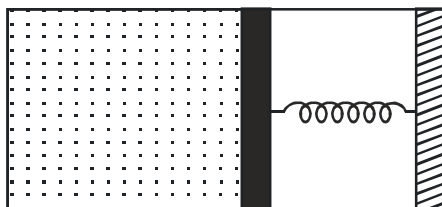
[JEE-Advance-2015]

- (A) The average energy per mole of the gas mixture is  $2RT$ .  
 (B) The ratio of speed of sound in the gas mixture to that in helium gas is  $\sqrt{6/5}$ .  
 (C) The ratio of the rms speed of helium atoms to that of hydrogen molecules is  $1/2$ .  
 (D) The ratio of the rms speed of helium atoms to that of hydrogen molecules is  $1/\sqrt{2}$ .

KT0134

17. An ideal monoatomic gas is confined in a horizontal cylinder by a spring loaded piston (as shown in the figure). Initially the gas is at temperature  $T_1$ , pressure  $P_1$  and volume  $V_1$  and the spring is in its relaxed state. The gas is then heated very slowly to temperature  $T_2$ , pressure  $P_2$  and volume  $V_2$ . During this process the piston moves out by a distance  $x$ . Ignoring the friction between the piston and the cylinder, the correct statement(s) is(are) :-

[JEE-Advance-2015]



- (A) If  $V_2 = 2V_1$  and  $T_2 = 3T_1$ , then the energy stored in the spring is  $\frac{1}{4}P_1V_1$   
 (B) If  $V_2 = 2V_1$  and  $T_2 = 3T_1$ , then the change in internal energy is  $3P_1V_1$   
 (C) If  $V_2 = 3V_1$  and  $T_2 = 4T_1$ , then the work done by the gas is  $\frac{7}{3}P_1V_1$   
 (D) If  $V_2 = 3V_1$  and  $T_2 = 4T_1$ , then the heat supplied to the gas is  $\frac{17}{6}P_1V_1$

KT0135

18. A gas is enclosed in a cylinder with a movable frictionless piston. Its initial thermodynamic state at pressure  $P_i = 10^5$  Pa and volume  $V_i = 10^{-3}$  m<sup>3</sup> changes to a final state at  $P_f = \left(\frac{1}{32}\right) \times 10^5$  Pa and  $V_f = 8 \times 10^{-3}$  m<sup>3</sup> in an adiabatic quasi-static process, such that  $P^3V^5 = \text{constant}$ . Consider another thermodynamic process that brings the system from the same initial state to the same final state in two steps: an isobaric expansion at  $P_i$  followed by an isochoric (isovolumetric) process at volumes  $V_f$ . The amount of heat supplied to the system in the two step process is approximately

[JEE-Advance 2016]

- (A) 112 J                      (B) 294 J                      (C) 588 J                      (D) 813 J

KT0136

19. A flat plate is moving normal to its plane through a gas under the action of a constant force  $F$ . The gas is kept at a very low pressure. The speed of the plate  $v$  is much less than the average speed  $u$  of the gas molecules. Which of the following options is/are true ? [JEE-Advance 2017]

- (A) The resistive force experienced by the plate is proportional to  $v$   
 (B) The pressure difference between the leading and trailing faces of the plate is proportional to  $uv$ .  
 (C) The plate will continue to move with constant non-zero acceleration, at all times  
 (D) At a later time the external force  $F$  balances the resistive force.

KT0137

Answer Q.20, Q.21 and Q.22 by appropriately matching the information given in the three columns of the following table.

An ideal gas is undergoing a cyclic thermodynamics process in different ways as shown in the corresponding  $P$ - $V$  diagrams in column 3 of the table. Consider only the path from state 1 to state 2.  $W$  denotes the corresponding work done on the system. The equations and plots in the table have standard notations as used in thermodynamics processes. Here  $\gamma$  is the ratio of heat capacities at constant pressure and constant volume. The number of moles in the gas is  $n$ . [JEE-Advance 2017]

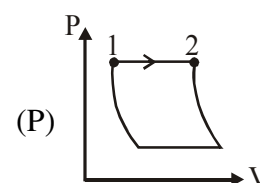
Column-1

Column-2

Column-3

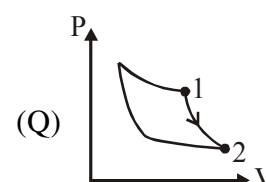
$$(I) W_{1 \rightarrow 2} = \frac{1}{\gamma - 1} (P_2 V_2 - P_1 V_1)$$

(i) Isothermal



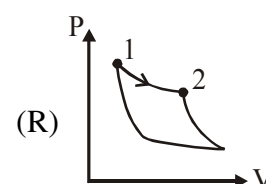
$$(II) W_{1 \rightarrow 2} = -PV_2 + PV_1$$

(ii) Isochoric



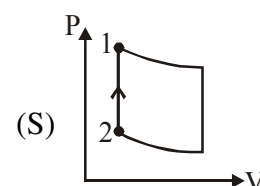
$$(III) W_{1 \rightarrow 2} = 0$$

(iii) Isobaric



$$(IV) W_{1 \rightarrow 2} = -nRT \ln \frac{V_2}{V_1}$$

(iv) Adiabatic





20. Which of the following options is the only correct representation of a process in which  $\Delta U = \Delta Q - P\Delta V$ ?

(A) (II) (iv) (R) (B) (II) (iii) (P) (C) (II) (iii) (S) (D) (III) (iii) (P)

KT0138

21. Which one of the following options is the correct combination ?

(A) (III) (ii) (S) (B) (II) (iv) (R) (C) (II) (iv) (P) (D) (IV) (ii) (S)

KT0138

22. Which one of the following options correctly represents a thermodynamics process that is used as a correction in the determination of the speed of sound in an ideal gas ?

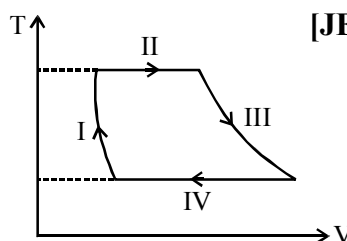
(A) (III) (iv) (R) (B) (I) (ii) (Q) (C) (IV) (ii) (R) (D) (I) (iv) (Q)

KT0138

23. One mole of a monatomic ideal gas undergoes a cyclic process as shown in the figure (where  $V$  is the volume and  $T$  is the temperature). Which of the statements below is (are) true ?

[JEE-Advance 2018]

- (A) Process I is an isochoric process  
(B) In process II, gas absorbs heat  
(C) In process IV, gas releases heat  
(D) Processes I and II are not isobaric



KT0139

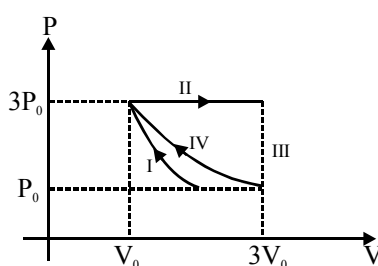
24. One mole of a monatomic ideal gas undergoes an adiabatic expansion in which its volume becomes eight times its initial value. If the initial temperature of the gas is 100 K and the universal gas constant  $R = 8.0 \text{ J mol}^{-1} \text{ K}^{-1}$ , the decrease in its internal energy, in Joule, is.....

[JEE-Advance 2018]

KT0140

25. One mole of a monatomic ideal gas undergoes four thermodynamic processes as shown schematically in the PV-diagram below. Among these four processes, one is isobaric, one is isochoric, one is isothermal and one is adiabatic. Match the processes mentioned in List-I with the corresponding statements in List-II.

[JEE-Advance 2018]



**List-I**

- P. In process I  
Q. In process II  
R. In process III  
S. In process IV

**List-II**

1. Work done by the gas is zero  
2. Temperature of the gas remains unchanged  
3. No heat is exchanged between the gas and its surroundings  
4. Work done by the gas is  $6 P_0 V_0$

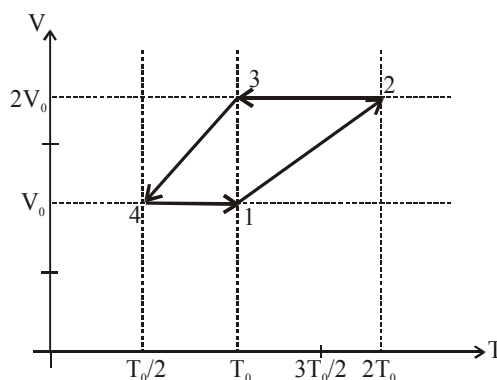
- (A)  $P \rightarrow 4$  ;  $Q \rightarrow 3$  ;  $R \rightarrow 1$  ;  $S \rightarrow 2$   
(B)  $P \rightarrow 1$  ;  $Q \rightarrow 3$  ;  $R \rightarrow 2$  ;  $S \rightarrow 4$   
(C)  $P \rightarrow 3$  ;  $Q \rightarrow 4$  ;  $R \rightarrow 1$  ;  $S \rightarrow 2$   
(D)  $P \rightarrow 3$  ;  $Q \rightarrow 4$  ;  $R \rightarrow 2$  ;  $S \rightarrow 1$

KT0141

26. One mole of a monoatomic ideal gas goes through a thermodynamic cycle, as shown in the volume versus temperature (V-T) diagram. The correct statement(s) is/are :

[R is the gas constant]

[JEE-Advance 2019]



- (1) Work done in this thermodynamic cycle ( $1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 1$ ) is  $|W| = \frac{1}{2}RT_0$
- (2) The ratio of heat transfer during processes  $1 \rightarrow 2$  and  $2 \rightarrow 3$  is  $\left| \frac{Q_{1 \rightarrow 2}}{Q_{2 \rightarrow 3}} \right| = \frac{5}{3}$
- (3) The above thermodynamic cycle exhibits only isochoric and adiabatic processes.
- (4) The ratio of heat transfer during processes  $1 \rightarrow 2$  and  $3 \rightarrow 4$  is  $\left| \frac{Q_{1 \rightarrow 2}}{Q_{3 \rightarrow 4}} \right| = \frac{1}{2}$

KT0142

27. A mixture of ideal gas containing 5 moles of monatomic gas and 1 mole of rigid diatomic gas is initially at pressure  $P_0$ , volume  $V_0$  and temperature  $T_0$ . If the gas mixture is adiabatically compressed to a volume  $V_0/4$ , then the correct statement(s) is/are,

(Give  $2^{1.2} = 2.3$  ;  $2^{3.2} = 9.2$ ; R is gas constant)

[JEE-Advance 2019]

- (1) The final pressure of the gas mixture after compression is in between  $9P_0$  and  $10P_0$
- (2) The average kinetic energy of the gas mixture after compression is in between  $18RT_0$  and  $19RT_0$
- (3) The work  $|W|$  done during the process is  $13RT_0$
- (4) Adiabatic constant of the gas mixture is 1.6

KT0143

28. Answer the following by appropriately matching the lists based on the information given in the paragraph. [JEE-Advance 2019]

In a thermodynamics process on an ideal monatomic gas, the infinitesimal heat absorbed by the gas is given by  $T\Delta X$ , where  $T$  is temperature of the system and  $\Delta X$  is the infinitesimal change in a thermodynamic quantity  $X$  of the system. For a mole of monatomic ideal gas

$$X = \frac{3}{2} R \ln \left( \frac{T}{T_A} \right) + R \ln \left( \frac{V}{V_A} \right). \text{ Here, } R \text{ is gas constant, } V \text{ is volume of gas, } T_A \text{ and } V_A \text{ are constants.}$$

constants.

The List-I below gives some quantities involved in a process and List-II gives some possible values of these quantities.

**List-I**

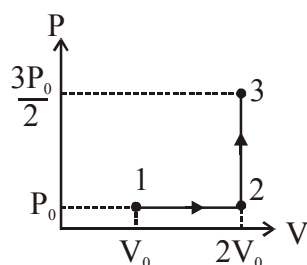
- (I) Work done by the system in process  $1 \rightarrow 2 \rightarrow 3$   
 (II) Change in internal energy in process  $1 \rightarrow 2 \rightarrow 3$   
 (III) Heat absorbed by the system in process  $1 \rightarrow 2 \rightarrow 3$   
 (IV) Heat absorbed by the system in process  $1 \rightarrow 2$

**List-II**

- (P)  $\frac{1}{3} RT_0 \ln 2$   
 (Q)  $\frac{1}{3} RT_0$   
 (R)  $RT_0$   
 (S)  $\frac{4}{3} RT_0$   
 (T)  $\frac{1}{3} RT_0 (3 + \ln 2)$   
 (U)  $\frac{5}{6} RT_0$

If the process carried out on one mole of monatomic ideal gas is as shown in figure in the PV-diagram

with  $P_0 V_0 = \frac{1}{3} RT_0$ , the correct match is,



- (1) I  $\rightarrow$  Q, II  $\rightarrow$  R, III  $\rightarrow$  P, IV  $\rightarrow$  U  
 (2) I  $\rightarrow$  S, II  $\rightarrow$  R, III  $\rightarrow$  Q, IV  $\rightarrow$  T  
 (3) I  $\rightarrow$  Q, II  $\rightarrow$  R, III  $\rightarrow$  S, IV  $\rightarrow$  U  
 (4) I  $\rightarrow$  Q, II  $\rightarrow$  S, III  $\rightarrow$  R, IV  $\rightarrow$  U

KT0144

29. Answer the following by appropriately matching the lists based on the information given in the paragraph. [JEE-Advance 2019]

In a thermodynamic process on an ideal monatomic gas, the infinitesimal heat absorbed by the gas is given by  $T\Delta X$ , where  $T$  is temperature of the system and  $\Delta X$  is the infinitesimal change in a thermodynamic quantity  $X$  of the system. For a mole of monatomic ideal gas

$$X = \frac{3}{2} R \ln \left( \frac{T}{T_A} \right) + R \ln \left( \frac{V}{V_A} \right). \text{ Here, } R \text{ is gas constant, } V \text{ is volume of gas, } T_A \text{ and } V_A \text{ are}$$

constants.

The List-I below gives some quantities involved in a process and List-II gives some possible values of these quantities.

**List-I**

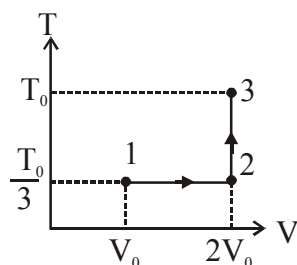
- (I) Work done by the system in process  $1 \rightarrow 2 \rightarrow 3$   
 (II) Change in internal energy in process  $1 \rightarrow 2 \rightarrow 3$   
 (III) Heat absorbed by the system in process  $1 \rightarrow 2 \rightarrow 3$   
 (IV) Heat absorbed by the system in process  $1 \rightarrow 2$

**List-II**

- (P)  $\frac{1}{3} RT_0 \ln 2$   
 (Q)  $\frac{1}{3} RT_0$   
 (R)  $RT_0$   
 (S)  $\frac{4}{3} RT_0$   
 (T)  $\frac{1}{3} RT_0 (3 + \ln 2)$   
 (U)  $\frac{5}{6} RT_0$

If the process on one mole of monatomic ideal gas is as shown in the TV-diagram

with  $P_0 V_0 = \frac{1}{3} RT_0$ , the correct match is



- (1) I  $\rightarrow$  S, II  $\rightarrow$  T, III  $\rightarrow$  Q, IV  $\rightarrow$  U  
 (2) I  $\rightarrow$  P, II  $\rightarrow$  R, III  $\rightarrow$  T, IV  $\rightarrow$  S  
 (3) I  $\rightarrow$  P, II  $\rightarrow$  T, III  $\rightarrow$  Q, IV  $\rightarrow$  T  
 (4) I  $\rightarrow$  P, II  $\rightarrow$  R, III  $\rightarrow$  T, IV  $\rightarrow$  P

KT0145

# ANSWER KEY

## EXERCISE (S-1)

1. Ans.  $1.25 \times 10^4 \text{ N/m}^2$       2. Ans.  $327^\circ\text{C}$
3. Ans. (i)  $P_1 < P_2$ ,  $T_1 < T_2$ ; (ii)  $T_1 = T_2 < T_3$ ; (iii)  $V_2 > V_1$ ; (iv)  $P_1 > P_2$       4. Ans. 40
5. Ans. (a)  $4/3 \text{ m}$ , (b)  $T_3 = 400\left(\frac{4}{3}\right)^{0.4} \text{ K}$       6. Ans. 3600 R      7. Ans.  $\Delta T = \frac{mv_0^2}{3R}$
8. Ans. 12600 J      9. Ans. 5R      10. Ans. 2.64      11. Ans. 1.5
12. Ans. 450 J      13. Ans. 16.9 J      14. Ans. R/2
15. Ans. (a) 0.5 atm (b) zero (c) zero (d) no
16. Ans. (a) A to B (b) C to D

$$(c) W_{AB} = \frac{1}{1-\gamma} (2^{1-\gamma} - 1)(P_B - P_A)V_A = \frac{3}{2} \left( 1 - \left( \frac{1}{2} \right)^{2/3} \right) (P_B - P_A)V_A$$

$$(d) \left[ 1 - \left( \frac{1}{2} \right)^{2/3} \right]$$

## EXERCISE (S-2)

1. Ans. (i) 74 cm, (ii) 73.94 cm, (iii) 69.52 cm
2. Ans.  $p_1 = p_{H_2} \simeq 1.25 \times 10^6 \text{ Pa}$ ;  $p_2 = p_{H_2} + p_{O_2} + p_{N_2} \simeq 2.8125 \times 10^6 \text{ Pa}$ ;  $p_3 = p_{H_2} + p_{N_2} \simeq 1.5625 \times 10^6 \text{ Pa}$
3. Ans. 1.6 m, 364 K
4. Ans. (a)  $P_i = P_a$  (b)  $P_f = P_a + k(V - V_0)/A^2$ ;       $V_0 = \ell A$   
 (c)  $\Delta Q = P_a(V - V_0) + \frac{1}{2} k(V - V_0)^2 + C_v(T - T_0)$ ;       $A = 1$
5. Ans. (a) Final pressure in A =  $\frac{27}{8} P_0$  = Final pressure in C, Final pressure in B =  $\frac{21}{4} P_0$   
 (b) Final temperature in A (and B) =  $\frac{21}{4} T_0$ , Final temperature in C =  $\frac{3}{2} T_0$ ,  
 (c)  $18 P_0 V_0$   
 (d) work done by gas in A =  $+ P_0 V_0$ , work done by gas in B = 0,  
 (e)  $\frac{17}{2} P_0 V_0$
6. Ans.  $C = 2R$       7. Ans. 400 J,  $2T_0$       8. Ans. 2000 N/m, 1295 J
9. Ans.  $Q_1 < Q_2$
10. Ans. (i)  $T_A = 120.33 \text{ K}$ ,  $T_B = 240.66 \text{ K}$ ,  $T_C = 481.32 \text{ K}$ ,  $T_D = 240.66 \text{ K}$ , (ii) No  
 (iii)  $\Delta Q_{ABC} = 3.25 \times 10^6 \text{ J}$ ;  $\Delta Q_{ADC} = 2.75 \times 10^6 \text{ J}$

$$11. \text{ Ans. } 1 - \frac{3 \left( 1 - \frac{1}{2^{1/3}} \right)}{\ln 2} \quad 12. \text{ Ans. } 2.515, 404.8 \text{ K}$$

**EXERCISE (O-1)**

1. Ans. (A) 2. Ans. (A) 3. Ans. (B) 4. Ans. (D) 5. Ans. (C) 6. Ans. (A)  
 7. Ans. (C) 8. Ans. (B) 9. Ans. (C) 10. Ans. (C) 11. Ans. (C) 12. Ans. (A)  
 13. Ans. (D) 14. Ans. (A) 15. Ans. (C) 16. Ans. (D) 17. Ans. (B) 18. Ans. (D)  
 19. Ans. (C) 20. Ans. (C) 21. Ans. (A) 22. Ans. (C) 23. Ans. (D) 24. Ans. (B)  
 25. Ans. (B) 26. Ans. (C) 27. Ans. (B) 28. Ans. (D) 29. Ans. (A) 30. Ans. (A)  
 31. Ans. (A) 32. Ans. (A) 33. Ans. (A) 34. Ans. (D) 35. Ans. (B) 36. Ans. (B)  
 37. Ans. (C) 38. Ans. (C) 39. Ans. (D) 40. Ans. (A) 41. Ans. (B) 42. Ans. (A,B,C)  
 43. Ans. (B,C) 44. Ans. (B,C) 45. Ans. (D) 46. Ans. (D) 47. Ans. (C, D)  
 48. Ans. (B) 49. Ans. (A) 50. Ans. (A) P (B) Q (C) Q

**EXERCISE (O-2)**

1. Ans. (C) 2. Ans. (A) 3. Ans. (D) 4. Ans. (D) 5. Ans. (B) 6. Ans. (C)  
 7. Ans. (D) 8. Ans. (C) 9. Ans. (D) 10. Ans. (C) 11. Ans. (D) 12. Ans. (C)  
 13. Ans. (A) 14. Ans. (C) 15. Ans. (C,D) 16. Ans. (B,D) 17. Ans. (A,B) 18. Ans. (A,B)  
 19. Ans. (B,C) 20. Ans. (B) 21. Ans. (A) 22. Ans. (B)  
 23. Ans. (A)  $\rightarrow$  (P,R) ; (B)  $\rightarrow$  (Q, S) ; (C)  $\rightarrow$  (Q, S) ; (D)  $\rightarrow$  (P, R)

**SUPPLEMENT FOR JEE-MAINS**

1. Ans. B & 1.8% more 2. Ans. 2400 joule per cycle 3. Ans. 900 cal 4. Ans. 5.7 J  
 5. Ans.  $3.36 \times 10^6$  J 6. Ans. (B) 7. Ans. (D) 8. Ans. (A) 9. Ans. (A)  
 10. Ans. (A) 11. Ans. (A) 12. Ans. (A) 13. Ans. (A) 14. Ans. (A)  
 15. Ans. (A) 16. Ans. (B) 17. Ans. (C) 18. Ans. (D) 19. Ans. (B)

**EXERCISE (J-M)**

1. Ans. (4) 2. Ans. (1) 3. Ans. (4) 4. Ans. (4) 5. Ans. (3) 6. Ans. (2)  
 7. Ans. (3) 8. Ans. (4) 9. Ans. (2) 10. Ans. (1) 11. Ans. (2) 12. Ans. (4)  
 13. Ans. (2) 14. Ans. (3) 15. Ans. (2) 16. Ans. (Bonus) 17. Ans. (1)  
 18. Ans. (1) 19. Ans. (2) 20. Ans. (3) 21. Ans. (2) 22. Ans. (1) 23. Ans. (2)

**EXERCISE (JA)**

1. Ans. (BD) 2. Ans. (B,D) 3. Ans. (D) 4. Ans. (A,B,C,D) 5. Ans. 4  
 6. Ans. (A) 7. Ans. (A)  $\rightarrow$  (P,R,T) ; (B)  $\rightarrow$  (P,R) ; (C)  $\rightarrow$  (Q,S) ; (D)  $\rightarrow$  (R,T)  
 8. Ans. (D) 9. Ans. (D) 10. Ans. (D) 11. Ans. (A, B, C, D) or (B, C, D)  
 12. Ans. (A) 13. Ans. 2 14. Ans. (D) 15. Ans. (D) 16. Ans. (A,B,D)  
 17. Ans. (A,B,C) 18. Ans. (C) 19. Ans. (A,B,D) 20. Ans. (B)  
 21. Ans. (A) 22. Ans. (D) 23. Ans. (B,C,D) 24. Ans. 900 [899.95, 900.05]  
 25. Ans. (C) 26. Ans. (1,2) 27. Ans. (1,3,4) 28. Ans. (3) 29. Ans. (4)



# Chapter Contents

## 03

## FLUID MECHANICS

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## Important Notes

[illegible]

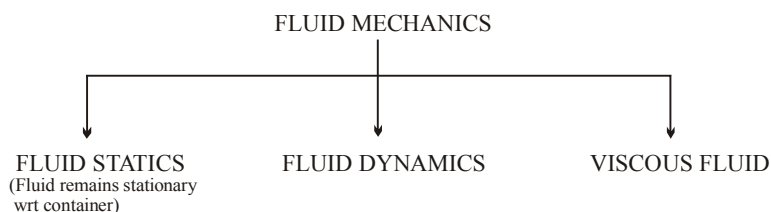


## FLUID MECHANICS

# KEY CONCEPTS

### What Is a Fluid?

A **fluid**, in contrast to a solid, is a substance that can flow. Fluids conform to the boundaries of any container in which we put them. They do so because a fluid cannot sustain a force that is tangential to its surface. a fluid is a substance that flows because it cannot withstand a shearing stress. It can, however, exert a force in the direction perpendicular to its surface.



Fluid includes property → (A) Density (B) Viscosity (C) Bulk modulus of elasticity (D) Pressure (E) Specific gravity.

### Assumptions used in fluid mechanics

1. Fluid is incompressible means density remains constant and volume also remains constant.
2. Fluid is non viscous. There is no tangential force between two layers.

### DENSITY ( $\rho$ )

Mass per unit volume is defined as density. So density at a point of a fluid is represented as

$$\rho = \lim_{\Delta V \rightarrow 0} \frac{\Delta m}{\Delta V} = \frac{dm}{dV} \quad \text{Density is a positive scalar quantity.}$$

SI UNIT :  $\text{kg/m}^3$

CGS UNIT :  $\text{g/cc}$

Dimensions :  $[\text{ML}^{-3}]$

### RELATIVE DENSITY

It is defined as the ratio of the density of the given fluid to the density of pure water at  $4^\circ\text{C}$ .

$$\text{Relative density (R.D.)} = \frac{\text{density of given liquid}}{\text{density of pure water at } 4^\circ\text{C}}$$

Relative density or specific gravity is a unitless and dimensionless positive scalar physical quantity. Being a dimensionless/unitless quantity R.D. of a substance is same in SI and CGS system.

### SPECIFIC GRAVITY

It is defined as the ratio of the specific weight of the given fluid to the specific weight of pure water at  $4^\circ\text{C}$ .

$$\text{Specific gravity} = \frac{\text{specific weight of given liquid}}{\text{specific weight of pure water at } 4^\circ\text{C} (9.81 \text{ kN/m}^3)} = \frac{\rho_\ell \times g}{\rho_w \times g} = \frac{\rho_\ell}{\rho_w} = \text{R.D. of the liquid}$$

Thus specific gravity of a liquid is numerically equal to the relative density of that liquid and for calculation purposes they are used interchangeably.

**Density of a Mixture of substance in the proportion of mass**

Let a number of substances of masses  $M_1, M_2, M_3$  etc., and densities  $\rho_1, \rho_2, \rho_3$  etc. respectively are mixed together. The total mass of the mixture  $= M_1 + M_2 + M_3 + \dots$

The total volume  $= \frac{M_1}{\rho_1} + \frac{M_2}{\rho_2} + \frac{M_3}{\rho_3} + \dots$  therefore, the density of the mixture is  $\rho = \frac{M_1 + M_2 + M_3 + \dots}{\frac{M_1}{\rho_1} + \frac{M_2}{\rho_2} + \frac{M_3}{\rho_3} + \dots}$

For two substances the density of the mixture  $\rho = \frac{\rho_1 \rho_2 (M_1 + M_2)}{\rho_1 M_2 + \rho_2 M_1}$

**Density of a mixture of substance in the proportion of volume**

Suppose that a number of substances of volume  $V_1, V_2, V_3$  etc. and densities  $\rho_1, \rho_2, \rho_3$  etc. respectively are mixed. The total mass of the mixture is  $= \rho_1 V_1 + \rho_2 V_2 + \rho_3 V_3 + \dots$

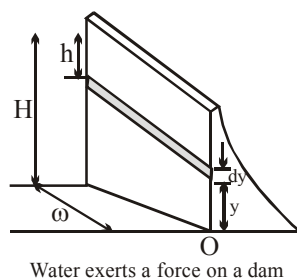
The total volume of the mixture is  $= V_1 + V_2 + V_3 + \dots$

Therefore, the density of the mixture is  $\rho = \frac{\rho_1 V_1 + \rho_2 V_2 + \rho_3 V_3}{V_1 + V_2 + V_3 + \dots}$

Therefore, for two substances we can write  $\rho = \frac{\rho_1 V_1 + \rho_2 V_2}{V_1 + V_2}$

**Example(a) :**

Water is filled to a height  $H$  behind a dam of width  $w$  (fig. ). Determine the resultant force exerted by the water on the dam.



Water exerts a force on a dam

**Solution :**

Let's consider a vertical  $y$  axis, starting from the bottom of the dam. Let's consider a thin horizontal strip at a height  $y$  above the bottom, such as shown in Fig. We need to consider force due to the pressure of the water only as atmospheric pressure acts on both sides of the dam.

The pressure due to the water at the depth  $h$  :

$$P = \rho gh = \rho g(H - y)$$

The force exerted on the shaded strip of area

$$dA = w dy :$$

$$dF = P dA = \rho g(H - y) w dy$$

Integrate to find the total force on the dam :

$$F = \int P dA = \int_0^H \rho g(H - y) w dy = 1/2 \rho g w H^2$$

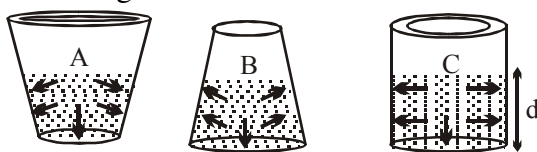
**Example(b) :**

In the previous example find the total torque exerted by the water on dam about a horizontal axis through  $O$ . Also find the effective line of action of the total force exerted by the water is at a distance  $1/3 H$  above  $O$ .



The force on the bottom of vessel C is equal to the weight of the water, as expected. The forces due to fluid pressure on the sides of the containers A and B have vertical components also. Hence the force between the fluid and the base of container will not be equal to the weight of the fluid. These containers support the fluid by exerting an upward force equal in magnitude to the weight of the fluid but some force is being applied by the sidewalls and the remaining by the bottom. Figure shows the forces acting on each container due to the water.

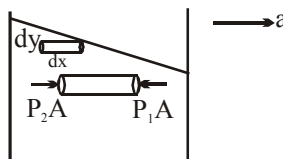
The force on the bottom of vessel A is less than the weight of the mercury in the container, while the force on the bottom of vessel B is greater than the weight of the mercury. In vessel A, the forces on the container walls have downward components as well as horizontal components. The sum of the downward components of the forces on the walls and the downward force on the bottom of the container is equal to the weight of the water. Similarly, the forces on the walls of vessel B have upward components. In each case, the total force on the bottom and sides of the container due to the water is equal to the weight of the water.



Forces exerted on the containers by the water.

### Linear Accelerated Motion :

We consider an open container of a liquid that is moving along a straight line with a constant acceleration  $a$  as shown in Fig.



Lets consider a small horizontal cylinder of length  $dx$  and cross-sectional area  $A$  located  $y$  below the free surface of the fluid. This cylinder is accelerating in ground frame with acceleration  $a$  hence the net horizontal force acting on it should be equal to the product of mass ( $dm$ ) and acceleration.

$$dm = A dx \rho$$

$$P_2 A - P_1 A = (A dx \rho) a$$

If we say that the right face of the cylinder is  $y$  below the free surface of the fluid then the left surface is  $y + dy$  below the surface of liquid. Thus

$$P_2 - P_1 = \rho g dy$$

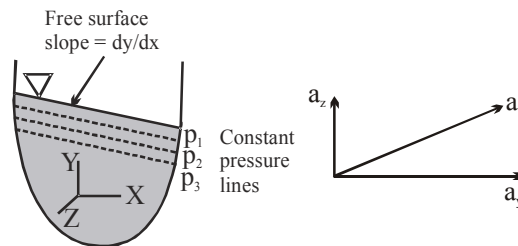
$$\therefore \frac{dy}{dx} = \frac{a}{g}$$

Since the slope of the free surface is coming out to be constant we can say that it must be straight line.

$$\tan \theta = \frac{a}{g}$$

If the container have acceleration along  $y$  also than the slope of this line is given by the relationship.

$$\frac{dy}{dx} = - \frac{a_x}{g + a_y}$$



Along a free surface the pressure is constant, so that for the accelerating mass shown in figure the free surface will be inclined if  $a_x \neq 0$ . In addition, all lines parallel to the free surface will have same pressure. For the special circumstance in which  $a_x = 0$ ,  $a_y \neq 0$ . which corresponds to the mass of fluid accelerating in the vertical direction, Equation indicates that the fluid surface will be horizontal. However, from below equation we see that the pressure variatio is not  $\rho g dy$ , but is given by the equation.

$$dP = \rho(g + a_v)dy$$

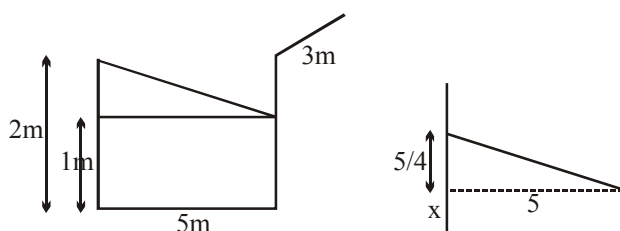
Thus, the pressure on the bottom of a liquid-filled tank which is resting on the floor of an elevator that is accelerating upward will be more than, if the, tank would have been at rest (or moving with a constant velocity). It is to be noted that for a freely falling fluid mass ( $a_y = -g$ ), the pressure variation in all three coordinate directions are zero, which means that the pressure throughout will be same. The pressure throughout a “blob” of a liquid floating in an orbiting space shuttle (a form of free fall) is zero.

**Ques :** The cross section of a tank kept on a vehicle is shown in Fig.. The rectangular tank is open to the atmosphere. During motion of the vehicle, the tank is subjected to a constant linear acceleration,  $a = 2.5 \text{ m/s}^2$ . How much fluid will be left inside the tank if initially the tank is half filled. The vessel is 5m wide and 2m high.

**Ans.**

If the height of the liquid on the left wall is greater than 2m the fluid will be spilled out. Now we can find the angle that the fluid will make with the horizontal.

$$\tan \theta = \frac{2.5}{10} = \frac{1}{4}$$



Lets assume that the dimension of tank in the plane perpendicular to the page is  $d$ .

From the geometry its easy to see that free surface on RHS will go down and will rise on LHS. Thus if we assume that fluid on RHS has not touched the floor, we will have fluid taking a shape as described in the diagram. The cuboid part will have volume  $x \times 5 \times d$ , where  $x$  is the height above the bottom.

The wedge part will have the volume  $\frac{1}{2} \times h \times 5 \times d$  where h can be found as following

$$(h/5) = \tan \theta = (1/4)$$

Thus total volume will be  $\frac{1}{2} \times (5/4) \times 5 \times d + x \times 5 \times d$  and if we assume there is no spilling than it must be equal to the final volume.

$$\frac{1}{2} \times (5/4) \times 5 \times d + x \times 5 \times d = 1 \times 5 \times d$$

solving we get  $x = \frac{3}{8}$

$$\therefore \text{Total length } \frac{5}{4} + \frac{3}{8} = \frac{10+3}{8} = \frac{13}{8} < 2$$

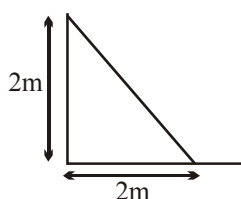
Thus, height is less than 2.

Hence water will not spill.

- Ex. How much fluid will be left inside the tank if the vehicle accelerates at acceleration,  $a = 10 \text{ m/s}^2$  ?  
If the height of the liquid on the left wall is greater than 2m the fluid will be spilled out.  
Now we can find the angle that the fluid will make with the horizontal.

$$\tan \theta = \frac{10}{10} = 1, \text{ thus } \theta = \frac{\pi}{4}$$

In this case fluid can not remain inside. Fluid having an inclined free surface at  $45^\circ$  angle, and covering the bottom of length 5m, will also be 5 m high. This will require the wall to be of 5 m height, which is just 2m for the given vessel. Instead if we think it other way round to keep in contact with the LHS wall, bottom will have to be covered only 2m with the fluid as shown in the diagram,



$$\text{Fluid Inside} = (1/2) \times 2 \times 2 \times d \text{ m}^3$$

$$\text{Remain inside} = 2d \text{ m}^3$$

$$\therefore \text{Thus volume of fluid gone Outside} = 3d \text{ m}^3$$

### Pascal's Principle

When you squeeze one end of a tube to get toothpaste out the other end, you are watching **Pascal's principle** in action. This principle is also the basis for the Heimlich maneuver, in which a sharp pressure increase properly applied to the abdomen is transmitted to the throat, forcefully ejecting food lodged there. The principle was first stated clearly in 1652 by Blaise Pascal (for whom the unit of pressure is named):

A change in the pressure applied to an enclosed incompressible fluid is transmitted undiminished to every portion of the fluid and to the walls of its container.

## Demonstrating Pascal's Principle

Consider the case in which the incompressible fluid is a liquid contained in a tall cylinder, as in Fig. The cylinder is fitted with a piston on which a container of lead shot rests. The atmosphere, container, and shot exert pressure  $p_{\text{ext}}$  on the piston and thus on the liquid. The pressure  $p$  at any point P in the liquid is then

$$p = p_{\text{ext}} + \rho_{\text{gh}}$$

Let us add a little more lead shot to the container to increase  $p_{\text{ext}}$  by an amount  $\Delta p_{\text{ext}}$ . The quantities  $\Delta p_{\text{ext}}$ ,  $g$  and  $h$  in Eq. are unchanged, so the pressure change at P is

$$\Delta p = \Delta p_{\text{ext}}$$

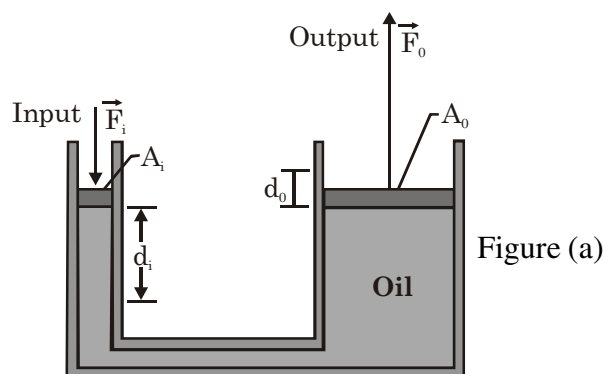
This pressure change is independent of  $h$ , so it must hold for all points within the liquid, as Pascal's principle states.

## Pascal's Principle and the Hydraulic Lever

Figure shows how Pascal's principle can be made the basis of a hydraulic lever. In operation, let an external force of magnitude  $F_i$  be directed downward on the left-hand (or input) piston, whose surface area is  $A_i$ . An incompressible liquid in the device then produces an upward force of magnitude  $F_o$  on the right-hand (or output) piston, whose surface area is  $A_o$ . To keep the system in equilibrium, there must be a downward force of magnitude  $F_o$  on the output piston from an external load (not shown). The force  $\vec{F}_i$  applied on the left and the downward force  $\vec{F}_o$  from the load on the right produce a change  $\Delta p$  in the pressure of the liquid that is given by

$$\Delta p = \frac{F_i}{A_i} = \frac{F_0}{A_0}$$

$$F_0 = F_1 \frac{A_0}{A_i}$$



Equation shows that the output force  $F_0$  on the load must be greater than the input force  $F_i$  if  $A_0 > A_p$  as is the case in figure.

If we move the input piston downward a distance  $d_i$ , the output piston moves upward a distance  $d_o$ , such that the same volume  $V$  of the incompressible liquid is displaced at both pistons. Then

$$V = A_i d_i = A_0 d_0$$

which we can write as

$$d_0 = d_i \frac{A_i}{A_0}$$

This shows that, if  $A_o > A_i$  (as in Figure), the output piston moves a smaller distance than the input piston moves.

From Eqs. we can write the output work as

$$W = F_o d_o = \left( F_i \frac{A_i}{A_o} \right) \left( d_i \frac{A_i}{A_o} \right) = F_i d_i$$

which shows that the work  $W$  done on the input piston by the applied force is equal to the work  $W$  done by the output piston in lifting the load placed on it.

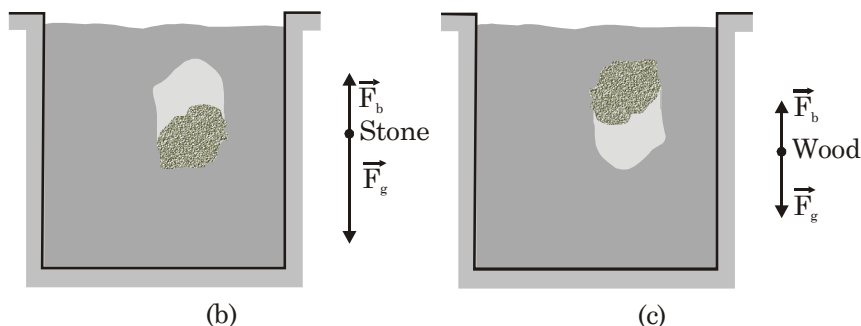
The advantage of a hydraulic lever is this:

With a hydraulic lever, a given force applied over a given distance can be transformed to a greater force applied over a smaller distance.

The product of force and distance remains unchanged so that the same work is done. However, there is often tremendous advantage in being able to exert the larger force. Most of us, for example, cannot lift an automobile directly but can with a hydraulic jack, even though we have to pump the handle farther than the automobile rises and in a series of small strokes.

### Archimedes' Principle

Figure shows a student in a swimming pool, manipulating a very thin plastic sack (of negligible mass) that is filled with water. She finds that the sack and its contained water are in static equilibrium, tending neither to rise nor to sink. The downward gravitational force  $\vec{F}_g$  on the contained water must be balanced by a net upward force from the water surrounding the sack.



This net upward force is a **buoyant force**  $\vec{F}_b$ . It exists because the pressure in the surrounding water increases with depth below the surface. Thus, the pressure near the bottom of the sack is greater than the pressure near the top, which means the forces on the sack due to this pressure are greater in magnitude near the bottom of the sack than near the top. Some of the forces are represented in figure (a) where the space occupied by the sack has been left empty. Note that the force vectors drawn near the bottom of that space (with upward components) have longer lengths than those drawn near the top of the sack (with downward components). If we vectorially add all the forces on the sack from the water, the horizontal components cancel and the vertical components add to yield the upward buoyant force  $\vec{F}_b$  on the sack. (Force  $\vec{F}_b$  is shown to the right of the pool in Figure (a))

Because the sack of water is in static equilibrium, the magnitude  $\vec{F}_b$  of is equal to the magnitude  $m_f g$  of the gravitational force  $\vec{F}_g$  on the sack of water:  $F_b = m_f g$ . (Subscript  $f$  refers to fluid, here the water.) In words, the magnitude of the buoyant force is equal to the weight of the water in the sack.



In Fig. (b), we have replaced the sack of water with a stone that exactly fills the hole in Fig. (a). The stone is said to displace the water, meaning that the stone occupies space that would otherwise be occupied by water. We have changed nothing about the shape of the hole, so the forces at the hole's surface must be the same as when the water-filled sack was in place. Thus, the same upward buoyant force that acted on the water-filled sack now acts on the stone; that is, the magnitude  $\vec{F}_b$  of the buoyant force is equal to  $m_f g$ , the weight of the water displaced by the stone.

Unlike the water-filled sack, the stone is not in static equilibrium. The downward gravitational force  $\vec{F}_g$  on the stone is greater in magnitude than the upward buoyant force, as is shown in the free-body diagram in Fig. (b). The stone thus accelerates downward, sinking to the bottom of the pool. Let us next exactly fill the hole in Fig. (a) with a block of lightweight wood, as in Fig. (c). Again, nothing has changed about the forces at the hole's surface, so the magnitude  $\vec{F}_b$  of the buoyant force is still equal to  $m_f g$ , the weight of the displaced water. Like the stone, the block is not in static equilibrium. However, this time the gravitational force  $\vec{F}_g$  is lesser in magnitude than the buoyant force (as shown to the right of the pool), and so the block accelerates upward, rising to the top surface of the water. Our results with the sack, stone, and block apply to all fluids and are summarized in **Archimedes' principle**:

When a body is fully or partially submerged in a fluid, a buoyant force  $\vec{F}_b$  from the surrounding fluid acts on the body. The force is directed upward and has a magnitude equal to the weight  $m_f g$  of the fluid that has been displaced by the body. The buoyant force on a body in a fluid has the magnitude

$$F_b = m_f g \text{ (buoyant force)}$$

where  $m_f$  is the mass of the fluid that is displaced by the body.

## Floating

When we release a block of lightweight wood just above the water in a pool, the block moves into the water because the gravitational force on it pulls it downward.

As the block displaces more and more water, the magnitude  $F_b$  of the upward buoyant force acting on it increases. Eventually,  $F_b$  is large enough to equal the magnitude  $F_g$  of the downward gravitational force on the block, and the block comes to rest. The block is then in static equilibrium and is said to be floating in the water. In general, When a body floats in a fluid, the magnitude  $F_b$  of the buoyant force on the body is equal to the magnitude  $F_g$  of the gravitational force on the body.

We can write this statement as

$$F_b = F_g \text{ (Floating)}$$

From Eq. we know that  $F_b = m_f g$ . Thus,

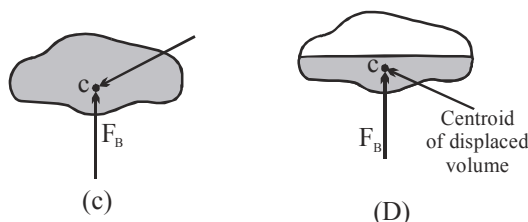
When a body floats in a fluid, the magnitude  $F_g$  of the gravitational force on the body is equal to the weight  $m_f g$  of the fluid that has been displaced by the body.

We can write this statement as

$$F_g = m_f g$$

In other words, a floating body displaces its own weight of fluid.

The location of the line of action of the buoyant force can be determined by adding torques of the forces due to pressure forces, with respect to some convenient axis. The buoyant force must pass through the center of mass of the displaced volume, as shown in Fig. (c), as it was in translational and rotational equilibrium. The point through which the buoyant force acts is called the center of buoyancy.



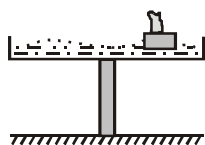
These same results apply to floating bodies which are only partially submerged, as shown in Fig.(d), if the density of the fluid above the liquid surface is very small compared with the liquid in which the body floats. Since the fluid above the surface is usually air, for practical purposes this condition is satisfied.

In the above discussion, the fluid is assumed to have a constant density. If a body is immersed in a fluid in which density varies with depth, such as having multiple layers of fluid, the magnitude of the buoyant force remains equal to the weight of the displaced fluid and the buoyant force passes through the center of mass of the displaced volume.

- Q.** A wooden block floats vertically in a glass filled with water. How will the level of the water in the glass change if the block is kept in a horizontal position ?

**Solution :** The level of the water will not change because the quantity of water displaced will remain the same.

- Q.** A vessel filled with water is placed exactly in middle of a thin wall (fig. ). Will the system topple if a small wooden boat carrying some weight is floated in the vessel?



**Solution :** The system will not topple, since according to Pascal's law the pressure on the bottom of the vessel will be the same everywhere thus the body will still remain in rotational equilibrium

- Ex.** A homogeneous piece of ice floats in a glass filled with water. How will the level of the water in the glass change when the ice melts ?

**Solution :** Since the piece of ice floats, the weight of the water displaced by it is equal to the weight of the ice itself or the weight of the water it produces upon melting. For this reason the water formed by the piece of ice will occupy a volume equal to that of the submerged portion, and the level of the water will not change.

- Q.** A piece of ice is floating in a tub filled with water. How will the level of the water in the tub change when the ice melts ? Consider the following cases :

- (1) a stone is frozen in the ice
- (2) the ice contains an air bubble

**Solution :**

- (1) The volume of the submerged portion of the piece with the stone is greater than the sum of the volumes of the stone and the water produced by the melting ice. Therefore, the level of the water in the glass will drop.
- (2) The weight of the displaced water is equal to that of the ice (the weight of the air in the bubble may be neglected). For this reason, as in conceptual eg., the level of the water will not change.

### Example :

A vessel with a body floating in it is kept in elevator accelerating downwards with acceleration  $a$  such that  $a < g$ . Will the body rise or sink further in the vessel?

**Solution :** The force of bouyancy on the body can be written as  $F = \rho V_2(g - a)$ , where  $V_2$  is the volume of the submerged portion of the body in the lift. As pressure at a point  $h$  below the surface will become  $\rho(g - a)h$  instead of  $\rho gh$ . Applying the Newton's second Law, remembering that the body was accelerating upwards at  $a$ .

$$Mg - \rho V_2 (g - a) = Ma$$

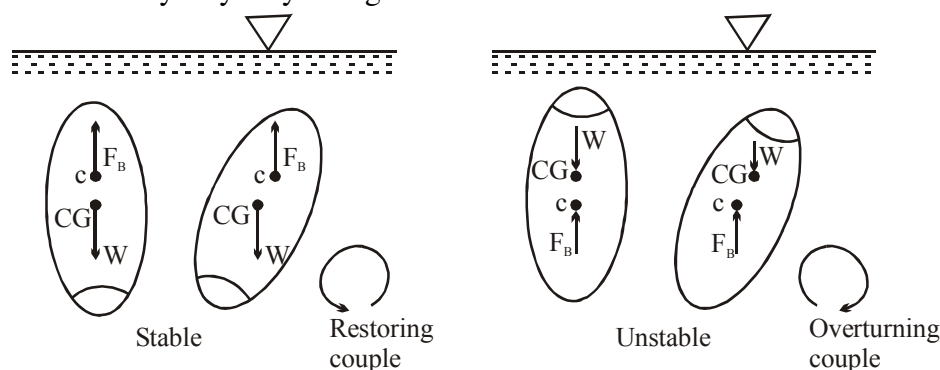
Hence,  $V_2 = \frac{M}{\rho}$  thus  $V_2 = V$ , as in a stationary vessel,  $V = \frac{M}{\rho}$ . Thus the body does not rise to the surface.

### Stability :

The center of buoyancy and center of gravity do not necessarily coincide so the floating or submerged body may not be in stable equilibrium. A small rotation can cause the buoyant force to produce either a restoring or overturning torque. For example, for the completely submerged body shown in Fig., which has a center of gravity below the center of buoyancy, a rotation from its equilibrium position will create a restoring torque by the buoyant force,  $F_B$ , which causes the body to rotate back to its original position. Thus, if the center of gravity falls below the center of buoyancy, the body is stable.

However, as shown in Fig., if the center of gravity of the completely submerged body is above the center of buoyancy, the resulting torque formed by the weight and the buoyant force will cause the body to overturn and move to a new equilibrium position. Thus, a completely submerged body with its center of gravity above its center of buoyancy is in an unstable equilibrium position.

For floating bodies the stability problem is more complicated, since as the body rotates the location of the center of buoyancy may change.



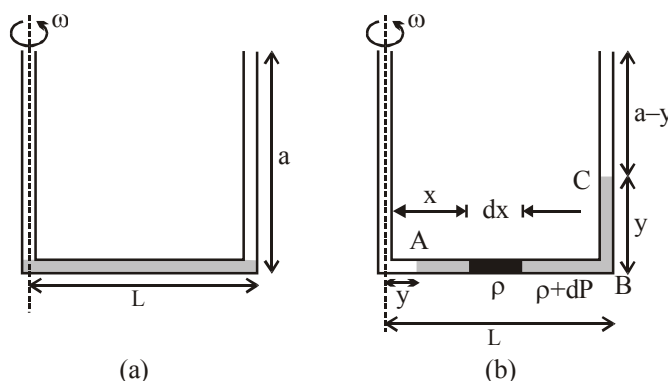
**Concept :** Length of a horizontal arm of a U-tube is  $L$  and ends of both the vertical arms are open to atmospheric pressure  $P_0$ . A liquid of density  $\rho$  is poured in the tube such that liquid just fills the horizontal part of the tube as shown in figure. Now one end of the opened ends is sealed and the tube is then rotated about a vertical axis passing through the other vertical arm with angular speed  $\omega$ . If length of each vertical arm is  $a$  and in the sealed end liquid rises to a height  $y$ , find pressure in the sealed tube during rotation.

The pressure difference across an element of width  $dx$ , which is given as

$$dP = dx\rho\omega^2x$$

Now integrating from A to B, we get

$$P_B - P_A = \int_y^L \rho\omega^2x dx$$



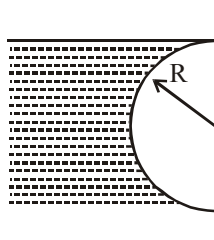
Thus pressure at point C can be given as

$$P_C = P_B - y\rho g$$

and at point A, pressure is atmospheric, thus we have

$$P_C = \frac{\rho\omega^2}{2}(L^2 - y^2) + P_A - y\rho g$$

- Q.** A hemisphere of radius  $R$  is just submerged is just sinking in water of density  $\rho$ . Find the
- horizontal thrust.
  - vertical thrust.
  - total hydrostatic force.
  - angle of orientation of total hydrostatic force acting on the hemisphere.
- Do not count atmospheric pressure

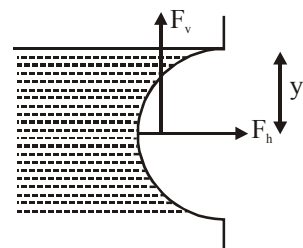


- Sol.** (a) Let the horizontal and vertical thrusts on the hemisphere be  $F_h$  and  $F_v$  respectively  
We know that

$$F_h = \rho g y_c A_y$$

where

$$y_c = R$$



and  $A_v = \pi R^2$

This gives  $F_h = \rho g \pi R^3$  (right)

(b) Similarly using the formula  $F_v = \rho Vg$

Where V = volume of the hemisphere =  $\frac{2}{3}\pi R^3$ ,

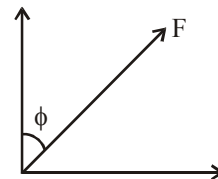
$$\text{we have } F_v = \frac{2}{3} \rho g \pi R^3 \text{ (up)}$$

(c) Hence the net hydrostatic force on the hemisphere is

$$F = \sqrt{F_h^2 + F_v^2}$$

$$F = \sqrt{(\rho g \pi R^3)^2 + \left(\frac{2}{3} \rho g \pi R^3\right)^2}$$

$$= \frac{\sqrt{13}}{3} \rho g \pi R^3$$



(d) The angle of orientation of the force  $F$  is

$$\phi = \tan^{-1} \frac{F_h}{F_v}$$

$$= \tan^{-1} \frac{\rho g \pi R^3}{\frac{2}{3} \rho g \pi R^3} = \tan^{-1} \frac{3}{2}$$

## FLUID DYNAMICS & VISCOSITY

## Ideal Fluids in Motion

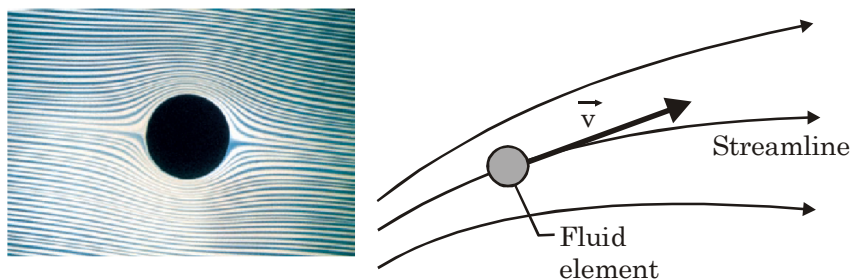
The motion of real fluids is very complicated and not yet fully understood. Instead, we shall discuss the motion of an **ideal fluid**, which is simpler to handle mathematically and yet provides useful results. Here are four assumptions that we make about our ideal fluid; they all are concerned with flow:

1. **Steady flow** In steady (or laminar) flow, the velocity of the moving fluid at any fixed point does not change with time, either in magnitude or in direction. The gentle flow of water near the center of a quiet stream is steady; the flow in a chain of rapids is not. Figure shows a transition from steady flow to nonsteady (or nonlaminar or turbulent) flow for a rising stream of smoke. The speed of the smoke particles increases as they rise and, at a certain critical speed, the flow changes from steady to nonsteady.
2. **Incompressible flow** We assume, as for fluids at rest, that our ideal fluid is incompressible; that is, its density has a constant, uniform value.

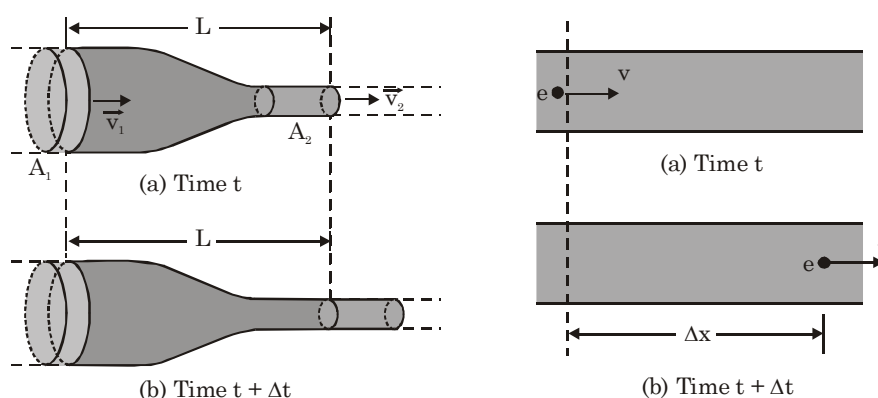
3. **Nonviscous flow** Roughly speaking, the viscosity of a fluid is a measure of how resistive the fluid is to flow. For example, thick honey is more resistive to flow than water, and so honey is said to be more viscous than water. Viscosity is the fluid analog of friction between solids; both are mechanisms by which the kinetic energy of moving objects can be transferred to thermal energy. In the absence of friction, a block could glide at constant speed along a horizontal surface. In the same way, an object moving through a nonviscous fluid would experience no viscous drag force—that is, no resistive force due to viscosity; it could move at constant speed through the fluid.
4. **Irrotational flow** : Although it need not concern us further, we also assume that the flow is irrotational. To test for this property, let a tiny grain of dust move with the fluid. Although this test body may (or may not) move in a circular path, in irrotational flow the test body will not rotate about an axis through its own center of mass. For a loose analogy, the motion of a Ferris wheel is rotational; that of its passengers is irrotational. That the velocity of a particle is always tangent to the path taken by the particle. Here the particle is the fluid element, and its velocity is  $\vec{v}$  always tangent to a streamline (Figure). For this reason, two streamlines can never intersect; if they did, then an element arriving at their intersection would have two different velocities simultaneously—an impossibility.

### The Equation of Continuity

You may have noticed that you can increase the speed of the water emerging from a garden hose by partially closing the hose opening with your thumb. Apparently the speed  $v$  of the water depends on the cross-sectional area  $A$  through which the water flows.



Here we wish to derive an expression that relates  $v$  and  $A$  for the steady flow of an ideal fluid through a tube with varying cross section, like that in Figure. The flow there is toward the right, and the tube segment shown (part of a longer tube) has length  $L$ . The fluid has speeds  $v_1$  at the left end of the segment and  $v_2$  at the right end. The tube has cross-sectional areas  $A_1$  at the left end and  $A_2$  at the right end. Suppose that in a time interval  $\Delta t$  a volume  $\Delta V$  of fluid enters the tube segment at its left end (that volume is colored purple in Figure). Then, because the fluid is incompressible, an identical volume  $\Delta V$  must emerge from the right end of the segment (it is colored green in Figure). We can use this common volume  $\Delta V$  to relate the speeds and areas. To do so, we first consider Fig. , which shows a side view of a tube of uniform crosssectional area  $A$ . In Fig.(a), a fluid element  $e$  is about to pass through the dashed line drawn across the tube width. The element's speed is  $v$ , so during a time interval  $\Delta t$ , the element moves along the tube a distance  $\Delta x = v \Delta t$ . The volume  $\Delta V$  of fluid that has passed through the dashed line in that time interval  $\Delta t$  is



$$\Delta V = A \Delta x = A v \Delta t.$$

Applying Eq. to both the left and right ends of the tube segment in Fig., we have

$$\Delta V = A_1 v_1 \Delta t = A_2 v_2 \Delta t$$

Or  $A_1 v_1 = A_2 v_2$  (equation of continuity)

This relation between speed and cross-sectional area is called the **equation of continuity** for the flow of an ideal fluid. It tells us that the flow speed increases when we decrease the cross-sectional area through which the fluid flows (as when we partially close off a garden hose with a thumb). Equation applies not only to an actual tube but also to any so-called tube of flow, or imaginary tube whose boundary consists of streamlines. Such a tube acts like a real tube because no fluid element can cross a streamline; thus, all the fluid within a tube of flow must remain within its boundary. Figure shows a tube of flow in which the cross-sectional area increases from area  $A_1$  to area  $A_2$  along the flow direction. From Eq. we know that, with the increase in area, the speed must decrease, as is indicated by the greater spacing between streamlines at the right in Fig. . Similarly, you can see that in Fig. the speed of the flow is greatest just above and just below the cylinder. We can rewrite Eq. as

$R_v = A_v =$  a constant (volume flow rate, equation of continuity), in which  $R_v$  is the **volume flow rate** of the fluid (volume past a given point per unit time). Its SI unit is the cubic meter per second ( $\text{m}^3/\text{s}$ ). If the density  $\rho$  of the fluid is uniform, we can multiply Eq. by that density to get the **mass flow rate**  $R_m$  (mass per unit time):

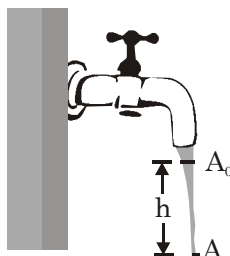
$$R_m = \rho R_v = A_v \rho = \text{a constant (mass flow rate)}.$$

The SI unit of mass flow rate is the kilogram per second ( $\text{kg/s}$ ). Equation says that the mass that flows into the tube segment of Fig. each second must be equal to the mass that flows out of that segment each second.

**Sample Problem**

Figure shows how the stream of water emerging from a faucet “necks down” as it falls. The indicated cross-sectional areas are  $A_0 = 1.2 \text{ cm}^2$  and  $A = 0.35 \text{ cm}^2$ . The two levels are separated by a vertical distance  $h = 45 \text{ mm}$ . What is the volume flow rate from the tap?

The volume flow rate through the higher cross section must be the same as that through the lower cross section.



where  $v_0$  and  $v$  are the water speeds at the levels corresponding to  $A_0$  and  $A$ . From Eq. we can also write, because the water is falling freely with acceleration  $g$ ,

$$v^2 = v_0^2 + 2gh.$$

Eliminating  $v$  between Eqs. and solving for  $v_0$ , we obtain

$$v_0 = \sqrt{\frac{2ghA^2}{A_0^2 - A^2}}$$

$$v_0 = \sqrt{\frac{(2) \times (9.8 \text{ m/s}^2)(0.045 \text{ m})(0.35 \text{ cm}^2)^2}{(1.2 \text{ cm}^2)^2 - (0.35 \text{ cm}^2)^2}}$$

$$v_0 = 0.286 \text{ m/s} = 28.6 \text{ cm/s}.$$

From Eq. , the volume flow rate  $R_V$  is then

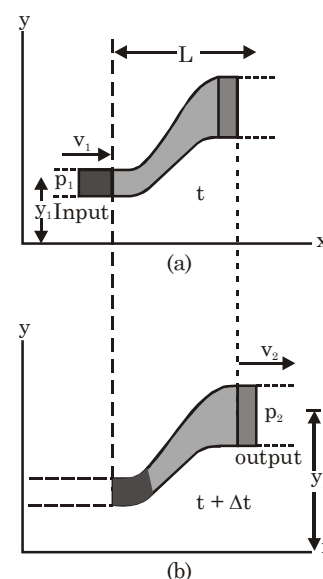
$$\begin{aligned} R_V &= A_0 v_0 = (1.2 \text{ cm}^2)(28.6 \text{ cm/s}) \\ &= 34 \text{ cm}^3/\text{s}. \text{ Ans. } \end{aligned}$$

**Bernoulli's Equation**

Figure represents a tube through which an ideal fluid is flowing at a steady rate. In a time interval  $\Delta t$ , suppose that a volume of fluid  $\Delta V$ , colored purple in Fig. , enters the tube at the left (or input) end and an identical volume, colored green in Fig. , emerges at the right (or output) end. The emerging volume must be the same as the entering volume because the fluid is incompressible, with an assumed constant density  $\rho$ .

$$p_1 + \frac{1}{2}\rho v_1^2 + \rho g y_1 = p_2 + \frac{1}{2}\rho v_2^2 + \rho g y_2$$

Let  $y_1$ ,  $v_1$ , and  $p_1$  be the elevation, speed, and pressure of the fluid entering at the left, and  $y_2$ ,  $v_2$ , and  $p_2$  be the corresponding quantities for the fluid emerging at the right. By applying the principle of conservation of energy to the fluid, we shall show that these quantities are related by





In general, the term  $\frac{1}{2}\rho v^2$  is called the fluid's **kinetic energy**

**density** (kinetic energy per unit volume). We can also write Eq. as

$$p + \frac{1}{2}\rho v^2 + \rho gy \text{ a constant (Bernoulli's equation).}$$

Equations are equivalent forms of **Bernoulli's equation**, after Daniel Bernoulli, who studied fluid flow in the 1700s.\* Like the equation of continuity Eq., Bernoulli's equation is not a new principle but simply the reformulation of a familiar principle in a form more suitable to fluid mechanics. As a check, let us apply Bernoulli's equation to fluids at rest, by putting  $v_1 = v_2 = 0$  in Eq. The result is

$$p_2 = p_1 + \rho g(y_1 - y_2)$$

Which is equation.

A major prediction of Bernoulli's equation emerges if we take  $y$  to be a constant ( $y = 0$ , say) so that the fluid does not change elevation as it flows. Equation then becomes

$$p_1 + \frac{1}{2}\rho v_1^2 = p_2 + \frac{1}{2}\rho v_2^2$$

Which tells us that :

If the speed of a fluid element increases as the element travels along a horizontal streamline, the pressure of the fluid must decrease, and conversely.

Put another way, where the streamlines are relatively close together (where the velocity is relatively great), the pressure is relatively low, and conversely. The link between a change in speed and a change in pressure makes sense if you consider a fluid element. When the element nears a narrow region, the higher pressure behind it accelerates it so that it then has a greater speed in the narrow region. When it nears a wide region, the higher pressure ahead of it decelerates it so that it then has a lesser speed in the wide region. Bernoulli's equation is strictly valid only to the extent that the fluid is ideal. If viscous forces are present, thermal energy will be involved. We take no account of this in the derivation that follows.

### Proof of Bernoulli's Equation

Let us take as our system the entire volume of the (ideal) fluid shown in Fig. . We shall apply the principle of conservation of energy to this system as it moves from its initial state (Fig. (a)) to its final state (Fig. (b)). The fluid lying between the two vertical planes separated by a distance  $L$  in Fig. does not change its properties during this process; we need be concerned only with changes that take place at the input and output ends. First, we apply energy conservation in the form of the work–kinetic energy theorem,

$$W = \Delta K,$$

which tells us that the change in the kinetic energy of our system must equal the net work done on the system. The change in kinetic energy results from the change in speed between the ends of the tube and is

$$\Delta K = \frac{1}{2}\Delta m v_2^2 - \frac{1}{2}\Delta m v_1^2 = \frac{1}{2}\rho \Delta V (v_2^2 - v_1^2)$$

in which  $\Delta m (= \rho \Delta V)$  is the mass of the fluid that enters at the input end and leaves at the output end during a small time interval  $\Delta t$ .

The work done on the system arises from two sources. The work  $W_g$  done by the gravitational force ( $\Delta m \vec{g}$ ) on the fluid of mass  $\Delta m$  during the vertical lift of the mass from the input level to the output level is

$$W_g = \Delta m g(y_2 - y_1) \\ = -\rho g \Delta V(y_2 - y_1)$$

This work is negative because the upward displacement and the downward gravitational force have opposite directions.

Work must also be done on the system (at the input end) to push the entering fluid into the tube and by the system (at the output end) to push forward the fluid that is located ahead of the emerging fluid. In general, the work done by a force of magnitude  $F$ , acting on a fluid sample contained in a tube of area  $A$  to move the fluid through a distance  $\Delta x$ , is

$$F \Delta x = (pA)(\Delta x) = p(A \Delta x) = p \Delta V$$

The work done on the system is then  $p_1 \Delta V$ , and the work done by the system is  $-p_2 \Delta V$ . Their sum  $W_p$  is  $W_p$

$$W_p = -p_2 \Delta V - p_1 \Delta V \\ = -(p_2 - p_1) \Delta V.$$

The work-kinetic energy theorem of Eq. now becomes

$$W = W_g + W_p = \Delta K.$$

Substituting from Eqs. yields

$$-\rho g \Delta V(y_2 - y_1) - \Delta V(p_2 - p_1) = \frac{1}{2} \rho \Delta V(v_2^2 - v_1^2)$$

This, after a slight rearrangement, matches Eq. , which we set out to prove.

**Sample Problem :** In the old West, a desperado fires a bullet into an open water tank (Fig. ), creating a hole a distance  $h$  below the water surface. What is the speed  $v$  of the water exiting the tank?

**Sol.** From Eq.  $R_v = av = Av_0$  and thus

$$v_0 = \frac{a}{A} v$$

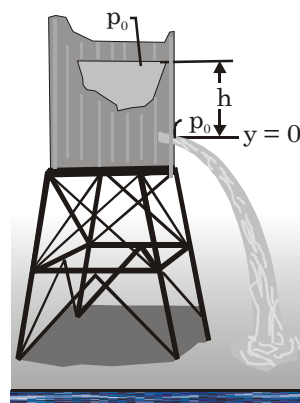
Because  $a \ll A$ , we see that  $v_0 \ll v$ . To apply Bernoulli's equation, we take the level of the hole as our reference level for measuring elevations (and thus gravitational potential energy). Noting that the pressure at the top of the tank and at the bullet hole is the atmospheric pressure  $p_0$  (because both places are exposed to the atmosphere), we write Eq. as

$$p_0 + \frac{1}{2} \rho v_0^2 + \rho gh = p_0 + \frac{1}{2} \rho v^2 + \rho g(0)$$

(Here the top of the tank is represented by the left side of the equation and the hole by the right side. The zero on the right indicates that the hole is at our reference level.) Before we solve Eq. for  $v$ , we can use our result that  $v_0 \ll v$  to simplify it: We assume that  $v_0^2$  and thus the

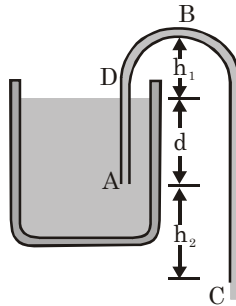
term  $\frac{1}{2} \rho v_0^2$  in Eq. , is negligible relative to the other terms, and we drop it. Solving the remaining equation for  $v$  then yields

$$v = \sqrt{2gh} \text{ Ans.}$$



This is the same speed that an object would have when falling a height  $h$  from rest. This is because the work done by atmospheric pressure is cancelling out at open surface and the hole.

**Sample Problem :** shows a siphon, which is a device for removing liquid from a container. Tube ABC must initially be filled, but once this has been done, liquid will flow through the tube until the liquid surface in the container is level with the tube opening at A. The liquid has density  $1000 \text{ kg/m}^3$  and negligible viscosity. The distances shown are  $h_1 = 25 \text{ cm}$ ,  $d = 12 \text{ cm}$ , and  $h_2 = 40 \text{ cm}$ . (a) With what speed does the liquid emerge from the tube at C? (b) If the atmospheric pressure is  $1.0 \times 10^5 \text{ Pa}$ , what is the pressure in the liquid at the topmost point B? (c) Theoretically, what is the greatest possible height  $h_1$  that a siphon can lift water?



**Solution:** You may have used siphon and you may recollect that lower the exit point of the fluid is, faster the fluid flows out.

We consider a point D on the surface of the liquid in the container, in the same tube of flow with points A, B and C. Applying Bernoulli's equation to points D and C, we obtain

$$p_D + \frac{1}{2}\rho v_D^2 + \rho gh_D = p_C + \frac{1}{2}\rho v_C^2 + \rho gh_C$$

$$v_C = \sqrt{\frac{2(p_D - p_C)}{\rho} + 2g(h_D - h_C) + v_D^2}$$

$$\approx \sqrt{2g(d + h_2)}$$

where in the last step we set  $p_D = p_C = p_{\text{air}}$  and  $v_D/v_C \approx 0$ . Plugging in the values, we obtain

$$v_c = \sqrt{2(9.8\text{m/s}^2)(0.40\text{m} + 0.12\text{m})} = 3.2\text{ m/s}$$

The result confirms our experience.

We now consider points B and C:

$$p_B + \frac{1}{2} \rho v_B^2 + \rho g h_B = p_C + \frac{1}{2} \rho v_C^2 + \rho g h_C$$

Since  $v_B = v_C$  by equation of continuity, and  $p_C = p_{air}$ , Bernoulli's equation becomes

$$\begin{aligned} p_B &= p_C + \rho g(h_C - h_g) = p_{\text{air}} - \rho g(h_1 + h_2 + d) \\ &= 1.0 \times 10^5 \text{ Pa} - (1.0 \times 10^3 \text{ kg/m}^3)(9.8 \text{ m/s}^2)(0.25 \text{ m} + 0.40 \text{ m} + 0.12 \text{ m}) \\ &= 9.2 \times 10^4 \text{ Pa.} \end{aligned}$$

Since  $p_B \geq 0$ , we must let  $p_{air} - \rho g (h_1 + d + h_2) \geq 0$ , which yields

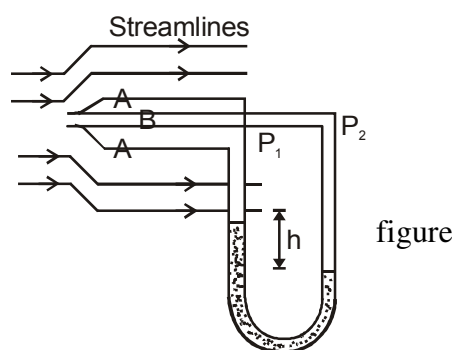
$$h_1 \leq h_{1\max} = \frac{p_{\text{air}}}{\rho} - d - h_2 \leq \frac{p_{\text{air}}}{\rho} = 10.3\text{m}$$

**Q.** In the example just above consider a small hole in the hose at location (1) as indicated. When the siphon is used, will water leak out of the hose, or will air leak into the hose and thus causing the siphon to stop ?

**Solution :** Whether air will leak out of the hose depends on whether the pressure within the hose at (1) is less than or greater than atmospheric.

Since the hose diameter is constant, it follows from the continuity equation ( $AV = \text{constant}$ ) that the water velocity in the hose is constant throughout. Also the pressure at the end of the hose is atmospheric so the pressure at all the points above it can be shown to be less than the atmospheric pressure. Thus air will leak into the hose stopping it.

**Example:** Fig. shows a device called pitot's tube. It measures the velocity of moving fluids. Determine the velocity of the fluid in terms of the density  $\rho$ , the density of the fluid in manometer (U-tube)  $\sigma$  and the height 'h'.



**Solution.**

The difference in the two tubes is that liquid will flow into the tube B with full Kinetic Energy while it will just pass over the tube A without directly entering into it.

This problem is based on the use of Bernoulli's principle, on two different situations.

The fluid inside the right tube must be at rest as the fluid exactly at the end is in contact with the fluid in pitot tube, which is at rest.

The velocity  $v_1$  is the fluid velocity  $v_f$ . The velocity  $v_2$  of the fluid at point B is zero and the pressure in the right arm is  $P_2$  (called stagnation pressure).

Thus using Bernoulli's principle 
$$P_1 + \frac{1}{2} \rho v_1^2 = P_2 + \frac{1}{2} \rho v_2^2$$

We get 
$$P_2 = P_1 + \frac{1}{2} \rho v_f^2$$

On the other hand the openings at point A is not along the flow lines, so we don't need to use Bernoulli's eqn. We can simply say that the pressure just outside the opening is same as that within the pitot tube.

Therefore the pressure at the left arm of the manometer is same as the fluid pressure  $P_f$ , i.e.,  $P_1 = P_f$ .

Also 
$$P_2 = P_1 + (\rho - \sigma) gh \quad \dots(3)$$

Generally  $\sigma \ll \rho$ , so it is ignored.

Thus 
$$P_2 = P_f + \rho gh \quad \dots(4)$$

From eqns. (4) and (3), 
$$\frac{1}{2} \sigma v_f^2 = \rho gh$$

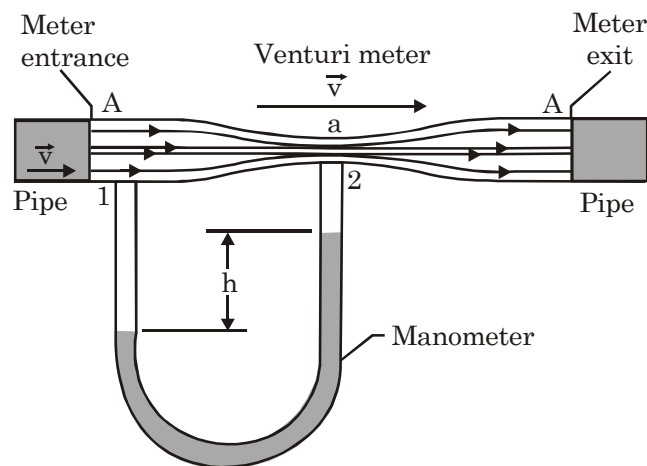
or 
$$v_f = \sqrt{\frac{2\rho gh}{\sigma}}$$

Thus we can see that we have measured the fluid velocity as this was the only difference between the two tubes leading to the pressure difference between the tubes.

**Conceptual example:** A venturi meter is used to measure the flow speed of a fluid in a pipe. The meter is connected between two sections A venturi meter is used to measure the flow speed of a fluid in a pipe. The meter is connected between two sections of the pipe (Fig. ); the cross-sectional area  $A$  of the entrance and exit of the meter matches the pipe's cross-sectional area. Between the entrance and exit, the fluid flows from the pipe with speed  $V$  and then through a narrow "throat" of cross-sectional area  $a$  with speed  $v$ . A manometer connects the wider portion of the meter to the narrower portion. The change in the fluid's speed is accompanied by a change  $\Delta p$  in the fluid's pressure, which causes a height difference  $h$  of the liquid in the two arms of the manometer. (Here  $\Delta p$  means pressure in the throat minus pressure in the pipe.) (a) By applying Bernoulli's equation and the equation of continuity to points 1 and 2 in Fig. , show that

$$v = \sqrt{\frac{2a^2\Delta p}{\rho(a^2 - A^2)}}$$

where  $\rho$  is the density of the fluid. (b) Suppose that the fluid is fresh water, that the cross-sectional areas are  $64 \text{ cm}^2$  in the pipe and  $32 \text{ cm}^2$  in the throat, and that the pressure is  $55 \text{ kPa}$  in the pipe and  $41 \text{ kPa}$  in the throat. What is the rate of water flow in cubic meters per second?



- (a) The continuity equation yields  $Av = aV$ , and Bernoulli's equation yields  $\Delta p + \frac{1}{2}\rho v^2 = \frac{1}{2}\rho V^2$  where  $\Delta p = p_1 - p_2$ . The first equation gives  $V = (A/a)v$ . We use this to substitute for  $V$  in the second equation, and obtain  $\Delta p + \frac{1}{2}\rho v^2 = \frac{1}{2}\rho(A/a)^2 v^2$ . We solve for  $v$ . The result is

$$v = \sqrt{\frac{2\Delta p}{\rho((A/a)^2 - 1)}} = \sqrt{\frac{2a^2\Delta p}{\rho(A^2 - a^2)}}$$

- (b) We substitute values to obtain

$$v = \sqrt{\frac{2(32 \times 10^{-4} \text{ m}^2)^2(55 \times 10^3 \text{ Pa} - 41 \times 10^3 \text{ Pa})}{(1000 \text{ kg/m}^3)((64 \times 10^{-4} \text{ m}^2)^2 - (32 \times 10^{-4} \text{ m}^2)^2)}} = 30.06 \text{ m/s}$$

Consequently, the flow rate is

$$Av = (64 \times 10^{-4} \text{ m}^2)(30.06 \text{ m/s}) = 2.0 \times 10^{-2} \text{ m}^3/\text{s}.$$

# SURFACE TENSION & VISCOSITY

## EXPLANATION OF SOME OBSERVED PHENOMENA

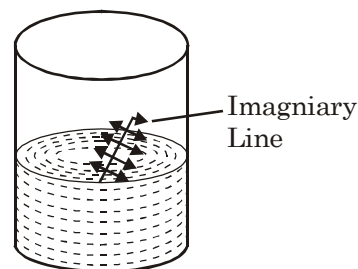
1. Lead balls are spherical in shape.
2. Rain drops and a globule of mercury placed on glass plate are spherical.
3. Hair of a shaving brush/painting brush, when dipped in water spread out, but as soon as it is taken out. Its hair stick together.
4. A greased needle placed gently on the free surface of water in a beaker does not sink.
5. Similarly, insects can walk on the free surface of water without drowning.
6. Bits of Camphor gum move irregularly when placed on water surface.

## SURFACE TENSION

Surface Tension is a property of liquid at rest by virtue of which a liquid surface gets contracted to a minimum area and behaves like a stretched membrane.

Surface Tension of a liquid is measured by force per unit length on either side of any imaginary line drawn tangentially over the liquid surface, force being normal to the imaginary line as shown in fig. i.e. Surface tension.

$$(T) = \frac{\text{Total force on either of the imaginary line (F)}}{\text{Length of the line (l)}}$$



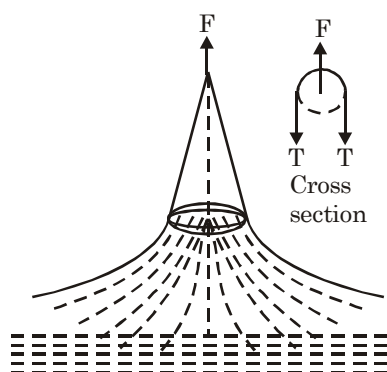
### Unit of Surface Tension

In C. G. S. system the unit of surface tension is dyne/cm ( $\text{dyne cm}^{-1}$ ) and SI system its units is  $\text{Nm}^{-1}$

### Solved Examples

**Ex** A ring is cut from a platinum tube of 8.5 cm internal and 8.7 cm external diameter. It is supported horizontally from a pan of a balance so that it comes in contact with the water in a glass vessel. What is the surface tension of water if an extra 3.97 g weight is required to pull it away from water? ( $g = 980 \text{ cm/s}^2$ ).

**Sol.**



The ring is in contact with water along its inner and outer circumference ; so when pulled out the total force on it due to surface tension will be

$$F = T(2\pi r_1 + 2\pi r_2)$$

$$\text{So, } T = \frac{mg}{2\pi(r_1 + r_2)} \quad [\because F = mg]$$

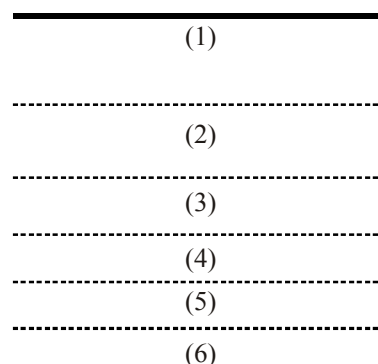
$$\text{i.e., } T = \frac{3.97 \times 980}{3.14 \times (8.5 + 8.7)} = 72.13 \text{ dyne/cm}$$

### Surface energy

The course of reasoning given below is usually followed to prove that the molecules of the surface layer of a liquid have surplus potential energy. A molecule inside the liquid is acted upon by the forces of attraction from the other molecules which compensate each other on the average. If a molecule is singled out on the surface, the resulting force of attraction from the other molecule is directed into the liquid. For this reason the molecule tends to move into the liquid, and definite work should be done to bring it to the surface. Therefore, each molecule of the surface layer has excess potential energy equal to this work. The average force that acts on any molecule from the side of all the others, however, is always equal to zero if the liquid is in equilibrium. This is why the work done to move the liquid from a depth to the surface should also be zero. What is the origin, in this case, of the surface energy ?

[Sol. The forces of attraction acting on a molecule in the surface layer from all the other molecules produce a resultant directed downward. The closest neighbours, however, exert a force of repulsion on the molecule which is therefore in equilibrium.

Owing to the forces of attraction and repulsion, the density of the liquid is smaller in the surface layer than inside. Indeed, molecule 1 (figure) is acted upon by the force of repulsion from molecule 2 and the forces of attraction from all the other molecules (3, 4, .....). Molecule 2 is acted upon by the forces of repulsion from 3 and 1 and the forces of attraction from the molecules in the deep layers. As a result, distance 1-2 should be greater than 2-3, etc.



This course of reasoning is quite approximate (thermal motion, etc. is disregarded), but nevertheless it gives a qualitatively correct result.

An increase in the surface of the liquid causes new sections of the rarefied surface layer to appear. Here work should be performed against the forces of attraction between the molecules. It is this work that constitutes the surface energy.

We know that the molecules on the liquid surface experience net downward force. So to bring a molecule from the interior of the liquid to the free surface, some work is required to be done against the intermolecular force of attraction, which will be stored as potential energy of the molecule on the surface. The potential energy of surface molecules per unit area of the surface is called surface energy. Unit of surface energy is  $\text{erg cm}^{-2}$  in C.G.S. system and  $\text{Jm}^{-2}$  in SI system. Dimensional formula of surface energy is  $[\text{ML}^0\text{T}^{-2}]$  surface energy depends on number of surfaces e.g. a liquid drop is having one liquid air surface while bubble is having two liquid air surface.

**Relation between surface tension and surface energy**

Consider a rectangular frame PQRS of wire, whose arm RS can slide on the arms PR and QS. If this frame is dipped in a soap solution, then a soap film is produced in the frame PQRS in fig. Due to surface tension ( $T$ ), the film exerts a force on the frame (towards the interior of the film). Let  $l$  be the length of the arm RS, then the force acting on the arm RS towards the film is  $F = T \times 2l$  [Since soap film has two surfaces, that is why the length is taken twice.]

$\therefore$  work done,  $W = Fx = 2Tlx$

Increase in potential energy of the soap film.

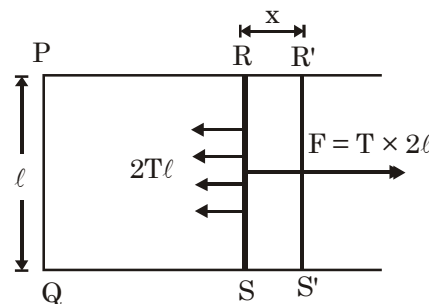
$= EA = 2Elx = \text{work done in increasing the area } (\Delta W)$

where  $E$  = surface energy of the soap film per unit area.

According to the law of conservation of energy, the work done must be equal to the increase in the potential energy.

$$\therefore 2Tlx = 2Elx \text{ or } T = E = \frac{\Delta W}{\Delta A}$$

Thus, surface tension is numerically equal to surface energy or work done per unit increase surface area.



**Ex** A mercury drop of radius 1 cm is sprayed into  $10^6$  droplets of equal size. Calculate the energy expanded if surface tension of mercury is  $35 \times 10^{-3}$  N/m.

**Sol.** If drop of radius  $R$  is sprayed into  $n$  droplets of equal radius  $r$ , then as a drop has only one surface, the initial surface area will be  $4\pi R^2$  while final area is  $n(4\pi r^2)$ . So the increase in area

$$\Delta S = n(4\pi r^2) - 4\pi R^2$$

So energy expended in the process.

$$W = T\Delta S = 4\pi T [nr^2 - R^2] \quad \dots (1)$$

Now since the total volume of  $n$  droplets is the same as that of initial drop, i.e.

$$\frac{4}{3} \pi R^3 = n \left[ \frac{4}{3} \pi r^3 \right] \text{ or } r = R/n^{1/3} \quad \dots (2)$$

Putting the value of  $r$  from equation (2) in (1)

$$W = 4\pi R^2 T (n^{1/3} - 1)$$

**Ex** If a number of little droplets of water, each of radius  $r$ , coalesce to form a single drop of radius  $R$ , show that the rise in temperature will be given by

$$\frac{3T}{J} \left( \frac{1}{r} - \frac{1}{R} \right)$$

where  $T$  is the surface tension of water and  $J$  is the mechanical equivalent of heat.

**Sol.** Let  $n$  be the number of little droplets.

Since volume will remain constant, hence volume of  $n$  little droplets = volume of single drop

$$\therefore n \times \frac{4}{3} \pi r^3 = \frac{4}{3} \pi R^3 \text{ or } nr^3 = R^3$$

Decrease in surface area =  $n \times 4\pi r^2 - 4\pi R^2$

$$\text{or } \Delta A = 4\pi [nr^2 - R^2] = 4\pi \left[ \frac{nr^3}{r} - R^2 \right] = 4\pi \left[ \frac{R^3}{r} - R^2 \right] = 4\pi R^3 \left[ \frac{1}{r} - \frac{1}{R} \right]$$

$$\text{Energy evolved } W = T \times \text{decrease in surface area} = T \times 4\pi R^3 \left[ \frac{1}{r} - \frac{1}{R} \right]$$



$$\text{Heat produced, } Q = \frac{W}{J} = \frac{4\pi TR^3}{J} \left[ \frac{1}{r} - \frac{1}{R} \right] \quad \text{But } Q = ms d\theta$$

where  $m$  is the mass of big drop,  $s$  is the specific heat of water and  $d\theta$  is the rise in temperature.

$$\therefore \frac{4\pi TR^3}{J} \left[ \frac{1}{r} - \frac{1}{R} \right] = \text{volume of big drop} \times \text{density of water} \times \text{sp. heat of water} \times d\theta$$

$$\text{or, } \frac{4}{3} \pi R^3 \times 1 \times 1 \times d\theta = \frac{4\pi TR^3}{J} \left( \frac{1}{r} - \frac{1}{R} \right) \quad \text{or, } d\theta = \frac{3T}{J} \left[ \frac{1}{r} - \frac{1}{R} \right]$$

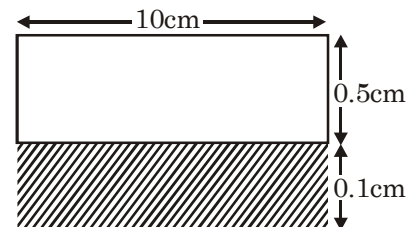
**Ex** A film of water is formed between two straight parallel wires each 10cm long and at a separation 0.5 cm. Calculate the work required to increase 1mm distance between them. Surface tension of water  $72 \times 10^{-3} \text{ N/m}$ .

**Sol.** Here the increase in area is shown by shaded portion in the figure.

Since this a water film, it has two surface, therefore increase in area,  $\Delta S = 2 \times 10 \times 0.1 = 2\text{cm}^2$

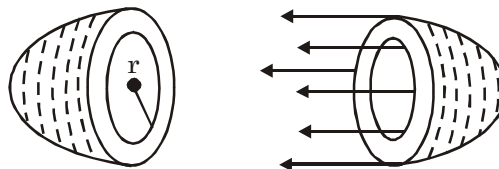
$\therefore$  Work required to be done

$$\begin{aligned} W &= \Delta S \times T \\ &= 2 \times 10^{-4} \times 72 \times 10^{-3} \\ &= 144 \times 10^{-7} \text{ joule} \\ &= 1.44 \times 10^{-5} \text{ joule} \end{aligned}$$



### Excess pressure inside A liquid drop and a bubble

1. Inside a bubble : Consider a soap bubble of radius  $r$ . Let  $p$  be the pressure inside the bubble and  $p_a$  outside. The excess pressure  $= p - p_a$ . Imagine the bubble broken into two halves, and consider one half of it as shown in fig. Since there are two surface, inner and outer, so the force due to surface tension is



$$F = \text{surface tension} \times \text{length} = T \times 2 (\text{circumference of the bubble}) = T \times 2 (2\pi r) \quad \dots (1)$$

The excess pressure  $(p - p_a)$  acts on a cross-sectional area  $\pi r^2$ , so the force due to excess pressure is

$$\Rightarrow F = (p - p_a) \pi r^2 \quad \dots (2)$$

The surface tension force given by equation (1) must balance the force due to excess pressure given by equation (2) to maintain the equilibrium, i.e.  $(p - p_a) \pi r^2 = T \times 2(2\pi r)$

$$\text{or} \quad (p - p_a) = \frac{4T}{r} = p_{\text{excess}}$$

above expression can also be obtained by equation of excess pressure of curve surface by putting  $R_1 = R_2$ .

2. Inside the drop : In a drop, there is only one surface and hence excess pressure can be written as

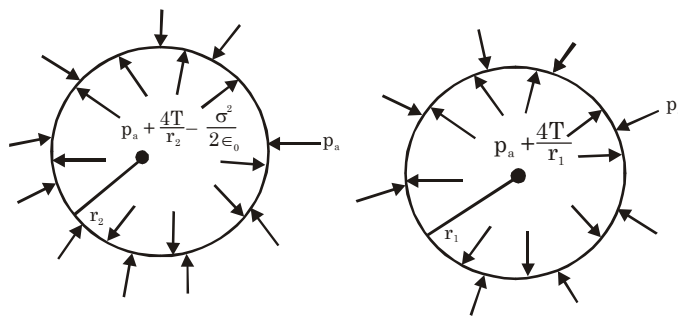
$$(p - p_a) = \frac{2T}{r} = p_{\text{excess}}$$

3. Inside air bubble in a liquid :

$$(p - p_a) = \frac{2T}{r} = p_{\text{excess}}$$

4. A charged bubble : If bubble is charged, it's radius increases. Bubble has pressure excess due to charge too. Initially pressure inside the bubble

$$= p_a + \frac{4T}{r_1}$$



for charge bubble, pressure inside  $= p_a + \frac{4T}{r_2} - \frac{\sigma^2}{2\epsilon_0}$ , where  $\sigma$  surface is surface charge density.

Taking temperature remains constant, then from Boyle's law

$$\left( p_a + \frac{4T}{r_1} \right) \frac{4}{3} \pi r_1^3 = \left[ p_a + \frac{4T}{r_2} - \frac{\sigma^2}{2\epsilon_0} \right] \frac{4}{3} \pi r_2^3$$

From above expression the radius of charged drop may be calculated. It can conclude that radius of charged bubble increases, i.e.  $r_2 > r_1$ .

### Solved example

**Ex** A minute spherical air bubble is rising slowly through a column of mercury contained in a deep jar. If the radius of the bubble at a depth of 100cm is 0.1 mm, calculate its depth where its radius is 0.126 mm, given that the surface tension of mercury is 567 dyne/cm. Assume that the atmospheric pressure is 76cm of mercury.

**Sol.** The total pressure inside the bubble at depth  $h_1$  is ( $P$  atmospheric pressure)

$$= (P + h_1 \rho g) + \frac{2T}{r_1} = P_1$$

and the total pressure inside the bubble at depth  $h_2$  is  $= (P + h_2 \rho g) + \frac{2T}{r_2} = P_2$

Now, according to Boyle's Law

$$P_1 V_1 = P_2 V_2 \quad \text{where } V_1 = \frac{4}{3} \pi r_1^3, \text{ and } V_2 = \frac{4}{3} \pi r_2^3$$

$$\text{Hence we get } \left[ (P + h_1 \rho g) + \frac{2T}{r_1} \right] \frac{4}{3} \pi r_1^3 = \left[ (P + h_2 \rho g) + \frac{2T}{r_2} \right] \frac{4}{3} \pi r_2^3$$

$$\text{or,} \quad \left[ (P + h_1 \rho g) + \frac{2T}{r_1} \right] r_1^3 = \left[ (P + h_2 \rho g) + \frac{2T}{r_2} \right] r_2^3$$

Given that :  $h_1 = 100$  cm,  $r_1 = 0.1$  mm = 0.01 cm,  $r_2 = 0.126$  cm,  $T = 567$  dyne / cm,  $P = 76$  cm of mercury. Substituting all the values, we get

$$h_2 = 9.48 \text{ cm.}$$

## THE FORCE OF COHESION

The force of attraction between the molecules of the same substance is called cohesion.

In case of solids, the force of cohesion is very large and due to this solids have definite shape and size. On the other hand, the force of cohesion in case of liquids is weaker than that of solids. Hence liquids do not have definite shape but have definite volume. The force of cohesion is negligible in case of gases. Because of this fact, gases have neither fixed shape nor volume.

### Example

- (i) Two drops of a liquid coalesce into one when brought in mutual contact because of the cohesive force.
- (ii) It is difficult to separate two sticky plates of glass wetted with water because a large force has to be applied against the cohesive force between the molecules of water.
- (iii) It is very difficult to break a drop of mercury into small droplets because of large cohesive force between mercury molecules.

## FORCE OF ADHESION

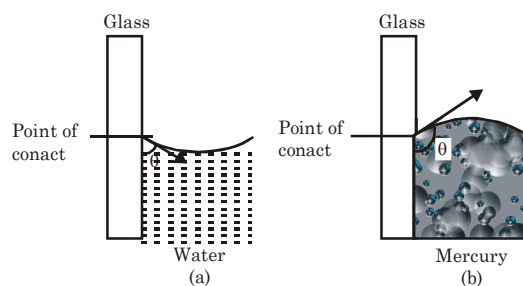
The force of attraction between molecules of different substance is called adhesion.

### Example

- (i) Adhesive force enables us to write on the black board with a chalk.
- (ii) Adhesive force helps us to write on the paper with ink.
- (iii) Large force of adhesion between cement and bricks helps us in construction work.
- (iv) Due to force of adhesive, water wets the glass plate.
- (v) Fevicol and gum are used in gluing two surfaces together because of adhesive force.

## ANGLE OF CONTACT

The angle which the tangent to the liquid surface at the point of contact makes with the solid surface inside the liquid is called angle of contact. Those liquids which wet the wall of the container (say in case of water and glass) have meniscus concave upwards and their value of angle of contact is less than  $90^\circ$  (also called acute angle). However, those liquids which don't wet the walls of the container (say in case of mercury and glass) have meniscus convex upwards and their value of angle of contact is greater than  $90^\circ$  (also called obtuse angle). The angle of contact of mercury with glass is about  $140^\circ$ , whereas the angle of contact of water with glass is about  $8^\circ$ . But, for pure water, the angle of contact  $\theta$  with glass is taken as  $0^\circ$ .



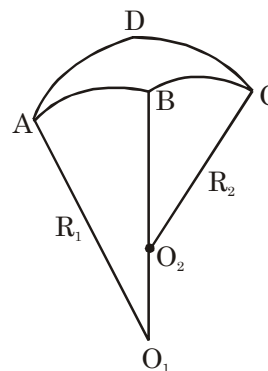
### RELATION BETWEEN SURFACE TENSION, RADII OF CURVATURE AND EXCESS PRESSURE ON A CURVED SURFACE.

Let us consider a small element ABCD (fig.) of a curved liquid surface which is convex on the upper side.  $R_1$  and  $R_2$  are the maximum and minimum radii of curvature respectively. They are called the 'principal radii of curvature' of the surface. Let  $p$  be the excess pressure on the concave side.

then  $p = T \left( \frac{1}{R_1} + \frac{1}{R_2} \right)$ . If instead of a liquid surface,

we have a liquid film, the above expression will be

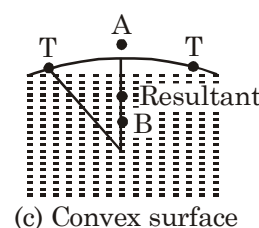
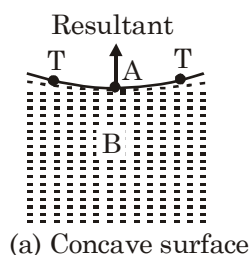
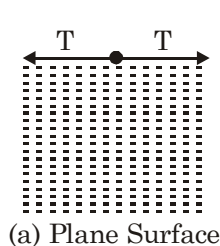
$p = 2T \left( \frac{1}{R_1} + \frac{1}{R_2} \right)$ , because a film has two surface.



### EXCESS OF PRESSURE INSIDE A CURVED SURFACE

1. Plane surface : If the surface of the liquid is plane [as shown in fig. (a)], the molecule on the liquid surface is attracted equally in all directions. The resultant force due to surface tension is zero. The pressure, therefore on the liquid surface is normal.
2. Concave surface : If the surface is concave upward [as shown in fig. (b)], there will be upward resultant force due to surface tension acting on the molecule. Since the molecule on the surface is in equilibrium, there must be an excess of pressure on the concave side in the downward direction to

balance the resultant force of surface tension  $p_A - p_B = \frac{2T}{r}$ .



3. Convex surface : If the surface is convex [as shown fig.(c)], the resultant force due to surface tension acts in the downward direction. Since the molecule on the surface are in equilibrium, there must be an excess of pressure on the concave side of the surface acting in the upward direction to balance the downward resultant force of surface tension. Hence there is always in excess of pressure on concave side of a curved surface over that on the convex side.

$$p_B - p_A = \frac{2T}{r}$$

### Solved Examples

**Ex** A barometer contains two uniform capillaries of radii  $1.44 \times 10^{-3}$  m and  $7.2 \times 10^{-4}$  m. if the height of the liquid in the tube is 0.2m more than that in the wide tube, calculate the true pressure difference. Density of liquid =  $10^3$  kg/m<sup>3</sup>, surface tension =  $72 \times 10^{-3}$  N/m and  $g = 9.8$  m/s<sup>2</sup>.

**Sol.** Let the pressure in the wide and narrow capillaries of radii  $r_1$  and  $r_2$  respectively be  $P_1$  and  $P_2$ . Then pressure just below the meniscus in the wide and narrow tubes respectively are :

$$\left( P_1 - \frac{2T}{r_1} \right) \text{ and } \left( P_2 - \frac{2T}{r_2} \right) \quad \left[ \text{excess pressure} = \frac{2T}{r} \right]$$

$$\text{Difference in these pressure} = \left( P_1 - \frac{2T}{r_1} \right) - \left( P_2 - \frac{2T}{r_2} \right) = h\rho g$$

$$\therefore \quad \text{True pressure difference} = P_1 - P_2$$

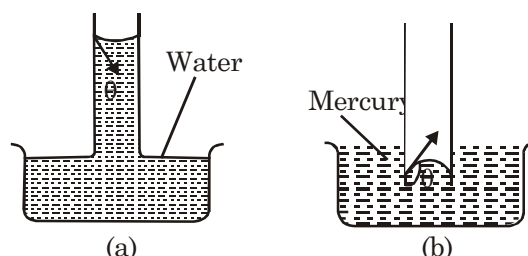
$$= h\rho g + 2T \left( \frac{1}{r_1} - \frac{1}{r_2} \right)$$

$$= 0.2 \times 10^3 \times 9.8 + 2 \times 72 \times 10^{-3} \left[ \frac{1}{1.44 \times 10^{-3}} - \frac{1}{7.2 \times 10^{-4}} \right]$$

$$= 1.86 \times 10^3 = 1860 \text{ N/m}^2$$

### Capillarity

A glass tube of very fine bore throughout the length of the tube is called capillary tube. If the capillary tube is dipped in water, the water wets the inner side of the tube and rises in it [shown in figure (a)]. If the same capillary tube is dipped in the mercury, then the mercury is depressed [shown in figure (b)]. The phenomenon of rise or fall of liquids in a capillary tube is called capillarity.



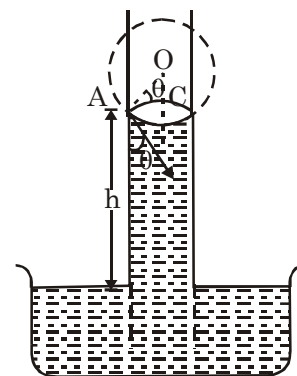
### Particle applications of capillarity

1. The oil in a lamp rises in the wick by capillary action.
2. The tip of nib of a pen is split up, to make a narrow capillary so that the ink rises upto the tin or nib continuously.
3. Sap and water rise upto the top of the leaves of the tree by capillary action.
4. If one end of the towel dips into a bucket of water and the Other end hangs over the bucket the towel soon becomes wet throughout due to capillary action.
5. Ink is absorbed by the blotter due to capillary action.
6. Sandy soil is more dry than clay. It is because the capillaries between sand particles are not so fine as to draw the water up by capillaries.
7. The moisture rises in the capillaries of soil to the surface, where it evaporates. To preserve the moisture in the soil, capillaries must be broken up. This is done by ploughing and leveling the fields.
8. Bricks are porous and behave like capillaries.

**Capillary rise (height of a liquid in a capillary tube) ascent formula**

consider the liquid which wets the wall of the tube, forms a concave meniscus shown in figure. Consider a capillary tube of radius  $r$  dipped in a liquid of surface tension  $T$  and density  $\rho$ . Let  $h$  be the height through which the liquid rises in the tube. Let  $p$  be the pressure on the concave side of the meniscus and  $p_a$  be the pressure on the convex side of the meniscus. The excess pressure

$$(p - p_a) \text{ is given by } (p - p_a) = \frac{2T}{R}$$



Where  $R$  is the radius of the meniscus. Due to this excess pressure, the liquid will rise in the capillary tube till it becomes equal to the hydrostatic pressure  $h\rho g$ . Thus in equilibrium state.

$$\text{Excess pressure} = \text{Hydrostatic pressure} \text{ or } \frac{2T}{R} = h\rho g$$

Let  $\theta$  be the angle of contact and  $r$  be the radius of the capillary tube shown in the fig.

$$\text{From } \triangle OAC, \frac{OC}{OA} = \cos \theta \text{ or } R = \frac{r}{\cos \theta} \Rightarrow h = \frac{2T \cos \theta}{r\rho g}$$

The expression is called Ascent formula.

**Discussion.**

- (i) For liquids which wet the glass tube or capillary tube, angle of contact  $\theta < 90^\circ$ . Hence  $\cos \theta =$  positive  $\Rightarrow h =$  positive. It means that these liquids rise in the capillary tube. Hence, the liquids which wet capillary tube rise in the capillary tube. For example, water, milk, kerosene oil, petrol etc.

**Solved Examples**

Ex. A liquid of specific gravity 1.5 is observed to rise 3.0 cm in a capillary tube of diameter 0.50 mm and the liquid wets the surface of the tube. Calculate the excess pressure inside a spherical bubble of 1.0 cm diameter blown from the same liquid. Angle of contact  $= 0^\circ$ .

Sol. The surface tension of the liquid is

$$T = \frac{r h \rho g}{2} = \frac{(0.025 \text{ cm})(3.0 \text{ cm})(1.5 \text{ gm/cm}^3)(980 \text{ cm/sec}^2)}{2}$$

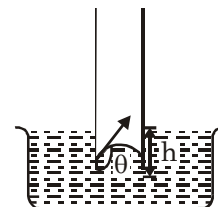
$$= 55 \text{ dyne/cm.}$$

Hence excess pressure inside a spherical bubble

$$p = \frac{4T}{R} = \frac{4 \times 55 \text{ dyne/cm}}{(0.5 \text{ cm})} = 440 \text{ dyne/cm}^2.$$

- (ii) For liquids which do not wet the glass tube or capillary tube, angle of contact  $\theta > 90^\circ$ . Hence  $\cos \theta =$  negative  $\Rightarrow h =$  negative. Hence, the liquids which do not wet capillary tube are depressed in the capillary tube. For example, mercury.

- (iii)  $T, \theta, \rho$  and  $g$  are constant and hence  $h \propto \frac{1}{r}$ . Thus, the liquid rises more in a narrow tube and less in a wider tube. This is called Jurin's Law.



- (iv) If two parallel plates with the spacing 'd' are placed in water reservoir, then height of rise

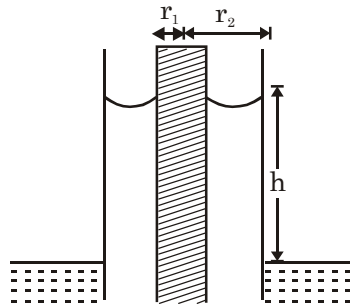
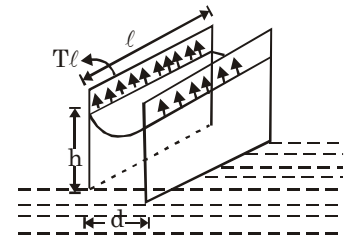
$$\Rightarrow 3T/\ell = \rho h d g$$

$$h = \frac{2T}{\rho d g}$$

- (v) If two concentric tube of radius 'r<sub>1</sub>' and 'r<sub>2</sub>' (inner one is solid) are placed in water reservoir, then height of rise

$$\Rightarrow T [2\pi r_1 + 2\pi r_2] = [\pi r_2^2 h - \pi r_1^2 h] \rho g$$

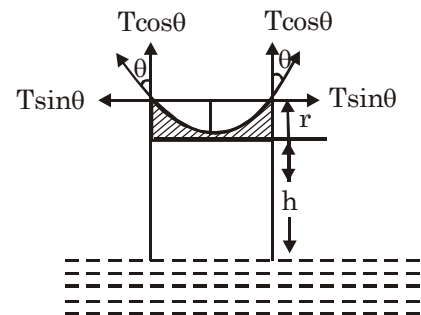
$$h = \frac{2T}{(r_2 - r_1)\rho g}$$



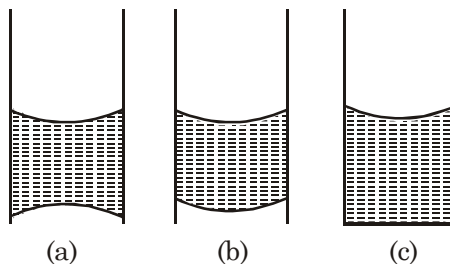
- (vi) If weight of the liquid in the meniscus is to be considered :

$$T \cos \theta \times 2\pi r = [\pi r^2 h + \frac{1}{3} \pi r^2 \times \pi r_1^2 h] \rho g$$

$$\left[ h + \frac{r}{3} \right] = \frac{2T \cos \theta}{r \rho g}$$



- (vii) When capillary tube (radius, 'r') is in vertical position, the upper meniscus is concave and pressure due to surface tension is directed vertically upward and is given by  $p_1 = 2T / R_1$  where  $R_1$  = radius of curvature of upper meniscus.



The hydrostatic pressure  $p_2 = h \rho g$  is always directed downwards.

If  $p_1 > p_2$  i.e. resulting pressure is directed upward. For equilibrium, the pressure due to lower meniscus should be downward. This makes lower meniscus concave downward (fig. (a)). The

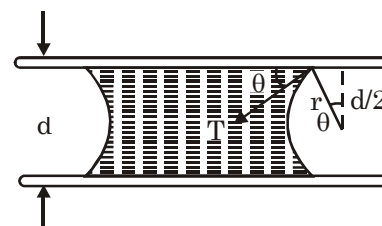
radius of lower meniscus  $R_2$  can be given by  $\frac{2T}{R_2} = (p_1 - p_2)$ .

If  $p_1 < p_2$  i.e. resulting pressure is directed downward for equilibrium, the pressure due to lower meniscus should be upward. This makes lower meniscus convex upward (fig. b).

The radius of lower meniscus can be given by  $\frac{2T}{R_2} = p_2 - p_1$ .

If  $p_1 = p_2$ , then is no resulting pressure, then  $p_1 - p_2 = \frac{2T}{R_2} = 0$  or  $R_2 = \infty$  i.e. lower surface will be FLAT. (fig.c).

- (viii) **Liquid between two plates :** When a small drop of water is placed between two glass plates put face to face, it forms a thin film which is concave outward along its boundary. Let 'R' and 'r' be the radii of curvature of the enclosed film in two perpendicular directions.



Hence the pressure inside the film is less than the atmospheric pressure outside it by an amount  $p$

given by  $p = T \left( \frac{1}{r} + \frac{1}{r = \infty} \right)$  and we have,  $p = \frac{T}{r}$ .

If  $d$  be the distance between the two plates and  $\theta$  the angle of contact for water and glass, then, from

the figure,  $\cos \theta = \frac{\frac{1}{2}d}{r}$  or  $\frac{1}{r} = \frac{2 \cos \theta}{d}$ .

Substituting for  $\frac{1}{r}$  in, we get  $p = \frac{2T}{d} \cos \theta$ .

$\theta$  can be taken zero water and glass, i.e.  $\cos \theta = 1$ . Thus the upper plate is pressed downward by the atmospheric pressure minus  $\frac{2T}{d}$ . Hence the resultant downward pressure acting on the upper plate is  $\frac{2T}{d}$ . If  $A$  be the area of the plate wetted by the film, the resultant force  $F$  pressing the upper plate

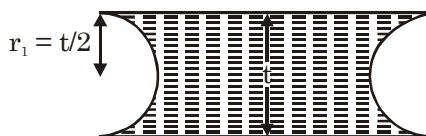
downward is given by  $F = \text{resultant pressure} \times \text{area} = \frac{2TA}{d}$ . For very nearly plane surface,  $d$  will be very small and hence the pressing force  $F$  very large. Therefore it will be difficult to separate the two plates normally.



### Solved Examples

**Ex** A drop of water volume  $0.05 \text{ cm}^3$  is pressed between two glass-plates, as a consequence of which, it spreads and occupies an area of  $40 \text{ cm}^2$ . If the surface tension of water is  $70 \text{ dyne/cm}$ , find the normal force required to separate out the two glass plates in newton.

**Sol.** Pressure inside the film is less than outside by an amount,  $P = T \left[ \frac{1}{r_1} + \frac{1}{r_2} \right]$ , where  $r_1$  and  $r_2$  are the radii of curvature of the meniscus. Here  $r_1 = 1/2$  and  $r_2 = \infty$ , then the force required to separate the two glass plates, between which a liquid film is enclosed (figure) is,  $F = P \times A = \frac{2AT}{t}$ , where  $t$  is the thickness of the film,  $A$  = area of film.



$$F = \frac{2A^2T}{At} = \frac{2A^2T}{V} = \frac{2 \times (40 \times 10^{-4})^2 \times (70 \times 10^{-3})}{0.05 \times 10^{-6}} = 45 \text{ N}$$

**Ex** A glass plate of length  $10 \text{ cm}$ , breadth  $1.54 \text{ cm}$  and thickness  $0.20 \text{ cm}$  weigh  $8.2 \text{ gm}$  in air. It is held vertically with the long side horizontal and the lower half under water. Find the apparent weight of the plate. Surface tension of water  $73 \text{ dyne per cm}$ ,  $g = 980 \text{ cm/sec}^2$ .

**Sol.** Volume of the portion of the plate immersed in water is

$$10 \times \frac{1}{2} (1.54) \times 0.2 = 1.54 \text{ cm}^3.$$

Therefore, if the density of water is taken as 1, then upthrust

= wt. of the water displaced

$$= 1.54 \times 1 \times 980 = 1509.2 \text{ dynes.}$$

Now, the total length of the plate in contact with the water surface is  $2 (10 + 0.2) = 20.4 \text{ cm}$ ,

$\therefore$  downward pull upon the plate due to surface tension

$$= 20.4 \times 73 = 1489.2 \text{ dynes}$$

$\therefore$  resultant upthrust

$$= 1509.2 - 1489.2$$

$$= 20.0 \text{ dynes} = \frac{20}{980}$$

$$= 0.0204 \text{ gm. wt.}$$

$\therefore$  apparent weight of the plate in water

= weight of the plate in air – resultant upthrust

$$= 8.2 - 0.0204 = 8.1796 \text{ gm Ans.}$$

Ex A glass tube of circular cross-section is closed at one end. This end is weighted and the tube floats vertically in water, heavy end down. How far below the water surface is the end of the tube? Give : Outer radius of the tube 0.14 cm, mass of weighted tube 0.2 gm, surface tension of water 73 dyne/cm and  $g = 980 \text{ cm/sec}^2$ .

Sol., Let  $l$  be the length of the tube inside water. The forces acting on the tube are :

(i) Upthrust of water acting upward

$$= \pi r^2 l \times 1 \times 980 = \frac{22}{7} \times (0.14)^2 l \times 980 = 60.368 l \text{ dyne.}$$

(ii) Weight of the system acting downward

$$= mg = 0.2 \times 980 = 196 \text{ dyne.}$$

(iii) Force of surface tension acting downward

$$= 2\pi rT$$

$$= 2 \times \frac{22}{7} \times 0.14 \times 73 = 64.24 \text{ dyne.}$$

Since the tube is in equilibrium, the upward force is balanced by the downward forces. That is,

$$60.368 l = 196 + 64.24 = 260.24.$$

$$\therefore l = \frac{260.24}{60.368} = 4.31 \text{ cm.}$$

Ex A glass U-tube is such that the diameter of one limb is 3.0 mm and that of the other is 6.00 mm. The tube is inverted vertically with the open ends below the surface of water in a beaker. What is the difference between the heights to which water rises in the two limbs? Surface tension of water is  $0.07 \text{ N m}^{-1}$ . Assume that the angle of contact between water and glass is  $0^\circ$ .

Sol. Suppose pressure at the points A, B, C and D be  $P_A$ ,  $P_B$ ,  $P_C$  and  $P_D$  respectively.

The pressure on the concave side of the liquid surface is greater than that on the other side by  $2T/R$ .

Angle of contact  $\theta$  is given to be  $0^\circ$ , hence  $R \cos 0^\circ = r$  or  $R = r$

$$\therefore P_A = P_B + 2T/r_1 \text{ and } P_C = P_D + 2T/r_2$$

where  $r_1$  and  $r_2$  are the radii of the two limbs

$$\text{But } P_A = P_C$$

$$\therefore P_B + \frac{2T}{r_1} = P_D + \frac{2T}{r_2}$$

$$\text{or } P_D - P_B = 2T \left( \frac{1}{r_1} - \frac{1}{r_2} \right)$$

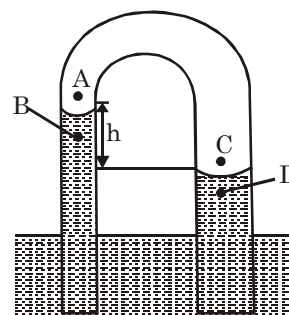
where  $h$  is the difference in water levels in the two limbs

$$\text{Now, } h = \frac{2T}{\rho g} \left( \frac{1}{r_1} - \frac{1}{r_2} \right)$$

Given that  $T = 0.07 \text{ N m}^{-1}$ ,  $\rho = 1000 \text{ kg m}^{-3}$

$$r_1 = \frac{3}{2} \text{ mm} = \frac{3}{20} \text{ cm} = \frac{3}{20 \times 100} \text{ m} = 1.5 \times 10^{-3} \text{ m}, r_2 = 3 \times 10^{-3} \text{ m}$$

$$\therefore h = \frac{2 \times 0.07}{1000 \times 9.8} \left( \frac{1}{1.5 \times 10^{-3}} - \frac{1}{3 \times 10^{-3}} \right) \text{ m} = 4.76 \times 10^{-3} \text{ m} = 4.76 \text{ mm}$$



**Ex** Two narrow bores of diameters 3.0 mm and 6.0 mm are joined together to form a U-shaped tube open at both ends. If the U-tube contains water, what is the difference in its levels in the two limbs of the tube? Surface tension of water at the temperature of the experiment is  $7.3 \times 10^{-2} \text{ Nm}^{-1}$ . Take the angle of contact to be zero.

**Sol.** Given that  $r_1 = \frac{3.0}{2} = 1.5 \text{ mm} = 1.5 \times 10^{-3} \text{ m}$ ,  $r_2 = \frac{6.0}{2} = 3.0 \times 10^{-3} \text{ m}$ ,

$$T = 7.3 \times 10^{-2} \text{ Nm}^{-1}, \theta = 0^\circ, \rho = 1.0 \times 10^3 \text{ kg m}^{-3}, g = 9.8 \text{ ms}^{-2}$$

When angle of contact is zero degree, the radius of the meniscus equals radius of bore.

$$\text{Excess pressure in the first bore, } P_1 = \frac{2T}{r_1} = \frac{2 \times 7.3 \times 10^{-2}}{1.5 \times 10^{-3}} = 97.3 \text{ Pascal}$$

$$\text{Excess pressure in the second bore, } P_2 = \frac{2T}{r_2} = \frac{2 \times 7.3 \times 10^{-2}}{3 \times 10^{-3}} = 48.7 \text{ Pascal}$$

Hence, pressure difference in the two limbs of the tube

$$\Delta P = P_1 - P_2 = h\rho g$$

$$\text{or } h = \frac{P_1 - P_2}{\rho g} = \frac{97.3 - 48.7}{1.0 \times 10^3 \times 9.8} = 5.0 \text{ mm.}$$

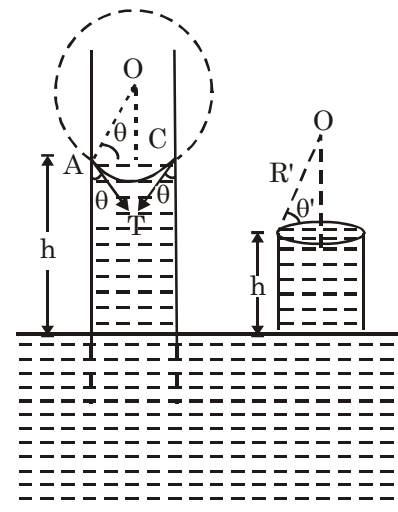
### Capillary rise in a tube of insufficient length

We know, the height through which a liquid rises in the capillary tube of radius  $r$  is given by

$$\therefore h = \frac{2T}{R\rho g} \text{ or } hR = \frac{2T}{\rho g} = \text{constant}$$

When the capillary tube is cut and its length is less than  $h$  (i.e.  $h'$ ). then the liquid rises upto the top of the tube and spreads in such a way that the radius ( $R'$ ) of the liquid meniscus increases and it becomes more flat so that  $hR = h'R' = \text{Constant}$ . Hence the liquid does not overflow.

$$\begin{aligned} \text{If } h' < h \text{ then } R' > R & \quad \text{or} \quad \frac{r}{\cos \theta'} > \frac{r}{\cos \theta} \\ \Rightarrow \cos \theta' < \cos \theta & \quad \Rightarrow \quad \theta' > \theta \end{aligned}$$



**Ex** If a 5cm long capillary tube with 0.1 mm internal diameter open at both ends is slightly dipped in water having surface tension  $75 \text{ dyne cm}^{-1}$ , state whether (i) water will rise half way in the capillary. (ii) Water will rise up to the upper end of capillary (iii) What will overflow out of the upper end of capillary. Explain your answer.

**Sol.** Given that surface tension of water,  $T = 75 \text{ dyne/cm}$

$$\text{Radius } r = \frac{0.1}{2} \text{ mm} = 0.05 \text{ mm} = 0.005 \text{ cm}$$

$$\text{density } \rho = 1 \text{ gm/cm}^3, \text{ angle of contact, } \theta = 0^\circ$$

Let  $h$  be the height to which water rise in the capillary tube. Then

$$h = \frac{2T \cos \theta}{r\rho g} = \frac{2 \times 75 \times \cos 0^\circ}{0.005 \times 1 \times 981} \text{ cm} = 30.58 \text{ cm}$$

But length of capillary tube,  $h' = 5\text{cm}$

- (i) Because  $h > \frac{h'}{2}$  therefore the first possibility does not exist.
- (ii) Because the tube is of insufficient length therefore the water will rise upto the upper end of the tube.
- (iii) The water will not overflow out of the upper end of the capillary. It will rise only upto the upper end of the capillary.

The liquid meniscus will adjust its radius of curvature  $R'$  in such a way that

$$R'h' = Rh \quad \left[ \because hR = \frac{2T}{\rho g} = \text{constant} \right]$$

where  $R$  is the radius of curvature that the liquid meniscus would possess if the capillary tube were of sufficient length.

$$\therefore R' = \frac{Rh}{h'} = \frac{rh}{h'} \quad \left[ \because R = \frac{r}{\cos\theta} = \frac{r}{\cos 0^\circ} = r \right] = \frac{0.005 \times 30.58}{5} = 0.0306 \text{ cm}$$

### Applications of surface tension

- (i) The wetting property is made use of in detergents and waterproofing. When the detergent materials are added to liquids, the angle of contact decreases and hence the wettability increases. On the other hand, when water proofing material is added to a fabric, it increases the angle of contact, making the fabric water-repellant.
- (ii) The antiseptics have very low value of surface tension. The low value of surface tension prevents the formation of drops that may otherwise block the entrance to skin or a wound. Due to low surface tension the antiseptics spreads properly over the wound. The lubricating oils and paints also have low surface tension. So they can spread properly.
- (iii) Surface tension of all lubricating oils and paints is kept low so that they spread over a large area.
- (iv) Oil spreads over the surface of water because the surface tension of oil is less than the surface tension of cold water.
- (v) A rough sea can be calmed by pouring oil on its surface.

### Effect of temperature and impurities on surface tension

The surface tension of a liquid decreases with the rise in temperature and vice versa. According to

$$\text{Ferguson, } T = T_0 \left( 1 - \frac{\theta}{\theta_c} \right)^n \text{ where } T_0 \text{ is surface tension at } 0^\circ\text{C, } \theta \text{ is absolute temperature of the}$$

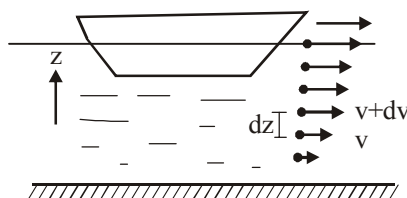
liquid,  $\theta_c$  is the critical temperature and  $n$  is a constant varies slightly from liquid and has means value 1.21. This formula shows that the surface tension becomes zero at the critical temperature, where the while machinery parts get jammed in winter.

The surface tension of a liquid change appreciably with addition of impurities. For example, surface tension of water increases with addition of highly soluble substances like  $\text{NaCl}$ ,  $\text{ZnSO}_4$  etc. On the other hand surface tension of water gets reduced with addition of sparingly soluble substances like phenol, soap etc.

## VISCOSITY

When a layer of a fluid slips or tends to slip on another layer in contact, the two layers exert tangential forces on each other. The directions are such that the relative motion between the layers is opposed. This property of a fluid to oppose relative motion between its layers is called viscosity. The forces between the layers opposing relative motion between them are known as the forces of viscosity. Thus, viscosity may be thought of as the internal friction of a fluid in motion.

If a solid surface is kept in contact with a fluid and is moved, forces of viscosity appear between the solid surface and the fluid layer in contact. The fluid in contact is dragged with the solid. If the viscosity is sufficient, the layer moves with the solid and there is no relative slipping. When a boat moves slowly on the water of a calm river, the water in contact with the boat is dragged with it, whereas the water in contact with the bed of the river remains at rest. Velocities of different layers are different. Let  $v$  be the velocity of the layer at a distance  $z$  from the bed and  $v + dv$  be the velocity at a distance  $z + dz$  (figure).



Thus, the velocity differs by  $dv$  in going through a distance  $dz$  perpendicular to it. The quantity  $dv/dz$  is called the velocity gradient.

The force of viscosity between two layers of a fluid is proportional to the velocity gradient in the direction perpendicular to the layers. Also the force is proportional to the area of the layer.

Thus, if  $F$  is the force exerted by a layer of area  $A$  on a layer in contact,

$$F \propto A \text{ and } F \propto dv/dz$$

or, 
$$F = -\eta A \, dv/dz$$

The negative sign is included as the force is frictional in nature and opposes relative motion. The constant of proportionality  $\eta$  is called the coefficient of viscosity.

The SI unit of viscosity can be easily worked out from equation. It is  $\text{N-s/m}^2$ . However, the corresponding CGS unit  $\text{dyne-s/cm}^2$  is in common use and is called a poise in honour of the French scientist Poiseuille. We have

$$1 \text{ poise} = 0.1 \text{ N-s/m}^2$$

## TERMINAL VELOCITY

The viscous force on a solid moving through a fluid is proportional to its velocity. When a solid is dropped in a fluid, the forces acting on it are

- (a) weight  $W$  acting vertically downward,
- (b) the viscous force  $F$  acting vertically upward and
- (c) the buoyancy force  $B$  acting vertically upward.

The weight  $W$  and the buoyancy  $B$  are constant but the force  $F$  is proportional to the velocity  $v$ , initially, the velocity and hence the viscous force  $F$  is zero and the solid is accelerated due to the force  $W - B$ . Because of the acceleration, the velocity increases. Accordingly, the viscous force also increases. At a certain instant the viscous force becomes equal to  $W - B$ . The net force then becomes zero and the solid falls with constant velocity. This constant velocity is known as the terminal velocity.

Consider a spherical body falling through a liquid. Suppose the density of the body =  $\rho$ , density of the liquid =  $\sigma$ , radius of the sphere =  $r$  and the terminal velocity =  $v_0$ . The viscous force is

$$F = 6\pi\eta r v_0$$

the weight

$$W = \frac{4}{3}\pi r^3 \rho g$$

and the buoyancy force

$$B = \frac{4}{3}\pi r^3 \sigma g$$

We have

$$6\pi\eta r v_0 = W - B = \frac{4}{3}\pi r^3 \rho g - \frac{4}{3}\pi r^3 \sigma g$$

or

$$v_0 = \frac{2r^2(\rho - \sigma)g}{9\eta}$$

- Q.** A large wooden plate of area  $10 \text{ m}^2$  floating on the surface of a river is made to move horizontally with a speed of  $2 \text{ m/s}$  by applying a tangential force. If the river is  $1 \text{ m}$  deep and the water in contact with the bed is stationary, find the tangential force needed to keep the plate moving. Coefficient of viscosity of water at the temperature of the river =  $10^{-2}$  poise.

- Sol.** The velocity decreases from  $2 \text{ m/s}$  to zero in  $1 \text{ m}$  of perpendicular length. Hence, velocity gradient.  
 $= dv/dx = 2 \text{ s}^{-1}$

Now,

$$\eta = \left| \frac{F/A}{dv/dx} \right|$$

or,

$$10^{-3} \frac{\text{N-s}}{\text{m}^2} = \frac{F}{(10\text{m})^2 (2\text{s}^{-1})}$$

or,

$$F = 0.02 \text{ N}.$$

- Q.** The velocity of water in a river is  $18 \text{ km/hr}$  near the surface. If the river is  $5 \text{ m}$  deep, find the shearing stress between the horizontal layers of water. The coefficient of viscosity of water =  $10^{-2}$  poise.

- Sol.** The velocity gradient in vertical direction is

$$\frac{dv}{dx} = \frac{18 \text{ km/hr}}{5 \text{ m}} = 1.0 \text{ s}^{-1}$$

The magnitude of the force of viscosity is

$$F = \eta A \frac{dv}{dx}$$

The shearing stress is

$$F/A = \eta \frac{dv}{dx} = (10^{-2} \text{ poise}) (1.0 \text{ s}^{-1}) = 10^{-3} \text{ N/m}^2$$

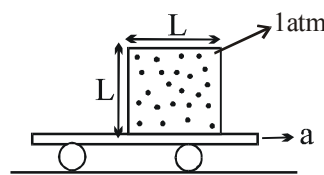
## EXERCISE (S-1)

### Fluid Statics

1. A spherical tank of 1.2 m radius is half filled with oil of relative density 0.8. If the tank is given a horizontal acceleration of  $10 \text{ m/s}^2$ . Calculate the inclination of the oil surface to horizontal and maximum pressure on the tank.

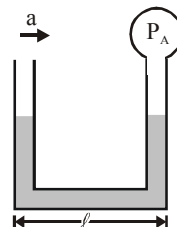
FM0001

2. A cubical sealed vessel with edge  $L$  is placed on a cart, which is moving horizontally with an acceleration 'a' as shown in figure. The cube is filled with an ideal fluid having density  $\rho$ . Find the gauge pressure at the centre of the cubical vessel.



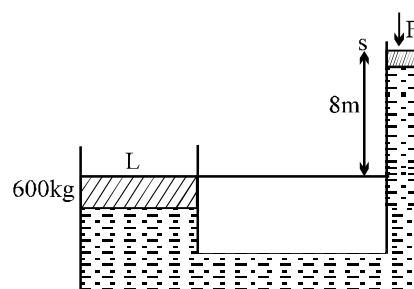
FM0002

3. A liquid of density  $\rho$  is filled in a U-tube, whose one end is open & at the other end a bulb is fitted whose pressure is  $P_A$ . Now this tube is moved horizontally with acceleration 'a' as shown in the figure. During motion it is found that liquid in both column is at same level at equilibrium. If atmospheric pressure is  $P_0$ , then find the value of  $P_A$ .



FM0003

4. For the system shown in the figure, the cylinder on the left at L has a mass of 600 kg and a cross sectional area of  $800 \text{ cm}^2$ . The piston on the right, at S, has cross sectional area  $25 \text{ cm}^2$  and negligible weight. If the apparatus is filled with oil. ( $\rho = 0.75 \text{ gm/cm}^3$ ) Find the force  $F$  required to hold the system in equilibrium.

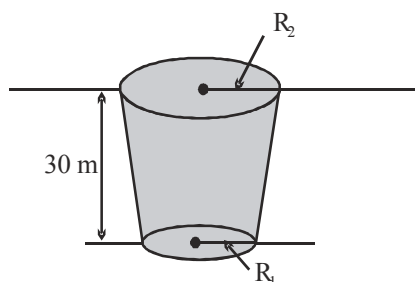


FM0004

5. Dams at two different locations are needed to form a lake. When the lake is filled, the water level will be at top of both dams. The second dam is twice as high and twice as wide as the first dam. The force of the water on the second dam is how much greater than the force on the first? (Ignore atmospheric pressure since it is pushing on both sides of the dams.)

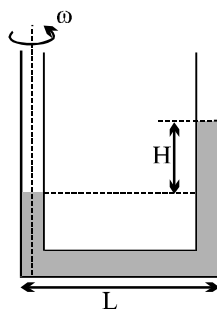
FM0005

6. As the drawing illustrates, a pond full of water has the shape of an inverted cone with the tip sliced off and has a depth of 30m. The atmospheric pressure above the pond is  $1.0 \times 10^5 \text{ Pa}$ . The circular top surface (radius =  $R_2$ ) and circular bottom surface (radius =  $R_1$ ) of the pond are both parallel to the ground. The magnitude of the force acting on the top surface by the liquid is the same as the magnitude of the force acting on the bottom surface by the liquid. Obtain  $\frac{R_2}{R_1}$ .



FM0006

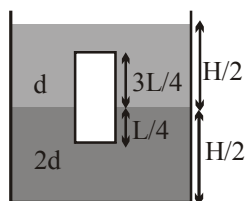
7. A U tube is rotated about one of its limbs with an angular velocity  $\omega$ . Find the difference in height  $H$  of the liquid (density  $\rho$ ) level, where diameter of the tube  $d \ll L$ . [IIT-JEE 2005]



FM0007

### Buoyancy

8. A container of a large uniform cross-sectional area  $A$  resting on a horizontal surface holds two immiscible, non viscous and incompressible liquids of densities  $d$  and  $2d$  each of height  $\frac{H}{2}$  as shown. The lower density liquid is open to atmosphere. A homogeneous solid cylinder of length  $L$  ( $< \frac{H}{2}$ ), cross-sectional area  $\frac{A}{5}$  is immersed such that it floats with its axis vertical to the liquid-liquid interface with length  $\frac{L}{4}$  in denser liquid. Find the density of the solid cylinder.



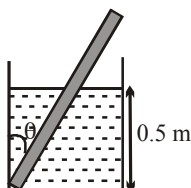
FM0008



9. In air an object weighs 15N, when immersed completely in water the same object weighs 12N. When immersed in another liquid completely, it weighs 13N. Find  
(a) the specific gravity of the object and  
(b) the specific gravity of the other liquid.

FM0009

10. A wooden plank of length 1 m and uniform cross-section is hinged at one end to the bottom of a tank as shown. The tank is filled with water upto a height of 0.5 m. The specific gravity of the plank is 0.5. Find the angle  $\theta$  made by the plank in equilibrium position



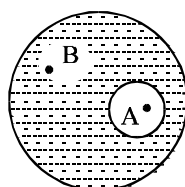
FM0010

11. The volume of an air bubble is doubled as it rises from the bottom of a lake to its surface. If the atmospheric pressure is  $H$  m of mercury & the density of mercury is  $n$  times that of lake water. Find the depth of the lake.

FM0011

### Surface Tension

12. There is an air bubble of radius  $R$  inside a drop of water of radius  $3R$ . Find the ratio of gauge pressure at point A to the gauge pressure at point B.

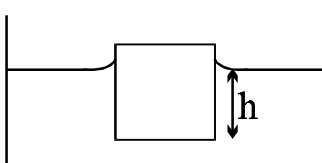


FM0012

13. Two arms of a U-tube have unequal diameters  $d_1 = 1.0$  mm and  $d_2 = 1.0$  cm. If water (surface tension  $7 \times 10^{-2}$  N/m) is poured into the tube held in the vertical position, find the difference of level of water in the U-tube. Assume the angle of contact to be zero.

FM0013

14. A cube with a mass ' $m$ ' completely wettable by water floats on the surface of water. Each side of the cube is ' $a$ '. What is the distance  $h$  between the lower face of cube and the surface of the water if surface tension is  $S$ . Take density of water as  $\rho_w$ . Take angle of contact as zero.



FM0014

**Fluid Dynamics**

15. A jet of water having velocity = 10 m/s and stream cross-section =  $2 \text{ cm}^2$  hits a flat plate perpendicularly, with the water splashing out parallel to plate. Find the force that the plate experiences.

FM0015

16. A laminar stream is flowing vertically down from a tap of cross-section area  $1 \text{ cm}^2$ . At a distance 10cm below the tap, the cross-section area of the stream has reduced to  $1/2 \text{ cm}^2$ . Find the volumetric flow rate of water from the tap.

FM0016

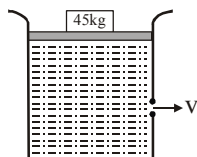
17. Calculate the rate of flow of glycerine of density  $1.25 \times 10^3 \text{ kg/m}^3$  through the conical section of a pipe if the radii of its ends are 0.1m & 0.04m and the pressure drop across its length is  $10 \text{ N/m}^2$ .

FM0017

18. A cylindrical vessel open at the top is 20cm high and 10cm in diameter. A circular hole whose cross-sectional area  $1 \text{ cm}^2$  is cut at the centre of the bottom of the vessel. Water flows from a tube above it into the vessel at the rate  $100 \text{ cm}^3 \text{ s}^{-1}$ . Find the height of water in the vessel under steady state.

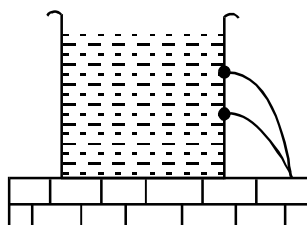
FM0018

19. A large cylindrical tank of cross-sectional area  $1 \text{ m}^2$  is filled with water. It has a small hole at a height of 1m from the bottom. A movable piston of mass 5 kg is fitted on the top of the tank such that it can slide in the tank freely. A load of 45 kg is applied on the top of water by piston, as shown in figure. Find the value of  $v$  when piston is 7m above the bottom ( $g = 10 \text{ m/s}^2$ )



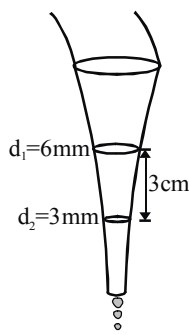
FM0019

20. In a cylindrical vessel containing liquid of density  $\rho$ , there are two holes in the side walls at heights  $h_1$  and  $h_2$  respectively such that the range of efflux at the bottom of the vessel is same. Find the height of a hole, for which the range of efflux would be maximum.



FM0020

21. The tap in the garden was closed in appropriately resulting in the water flowing freely out of it which forms a downward narrowing beam. The beam of water has a circular cross-section, the diameter of the circle is 6 mm at one point and 3 cm below it is only 3 mm as shown in figure. If the rate of water wasted is  $(x \times \pi)$  mL/minute then find the value of  $x$ . (Neglect the effect of viscosity and surface tension of the flowing water.)



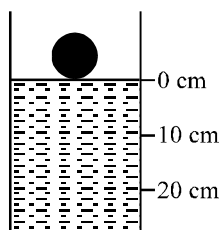
FM0021

### Viscosity

22. A spherical ball of radius  $1 \times 10^{-4}$  m and density  $10^4$  kg/m<sup>3</sup> falls freely under gravity through a distance  $h$  before entering a tank of water. If after entering the water the velocity of the ball does not change, find  $h$ . The viscosity of water is  $9.8 \times 10^{-6}$  N-s/m<sup>2</sup>.

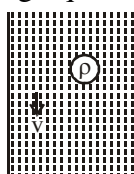
FM0022

23. A spherical ball of density  $\rho$  and radius 0.003m is dropped into a tube containing a viscous fluid filled up to the 0 cm mark as shown in the figure. Viscosity of the fluid =  $1.260$  N.m<sup>-2</sup> and its density  $\rho_L = \rho/2 = 1260$  kg.m<sup>-3</sup>. Assume the ball reaches a terminal speed by the 10 cm mark. Find the time taken by the ball to traverse the distance between the 10 cm and 20 cm mark. ( $g$  = acceleration due to gravity =  $10$  ms<sup>-2</sup>)



FM0023

24. A liquid of density  $\rho = 1000$  kg/m<sup>3</sup> and coefficient of viscosity  $\eta = 0.1$  Ns/m<sup>2</sup> is flowing down in vertical pipe of large cross section. A small ball of density  $\rho_0 = 100$  kg/m<sup>3</sup> and  $r = 5$  cm will be at rest in flowing liquid, if velocity of flowing liquid is  $10$  k m/s. Then find the value of  $k$ .



FM0024

25. A small sphere falls from rest in a viscous liquid. Due to friction, heat is produced. Find the relation between the rate of production of heat and the radius of the sphere at terminal velocity.

[IIT-JEE 2004]

FM0025

## EXERCISE (S-2)

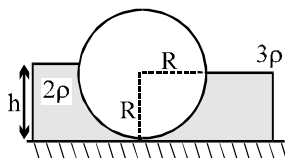
1. Place a glass beaker, partially filled with water, in a sink. The beaker has a mass 390 gm and an interior volume of  $500\text{cm}^3$ . You now start to fill the sink with water and you find, by experiment, that if the beaker is less than half full, it will float; but if it is more than half full, it remains on the bottom of the sink as the water rises to its rim. What is the density of the material of which the beaker is made?

FM0026

2. A hollow cone floats with its axis vertical upto one-third of its height in a liquid of relative density 0.8 and with its vertex submerged. When another liquid of relative density  $\rho$  is filled in it upto one-third of its height, the cone floats upto half its vertical height. The height of the cone is 0.10 m and the radius of the circular base is 0.05 m. Find the specific gravity  $\rho$ .

FM0027

3. In the figure shown, the heavy cylinder (radius  $R$ ) resting on a smooth surface separates two liquids of densities  $2\rho$  and  $3\rho$ . Find the height ' $h$ ' for the equilibrium of cylinder.

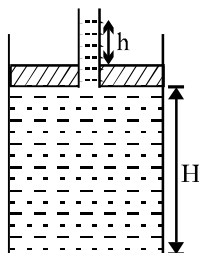


FM0028

4. A small wooden ball of density ' $\rho$ ' is immersed in water of density  $\sigma$  to depth  $h$  and then released. The height  $H$  above the surface of water up to which the ball jumps out of water is  $\left(\frac{\sigma}{\rho} - \frac{n}{2}\right)h$ . Value of  $n$  is (Neglect the effect of viscosity and surface tension and assume all quantities in SI unit)

FM0029

5. A piston of mass  $M = 3\text{kg}$  and radius  $R = 4\text{cm}$  has a hole into which a thin pipe of radius  $r = 1\text{cm}$  is inserted. The piston can enter a cylinder tightly and without friction, and initially it is at the bottom of the cylinder. 750gm of water is now poured into the pipe so that the piston & pipe are lifted up as shown. Find the height  $H$  of water in the cylinder and height  $h$  of water in the pipe.

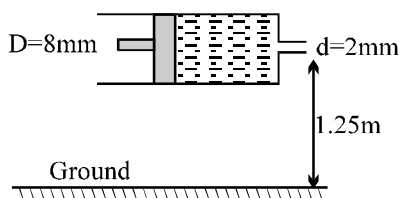


FM0030

6. A cylindrical wooden float whose base area is  $4 \text{ m}^2$  and height  $1 \text{ m}$  drifts on the water surface in vertical position. Density of wood is  $500 \text{ kg/m}^3$  and that of water is  $1000 \text{ kg/m}^3$ . What minimum work (in kJ) must be performed to take the float out of the water?

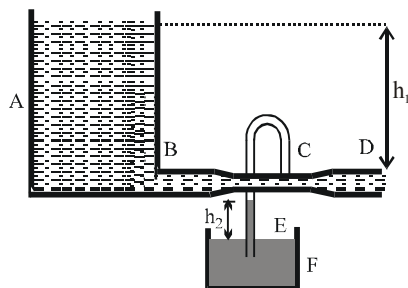
**FM0031**

7. Consider a horizontally oriented syringe containing water located at a height of 1.25 m above the ground. The diameter of the plunger is 8 mm and the diameter of the nozzle is 2 mm. The plunger is pushed with a constant speed of 0.25 m/s. Find the horizontal range of water stream on the ground. Take  $g = 10 \text{ m/s}^2$ . **[IIT-JEE 2004]**

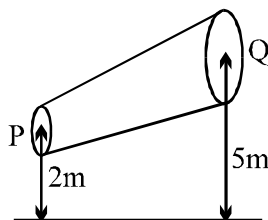


FM0032

8. Two very large open tanks A and F both contain the same liquid. A horizontal pipe BCD, having a constriction at C leads out of the bottom of tank A, and a vertical pipe E opens into the constriction at C and dips into the liquid in tank F. Assume streamline flow and no viscosity. If the cross section at C is one half that at D and if D is at a distance  $h_1$  below the level of liquid in A, to what height  $h_2$  (in terms of  $h_1$ ) will liquid rise in pipe E?

**FM0033**

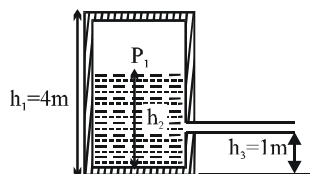
9. A non-viscous liquid of constant density  $1000 \text{ kg/m}^3$  flows in a streamline motion along a tube of variable cross section. The tube is kept inclined in the vertical plane as shown in the figure. The area of cross section of the tube at two points P and Q at heights of 2 meters and 5 meters are respectively  $4 \times 10^{-3} \text{ m}^2$  and  $8 \times 10^{-3} \text{ m}^2$ . The velocity of the liquid at point P is  $1 \text{ m/s}$ . Find the work done per unit volume by the pressure and the gravity forces as the fluid flows from point P to Q.



FM0034

10. A large tank of (height  $h_1 = 4\text{m}$ ) water has a hose connected to it, as shown in figure. The tank is sealed at the top and has compressed air between the water surface & the top. When the water height  $h_2$  is 3m, the gauge pressure of air  $P_1 = 1 \times 10^5 \text{N/m}^2$ . Assume that the air above the water surface expands isothermally, the velocity of flow out of the hose when  $h_2$  has decreased to 2m is

given by  $\alpha \text{ m/s}$  ? Fill  $\frac{\alpha^2}{5}$  in OMR sheet. (Assume ideal fluid flow &  $P_{\text{atm}} = 10^5 \text{N/m}^2$ )

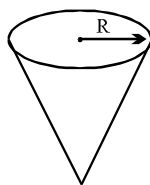


FM0035

11. Two large vertical parallel plates separated by a gap of  $d = 1\text{mm}$  have a highly viscous liquid of density  $800 \text{ kg/m}^3$  and viscosity coefficient  $\eta = 5 \text{ poise}$  flowing steadily under gravity in between the gap. Find the velocity gradient of flow near plates surface in  $\text{sec}^{-1}$ . (1 poise =  $0.1 \text{ NS/m}^2$ ).

FM0036

12. An inverted hollow cone, of height  $H$  and radius at the top =  $R$ , is filled completely with a liquid of density  $d$ . Neglect atmosphere pressure.



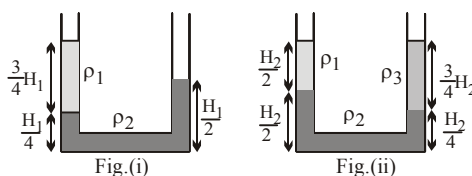
- (a) Find horizontal component of the force which the liquid exerts on half curved side wall of the cone.  
(b) Find the vertical downward component of the thrust force due to water pressure which the liquid exerts on half side curved wall of the cone.

FM0037

13. A U-tube filled with a liquid of volumetric coefficient of  $10^{-5}/^\circ\text{C}$  lies in a vertical plane. The height of liquid column in the left vertical limb is 100 cm. The liquid in the left vertical limb is maintained at a temperature =  $0^\circ\text{C}$  while the liquid in the right limb is maintained at a temperature =  $100^\circ\text{C}$ . Find the difference in levels in the two limbs.

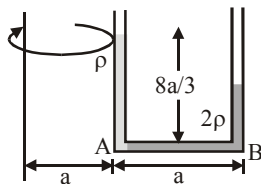
FM0038

14. There are three immiscible liquids of densities  $\rho_1$ ,  $\rho_2$  and  $\rho_3$ . If the first two liquids are taken in a U-tube then they stand like fig(i). When all the three liquids are taken in U-tube, they stand like fig(ii). Find the ratio of the densities of the three liquids.



FM0039

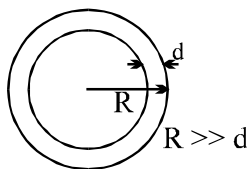
15. The interface of two liquids of densities  $\rho$  and  $2\rho$  respectively lies at the point A in a U tube at rest. The height of liquid column above A is  $8a/3$  where  $AB = a$ . The cross sectional area of the tube is  $S$ . With what angular velocity the tube must be whirled about a vertical axis at a distance 'a' such that the interface of the liquids shifts towards B by  $2a/3$ .

**FM0040**

- 16.** An expansible balloon filled with air floats on the surface of a lake with  $2/3$  of its volume submerged. How deep must it be sunk in the water so that it is just in equilibrium neither sinking further nor rising? It is assumed that the temperature of the water is constant & that the height of the water barometer is 9 meters.

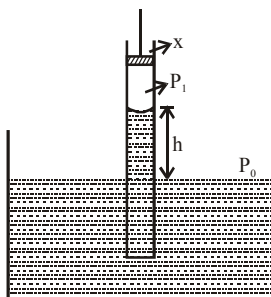
**FM0041**

17. A soap bubble has radius  $R$  and thickness  $d$  ( $d \ll R$ ) as shown. It collapses into a spherical drop. Find the ratio of excess pressure in the drop to the excess pressure inside the bubble.



FM0042

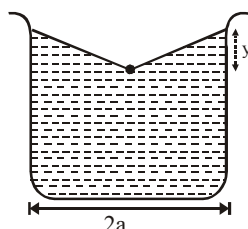
- 18.** A capillary of radius  $r = 0.2$  mm is dipped vertically in a liquid of density, ( $\rho = 1 \text{ gm/cc}$ ). A small piston (x) is inserted inside the capillary which maintains the constant pressure of air ( $P_1 = 2 \times 10^4 \text{ dynes/cm}^2$ ) above the hemispherical meniscus. If ambient pressure is  $P_0 = 10^5 \text{ dynes/cm}^2$  and surface tension of liquid is  $T = 72 \text{ dynes/cm}$ , then find the height  $h$  of liquid in cm. (weight of curved part of liquid is ignored)



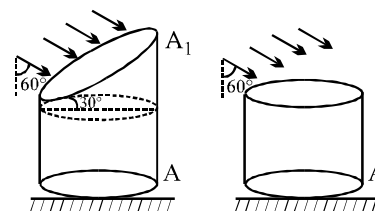
FM0043

19. A container of width  $2a$  is filled with a liquid. A thin wire of weight per unit length  $\lambda$  is gently placed over the liquid surface in the middle of the surface as shown in the figure. As a result, the liquid surface is depressed by a distance  $y$  ( $y \ll a$ ). Determine the surface tension of the liquid.

[2004, 2M]



20. A vertical cylindrical container of base area  $A$  and upper cross-section area  $A_1$  making an angle  $30^\circ$  with the horizontal is placed in an open rainy field as shown near another cylindrical container having same base area  $A$ . Find the ratio of rates of collection of water in the two containers.

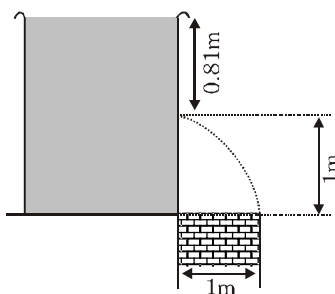


FM0044

21. A wide vessel with small hole in the bottom is filled with water and kerosene. Neglecting viscosity, find the velocity of water flow  $v$ , if the thickness of water layer is  $h_1$  and that of kerosene layer is  $h_2$  is (density of water  $\rho_1$  gm/cc and that of kerosene is  $\rho_2$  gm/cc).

FM0045

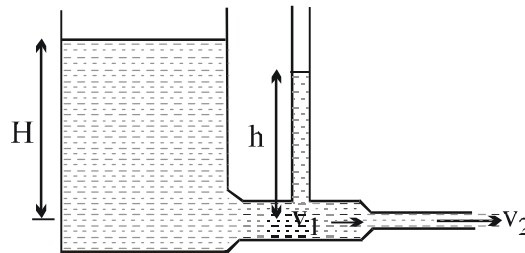
22. For the arrangement shown in the figure. The time interval in seconds after which the water jet ceases to cross the wall is found to be  $(\alpha)^3$ . Area of the cross section of the tank  $A = \sqrt{5} \text{ m}^2$  and area of the orifice  $a = 32 \text{ cm}^2$ . [Assume that the container remaining fixed]. Find the value of  $\alpha$ . (Take  $g = 10 \text{ m/s}^2$ )



FM0047



23. Water fills a reservoir, open to the atmosphere, to height  $H = 5\text{m}$  above the centerline of a horizontal exit pipe at the bottom of the reservoir. The first section of the pipe has radius  $r_1 = 2\text{ cm}$ , the unknown velocity there is  $v_1$  and this section of the pipe has a manometer in which the water rises to an unknown height  $h$  above the centerline of the pipe. The water leaves the second section of radius  $r_2 = 1\text{ cm}$ , with speed  $v_2$ . Assume the flow is incompressible, frictionless, irrotational, and steady. Also, assume the reservoir is so large compared to the pipe, that the water level in the reservoir is almost constant.

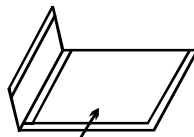


- What is the speed,  $v_2$ , of the water leaving the exit pipe?
- What is the speed,  $v_1$ , in the first section of the pipe?
- What is the height,  $h$ , of the water in the manometer?
- What is the volume flow rate of the water flowing through the exit pipe?

FM0048

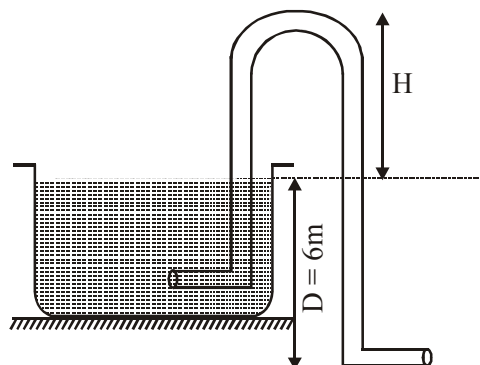
24. A glass plate of area  $A$  and mass  $m$  is hinged along one of its sides. At what speed the air should be blown parallel to its upper surface to hold the plate horizontal? The density of air is  $\rho$ . Give the

answer in term of  $\sqrt{\frac{mg}{18\rho A}}$



FM0049

25. Figure shows a siphon, the vessel area is very large as compared to cross section of tube. Tube has a uniform cross section, its lower end is  $6\text{m}$  below the surface of water. What is maximum height  $H$  (in m) of the upper end for siphon to work? Take density of water  $= 10^3\text{ kg/m}^3$ , atmospheric pressure  $= 10^5\text{ N/m}^2$ .



FM0050

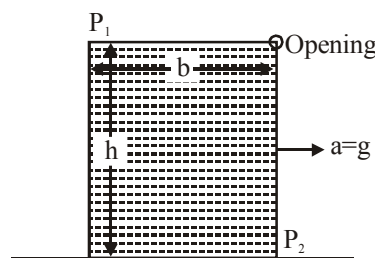
## EXERCISE (O-1)

### SINGLE CORRECT TYPE QUESTIONS

#### Fluid Statics

1. A vessel filled with a liquid of density  $\rho$  falls vertically downwards with an acceleration  $a (< g)$ . The gauge pressure  $P$  at depth  $h$  below the free surface of liquid is :  
 (A)  $P = h\rho(g + a)$       (B)  $P = h\rho g$       (C)  $P = h\rho(g - a)$       (D)  $P = h\rho a$
2. A closed rectangular vessel completely filled with a liquid of density  $\rho$  moves with an acceleration  $a = g$ . The value of the pressure difference ( $P_1 - P_2$ ) is :

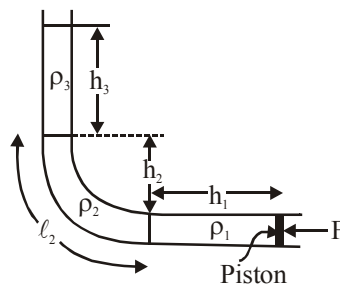
FM0051



- (A)  $\rho gb$       (B)  $\frac{\rho g(b+h)}{2}$       (C)  $\rho(ab - gh)$       (D)  $\rho gh$

FM0052

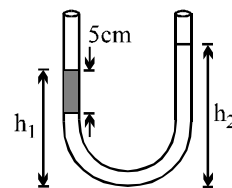
3. A tube is bent into L shape and kept in a vertical plane. If these three liquids are kept in equilibrium by the piston of area  $A$ , the value of  $\frac{F}{A}$  is :



- (A)  $(\rho_1 h_1 + \rho_2 h_2 + \rho_3 h_3)g$       (B)  $(\rho_1 h_1 + \rho_2 l_2 + \rho_3 h_3)g$   
 (C)  $(\rho_2 h_2 + \rho_3 h_3)g$       (D)  $(\rho_2 l_2 + \rho_3 h_3)g$

FM0053

4. An open-ended U-tube of uniform cross-sectional area contains water (density  $1.0 \text{ gram/centimeter}^3$ ) standing initially 20 centimeters from the bottom in each arm. An immiscible liquid of density  $4.0 \text{ grams/centimeter}^3$  is added to one arm until a layer 5 centimeters high forms, as shown in the figure above. What is the ratio  $h_2/h_1$  of the heights of the liquid in the two arms?
- (A)  $3/1$                       (B)  $5/2$                       (C)  $2/1$                       (D)  $3/2$



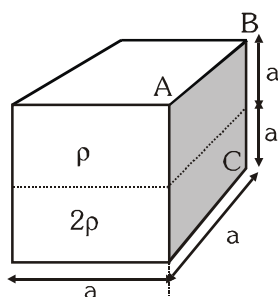
FM0054

5. A tube of length  $L$  is filled completely with incompressible liquid of mass  $M$  and closed at both the ends. The tube is then rotated in a horizontal plane about one of its ends with a uniform angular velocity  $\omega$ . The force exerted by the liquid on the tube at other end is :-

- (A)  $\frac{M\omega^2 L}{2}$       (B)  $M\omega^2 L$       (C)  $\frac{M\omega^2 L}{4}$       (D)  $\frac{3M\omega^2 L}{4}$

**FM0055**

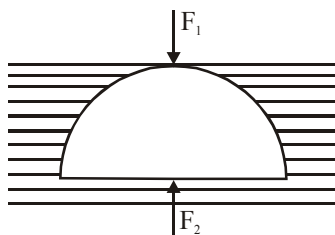
6. A cuboid ( $a \times a \times 2a$ ) is filled with two immiscible liquids of density  $2\rho$  &  $\rho$  as shown in the figure. Neglecting atmospheric pressure, ratio of force on base & side wall of the cuboid is :-



- (A) 2 : 3                      (B) 1 : 3                      (C) 5 : 6                      (D) 6 : 5

FM0056

7. A solid hemisphere is just pressed below the liquid, the value of  $\frac{F_1}{F_2}$  is (where  $F_1$  and  $F_2$  are the hydrostatic forces acting on the curved and flat surfaces of the hemisphere) (Neglect atmospheric pressure).



- (A)  $\frac{1}{2}$                       (B)  $\frac{2}{3}$                       (C)  $\frac{1}{3}$                       (D) none of these

FM0057

8. A thin walled cylindrical metal vessel of linear coefficient of expansion  $10^{-3} \text{ }^{\circ}\text{C}^{-1}$  contains benzene of volume expansion coefficient  $10^{-3} \text{ }^{\circ}\text{C}^{-1}$ . If the vessel and its contents are now heated by  $10^{\circ}\text{C}$ , the pressure due to the liquid at the bottom.

- (A) increases by 2%                      (B) decreases by 1%  
(C) decreases by 2%                      (D) remains unchanged

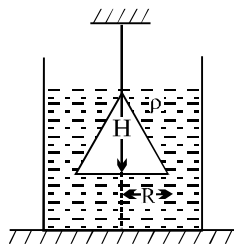
FM0058

9. A cork of density  $0.5\text{gcm}^{-3}$  floats on a calm swimming pool. The fraction of the cork's volume which is under water is :-  
 (A) 0% (B) 25% (C) 10% (D) 50%

FM0059

**Buoyancy**

10. A cone of radius  $R$  and height  $H$ , is hanging inside a liquid of density  $\rho$  by means of a string as shown in the figure. The force, due to the liquid acting on the slant surface of the cone is (Neglect atmosphere pressure)

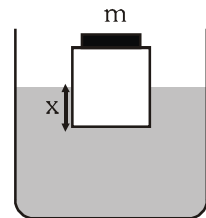


- (A)  $\rho\pi gHR^2$  (B)  $\pi\rho HR^2$  (C)  $\frac{4}{3}\pi\rho gHR^2$  (D)  $\frac{2}{3}\pi\rho gHR^2$

FM0060

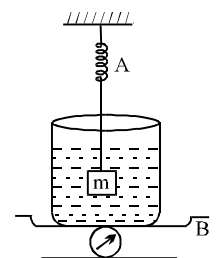
11. A cubical block ( $a \times a \times a$ ), with a coin of mass ' $m$ ' over it is floating over a liquid of density  $\rho$ . In this case  $x_1$  depth of the block is immersed. Now the coin is removed & it is found that  $x_2$  depth is immersed in liquid. Value of  $(x_1 - x_2)$  is :-

- (A)  $\frac{m}{\rho a^2}$  (B)  $\frac{\rho a^4}{m}$   
 (C)  $\frac{m}{2\rho a^2}$  (D) data insufficient



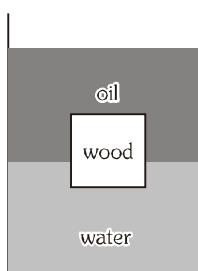
FM0061

12. The spring balance A reads 2 kg with a block  $m$  suspended from it. A balance B reads 5 kg when a beaker with liquid is put on the pan of the balance. The two balances are now so arranged that the hanging mass is inside the liquid in the beaker as shown in the figure in this situation:  
 (A) the balance A will read more than 2 kg  
 (B) the balance B will read more than 5 kg  
 (C) the balance A will read less than 2 kg and B will read less than 5 kg  
 (D) the balances A and B will read 2 kg and 5 kg respectively.



FM0062

13. A cubical block of wood of side of 10 cm, floats at the interface between oil and water as shown in figure with its lower face 2 cm below the interface. The density of oil is  $0.6 \text{ g/cm}^3$ . The mass of the block is



- (A) 600 g (B) 680 g (C) 800 g (D) 200 g

FM0063

14. A solid floats with  $2/3$  of its volume immersed in a liquid and with  $3/4$  of its volume immersed in another liquid. What fraction of its volume will be immersed if it floats in a homogenous mixture formed of equal volumes of the liquids?

- (A)  $6/7$  (B)  $8/11$  (C)  $11/16$  (D)  $12/17$

FM0064

15. A small ball of relative density 0.8 falls into water from a height of 2m. The depth to which the ball will sink is (neglect viscous forces):

- (A) 8 m (B) 2 m (C) 6 m (D) 4 m

FM0065

16. An object of density  $d_0$  kept deep inside water of density  $d_w$  and released. During the time it moves a vertical distance  $h$  within the water :-

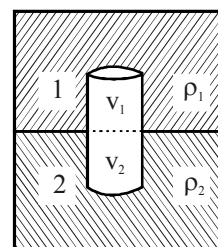
- (A) The gravitational potential energy of the water in the vessel increases if  $d_0 < d_w$   
 (B) The gravitational potential energy of the water in the vessel decreases if  $d_0 < d_w$   
 (C) The gravitational potential energy of the object increases if  $d_0 > d_w$   
 (D) The gravitational potential energy of the object decreases if  $d_0 < d_w$

FM0066

17. **Statement-1** : When a body floats such that its parts are immersed into two immiscible liquids then force exerted by liquid-1 is of magnitude  $\rho_1 v_1 g$ .

**Statement-2** : Total Buoyant force  $= \rho_1 v_1 g + \rho_2 v_2 g$

- (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.  
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.  
 (C) Statement-1 is true, statement-2 is false.  
 (D) Statement-1 is false, statement-2 is true.



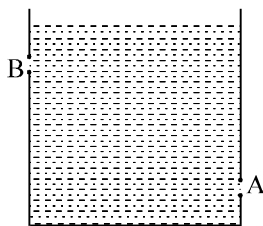
FM0067

18. **Statement-1** : Submarine sailors are advised that they should not allow it to rest on floor of the ocean.  
**Statement-2** : The force exerted by a liquid on a submerged body may be downwards.  
 (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.  
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.  
 (C) Statement-1 is true, statement-2 is false.  
 (D) Statement-1 is false, statement-2 is true.

FM0068

**Bernoulli**

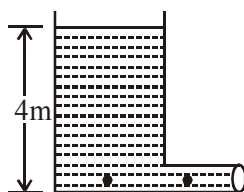
19. Water is flowing steadily through a horizontal tube of non uniform cross-section. If the pressure of water is  $4 \times 10^4 \text{ N/m}^2$  at a point where cross-section is  $0.02 \text{ m}^2$  and velocity of flow is  $2 \text{ m/s}$ , what is pressure at a point where cross-section reduces to  $0.01 \text{ m}^2$ .  
 (A)  $1.4 \times 10^4 \text{ N/m}^2$  (B)  $3.4 \times 10^4 \text{ N/m}^2$  (C)  $2.4 \times 10^4 \text{ N/m}^2$  (D) none of these
20. In the case of a fluid, Bernoulli's theorem expresses the application of the principle of conservation of :-  
 (A) linear momentum (B) energy (C) mass (D) angular momentum
21. Two water pipes P and Q having diameters  $2 \times 10^{-2} \text{ m}$  and  $4 \times 10^{-2} \text{ m}$ , respectively, are joined in series with the main supply line of water. The velocity of water flowing in pipe P is :-  
 (A) 4 times that of Q (B) 2 times that of Q  
 (C)  $1/2$  times of that of Q (D)  $1/4$  times that of Q
22. A cylinder of height  $20 \text{ m}$  is completely filled with water. The velocity of efflux of water (in  $\text{ms}^{-1}$ ) through a small hole on the side wall of the cylinder near its bottom, is- [AIEEE - 2002]  
 (A) 10 (B) 20 (C) 25.5 (D) 5
23. Two identical holes each of cross-sectional area  $10^{-3} \text{ m}^2$  are made on the opposite sides of a tank containing water as shown in the figure. As the water comes out of the holes, the tank will experience a net horizontal force of  $20 \text{ N}$ . The difference in height between the holes A and B is.



- (A)  $1 \text{ m}$  (B)  $0.5 \text{ m}$  (C)  $2 \text{ m}$  (D)  $0.25 \text{ m}$

FM0073

24. A vent tank of large cross-sectional area has a horizontal pipe 0.12 m in diameter at the bottom. This holds a liquid whose density is  $1500 \text{ kg/m}^3$  to a height of 4.0 m. Assume the liquid is an ideal fluid in laminar flow. In figure, the velocity with which fluid flows out is :-

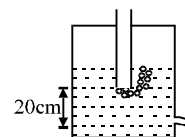


- (A)  $2\sqrt{5} \text{ m/s}$  (B)  $\sqrt{5} \text{ m/s}$  (C)  $4\sqrt{5} \text{ m/s}$  (D)  $\sqrt{10} \text{ m/s}$

FM0074

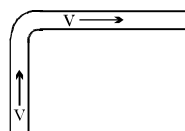
25. A tube is attached as shown in closed vessel containing water. The velocity of water coming out from a small hole is :

- (A)  $\sqrt{2} \text{ m/s}$   
 (B)  $2 \text{ m/s}$   
 (C) depends on pressure of air inside vessel  
 (D) None of these



FM0075

26. A fire hydrant delivers water of density  $\rho$  at a volume rate  $L$ . The water travels vertically upward through the hydrant and then does  $90^\circ$  turn to emerge horizontally at speed  $V$ . The pipe and nozzle have uniform cross-section throughout. The force exerted by the water on the corner of the hydrant is :-



- (A)  $\rho VL$  (B) zero (C)  $2\rho VL$  (D)  $\sqrt{2}\rho VL$

FM0076

27. A large tank is filled with water to a height  $H$ . A small hole is made at the base of the tank. It takes  $T_1$  time to decrease the height of water to  $H/\eta$ , ( $\eta > 1$ ) and it takes  $T_2$  time to take out the rest of water. If  $T_1 = T_2$ , then the value of  $\eta$  is :

- (A) 2 (B) 3 (C) 4 (D)  $2\sqrt{2}$

FM0077

28. Water is filled in a container upto height 3m. A small hole of area 'a' is punched in the wall of the container at a height 52.5 cm from the bottom. The cross sectional area of the container is A. If

$\frac{a}{A} = 0.1$  then  $v^2$  is (where  $v$  is the velocity of water coming out of the hole) [IIT-JEE' 2005 (Scr)]

- (A) 48 (B) 51 (C) 50 (D) 51.5

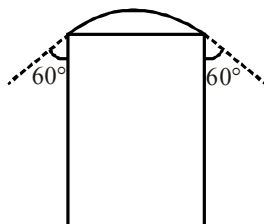
FM0078

Surface Tension

29. If two soap bubbles of different radii are connected by a tube,  
 (A) air flows from the bigger bubble to the smaller bubble till the sizes become equal  
 (B) air flows from bigger bubble to the smaller bubble till the sizes are interchanged  
 (C) air flows from the smaller bubble to the bigger  
 (D) there is no flow of air.

FM0079

30. A soap bubble is being blown on a tube of radius 1 cm. The surface tension of the soap solution is 0.05 N/m and the bubble makes an angle of  $60^\circ$  with the tube as shown. The excess of pressure over the atmospheric pressure in the tube is :



- (A) 5 Pa (B) 1 Pa (C) 10 Pa (D) 20 Pa

FM0080

31. When an air bubble rises from the bottom of a deep lake to a point just below the water surface, the pressure of air inside the bubble :-

- (A) is greater than the pressure outside it (B) is less than the pressure outside it  
 (C) increases as the bubble moves up (D) remains same as the bubble moves up

FM0081

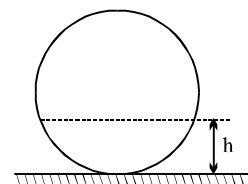
32. Two mercury drops (each of radius 'r') merge to form a bigger drop. The surface energy of the bigger drop, if  $\frac{1}{\pi}$  is the surface tension (in SI unit), is :

- (A)  $2^{5/3} r^2$  (B)  $4r^2$  (C)  $2r^2$  (D)  $2^{8/3} r^2$

FM0082

33. A liquid is filled in a spherical container of radius R till a height h. At this position the liquid surface at the edges is also horizontal. The contact angle is :-

- (A) 0 (B)  $\cos^{-1}\left(\frac{R-h}{R}\right)$   
 (C)  $\cos^{-1}\left(\frac{h-R}{R}\right)$  (D)  $\sin^{-1}\left(\frac{R-h}{R}\right)$



FM0083

34. An open capillary tube is lowered in a vessel with mercury. The difference between the levels of the mercury in the vessel and in the capillary tube  $\Delta h = 4.6\text{mm}$ . What is the radius of curvature of the mercury meniscus in the capillary tube? Surface tension of mercury is 0.46 N/m, density of mercury is 13.6 gm/cc.

- (A)  $\frac{1}{340} \text{ m}$  (B)  $\frac{1}{680} \text{ m}$   
 (C)  $\frac{1}{1020} \text{ m}$  (D) Information insufficient

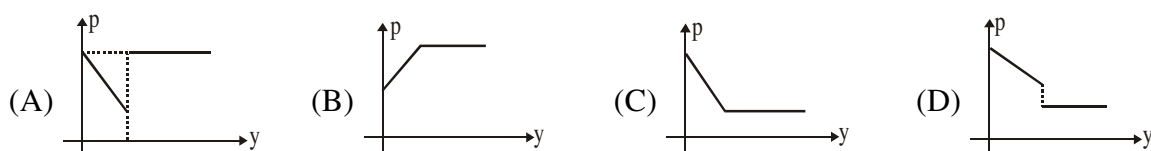
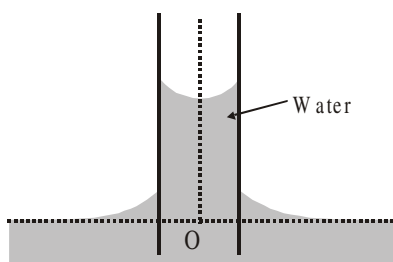
FM0084



35. A container, whose bottom has round holes with diameter 0.1 mm is filled with water. The maximum height in cm upto which water can be filled without leakage will be what?  
Surface tension =  $75 \times 10^{-3}$  N/m and  $g = 10$  m/s<sup>2</sup>:  
(A) 20 cm (B) 40 cm (C) 30 cm (D) 60 cm

FM0085

36. Water rises in a capillary as shown. The correct graph of pressure ( $p$ ) vs height ( $y$ ) above point O along the axis of capillary is :-



FM0086

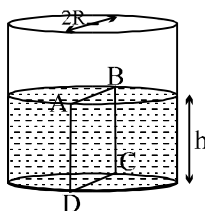
37. If two soap bubbles of different radii are connected by a tube- [AIEEE - 2004]  
(A) air flows from the bigger bubble to the smaller bubble till the sizes become equal  
(B) air flows from bigger bubble to the smaller bubble till the sizes are interchanged  
(C) air flows from the smaller bubble to the bigger bubble  
(D) there is no flow of air

FM0087

38. A 20 cm long capillary tube is dipped in water. The water rises upto 8 cm. If the entire arrangement is put in a freely falling elevator, the length of water column in the capillary tube will be- [AIEEE - 2005]  
(A) 8 cm (B) 10 cm (C) 4 cm (D) 20 cm

FM0088

39. Water is filled up to a height  $h$  in a beaker of radius  $R$  as shown in the figure. The density of water is  $\rho$ , the surface tension of water is  $T$  and the atmospheric pressure is  $P_0$ . Consider a vertical section ABCD of the water column through a diameter of the beaker. The force on water on one side of this section by water on the other side of this section has magnitude [IIT-JEE 2007]



- (A)  $|2P_0Rh + \pi R^2 \rho gh - 2RT|$  (B)  $|2P_0Rh + R\rho gh^2 - 2RT|$   
(C)  $|P_0\pi R^2 + R\rho gh^2 - 2RT|$  (D)  $|P_0\pi R^2 + R\rho gh^2 + 2RT|$

FM0089

**Viscosity**

40. Two drops of same radius are falling through air with steady velocity of  $v$  cm/s. If the two drops coalesce, what would be the terminal velocity?  
 (A)  $4v$  (B)  $(4)^{1/3}v$  (C)  $2v$  (D)  $64v$

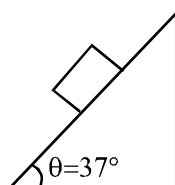
FM0090

41. Spherical balls of radius  $R$  are falling in a viscous fluid of viscosity  $\eta$  with a velocity  $v$ . the retarding viscous force acting on the spherical ball is-  
 (A) directly proportional to  $R$  but inversely proportional to  $v$   
 (B) directly proportional to both radius  $R$  and velocity  $v$   
 (C) inversely proportional to both radius  $R$  and velocity  $v$   
 (D) inversely proportional to  $R$  but directly proportional to velocity  $v$

[AIEEE - 2004]

FM0091

42. A cubical block of side ' $a$ ' and density ' $\rho$ ' slides over a fixed inclined plane with constant velocity ' $v$ '. There is a thin film of viscous fluid of thickness ' $t$ ' between the plane and the block. Then the coefficient of viscosity of the thin film will be:



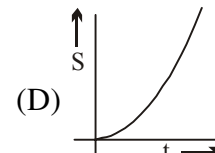
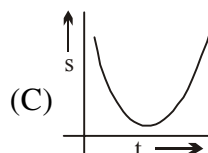
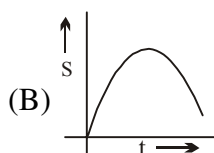
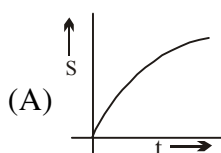
- (A)  $\frac{3\rho a g t}{5v}$  (B)  $\frac{4\rho a g t}{5v}$  (C)  $\frac{\rho a g t}{v}$  (D) none of these

FM0092

43. There is a 1mm thick layer of glycerine between a flat plate of area  $100 \text{ cm}^2$  & a big fixed plate. If the coefficient of viscosity of glycerine is  $1.0 \text{ kg/m-s}$  then how much force is required to move the plate with a velocity of  $7 \text{ cm/s}$ ?  
 (A)  $3.5 \text{ N}$  (B)  $0.7 \text{ N}$  (C)  $1.4 \text{ N}$  (D) None

FM0093

44. The displacement of a ball falling from rest in a viscous medium is plotted against time. Choose a possible option



FM0094

45. An air bubble of radius  $1 \text{ cm}$  is found to rise in a cylindrical vessel of large radius at a steady rate of  $0.2 \text{ cm per second}$ . If the density of the liquid is  $1470 \text{ kg m}^{-3}$ , then coefficient of viscosity of liquid is approximately equal to  
 (A)  $163 \text{ poise}$  (B)  $163 \text{ centi-poise}$  (C)  $140 \text{ poise}$  (D)  $140 \text{ centi-poise}$

FM0095

46. **Statement-1** : The free surface of a liquid at rest with respect to stationary container is always normal to the  $\vec{g}$ .

**Statement-2** : Liquids at rest cannot have shear stress.

- (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.  
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.  
 (C) Statement-1 is true, statement-2 is false.  
 (D) Statement-1 is false, statement-2 is true.

FM0096

47. If the terminal speed of a sphere of gold (density =  $19.5 \text{ kg/m}^3$ ) is  $0.2 \text{ m/s}$  in a viscous liquid (density =  $1.5 \text{ kg/m}^3$ ), find the terminal speed of a sphere of silver (density =  $10.5 \text{ kg/m}^3$ ) of the same size in the same liquid.

[AIEEE- 2006]

- (A)  $0.4 \text{ m/s}$  (B)  $0.133 \text{ m/s}$  (C)  $0.1 \text{ m/s}$  (D)  $0.2 \text{ m/s}$

FM0097

### MULTIPLE CORRECT TYPE QUESTIONS

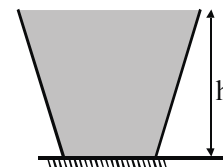
#### Fluid Statics

48.  $M \text{ gm}$  of a liquid of density  $\rho$  is filled in a light beaker and kept on a horizontal table as shown in the figure. The height of the liquid in the beaker is  $h$ . The beaker is wider on top than at its base and the cross-sectional area of the base is  $A$ . Neglect the effect of atmospheric pressure. Now, choose the **CORRECT** statement(s) from the following.

- (A) The pressure of liquid at the bottom surface is  $\rho gh$ .  
 (B) The normal reaction exerted by the table on the beaker is  $\rho ghA$ .

- (C) The pressure of the liquid at the bottom surface is  $\frac{Mg}{A}$ .

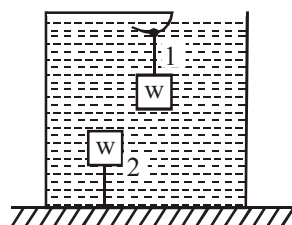
- (D) The normal reaction exerted by the table on the beaker is  $Mg$ .



FM0098

49. An iron block and a wooden block are positioned in a vessel containing water as shown in the figure. The iron block (1) hangs from a massless string with a rigid support from the top while the wooden block (2) floats being tied to the bottom through a massless string. If now the vessel starts acceleration towards right.

- (A) iron block gets deflected towards right.  
 (B) wooden block gets deflected towards right.  
 (C) iron block gets deflected towards left.  
 (D) wooden block gets deflected towards left.



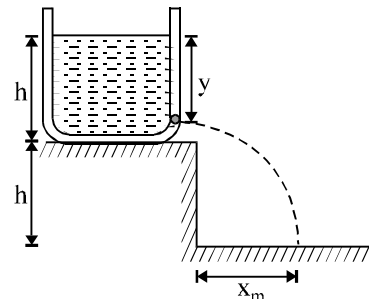
FM0099

50. A beaker is filled in with water is accelerated  $a$   $\text{m/s}^2$  in  $+x$  direction. The surface of water shall make an angle
- (A)  $\tan^{-1}(a/g)$  backwards (B)  $\tan^{-1}(a/g)$  forwards  
(C)  $\cot^{-1}(g/a)$  backwards (D)  $\cot^{-1}(g/a)$  forwards

FM0100

**Fluid Dynamics**

51. A tank is filled upto a height  $h$  with a liquid and is placed on a platform of height  $h$  from the ground. To get maximum range  $x_m$  a small hole is punched at a distance of  $y$  from the free surface of the liquid. Then
- (A)  $x_m = 2h$  (B)  $x_m = 1.5h$   
(C)  $y = h$  (D)  $y = 0.75h$



FM0101

**MATRIX MATCH TYPE QUESTION****Fluid Statics**

52. Bucket A contains only water ; an identical bucket B contains water, but also contains a solid object in the water. Consider the following four situations. Which bucket weighs more ?

**Column-I**

- (A) The object floats in bucket B, and the buckets have the same water level  
(B) The object floats in bucket B, and the buckets have the same volume of water  
(C) The object sinks completely in bucket B, and the buckets have the same water level  
(D) The object sinks completely in bucket B, and the buckets have the same volume of water

**Column-II**

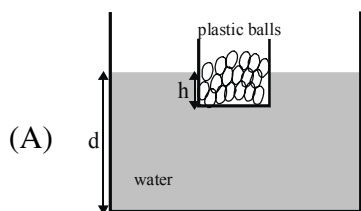
- (P) Bucket A  
(Q) Bucket B  
(R) Both buckets have the same weight  
(S) The answer cannot be determined from the information given.

FM0102

**53.** Column I shows different system as describe, with some parameter while column-II gives the change in the parameter.

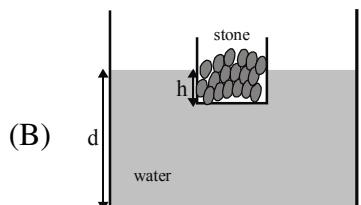
### Column-I

**Column-II**



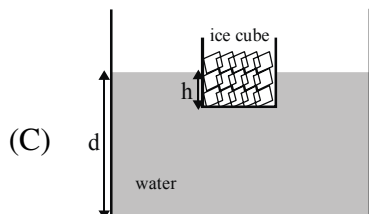
Plastic ball is thrown from the container  
in the water.

[d is the height of water level while h is the depth to which the container is submerged]



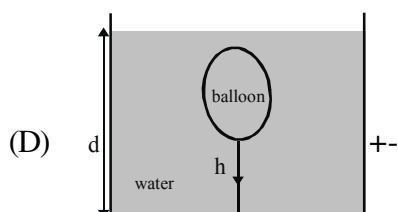
Stone is thrown from the container  
in the water.

[d is the height of water level while h is the depth to which the container is submerged]



Ice melts and remain in the container.

[d is the height of water level while h is the depth to which the container is submerged]



The water is heated slowly.

[d is the height of water level while h is the tension in the string.  $\gamma_{\text{air}} > \gamma_{\text{water}}$ ]

(P)  $h$  decreases

(Q) h remain same

(R)  $h$  increases

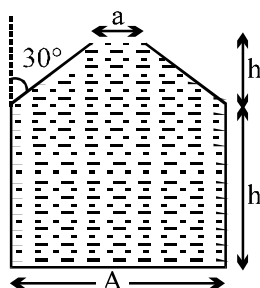
(S)  $d$  decreases

(T) d remain same

## EXERCISE (O-2)

## SINGLE CORRECT TYPE QUESTIONS

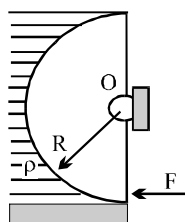
1. The vessel shown in the figure has two sections. The lower part is a rectangular vessel with area of cross-section  $A$  and height  $h$ . The upper part is a conical vessel of height  $h$  with base area ' $A$ ' and top area ' $a$ ' and the walls of the vessel are inclined at an angle  $30^\circ$  with the vertical. A liquid of density  $\rho$  fills both the sections up to a height  $2h$ . Neglecting atmospheric pressure.



- (A) The force  $F$  exerted by the liquid on the base of the vessel is  $2h\rho g \frac{(A+a)}{2}$   
 (B) the pressure  $P$  at the base of the vessel is  $2h\rho g \frac{A}{a}$   
 (C) the weight of the liquid  $W$  is greater than the force exerted by the liquid on the base  
 (D) the walls of the vessel exert a downward force  $(F-W)$  on the liquid.

FM0104

2. A light semi cylindrical gate of radius  $R$  is pivoted at its mid point  $O$ , of the diameter as shown in the figure holding liquid of density  $\rho$ . The force  $F$  required to prevent the rotation of the gate is equal to



- (A)  $2\pi R^3 \rho g$  (B)  $2\rho g R^3 l$  (C)  $\frac{2R^2 l \rho g}{3}$  (D) none of these

FM0105

3. A heavy hollow cone of radius  $R$  and height  $h$  is placed on a horizontal table surface, with its flat base on the table. The whole volume inside the cone is filled with water of density  $\rho$ . The circular rim of the cone's base has a watertight seal with the table's surface and the top apex of the cone has a small hole. Neglecting atmospheric pressure find the total upward force exerted by water on the cone is

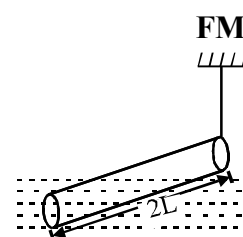
(A)  $(2/3)\pi R^2 h \rho g$  (B)  $(1/3)\pi R^2 h \rho g$  (C)  $\pi R^2 h \rho g$  (D) None

FM0106

4. Two cubes of size 1.0 m sides, one of relative density 0.60 and another of relative density = 1.15 are connected by weightless wire and placed in a large tank of water. Under equilibrium the lighter cube will project above the water surface to a height of

(A) 50 cm (B) 25 cm (C) 10 cm (D) zero

5. A slender homogeneous rod of length  $2L$  floats partly immersed in water, being supported by a string fastened to one of its ends, as shown. The specific gravity of the rod is 0.75. The length of rod that extends out of water is :



FM0107

(A)  $L$  (B)  $\frac{1}{2} L$  (C)  $\frac{1}{4} L$  (D)  $3 L$

FM0108

6. Mercury is poured into a uniform vertical U-tube and water is poured in above it. The level of water is the same in both arms. A piece of wood is dropped into one arm and some water equal in weight to the piece of wood is added to the other. Then, consider the statements :

(I) The level of mercury will be same in both the arms.

(II) The level of water will be same in both the arms.

(A) Both I and II are true

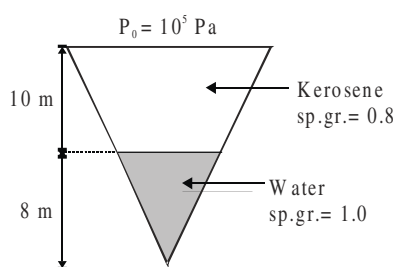
(B) I is true but II is false

(C) I is false but II is true

(D) Both I and II are false.

FM0109

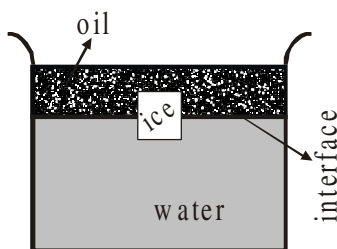
7. Shown in figure, a conical container of half-apex angle  $37^\circ$  filled with certain quantities of kerosene and water. The force exerted by the water on the kerosene is approximately, (Take atmospheric pressure =  $10^5$  Pa)



(A)  $3 \times 10^7$  N (B)  $4 \times 10^7$  N (C)  $2 \times 10^7$  N (D)  $5 \times 10^7$  N

FM0110

8. An ice cube is floating in water above which a layer of a lighter oil is poured. As the ice melts completely, the level of interface and the upper most level of oil will respectively :-

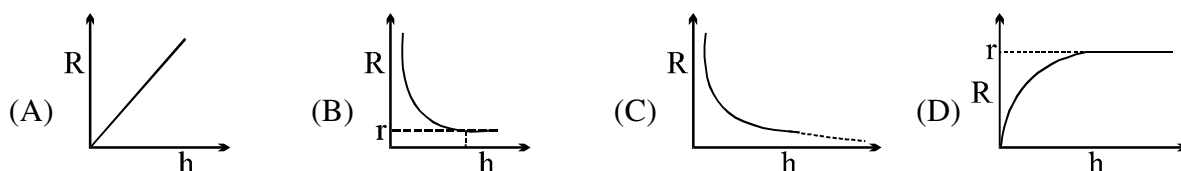


- (A) rise and fall      (B) fall and rise      (C) rise and not change      (D) not change and fall

FM0111

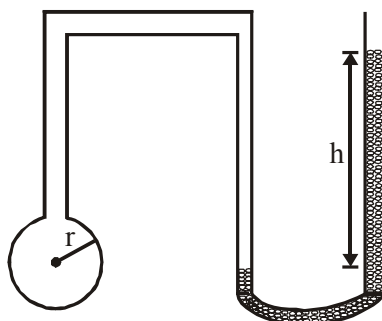
### Statics

9. A long capillary tube of radius 'r' is initially just vertically completely immersed inside a liquid of angle of contact  $0^\circ$ . If the tube is slowly raised then relation between radius of curvature of meniscus inside the capillary tube and displacement (h) of tube can be represented by



FM0112

10. If the radius of the Soap-bubble on one side of tube is r and difference in height of liquid of density  $\rho$  in manometer is h, then surface tension of liquid used to make the bubble is :-



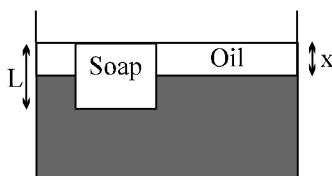
- (A)  $T = 2r\rho hg$       (B)  $T = \frac{r\rho hg}{4}$       (C)  $T = \frac{2\pi r\rho hg}{2}$       (D)  $T = \frac{r\rho hg}{2}$

FM0113



### Surface tension

11. A rectangular bar of soap has density  $800 \text{ kg/m}^3$  floats in water of density  $1000 \text{ kg/m}^3$ . Oil of density  $300 \text{ kg/m}^3$  is slowly added, forming a layer that does not mix with the water. When the top surface of the oil is at the same level as the top surface of the soap. What is the ratio of the oil layer thickness to the soap's thickness,  $x/L$  ?



- (A)  $\frac{2}{10}$                       (B)  $\frac{2}{7}$                       (C)  $\frac{3}{10}$                       (D)  $\frac{3}{8}$

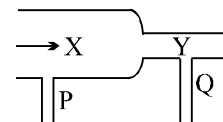
FM0114

12. Water coming out of a horizontal tube at a speed  $v$  strikes normally a vertically wall close to the mouth of the tube and falls down vertically after impact. When the speed of water is increased to  $2v$ .
- (A) the thrust exerted by the water on the wall will be doubled
- (B) the thrust exerted by the water on the wall will be unchanged
- (C) the energy lost per second by water striking the wall will also be four times
- (D) the energy lost per second by water striking the wall be increased eight times.

FM0115

13. A steady flow of water passes along a horizontal tube from a wide section X to the narrower section Y, see figure. Manometers are placed at P and Q at the sections. Which of the statements A, B, C, D, is most correct?

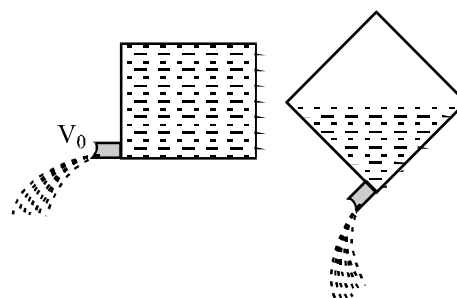
- (A) water velocity at X is greater than at Y  
(B) the manometer at P shows lower pressure than at Q  
(C) kinetic energy per  $\text{m}^3$  of water at X = kinetic energy per  $\text{m}^3$  at Y  
(D) the manometer at P shows greater pressure than at Q



FM0116

14. A cubical box of wine has a small spout located in one of the bottom corners. When the box is full and placed on a level surface, opening the spout results in a flow of wine with a initial speed of  $v_0$  (see figure). When the box is half empty, someone tilts it at  $45^\circ$  so that the spout is at the lowest point (see figure). When the spout is opened the wine will flow out with a speed of

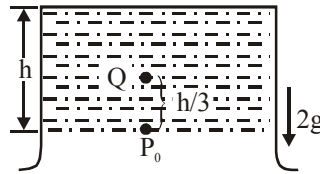
- (A)  $v_0$                       (B)  $v_0/2$                       (C)  $v_0/\sqrt{2}$                       (D)  $v_0/\sqrt[4]{2}$



FM0117

Bernoulli's

15. An inverted test tube having a liquid of density  $\rho$  accelerates down with  $a = 2g$  then,  $P_Q - P_0$  is:



- (A)  $-\frac{\rho gh}{3}$  (B)  $\frac{2\rho gh}{3}$  (C)  $\frac{\rho gh}{3}$  (D) indeterminate

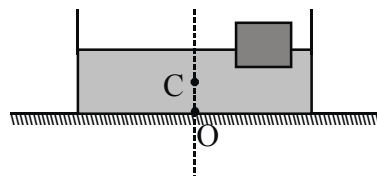
FM0118

16. Some liquid is filled in a cylindrical vessel of radius  $R$ . Let  $F_1$  be the force applied by the liquid on the bottom of the cylinder. Now the same liquid is poured into a vessel of uniform square cross-section of side  $R$ . Let  $F_2$  be the force applied by the liquid on the bottom of this new vessel. (Neglect atmospheric pressure) Then:

- (A)  $F_1 = \pi F_2$  (B)  $F_1 = \frac{F_2}{\pi}$  (C)  $F_1 = \sqrt{\pi} F_2$  (D)  $F_1 = F_2$

FM0119

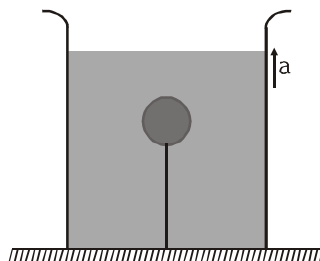
17. There is water in container with center of mass at  $C$ . Now a small wooden piece is placed towards right as shown in the figure. After putting the wooden piece.



- (A) Pressure at base remains same and centre of mass of water and wooden piece will be right of line OC.  
 (B) Pressure at base remains same and centre of mass of water and wooden piece will be on line OC.  
 (C) Pressure at base changes and centre of mass of water and wooden piece will be right of line OC.  
 (D) Pressure at base changes and centre of mass of water and wooden piece will be on line OC.

FM0120

18. The tension in a string holding a solid block below the surface of a liquid (of density greater than that of solid) as shown in figure is  $T_0$  when the system is at rest. What will be the tension in the string if the system has an upward acceleration  $a$ ?



- (A)  $T_0 \left(1 + \frac{a}{g}\right)$  (B)  $T_0 \left(1 - \frac{a}{g}\right)$  (C)  $2T_0 \left(1 - \frac{a}{g}\right)$  (D)  $3T_0 \left(1 + \frac{2a}{g}\right)$

FM0121

19. A wooden block floats in water in a sealed container. When the container is at rest, 25% of the block is above the water. Consider the following five situation :

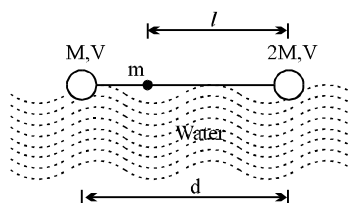
- (i) the container is lifted up at constant speed.
- (ii) the container is lowered at constant speed.
- (iii) the container is lifted up at an increasing speed.
- (iv) the container is lowered at a decreasing speed.
- (v) the air pressure above the water in the container is increased.

What happens in each situation?

- (A) 75% of block is submerged in situation (i) & (ii) and higher fraction is submerged in situations (iii), (iv) & (v)
- (B) 75% of block is submerged in situation (i), (ii) & (v) and higher fraction is submerged in situations (iii), (iv).
- (C) 75% of block is submerged in situation (i), (ii), (iii) & (iv) and higher fraction is submerged in situations (v).
- (D) 75% of block is submerged in all situation.

FM0122

20. A dumbbell is placed in water of density  $\rho$ . It is observed that by attaching a mass  $m$  to the rod, the dumbbell floats with the rod horizontal on the surface of water and each sphere exactly half submerged as shown in the figure. The volume of the mass  $m$  is negligible. The value of length  $l$  is :-



- (A)  $\frac{d(V\rho - 3M)}{2(V\rho - 2M)}$
- (B)  $\frac{d(V\rho - 2M)}{2(V\rho - 3M)}$
- (C)  $\frac{d(V\rho + 2M)}{2(V\rho - 3M)}$
- (D)  $\frac{d(V\rho - 2M)}{2(V\rho + 3M)}$

FM0123

21. Fountains usually seen in gardens are generated by a wide pipe with an enclosure at one end having many small holes. Consider one such fountain which is produced by a pipe of internal diameter 2 cm in which water flows at a rate  $3 \text{ ms}^{-1}$ . The enclosure has 100 holes each of diameter 0.05 cm. The velocity of water coming out of the holes is ( in  $\text{ms}^{-1}$ ):

- (A) 0.48
- (B) 96
- (C) 24
- (D) 48

FM0124

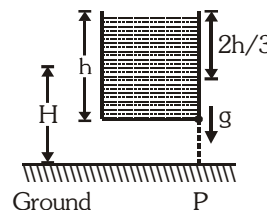
22. An open vessel full of water is falling freely under gravity. There is a small hole in one face of the vessel, as shown in the figure. The water which comes out from the hole at the instant when hole is at height  $H$  above the ground, strikes the ground at a distance of  $x$  from  $P$ . Which of the following is correct for the situation described ?

(A) The value of  $x$  is  $2\sqrt{\frac{2hH}{3}}$

(B) The value of  $x$  is  $\sqrt{\frac{4hH}{3}}$

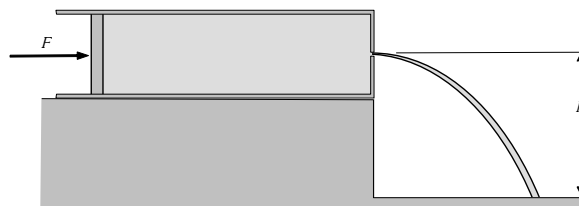
(C) The value of  $x$  can't be computed from information provided.

(D) The question is irrelevant as no water comes out from the hole.



FM0125

23. An ideal liquid of density  $\rho$  is filled in a horizontally fixed syringe fitted with piston. There is no friction between the piston and the inner surface of the syringe. Cross-section area of the syringe is  $A$ . At one end of the syringe, an orifice of negligible cross-section area is made. When the piston is pushed into the syringe, the liquid comes out of the orifice following parabolic path and falls on the ground. With what speed the liquid strikes the ground? Neglect the air drag :-



(A)  $\sqrt{\frac{F + \rho ghA}{\rho A}}$

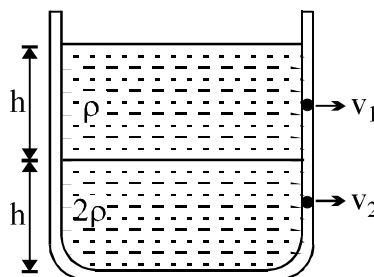
(B)  $\sqrt{\frac{F + 2\rho ghA}{\rho A}}$

(C)  $\sqrt{\frac{2F + \rho ghA}{\rho A}}$

(D)  $\sqrt{\frac{2(F + \rho ghA)}{\rho A}}$

FM0126

24. Equal volumes of two immiscible liquids of densities  $\rho$  and  $2\rho$  are filled in a vessel as shown in figure. Two small holes are punched at depth  $h/2$  and  $3h/2$  from the surface of lighter liquid. If  $v_1$  and  $v_2$  are the velocities of a flux at these two holes, then  $v_1/v_2$  is :



(A)  $\frac{1}{2\sqrt{2}}$

(B)  $\frac{1}{2}$

(C)  $\frac{1}{4}$

(D)  $\frac{1}{\sqrt{2}}$

FM0127

25. A vertical tank, open at the top, is filled with a liquid and rests on a smooth horizontal surface. A small hole is opened at the centre of one side of the tank. The area of cross-section of the tank is  $N$  times the area of the hole, where  $N$  is a large number. Neglect mass of the tank itself. The initial acceleration of the tank is :-

(A)  $\frac{g}{2N}$  (B)  $\frac{g}{\sqrt{2N}}$  (C)  $\frac{g}{N}$  (D)  $\frac{g}{2\sqrt{N}}$

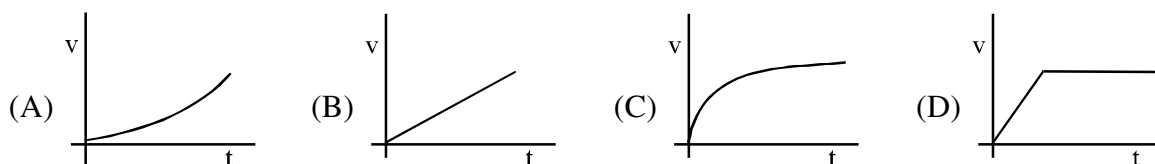
FM0128

26. A solid metallic sphere of radius  $r$  is allowed to fall freely through air. If the frictional resistance due to air is proportional to the cross-sectional area and to the square of the velocity, then the terminal velocity of the sphere is proportional to which of the following?

(A)  $r^2$  (B)  $r$  (C)  $r^{3/2}$  (D)  $r^{1/2}$

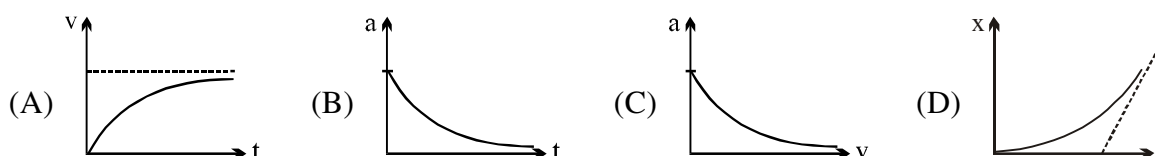
FM0129

27. Which of the following graphs best represents the motion of a raindrop?



FM0130

28. Which of the following is the **incorrect** graph for a sphere falling in a viscous liquid? (Given at  $t = 0$ , velocity  $v = 0$  and displacement  $x = 0$ .)



FM0131

### MULTIPLE CORRECT TYPE QUESTIONS

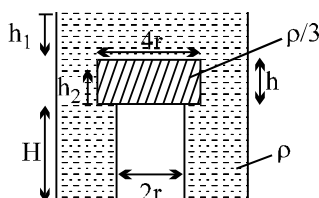
29. A cubical block is floating in a liquid with one third of its volume immersed in the liquid. When the whole system accelerates upwards with acceleration of  $g/2$  :-
- (A) the fraction of volume immersed in the liquid will change.  
 (B) the buoyancy force on the block will change.  
 (C) the buoyancy force will increase by 50 percent.  
 (D) the pressure in the liquid will increased.

FM0132

## COMPREHENSION TYPE QUESTIONS

## Paragraph for Question no. 30 to 32

A wooden cylinder of diameter  $4r$ , height  $h$  and density  $\rho/3$  is kept on a hole of diameter  $2r$  of a tank, filled with water of density  $\rho$  as shown in the figure. The height of the base of cylinder from the base of tank is  $H$ . [IIT-JEE 2006]



30. If level of liquid starts decreasing slowly when the level of liquid is at a height  $h_1$  above the cylinder, the block just starts moving up. Then, value of  $h_1$  is :-

(A)  $\frac{2h}{3}$  (B)  $\frac{5h}{4}$  (C)  $\frac{5h}{3}$  (D)  $\frac{5h}{2}$

FM0133

31. Let the cylinder is prevented from moving up, by applying a force and water level is further decreased. Then, height of water level ( $h_2$  in figure) for which the cylinder remains in original position without application of force is

(A)  $\frac{h}{3}$  (B)  $\frac{4h}{9}$  (C)  $\frac{2h}{3}$  (D)  $h$

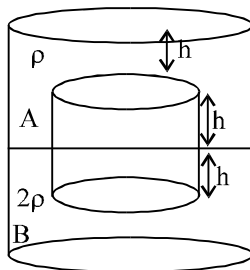
FM0133

32. If height  $h_2$  of water level is further decreased, then
- (A) cylinder will not move up and remains at its original position.
- (B) for  $h_2 = h/3$ , cylinder again starts moving up
- (C) for  $h_2 = h/4$ , cylinder again starts moving up
- (D) for  $h_2 = h/5$  cylinder again starts moving up

FM0133

MATRIX MATCH TYPE QUESTION

33. Shown below is a cylinder of radius  $R$  floating in vessel containing liquids A and B. Neglecting atmospheric pressure match the quantities mentioned in column-I with corresponding expression in column-II.



Column-I

- (A) Net force exerted by liquid A of density  $\rho$  on the cylinder.  
 (B) Net force exerted by liquid B of density  $2\rho$  on the cylinder.  
 (C) Net force exerted by liquids A and B on the left half of the curved part of cylinder.  
 (D) Net force exerted by liquid A and B on the cylinder.

Column-II

- (P)  $9 \rho g R h^2$   
 (Q)  $\pi \rho g R^2 h$   
 (R)  $4 \pi \rho g R^2 h$   
 (S)  $3 \pi \rho g R^2 h$

FM0134

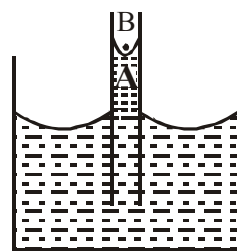
34. Capillary rise and shape of droplets on a plate due to surface tension are shown in column-II. Point A and B are just inside and outside the surface. Match the following

## Column-I

## Column-II

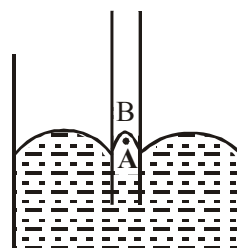
(A) liquid is wetting the solid

(P)



(B) liquid is not wetting the solid

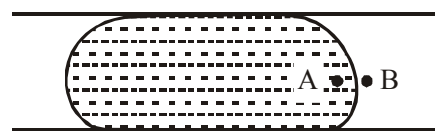
(Q)



(C) Pressure at A > pressure at B

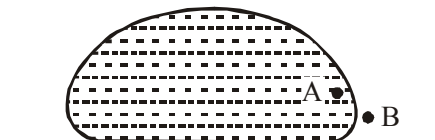
(R)

A liquid drop is pressed between two parallel glass plates.



(D) Pressure at B > Pressure at A

(S)



(T)



both end open cylindrical pipe

FM0166



# EXERCISE (JM)

1. A spherical solid ball of volume  $V$  is made of a material of density  $\rho_1$ . It is falling through a liquid of density  $\rho_2$  ( $\rho_2 < \rho_1$ ). Assume that the liquid applies a viscous force on the ball that is proportional to the square of its speed  $v$ , i.e.,  $F_{\text{viscous}} = -kv^2$  ( $k > 0$ ). Then terminal speed of the ball is

[AIEEE - 2008]

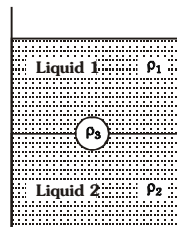
(1)  $\sqrt{\frac{Vg(\rho_1 - \rho_2)}{k}}$  (2)  $\frac{Vg\rho_1}{k}$  (3)  $\sqrt{\frac{Vg\rho_1}{k}}$  (4)  $\frac{Vg(\rho_1 - \rho_2)}{k}$

FM0135

2. A jar is filled with two non-mixing liquids 1 and 2 having densities  $\rho_1$  and  $\rho_2$ , respectively. A solid ball, made of a material of density  $\rho_3$ , is dropped in the jar. It comes to equilibrium in the position shown in the figure. Which of the following is true for  $\rho_1$ ,  $\rho_2$  and  $\rho_3$

[AIEEE - 2008]

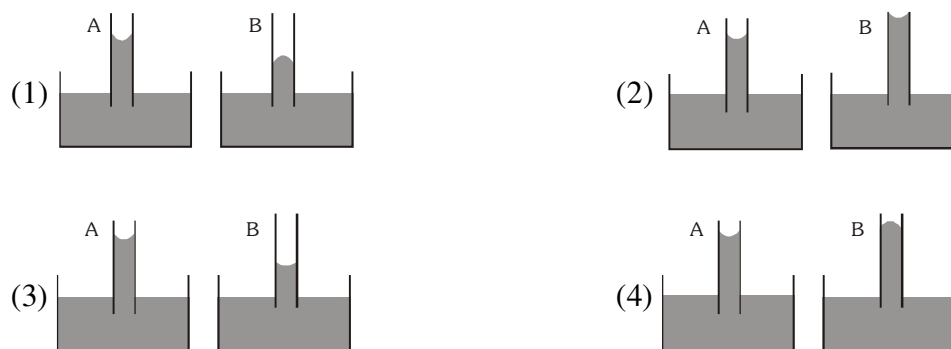
- (1)  $\rho_3 < \rho_1 < \rho_2$   
(2)  $\rho_1 > \rho_3 > \rho_2$   
(3)  $\rho_1 < \rho_2 < \rho_3$   
(4)  $\rho_1 < \rho_3 < \rho_2$



FM0136

3. A capillary tube (1) is dipped in water. Another identical tube (2) is dipped in a soap-water solution. Which of the following shows the relative nature of the liquid columns in the two tubes?

[AIEEE - 2008]



FM0137

4. A ball is made of a material of density  $\rho$  where  $\rho_{\text{oil}} < \rho < \rho_{\text{water}}$  with  $\rho_{\text{oil}}$  and  $\rho_{\text{water}}$  representing the densities of oil and water, respectively. The oil and water are immiscible. If the above ball is in equilibrium in a mixture of this oil and water, which of the following pictures represents its equilibrium position?

[AIEEE-2010]



FM0138

5. Water is flowing continuously from a tap having an internal diameter  $8 \times 10^{-3}$  m. The water velocity as it leaves the tap is  $0.4 \text{ ms}^{-1}$ . The diameter of the water stream at a distance  $2 \times 10^{-1}$  m below the tap is close to :- [AIEEE-2011]

(1)  $9.6 \times 10^{-3}$  m      (2)  $3.6 \times 10^{-3}$  m      (3)  $5.0 \times 10^{-3}$  m      (4)  $7.5 \times 10^{-3}$  m

FM0139

6. Work done in increasing the size of a soap bubble from a radius of 3 cm to 5 cm is nearly (Surface tension of soap solution =  $0.03 \text{ Nm}^{-1}$ ) :- [AIEEE-2011]

(1)  $2\pi \text{ mJ}$       (2)  $0.4 \pi \text{ mJ}$       (3)  $4\pi \text{ mJ}$       (4)  $0.2 \pi \text{ mJ}$

FM0140

7. Two mercury drops (each of radius 'r') merge to form a bigger drop. The surface energy of the bigger drop, if  $T$  is the surface tension, is : [AIEEE-2011]

(1)  $2^{5/3} \pi r^2 T$       (2)  $4\pi r^2 T$       (3)  $2\pi r^2 T$       (4)  $2^{8/3} \pi r^2 T$

FM0141

8. If a ball of steel (density  $\rho = 7.8 \text{ g cm}^{-3}$ ) attains a terminal velocity of  $10 \text{ cm s}^{-1}$  when falling in a tank of water (coefficient of viscosity  $\eta_{\text{water}} = 8.5 \times 10^{-4} \text{ Pa.s}$ ) then its terminal velocity in glycerine ( $\rho = 1.2 \text{ g cm}^{-3}$ ,  $\eta = 13.2 \text{ Pa.s}$ ) would be nearly :- [AIEEE-2011]

(1)  $1.6 \times 10^{-5} \text{ cm s}^{-1}$       (2)  $6.25 \times 10^{-4} \text{ cm s}^{-1}$   
(3)  $6.45 \times 10^{-4} \text{ cm s}^{-1}$       (4)  $1.5 \times 10^{-5} \text{ cm s}^{-1}$

FM0142

9. A thin liquid film formed between a U-shaped wire and a light slider supports a weight of  $1.5 \times 10^{-2} \text{ N}$  (see figure). The length of the slider is 30 cm and its weight negligible. The surface tension of the liquid film is :- [AIEEE-2012]

(1)  $0.025 \text{ Nm}^{-1}$   
(2)  $0.0125 \text{ Nm}^{-1}$   
(3)  $0.1 \text{ Nm}^{-1}$   
(4)  $0.05 \text{ Nm}^{-1}$



FM0143

10. A uniform cylinder of length  $L$  and mass  $M$  having cross-sectional area  $A$  is suspended, with its length vertical, from a fixed point by a massless spring, such that it is half submerged in a liquid of density  $\sigma$  at equilibrium position. The extension  $x_0$  of the spring when it is in equilibrium is : [AIEEE-2013]

(1)  $\frac{Mg}{k}$       (2)  $\frac{Mg}{k} \left( 1 - \frac{LA\sigma}{M} \right)$       (3)  $\frac{Mg}{k} \left( 1 - \frac{LA\sigma}{2M} \right)$       (4)  $\frac{Mg}{k} \left( 1 + \frac{LA\sigma}{M} \right)$

(Here  $k$  is spring constant)

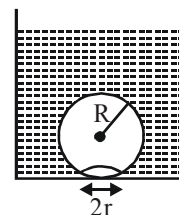
FM0144

11. Assume that a drop of liquid evaporates by decrease in its surface energy, so that its temperature remains unchanged. What should be the minimum radius of the drop for this to be possible? The surface tension is  $T$ , density of liquid is  $\rho$  and  $L$  is its latent heat of vaporization. [AIEEE-2013]

(1)  $\frac{\rho L}{T}$       (2)  $\sqrt{\frac{T}{\rho L}}$       (3)  $\frac{T}{\rho L}$       (4)  $\frac{2T}{\rho L}$

FM0145

12. On heating water, bubbles being formed at the bottom of the vessel detach and rise. Take the bubbles to be spheres of radius  $R$  and making a circular contact of radius  $r$  with the bottom of the vessel. If  $r \ll R$ , and the surface tension of water is  $T$ , value of  $r$  just before bubbles detach is:-  
(density of water is  $\rho_w$ )



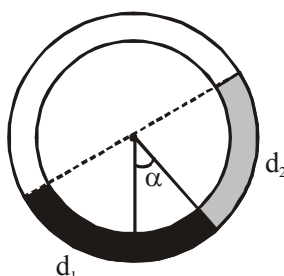
[JEE Mains-2014]

- (1)  $R^2 \sqrt{\frac{\rho_w g}{T}}$  (2)  $R^2 \sqrt{\frac{3\rho_w g}{T}}$  (3)  $R^2 \sqrt{\frac{\rho_w g}{3T}}$  (4)  $R^2 \sqrt{\frac{\rho_w g}{6T}}$

FM0146

13. There is a circular tube in a vertical plane. Two liquids which do not mix and of densities  $d_1$  and  $d_2$  are filled in the tube. Each liquid subtends  $90^\circ$  angle at centre. Radius joining their interface makes an angle  $\alpha$  with vertical. Ratio  $\frac{d_1}{d_2}$  is :

[JEE Mains-2014]



- (1)  $\frac{1 + \tan \alpha}{1 - \tan \alpha}$  (2)  $\frac{1 + \sin \alpha}{1 - \cos \alpha}$  (3)  $\frac{1 + \sin \alpha}{1 - \sin \alpha}$  (4)  $\frac{1 + \cos \alpha}{1 - \cos \alpha}$

FM0147

## EXERCISE (JA)

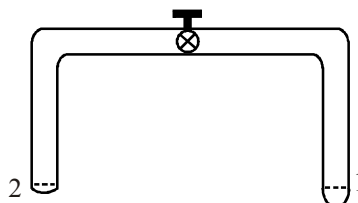
1. **Statement-1** : The stream of water flowing at high speed from a garden hose pipe tends to spread like a fountain when held vertically up, but tends to narrow down when held vertically down.  
and

**Statement-2** : In any steady flow of an incompressible fluid, the volume flow rate of the fluid remains constant. [IIT-JEE 2008]

- (A) Statement-1 is True, Statement-2 is True ; statement-2 is a correct explanation for statement-1  
(B) Statement-1 is True, Statement-2 is True ; statement-2 is NOT a correct explanation for statement-1  
(C) Statement-1 is True, Statement-2 is False  
(D) Statement-1 is False, Statement-2 is True

FM0148

2. A glass tube of uniform internal radius ( $r$ ) has a valve separating the two identical ends. Initially, the valve is in a tightly closed position. End 1 has a hemispherical soap bubble of radius  $r$ . End 2 has sub-hemispherical soap bubble as shown in figure. Just after opening the valve, [IIT-JEE 2008]

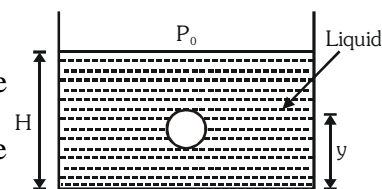


- (A) air from end 1 flows towards end 2. No change in the volume of the soap bubbles  
(B) air from end 1 flows towards end 2. Volume of the soap bubble at end 1 decreases  
(C) no change occurs  
(D) air from end 2 flows towards end 1. Volume of the soap bubble at end 1 increases

## Paragraph for Question Nos. 3 to 5

A small spherical monoatomic ideal gas bubble  $\left(\gamma = \frac{5}{3}\right)$  is

trapped inside a liquid of density  $\rho_\ell$  (see figure). Assume that the bubble does not exchange any heat with the liquid. The bubble contains  $n$  moles of gas. The temperature of the gas when the bubble is at the bottom is  $T_0$ , the height of the liquid is  $H$  and the atmospheric pressure is  $P_0$  (Neglect surface tension). [IIT-JEE 2008]



FM0149

3. As the bubble moves upwards, besides the buoyancy force the following forces are acting on it.
- (A) Only the force of gravity  
 (B) The force due to gravity and the force due to the pressure of the liquid  
 (C) The force due to gravity, the force due to the pressure of the liquid and the force due to viscosity of the liquid  
 (D) The force due to gravity and the force due to viscosity of the liquid.

FM0150

4. When the gas bubble is at a height  $y$  from the bottom, its temperature is :-

(A)  $T_0 \left( \frac{P_0 + \rho_l g H}{P_0 + \rho_l g y} \right)^{2/5}$  (B)  $T_0 \left( \frac{P_0 + \rho_l g (H - y)}{P_0 + \rho_l g H} \right)^{2/5}$   
 (C)  $T_0 \left( \frac{P_0 + \rho_l g H}{P_0 + \rho_l g y} \right)^{3/5}$  (D)  $T_0 \left( \frac{P_0 + \rho_l g (H - y)}{P_0 + \rho_l g H} \right)^{3/5}$

FM0150

5. The buoyancy force acting on the gas bubble is (Assume  $R$  is the universal gas constant)

(A)  $p_l n R g T_0 \frac{(P_0 + \rho_l g H)^{2/5}}{(P_0 + \rho_l g y)^{7/5}}$  (B)  $\frac{\rho_l n R g T_0}{(P_0 + \rho_l g H)^{2/5} [P_0 + \rho_l g (H - y)]^{3/5}}$   
 (C)  $p_l n R g T_0 \frac{(P_0 + \rho_l g H)^{3/5}}{(P_0 + \rho_l g y)^{8/5}}$  (D)  $\frac{\rho_l n R g T_0}{(P_0 + \rho_l g H)^{3/5} [P_0 + \rho_l g (H - y)]^{2/5}}$

FM0150

6. A cylindrical vessel of height 500 mm has an orifice (small hole) at its bottom. The orifice is initially closed and water is filled in it up to height  $H$ . Now the top is completely sealed with a cap and the orifice at the bottom is opened. Some water comes out from the orifice and the water level in the vessel becomes steady with height of water column being 200 mm. Find the fall in height (in mm) of water level due to opening of the orifice. [Take atmospheric pressure =  $1.0 \times 10^5$  N/m<sup>2</sup>, density of water = 1000 kg/m<sup>3</sup> and  $g = 10$  m/s<sup>2</sup>. Neglect any effect of surface tension.] [IIT-JEE-2009]

FM0151

7. Two soap bubbles A and B are kept in a closed chamber where the air is maintained at pressure 8 N/m<sup>2</sup>. The radii of bubbles A and B are 2 cm and 4 cm, respectively. Surface tension of the soap-water used to make bubbles is 0.04 N/m. Find the ratio  $n_B/n_A$ , where  $n_A$  and  $n_B$  are the number of moles of air in bubbles A and B, respectively. [Neglect the effect of gravity] [IIT-JEE-2009]

FM0152

## Paragraph for Questions no. 8 to 10

When liquid medicine of density  $\rho$  is to be put in the eye, it is done with the help of a dropper. As the bulb on the top of the dropper is pressed, a drop forms at the opening of the dropper. We wish to estimate the size of the drop. We first assume that the drop formed at the opening is spherical because that requires a minimum increase in its surface energy. To determine the size, we calculate the net vertical force due to the surface tension  $T$  when the radius of the drop is  $R$ . When this force becomes smaller than the weight of the drop, the drop gets detached from the dropper.

[IIT-JEE-2010]

8. If the radius of the opening of the dropper is  $r$ , the vertical force due to the surface tension on the drop of radius  $R$  (assuming  $r \ll R$ ) is

(A)  $2\pi rT$                       (B)  $2\pi RT$                       (C)  $\frac{2\pi r^2 T}{R}$                       (D)  $\frac{2\pi R^2 T}{r}$

FM0153

9. If  $r = 5 \times 10^{-4} \text{ m}$ ,  $\rho = 10^3 \text{ kg m}^{-3}$ ,  $g = 10 \text{ ms}^{-2}$ ,  $T = 0.11 \text{ Nm}^{-1}$ , the radius of the drop when it detaches from the dropper is approximately

(A)  $1.4 \times 10^{-3} \text{ m}$                       (B)  $3.3 \times 10^{-3} \text{ m}$                       (C)  $2.0 \times 10^{-3} \text{ m}$                       (D)  $4.1 \times 10^{-3} \text{ m}$

FM0153

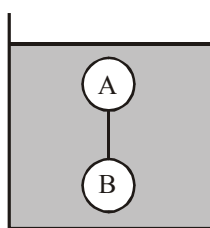
10. After the drop detaches, its surface energy is

(A)  $1.4 \times 10^{-6} \text{ J}$                       (B)  $2.7 \times 10^{-6} \text{ J}$                       (C)  $5.4 \times 10^{-6} \text{ J}$                       (D)  $8.1 \times 10^{-6} \text{ J}$

FM0153

11. Two solid spheres A and B of equal volumes but of different densities  $d_A$  and  $d_B$  are connected by a string. They are fully immersed in a fluid of density  $d_F$ . They get arranged into an equilibrium state as shown in the figure with a tension in the string. The arrangement is possible only if

[IIT-JEE-2011]



(A)  $d_A < d_F$                       (B)  $d_B > d_F$                       (C)  $d_A > d_F$                       (D)  $d_A + d_B = 2d_F$

FM0154

12. A thin uniform cylindrical shell, closed at both ends, is partially filled with water. It is floating vertically in water in half-submerged state. If  $\rho_c$  is the relative density of the material of the shell with respect to water, then the correct statement is that the shell is

[IIT-JEE-2012]

- (A) more than half-filled if  $\rho_c$  is less than 0.5  
 (B) more than half-filled if  $\rho_c$  is more than 1.0  
 (C) half-filled if  $\rho_c$  is more than 0.5  
 (D) less than half-filled if  $\rho_c$  is less than 0.5

FM0155

13. A solid sphere of radius  $R$  and density  $\rho$  is attached to one end of a mass-less spring of force constant  $k$ . The other end of the spring is connected to another solid sphere of radius  $R$  and density  $3\rho$ . The complete arrangement is placed in a liquid of density  $2\rho$  and is allowed to reach equilibrium. The correct statement(s) is (are) [IIT-JEE-2013]

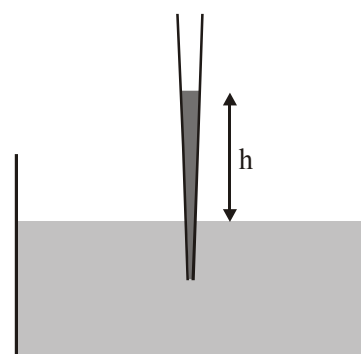
- (A) the net elongation of the spring is  $\frac{4\pi R^3 \rho g}{3k}$   
 (B) the net elongation of the spring is  $\frac{8\pi R^3 \rho g}{3k}$   
 (C) the light sphere is partially submerged.  
 (D) the light sphere is completely submerged.

FM0156

14. A glass capillary tube is of the shape of truncated cone with an apex angle  $\alpha$  so that its two ends have cross sections of different radii. When dipped in water vertically, water rises in it to height  $h$ , where the radius of its cross section is  $b$ . If the surface tension of water is  $S$ , its density is  $\rho$ , and its contact angle with glass is  $\theta$ , the value of  $h$  will be ( $g$  is the acceleration due to gravity)

[JEE Advanced-2014]

- (A)  $\frac{2S}{b\rho g} \cos(\theta - \alpha)$  (B)  $\frac{2S}{b\rho g} \cos(\theta + \alpha)$   
 (C)  $\frac{2S}{b\rho g} \cos(\theta - \alpha/2)$  (D)  $\frac{2S}{b\rho g} \cos(\theta + \alpha/2)$

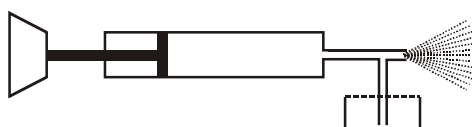


FM0157

### Paragraph for Questions 15 to 17

A spray gun is shown in the figure where a piston pushes air out of a nozzle. A thin tube of uniform cross section is connected to the nozzle. The other end of the tube is in a small liquid container. As the piston pushes air through the nozzle, the liquid from the container rises into the nozzle and is sprayed out. For the spray gun shown, the radii of the piston and the nozzle are 20 mm and 1 mm respectively. The upper end of the container is open to the atmosphere.

[JEE Advanced-2014]



15. If the piston is pushed at a speed of  $5 \text{ mms}^{-1}$ , the air comes out of the nozzle with a speed of  
 (A)  $0.1 \text{ ms}^{-1}$  (B)  $1 \text{ ms}^{-1}$  (C)  $2 \text{ ms}^{-1}$  (D)  $8 \text{ ms}^{-1}$

FM0158

16. If the density of air is  $\rho_a$  and that of the liquid  $\rho_\ell$ , then for a given piston speed the rate (volume per unit time) at which the liquid is sprayed will be proportional to

(A)  $\sqrt{\frac{\rho_a}{\rho_\ell}}$  (B)  $\sqrt{\rho_a \rho_\ell}$  (C)  $\sqrt{\frac{\rho_\ell}{\rho_a}}$  (D)  $\rho_\ell$

FM0158

17. A person in a lift is holding a water jar, which has a small hole at the lower end of its side. When the lift is at rest, the water jet coming out of the hole hits the floor of the lift at a distance  $d$  of 1.2 m from the person. In the following, state of the lift's motion is given in List I and the distance where the water jet hits the floor of the lift is given in List II. Match the statements from List I with those in List II and select the correct answer using the code given below the lists.

**List - I**

- (P) Lift is accelerating vertically up.  
 (Q) Lift is accelerating vertically down with an acceleration less than the gravitational acceleration.  
 (R) Lift is moving vertically up with constant speed.  
 (S) Lift is falling freely.

**List-II**

[JEE Advanced-2014]

- (1)  $d = 1.2 \text{ m}$   
 (2)  $d > 1.2 \text{ m}$   
 (3)  $d < 1.2 \text{ m}$   
 (4) No water leaks out of the jar

**Code :**

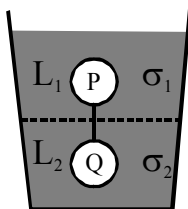
- (A) P-2, Q-3, R-2, S-4  
 (C) P-1, Q-1, R-1, S-4

- (B) P-2, Q-3, R-1, S-4  
 (D) P-2, Q-3, R-1, S-1

FM0159

18. Two spheres P and Q of equal radii have densities  $\rho_1$  and  $\rho_2$ , respectively. The spheres are connected by a massless string and placed in liquids  $L_1$  and  $L_2$  of densities  $\sigma_1$  and  $\sigma_2$  and viscosities  $\eta_1$  and  $\eta_2$ , respectively. They float in equilibrium with the sphere P in  $L_1$  and sphere Q in  $L_2$  and the string being taut (see figure). If sphere P alone in  $L_2$  has terminal velocity  $\vec{V}_P$  and Q alone in  $L_1$  has terminal velocity  $\vec{V}_Q$ , then

[JEE Advanced-2015]



(A)  $\frac{|\vec{V}_P|}{|\vec{V}_Q|} = \frac{\eta_1}{\eta_2}$  (B)  $\frac{|\vec{V}_P|}{|\vec{V}_Q|} = \frac{\eta_2}{\eta_1}$  (C)  $\vec{V}_P \cdot \vec{V}_Q > 0$  (D)  $\vec{V}_P \cdot \vec{V}_Q < 0$

FM0160



19. Consider two solid spheres P and Q each of density  $8 \text{ gm cm}^{-3}$  and diameters 1 cm and 0.5 cm, respectively. Sphere P is dropped into a liquid of density  $0.8 \text{ gm cm}^{-3}$  and viscosity  $\eta = 3$  poiseulles. Sphere Q is dropped into a liquid of density  $1.6 \text{ gm cm}^{-3}$  and viscosity  $\eta = 2$  poiseulles. The ratio of the terminal velocities of P and Q is. [JEE Advanced-2016]

FM0161

20. A drop of liquid of radius  $R = 10^{-2} \text{ m}$  having surface tension  $S = \frac{0.1}{4\pi} \text{ Nm}^{-1}$  divides itself into K identical drops. In this process the total change in the surface energy  $\Delta U = 10^{-3} \text{ J}$ . If  $K = 10^\alpha$  then the value of  $\alpha$  is [JEE Advanced-2017]

FM0162

21. A uniform capillary tube of inner radius  $r$  is dipped vertically into a beaker filled with water. The water rises to a height  $h$  in the capillary tube above the water surface in the beaker. The surface tension of water is  $\sigma$ . The angle of contact between water and the wall of the capillary tube is  $\theta$ . Ignore the mass of water in the meniscus. Which of the following statements is (are) true? [JEE Advanced-2018]

- (A) For a given material of the capillary tube,  $h$  decreases with increase in  $r$
- (B) For a given material of the capillary tube,  $h$  is independent of  $\sigma$ .
- (C) If this experiment is performed in a lift going up with a constant acceleration, then  $h$  decreases.,
- (D)  $h$  is proportional to contact angle  $\theta$ .

FM0163

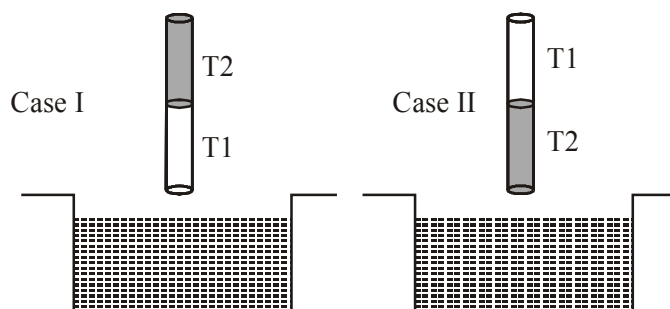
22. Consider a thin square plate floating on a viscous liquid in a large tank. The height  $h$  of the liquid in the tank is much less than the width of the tank. The floating plate is pulled horizontally with a constant velocity  $u_0$ . Which of the following statements is (are) true ? [JEE Advanced-2018]

- (A) The resistive force of liquid on the plate is inversely proportional to  $h$
- (B) The resistive force of liquid on the plate is independent of the area of the plate
- (C) The tangential (shear) stress on the floor of the tank increases with  $u_0$ .
- (D) The tangential (shear) stress on the plate varies linearly with the viscosity  $\eta$  of the liquid.

FM0164

23. A cylindrical capillary tube of 0.2 mm radius is made by joining two capillaries T1 and T2 of different materials having water contact angles of  $0^\circ$  and  $60^\circ$ , respectively. The capillary tube is dipped vertically in water in two different configurations, case I and II as shown in figure. Which of the following option(s) is(are) correct ? [JEE Advanced-2019]

(Surface tension of water = 0.075 N/m, density of water =  $1000 \text{ kg/m}^3$ , take  $g = 10 \text{ m/s}^2$ )



- (1) The correction in the height of water column raised in the tube, due to weight of water contained in the meniscus, will be different for both cases.
- (2) For case I, if the capillary joint is 5 cm above the water surface, the height of water column raised in the tube will be more than 8.75 cm. (Neglect the weight of the water in the meniscus)
- (3) For case I, if the joint is kept at 8 cm above the water surface, the height of water column in the tube will be 7.5 cm. (Neglect the weight of the water in the meniscus)
- (4) For case II, if the capillary joint is 5 cm above the water surface, the height of water column raised in the tube will be 3.75 cm. (Neglect the weight of the water in the meniscus)

FM0165

## ANSWER KEY

### EXERCISE (S-1)

1. Ans.  $45^\circ$ ,  $9600\sqrt{2}$  Pa      2. Ans.  $\frac{L}{2}\rho(g+a)$       3. Ans.  $P_0 - \rho a \ell$       4. Ans. 37.5 N  
 5. Ans. 8      6. Ans. 2      7. Ans.  $H = \frac{L^2\omega^2}{2g}$       8. Ans.  $\frac{5d}{4}$       9. Ans. (a) 5, (b)  $\frac{2}{3}$   
 10. Ans.  $45^\circ$       11. Ans.  $nH$       12. Ans. 4      13. Ans. 2.52 cm      14. Ans.  $h = \frac{mg + 4sa}{\rho_w a^2 g}$   
 15. Ans. 20 N      16. Ans. 4.9 litre/min      17. Ans.  $6.43 \times 10^{-4}$  m<sup>3</sup>/s      18. Ans. 5 cm  
 19. Ans. 11 m/s      20. Ans.  $\frac{h_2 + h_1}{2}$       21. Ans. 108      22. Ans. 20.4 m      23. Ans. 5 s      24. Ans. 5  
 25. Ans.  $\frac{dQ}{dt} \propto r^5$

### EXERCISE (S-2)

1. Ans. 2.79 gm/cc      2. Ans. 1.9      3. Ans.  $R\sqrt{\frac{3}{2}}$       4. Ans. 2  
 5. Ans.  $h = \frac{2m}{\pi}$ ,  $H = \frac{11}{32\pi} m$       6. Ans. 5      7. Ans.  $x = 2 m$   
 8. Ans.  $h_2 = 3 h_1$       9. Ans.  $+ 29625 \text{ J/m}^3, - 30000 \text{ J/m}^3$       10. Ans. 4  
 11. Ans. 8      12. Ans. (a)  $\left[ \frac{dgRH^2}{3} \right]$ , (b)  $\frac{dg\pi R^2 H}{6}$       13. Ans. 0.1 cm  
 14. Ans. 3 : 9 : 5      15. Ans.  $\sqrt{\frac{18g}{19a}}$       16. Ans. 4.5m      17. Ans.  $\left( \frac{R}{24d} \right)^{\frac{1}{3}}$   
 18. Ans. 80.72      19. Ans.  $\frac{\lambda_a}{2y} g$       20. Ans. 2      21. Ans.  $v = \sqrt{2g \left[ h_1 + h_2 \left( \frac{\rho_2}{\rho_1} \right) \right]}$   
 22. Ans. 5  
 23. Ans. (a) 10 m/s, (b) 2.5 m/s, (c) 4.688 m, (d)  $3.14 \times 10^{-3} \text{ m}^3/\text{s}$       24. Ans. 6  
 25. Ans. 4

**EXERCISE (O-1)****SINGLE CORRECT TYPE QUESTIONS**

- |              |              |              |              |              |              |
|--------------|--------------|--------------|--------------|--------------|--------------|
| 1. Ans. (C)  | 2. Ans. (C)  | 3. Ans. (C)  | 4. Ans. (C)  | 5. Ans. (A)  | 6. Ans. (D)  |
| 7. Ans. (C)  | 8. Ans. (C)  | 9. Ans. (D)  | 10. Ans. (D) | 11. Ans. (A) | 12. Ans. (B) |
| 13. Ans. (B) | 14. Ans. (D) | 15. Ans. (A) | 16. Ans. (B) | 17. Ans. (D) | 18. Ans. (A) |
| 19. Ans. (B) | 20. Ans. (B) | 21. Ans. (A) | 22. Ans. (B) | 23. Ans. (A) | 24. Ans. (C) |
| 25. Ans. (B) | 26. Ans. (D) | 27. Ans. (C) | 28. Ans. (C) | 29. Ans. (C) | 30. Ans. (C) |
| 31. Ans. (A) | 32. Ans. (D) | 33. Ans. (B) | 34. Ans. (B) | 35. Ans. (C) | 36. Ans. (A) |
| 37. Ans. (C) | 38. Ans. (D) | 39. Ans. (B) | 40. Ans. (B) | 41. Ans. (B) | 42. Ans. (A) |
| 43. Ans. (B) | 44. Ans. (D) | 45. Ans. (A) | 46. Ans. (A) | 47. Ans. (C) |              |

**MULTIPLE CORRECT TYPE QUESTIONS**

48. Ans. (A,D)    49. Ans. (B, C)    50. Ans. (A,C)    51. Ans. (A,C)

**MATRIX MATCH TYPE QUESTION**

52. Ans. (A)-R; (B)-Q; (C)-Q; (D)-Q    53. Ans. (A)-P,T; (B)-P,S; (C)-Q,T; (D)-R

**EXERCISE (O-2)****SINGLE CORRECT TYPE QUESTIONS**

- |              |              |              |              |              |              |
|--------------|--------------|--------------|--------------|--------------|--------------|
| 1. Ans. (D)  | 2. Ans. (D)  | 3. Ans. (A)  | 4. Ans. (B)  | 5. Ans. (A)  | 6. Ans. (A)  |
| 7. Ans. (C)  | 8. Ans. (A)  | 9. Ans. (B)  | 10. Ans. (B) | 11. Ans. (B) | 12. Ans. (D) |
| 13. Ans. (D) | 14. Ans. (D) | 15. Ans. (C) | 16. Ans. (D) | 17. Ans. (D) | 18. Ans. (A) |
| 19. Ans. (D) | 20. Ans. (B) | 21. Ans. (D) | 22. Ans. (D) | 23. Ans. (D) | 24. Ans. (D) |
| 25. Ans. (C) | 26. Ans. (D) | 27. Ans. (C) | 28. Ans. (C) |              |              |

**MULTIPLE CORRECT TYPE QUESTIONS**

29. Ans. (B, C, D)

**COMPREHENSION TYPE QUESTIONS**

30. Ans. (C)    31. Ans. (B)    32. Ans. (A)

**MATRIX MATCH TYPE QUESTION**

33. Ans. (A)-Q, (B)-R, (C)-P, (D)-S    34. Ans. (A)-P,T; (B)-Q,R,S; (C)-Q,R,S,T; (D)-P

**EXERCISE (JM)**

- |              |             |             |              |              |              |
|--------------|-------------|-------------|--------------|--------------|--------------|
| 1. Ans. (1)  | 2. Ans. (4) | 3. Ans. (3) | 4. Ans. (3)  | 5. Ans. (2)  | 6. Ans. (2)  |
| 7. Ans. (4)  | 8. Ans. (2) | 9. Ans. (1) | 10. Ans. (3) | 11. Ans. (4) | 12. Ans. (3) |
| 13. Ans. (1) |             |             |              |              |              |

**EXERCISE (JA)**

- |                  |                  |              |              |                  |           |
|------------------|------------------|--------------|--------------|------------------|-----------|
| 1. Ans. (A)      | 2. Ans. (B)      | 3. Ans. (D)  | 4. Ans. (B)  | 5. Ans. (B)      | 6. Ans. 6 |
| 7. Ans. 6        | 8. Ans. (C)      | 9. Ans. (A)  | 10. Ans. (B) | 11. Ans. (A,B,D) |           |
| 12. Ans. (A)     | 13. Ans. (A, D)  | 14. Ans. (D) | 15. Ans. (C) | 16. Ans. (A)     |           |
| 17. Ans. (C)     | 18. Ans. (A, D)  | 19. Ans. 3   | 20. Ans. 6   | 21. Ans. (A,C)   |           |
| 22. Ans. (A,C,D) | 23. Ans. (1,3,4) |              |              |                  |           |

# Chapter Contents

## Simple Harmonic Motion

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## SIMPLE HARMONIC MOTION

### KEY CONCEPTS

#### PERIODIC MOTION

- Any motion which repeats itself after regular interval of time is called periodic motion.
- The constant interval of time after which the motion is repeated is called time period.

**Examples :** (i) Motion of planets around the sun. (ii) Motion of the pendulum of wall clock.

#### OSCILLATORY MOTION

- The motion of body is said to be oscillatory if it moves back and forth (to and fro) about a fixed point. Oscillation of very high frequency & small amplitude is called vibration.
- The fixed point about which the body oscillates is called mean position or equilibrium position.

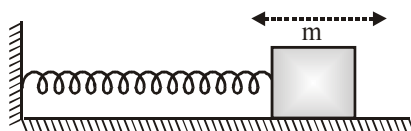
**Examples :** (i) Vibration of the wire of 'Sitar'. (ii) Oscillation of the mass suspended from spring.

#### SIMPLE HARMONIC MOTION (S.H.M.)

Simple harmonic motion is the simplest form of oscillatory motion.

##### (i) S.H.M. are of two types

- Linear S.H.M. :** When a particle moves to and fro about a fixed point (called equilibrium position) along a straight line then its motion is called linear simple harmonic motion.

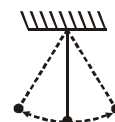


Example : Motion of a mass connected to spring.

##### Angular S.H.M.

When a system oscillates angularly with respect to a fixed axis then its motion is called angular simple harmonic motion.

Example :- Motion of a bob of simple pendulum.



##### (ii) Necessary Condition to execute S.H.M.

- In linear S.H.M. :** The restoring force (or acceleration) acting on the particle should always be proportional to the displacement of the particle and directed towards the equilibrium position

$$\therefore F \propto -x \quad \text{or} \quad a \propto -x$$

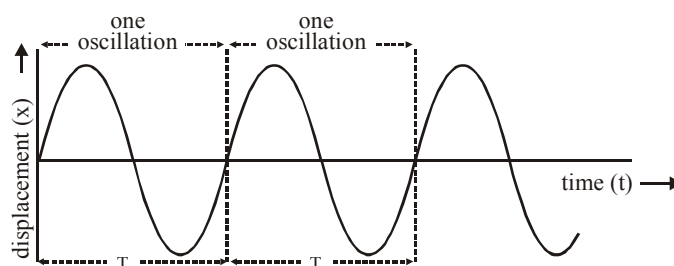
Negative sign shows that direction of force and acceleration is towards equilibrium position and  $x$  is displacement of particle from equilibrium position.

- In angular S.H.M. :** The restoring torque (or angular acceleration) acting on the particle should always be proportional to the angular displacement of the particle and directed towards the equilibrium position

$$\therefore \tau \propto -\theta \quad \text{or} \quad \alpha \propto -\theta$$

## SOME BASIC TERMS

- **Mean Position :** The point at which the restoring force on the particle is zero and potential energy is minimum, is known as its mean position.
- **Restoring Force**
  - The force acting on the particle which tends to bring the particle towards its mean position, is known as restoring force.
  - Restoring force always acts in a direction opposite to that of displacement. Displacement is measured from the mean position.
- **Amplitude :** The maximum value of displacement of particle from mean position is define as amplitude.
- **Time period (T)**
  - The minimum time after which the particle keeps on repeating its motion is known as time period.
  - The smallest time taken to complete one oscillation or vibration is also define as time period.
  - It is given by  $T = \frac{2\pi}{\omega} = \frac{1}{n}$  where  $\omega$  is angular frequency and  $n$  is frequency.
- **Oscillation :** When a particle goes on one side from mean position and returns back and then it goes to other side and again returns back to mean position, then this process is known as one oscillation.



- **Frequency (n or f)**
  - The number of oscillations per second is define as frequency.
  - It is given by  $n = \frac{1}{T}$ ,  $n = \frac{\omega}{2\pi}$
  - **SI UNIT :** hertz (Hz), 1 hertz = 1 cycle per second.
  - **Dimensions :**  $M^0 L^0 T^{-1}$ .
- **Phase :**
  - Phase of a vibrating particle at any instant is the state of the vibrating particle regarding its displacement and direction of vibration at that particular instant.
  - In the equation  $x = A \sin (\omega t + \theta)$ ,  $(\omega t + \theta)$  is the phase of the particle.
  - The phase angle at time  $t = 0$  is known as initial phase.
  - The difference of total phase angles of two particles executing S.H.M. with respect to the mean position is known as phase difference.
  - Two vibrating particles are said to be in same phase if the phase difference between them is an even multiple of  $\pi$ , i.e.,  $\Delta\phi = 2n\pi$  Where  $n = 0, 1, 2, 3, \dots$
  - Two vibrating particle are said to be in opposite phase if the phase difference between them is an odd multiple of  $\pi$  i.e.,  $\Delta\phi = (2n + 1)\pi$  Where  $n = 0, 1, 2, 3, \dots$



• **Angular frequency ( $\omega$ ) :**

The rate of change of phase angle of a particle with respect to time is define as its angular frequency.

*SI unit* : radian/second, *Dimensions* :  $M^0 L^0 T^{-1}$ ,

**DISPLACEMENT IN S.H.M.**

(i) The displacement of a particle executing linear S.H.M. at any instant is defined as the distance of the particle from the mean position at that instant.

(ii) It can be given by relation  $x = A \sin \omega t$  or  $x = A \cos \omega t$ .

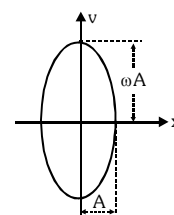
The first relation is valid when the time is measured from the mean position and the second relation is valid when the time is measured from the extreme position of the particle executing S.H.M. along a straight line path.

**VELOCITY IN SHM**

(i) It is define as the time rate of change of the displacement of the particle at the given instant.

(ii) Velocity in S.H.M. is given by  $v = \frac{dx}{dt} = \frac{d}{dt}(A \sin \omega t) \Rightarrow v = A\omega \cos \omega t$

$$v = \pm A\omega \sqrt{1 - \sin^2 \omega t} \Rightarrow v = \pm A\omega \sqrt{1 - \frac{x^2}{A^2}} = \pm \omega \sqrt{(A^2 - x^2)} \quad [\because x = A \sin \omega t]$$



$$\text{Squaring both the sides } v^2 = \omega^2 (A^2 - x^2) \Rightarrow \frac{v^2}{\omega^2} = A^2 - x^2 \Rightarrow \frac{v^2}{\omega^2 A^2} = 1 - \frac{x^2}{A^2} \Rightarrow \frac{x^2}{A^2} + \frac{v^2}{A^2 \omega^2} = 1$$

This is equation of ellipse. So curve between displacement and velocity of particle executing S.H.M. is ellipse.

(iii) The graph between velocity and displacement is shown in figure. If particle oscillates with unit angular frequency ( $\omega = 1$ ) then curve between  $v$  and  $x$  will be circular.

**Note :**

(i) The direction of velocity of a particle in S.H.M. is either towards or away from the mean position.

(ii) At mean position ( $x = 0$ ), velocity is maximum ( $= A\omega$ ) and at extreme position

( $x = \pm A$ ), the velocity of particle executing S.H.M. is zero (minimum).

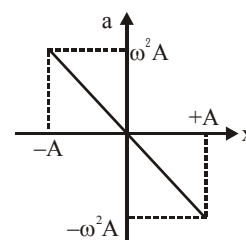
**ACCELERATION IN SHM**

(i) It is define as the time rate of change of the velocity of the particle at given instant.

(ii) Acceleration in S.H.M. is given by  $a = \frac{dv}{dt} = \frac{d}{dt}(A\omega \cos \omega t)$

$$a = -\omega^2 A \sin \omega t \Rightarrow a = -\omega^2 x$$

(iii) The graph between acceleration and displacement as shown in figure



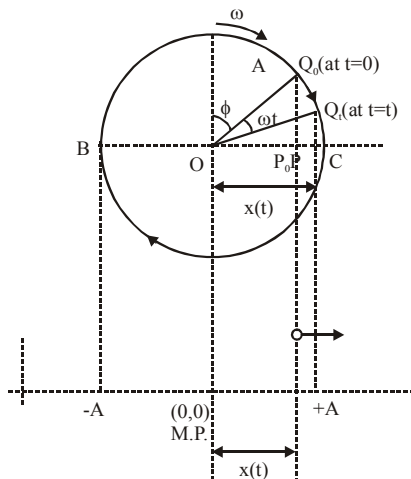
**Note**

(i) The acceleration of a particle executing S.H.M. is always directed towards the mean position.

(ii) The acceleration of the particle executing S.H.M. is maximum at extreme position ( $= \omega^2 A$ ) and minimum at mean position ( $= \text{zero}$ )

### SHM AS A PROJECTION OF UNIFORM CIRCULAR MOTION

Consider a particle Q, moving on a circle of radius A with constant angular velocity  $\omega$ . The projection of Q on a diameter BC is P. It is clear from the figure that as Q moves around the circle the projection P executes a simple harmonic motion on the x-axis between B and C. The angle that the radius OQ makes with the +ve vertical in clockwise direction in at  $t = 0$  is equal to phase constant ( $\phi$ ).



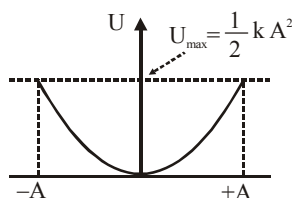
Let the radius  $OQ_0$  makes an angle  $\omega t$  with the  $OQ_t$  at time  $t$ . Then  $x(t) = A \sin(\omega t + \phi)$

In the above discussion the foot of projection is x-axis so it is called horizontal phasor. Similarly the foot of perpendicular on y-axis will also execute SHM of amplitude A and angular frequency  $\omega$  [ $y(t) = A \cos \omega t$ ]. This is called vertical phasor. The phasor of the two SHM differ by  $\pi/2$ .

### ENERGY OF PARTICLE IN S.H.M.

- **Potential Energy (U or P.E.)**
  - In terms of displacement**

The potential energy is related to force by the relation  $F = -\frac{dU}{dx} \Rightarrow \int dU = -\int F dx$

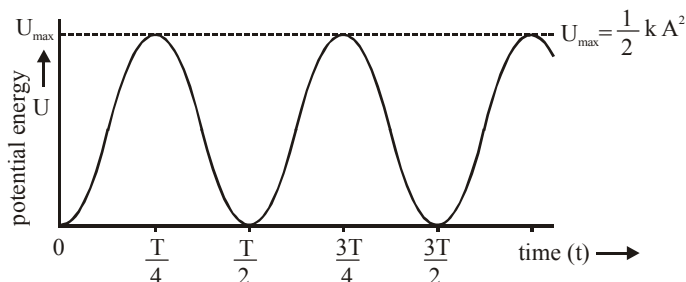


For S.H.M.  $F = -kx$  so  $\int dU = -\int (-kx) dx = \int kx dx \Rightarrow U = \frac{1}{2} kx^2 + C$

At  $x = 0$ ,  $U = U_0 \Rightarrow C = U_0$  So  $U = \frac{1}{2} kx^2 + U_0$

Where the potential energy at equilibrium position  $= U_0$  when  $U_0 = 0$  then  $U = \frac{1}{2} kx^2$

(ii) In terms of time



Since  $x = A \sin(\omega t + \phi)$ ,  $U = \frac{1}{2} k A^2 \sin^2(\omega t + \phi)$

If initial phase ( $\phi$ ) is zero then  $U = \frac{1}{2} k A^2 \sin^2 \omega t = \frac{1}{2} m \omega^2 A^2 \sin^2 \omega t$

**Note :**

- (i) In S.H.M. the potential energy is a parabolic function of displacement, the potential energy is minimum at the mean position ( $x = 0$ ) and maximum at extreme position ( $x = \pm A$ )
- (ii) The potential energy is the periodic function of time. For  $x = A \sin(\omega t)$ , it is minimum at

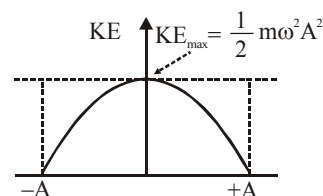
$t = 0, \frac{T}{2}, T, \frac{3T}{2}, \dots$  and maximum at  $t = \frac{T}{4}, \frac{3T}{4}, \frac{5T}{4}, \dots$

**Kinetic Energy (K)**

(i) In terms of displacement

If mass of the particle executing S.H.M. is  $m$  and its velocity is  $v$  then kinetic energy at any instant.

$$K = \frac{1}{2} m v^2 = \frac{1}{2} m \omega^2 (A^2 - x^2) = \frac{1}{2} k (A^2 - x^2)$$



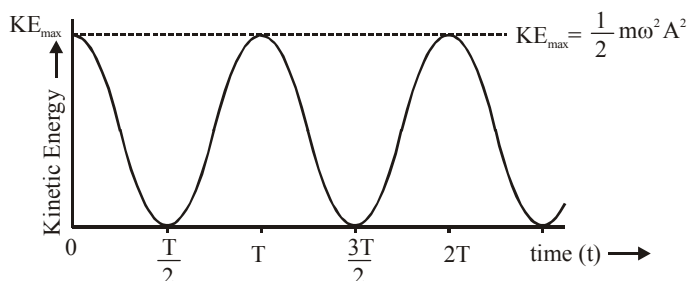
(ii) In terms of time

$$v = A \omega \cos(\omega t + \phi)$$

$$K = \frac{1}{2} m \omega^2 A^2 \cos^2(\omega t + \phi)$$

If initial phase  $\phi$  is zero

$$K = \frac{1}{2} m \omega^2 A^2 \cos^2 \omega t$$



**Note :**

- (i) In S.H.M. the kinetic energy is an inverted parabolic function of displacement. The kinetic energy is maximum ( $\frac{1}{2} k A^2$ ) at mean position ( $x = 0$ ) and minimum (zero) at extreme position ( $x = \pm A$ )
- (ii) The kinetic energy is the periodic function of time. For  $x = A \sin(\omega t)$ , it is maximum at  $t = 0,$

$T, 2T, 3T, \dots$  and minimum at  $t = \frac{T}{2}, \frac{3T}{2}, \frac{5T}{2}, \dots$

- Total energy (E)**

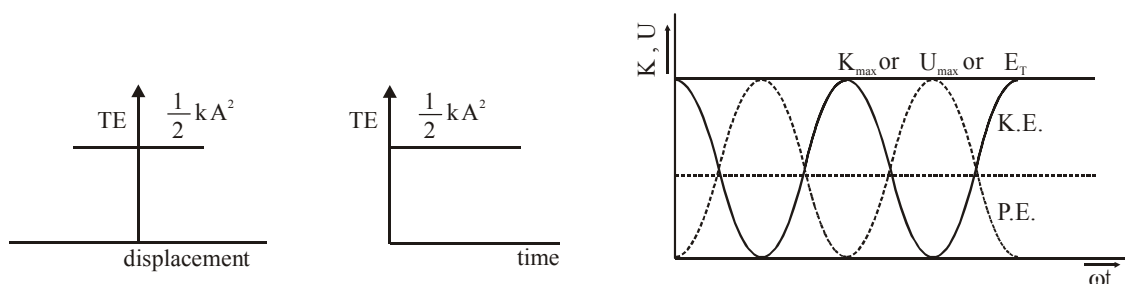
Total energy in S.H.M. is given by ;  $E = \text{potential energy} + \text{kinetic energy} = U + K$

(i) w.r.t. position  $E = \frac{1}{2} kx^2 + \frac{1}{2} k (A^2 - x^2) \Rightarrow E = \frac{1}{2} kA^2 = \text{constant}$

(ii) w.r.t. time

$$E = \frac{1}{2} m\omega^2 A^2 \sin^2 \omega t + \frac{1}{2} m\omega^2 A^2 \cos^2 \omega t = \frac{1}{2} m\omega^2 A^2 (\sin^2 \omega t + \cos^2 \omega t) = \frac{1}{2} m\omega^2 A^2$$

$$= \frac{1}{2} kA^2 = \text{constant}$$



**Note :**

- (i) Total energy of a particle in S.H.M. is same at all instant and at all displacement.
- (ii) Total energy depends upon mass, amplitude and frequency of vibration of the particle executing S.H.M.

### SIMPLE PENDULUM

If a heavy point mass is suspended by a weightless, inextensible string from a rigid support, then this arrangement is called a simple pendulum

- Second's pendulum**

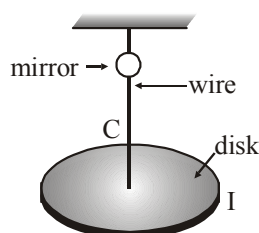
If the time period of a simple pendulum is 2 second then it is called second's pendulum. Second's pendulum take one second to go from one extreme position to other extreme position.

### COMPOUND PENDULUM

Any rigid body which is free to oscillate in a vertical plane about a horizontal axis passing through a point, is define compound pendulum

- Torsional Oscillator : (Angular SHM)**

$$T = 2\pi \sqrt{\frac{I}{C}} \quad \text{where } C = \frac{\eta \pi r^4}{2\ell}$$



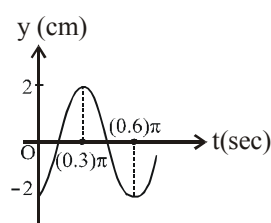
$\eta$  = modulus of elasticity of the wire ;  
 $L$  = length of the wire

$r$  = radius of the wire  
 $I$  = Moment of inertia of the disc

## EXERCISE (S-1)

### Kinematics of SHM :

- Part of a simple harmonic motion is graphed in the figure, where  $y$  is the displacement from the mean position. The correct equation describing this S.H.M is :-



SH0001

- The displacement of a body executing SHM is given by  $x = A \sin (2\pi t + \pi/3)$ . The first time from  $t = 0$  when the velocity is maximum is.

SH0002

- A body undergoing SHM about the origin has its equation given by  $x = 0.2 \cos 5\pi t$ . Find its average speed from  $t = 0$  to  $t = 0.7$  sec.

SH0003

### Energy of SHM :

- An object of mass 0.2 kg executes SHM along the  $x$ -axis with frequency of  $(25/\pi)$  Hz. At the point  $x = 0.04$  m the object has KE 0.5 J and PE 0.4 J. The amplitude of oscillation is \_\_\_\_\_.

SH0004

- A point particle of mass 0.1 kg is executing SHM with amplitude of 0.1 m. When the particle passes through the mean position, its K.E. is  $8 \times 10^{-3}$  J. Obtain the equation of motion of this particle if the initial phase of oscillation is  $45^\circ$ .

SH0005

- Potential Energy ( $U$ ) of a body of unit mass moving in a one-dimension conservative force field is given by  $U = (x^2 - 4x + 3)$ . All units are in S.I.

(i) Find the equilibrium position of the body.

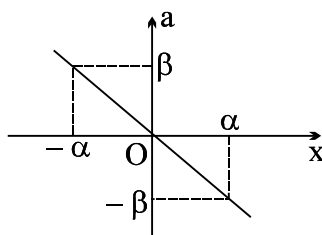
(ii) Show that oscillations of the body about this equilibrium position is simple harmonic motion & find its time period.

(iii) Find the amplitude of oscillations if speed of the body at equilibrium position is  $2\sqrt{6}$  m/s.

SH0006

**Time Period :**

7. The acceleration-displacement ( $a-x$ ) graph of a particle executing simple harmonic motion is shown in the figure. Find the frequency of oscillation.

**SH0007**

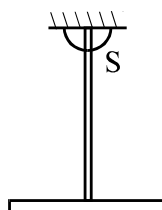
8. A small body of mass  $m$  is fixed to the middle of a stretched string of length  $2\ell$ . In the equilibrium position the string tension is equal to  $T_0$ . Find the angular frequency of small oscillations of the body in the transverse direction. The mass of the string is negligible, the gravitational field is absent. Assume tension in string to be constant.

**SH0008**

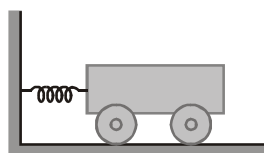
9. A body is in SHM with period  $T$  when oscillated from a freely suspended spring. If this spring is cut in two parts of length ratio  $1 : 3$  & again oscillated from the two parts separately, then the periods are  $T_1$  &  $T_2$  then find  $T_1/T_2$ .

**SH0009**

10. Two identical rods each of mass  $m$  and length  $L$ , are rigidly joined and then suspended in a vertical plane so as to oscillate freely about an axis normal to the plane of paper passing through 'S' (point of suspension). Find the time period of such small oscillations.

**SH0010**

11. A cart consists of a body and four wheels on frictionless axles. The body has a mass  $m$ . The wheels are uniform disks of mass  $M$  and radius  $R$ . The cart rolls, without slipping, back and forth on a horizontal plane under the influence of a spring attached to one end of the cart (figure). The spring constant is  $k$ . Taking into account the moment of inertia of the wheels, find a formula for the frequency of the back and forth motion of the cart.

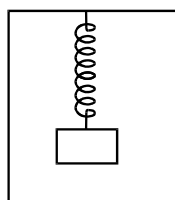
**SH0011**

12. A mass  $M$  attached to a spring, oscillates with a period of 2 sec. If the mass is increased by 2 kg the period increases by one sec. Find the initial mass  $M$  assuming that Hooke's Law is obeyed.

SH0012

**Complex situations :**

13. A spring mass system is hanging from the ceiling of an elevator in equilibrium. Elongation of spring is  $l$ . The elevator suddenly starts accelerating downwards with acceleration  $g/3$ , find  
(a) the frequency and (b) the amplitude of the resulting SHM.



SH0013

14. (a) Find the time period of oscillations of a torsional pendulum, if the torsional constant of the wire is  $K = 10\pi^2 \text{ J/rad}$ . The moment of inertia of rigid body is  $10 \text{ kg m}^2$  about the axis of rotation.  
(b) A simple pendulum of length  $l = 0.5 \text{ m}$  is hanging from ceiling of a car. The car is kept on a horizontal plane. The car starts accelerating on the horizontal road with acceleration of  $5 \text{ m/s}^2$ . Find the time period of oscillations of the pendulum for small amplitudes about the mean position.

SH0014

15. The motion of a simple pendulum is given by

$$\theta = A \cos \left( \sqrt{\frac{g}{\ell}} t \right) \text{ (symbols have their usual meaning)}$$

- (a) Find the tension in the string of this pendulum as a function of time assume that  $\theta \ll \ell$ .  
(b) At what time is the tension maximum? What is the value of this maximum tension?

SH0015

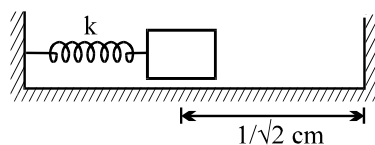
16. A physical pendulum has the shape of a disk of radius  $R$ . The pendulum swings about an axis perpendicular to the plane of the disk and at distance  $\ell$  from the center of the disk.

(a) Show that the frequency of the oscillations of this pendulum is  $\omega = \sqrt{\frac{g\ell}{\frac{1}{2}R^2 + \ell^2}}$

- (b) For what value of  $\ell$  is this frequency at a maximum ?

SH0016

17. A block of mass  $0.9 \text{ kg}$  attached to a spring of force constant  $k$  is lying on a frictionless floor. The spring is compressed to  $\sqrt{2} \text{ cm}$  and the block is at a distance  $1/\sqrt{2} \text{ cm}$  from the wall as shown in the figure. When the block is released, it makes elastic collision with the wall and its period of motion is  $0.2 \text{ sec}$ . Find the approximate value of  $k$ .

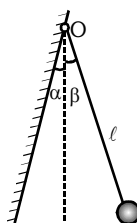


SH0017

18. A body of mass  $1 \text{ kg}$  is suspended from a weightless spring having force constant  $600 \text{ N/m}$ . Another body of mass  $0.5 \text{ kg}$  moving vertically upwards hits the suspended body with a velocity of  $3.0 \text{ m/s}$  and get embedded in it. Find the frequency of oscillations and amplitude of motion.

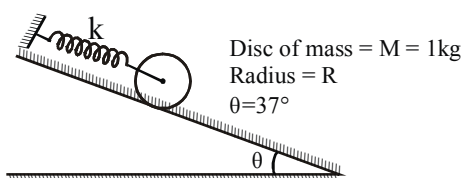
SH0018

19. A ball is suspended by a thread of length  $\ell$  at the point  $O$  on the wall, forming a small angle  $\alpha$  with the vertical as shown in figure. Then the thread with the ball was deviated through a small angle  $\beta$  ( $\beta > \alpha$ ) and set free. Assuming the collision of the ball against the wall to be perfectly elastic, find the oscillation period of such a pendulum.



SH0019

20. A disc of mass  $m$  is connected to an ideal spring of force constant ' $k$ '. If disc is released from rest, then what is maximum friction force on disc (in  $\text{N}$ ). Assuming friction is sufficient for pure rolling



SH0020

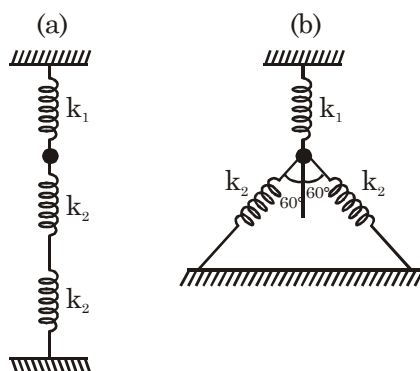
21. A particle of mass  $5 \times 10^{-5} \text{ kg}$  is placed at the lowest point of a smooth parabola having the equation  $20x^2 = y$  ( $x, y$  in  $\text{m}$ ). Here  $y$  is the vertical height. If it is displaced slightly and it moves such that it is constrained to move along the parabola, the angular frequency of oscillation will be, (in  $\text{rad/s}$ ). If your answer is  $N$  fill value  $N/4$ .

SH0021



22. Mass  $m$  is suspended by ideal massless springs in two different ways, indicated by (a) and (b) in the figure. The mass is displaced upwards by a small amount from equilibrium position and is then released resulting in a SHM of the mass in the vertical direction. We denote the oscillation frequencies

associated with the two cases (a) and (b) by  $f_a$  and  $f_b$  respectively. Find  $\frac{f_a}{f_b}$ . Given  $k_2 = 2k_1$ .



SH0022

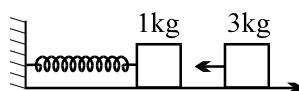
23. A solid sphere of radius  $R$  is floating in a liquid of density  $\rho$  with half of its volume submerged. If the sphere is slightly pushed and released, it starts performing simple harmonic motion. Find the frequency of these oscillations.

[IIT-JEE 2004]

SH0023

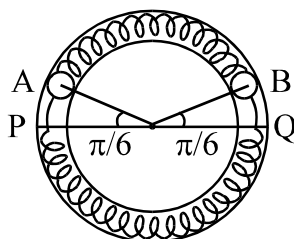
## EXERCISE (S-2)

1. One end of an ideal spring is fixed to a wall at origin O and the axis of spring is parallel to x-axis. A block of mass  $m = 1 \text{ kg}$  is attached to free end of the spring and it is performing SHM. Equation of position of block in coordinate system shown is  $x = 10 + 3\sin 10t$ ,  $t$  is in second and  $x$  in cm. Another block of mass  $M = 3 \text{ kg}$ , moving towards the origin with velocity  $30 \text{ cm/s}$  collides with the block performing SHM at  $t = 0$  and gets struck to it, calculate :
- new amplitude of oscillations.
  - new equation for position of the combined body.
  - loss of energy during collision. Neglect friction.



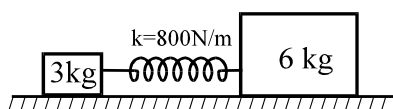
SH0024

2. Two identical balls A and B each of mass  $0.1 \text{ kg}$  are attached to two identical massless springs. The spring mass system is constrained to move inside a rigid smooth pipe in the form of a circle as in fig. The pipe is fixed in a horizontal plane. The centres of the ball can move in a circle of radius  $0.06 \text{ m}$ . Each spring has a natural length  $0.06\pi \text{ m}$  and force constant  $0.1 \text{ N/m}$ . Initially both the balls are displaced by an angle of  $\theta = \pi/6$  radian with respect to diameter PQ of the circle and released from rest
- Calculate the frequency of oscillation of the ball B.
  - What is the total energy of the system.
  - Find the speed of the ball A when A and B are at the two ends of the diameter PQ.



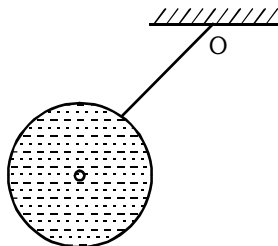
SH0025

3. The system shown in the figure can move on a smooth surface. The spring is initially compressed by  $6 \text{ cm}$  and then released. Find
- time period;
  - amplitude of  $3 \text{ kg}$  block and
  - maximum momentum of  $6 \text{ kg}$  block



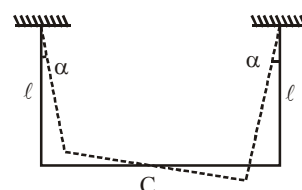
SH0026

4. A pendulum is constructed as a light thin-walled sphere of radius  $R$  filled up with water and suspended at the point  $O$  from a light rigid rod. The distance between the point  $O$  and the centre of the sphere is equal to  $\ell$ . How many times will the small oscillations of such a pendulum change after the water freezes? The viscosity of water and the change of its volume on freezing are to be neglected.



SH0027

5. A uniform rod of mass  $m = 1.5$  kg suspended by two identical threads  $\ell = 90$  cm in length was turned through a small angle about the vertical axis passing through its middle point  $C$ . The threads deviated in the process through an angle  $\alpha = 5.0^\circ$ . Then the rod was released to start performing small oscillations. Find:  
(a) the oscillation period;  
(b) the rod's oscillation energy.

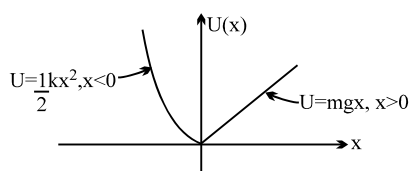


SH0028

6. Two physical pendulums perform small oscillations about the same horizontal axis with frequencies  $\omega_1$  and  $\omega_2$ . Their moments of inertia relative to the given axis are equal to  $I_1$  and  $I_2$  respectively. In a state of stable equilibrium the pendulums were fastened rigidly together. What will be the frequency of small oscillations of the compound pendulum?

SH0029

7. A particle of mass  $m$  moves in the potential energy  $U$  shown above. Find the period of the motion when the particle has total energy  $E$ .



SH0030

8. A particle of mass 4 kg moves between two points  $A$  and  $B$  on a smooth horizontal surface under the action of two forces such that when it is at a point  $P$ , the forces are  $2\overrightarrow{PA}$  N and  $2\overrightarrow{PB}$  N. If the particle is released from rest at  $A$ , find the time it takes to travel a quarter of the way from  $A$  to  $B$ .

SH0031

9. Two masses  $m_1$  and  $m_2$  connected by a light spring of natural length  $\ell_0$  is compressed completely and tied by a string. This system while moving with a velocity  $v_0$  along +ve  $x$ -axis pass through the origin at  $t = 0$ . At this position the string snaps. Position of mass  $m_1$  at time  $t$  is given by the equation

$$x_1(t) = v_0 t - A(1 - \cos \omega t)$$

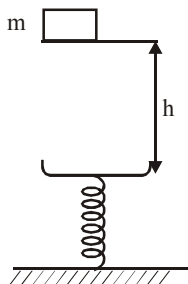
Calculate :

[IIT JEE 2003]

- (i) Position of the particle  $m_2$  as a function of time. (ii)  $\ell_0$  in terms of  $A$ .

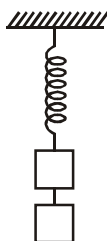
SH0032

10. A body of mass  $m$  fell from a height  $h$  onto the pan of a spring balance. The masses of the pan and the spring are negligible, the stiffness of the latter is  $x$ . Having stuck to the pan, the body starts performing harmonic oscillations in the vertical direction. Find the amplitude and the energy of these oscillations.



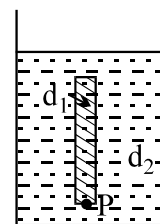
SH0033

11. Two small bodies of mass  $2\text{ kg}$  each hang on a spring of force constant of  $200\text{ N/m}$ , attached to each other using a thread with a length of  $10\text{ cm}$ , as shown in the figure. We burn the thread. Find the distance between the two bodies (in  $\text{cm}$ ) when the top body first arrives at its highest position? If the distance comes out to be  $16 \times n$ . Fill  $n$  [Take:  $g = 10\text{ m/s}^2$  and  $\pi^2 = 10$ ]



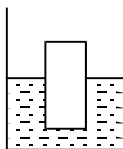
SH0034

12. A thin rod of length  $L$  & area of cross-section  $S$  is pivoted at its lowest point  $P$  inside a stationary, homogeneous & non-viscous liquid (Figure). The rod is free to rotate in a vertical plane about a horizontal axis passing through  $P$ . The density  $d_1$  of the material of the rod is smaller than the density  $d_2$  of the liquid. The rod is displaced by a small angle  $\theta$  from its equilibrium position and then released. Show that the motion of the rod is simple harmonic and determine its angular frequency in terms of the given parameters.



SH0035

13. A liquid of density  $2\rho$  is filled in a cylindrical vessel, whose cross-sectional area is  $2A$ . A wooden cylinder of height  $H$ , cross-sectional area  $A$  and density  $\rho$  is floating in the liquid at equilibrium with its axis vertical. The cylinder is pushed down by a small distance  $x$  from its equilibrium position and released. Find its initial acceleration.



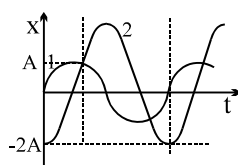
SH0036

## EXERCISE (O-1)

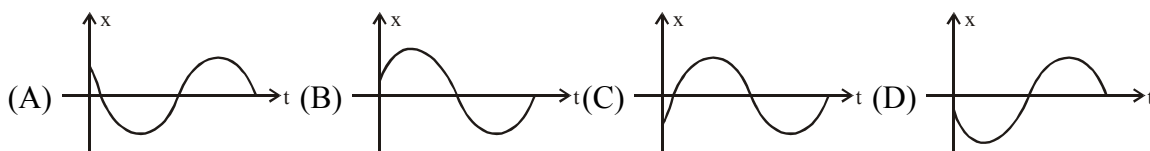
### SINGLE CORRECT TYPE QUESTIONS

#### Kinematics of SHM :

- The equation of motion of a particle is  $x = a \cos(\omega t)^2$ . The motion is  
 (A) periodic but not oscillatory. (B) periodic and oscillatory.  
 (C) oscillatory but not periodic. (D) neither periodic nor oscillatory. SH0037
- Statement 1 :** Position–time equation of a particle moving along x–axis is  $x = 4 + 6 \sin \omega t$ . Under this situation, motion of particle is not simple harmonic.  
**and**  
**Statement 2 :**  $\frac{d^2x}{dt^2}$  for the given equation is not proportional to  $-x$ .  
 (A) Statement–1 is True, Statement–2 is True ; Statement–2 is a correct explanation for Statement–1.  
 (B) Statement–1 is True, Statement–2 is True ; Statement–2 is not a correct explanation for Statement–1.  
 (C) Statement–1 is True, Statement–2 is False.  
 (D) Statement–1 is False, Statement–2 is True. SH0038
- A simple harmonic motion having an amplitude A and time period T is represented by the equation :  $y = 5 \sin \pi(t + 4)$  m. Then the values of A (in m) and T (in sec) are :  
 (A) A = 5 ; T = 2 (B) A = 10 ; T = 1 (C) A = 5 ; T = 1 (D) A = 10 ; T = 2 SH0039
- The oscillations represented by curve 1 in the graph are expressed by equation  $x = A \sin \omega t$ . The equation for the oscillations represented by curve 2 is expressed as:



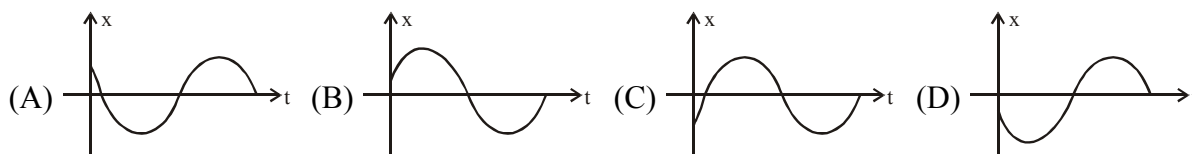
- (A)  $x = 2A \sin(\omega t - \pi/2)$  (B)  $x = 2A \sin(\omega t + \pi/2)$   
 (C)  $x = -2A \sin(\omega t - \pi/2)$  (D)  $x = A \sin(\omega t - \pi/2)$  SH0040
- A particle performing S.H.M. about mean position  $x = 0$  and at  $t = 0$  it is at position  $x = \frac{A}{\sqrt{2}}$  and moving towards the origin. Then which of the following is its possible graph between position (x) and time (t)



SH0041

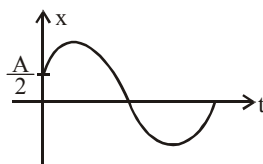
6. A particle is performing S.H.M. and at  $t = \frac{3T}{4}$ , is at position  $= \frac{A}{\sqrt{2}}$  and moving towards the origin.

Equilibrium position of the particle is at  $x = 0$ . After  $t = \frac{3T}{2}$  what will be the graph of the particle :-



SH0042

7. Following graph shows a particle performing S.H.M. about mean position  $x = 0$ . The equation of particle if  $t = \frac{T}{4}$  is taken as starting time is (Notations have usual meanings)



- (A)  $A \sin\left(\omega t + \frac{2\pi}{3}\right)$  (B)  $A \sin\left(\omega t + \frac{\pi}{3}\right)$  (C)  $A \sin\left(\omega t + \frac{\pi}{6}\right)$  (D)  $A \cos\left(\omega t + \frac{2\pi}{3}\right)$

SH0043

8. A particle is performing S.H.M. and at  $t = \frac{3T}{4}$ , is at position  $\frac{\sqrt{3}A}{2}$  and moving towards the origin.

Equilibrium position of the particle is at  $x = 0$ . Then what was the position and direction of particle at  $t = 0$  ?

- (A)  $-\frac{A}{2}$ , away from mean position (B)  $\frac{A}{2}$ , away from mean position  
(C)  $\frac{A}{2}$ , towards mean position (D)  $-\frac{A}{2}$ , towards mean position

SH0044

9. The time taken by a particle performing SHM to pass from point A to B where its velocities are same is 2 seconds. After another 2 seconds it returns to B. The time period of oscillation is (in seconds)

- (A) 2 (B) 8 (C) 6 (D) 4

SH0045

10. A bob is attached to a long, light string. The string is deflected by  $3^\circ$  initially with respect to vertical. The length of the string is 1 m. The value of  $\theta$  at any time  $t$  after the bob released can be approximately written as (Use :  $g = \pi^2$ )

- (A)  $3^\circ \cos \pi t$  (B)  $3^\circ \sin \pi t$  (C)  $3^\circ \sin\left(\pi t + \frac{\pi}{6}\right)$  (D)  $3^\circ \cos\left(\pi t + \frac{\pi}{6}\right)$

SH0046

11. A particle performing SHM is found at its equilibrium at  $t = 1$  sec. and it is found to have a speed of  $0.25$  m/s at  $t = 2$  sec. If the period of oscillation is  $6$  sec. Calculate amplitude of oscillation

(A)  $\frac{3}{2\pi}$  m      (B)  $\frac{3}{4\pi}$  m      (C)  $\frac{6}{\pi}$  m      (D)  $\frac{3}{8\pi}$

SH0047

12. A particle performs SHM with a period  $T$  and amplitude  $a$ . The mean velocity of the particle over the time interval during which it travels a distance  $a/2$  from the extreme position is

(A)  $a/T$       (B)  $2a/T$       (C)  $3a/T$       (D)  $a/2T$

SH0048

13. Two particles execute SHM of same amplitude of  $20$  cm with same period along the same line about the same equilibrium position. The maximum distance between the two is  $20$  cm. Their phase difference in radians is :-

(A)  $\frac{2\pi}{3}$       (B)  $\frac{\pi}{2}$       (C)  $\frac{\pi}{3}$       (D)  $\frac{\pi}{4}$

SH0049

14. Two particles are in SHM on same straight line with amplitude  $A$  and  $2A$  and with same angular frequency  $\omega$ . It is observed that when first particle is at a distance  $A/\sqrt{2}$  from origin and going toward mean position, other particle is at extreme position on other side of mean position. Find phase difference between the two particles

(A)  $45^\circ$       (B)  $90^\circ$       (C)  $135^\circ$       (D)  $180^\circ$

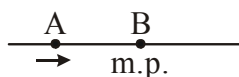
SH0050

15. Two pendulums have time periods  $T$  and  $5T/4$ . They start SHM at the same time from the mean position. After how many oscillations of the smaller pendulum they will be again in the same phase:

(A)  $5$       (B)  $4$       (C)  $11$       (D)  $9$

SH0051

16. A particle is oscillating simple harmonically with angular frequency  $\omega$  and amplitude  $A$ . It is at a point (A) at certain instant (shown in figure). At this instant it is moving towards mean position (B). It takes time  $t$  to reach mean position (B). If time period of oscillation is  $T$ , the average speed between A and B is :-



(A)  $\frac{A \sin \omega t}{t}$       (B)  $\frac{A \cos \omega t}{t}$       (C)  $\frac{A \sin \omega t}{T}$       (D)  $\frac{A \cos \omega t}{T}$

SH0052

17. A particle executes SHM on a straight line path. The amplitude of oscillation is  $2$  cm. When the displacement of the particle from the mean position is  $1$  cm, the numerical value of magnitude of acceleration is equal to the numerical value of magnitude of velocity. The frequency of SHM (in  $\text{second}^{-1}$ ) is:

(A)  $2\pi\sqrt{3}$       (B)  $\frac{2\pi}{\sqrt{3}}$       (C)  $\frac{\sqrt{3}}{2\pi}$       (D)  $\frac{1}{2\pi\sqrt{3}}$

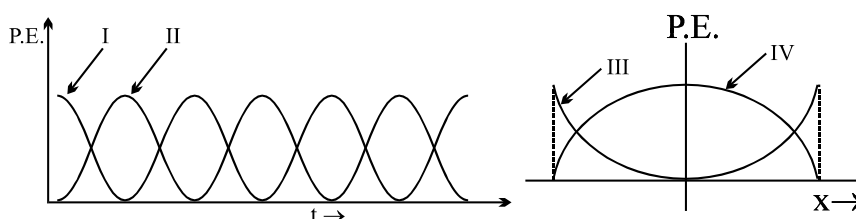
SH0053

18. Speed  $v$  of a particle moving along a straight line, when it is at a distance  $x$  from a fixed point on the line is given by  $v^2 = 108 - 9x^2$  (all quantities in S.I. unit). Then
- (A) The motion is uniformly accelerated along the straight line
- (B) The magnitude of the acceleration at a distance 3 cm from the fixed point is  $0.27 \text{ m/s}^2$ .
- (C) The motion is simple harmonic about  $x = \sqrt{12} \text{ m}$ .
- (D) The maximum displacement from the fixed point is 4 cm.

SH0054

### Energy of SHM :

19. A particle is executing SHM according to  $x = a \cos \omega t$ . Then which of the graphs represents variations of potential energy :
- [IIT JEE (Scr) 2003]



- (A) (I) & (III)      (B) (II) & (IV)      (C) (I) & (IV)      (D) (II) & (III)

SH0055

20. Potential energy of a particle is given as  $U(x) = 2x^3 - 9x^2 + 12x$  where  $U$  is in joule and  $x$  is in metre. If the motion of a particle is S.H.M., then find the potential energy of the particle at mean position :-
- (A)  $-36 \text{ J}$       (B)  $4 \text{ J}$       (C)  $5 \text{ J}$       (D) None of these

SH0056

21. If the potential energy of a harmonic oscillator of mass  $2 \text{ kg}$  on its equilibrium position is  $5 \text{ joules}$  and the total energy is  $9 \text{ joules}$ . If the amplitude is one meter then period of the oscillator (in sec) is:
- (A)  $1.5$       (B)  $3.14$       (C)  $6.28$       (D)  $4.67$

SH0057

22. Kinetic energy of a particle executing simple harmonic motion in straight line is  $pv^2$  and potential energy is  $qx^2$ , where  $v$  is speed at distance  $x$  from the mean position. Its time period is given by the expression :-

- (A)  $2\pi\sqrt{\frac{q}{p}}$       (B)  $2\pi\sqrt{\frac{p}{q}}$       (C)  $2\pi\sqrt{\frac{q}{p+q}}$       (D)  $2\pi\sqrt{\frac{p}{p+q}}$

SH0058

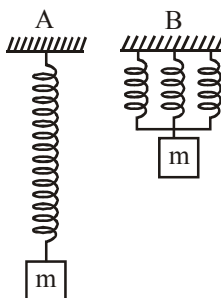
### Time Period :

23. The angular frequency of motion whose equation is  $4\frac{d^2y}{dt^2} + 9y = 0$  is ( $y$  = displacement and  $t$  = time)
- (A)  $\frac{9}{4}$       (B)  $\frac{4}{9}$       (C)  $\frac{3}{2}$       (D)  $\frac{2}{3}$

SH0059



24. The springs in figure A and B are identical but length of spring in A is three times than the length of each spring in B. The ratio of period  $T_A/T_B$  is :-



- (A)  $\sqrt{3}$  (B)  $1/3$  (C) 3 (D)  $1/\sqrt{3}$

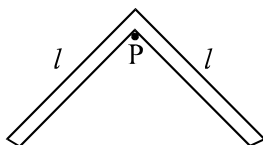
SH0060

25. A rod whose ends are A & B of length 25 cm is hanged in vertical plane. When hanged from point A and point B the time periods calculated are 3 sec & 4 sec respectively. Given the moment of inertia of rod about axis perpendicular to the rod is in ratio 9 : 4 at points A and B. Find the distance of the centre of mass from point A.

- (A) 9 cm (B) 5 cm (C) 25 cm (D) 20 cm

SH0061

26. A system of two identical rods (L-shaped) of mass  $m$  and length  $l$  are resting on a peg P as shown in the figure. If the system is displaced in its plane by a small angle  $\theta$ , find the period of oscillations:



- (A)  $2\pi\sqrt{\frac{\sqrt{2}l}{3g}}$  (B)  $2\pi\sqrt{\frac{2\sqrt{2}l}{3g}}$  (C)  $2\pi\sqrt{\frac{2l}{3g}}$  (D)  $3\pi\sqrt{\frac{l}{3g}}$

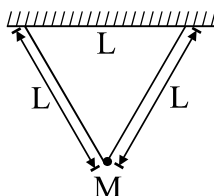
SH0062

27. A ring of diameter  $2m$  oscillates as a compound pendulum about a horizontal axis passing through a point at its rim. It oscillates such that its centre move in a plane which is perpendicular to the plane of the ring. The equivalent length of the simple pendulum is :-

- (A)  $2m$  (B)  $4m$  (C)  $1.5m$  (D)  $3m$

SH0063

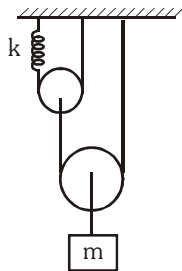
28. A man is swinging on a swing made of 2 ropes of equal length  $L$  and in direction perpendicular to the plane of paper. The time period of the small oscillations about the mean position is :-



- (A)  $2\pi\sqrt{\frac{L}{2g}}$  (B)  $2\pi\sqrt{\frac{\sqrt{3}L}{2g}}$  (C)  $2\pi\sqrt{\frac{L}{2\sqrt{3}g}}$  (D)  $\pi\sqrt{\frac{L}{g}}$

SH0064

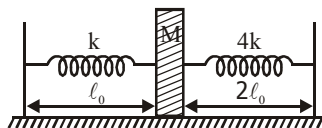
29. What is the period of small oscillations of the block of mass  $m$  if the springs are ideal and pulleys are massless ?



- (A)  $\frac{\pi}{2} \sqrt{\frac{m}{k}}$  (B)  $\frac{\pi}{2} \sqrt{\frac{m}{2k}}$  (C)  $\frac{\pi}{2} \sqrt{\frac{2m}{k}}$  (D)  $\pi \sqrt{\frac{m}{2k}}$

SH0065

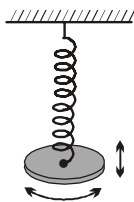
30. A block of mass  $M$  is kept in gravity free space and touches the two springs as shown in the figure. Initially springs are in their natural lengths. Now, the block is shifted  $(\ell_0/2)$  from the given position in such a way that it compresses a spring and is released. The time-period of oscillation of mass will be:-



- (A)  $\frac{\pi}{2} \sqrt{\frac{M}{K}}$  (B)  $2\pi \sqrt{\frac{m}{5K}}$  (C)  $\frac{3\pi}{2} \sqrt{\frac{M}{K}}$  (D)  $\pi \sqrt{\frac{M}{2K}}$

SH0066

31. A solid disk of radius  $R$  is suspended from a spring of linear spring constant  $k$  and torsional constant  $c$ , as shown in figure. In terms of  $k$  and  $c$ , what value of  $R$  will give the same period for the vertical and torsional oscillations of this system?



- (A)  $\sqrt{\frac{2c}{k}}$  (B)  $\sqrt{\frac{c}{2k}}$  (C)  $2\sqrt{\frac{c}{k}}$  (D)  $\frac{1}{2} \sqrt{\frac{c}{k}}$

SH0067

### Superposition :

32. The amplitude of the vibrating particle due to superposition of two SHMs,

$$y_1 = \sin \left( \omega t + \frac{\pi}{3} \right) \text{ and } y_2 = \sin \omega t \text{ is :}$$

- (A) 1 (B)  $\sqrt{2}$  (C)  $\sqrt{3}$  (D) 2

SH0068

33. Two simple harmonic motions  $y_1 = A \sin \omega t$  and  $y_2 = A \cos \omega t$  are superimposed on a particle of mass  $m$ . The total mechanical energy of the particle is:

(A)  $\frac{1}{2} m \omega^2 A^2$  (B)  $m \omega^2 A^2$  (C)  $\frac{1}{4} m \omega^2 A^2$  (D) zero

SH0069

34. The displacement of a particle varies with time according to the relation  $y = a \sin \omega t + b \cos \omega t$ .

(A) The motion is oscillatory but not S.H.M.  
 (B) The motion is S.H.M. with amplitude  $a + b$ .  
 (C) The motion is S.H.M. with amplitude  $a^2 + b^2$ .  
 (D) The motion is S.H.M. with amplitude  $\sqrt{a^2 + b^2}$

SH0070

35. Equations  $y = 2A \cos^2 \omega t$  and  $y = A (\sin \omega t + \sqrt{3} \cos \omega t)$  represent the motion of two particles.

(A) Only one of these is S.H.M. (B) Ratio of maximum speeds is 2 : 1  
 (C) Ratio of maximum speeds is 1 : 1 (D) Ratio of maximum accelerations is 1 : 4

SH0071

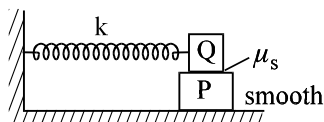
### Complex situations :

36. A plank with a small block on top of it is under going vertical SHM. Its period is 2 sec. The minimum amplitude at which the block will separate from plank is :

(A)  $\frac{10}{\pi^2}$  (B)  $\frac{\pi^2}{10}$  (C)  $\frac{20}{\pi^2}$  (D)  $\frac{\pi}{10}$

SH0072

37. A block P of mass  $m$  is placed on a frictionless horizontal surface. Another block Q of same mass is kept on P and connected to the wall with the help of a spring of spring constant  $k$  as shown in the figure.  $\mu_s$  is the coefficient of friction between P and Q. The blocks move together performing SHM of amplitude  $A$ . The maximum value of the friction force between P and Q is :- [IIT JEE 2004]



(A)  $kA$  (B)  $\frac{kA}{2}$  (C) zero (D)  $\mu_s mg$

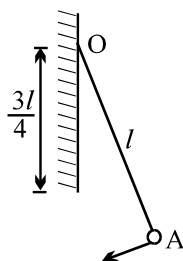
SH0073

38. Two bodies P & Q of equal mass are suspended from two separate massless springs of force constants  $k_1$  &  $k_2$  respectively. If the maximum velocity of them are equal during their motion, the ratio of amplitude of P to Q is :

(A)  $\frac{k_1}{k_2}$  (B)  $\sqrt{\frac{k_2}{k_1}}$  (C)  $\frac{k_2}{k_1}$  (D)  $\sqrt{\frac{k_1}{k_2}}$

SH0074

39. A small bob attached to a light inextensible thread of length  $l$  has a periodic time  $T$  when allowed to vibrate as a simple pendulum. The thread is now suspended from a fixed end  $O$  of a vertical rigid rod of length  $\frac{3l}{4}$  (as in figure). If now the pendulum performs periodic oscillations in this arrangement, the periodic time will be



- (A)  $\frac{3T}{4}$  (B)  $\frac{T}{2}$  (C)  $T$  (D)  $2T$

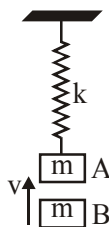
SH0075

40. Vertical displacement of a plank with a body of mass ' $m$ ' on it is varying according to law  $y = \sin \omega t + \sqrt{3} \cos \omega t$ . The minimum value of  $\omega$  for which the mass just breaks off the plank and the moment it occurs first after  $t = 0$  are given by: ( $y$  is positive vertically upwards)

- (A)  $\sqrt{\frac{g}{2}}, \frac{\sqrt{2}}{6} \frac{\pi}{\sqrt{g}}$  (B)  $\frac{g}{\sqrt{2}}, \frac{2}{3} \sqrt{\frac{\pi}{g}}$  (C)  $\sqrt{\frac{g}{2}}, \frac{\pi}{3} \sqrt{\frac{2}{g}}$  (D)  $\sqrt{2g}, \sqrt{\frac{2\pi}{3g}}$

SH0076

41. Block A is hanging from a vertical spring and is at rest. Block B strikes the block A with velocity  $v$  and sticks to it. Then the value of  $v$  for which the spring just attains natural length is



- (A)  $\sqrt{\frac{60mg^2}{k}}$  (B)  $\sqrt{\frac{6mg^2}{k}}$  (C)  $\sqrt{\frac{10mg^2}{k}}$  (D)  $\sqrt{\frac{8mg^2}{k}}$

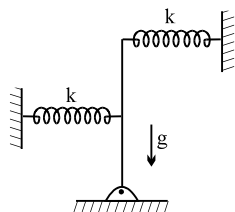
SH0077

42. A particle of mass 10 gm moves in a field where potential energy per unit mass is given by expression  $v = 8 \times 10^4 x^2$  erg/gm. If the total energy of the particle is  $8 \times 10^7$  erg then the relation between  $x$  and time  $t$  is : [ $\phi = \text{constant}$ ]

- (A)  $x = 10 \sin (400 t + \phi)$  cm (B)  $x = \sin (400 t + \phi)$  m  
(C)  $x = 10 \sin (40 t + \phi)$  cm (D)  $x = 100 \sin (4 t + \phi)$  m

SH0078

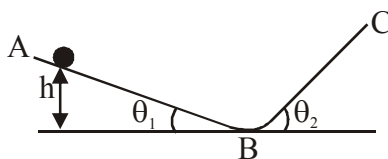
43. In the figure shown, the spring are connected to the rod at one end and at the midpoint. The rod is hinged at its lower end. Rotational SHM of the rod (Mass  $m$ , length  $L$ ) will occur only if



- (A)  $k > mg/3L$       (B)  $k > 2mg/3L$       (C)  $k > 2mg/5L$       (D)  $k > 0$

SH0079

44. A small glass bead of mass  $m$  initially at rest starts from a point at height  $h$  above the horizontal and rolls down the inclined plane AB as shown. Then it rises along the inclined plane BC. Assuming no loss of energy and sufficient friction for pure rolling, the time period of oscillation of the glass bead is:



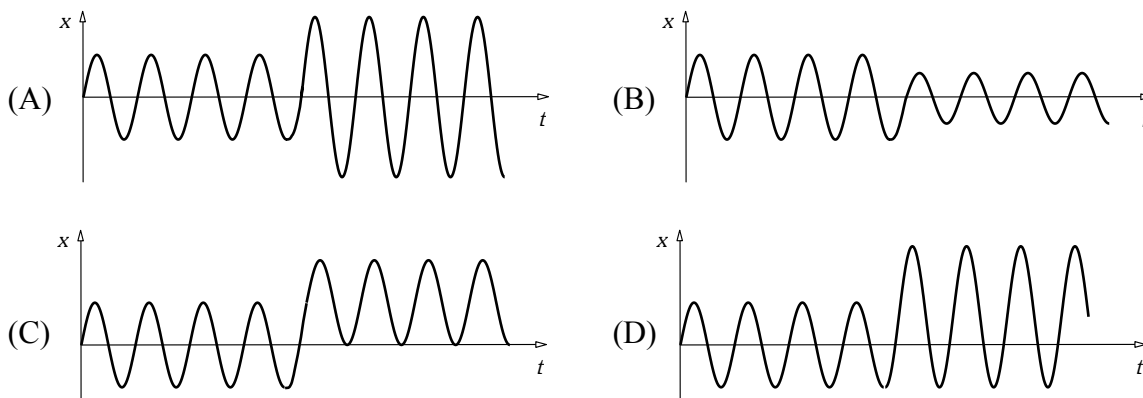
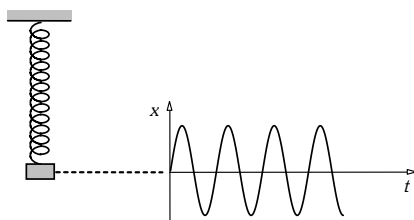
- (A)  $\sqrt{\frac{8h}{g}}(\sin \theta_1 + \sin \theta_2)$       (B)  $2\sqrt{\frac{14h}{5g}}\left(\frac{1}{\sin \theta_1} + \frac{1}{\sin \theta_2}\right)$   
 (C)  $\sqrt{\frac{8h}{g}}\left(\frac{1}{\sin \theta_1} + \frac{1}{\sin \theta_2}\right)$       (D)  $\sqrt{\frac{8h}{5g}}\left(\frac{1}{\sin \theta_1} + \frac{1}{\sin \theta_2}\right)$

SH0080

## EXERCISE (O-2)

## SINGLE CORRECT TYPE QUESTIONS

1. In the figure is shown a spring-mass system oscillating in uniform gravity. If we neglect all dissipative forces, it will keep on oscillating endlessly with constant amplitude and frequency. Accompanying graph shows how displacement  $x$  of the block from the equilibrium position varies with time  $t$ . Now at a certain instant  $t = t_0$  when the block reaches its lowest position, gravity is switched off by some unknown mechanism. Which of the following graphs would correctly describes the changes taking place due to switching off the gravity?



SH0081

2. Time period of a particle executing SHM is 8 sec. At  $t = 0$  it is at the mean position. The ratio of the distance covered by the particle in the 1st second to the 2nd second is :

- (A)  $\frac{1}{\sqrt{2}+1}$  (B)  $\sqrt{2}$  (C)  $\frac{1}{\sqrt{2}}$  (D)  $\sqrt{2}+1$

SH0082

3. A particle executes SHM with time period  $T$  and amplitude  $A$ . The maximum possible average velocity in time  $\frac{T}{4}$  is

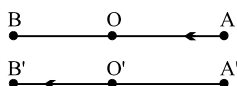
- (A)  $\frac{2A}{T}$  (B)  $\frac{4A}{T}$  (C)  $\frac{8A}{T}$  (D)  $\frac{4\sqrt{2}A}{T}$

SH0083

4. A particle moves along the x-axis according to :  $x = A.[1 + \sin \omega t]$ . What distance does it travel between  $t = 0$  and  $t = 2.5\pi/\omega$ ?
- (A)  $4A$  (B)  $6A$  (C)  $5A$  (D) None

SH0084

5. Two particles undergo SHM along parallel lines with the same time period ( $T$ ) and equal amplitudes. At a particular instant, one particle is at its extreme position while the other is at its mean position. They move in the same direction. They will cross each other after a further time



- (A)  $T/8$  (B)  $3T/8$  (C)  $T/6$  (D)  $4T/3$

SH0085

6. Two particles are in SHM in a straight line about same equilibrium position. Amplitude  $A$  and time period  $T$  of both the particles are equal. At time  $t=0$ , one particle is at displacement  $y_1 = +A$  and the other at  $y_2 = -A/2$ , and they are approaching towards each other. After what time they cross each other ?

- (A)  $T/3$  (B)  $T/4$  (C)  $5T/6$  (D)  $T/6$

SH0086

7. Two particles A and B perform SHM along the same straight line with the same amplitude ' $a$ ', same frequency ' $f$ ' and same equilibrium position ' $O$ '. The greatest distance between them is found to be  $\frac{3a}{2}$ . At some instant of time they have the same displacement from mean position. What is the displacement?

- (A)  $a/2$  (B)  $a\sqrt{7}/4$  (C)  $\sqrt{3}a/2$  (D)  $3a/4$

SH0087

8. A wire frame in the shape of an equilateral triangle is hinged at one vertex so that it can swing freely in a vertical plane, with the plane of the  $\Delta$  always remaining vertical. The side of the frame is  $1/\sqrt{3}$  m. The time period in seconds of small oscillations of the frame will be

- (A)  $\frac{\pi}{\sqrt{2}}$  (B)  $\pi\sqrt{2}$  (C)  $\frac{\pi}{\sqrt{6}}$  (D)  $\frac{\pi}{\sqrt{5}}$

SH0088

9. For a particle acceleration is defined as  $\vec{a} = \frac{-5x\hat{i}}{|x|}$  for  $x \neq 0$  and  $\vec{a} = 0$  for  $x = 0$ . If the particle is initially at rest at  $(a, 0)$ , what is period of motion of the particle.

- (A)  $4\sqrt{2a/5}$  sec. (B)  $8\sqrt{2a/5}$  sec. (C)  $2\sqrt{2a/5}$  sec. (D) cannot be determined

SH0089

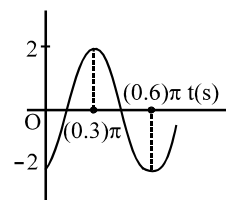
10. Part of a simple harmonic motion is graphed in the figure, where  $y$  is the displacement from the mean position. The correct equation describing this S.H.M is

(A)  $y = 4 \cos(0.6t)$

(B)  $y = 2 \sin\left(\frac{10}{3}t - \frac{\pi}{2}\right)$

(C)  $y = 4 \sin\left(\frac{10}{3}t + \frac{\pi}{2}\right)$

(D)  $y = 2 \cos\left(\frac{10}{3}t + \frac{\pi}{2}\right)$



SH0090

11. The angular frequency of a spring block system is  $\omega_0$ . This system is suspended from the ceiling of an elevator moving downwards with a constant speed  $v_0$ . The block is at rest relative to the elevator. Lift is suddenly stopped. Assuming the downwards as a positive direction, choose the wrong statement:

(A) The amplitude of the block is  $\frac{v_0}{\omega_0}$

(B) The initial phase of the block is  $\pi$ .

(C) The equation of motion for the block is  $\frac{v_0}{\omega_0} \sin \omega_0 t$ .

(D) The maximum speed of the block is  $v_0$ .

SH0091

12. Two particles are performing SHM with same angular frequency and amplitudes  $A$  and  $2A$  respectively along same straight line with same mean position. They cross each other at position  $A/2$  distance from mean position in opposite direction. The phase between them is :

(A)  $\frac{5\pi}{6} - \sin^{-1}\left(\frac{1}{4}\right)$  (B)  $\frac{\pi}{6} - \sin^{-1}\left(\frac{1}{4}\right)$  (C)  $\frac{5\pi}{6} - \cos^{-1}\left(\frac{1}{4}\right)$  (D)  $\frac{\pi}{6} - \cos^{-1}\left(\frac{1}{4}\right)$

SH0092

13. A particle executing a simple harmonic motion of period  $2s$ . When it is at its extreme displacement from its mean position, it receives an additional energy equal to what it had in its mean position. Due to this, in its subsequent motion,

(A) its amplitude will change and become equal to  $\sqrt{2}$  times its previous amplitude

(B) its periodic time will become doubled i.e.  $4s$

(C) its potential energy will be decreased

(D) it will continue to execute simple harmonic motion of the same amplitude and period as before receiving the additional energy.

SH0093

14. A particle is executing SHM of amplitude  $A$ , about the mean position  $x = 0$ . Which of the following cannot be a possible phase difference between the positions of the particle at  $x = +A/2$  and  $x = -A/\sqrt{2}$ .

(A)  $75^\circ$

(B)  $165^\circ$

(C)  $135^\circ$

(D)  $195^\circ$

SH0094



15. A particle free to move along the  $x$ -axis has potential energy given by  $U(x) = k[1 - \exp(-x^2)]$  for  $-\infty < x < +\infty$ , where  $k$  is a positive constant of appropriate dimensions. Then  
 (A) at point away from the origin, the particle is in unstable equilibrium.  
 (B) for any finite non-zero value of  $x$ , there is a force directed away from the origin.  
 (C) if its total mechanical energy is  $k/2$ , it has its minimum kinetic energy at the origin.  
 (D) for small displacements from  $x = 0$ , the motion is simple harmonic.

SH0095

16. The time period of a bar pendulum when suspended at distances 30 cm and 50 cm from its centre of gravity comes out to be the same. If the mass of the body is 2 kg. Find out its moment of inertia about an axis passing through first point.  
 (A)  $0.24 \text{ kg-m}^2$  (B)  $0.72 \text{ kg-m}^2$  (C)  $0.48 \text{ kg-m}^2$  (D) Data insufficient

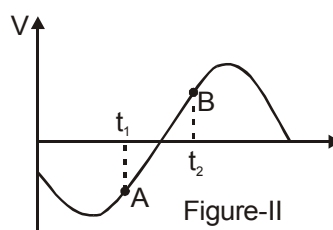
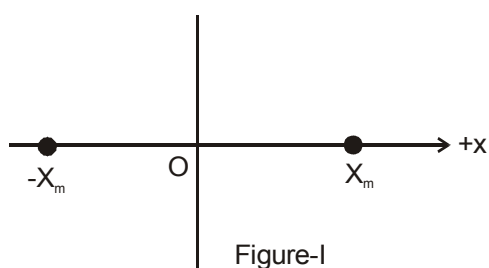
SH0096

### MULTIPLE CORRECT TYPE QUESTIONS

17. For a SHM with given angular frequency, two arbitrary initial conditions are necessary and sufficient to determine the motion completely. These initial conditions may be-  
 (A) Amplitude and initial phase  
 (B) Amplitude and total energy of oscillation  
 (C) Initial phase and total energy of oscillation  
 (D) Initial position and initial velocity

SH0097

18. A particle is executing SHM between points  $-X_m$  and  $X_m$ , as shown in figure-I. The velocity  $V(t)$  of the particle is partially graphed and shown in figure-II. Two points A and B corresponding to time  $t_1$  and time  $t_2$  respectively are marked on the  $V(t)$  curve.



- (A) At time  $t_1$ , it is going towards  $X_m$ .  
 (B) At time  $t_1$ , its speed is decreasing.  
 (C) At time  $t_2$ , its position lies in between  $-X_m$  and O.  
 (D) The phase difference  $\Delta\phi$  between points A and B must be expressed as  $90^\circ < \Delta\phi < 180^\circ$ .

SH0098

19. For a body executing SHM with amplitudes  $A$ , time period  $T$ , max velocity  $v_{\max}$  and phase constant zero, which of the following statements are correct?  
 (A) At  $y = (A/2)$ ,  $v > (v_{\max}/2)$  (B)  $v = (v_{\max}/2)$  for  $|y| > (A/2)$   
 (C) For  $t = (T/8)$ ,  $y > (A/2)$  (D) For  $y = (A/2)$ ,  $t < (T/8)$

SH0099

20. The amplitude of a particle executing SHM about O is 10 cm. Then :
- (A) When the K.E. is 0.64 of its max. K.E. its displacement is 6cm from O.
- (B) When the displacement is 5 cm from O its K.E. is 0.75 of its max.P.E.
- (C) Its total energy at any point is equal to its maximum K.E.
- (D) Its velocity is half the maximum velocity when its displacement is half the maximum displacement.

SH0100

21. The position vector of a particle that is moving in space is given by

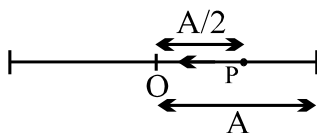
$$\vec{r} = (1 + 2 \cos 2\omega t)\hat{i} + (3 \sin^2 \omega t)\hat{j} + (3)\hat{k}$$

in the ground frame. All units are in SI. Choose the correct statement (s) :

- (A) The particle executes SHM in the ground frame about the mean position  $\left(1, \frac{3}{2}, 3\right)$
- (B) The particle executes SHM in a frame moving along the z-axis with a velocity of 3 m/s.
- (C) The amplitude of the SHM of the particle is  $\frac{5}{2}$  m.
- (D) The direction of the SHM of the particle is given by the vector  $\left(\frac{4}{5}\hat{i} - \frac{3}{5}\hat{j}\right)$

SH0101

22. A particle starts from a point P at a distance of  $A/2$  from the mean position O & travels towards left as shown in the figure. If the time period of SHM, executed about O is T and amplitude A then the equation of motion of particle is :



- (A)  $x = A \sin \left( \frac{2\pi}{T} t + \frac{\pi}{6} \right)$
- (B)  $x = A \sin \left( \frac{2\pi}{T} t + \frac{5\pi}{6} \right)$
- (C)  $x = A \cos \left( \frac{2\pi}{T} t + \frac{\pi}{6} \right)$
- (D)  $x = A \cos \left( \frac{2\pi}{T} t + \frac{\pi}{3} \right)$

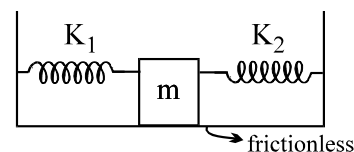
SH0102

23. A spring has natural length 40 cm and spring constant 500 N/m. A block of mass 1 kg is attached at one end of the spring and other end of the spring is attached to ceiling. The block released from the position, where the spring has length 45 cm.
- (A) the block will perform SHM of amplitude 5 cm.
- (B) the block will have maximum velocity  $30\sqrt{5}$  cm/sec.
- (C) the block will have maximum acceleration  $15 \text{ m/s}^2$
- (D) the minimum potential energy of the spring will be zero.

SH0103

24. Two springs with negligible masses and force constant of  $K_1 = 200 \text{ Nm}^{-1}$  and  $K_2 = 160 \text{ Nm}^{-1}$  are attached to the block of mass  $m = 10 \text{ kg}$  as shown in the figure. Initially the block is at rest, at the equilibrium position in which both springs are neither stretched nor compressed. At time  $t = 0$ , a sharp impulse of  $50 \text{ Ns}$  is given to the block with a hammer.

- (A) Period of oscillations for the mass  $m$  is  $\frac{\pi}{3} \text{ s}$ .  
 (B) Maximum velocity of the mass  $m$  during its oscillation is  $5 \text{ ms}^{-1}$ .  
 (C) Data are insufficient to determine maximum velocity.  
 (D) Amplitude of oscillation is  $0.42 \text{ m}$ .



SH0104

25. A mass of  $0.2 \text{ kg}$  is attached to the lower end of a massless spring of force-constant  $200 \text{ N/m}$ , the upper end of which is fixed to a rigid support. Which of the following statements is/are true?

- (A) In equilibrium, the spring will be stretched by  $1 \text{ cm}$ .  
 (B) If the mass is raised till the spring is unstretched state and then released, it will go down by  $2 \text{ cm}$  before moving upwards.  
 (C) The frequency of oscillation will be nearly  $5 \text{ Hz}$ .  
 (D) If the system is taken to the moon, the frequency of oscillation will be the same as on the earth.

SH0105

26. A particle moves in  $x$ - $y$  plane according to equation  $\vec{S} = (2\hat{i} + \hat{j}) \cos \omega t$ . The motion of particle is:-

- (A) On straight line (B) On ellipse (C) Periodic (D) SHM

SH0106

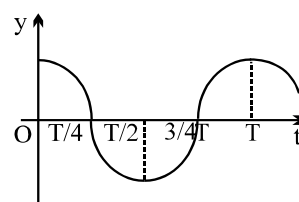
27. Two particles are in SHM with same amplitude  $A$  and same angular frequency  $\omega$ . At time  $t = 0$ , one is at  $x = +\frac{A}{2}$  and other is at  $x = -\frac{A}{2}$ . Both are moving in same direction.

- (A) Phase difference between the two particle is  $\frac{\pi}{3}$   
 (B) Phase difference between the two particle is  $\frac{2\pi}{3}$   
 (C) They will collide after time  $t = \frac{\pi}{2\omega}$   
 (D) They will collide after time  $t = \frac{3\pi}{\omega}$

SH0107

28. The displacement-time graph of a particle executing SHM is shown. Which of the following statements is/are true?

- (A) The velocity is maximum at  $t = T/2$   
 (B) The acceleration is maximum at  $t = T$   
 (C) The force is zero at  $t = 3T/4$   
 (D) The potential energy equals the oscillation energy at  $t = T/2$ .



SH0108

29. A particle is executing SHM with amplitude  $A$ , time period  $T$ , maximum acceleration  $a_0$  and maximum velocity  $v_0$ . It starts from mean position at  $t = 0$  and at time  $t$ , it has the displacement  $A/2$ , acceleration  $a$  and velocity  $v$  then :-

(A)  $t = T/12$                       (B)  $a = a_0/2$                       (C)  $v = v_0/2$                       (D)  $t = T/8$

SH0109

30. Three simple harmonic motions in the same direction having the same amplitude  $a$  and same period are superposed. If each differs in phase from the next by  $45^\circ$ , then :- (1999S)

(A) the resultant amplitude is  $(1 + \sqrt{2})a$   
 (B) the phase of the resultant motion relative to the first is  $90^\circ$   
 (C) the energy associated with the resulting motion is  $(3 + 2\sqrt{2})$  times the energy associated with any single motion  
 (D) the resulting motion is not simple harmonic

SH0110

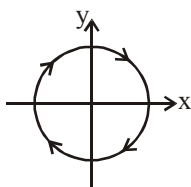
### COMPREHENSION TYPE QUESTIONS

#### Paragraph for question no. 31 and 32

Lissajous figures are produced by superposition of 2 SHM's in mutually perpendicular directions. If both SHM have same frequency, the lissajous figure is simple. It may be a circle, ellipse or a straight line depending on the phase difference between two SHMs. Interesting case occurs when one of the SHM is at double frequency of another. Suppose a body executes SHM vertically with frequency  $f_0$ , but horizontally with a frequency  $2f_0$  and is initially at  $A$ , mean position of vertical as well as horizontal SHM. It can be seen to trace out a figure of 8 in space as shown. Since horizontal SHM has half the time period, it executes two horizontal oscillations by the time it completes a vertical oscillation.



31. A lissajous figure as shown here can be produced by



(A)  $x = A \sin \omega t$  ;  $y = A \cos \omega t$                       (B)  $x = A \cos \omega t$  ;  $y = A \sin \omega t$   
 (C)  $x = A \sin \omega t$  ;  $y = A \sin \left( \omega t + \frac{\pi}{4} \right)$                       (D)  $x = A \sin \omega t$  ;  $y = A \sin \left( \omega t + \frac{3\pi}{4} \right)$

SH0111

32. For the Lissajous figure shown here, the frequency in vertical direction is \_\_\_\_\_ times the frequency in horizontal direction :-

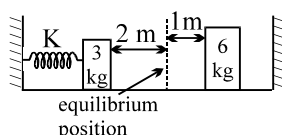


- (A) 3 (B)  $\frac{1}{3}$  (C) 2 (D)  $\frac{1}{2}$

SH0111

**Paragraph for Question No. 33 and 34**

Two blocks of masses 3 kg and 6 kg rest on a horizontal frictionless surface. The 3 kg block is attached to a spring with a force constant  $K = 900 \text{ N/m}$  which is compressed 2 m initially from its equilibrium position. When 3 kg mass is released, it strikes the 6 kg mass and the two stick together.



33. The common velocity of the blocks after collision is :-  
(A) 10 m/s (B) 30 m/s (C) 15 m/s (D) 2 m/s

SH0112

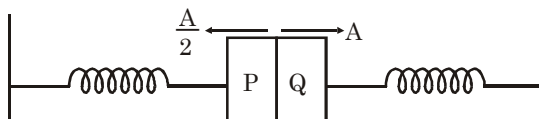
34. The amplitude of resulting oscillation after the collision is :-

- (A)  $\frac{1}{\sqrt{2}} \text{ m}$  (B)  $\frac{1}{\sqrt{3}} \text{ m}$  (C)  $\sqrt{2} \text{ m}$  (D)  $\sqrt{3} \text{ m}$

SH0112

**Paragraph for Question no. 35 and 36**

Two identical blocks P and Q have mass  $m$  each. They are attached to two identical springs initially unstretched. Now the left spring (alongwith P) is compressed by  $\frac{A}{2}$  and the right spring (alongwith Q) is compressed by  $A$ . Both the blocks are released simultaneously. They collide perfectly inelastically. Initially time period of both the block was  $T$ .



35. The time period of oscillation of combined mass is :

- (A)  $\frac{T}{\sqrt{2}}$  (B)  $\sqrt{2}T$  (C)  $T$  (D)  $\frac{T}{2}$

SH0113

36. The amplitude of combined mass is :

- (A)  $\frac{A}{4}$  (B)  $\frac{A}{2}$  (C)  $\frac{2A}{3}$  (D)  $\frac{3A}{4}$

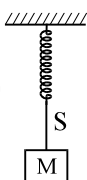
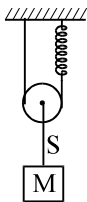
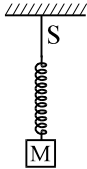
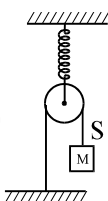
SH0113

## MATRIX MATCH TYPE QUESTION

37. In the following four situations, mass  $M$  of  $1\text{ kg}$  is kept in equilibrium.  $k = 100\text{ N/m}$  in all cases. What speed can be given to mass  $M$  vertically so that inextensible string  $S$  does not become slack in subsequent motion. Consider that pulley is ideal and string is massless :

## Column-I

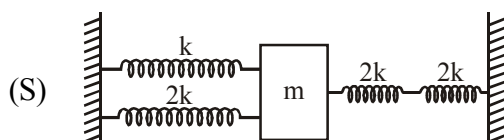
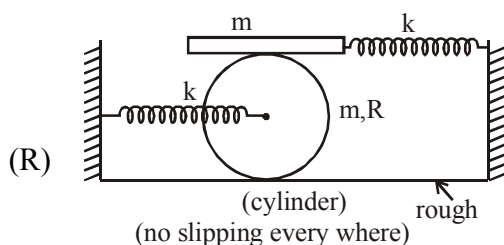
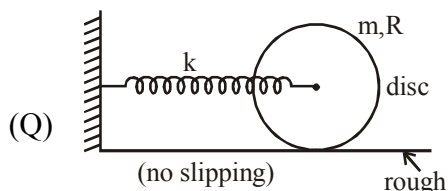
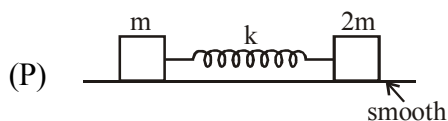
## Column-II

- (A)  (the block is attached to spring by an inextensible thread) (P)  $1\text{ ms}^{-1}$
- (B)  (Q)  $0.5\text{ ms}^{-1}$
- (C)  (block is attached to the spring) (R)  $0.25\text{ ms}^{-1}$
- (D)  (S)  $2\text{ ms}^{-1}$

SH0114

38. In list-I, the systems are performing SHM and in list-II, the time period of SHM is shown then match list-I with list-II.

**List-I**



**List-II**

(1)  $\left[ 2\pi\sqrt{\frac{11m}{10k}} \right]$

(2)  $2\pi\sqrt{\frac{m}{4k}}$

(3)  $2\pi\sqrt{\frac{3m}{2k}}$

(4)  $2\pi\sqrt{\frac{2m}{3k}}$

**Codes :**

|     | P | Q | R | S |
|-----|---|---|---|---|
| (A) | 3 | 2 | 1 | 4 |
| (B) | 4 | 3 | 2 | 1 |
| (C) | 3 | 4 | 2 | 1 |
| (D) | 4 | 3 | 1 | 2 |

SH0115

## EXERCISE (JM)

1. A mass  $M$ , attached to a horizontal spring, executes S.H.M. with amplitude  $A_1$ . When the mass  $M$  passes through its mean position then a smaller mass  $m$  is placed over it and both of them move

together with amplitude  $A_2$ . The ratio of  $\left(\frac{A_1}{A_2}\right)$  is :- [AIEEE-2011]

(1)  $\left(\frac{M}{M+m}\right)^{1/2}$       (2)  $\left(\frac{M+m}{M}\right)^{1/2}$       (3)  $\frac{M}{M+m}$       (4)  $\frac{M+m}{M}$

SH0116

2. Two particles are executing simple harmonic motion of the same amplitude  $A$  and frequency  $\omega$  along the  $x$ -axis. Their mean position is separated by distance  $X_0$  ( $X_0 > A$ ). If the maximum separation between them is  $(X_0 + A)$ , the phase difference between their motion is :- [AIEEE-2011]

(1)  $\frac{\pi}{4}$       (2)  $\frac{\pi}{6}$       (3)  $\frac{\pi}{2}$       (4)  $\frac{\pi}{3}$

SH0117

3. A wooden cube (density of wood ' $d$ ') of side ' $\ell$ ' floats in a liquid of density ' $\rho$ ' with its upper and lower surfaces horizontal. If the cube is pushed slightly down and released, it performs simple harmonic motion of period ' $T$ '. Then, ' $T$ ' is equal to :- [AIEEE-2011]

(1)  $2\pi\sqrt{\frac{\ell\rho}{(\rho-d)g}}$       (2)  $2\pi\sqrt{\frac{\ell d}{\rho g}}$   
 (3)  $2\pi\sqrt{\frac{\ell\rho}{dg}}$       (4)  $2\pi\sqrt{\frac{\ell d}{(\rho-d)g}}$

SH0118

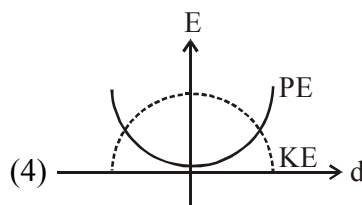
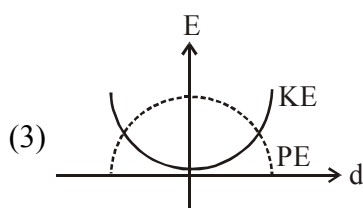
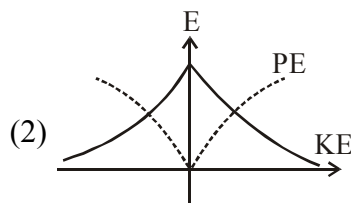
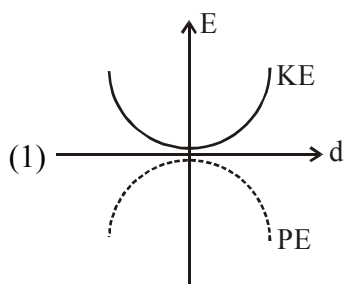
4. A particle moves with simple harmonic motion in a straight line. In first  $\tau$  s, after starting from rest it travels a distance  $a$ , and in next  $\tau$  s it travels  $2a$ , in same direction, then : [JEE Mains-2014]

- (1) Amplitude of motion is  $4a$   
 (2) Time period of oscillation is  $6\tau$   
 (3) Amplitude of motion is  $3a$   
 (4) Time period of oscillation is  $8\tau$

SH0119



5. For a simple pendulum, a graph is plotted between its kinetic energy (KE) and potential energy (PE) against its displacement  $d$ . Which one of the following represents these correctly? (graphs are schematic and not drawn to scale) [JEE Mains-2015]



SH0120

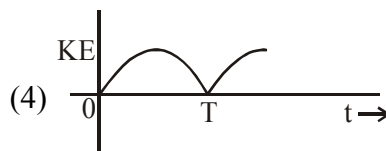
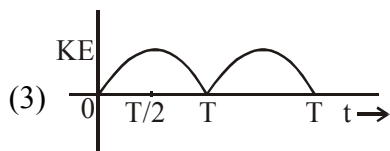
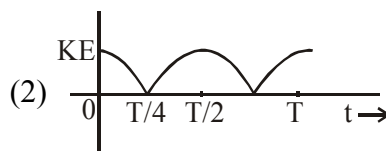
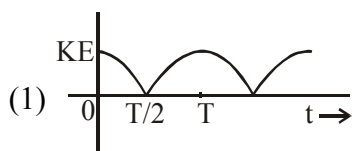
6. A particle performs simple harmonic motion with amplitude  $A$ . Its speed is trebled at the instant that it is at a distance  $\frac{2A}{3}$  from equilibrium position. The new amplitude of the motion is :-

[JEE Mains-2016]

- (1)  $\frac{7A}{3}$       (2)  $\frac{A}{3}\sqrt{41}$       (3)  $3A$       (4)  $A\sqrt{3}$

SH0121

7. A particle is executing simple harmonic motion with a time period  $T$ . At time  $t = 0$ , it is at its position of equilibrium. The kinetic energy-time graph of the particle will look like [JEE Main-2017]



SH0122

8. A silver atom in a solid oscillates in simple harmonic motion in some direction with a frequency of  $10^{12}/\text{sec}$ . What is the force constant of the bonds connecting one atom with the other? (Mole wt. of silver = 108 and Avagadro number =  $6.02 \times 10^{23} \text{ gm mole}^{-1}$ ) [JEE Main-2018]

- (1) 7.1 N/m      (2) 2.2 N/m      (3) 5.5 N/m      (4) 6.4 N/m

SH0123

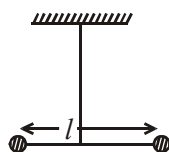
## SELECTIVE PROBLEMS FROM JEE-MAINS ONLINE PAPERS

9. A rod of mass 'M' and length '2L' is suspended at its middle by a wire. It exhibits torsional oscillations; If two masses each of 'm' are attached at distance 'L/2' from its centre on both sides, it reduces the oscillation frequency by 20%. The value of ratio m/M is close to : [JEE-Main-2019\_Jan]
- (1) 0.17 (2) 0.37 (3) 0.57 (4) 0.77

SH0142

10. Two masses m and  $\frac{m}{2}$  are connected at the two ends of a massless rigid rod of length l. The rod is suspended by a thin wire of torsional constant k at the centre of mass of the rod-mass system (see figure). Because of torsional constant k, the restoring torque is  $\tau = k\theta$  for angular displacement  $\theta$ . If the rod is rotated by  $\theta_0$  and released, the tension in it when it passes through its mean position will be:

[JEE-Main-2019\_Jan]



- (1)  $\frac{3k\theta_0^2}{l}$  (2)  $\frac{k\theta_0^2}{2l}$  (3)  $\frac{2k\theta_0^2}{l}$  (4)  $\frac{k\theta_0^2}{l}$

SH0143

11. A particle executes simple harmonic motion with an amplitude of 5 cm. When the particle is at 4 cm from the mean position, the magnitude of its velocity in SI units is equal to that of its acceleration. Then, its periodic time in seconds is : [JEE-Main-2019\_Jan]

- (1)  $\frac{7}{3}\pi$  (2)  $\frac{3}{8}\pi$  (3)  $\frac{4\pi}{3}$  (4)  $\frac{8\pi}{3}$

SH0144

12. A simple pendulum of length 1 m is oscillating with an angular frequency 10 rad/s. The support of the pendulum starts oscillating up and down with a small angular frequency of 1 rad/s and an amplitude of  $10^{-2}$  m. The relative change in the angular frequency of the pendulum is best given by :-

[JEE-Main-2019\_Jan]

- (1)  $10^{-3}$  rad/s (2)  $10^{-1}$  rad/s (3) 1 rad/s (4)  $10^{-5}$  rad/s

SH0145

13. A pendulum is executing simple harmonic motion and its maximum kinetic energy is  $K_1$ . If the length of the pendulum is doubled and it performs simple harmonic motion with the same amplitude as in the first case, its maximum kinetic energy is  $K_2$ . Then :- [JEE-Main-2019\_Jan]

- (1)  $K_2 = \frac{K_1}{4}$  (2)  $K_2 = \frac{K_1}{2}$  (3)  $K_2 = 2K_1$  (4)  $K_2 = K_1$

SH0146

14. A simple harmonic motion is represented by:

$$y = 5(\sin 3\pi t + \sqrt{3} \cos 3\pi t) \text{ cm}$$

The amplitude and time period of the motion are:

[JEE-Main-2019\_Jan]

- (1) 5cm,  $\frac{3}{2}$ s                      (2) 5cm,  $\frac{2}{3}$ s                      (3) 10cm,  $\frac{3}{2}$ s                      (4) 10cm,  $\frac{2}{3}$ s

SH0147

15. A simple pendulum oscillating in air has period T. The bob of the pendulum is completely immersed in a non-viscous liquid. The density of the liquid is  $\frac{1}{16}$ th of the material of the bob. If the bob is inside liquid all the time, its period of oscillation in this liquid is :

[JEE-Main-2019\_April]

- (1)  $4T\sqrt{\frac{1}{15}}$                       (2)  $2T\sqrt{\frac{1}{10}}$                       (3)  $4T\sqrt{\frac{1}{14}}$                       (4)  $2T\sqrt{\frac{1}{14}}$

SH0148

16. A ring is hung on a nail. It can oscillate, without slipping or sliding (i) in its plane with a time period  $T_1$  and, (ii) back and forth in a direction perpendicular to its plane, with a period  $T_2$ . the ratio  $\frac{T_1}{T_2}$  will be :

[JEE-Main-2020\_Sep]

- (1)  $\frac{2}{\sqrt{3}}$                       (2)  $\frac{\sqrt{2}}{3}$                       (3)  $\frac{2}{3}$                       (4)  $\frac{3}{\sqrt{2}}$

SH0149

17. When a particle of mass m is attached to a vertical spring of spring constant k and released, its motion is described by  $y(t) = y_0 \sin^2 \omega t$ , where 'y' is measured from the lower end of unstretched spring. Then  $\omega$  is :

[JEE-Main-2020\_Sep]

- (1)  $\sqrt{\frac{g}{y_0}}$                       (2)  $\sqrt{\frac{g}{2y_0}}$                       (3)  $\frac{1}{2}\sqrt{\frac{g}{y_0}}$                       (4)  $\sqrt{\frac{2g}{y_0}}$

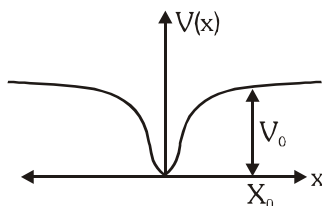
SH0150

## EXERCISE (JA)

## Paragraph for Question No. 1 to 3

When a particle of mass  $m$  moves on the  $x$ -axis in a potential of the form  $V(x) = kx^2$ , it performs simple harmonic motion. The corresponding time period is proportional to  $\sqrt{\frac{m}{k}}$ , as can be seen easily using dimensional analysis. However, the motion of a particle can be periodic even when its potential energy increases on both sides of  $x = 0$  in a way different from  $kx^2$  and its total energy is such that the particle does not escape to infinity. Consider a particle of mass  $m$  moving on the  $x$ -axis. Its potential energy is  $V(x) = \alpha x^4$  ( $\alpha > 0$ ) for  $|x|$  near the origin and becomes a constant equal to  $V_0$  for  $|x| \geq X_0$  (see figure)

[IIT-JEE 2010]



1. If the total energy of the particle is  $E$ , it will perform periodic motion only if :-  
 (A)  $E < 0$  (B)  $E > 0$  (C)  $V_0 > E > 0$  (D)  $E > V_0$

SH0124

2. For periodic motion of small amplitude  $A$ , the time period  $T$  of this particle is proportional to :-

- (A)  $A\sqrt{\frac{m}{\alpha}}$  (B)  $\frac{1}{A}\sqrt{\frac{m}{\alpha}}$  (C)  $A\sqrt{\frac{\alpha}{m}}$  (D)  $\frac{1}{A}\sqrt{\frac{\alpha}{m}}$

SH0124

3. The acceleration of this particle for  $|x| > X_0$  is :-

- (A) proportional to  $V_0$  (B) proportional to  $\frac{V_0}{mX_0}$   
 (C) proportional to  $\sqrt{\frac{V_0}{mX_0}}$  (D) Zero

SH0124

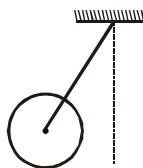
4. A point mass is subjected to two simultaneous sinusoidal displacements in  $x$ -direction,  $x_1(t) = A \sin \omega t$  and  $x_2(t) = A \sin\left(\omega t + \frac{2\pi}{3}\right)$ . Adding a third sinusoidal displacement  $x_3(t) = B \sin(\omega t + \phi)$  brings the mass to a complete rest. The values of  $B$  and  $\phi$  are :-

[IIT-JEE 2011]

- (A)  $\sqrt{2}A, \frac{3\pi}{4}$  (B)  $A, \frac{4\pi}{3}$  (C)  $\sqrt{3}A, \frac{5\pi}{6}$  (D)  $A, \frac{\pi}{3}$

SH0125

5. A metal rod of length 'L' and mass 'm' is pivoted at one end. A thin disk of mass 'M' and radius 'R' ( $< L$ ) is attached at its center to the free end of the rod. Consider two ways the disc is attached: (case A). The disc is not free to rotate about its center and (case B) the disc is free to rotate about its center. The rod-disc system performs SHM in vertical plane after being released from the same displaced position. Which of the following statement(s) is(are) true? [IIT-JEE 2011]

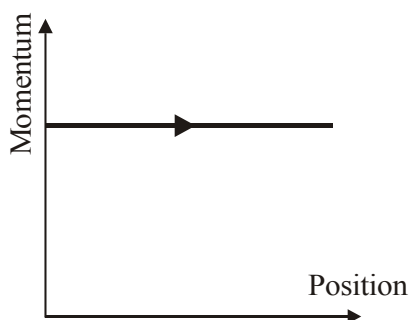


- (A) Restoring torque in case A = Restoring torque in case B  
 (B) Restoring torque in case A < Restoring torque in case B  
 (C) Angular frequency for case A > Angular frequency for case B  
 (D) Angular frequency for case A < Angular frequency for case B

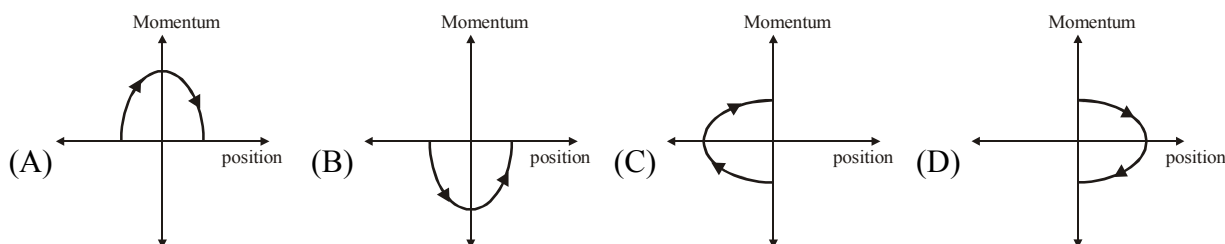
SH0126

### Paragraph for Questions Nos. 6 to 8

Phase space diagrams are useful tools in analyzing all kinds of dynamical problems. They are especially useful in studying the changes in motion as initial position and momentum are changed. Here we consider some simple dynamical systems in one-dimension. For such systems, phase space is a plane in which position is plotted along horizontal axis and momentum is plotted along vertical axis. The phase space diagram is  $x(t)$  vs.  $p(t)$  curve in this plane. The arrow on the curve indicates the time flow. For example, the phase space diagram for a particle moving with constant velocity is a straight line as shown in the figure. We use the sign convention in which position or momentum upwards (or to right) is positive and downwards (or to left) is negative. [IIT-JEE 2011]

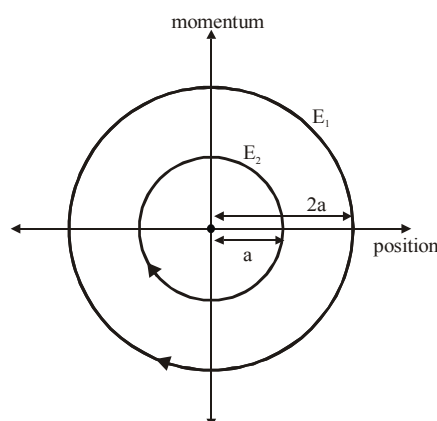


6. The phase space diagram for a ball thrown vertically up from ground is



SH0127

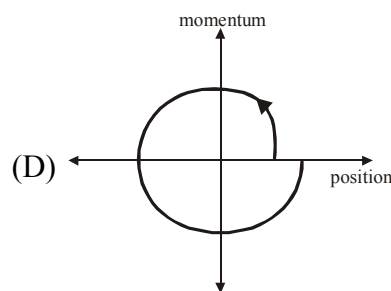
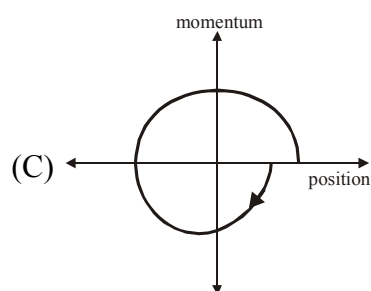
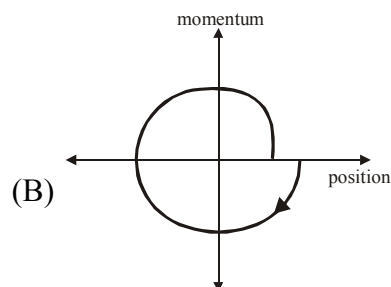
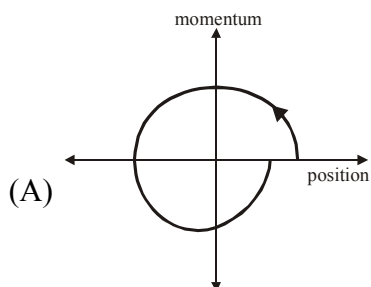
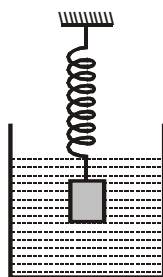
7. The phase space diagram for simple harmonic motion is a circle centered at the origin. In the figure, the two circles represent the same oscillator but for different initial conditions, and  $E_1$  and  $E_2$  are the total mechanical energies respectively. Then



- (A)  $E_1 = \sqrt{2}E_2$       (B)  $E_1 = 2E_2$       (C)  $E_1 = 4E_2$       (D)  $E_1 = 16E_2$

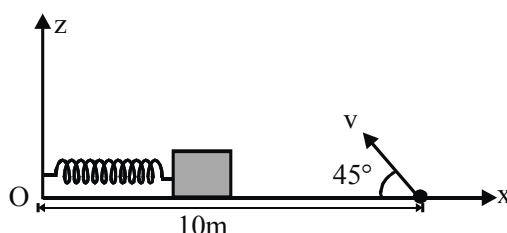
SH0127

8. Consider the spring-mass system, with the mass submerged in water, as shown in the figure. The phase space diagram for one cycle of this system is



SH0127

9. A small block is connected to one end of a massless spring of un-stretched length 4.9 m. The other end of the spring (see the figure) is fixed. The system lies on a horizontal frictionless surface. The block is stretched by 0.2 m and released from rest at  $t = 0$ . It then executes simple harmonic motion with angular frequency  $\omega = \frac{\pi}{3}$  rad/s. Simultaneously at  $t = 0$ , a small pebble is projected with speed  $v$  from point P at an angle of  $45^\circ$  as shown in the figure. Point P is at a horizontal distance of 10m from O. If the pebble hits the block at  $t = 1$ s, the value of  $v$  is:- (take  $g = 10 \text{ m/s}^2$ ) [IIT-JEE 2012]



- (A)  $\sqrt{50}$  m/s (B)  $\sqrt{51}$  m/s  
(C)  $\sqrt{52}$  m/s (D)  $\sqrt{53}$  m/s

SH0128

10. A particle of mass  $m$  is attached to one end of a mass-less spring of force constant  $k$ , lying on a frictionless horizontal plane. The other end of the spring is fixed. The particle starts moving horizontally from its equilibrium position at time  $t = 0$  with an initial velocity  $u_0$ . When the speed of the particle is  $0.5 u_0$ , it collides elastically with a rigid wall. After this collision :- [JEE-Advanced-2013]
- (A) the speed of the particle when it returns to its equilibrium position is  $u_0$

(B) the time at which the particle passes through the equilibrium position for the first time is  $t = \pi \sqrt{\frac{m}{k}}$

(C) the time at which the maximum compression of the spring occurs is  $t = \frac{4\pi}{3} \sqrt{\frac{m}{k}}$

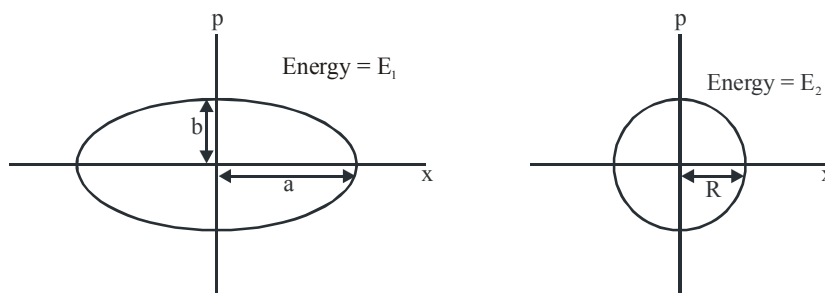
(D) the time at which the particle passes through the equilibrium position for the second time is

$$t = \frac{5\pi}{3} \sqrt{\frac{m}{k}}$$

SH0129

11. Two independent harmonic oscillators of equal mass are oscillating about the origin with angular frequencies  $\omega_1$  and  $\omega_2$  and have total energies  $E_1$  and  $E_2$ , respectively. The variations of their momenta  $p$  with positions  $x$  are shown in the figures. If  $\frac{a}{b} = n^2$  and  $\frac{a}{R} = n$ , then the correct equation (s) is (are)

[JEE-Advanced-2015]



- (A)  $E_1 \omega_1 = E_2 \omega_2$       (B)  $\frac{\omega_2}{\omega_1} = n^2$       (C)  $\omega_1 \omega_2 = n^2$       (D)  $\frac{E_1}{\omega_1} = \frac{E_2}{\omega_2}$

SH0130

12. A particle of unit mass is moving along the  $x$ -axis under the influence of a force and its total energy is conserved. Four possible forms of the potential energy of the particle are given in column I ( $a$  and  $U_0$  are constants). Match the potential energies in column I to the corresponding statement(s) in column-II

[JEE-Advanced-2015]

## Column-I

## Column-II

(A)  $U_1(x) = \frac{U_0}{2} \left[ 1 - \left( \frac{x}{a} \right)^2 \right]^2$

(P) The force acting on the particle is zero at  $x = a$ .

(B)  $U_2(x) = \frac{U_0}{2} \left( \frac{x}{a} \right)^2$

(Q) The force acting on the particle is zero at  $x = 0$ .

(C)  $U_3(x) = \frac{U_0}{2} \left( \frac{x}{a} \right)^2 \exp \left[ - \left( \frac{x}{a} \right)^2 \right]$

(R) The force acting on the particle is zero at  $x = -a$ .

(D)  $U_4(x) = \frac{U_0}{2} \left[ \frac{x}{a} - \frac{1}{3} \left( \frac{x}{a} \right)^3 \right]$

(S) The particle experiences an attractive force towards  $x = 0$  in the region  $|x| < a$ .

(T) The particle with total energy  $\frac{U_0}{4}$  can oscillate about the point  $x = -a$ .

SH0131



13. A block with mass  $M$  is connected by a massless spring with stiffness constant  $k$  to a rigid wall and moves without friction on a horizontal surface. The block oscillates with small amplitude  $A$  about an equilibrium position  $x_0$ . Consider two cases : (i) when the block is at  $x_0$ ; and (ii) when the block is at  $x = x_0 + A$ . In both the cases, a particle with mass  $m$  ( $< M$ ) is softly placed on the block after which they stick to each other. Which of the following statement(s) is(are) true about the motion after the mass  $m$  is placed on the mass  $M$ ? **[JEE-Advanced-2016]**

- (A) The amplitude of oscillation in the first case changes by a factor of  $\sqrt{\frac{M}{m+M}}$ , whereas in the second case it remains unchanged
- (B) The final time period of oscillation in both the cases is same
- (C) The total energy decreases in both the cases
- (D) The instantaneous speed at  $x_0$  of the combined masses decreases in both the cases.

SH0132

## ANSWER KEY

## EXERCISE (S-1)

1. Ans.  $y = 2 \sin \left( \frac{10}{3}t - \frac{\pi}{2} \right)$  2. Ans. 0.33 sec 3. Ans. 2 m/s 4. Ans. 0.06m
5. Ans.  $y = 0.1 \sin(4t + \pi/4)$  6. Ans. (i)  $x_0 = 2m$ ; (ii)  $T = \sqrt{2} \pi \text{ sec.}$ ; (iii)  $2\sqrt{3}$
7. Ans.  $\frac{1}{2\pi} \sqrt{\frac{\beta}{\alpha}}$  8. Ans.  $\omega = \sqrt{\frac{2T_0}{m\ell}}$  9. Ans.  $1/\sqrt{3}$  10. Ans.  $2\pi \sqrt{\frac{17L}{18g}}$
11. Ans.  $\omega = \sqrt{\frac{k}{(m+6M)}}$  12. Ans.  $M = 1.6 \text{ kg}$  13. Ans. (a)  $\frac{1}{T} = \frac{1}{2\pi} \sqrt{\frac{g}{L}}$ , (b)  $\frac{L}{3}$
14. Ans. (a) 2 sec, (b)  $T = \frac{2}{5^{1/4}} \text{ sec}$
15. Ans. (a)  $T = mg + mgA^2 \sin^2 \sqrt{\frac{g}{\ell}} t$  (b)  $t = (2n+1) \frac{\pi}{2} \sqrt{\frac{\ell}{g}}$ ;  $n \in \mathbb{I}$   $T_{\max} = mg + mgA^2$
16. Ans. (b)  $\frac{R}{\sqrt{2}}$  17. Ans. 100 Nm<sup>-1</sup> 18. Ans. 10/π Hz,  $\frac{5\sqrt{37}}{6} \text{ cm}$
19. Ans.  $T = 2\sqrt{\ell/g} [\pi/2 + \sin^{-1}(\alpha/\beta)]$  20. Ans. 2 21. Ans. 5
22. Ans. 1 23. Ans.  $f = \frac{1}{2\pi} \sqrt{\frac{3g}{2R}}$

## EXERCISE (S-2)

1. Ans. 3cm,  $x = 10 - 3\sin 5t$ ;  $\Delta E = 0.135 \text{ J}$  2. Ans.  $f = \frac{1}{\pi}$ ;  $E = 4\pi^2 \times 10^{-5} \text{ J}$ ;  $v = 2\pi \times 10^{-2} \text{ m/s}$
3. Ans. (a)  $\frac{\pi}{10} \text{ sec}$ , (b) 4 cm, (c) 2.40 kg m/s. 4. Ans. Will increase  $\sqrt{1 + \frac{2}{5} \left( \frac{R}{\ell} \right)^2}$  times.
5. Ans. (a)  $T = 2\pi \sqrt{\ell/3g} = 1.1 \text{ s}$ ; (b)  $E = \frac{1}{2} mg\ell\alpha^2 = 0.05 \text{ J}$
6. Ans.  $\omega = \sqrt{(I_1\omega_1^2 + I_2\omega_2^2)/(I_1 + I_2)}$  7. Ans.  $\pi\sqrt{m/k} + 2\sqrt{2E/mg^2}$
8. Ans.  $\frac{\pi}{3} \text{ s}$  9. Ans. (i)  $x_2 = v_0 t + \frac{m_1}{m_2} A(1 - \cos \omega t)$  (ii)  $\ell_0 = \left( \frac{m_1}{m_2} + 1 \right) A$
10. Ans.  $a = (mg/x) \sqrt{1 + 2hx/mg}$ ,  $\frac{1}{2} xA^2$  11. Ans. 5
12. Ans.  $\omega = \sqrt{\frac{3g}{2L} \left( \frac{d_2 - d_1}{d_1} \right)}$  13. Ans.  $\left( \frac{4g}{H} \right) x$

**EXERCISE (O-1)**

- |              |              |              |              |              |              |
|--------------|--------------|--------------|--------------|--------------|--------------|
| 1. Ans. (C)  | 2. Ans. (D)  | 3. Ans. (A)  | 4. Ans. (A)  | 5. Ans. (A)  | 6. Ans. (B)  |
| 7. Ans. (A)  | 8. Ans. (A)  | 9. Ans. (B)  | 10. Ans. (A) | 11. Ans. (A) | 12. Ans. (C) |
| 13. Ans. (C) | 14. Ans. (C) | 15. Ans. (A) | 16. Ans. (A) | 17. Ans. (C) | 18. Ans. (B) |
| 19. Ans. (A) | 20. Ans. (B) | 21. Ans. (B) | 22. Ans. (B) | 23. Ans. (C) | 24. Ans. (C) |
| 25. Ans. (D) | 26. Ans. (B) | 27. Ans. (C) | 28. Ans. (B) | 29. Ans. (A) | 30. Ans. (C) |
| 31. Ans. (A) | 32. Ans. (C) | 33. Ans. (B) | 34. Ans. (D) | 35. Ans. (C) | 36. Ans. (A) |
| 37. Ans. (B) | 38. Ans. (B) | 39. Ans. (A) | 40. Ans. (A) | 41. Ans. (B) | 42. Ans. (A) |
| 43. Ans. (C) | 44. Ans. (B) |              |              |              |              |

**EXERCISE (O-2)**

- |                                                |                  |                    |                  |                |                 |
|------------------------------------------------|------------------|--------------------|------------------|----------------|-----------------|
| 1. Ans. (D)                                    | 2. Ans. (D)      | 3. Ans. (D)        | 4. Ans. (C)      | 5. Ans. (B)    | 6. Ans. (D)     |
| 7. Ans. (B)                                    | 8. Ans. (D)      | 9. Ans. (A)        | 10. Ans. (B)     | 11. Ans. (B)   | 12. Ans. (A)    |
| 13. Ans. (A)                                   | 14. Ans. (C)     | 15. Ans. (D)       | 16. Ans. (C)     | 17. Ans. (A,D) | 18. Ans. (B, C) |
| 19. Ans. (A,B,C)                               | 20. Ans. (A,B,C) | 21. Ans. (A,C,D)   | 22. Ans. (B,D)   |                |                 |
| 23. Ans. (B,C,D)                               | 24. Ans. (A,B)   | 25. Ans. (A,B,C,D) | 26. Ans. (A,C,D) |                |                 |
| 27. Ans. (A, C)                                | 28. Ans. (B,C,D) | 29. Ans. (A,B)     | 30. Ans. (A, C)  |                |                 |
| 31. Ans. (A)                                   | 32. Ans. (A)     | 33. Ans. (A)       | 34. Ans. (C)     | 35. Ans. (C)   | 36. Ans. (A)    |
| 37. Ans. (A)-Q,R ; (B)-R ; (C)-Q,R ; (D)-P,Q,R | 38. Ans. (D)     |                    |                  |                |                 |

**EXERCISE (JM)**

- |              |              |              |              |              |              |
|--------------|--------------|--------------|--------------|--------------|--------------|
| 1. Ans. (2)  | 2. Ans. (4)  | 3. Ans. (2)  | 4. Ans. (2)  | 5. Ans. (4)  | 6. Ans. (1)  |
| 7. Ans. (2)  | 8. Ans. (1)  | 9. Ans. (2)  | 10. Ans. (4) | 11. Ans. (4) | 12. Ans. (1) |
| 13. Ans. (2) | 14. Ans. (4) | 15. Ans. (1) | 16. Ans. (1) | 17. Ans. (2) |              |

**EXERCISE (JA)**

- |                                                       |                  |             |                |                |             |
|-------------------------------------------------------|------------------|-------------|----------------|----------------|-------------|
| 1. Ans. (C)                                           | 2. Ans. (B)      | 3. Ans. (D) | 4. Ans. (B)    | 5. Ans. (A,D)  | 6. Ans. (D) |
| 7. Ans. (C)                                           | 8. Ans. (B)      | 9. Ans. (A) | 10. Ans. (A,D) | 11. Ans. (B,D) |             |
| 12. Ans. (A)-P,Q,R,T; (B)-Q,S; (C)-P,Q,R,S; (D)-P,R,T | 13. Ans. (A,B,D) |             |                |                |             |

# WAVES ON STRING

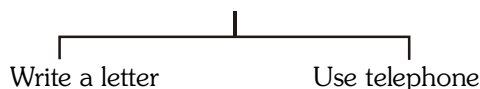
## KEY CONCEPTS

### INTRODUCTION OF WAVES

#### What is wave motion ?

- When a particle moves through space, it carries KE with itself. Wherever the particle goes, the energy goes with it. (One way of transport energy from one place to another place)
- There is another way (wave motion) to transport energy from one part of space to other without any bulk motion of material together with it. Sound is transmitted in air in this manner.

**Ex.** You (Kota) want to communicate your friend (Delhi)



1<sup>st</sup> option involves the concept of particle & the second choice involves the concept of wave.

**Ex.** When you say "Namaste" to your friend no material particle is ejected from your lips to fall on your friend's ear. Basically you create some disturbance in the part of the air close to your lips. Energy is transferred to these air particles either by pushing them ahead or pulling them back. The density of the air in this part temporarily increases or decreases. These disturbed particles exert force on the next layer of air, transferring the disturbance to that layer. In this way, the disturbance proceeds in air and finally the air near the ear of the listener gets disturbed.

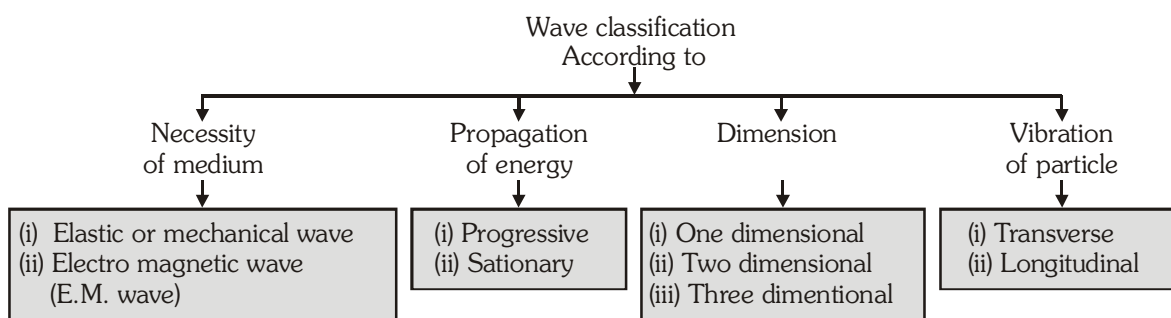
**Note :-** In the above example air itself does not move.

A **wave** is a disturbance that propagates in space, transports energy and momentum from one point to another without the transport of matter.

#### Few examples of waves :

The ripples on a pond (water waves), the sound we hear, visible light, radio and TV signals etc.

### CLASSIFICATION OF WAVES



- Based on medium necessity :-** A wave may or may not require a medium for its propagation. The waves which do not require medium for their propagation are called non-mechanical, e.g. light, heat (infrared), radio waves etc. On the other hand the waves which require medium for their propagation are called mechanical waves. In the propagation of mechanical waves elasticity and density of the medium play an important role therefore mechanical waves are also known as **elastic waves**.

**Example :** Sound waves in water, seismic waves in earth's crust.

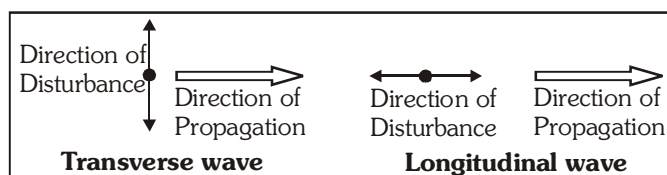
2. **Based on energy propagation :-** Waves can be divided into two parts on the basis of energy propagation (i) Progressive wave (ii) Stationary waves. The progressive wave propagates with fixed velocity in a medium. In stationary waves particles of the medium vibrate with different amplitude but energy does not propagate.
3. **Based on direction of propagation :-** Waves can be one, two or three dimensional according to the number of dimensions in which they propagate energy. Waves moving along strings are one-dimensional. Surface waves or ripples on water are two dimensional, while sound or light waves from a point source are three dimensional.
4. **Based on the motion of particles of**

**medium :**

Waves are of two types on the basis of motion of particles of the medium.

- (i) Longitudinal waves
- (ii) Transverse waves

In the transverse wave the direction associated with the disturbance (i.e. motion of particles of the medium) is at right angle to the direction of propagation of wave while in the longitudinal wave the direction of disturbance is along the direction of propagation.



## TRANSVERSE WAVE MOTION

Mechanical transverse waves produce in such type of medium which have shearing property, so they are known as shear wave or S-wave

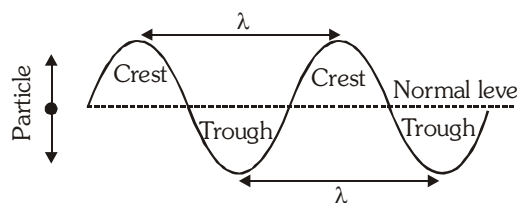
**Note :-** Shearing is the property of a body by which it changes its shape on application of force.

⇒ Mechanical transverse waves are generated only in solids & surface of liquid.

In this individual particles of the medium execute SHM about their mean position in direction  $\perp$  to the direction of propagation of wave motion.

A **crest** is a portion of the medium, which is raised temporarily above the normal position of rest of particles of the medium, when a transverse wave passes.

A **trough** is a portion of the medium, which is depressed temporarily below the normal position of rest of particles of the medium, when a transverse wave passes.

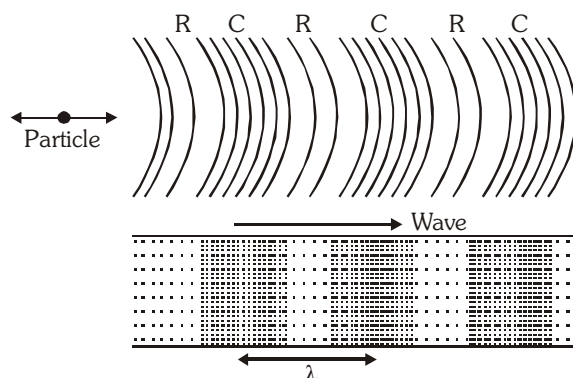


## LONGITUDINAL WAVE MOTION

In this type of waves, oscillatory motion of the medium particles produces regions of compression (high pressure) and rarefaction (low pressure) which propagated in space with time (see figure).

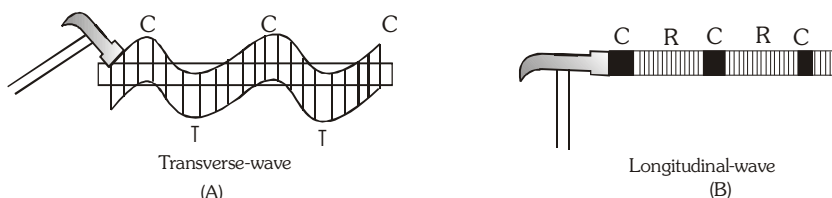
**Note :** The regions of high particle density are called compressions and regions of low particle density are called rarefactions.

- The propagation of sound waves in air is visualized as the propagation of pressure or density fluctuations. The pressure fluctuations are of the order of 1 Pa, whereas atmospheric pressure is  $10^5$  Pa.



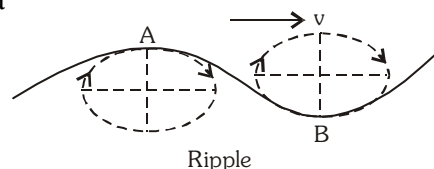
## Mechanical Waves in Different Media

- A mechanical wave will be transverse or longitudinal depends on the nature of medium and mode of excitation.
- In strings mechanical waves are always transverse when string is under a tension. In gases and liquids mechanical waves are always longitudinal e.g. sound waves in air or water. This is because fluids cannot sustain shear.
- In solids, mechanical waves (may be sound) can be either transverse or longitudinal depending on the mode of excitation. The speed of the two waves in the same solid are different. (Longitudinal waves travels faster than transverse waves). e.g., if we struck a rod at an angle as shown in fig. (A) the waves in the rod will be transverse while if the rod is struck at the side as shown in fig. (B) or is rubbed with a cloth the waves in the rod will be longitudinal. In case of vibrating tuning fork waves in the prongs are transverse while in the stem are longitudinal.



Further more in case of seismic waves produced by Earthquakes both S (shear) and P (pressure) waves are produced simultaneously which travel through the rock in the crust at different speeds [ $v_s \cong 5 \text{ km/s}$  while  $v_p \cong 9 \text{ km/s}$ ] S-waves are transverse while P-waves longitudinal.

Some waves in nature are neither transverse nor longitudinal but a combination of the two. These waves are called 'ripple' and waves on the surface of a liquid are of this type. In these waves particles of the medium vibrate up and down and back and forth simultaneously describing ellipses in a vertical plane [Fig.]



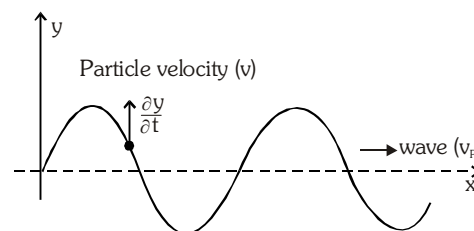
## CHARACTERISTICS OF WAVE MOTION

Some of the important characteristics of wave motion are as follows :

- In a wave motion, the disturbance travels through the medium due to repeated periodic oscillations of the particles of the medium about their mean positions.
- The energy is transferred from place to another without any actual transfer of the particles of the medium.
- Each particle receives disturbance a little later than its preceding particle i.e., there is a regular phase difference between one particle and the next.
- The velocity with which a wave travels is different from the velocity of the particles with which they vibrate about their mean positions.
- The wave velocity remains constant in a given medium while the particle velocity changes continuously during its vibration about the mean position. It is maximum at the mean position and zero at the extreme position.
- For the propagation of a mechanical wave, the medium must possess the properties of inertia, elasticity and minimum friction amongst its particles.

### SOME IMPORTANT TERMS CONNECTED WITH WAVE MOTION

- **Wavelength ( $\lambda$ ) [length of one wave]**  
Distance travelled by the wave during the time, any one particle of the medium completes one vibration about its mean position. We may also define wavelength as the distance between any two nearest particles of the medium, vibrating in the same phase.
- **Frequency ( $n$ )** : Number of vibrations (Number of complete wavelengths) complete by a particle in one second.
- **Time period ( $T$ )** : Time taken by wave to travel a distance equal to one wavelength.
- **Amplitude ( $A$ )** : Maximum displacement of vibrating particle from its equilibrium position.
- **Angular frequency ( $\omega$ )** : It is defined as  $\omega = \frac{2\pi}{T} = 2\pi n$
- **Phase** : Phase is a quantity which contains all information related to any vibrating particle in a wave. For equation  $y = A \sin (\omega t - kx)$ ;  $(\omega t - kx) = \text{phase}$ .
- **Angular wave number ( $k$ )** : It is defined as  $k = \frac{2\pi}{\lambda}$
- **Wave number ( $\vec{v}$ )** : It is defined as  $\vec{v} = \frac{1}{\lambda} = \frac{k}{2\pi} = \text{number of waves in a unit length of the wave pattern}$ .
- **Particle velocity, wave velocity and particle's acceleration** : In plane progressive harmonic wave particles of the medium oscillate simple harmonically about their mean position. Therefore, all the formulae what we have read in SHM apply to the particles here also. For example, maximum particle velocity is  $\pm A\omega$  at mean position and it is zero at extreme positions etc. Similarly maximum particle acceleration is  $\pm \omega^2 A$  at extreme positions and zero at mean position. However the wave velocity is different from the particle velocity. This depends on certain characteristics of the medium. Unlike the particle velocity which oscillates simple harmonically (between  $+A\omega$  and  $-A\omega$ ) the wave velocity is constant for given characteristics of the medium.
- **Particle velocity in wave motion** :  
The individual particles which make up the medium do not travel through the medium with the waves. They simply oscillate about their equilibrium positions. The instantaneous velocity of an oscillating particle of the medium, through which a wave is travelling, is known as "Particle velocity".
- **Wave velocity** : The velocity with which the disturbance, or planes of equal phase (wave front), travel through the medium is called wave (or phase) velocity.
- **Relation between particle velocity and wave velocity** :



Wave equation :-  $y = A \sin (\omega t - kx)$ , Particle velocity  $v = \frac{\partial y}{\partial t} = A\omega \cos (\omega t - kx)$ .

$$\text{Wave velocity} = v_p = \frac{\lambda}{T} = \lambda \frac{\omega}{2\pi} = \frac{\omega}{k}, \quad \frac{\partial y}{\partial x} = -Ak \cos (\omega t - kx) = -\frac{A}{\omega} \omega k \cos (\omega t - kx) = -\frac{1}{v_p} \frac{\partial y}{\partial t}$$

$$\Rightarrow \frac{\partial y}{\partial x} = -\frac{1}{v_p} \frac{\partial y}{\partial t}$$

**Note :**  $\frac{\partial y}{\partial x}$  represent the slope of the string (wave) at the point x.

Particle velocity at a given position and time is equal to negative of the product of wave velocity with slope of the wave at that point at that instant.

• **Differential equation of harmonic progressive waves :**

$$\frac{\partial^2 y}{\partial t^2} = -A\omega^2 \sin(\omega t - kx) \Rightarrow \frac{\partial^2 y}{\partial x^2} = -Ak^2 \sin(\omega t - kx) \Rightarrow \frac{\partial^2 y}{\partial x^2} = \frac{1}{v_p^2} \frac{\partial^2 y}{\partial t^2}$$

• **Particle velocity ( $v_p$ ) and acceleration ( $a_p$ ) in a sinusoidal wave :**

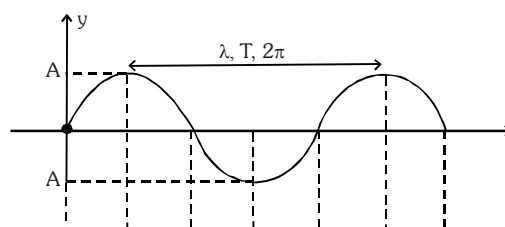
The acceleration of the particle is the second partial derivative of  $y(x, t)$  with respect to  $t$ ,

$$\therefore a_p = \frac{\partial^2 y(x, t)}{\partial t^2} = \omega^2 A \sin(kx - \omega t) = -\omega^2 y(x, t)$$

i.e., the acceleration of the particle equals  $-\omega^2$  times its displacement, which is the result we obtained for SHM. Thus,  $a_p = -\omega^2$  (displacement)

• **Relation between Phase difference, Path difference & Time difference**

|                           |   |                     |                     |                      |           |                      |                      |
|---------------------------|---|---------------------|---------------------|----------------------|-----------|----------------------|----------------------|
| Phase ( $\phi$ )          | 0 | $\frac{\pi}{2}$     | $\pi$               | $\frac{3\pi}{2}$     | $2\pi$    | $\frac{5\pi}{2}$     | $3\pi$               |
| Wave length ( $\lambda$ ) | 0 | $\frac{\lambda}{4}$ | $\frac{\lambda}{2}$ | $\frac{3\lambda}{4}$ | $\lambda$ | $\frac{5\lambda}{4}$ | $\frac{3\lambda}{2}$ |
| Time-period (T)           | 0 | $\frac{T}{4}$       | $\frac{T}{2}$       | $\frac{3T}{4}$       | T         | $\frac{5T}{4}$       | $\frac{3T}{2}$       |



$$\Rightarrow \frac{\Delta\phi}{2\pi} = \frac{\Delta\lambda}{\lambda} = \frac{\Delta T}{T} \Rightarrow \text{Path difference} = \left(\frac{\lambda}{2\pi}\right) \text{Phase difference}$$

**Ex.** A progressive wave of frequency 500 Hz is travelling with a velocity of 360 m/s. How far apart are two points  $60^\circ$  out of phase.

**Sol.** We know that for a wave  $v = f\lambda$  So  $\lambda = \frac{v}{f} = \frac{360}{500} = 0.72$  m

Phase difference  $\Delta\phi = 60^\circ = (\pi/180) \times 60 = (\pi/3)$  rad,

so path difference  $\Delta x = \frac{\lambda}{2\pi} (\Delta\phi) = \frac{0.72}{2\pi} \times \frac{\pi}{3} = 0.12$  m

**THE GENERAL EQUATION OF WAVE MOTION**

Some physical quantity (say  $y$ ) is made to oscillate at one place and these oscillations of  $y$  propagate to other places. The  $y$  may be,

- displacement of particles from their mean position in case of transverse wave in a rope or longitudinal sound wave in a gas.
- pressure difference ( $dP$ ) or density difference ( $dp$ ) in case of sound wave or
- electric and magnetic fields in case of electromagnetic waves.



The oscillations of  $y$  may or may not be simple harmonic in nature. Consider one-dimensional wave travelling along  $x$ -axis. In this case  $y$  is a function of  $x$  and  $t$ . i.e.  $y = f(x, t)$  But only those function of

$x$  &  $t$ , represent a wave motion which satisfy the differential equation.  $\frac{\partial^2 y}{\partial t^2} = v^2 \frac{\partial^2 y}{\partial x^2}$  ... (i)

The general solution of this equation is of the form  $y(x, t) = f(ax \pm bt)$  ... (ii)

Thus, any function of  $x$  and  $t$  and which satisfies equation (i) or which can be written as equation (ii) represents a wave. The only condition is that it should be finite everywhere and at all times, Further,

if these conditions are satisfied, then speed of wave ( $v$ ) is given by  $v = \frac{\text{coefficient of } t}{\text{coefficient of } x} = \frac{b}{a}$

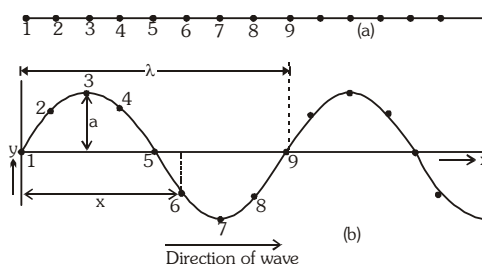
**Ex.** Which of the following functions represent a travelling wave ?

- (a)  $(x - vt)^2$                       (b)  $\ell n(x + vt)$                       (c)  $e^{-(x-vt)^2}$                       (d)  $\frac{1}{x + vt}$

**Sol.** Although all the four functions are written in the form  $f(ax \pm bt)$ , only (c) among the four functions is finite everywhere at all times. Hence only (c) represents a travelling wave.

### Equation of a Plane Progressive Wave

If, on the propagation of wave in a medium, the particles of the medium perform simple harmonic motion then the wave is called a 'simple harmonic progressive wave'. Suppose, a simple harmonic progressive wave is propagating in a medium along the positive direction of the  $x$ -axis (from left to right). In fig. (a) are shown the equilibrium positions of the particles 1, 2, 3 .....



When the wave propagates, these particles oscillate about their equilibrium positions. In Fig. (b) are shown the instantaneous positions of these particles at a particular instant. The curve joining these positions represents the wave. Let the time be counted from the instant when the particle 1 situated at the origin starts oscillating. If  $y$  be the displacement of this particle after  $t$  seconds, then  $y = a \sin \omega t$  ... (i)

where  $a$  is the amplitude of oscillation and  $\omega = 2\pi n$ , where  $n$  is the frequency. As the wave reaches the particles beyond the particle 1, the particles start oscillating. If the speed of the wave be  $v$ , then it will reach particle 6, distant  $x$  from the particle 1, in  $x/v$  sec. Therefore, the particle 6 will start oscillating  $x/v$  sec after the particle 1. It means that the displacement of the particle 6 at a time  $t$  will be the same as that of the particle 1 at a time  $x/v$  sec earlier i.e. at time  $t - (x/v)$ . The displacement of particle 1 at time  $t - (x/v)$  can be the particle 6, distant  $x$  from the origin (particle 1), at time  $t$  is given by

$$y = a \sin \omega \left( t - \frac{x}{v} \right) \quad \text{But } \omega = 2\pi n, y = a \sin (\omega t - kx) \left( k = \frac{\omega}{v} \right) \quad \dots (ii)$$

$$y = a \sin \left[ \frac{2\pi}{T} t - \frac{2\pi}{\lambda} x \right] \quad \text{Also } k = \frac{2\pi}{\lambda} \quad \dots (iii) \quad y = a \sin 2\pi \left[ \frac{t}{T} - \frac{x}{\lambda} \right] \quad \dots (iv)$$

This is the equation of a simple harmonic wave travelling along +x direction. If the wave is travelling along the -x direction then inside the brackets in the above equations, instead of minus sign there will

be plus sign. For example, equation (iv) will be of the following form :  $y = a \sin 2\pi \left( \frac{t}{T} + \frac{x}{\lambda} \right)$ . If  $\phi$  be the phase difference between the above wave travelling along the +x direction and an other wave, then the equation of that wave will be

$$y = a \sin \left\{ 2\pi \left( \frac{t}{T} - \frac{x}{\lambda} \right) \pm \phi \right\}$$

**Ex.** The equation of a wave is,  $y(x, t) = 0.05 \sin \left[ \frac{\pi}{2} (10x - 40t) - \frac{\pi}{4} \right] \text{m}$

- Find:** (a) The wavelength, the frequency and the wave velocity  
(b) The particle velocity and acceleration at  $x=0.5 \text{ m}$  and  $t=0.05 \text{ s}$ .

**Sol. :** (a) The equation may be rewritten as,  $y(x, t) = 0.05 \sin \left( 5\pi x - 20\pi t - \frac{\pi}{4} \right) \text{m}$

Comparing this with equation of plane progressive harmonic wave,

$$y(x, t) = A \sin(kx - \omega t + \phi) \text{ we have, wave number } k = \frac{2\pi}{\lambda} = 5\pi \text{ rad/m} \quad \therefore \lambda = 0.4 \text{m}$$

$$\text{The angular frequency is, } \omega = 2\pi f = 20\pi \text{ rad/s} \quad \therefore f = 10 \text{Hz}$$

$$\text{The wave velocity is, } v = f \lambda = \frac{\omega}{k} = 4 \text{ms}^{-1} \text{ in } +x \text{ direction}$$

- (b) The particle velocity and acceleration are,  $v_p = \frac{\partial y}{\partial t} = -(20\pi)(0.05) \cos \left( \frac{5\pi}{2} - \pi - \frac{\pi}{4} \right) = 2.22 \text{m/s}$

$$a_p = \frac{\partial^2 y}{\partial t^2} = -(20\pi)^2 (0.05) \sin \left( \frac{5\pi}{2} - \pi - \frac{\pi}{4} \right) = 140 \text{ m/s}^2$$

## INTENSITY OF WAVE

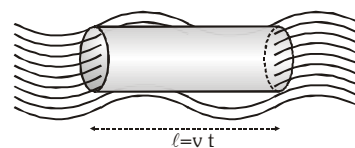
The amount of energy flowing per unit area and per unit time is called the intensity of wave. It is represented by  $I$ . Its units are  $\text{J/m}^2\text{s}$  or  $\text{watt/metre}^2$ .  $I = 2\pi^2 f^2 A^2 \rho v$  i.e.  $I \propto A^2$  and  $I \propto A^2$ . If  $P$  is the power of an isotropic point source, then intensity at a distance  $r$  is given by,

$$I = \frac{P}{4\pi r^2} \text{ or } I \propto \frac{1}{r^2} \text{ (for a point source)}$$

If  $P$  is the power of a line source, then intensity at a distance  $r$  is given by,

$$I = \frac{P}{2\pi r \ell} \text{ or } I \propto \frac{1}{r} \text{ (for a line source) As, } I \propto A^2$$

Therefore,  $A \propto \frac{1}{r}$  (for a point source) and  $A \propto \frac{1}{\sqrt{r}}$  (for a line source)



## SUPERPOSITION PRINCIPLE

Two or more waves can propagate in the same medium without affecting the motion of one another. If several waves propagate in a medium simultaneously, then the resultant displacement of any particle of the medium at any instant is equal to the vector sum of the displacements produced by individual wave. The phenomenon of intermixing of two or more waves to produce a new wave is called Superposition of waves. Therefore according to superposition principle.

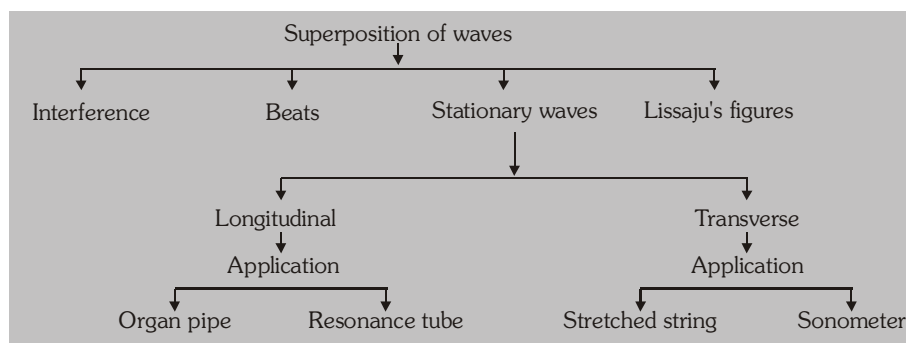
The resultant displacement of a particle at any point of the medium, at any instant of time is the vector sum of the displacements caused to the particle by the individual waves.

If  $\bar{y}_1, \bar{y}_2, \bar{y}_3, \dots$  are the displacement of particle at a particular time due to individual waves, then the resultant displacement is given by  $\bar{y} = \bar{y}_1 + \bar{y}_2 + \bar{y}_3 + \dots$

Principle of superposition holds for all types of waves, i.e., mechanical as well as electromagnetic waves. But this principle is not applicable to the waves of very large amplitude.

Due to superposition of waves the following phenomenon can be seen

- **Interference** : Superposition of two waves having equal frequency and nearly equal amplitude.
- **Beats** : Superposition of two waves of nearly equal frequency in same direction.
- **Stationary waves** : Superposition of equal wave from opposite direction.
- **Lissajous' figure** : Superposition of perpendicular waves.



## INTERFERENCE OF WAVES :

When two waves of equal frequency and nearly equal amplitude travelling in same direction having same state of polarisation in medium superimpose, then intensity is different at different points. At some points intensity is large, whereas at other points it is nearly zero.

Consider two waves  $y_1 = A_1 \sin(\omega t - kx)$  and  $y_2 = A_2 \sin(\omega t - kx + \phi)$

By principle of superposition  $y = y_1 + y_2 = A \sin(\omega t - kx + \delta)$

$$\text{where } A^2 = A_1^2 + A_2^2 + 2A_1A_2 \cos \phi \text{ and } \tan \delta = \frac{A_2 \sin \phi}{A_1 + A_2 \cos \phi}$$

$$\text{As intensity } I \propto A^2 \text{ so } I = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos \phi$$

- **Constructive interference (maximum intensity) :**

Phase difference  $\phi = 2n\pi$  or path difference  $= n\lambda$  where  $n = 0, 1, 2, 3, \dots$

$$\Rightarrow A_{\max} = A_1 + A_2 \quad \text{and} \quad I_{\max} = I_1 + I_2 + 2\sqrt{I_1 I_2}$$

• **Destructive interference (minimum intensity) :**

Phase difference  $\phi = (2n+1)\pi$ , or path difference  $= (2n-1) \frac{\lambda}{2}$  where  $n = 0, 1, 2, 3, \dots$

$$\Rightarrow A_{\min} = A_1 - A_2 \quad \text{and} \quad I_{\min} = I_1 + I_2 - 2\sqrt{I_1 I_2}$$

**KEY POINTS**

- Maximum and minimum intensities in any interference wave form.  $\frac{I_{\max}}{I_{\min}} = \left( \frac{\sqrt{I_1} + \sqrt{I_2}}{\sqrt{I_1} - \sqrt{I_2}} \right)^2 = \left( \frac{a_1 + a_2}{a_1 - a_2} \right)^2$

- Average intensity of interference wave form :-  $\langle I \rangle$  or  $I_{\text{av}} = \frac{I_{\max} + I_{\min}}{2} = I_1 + I_2$

if  $a = a_1 = a_2$  and  $I_1 = I_2 = I$  then  $I_{\max} = 4I$ ,  $I_{\min} = 0$  and  $I_{\text{av}} = 2I$

- Degree of interference Pattern (f) : Degree of hearing (Sound Wave) or

$$\text{Degree of visibility (Light Wave)} \quad f = \frac{I_{\max} - I_{\min}}{I_{\max} + I_{\min}} \times 100$$

In condition of perfect interference degree of interference pattern is maximum  $f_{\max} = 1$  or 100%

- Condition of maximum contrast in interference wave form  $a_1 = a_2$  and  $I_1 = I_2$  then  $I_{\max} = 4I$  and  $I_{\min} = 0$

For perfect destructive interference we have a maximum contrast in interference wave form.

**VELOCITY OF TRANSVERSE WAVE**

Mass of per unit length  $m = \frac{\pi r^2 \ell \times d}{\ell}$ ,  $m = \pi r^2 d$ , where  $d$  = Density of matter

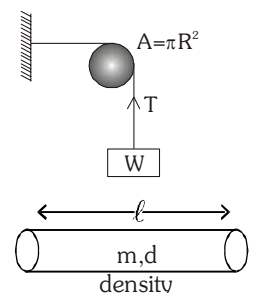
Velocity of transverse wave in any wire  $v = \sqrt{\frac{T}{m}}$  or  $\sqrt{\frac{T}{\pi r^2 d}} = \sqrt{\frac{T}{Ad}} \because \pi r^2 = A$

- If  $m$  is constant then,  $v \propto \sqrt{T}$  it is called tension law.

- If tension is constant then  $v \propto \sqrt{\frac{1}{m}} \leftarrow$  it is called law of mass

- If  $T$  is constant & take wire of different radius for same material then  $v \propto \frac{1}{r} \leftarrow$  it is called law of radius

- If  $T$  is constant & take wire of same radius for different material. Then  $v \propto \sqrt{\frac{1}{d}} \leftarrow$  law of density

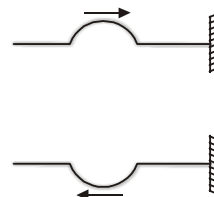


**REFLECTION FROM RIGID END**

When the pulse reaches the right end which is clamped at the wall, the element at the right end exerts a force on the clamp and the clamp exerts equal and opposite force on the element. The element at the right end is thus acted upon by the force from the clamp. As this end remains fixed, the two forces are opposite to each other. The force from the left part of the string transmits the forward wave pulse and hence, the force exerted by the clamp sends a return pulse on the string whose shape is similar to a return pulse but is inverted. The original pulse tries to pull the element at the fixed end up and the return pulse sent by the clamp tries to pull it down, so the resultant displacement is zero. Thus, the wave is reflected from the fixed end and the reflected wave is inverted with respect to the original wave. The shape of the string at any time, while the pulse is being reflected, can be found by adding an inverted image pulse to the incident pulse.

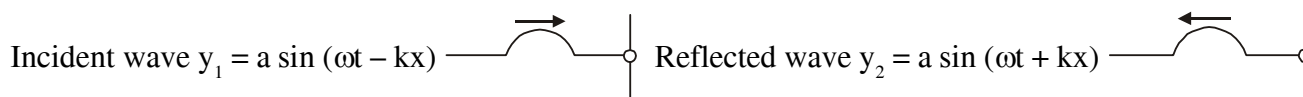
Equation of wave propagating in +ve x-axis

Incident wave  $y_1 = a \sin(\omega t - kx)$   
 Reflected wave  $y_2 = a \sin(\omega t + kx + \pi)$   
 or  $y_2 = -a \sin(\omega t + kx)$



### REFLECTION FROM FREE END

The right end of the string is attached to a light frictionless ring which can freely move on a vertical rod. A wave pulse is sent on the string from left. When the wave reaches the right end, the element at this end is acted on by the force from the left to go up. However, there is no corresponding restoring force from the right as the rod does not exert a vertical force on the ring. As a result, the right end is displaced in upward direction more than the height of the pulse i.e., it overshoots the normal maximum displacement. The lack of restoring force from right can be equivalent described in the following way. An extra force acts from right which sends a wave from right to left with its shape identical to the original one. The element at the end is acted upon by both the incident and the reflected wave and the displacements add. Thus, a wave is reflected by the free end without inversion.



### STATIONARY WAVES

- \* **Definition :** The wave propagating in such a medium will be reflected at the boundary and produce a wave of the same kind travelling in the opposite direction. The superposition of the two waves will give rise to a stationary wave. Formation of stationary wave is possible only and only in bounded medium.

### ANALYTICAL METHOD FOR STATIONARY WAVES

- **From rigid end :** We know equation for progressive wave in positive x-direction  $y_1 = a \sin(\omega t - kx)$   
 After reflection from rigid end  $y_2 = a \sin(\omega t + kx + \pi) = -a \sin(\omega t + kx)$   
 By principle of super position.  $y = y_1 + y_2 = a \sin(\omega t - kx) - a \sin(\omega t + kx) = -2a \sin kx \cos \omega t$   
 This is equation of stationary wave reflected from rigid end

**Amplitude**  $= 2a \sin kx$

**Velocity of particle**  $v_{pa} = \frac{dy}{dt} = 2a \omega \sin kx \sin \omega t$

**Strain**  $\frac{dy}{dx} = -2ak \cos kx \cos \omega t$

**Elasticity**  $E = \frac{\text{stress}}{\text{strain}} = \frac{dp}{\frac{dy}{dx}}$

**Change in pressure**  $dp = E \frac{dy}{dx}$

• **Node**  $x = 0, \frac{\lambda}{2}, \lambda, \dots$

$A = 0, V_{pa} = 0, \text{strain} \rightarrow \text{max}$

**Change in pressure**  $\rightarrow \text{max}$

• **Antinode**  $x = \frac{\lambda}{4}, \frac{3\lambda}{4}, \dots$

$A \rightarrow \text{max}, -V_{pa} \rightarrow \text{max}, \text{strain} = 0$

**Change in pressure**  $= 0$

- **From free end :** we know equation for progressive wave in positive x-direction  $y_1 = a \sin(\omega t - kx)$   
 After reflection from free end  $y_2 = a \sin(\omega t + kx)$

By Principle of superposition  $y = y_1 + y_2 = a \sin(\omega t - kx) + a \sin(\omega t + kx) = 2a \sin \omega t \cos kx$

**Amplitude** =  $2a \cos kx$ ,

**Velocity of particle** =  $v_{pa} = \frac{dy}{dt} = 2a \omega \cos \omega t \cos kx$

**Strain**  $\frac{dy}{dx} = -2ak \sin \omega t \sin kx$

**Change in pressure**  $dp = E \frac{dy}{dx}$

• **Antinode** :  $x = 0, \frac{\lambda}{2}, \lambda, \dots$   $A \rightarrow \text{Max}$ ,  $V_{pa} = \frac{dy}{dt} \rightarrow \text{max}$ .

Strain = 0,  $dp = 0$

• **Node** :  $x = \frac{\lambda}{4}, \frac{3\lambda}{4}, \frac{5\lambda}{4}, \dots$   $A = 0$ ,  $V_{pa} = \frac{dy}{dt} = 0$ , strain  $\rightarrow \text{max}$ ,  $dp \rightarrow \text{max}$

## PROPERTIES OF STATIONARY WAVES

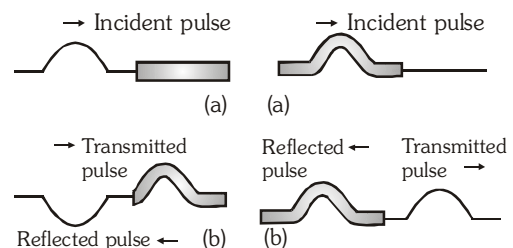
The stationary waves are formed due to the superposition of two identical simple harmonic waves travelling in opposite direction with the same speed.

### Important characteristics of stationary waves are:-

- Stationary waves are produced in the bounded medium and the boundaries of bounded medium may be rigid or free.
- In stationary waves nodes and antinodes are formed alternately. Nodes are the points which are always in rest having maximum strain. Antinodes are the points where the particles vibrate with maximum amplitude having minimum strain.
- All the particles except at the nodes vibrate simple harmonically with the same period.
- The distance between any two successive nodes or antinodes is  $\lambda/2$ .
- The amplitude of vibration gradually increases from zero to maximum value from node to antinode.
- All the particles in one particular segment vibrate in the same phase, but the particle of two adjacent segments differ in phase by  $180^\circ$
- All points of the medium pass through their mean position simultaneously twice in each period.
- Velocity of the particles while crossing mean position varies from maximum at antinodes to zero at nodes.
- In a stationary wave the medium is split into segments and each segment is vibrating up and down as a whole.
- In longitudinal stationary waves, condensation (compression) and rarefaction do not travel forward as in progressive waves but they appear and disappear alternately at the same place.
- These waves do not transfer energy in the medium. Transmission of energy is not possible in a stationary wave.

## TRANSMISSION OF WAVES

We may have a situation in which the boundary is intermediate between these two extreme cases, that is, one in which the boundary is neither rigid nor free. In this case, part of the incident energy is transmitted and part is reflected. For instance, suppose a light string is attached to a heavier string as in (figure). When a pulse travelling on the light reaches the knot, same part of it is reflected and inverted and same part of it is transmitted to the heavier string.



As one would expect, the reflected pulse has a smaller amplitude than the incident pulse, since part of the incident energy is transferred to the pulse in the heavier string. The inversion in the reflected wave is similar to the behaviour of a pulse meeting a rigid boundary, when it is totally reflected. When a pulse travelling on a heavy string strikes the boundary of a lighter string, as in (figure), again part is reflected and part is transmitted. However, in this case the reflected pulse is not inverted. In either case, the relative height of the reflected and transmitted pulses depend on the relative densities of the two string. In the previous section, we found that the speed of a wave on a string increases as the density of the string decreases. That is, a pulse travels more slowly on a heavy string than on a light string, if both are under the same tension. The following general rules apply to reflected waves. When a wave pulse travels from medium A to medium B and  $v_A > v_B$  (that is, when B is denser than A), the pulse will be inverted upon reflection. When a wave pulse travels from medium A to medium B and  $v_A < v_B$  (A is denser than B), it will not be inverted upon reflection.

### KEY POINTS

Phenomenon of reflection and transmission of waves obeys the laws of reflection and refraction.

The frequency of these wave remains constant i.e. does not change.  $\omega_i = \omega_r = \omega_t = \omega$

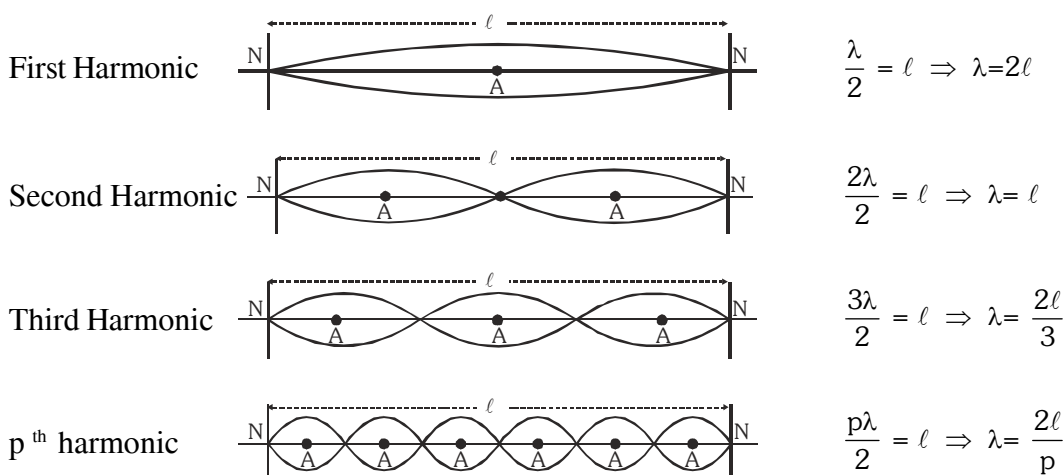
From rarer to denser medium  $y_i = a_i \sin(\omega t - k_1 x)$   $y_r = -a_i \sin(\omega t + k_1 x)$   $y_t = a_t \sin(\omega t - k_2 x)$

From denser to rarer medium  $y_i = a_i \sin(\omega t - k_1 x)$   $y_r = a_i \sin(\omega t + k_1 x)$   $y_t = a_t \sin(\omega t - k_2 x)$

### STATIONARY WAVE ARE OF TWO TYPES :

(i) Transverse st. wave (stretched string) (ii) Longitudinal st. wave (organ pipes)

#### (i) Transverse Stationary wave

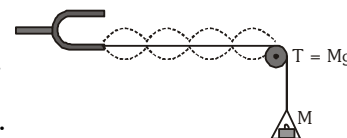


- **Law of length :** For a given string, under a given tension, the fundamental frequency of vibration is inversely proportional to the length of the string, i.e.,  $n \propto \frac{1}{l}$  (T and m are constant)
- **Law of tension :** The fundamental frequency of vibration of stretched string is directly proportional to the square root of the tension in the string, provided that length and mass per unit length of the string are kept constant.  $n \propto \sqrt{T}$  ( $l$  and m are constant)

- **Law of mass :** The fundamental frequency of vibration of a stretched string is inversely proportional to the square root of its mass per unit length provided that length of the string and tension in the string are kept constant, i.e.,  $n \propto \frac{1}{\sqrt{m}}$  ( $\ell$  and  $T$  are constant)
- **Melde's experiment :** In Melde's experiment, one end of a flexible piece of thread is tied to the end of a tuning fork. The other end passed over a smooth pulley carries a pan which can be loaded. There are two arrangements to vibrate the tied fork with thread.

**Transverse arrangement :**

**Case 1.** In a vibrating string of fixed length, the product of number of loops and square root of tension are constant or  $p \sqrt{T} = \text{constant}$ .



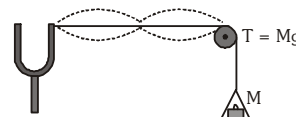
**Case 2.** When the tuning fork is set vibrating as shown in fig. then the prong vibrates at right angles to the thread. As a result the thread is set into motion. The frequency of vibration of the thread (string) is equal to the frequency of the tuning fork. If length and tension are properly adjusted then, standing waves are formed in the string. (This happens when frequency of one of the normal modes of the string matched with the frequency of the tuning fork). Then, if  $p$  loops

are formed in the thread, then the frequency of the tuning fork is given by  $n = \frac{p}{2\ell} \sqrt{\frac{T}{m}}$

**Case 3.** If the tuning fork is turned through a right angle, so that the prong vibrates along the length of the thread, then the string performs only a half oscillation for each complete vibrations of the prong. This is because the thread only makes node at the midpoint when the prong moves towards the pulley i.e. only once in a vibration.

**Longitudinal arrangement :**

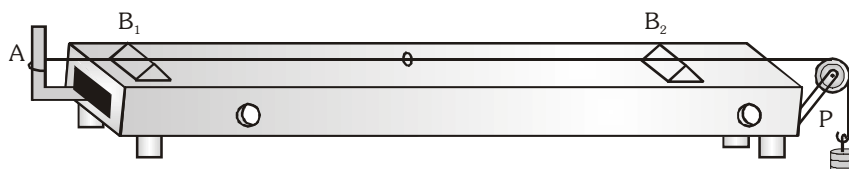
The thread performs sustained oscillations when the natural frequency of the given length of the thread under tension is half that of the fork.



Thus if  $p$  loops are formed in the thread, then the frequency of the tuning fork is  $n = \frac{2p}{2\ell} \sqrt{\frac{T}{m}}$

**SONOMETER :**

Sonometer consists of a hollow rectangular box of light wood. One end of the experimental wire is fastened to one end of the box. The wire passes over a frictionless pulley  $P$  at the other end of the box. The wire is stretched by a tension  $T$ .



The box serves the purpose of increasing the loudness of the sound produced by the vibrating wire. If

the length the wire between the two bridges is  $\ell$ , then the frequency of vibration is  $n = \frac{1}{2\ell} \sqrt{\frac{T}{m}}$



To test the tension of a tuning fork and string, a small paper rider is placed on the string. When a vibrating tuning fork is placed on the box, and if the length between the bridges is properly adjusted, then when the two frequencies are exactly equal, the string quickly picks up the vibrations of the fork and the rider is thrown off the wire. There are three laws of vibration of a wire.

### COMPARISON OF PROGRESSIVE AND STATIONARY WAVES

#### Progressive waves

1. These waves travel in a medium with definite velocity.
2. These waves transmit energy in the medium.
3. The phase of vibration varies continuously from particle to particle.
4. No particle of medium is permanently at rest.
5. All particles of the medium vibrate and amplitude of vibration is same.
6. All the particles do not attain the maximum displacement position simultaneously.

#### Stationary waves

- These waves do not travel and remain confined between two boundaries in the medium.
- These waves do not transmit energy in the medium.
- The phase of all the particles in between two nodes is always same. But particles of two adjacent nodes differ in phase by  $180^\circ$
- Particles at nodes are permanently at rest.
- The amplitude of vibration changes from particle to particle. The amplitude is zero for all at nodes and maximum at antinodes.
- All the particles attain the maximum displacement

### WORKED OUT EXAMPLES

**Ex.1** A wave is propagating along x-axis. The displacement of particles of the medium in z-direction at  $t = 0$  is given by:  $z = \exp[-(x + 2)^2]$ , where 'x' is in meters. At  $t = 1$  s, the same wave disturbance is given by:  $z = \exp[-(2 - x)^2]$ . Then, the wave propagation velocity is

- (A) 4 m/s in + x direction                      (B) 4 m/s in -x direction  
(C) 2 m/s in + x direction                      (D) 2 m/s in - x direction

**Ans. (D)**

**Ex.2** A transverse wave is propagating along +x direction. At  $t = 2$  sec, the particle at  $x = 4$  m is at  $y = 2$  mm. With the passage of time its y coordinate increases and reaches to a maximum of 4 mm. The wave equation may be (using  $\omega$  and  $k$  with their usual meanings)

- (A)  $y = 4 \sin(\omega(t + 2) + k(x - 2) + \frac{\pi}{6})$                       (B)  $y = 4 \sin(\omega(t + 2) + k(x) + \frac{\pi}{6})$   
(C)  $y = 4 \sin(\omega(t - 2) - k(x - 4) + \frac{5\pi}{6})$                       (D)  $y = 4 \sin(\omega(t - 2) - k(x - 4) + \frac{\pi}{6})$

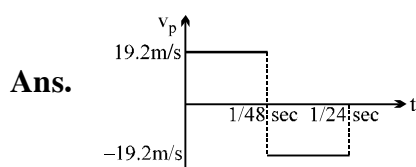
**Ans. (D)**

**Ex.3**  $Y(x,t) = 0.05 / [(4x + 2t)^2 + 5]$  represents a moving wave pulse, where  $x$  and  $y$  are in meters and  $t$  is in seconds. Then which statement(s) are **CORRECT**:

- (A) Pulse is moving in  $-x$  direction (B) Wave speed is  $0.5$  m/s  
(C) Maximum particle displacement is  $1$  cm (D) It is a symmetric pulse

**Ans.** (A,B,C,D)

**Ex.4** A symmetrical triangular pulse of maximum height  $0.4$  m and total length  $1$  m is moving in the positive  $x$ -direction on a string on which the wave speed is  $24$  m/s. At  $t = 0$  the pulse is entirely located between  $x = 0$  and  $x = 1$  m. Draw a graph of the transverse velocity of particle of string versus time at  $x = +1$  m.



**Ex.5** A transverse harmonic disturbance is produced in a string. The maximum transverse velocity is  $3$  m/s and maximum transverse acceleration is  $90$  m/s<sup>2</sup>. If the wave velocity is  $20$  m/s then find the waveform. **[IIT-JEE 2005]**

**Ans.**  $y = (10 \text{ cm}) \sin \left( 30t \pm \frac{3}{2}x + f \right)$

**Ex.6** A non-uniform rope of mass  $M$  and length  $L$  has a variable linear mass density given by  $\mu = kx$  where  $x$  is the distance from one end of the wire and  $k$  is a constant.

- (a) Show that  $M = kL^2/2$   
(b) Show that the time required for a pulse generated at one end of the wire to travel to the other end is given by  $t = \sqrt{8ML/9F}$  where  $F$  (constant) is the tension in the wire.

**Ex.7** One end of a long string of linear mass density  $10^{-2}$  kg m<sup>-1</sup> is connected to an electrically driven tuning fork of frequency  $150$  Hz. The other end passes over a pulley and is tied to a pan containing a mass of  $90$  kg. The pulley end absorbs all the incoming energy so that reflected waves from this end have negligible amplitude. At  $t = 0$ , the left end (fork end) of the string is at  $x = 0$  has a transverse displacement of  $2.5$  cm and is moving along positive  $y$ -direction. The amplitude of the wave is  $5$  cm. Write down the transverse displacement  $y$  (in cm) as function of  $x$  (in m) and  $t$  (in sec) that describes the wave on the string.

**Ans.**  $y = 5 \sin \left\{ \pi(300t - x) + \frac{\pi}{6} \right\}$

**Ex.8** If the tension in a stretched string fixed at both ends is changed by  $20\%$ , the fundamental frequency is found to increase by  $15\text{Hz}$ . then the

- (A) original frequency is  $157$  Hz  
(B) velocity of propagation of the transverse wave along the string changes by  $5\%$   
(C) velocity of propagation of the transverse wave along the string changes by  $10\%$ .  
(D) fundamental wave length on the string does not change.

**Ans.** (A,C,D)

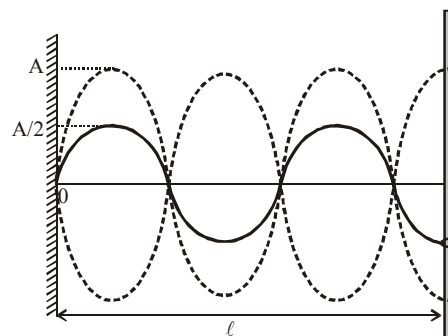
**Ex.9** Here given snap shot at  $t = \frac{T}{12}$  of a standing wave. Then the equations of the wave will be when particles are moving towards their extreme and when particles are moving towards the mean position respectively  $\left( \text{Here, } T = \frac{2\pi}{\omega} \right)$ .

(A)  $y = A \sin kx \sin \omega t$ ,  $y = A \sin \left( \omega t + \frac{2\pi}{3} \right) \sin kx$

(B)  $y = A \sin kx \cos \omega t$ ,  $y = A \sin \left( \omega t + \frac{2\pi}{3} \right) \sin kx$

(C)  $y = A \cos kx \cos \omega t$ ,  $y = A \cos \left( \omega t + \frac{2\pi}{3} \right) \sin kx$

(D)  $y = A \cos kx \sin \omega t$ ,  $y = A \cos \left( \omega t + \frac{2\pi}{3} \right) \cos kx$



**Ans. (A)**

**Ex.10** A standing wave of time period  $T$  is set up in a string clamped between two rigid supports. At  $t = 0$  antinode is at its maximum displacement  $2A$ .

(A) The energy density of a node is equal to energy density of an antinode for the first time at  $t = T/4$ .

(B) The energy density of node and antinode becomes equal after  $T/2$  second.

(C) The displacement of the particle at antinode at  $t = \frac{T}{8}$  is  $\sqrt{2}A$

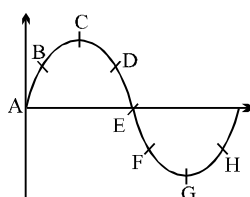
(D) The displacement of the particle at node is zero

**Ans. (C,D)**

## EXERCISE (S-1)

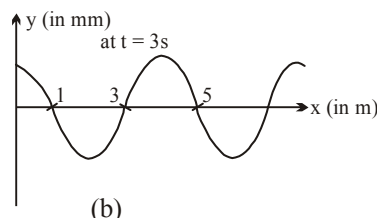
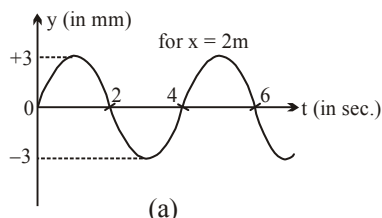
### Wave equation

1. A transverse wave is travelling along a string from left to right. The fig. represents the shape of the string (snap-shot) at a given instant. At this instant (a) which points have an upward velocity (b) which points will have downward velocity (c) which points have zero velocity (d) which points have maximum magnitude of velocity.



WA0001

2. A sinusoidal wave propagates along a string. In figure (a) and (b) 'y' represents displacement of particle from the mean position. 'x' & 't' have usual meanings. Find :  
(a) wavelength, frequency and speed of the wave.  
(b) maximum velocity and maximum acceleration of the particles  
(c) the magnitude of slope of the string for  $x = 2$  m at  $t = 4$  sec.



WA0002

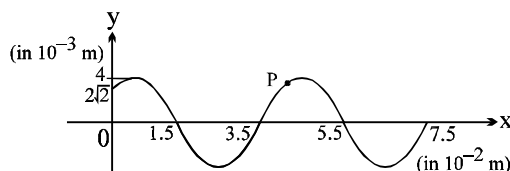
3. A light pointer fixed to one prong of a tuning fork touches a vertical plate. The fork is set vibrating and plate is allowed to fall freely. 8 complete oscillations are counted when the plate falls through 10 cm. What is the frequency of the tuning fork ?

WA0003

4. A long uniform string of mass density  $0.1 \text{ kg/m}$  is stretched with a force of  $40 \text{ N}$ . One end of the string ( $x = 0$ ) is oscillated transversely (sinusoidally) with an amplitude of  $0.02 \text{ m}$  and a period of  $0.1 \text{ sec}$ , so that travelling waves in the  $+x$  direction are set up.  
(a) What is the velocity of the waves?  
(b) What is their wavelength?  
(c) If at the driving end ( $x = 0$ ) the displacement ( $y$ ) at  $t = 0$  is  $0.01 \text{ m}$  with  $dy/dt$  negative, what is the equation of the travelling waves?

WA0004

5. The figure shows a snap photograph of a vibrating string at  $t = 0$ . The particle P is observed moving up with velocity  $20\pi$  cm/s. The angle made by string with x-axis at P is  $6^\circ$ .
- Find the direction in which the wave is moving
  - the equation of the wave
  - the total energy carried by the wave per cycle of the string, assuming that  $\mu$ , the mass per unit length of the string = 50 gm/m.



WA0005

### Velocity of wave

6. The extension in a string, obeying Hooke's law is  $x$ . The speed of wave in the stretched string is  $v$ . If the extension in the string is increased to  $1.5x$  find the new speed of wave.
7. A uniform rope of length  $L$  and mass  $m$  is held at one end and whirled in a horizontal circle with angular velocity  $\omega$ . Ignore gravity. Find the time required for a transverse wave to travel from one end of the rope to the other.
8. A copper wire is held at the two ends by rigid supports. At  $30^\circ\text{C}$ , the wire is just taut, with negligible. Find the speed of transverse waves in this wire at  $10^\circ\text{C}$ .  
 Given : Young modulus of copper =  $1.3 \times 10^{11}$  N/m<sup>2</sup>.  
 Coefficient of linear expansion of copper =  $1.7 \times 10^{-5} \text{ }^\circ\text{C}^{-1}$ .  
 Density of copper =  $9 \times 10^3$  kg/m<sup>3</sup>.

WA0006

WA0007

[IIT-1979]

WA0008

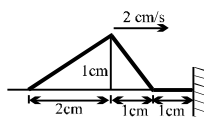
### Energy of wave

9. A steel wire has a mass of 5g/m and is under tension 450 N. Find the maximum average power that can be carried by the transverse wave in the wire if the amplitude is not to exceed 20% of the wavelength.

WA0009

### Superposition of waves

10. The figure shown a triangular pulse on a rope at  $t = 0$ . It is approaching a fixed end at 2 cm/s
- Draw the pulse at  $t = 2$  sec.
  - The particle speed on the leading edge at the instant depicted is \_\_\_\_\_.



WA0010

11. A 40 cm long wire having a mass 3.2 gm and area of cross-section  $1 \text{ mm}^2$  is stretched between the supports 40.05 cm apart. In its fundamental mode it vibrates with a frequency  $1000/64$  Hz. Find the young's modulus of the wire.

WA0011

12. A plane wave given by equation  $y = 0.04 \sin (0.5\pi x - 100\pi t)$ , where  $x$  and  $y$  are in meter and  $t$  in sec is incident normally on a boundary between two media beyond which wave speed becomes doubled. State boundary condition and find the equation of the reflected and transmitted waves. Take  $x = 0$  as the boundary between two media.

WA0012

13. A string between  $x = 0$  and  $x = l$  vibrates in fundamental mode. The amplitude  $A$ , tension  $T$  and mass per unit length  $\mu$  is given. Find the total energy of the string. [IIT-JEE 2003(Scr)]



WA0013

## EXERCISE (S-2)

1. A long string under tension of 100 N has one end at  $x = 0$ . A sinusoidal wave is generated at  $x = 0$

whose equation is given by  $y = (0.01 \text{ cm}) \sin \left[ \left( \frac{\pi x}{10} \right) - 50\pi t (\text{sec}) \right]$

- (i) Sketch the shape of the string at  $t = \frac{1}{50}$  sec.  
 (ii) Find the average power transmitted by the wave.  
 (iii) Draw velocity time graph of particle at  $x = 5 \text{ m}$ .

WA0014

2. A string fixed at both ends is vibrating in the lowest mode of vibration for which a point at quarter of its length from one end is a point of maximum displacement. The frequency of vibration in this mode is 100 Hz. What will be the frequency emitted when it vibrates in the next mode such that this point is again a point of maximum displacement?

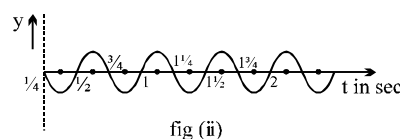
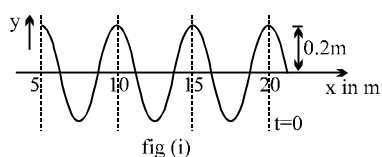
WA0015

3. A uniform string of length  $L$  and total mass  $M$  is suspended vertically and a transverse pulse is given at the top end of it. At the same moment a body is released from rest and falls freely from the top of the string. How far from the bottom does the body pass the pulse.

WA0016

4. A sinusoidal wave is moving along the positive  $x$ -direction as shown in figure (i) and (ii).

- (i) Write the complete expression for the wave  $y(x, t)$   
 (ii) Find the possible values of  $x_0$  for which figure (ii) refers.

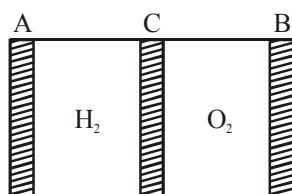


WA0017

5. A plane progressive wave of frequency 25 Hz, amplitude  $2.5 \times 10^{-5}$  m and initial phase zero propagates along the negative x-direction with a velocity of 300 m/s. At any instant, the phase difference between the oscillations at two points 6m apart along the line of propagation is ....., and the corresponding amplitude difference is ..... m. [IIT-1997C]

WA0018

6. AB is a cylinder of length 1m fitted a thin flexible diaphragm C at the middle and other thin flexible diaphragms A and B at the ends. The portions AC and BC contain hydrogen and oxygen gases respectively. The diaphragms A and B are set into vibrations of same frequency. What is the minimum frequency of these vibration for which diaphragm C is a not? (Under the conditions of experiment  $v_{H_2} = 1100$  m/s,  $v_{O_2} = 300$  m/s). [IIT-1978]



WA0019

7. A steel wire  $8 \times 10^{-4}$  m in diameter is fixed to a support at one end and is wrapped round a cylindrical tuning peg 5 mm in diameter at the other end. The length of the wire between the peg and the support is 0.06 m. The wire is initially kept taut but without any tension. What will be the fundamental frequency of vibration of the wire if it is tightened by giving the peg a quarter of a turn? Density of steel =  $7800 \text{ kg/m}^3$ ,  $Y$  of steel =  $20 \times 10^{10} \text{ N/m}^2$ .

WA0020

8. A steel of wire of length 25 cm is fixed at its ends to rigid walls. Young's modulus of steel = 200 GPa, coefficient of linear thermal expansion =  $10^{-5} / ^\circ\text{C}$ . Initially, the wire is just taut at  $20^\circ\text{C}$  temperature. The density of steel =  $8.0 \text{ g/cc}$ . A tuning fork of frequency 200 Hz is touched to the wire, to execute oscillations. Simultaneously, the temperature is slowly lowered. At what temperature will resonance occur corresponding to the third overtone?

WA0021

9. A non-uniform rope of mass  $M$  and length  $L$  has a variable linear mass density given by  $\mu = kx$  where  $x$  is the distance from one end of the wire and  $k$  is a constant.
- (a) Show that  $M = kL^2/2$
- (b) Show that the time required for a pulse generated at one end of the wire to travel to the other end is given by  $t = \sqrt{8ML/9F}$  where  $F$  (constant) is the tension in the wire.

WA0022



## EXERCISE (O-1)

Wave equation

1. The function of  $x$  and  $t$  that does not represent a progressive wave is :-

(A)  $y = 2 \sin(4t - 3x)$  (B)  $y = e^{(4t + (4t - 3x))}$  (C)  $y = [4t - 3x]^{-1}$  (D)  $y = [4t - 3x]$

WA0023

2. At  $x = 0$  particle oscillate by law  $y = \frac{3}{2t^2 + 1}$ . If wave is propagating along  $-ve$   $x$  axis with velocity  $2\text{m/s}$ . Find equation of wave

(A)  $y = \frac{3}{2\left(t - \frac{x}{2}\right)^2 + 1}$  (B)  $y = \frac{3}{2\left(t + \frac{x}{2}\right)^2 + 1}$  (C)  $y = \frac{3}{2\left(t - \frac{z}{2}\right)^2 + 1}$  (D)  $y = \frac{3}{2\left(t + \frac{z}{2}\right)^2 + 1}$

WA0024

3. The shape of a wave propagating in the positive  $x$  or negative  $x$ - direction is given  $y = \frac{1}{\sqrt{1+x^2}}$  at

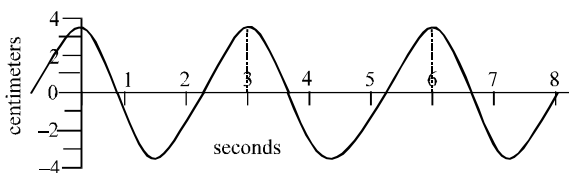
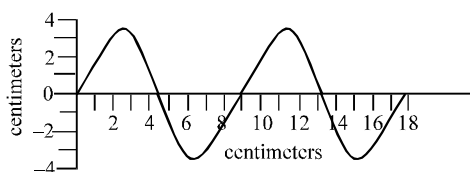
$t = 0$  and  $y = \frac{1}{\sqrt{2-2x+x^2}}$  at  $t = 1\text{s}$  where  $x$  and  $y$  are in meters. The shape the wave disturbance

does not change during propagation. Find the velocity of the wave.

- (A)  $1\text{ m/s}$  in positive  $x$  direction (B)  $1\text{ m/s}$  in negative  $x$  direction  
(C)  $\frac{1}{2}\text{ m/s}$  in positive  $x$  direction (D)  $\frac{1}{2}\text{ m/s}$  in negative  $x$  direction

WA0025

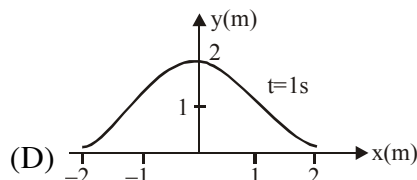
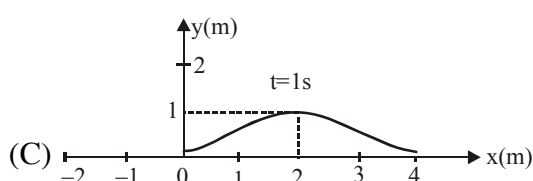
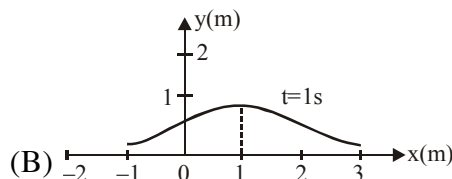
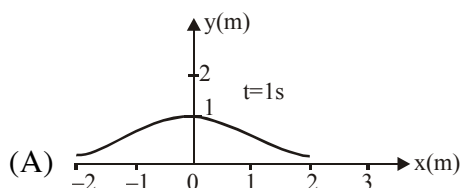
4. A transverse wave is travelling along a horizontal string. The first picture shows the shape of the string at an instant of time. This picture is superimposed on a coordinate system to help you make any necessary measurements. The second picture is a graph of the vertical displacement of *one* point along the string as a function of time. How far does this wave travel along the string in one second?



- (A)  $0.3\text{ cm}$  (B)  $3.0\text{ cm}$  (C)  $9.0\text{ cm}$  (D)  $27\text{ cm}$

WA0026

5. A wave pulse is given by the equation  $y = f(x, t) = A \exp(-B(x-vt)^2)$ . Given  $A = 1.0\text{m}$ ,  $B = 1.0\text{ m}^{-2}$  and  $v = + 2.0\text{ m/s}$ , which of the following graph shows the correct wave profile at the instant  $t = 1\text{ s}$ ?



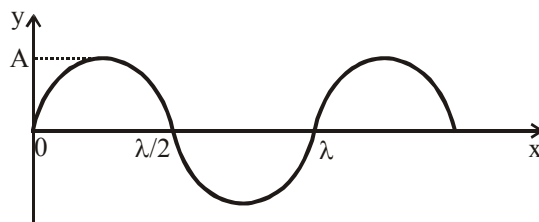
WA0027

6. The displacement from the position of equilibrium of a point 4 cm from a source of sinusoidal oscillations is half the amplitude at the moment  $t = T/6$  ( $T$  is the time period). Assume that the source was at mean position at  $t = 0$ . The wavelength of the running wave is  
(A) 0.96 m (B) 0.48 m (C) 0.24 m (D) 0.12 m

WA0028

7. Here given snap shot of a progressive wave at  $t = 0$  with time period  $= T$ . Then the equation of the wave if wave is going in +ve x-direction and if wave is going in -ve x-direction will be respectively.

(Here,  $T = \frac{2\pi}{\omega}$ )



- (A)  $y = A \sin(kx + \omega t)$ ,  $y = A \sin(kx - \omega t)$  (B)  $y = A \cos(kx + \omega t)$ ,  $y = A \cos(kx - \omega t)$   
(C)  $y = A \sin(\omega t - kx)$ ,  $y = A \sin(\omega t + kx)$  (D)  $y = A \sin(kx - \omega t)$ ,  $y = A \sin(kx + \omega t)$

WA0029

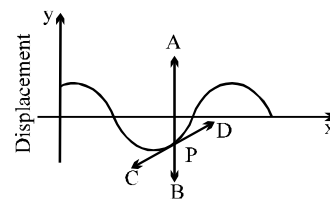
8. A sinusoidal wave travelling in the positive direction of  $x$  on a stretched string has amplitude 2.0 cm, wavelength 1 m and wave velocity 5.0 m/s. At  $x = 0$  and  $t = 0$ , it is given that displacement  $y = 0$  and

$\frac{\partial y}{\partial t} < 0$ . Express the wave function correctly in the form  $y = f(x, t)$  :-

- (A)  $y = (0.02\text{ m}) \sin 2\pi(x - 5t)$  (B)  $y = (0.02\text{ cm}) \cos 2\pi(x - 5t)$   
(C)  $y = (0.02\text{ m}) \sin 2\pi\left(x - 5t + \frac{1}{4}\right)$  (D)  $y = (0.02\text{ cm}) \cos 2\pi\left(x - 5t + \frac{1}{4}\right)$

WA0030

9. The figure below shows a snap photograph of a simple harmonic progressive wave, progressing in the negative X-axis, at a given instant. The direction of the velocity of the particle at the stage P on the figure is best represented by the arrow.

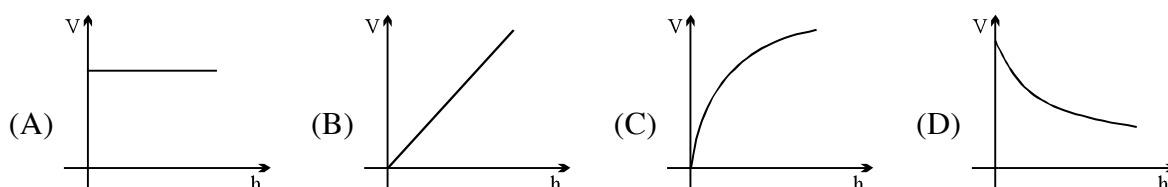


- (A)  $\vec{PA}$  (B)  $\vec{PB}$   
(C)  $\vec{PC}$  (D)  $\vec{PD}$

WA0031

### Velocity of wave

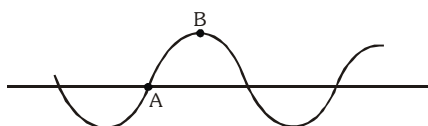
10. A uniform rope having some mass hangs vertically from a rigid support. A transverse wave pulse is produced at the lower end. The speed ( $v$ ) of the wave pulse varies with height ( $h$ ) from the lower end as:



WA0032

### Energy of wave

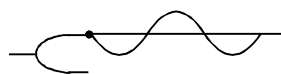
11. A progressive wave is travelling in a string as shown. Then which of the following statement about KE and potential energy of the elements A and B is true?



- (A) For point A : kinetic energy is maximum and potential energy is min.  
(B) For point B : kinetic energy is minimum and potential energy is min.  
(C) For point A : kinetic energy is minimum and potential energy is max.  
(D) For point B : kinetic energy is minimum and potential energy is max.

WA0033

12. The prong of a electrically operated tuning fork is connected to a long string of  $\mu = 1 \text{ kg/m}$  and tension  $25\text{N}$ . The maximum velocity of the prong is  $1 \text{ cm/s}$ , then the average power needed to drive the prong is:

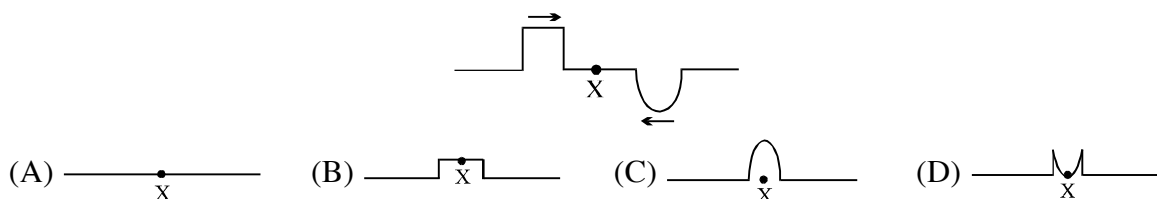


- (A)  $5 \times 10^{-4} \text{ W}$  (B)  $2.5 \times 10^{-4} \text{ W}$  (C)  $1 \times 10^{-4} \text{ W}$  (D)  $10^{-3} \text{ W}$

WA0034

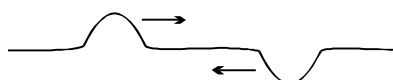
# Superposition of waves

13. The diagram below shows two pulses traveling towards each other in a uniform medium with same speed. Pulses in the figure are at the same distance from X and has same height & width. Which diagram best represents the medium when the pulses meet at point X ?



WA0035

14. Two symmetric, identical pulses of opposite amplitude travel along a stretched string in opposite directions as shown in the figure below. Which one of the following statements most fully describes the situation ?

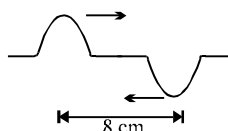


- (A) There is an instant when the string is straight  
(B) When the two pulses interfere completely, the energy of the wave is zero  
(C) There is a point on the string that does not move up or down  
(D) Both A and C  
(E) Both A and B

WA0036

15. Two pulses in a stretched string whose centres are initially 8 cm apart are moving towards each other as shown in figure. The speed of each pulse is 2 cm/s. After 2 seconds, the total energy of the pulses will be :-

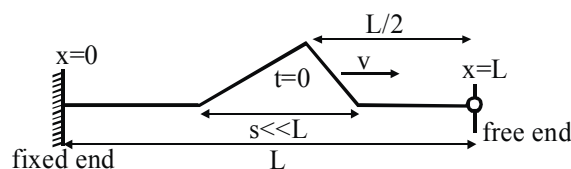
[IIT-JEE 2001(Scr)]



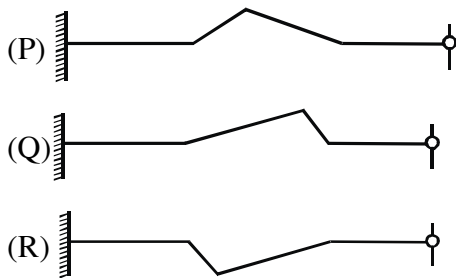
- (A) zero  
(B) purely kinetic  
(C) purely potential  
(D) partly kinetic and partly potential

WA0037

16. A small pulse travelling with speed  $v$  in a string is shown at  $t = 0$ , moving towards free end. Which of these is not **CORRECTLY** matched.



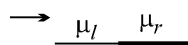
- (i)  $t = \frac{L}{v}$   
(ii)  $t = \frac{2L}{v}$   
(iii)  $t = \frac{3L}{v}$



- (A) (i) (B) (ii) (C) (iii) (D) None of these

WA0038

17. A string consists of two parts attached at  $x = 0$ . The right part of the string ( $x > 0$ ) has mass  $\mu_r$  per unit length and the left part of the string ( $x < 0$ ) has mass  $\mu_l$  per unit length. The string tension is  $T$ . If a wave of unit amplitude travels along the left part of the string, as shown in the figure, what is the amplitude of the wave that is transmitted to the right part of the string ?



- (A) 1 (B)  $\frac{2}{1 + \sqrt{\mu_l / \mu_r}}$  (C)  $\frac{2\sqrt{\mu_l / \mu_r}}{1 + \sqrt{\mu_l / \mu_r}}$  (D)  $\frac{\sqrt{\mu_l / \mu_r} - 1}{\sqrt{\mu_l / \mu_r} + 1}$

WA0039

18. A wave travels on a light string. The equation of the wave is  $Y = A \sin(kx - \omega t + 30^\circ)$ . It is reflected from a heavy string tied to an end of the light string at  $x = 0$ . If 64% of the incident energy is reflected the equation of the reflected wave is

- (A)  $Y = 0.8A \sin(kx - \omega t + 30^\circ + 180^\circ)$  (B)  $Y = 0.8A \sin(kx + \omega t + 30^\circ + 180^\circ)$   
(C)  $Y = 0.8A \sin(kx + \omega t - 30^\circ)$  (D)  $Y = 0.8A \sin(kx + \omega t + 30^\circ)$

WA0040

19. A wave is represented by the equation  $y = 10 \sin 2\pi(100t - 0.02x) + 10 \sin 2\pi(100t + 0.02x)$ . The maximum amplitude and loop length are respectively

- (A) 20 units and 30 units (B) 20 units and 25 units  
(C) 30 units and 20 units (D) 25 units and 20 units

WA0041

20. A wave represented by the equation  $y = A \cos(kx - \omega t)$  is superimposed with another wave to form a stationary wave such that the point  $x = 0$  is a node. The equation of the other wave is:

- (A)  $-A \sin(kx + \omega t)$  (B)  $-A \cos(kx + \omega t)$  (C)  $A \sin(kx + \omega t)$  (D)  $A \cos(kx + \omega t)$

WA0042

21. Five waveforms moving with equal speeds on the  $x$ -axis

$$y_1 = 8 \sin(\omega t + kx); y_2 = 6 \sin(\omega t + \frac{\pi}{2} + kx); y_3 = 4 \sin(\omega t + \pi + kx); y_4 = 2 \sin(\omega t + \frac{3\pi}{2} + kx);$$

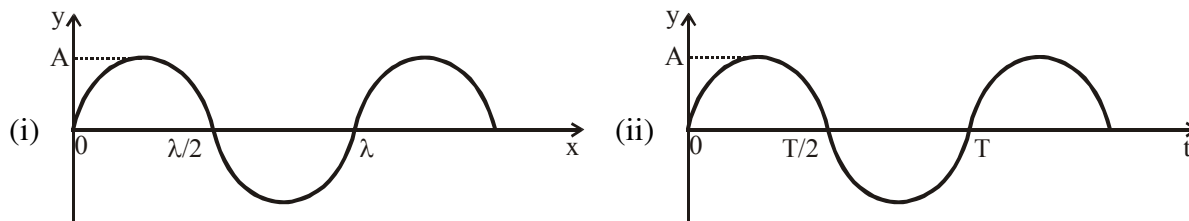
$$y_5 = 4\sqrt{2} \sin(\omega t - kx + \frac{\pi}{4}) \text{ are superimposed on each other. The resulting wave is :}$$

- (A)  $8\sqrt{2} \cos kx \sin(\omega t + \frac{\pi}{4})$  (B)  $8\sqrt{2} \sin(\omega t - kx + \frac{\pi}{4})$   
(C)  $8\sqrt{2} \sin kx \cos(\omega t + \frac{\pi}{4})$  (D)  $8 \sin(\omega t + kx)$

WA0043

22. Here given figure (i) shows snap shot at  $t = T/4$  and figure (ii) shows motion of particle at  $x = \lambda/4$ .

Then the possible equations of the wave will be (Here,  $T = \frac{2\pi}{\omega}$ ) :-



(A)  $y = A \sin \left( \omega t + kx - \frac{\pi}{2} \right)$

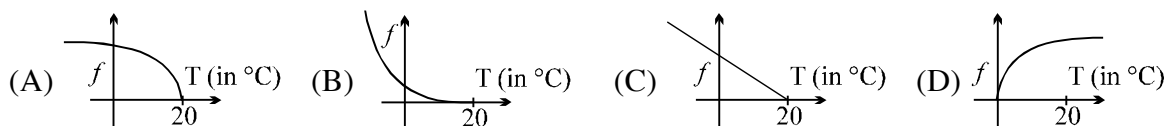
(B)  $y = A \sin \left( \omega t - kx + \frac{\pi}{2} \right)$

(C) Both  $y = A \cos \left( \omega t + kx - \frac{\pi}{2} \right)$  and  $y = A \cos \left( \omega t - kx + \frac{\pi}{2} \right)$

(D) Both  $y = A \sin \left( \omega t + kx - \frac{\pi}{2} \right)$  and  $y = A \sin \left( \omega t - kx + \frac{\pi}{2} \right)$

WA0044

23. A metal wire is clamped between two vertical walls. At  $20^\circ\text{C}$  the unstrained length of the wire is exactly equal to the separation between walls. If the temperature of the wire is decreased the graph between fundamental frequency ( $f$ ) and temperature ( $T$ ) of the wire is



WA0045

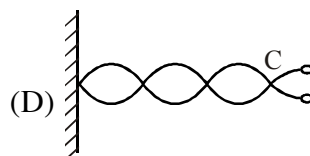
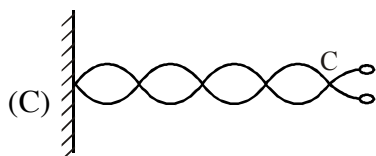
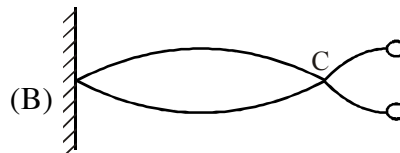
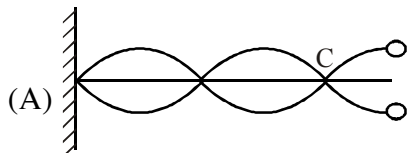
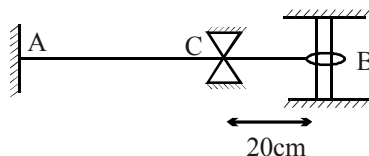
24. What is the fractional change in the tension necessary in a sonometer of fixed length to produce a note one octave lower (half of original frequency) than before
- (A)  $1/4$  (B)  $1/2$  (C)  $2/3$  (D)  $3/4$   
(E)  $2/1$

WA0046

25. A string clamped at both ends is vibrating. At the moment the string looks flat, the instantaneous transverse velocity of points along the string, excluding its end-points, must be
- (A) zero everywhere (B) dependent on the location along the string  
(C) non zero everywhere (D) non-zero and in the same direction everywhere

WA0047

26. A 1m long wire having tension  $T$  is fixed at A and free at B. The point C, 20 cm from B is constrained to be stationary. What is shape of string for fundamental mode?



WA0048

27. The ends of a stretched wire of length  $L$  are fixed at  $x = 0$  and  $x = L$ . In one experiment, the displacement of the wire is  $y_1 = A \sin(\pi x/L) \sin \omega t$  and energy is  $E_1$  and in another experiment its displacement is  $y_2 = A \sin(2\pi x/L) \sin 2\omega t$  and energy is  $E_2$ . Then

[JEE 2001 (Scr)]

- (A)  $E_2 = E_1$       (B)  $E_2 = 2E_1$       (C)  $E_2 = 4E_1$       (D)  $E_2 = 16E_1$

WA0049

## EXERCISE (O-2)

1. Consider a hypothetical wave pulse (at time  $t = 0$ ) given by the following ( $y, x$  in meter)

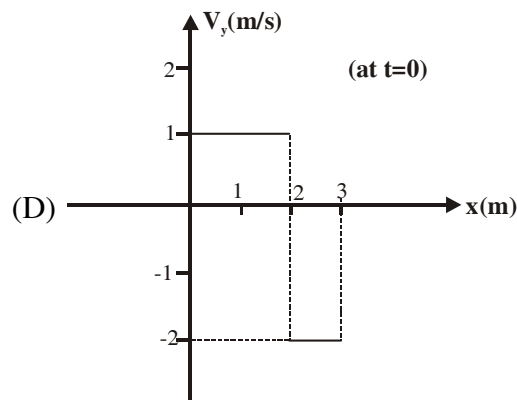
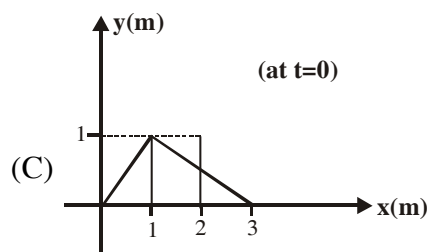
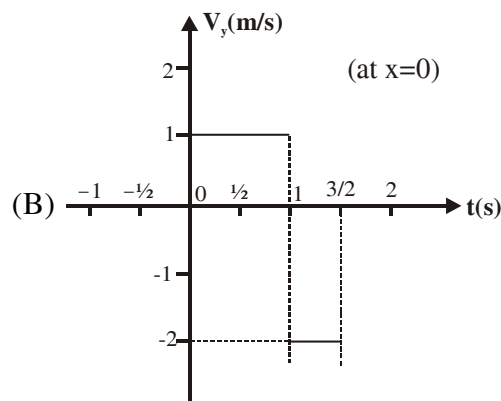
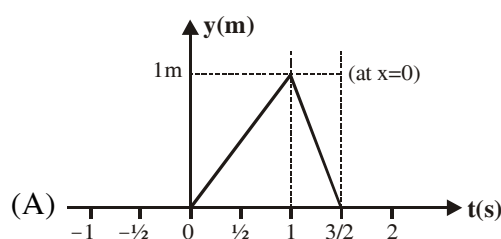
$$y = 0, x < 0$$

$$y = x/2, 2 > x \geq 0$$

$$y = 3 - x, 3 \geq x \geq 2$$

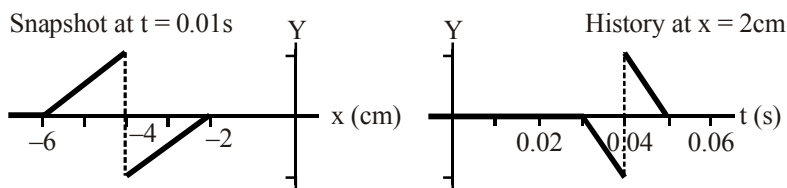
$$y = 0, x > 3$$

The pulse travels leftwards (negative  $x$  direction) at speed  $v = 2$  m/s. Which of the following plots are correct? [ $V_y$  is the velocity of particle]





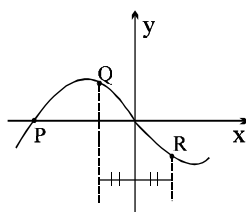
2. Figure shows a snapshot graph and a history graph for a wave pulse on a stretched string. They describe the same wave from two perspectives.



- (A) the wave is travelling in positive x-direction  
 (B) the wave is travelling in negative x-direction  
 (C) the speed of the wave is 2 m/s.  
 (D) the peak is located at  $x = -6$  cm at  $t = 0$ .

WA0051

3. At a certain moment, the photograph of a string on which a harmonic wave is travelling to the right is shown. Then, which of the following is true regarding the velocities of the points P, Q and R on the string.



- (A)  $v_P$  is upwards      (B)  $v_Q = -v_R$       (C)  $|v_P| > |v_Q| = |v_R|$       (D)  $v_Q = v_R$

WA0052

4. An string has resonant frequencies given by 1001 Hz and 2639 Hz.  
 (A) If the string is fixed at one end only, 910 Hz can be a resonance frequency.  
 (B) If the string is fixed at one end only, 1911 Hz can be a resonance frequency.  
 (C) If the string is fixed at both the ends, 364 Hz can be one of the resonant frequency.

WA0053

- (D) 1001 Hz is definitely not the fundamental frequency of the string.

5. In a travelling one dimensional mechanical sinusoidal, wave  
 (A) potential energy and kinetic energy of an element become maximum simultaneously.  
 (B) all particles oscillate with the same frequency and the same amplitude  
 (C) all particles may come to rest simultaneously  
 (D) we can find two particles, in a length equal to half of a wavelength, which have the same non zero acceleration simultaneously.

WA0054

**Paragraph for Question Nos. 6 to 8**

A wave represented by equation  $y = 2(\text{mm}) \sin [4\pi(\text{sec}^{-1})t - 2\pi(\text{m}^{-1})x]$  is superimposed with another wave  $y = 2(\text{mm}) \sin [4\pi(\text{sec}^{-1})t + 2\pi(\text{m}^{-1})x + \pi/3]$  on a tight string.

6. Phase difference between two particles which are located at  $x_1 = 1/7$  and  $x_2 = 5/12$  is :-

- (A) 0 (B)  $\frac{5\pi}{6}$  (C)  $\pi$  (D)  $\frac{5\pi}{3}$

WA0055

7. Which of the following is not a location of antinode?

- (A)  $\frac{2}{3}$  (B)  $\frac{11}{12}$  (C)  $\frac{5}{12}$  (D)  $\frac{17}{12}$

WA0055

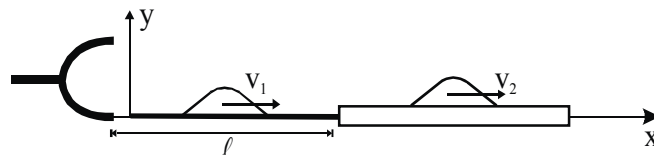
8. The location having maximum potential energy is

- (A)  $1/7$  (B)  $1/6$  (C)  $5/12$  (D)  $23/12$

WA0055

**Paragraph for Question Nos. 9 to 11**

A harmonic oscillator at  $x = 0$ , oscillates with a frequency  $\frac{\omega}{2\pi}$  and amplitude  $a$ . It is generating waves at end of a thin string in which velocity of wave is  $v_1$  and which is connected to another heavier string in which velocity of wave is  $v_2$  as shown, length of first string is  $\ell$ .



9. If harmonic oscillator oscillates by an equation  $y = a \sin \omega t$ . The equation of incident wave in first string is

- (A)  $y = a \sin \omega \left( t - \frac{x}{v_1} \right)$  (B)  $y = a \sin \omega \left( t + \frac{x}{v_1} \right)$   
 (C)  $y = a \sin \left[ \omega \left( t - \frac{x}{v_1} \right) + \pi \right]$  (D)  $y = a \sin \left[ \omega \left( t + \frac{x}{v_1} \right) + \pi \right]$

WA0056

10. Equation of transmitted wave in second string if its amplitude is  $a_t$  is

(A)  $y = a_t \sin \omega \left( t - \frac{x}{v_2} \right)$

(B)  $y = a_t \sin \omega \left( t - \frac{\ell}{v_1} \right)$

(C)  $y = a_t \sin \omega \left( t - \frac{\ell}{v_1} - \frac{x - \ell}{v_2} \right)$

(D)  $y = a_t \sin \omega \left( t - \frac{x}{v_2} \right)$

WA0056

11. Equation of reflected wave, if it is reflecting at the joint and amplitude of reflected wave is  $a_r$

(A)  $y = a_r \sin \omega \left( t - \frac{x}{v_2} \right)$

(B)  $y = a_r \sin \left[ \omega \left( t - \frac{\ell}{v_1} - \frac{\ell - x}{v_1} \right) + \pi \right]$

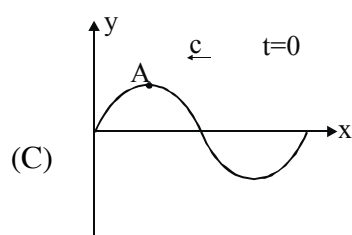
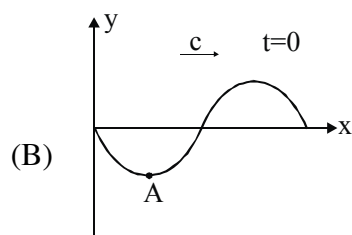
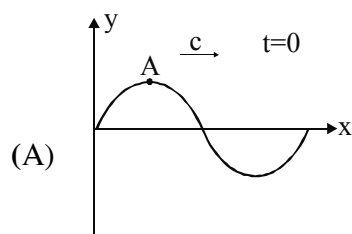
(C)  $y = a_r \sin \left[ \omega \left( t + \frac{x}{v_1} \right) + \pi \right]$

(D)  $y = a_r \sin \left[ \omega \left( t + \frac{2\ell + x}{v_1} \right) + \pi \right]$

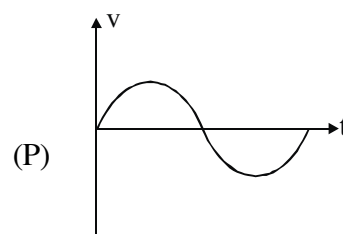
WA0056

12. In column-I transverse waves travelling on a string at  $t = 0$  is shown. Wave velocity is indicated by 'c'. Column-II describes variation of different parameters for particle A or for all the particles.

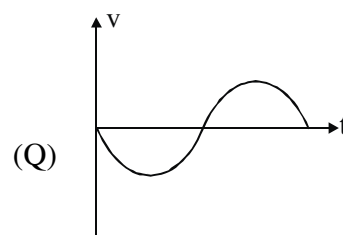
## Column-I



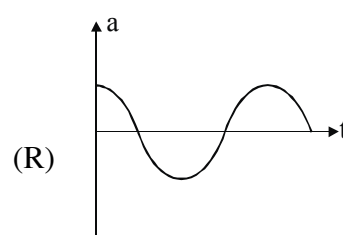
## Column-II

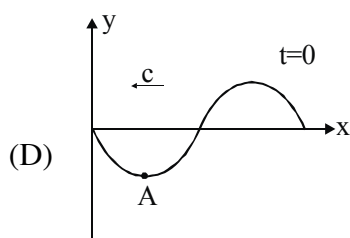


For particle A

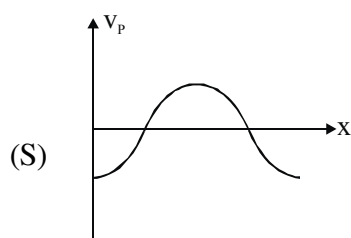


For particle A

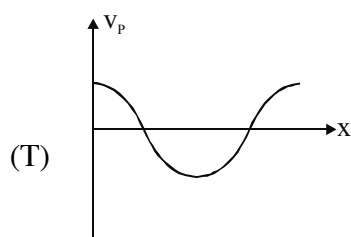




For particle A



At  $t = 0$  for all the particles



At  $t = 0$  for all the particles

**WA0057**

13. In a string a standing wave is set up whose equation is given as  $y = 2A \sin kx \cos \omega t$ . The mass per unit length of the string is  $\mu$ .

**Column-I**

- (A) at  $t = 0$   
 (B) at  $t = \frac{T}{8}$   
 (C) at  $t = \frac{T}{4}$   
 (D) at  $t = \frac{T}{2}$

**Column-II**

- (P) Total energy per unit length at  $x = 0$  is  $2\mu A^2 \omega^2$ .  
 (Q) Total energy per unit length at  $x = \lambda/4$  is  $2\mu A^2 \omega^2$ .  
 (R) Total energy per unit length at  $x = \lambda$  is  $2\mu A^2 \omega^2$ .  
 (S) power transmitted through a point at  $x = \lambda$  is 0.  
 (T) power transmitted through a point at  $x = \lambda/4$  is 0.

**WA0058**

## EXERCISE-JM

1. The equation of a wave on a string of linear mass density  $0.04 \text{ kg m}^{-1}$  is given by

$$y = 0.02(\text{m}) \sin \left[ 2\pi \left( \frac{t}{0.04(\text{s})} - \frac{x}{0.50(\text{m})} \right) \right]. \text{ The tension in the string is : } \quad [\text{AIEEE - 2010}]$$

- (1) 6.25 N                      (2) 4.0 N                      (3) 12.5 N                      (4) 0.5 N

WA0059

2. The transverse displacement  $y(x, t)$  of a wave on a string is given by  $y(x, t) = e^{-(ax^2 + bt^2 + 2\sqrt{ab}xt)}$ . This represents a :- [AIEEE - 2011]

(1) standing wave of frequency  $\sqrt{b}$

(2) standing wave of frequency  $\frac{1}{\sqrt{b}}$

(3) wave moving in  $+x$  direction with speed  $\sqrt{\frac{a}{b}}$

(4) wave moving in  $-x$  direction with speed  $\sqrt{\frac{b}{a}}$

WA0060

3. A travelling wave represented by  $y = A \sin(\omega t - kx)$  is superimposed on another wave represented by  $y = A \sin(\omega t + kx)$ . The resultant is :- [AIEEE-2011]

(1) A standing wave having nodes at  $x = \left(n + \frac{1}{2}\right) \frac{\lambda}{2}, n = 0, 1, 2$

(2) A wave travelling along  $+x$  direction

(3) A wave travelling along  $-x$  direction

(4) A standing wave having nodes at  $x = \frac{n\lambda}{2}; n = 0, 1, 2$

WA0061

4. A sonometer wire of length 1.5m is made of steel. The tension in it produces an elastic strain of 1%. What is the fundamental frequency of steel if density and elasticity of steel are  $7.7 \times 10^3 \text{ kg/m}^3$  and  $2.2 \times 10^{11} \text{ N/m}^2$  respectively ? [JEE-Main-2013]

- (1) 188.5 Hz                      (2) 178.2 Hz                      (3) 200.5 Hz                      (4) 770 Hz

WA0062

5. A uniform string of length 20m is suspended from a rigid support. A short wave pulse is introduced at its lowest end. It starts moving up the string. The time taken to reach the support is :-  
(take  $g = 10 \text{ ms}^{-2}$ ) [JEE-Main-2016]

(1)  $\sqrt{2} \text{ s}$  (2)  $2\pi\sqrt{2} \text{ s}$  (3)  $2 \text{ s}$  (4)  $2\sqrt{2} \text{ s}$

WA0063

6. A granite rod of 60 cm length is clamped at its middle point and is set into longitudinal vibrations. The density of granite is  $2.7 \times 10^3 \text{ kg/m}^3$  and its Young's modulus is  $9.27 \times 10^{10} \text{ Pa}$ . What will be the fundamental frequency of the longitudinal vibrations ? [JEE-Main-2018]

(1) 2.5 kHz (2) 10 kHz (3) 7.5 kHz (4) 5 kHz

WA0064

## EXERCISE (JA)

1. A horizontal stretched string, fixed at two ends, is vibrating in its fifth harmonic according to the equation,  $y(x, t) = (0.01\text{m}) \sin [(62.8 \text{ m}^{-1})x] \cos [(628 \text{ s}^{-1})t]$ . Assuming  $\pi = 3.14$ , the correct statement(s) is (are) **[JEE-Advance-2013]**
- (A) The number of nodes is 5.  
 (B) The length of the string is 0.25 m.  
 (C) The maximum displacement of the midpoint of the string, from its equilibrium position is 0.01m.  
 (D) The fundamental frequency is 100 Hz.

WA0065

2. One end of a taut string of length 3m along the x-axis is fixed at  $x = 0$ . The speed of the waves in the string is  $100 \text{ ms}^{-1}$ . The other end of the string is vibrating in the y direction so that stationary waves are set up in the string. The possible waveform (s) of these stationary waves is(are) :- **[JEE-Advance-2014]**

$$(A) y(t) = A \sin \frac{\pi x}{6} \cos \frac{50\pi t}{3}$$

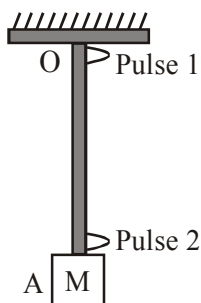
$$(B) y(t) = A \sin \frac{\pi x}{3} \cos \frac{100\pi t}{3}$$

$$(C) y(t) = A \sin \frac{5\pi x}{6} \cos \frac{250\pi t}{3}$$

$$(D) y(t) = A \sin \frac{5\pi x}{2} \cos 250\pi t$$

WA0066

3. A block M hangs vertically at the bottom end of a uniform rope of constant mass per unit length. The top end of the rope is attached to a fixed rigid support at O. A transverse wave pulse (Pulse 1) of wavelength  $\lambda_0$  is produced at point O on the rope. The pulse takes time  $T_{OA}$  to reach point A. If the wave pulse of wavelength  $\lambda_0$  is produced at point A (Pulse 2) without disturbing the position of M it takes time  $T_{AO}$  to reach point O. Which of the following options is/are correct ? **[JEE-Advance-2017]**



- (A) The time  $T_{AO} = T_{OA}$   
 (B) The velocities of the two pulses (Pulse 1 and Pulse 2) are the same at the midpoint of rope  
 (C) The wavelength of Pulse 1 becomes longer when it reaches point A  
 (D) The velocity of any pulse along the rope is independent of its frequency and wavelength.

WA0067

4. Answer the following by appropriately matching the lists based on the information given in the paragraph.

A musical instrument is made using four different metal strings, 1,2,3 and 4 with mass per unit length  $\mu$ ,  $2\mu$ ,  $3\mu$  and  $4\mu$  respectively. The instrument is played by vibrating the strings by varying the free length in between the range  $L_0$  and  $2L_0$ . It is found that in string-1 ( $\mu$ ) at free length  $L_0$  and tension  $T_0$  the fundamental mode frequency is  $f_0$ .

List-I gives the above four strings while list-II lists the magnitude of some quantity.

If the tension in each string is  $T_0$ , the correct match for the highest fundamental frequency in  $f_0$  units will be, [JEE-Advance-2019]

**List-I**

- (I) String-1 ( $\mu$ )  
(II) String-2 ( $2\mu$ )  
(III) String-3 ( $3\mu$ )  
(IV) String-4 ( $4\mu$ )

**List-II**

- (P) 1  
(Q)  $1/2$   
(R)  $1/\sqrt{2}$   
(S)  $1/\sqrt{3}$   
(T)  $3/16$   
(U)  $1/16$   
(1) I  $\rightarrow$  P, II  $\rightarrow$  R, III  $\rightarrow$  S, IV  $\rightarrow$  Q  
(2) I  $\rightarrow$  P, II  $\rightarrow$  Q, III  $\rightarrow$  T, IV  $\rightarrow$  S  
(3) I  $\rightarrow$  Q, II  $\rightarrow$  S, III  $\rightarrow$  R, IV  $\rightarrow$  P  
(4) I  $\rightarrow$  Q, II  $\rightarrow$  P, III  $\rightarrow$  R, IV  $\rightarrow$  T

WA0068

5. Answer the following by appropriately matching the lists based on the information given in the paragraph.

A musical instrument is made using four different metal strings, 1,2,3 and 4 with mass per unit length  $\mu$ ,  $2\mu$ ,  $3\mu$  and  $4\mu$  respectively. The instrument is played by vibrating the strings by varying the free length in between the range  $L_0$  and  $2L_0$ . It is found that in string-1 ( $\mu$ ) at free length  $L_0$  and tension  $T_0$  the fundamental mode frequency is  $f_0$ .

List-I gives the above four strings while list-II lists the magnitude of some quantity.

[JEE-Advance-2019]

**List-I**

- (I) String-1 ( $\mu$ )  
(II) String-2 ( $2\mu$ )  
(III) String-3 ( $3\mu$ )  
(IV) String-4 ( $4\mu$ )

**List-II**

- (P) 1  
(Q)  $1/2$   
(R)  $1/\sqrt{2}$   
(S)  $1/\sqrt{3}$   
(T)  $3/16$   
(U)  $1/16$

The length of the string 1,2,3 and 4 are kept fixed at  $L_0$ ,  $\frac{3L_0}{2}$ ,  $\frac{5L_0}{4}$  and  $\frac{7L_0}{4}$ , respectively. Strings

1,2,3 and 4 are vibrated at their 1<sup>st</sup>, 3<sup>rd</sup>, 5<sup>th</sup> and 14<sup>th</sup> harmonics, respectively such that all the strings have same frequency. The correct match for the tension in the four strings in the units of  $T_0$  will be.

- (1) I  $\rightarrow$  P, II  $\rightarrow$  Q, III  $\rightarrow$  T, IV  $\rightarrow$  U  
(2) I  $\rightarrow$  T, II  $\rightarrow$  Q, III  $\rightarrow$  R, IV  $\rightarrow$  U  
(3) I  $\rightarrow$  P, II  $\rightarrow$  Q, III  $\rightarrow$  R, IV  $\rightarrow$  T  
(4) I  $\rightarrow$  P, II  $\rightarrow$  R, III  $\rightarrow$  T, IV  $\rightarrow$  U

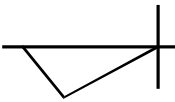
WA0069



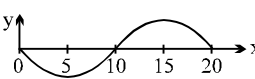
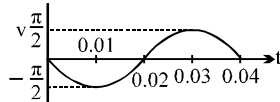
## ANSWER KEY

## EXERCISE (S-1)

1. Ans. (a) DEF, (b) ABH, (c) CG, (D) AE

2. Ans. (a)  $\lambda = 4\text{m}$ ,  $f = \frac{1}{4} \text{ Hz}$ ,  $1 \text{ m/s}$ , (b)  $\frac{3\pi}{2} \text{ mm/s}$ ,  $\frac{3\pi^2}{4} \text{ mm/s}^2$ , (c)  $\frac{3\pi}{2} \times 10^{-3}$ 3. Ans.  $40\sqrt{2}$ 4. Ans. (a)  $20 \text{ m/s}$ ; (b)  $2 \text{ m}$ ; (c)  $y(x,t) = 0.02\sin(\pi x - 20\pi t + \pi/6)$ 5. Ans. (a) negative  $x$ ; (b)  $y = 4 \times 10^{-3} \sin 100\pi \left( 3t + 0.5x + \frac{1}{400} \right)$  ( $x, y$  in meter); (c)  $144\pi^2 \times 10^{-5} \text{ J}$ 6. Ans.  $1.22 \text{ v}$ 7. Ans.  $\frac{\pi}{\sqrt{2}\omega}$ 8. Ans.  $70 \text{ m/s}$ 9. Ans.  $10.8 \times 10^4 \text{ W}$ 10. Ans. (a) , (b)  $2 \text{ cm/s}$ 11. Ans.  $1 \times 10^9 \text{ Nm}^2$ 12. Ans.  $A_t = \frac{4}{3} A_i$ ,  $A_r = \frac{1}{3} A_i$ ,  $y_r = -\frac{0.04}{3} \sin(0.5\pi x + 100\pi t)$ ;  $y_t = +\frac{0.16}{3} \sin(0.25\pi x - 100\pi t)$ 13. Ans.  $E = \frac{A^2 \pi^2 T}{4l}$ 

## EXERCISE (S-2)

1. Ans. (i)  (ii)  $25 \times 10^{-6} \text{ W}$  (iii) , solved2. Ans.  $300 \text{ Hz}$ 3. Ans.  $L/9$ 4. Ans.  $0.2 \cos[2\pi/5(x-10t)]$ , (ii)  $5n - (15/4)$ 5. Ans.  $\pi, 0$ 6. Ans.  $1650 \text{ Hz}$ 7. Ans.  $10800 \text{ Hz}$ 8. Ans.  $17.5^\circ$ 

## EXERCISE (O-1)

1. Ans. (C)

2. Ans. (B)

3. Ans. (A)

4. Ans. (B)

5. Ans. (C)

6. Ans. (B)

7. Ans. (D)

8. Ans. (A)

9. Ans. (A)

10. Ans. (C)

11. Ans. (B)

12. Ans. (B)

13. Ans. (D)

14. Ans. (D)

15. Ans. (B)

16. Ans. (B)

17. Ans. (C)

18. Ans. (C)

19. Ans. (B)

20. Ans. (B)

21. Ans. (A)

22. Ans. (D)

23. Ans. (A)

24. Ans. (D)

25. Ans. (B)

26. Ans. (A)

27. Ans. (C)

**EXERCISE (O-2)**

1. Ans. (A,B,D)      2. Ans. (A,C,D)      3. Ans. (C,D)      4. Ans. (B,C,D)  
 5. Ans. (A,B,D)      6. Ans. (C)      7. Ans. (A)      8. Ans. (B)      9. Ans. (A)  
 10. Ans. (C)      11. Ans. (B)  
 12. Ans. (A) – (Q,S) ; (B) – (P,R,T) ; (C) – (Q,T) ; (D) – (P,R,S)  
 13. Ans. (A) – (P,R,S,T) ; (B) – (S,T) ; (C) – (Q,S,T) ; (D) – (P,R,S,T)

**EXERCISE-JM**

1. Ans. (1)      2. Ans. (4)      3. Ans. (1)      4. Ans. (2)      5. Ans. (4)      6. Ans. (4)

**EXERCISE (JA)**

1. Ans. (B,C)      2. Ans. (A,C,D)      3. Ans. (A,D)      4. Ans. (1)      5. Ans. (1)

## SOUND WAVES

# KEY CONCEPTS

Sound is a mechanical three dimensional and longitudinal wave that is created by a vibrating source such as a guitar string, the human vocal cords, the prongs of a tuning fork or the diaphragm of a loudspeaker.

Sound waves are the most common example of longitudinal waves. They travel through any material waves. They travel through any material medium with a speed that depends on the properties of the medium. As the waves travel through air, the elements of air vibrate to produce changes in density and pressure along the direction of motion of the wave. If the source of the sound waves vibrates sinusoidally, the pressure variations are also sinusoidal. The mathematical description of sinusoidally, the pressure variations are also sinusoidal sound waves is very similar to that of sinusoidal string waves.

### EQUATION OF SOUND WAVES

As the piston oscillates sinusoidally, regions of compression and rarefaction are continuously set up. The distance between two successive compressions (or two successive rarefactions) equals the wavelength  $\lambda$ . As these regions travel through the tube, any small element of the medium moves with simple harmonic motion parallel to the direction of the wave. If  $s(x, t)$  is the position of a small element relative to its equilibrium position, We can express this harmonic position function as

$$s(x, t) = s_{\max} \cos(kx - \omega t)$$

Where  $s_{\max}$  is the maximum position of the element relative to equilibrium. This is often called the displacement amplitude of the wave. The parameter  $k$  is the wave number and  $\omega$  is the angular frequency of the piston. Note that the displacement of the element is along  $s$ , in the direction of propagation of the sound wave, which means we are describing a longitudinal wave.

Consider a thin disk-shaped element of gas whose circular cross section is parallel to the piston in figure. This element will undergo changes in position, pressure, and density as a sound wave propagates through the gas. From the definition of bulk modulus, the pressure variation in the gas is

$$\Delta P = -B \frac{\Delta V}{V_i}$$

The element has a thickness  $\Delta x$  in the horizontal direction and a cross-sectional area  $A$ , so its volume is  $V_i = A\Delta x$ . The change in volume  $\Delta V$  accompanying the pressure change is equal to  $A\Delta s$ , where  $\Delta s$  is the difference between the value of  $s$  at  $x + \Delta x$  and the value of  $s$  at  $x$ . Hence, we can express  $\Delta P$  as

$$\Delta P = -B \frac{\Delta V}{V_i} = -B \frac{A\Delta s}{A\Delta x} = -B \frac{\Delta s}{\Delta x}$$

As  $\Delta x$  approaches zero, the ratio  $\Delta s/\Delta x$  becomes  $\partial s/\partial x$  (The partial derivative indicates that we are interested in the variation of  $s$  with position at a fixed time.) Therefore,

$$\Delta P = -B \frac{\partial s}{\partial x}$$

If the position function is the simple sinusoidal function given by Equation , we find that

$$\Delta P = -B \frac{\partial}{\partial x} [s_{\max} \cos(kx - \omega t)] = Bs_{\max} k \sin(kx - \omega t)$$

$$\Delta P = \Delta P_{\max} \sin(kx - \omega t)$$

Thus we can describe sound waves either in terms of excess pressure (equation 1.1) or in terms of the longitudinal displacement suffered by the particles of the medium.

If  $s = s_0 \sin \omega(t - x/v)$  represents a sound wave where  $s$  = displacement of medium particle from its mean position at  $x$ , then it can be proved that

$$P = P_0 \sin \{ \omega(t - x/v) + \pi/2 \} \quad \dots (3.2)$$

represents that same sound wave where,  $P$  is excess pressure at position  $x$ , over and above the average atmospheric pressure and the pressure amplitude  $P_0$  is given by

$$P_0 = \frac{B\omega s_0}{v} = BKs_0 \quad \dots (3.3)$$

( $B$  = Bulk modulus of the medium,  $K$  = angular wave number)

Note from equation (3.1) and (3.2) that the displacement of a medium particle and excess pressure

at any position are out of phase by  $\frac{\pi}{2}$ . Hence a displacement maxima corresponds to a pressure minima and vice-versa.

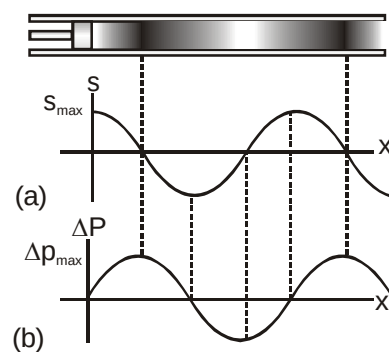


Figure : (a) Displacement amplitude and (b) pressure amplitude versus position for a sinusoidal longitudinal wave

**Ex.** The equation of a sound wave in air is given by  $P = 0.2 \sin [3000 t - 9x]$ , where all variables are in S.I. units.

(a) Find the frequency, wavelength and the speed of sound wave in air.

(b) If the equilibrium pressure of air is  $1.0 \times 10^5 \text{ N/m}^2$ , what are the maximum and minimum pressures at a point as the wave passes through that point?

**Sol.** (a) Comparing with the standard form of a travelling wave

$$P = P_0 \sin [\omega(t - x/v)]$$

we see that  $\omega = 3000 \text{ s}^{-1}$ . The frequency is

$$f = \frac{\omega}{2\pi} = \frac{3000}{2\pi} \text{ Hz}$$

Also from the same comparison,  $\omega/v = 9.0 \text{ m}^{-1}$ .

$$\text{or, } v = \frac{\omega}{9.0 \text{ m}^{-1}} = \frac{3000 \text{ s}^{-1}}{9.0 \text{ m}^{-1}} = \frac{1000}{3} \text{ m/s}^{-1}$$

$$\text{The wavelength is } \lambda = \frac{v}{f} = \frac{1000/3 \text{ m/s}}{3000/2\pi \text{ Hz}} = \frac{2\pi}{9} \text{ m}$$

(b) The pressure amplitude is  $P_0 = 0.02 \text{ N/m}^2$ . Hence, the maximum and minimum pressures at a point in the wave motion will be  $(1.01 \times 10^5 \pm 0.02) \text{ N/m}^2$ .

**Ex.** A sound wave of wavelength 40cm travels in air. If the difference between the maximum and minimum pressure at a given point is  $4.0 \times 10^{-3} \text{ N/m}^2$ , find the amplitude of vibration of the particles of the medium. The bulk modulus of air is  $1.4 \times 10^5 \text{ N/m}^2$ .

**Sol.** The pressure amplitude is

$$P_0 = \frac{4.0 \times 10^{-3} \text{ N/m}^2}{2} = 2 \times 10^{-3} \text{ N/m}^2$$

The displacement amplitude  $s_0$  is given by

$$P_0 = Bks_0$$

or 
$$s_0 = \frac{P_0}{Bk} = \frac{P_0 \lambda}{2\pi B}$$

$$= \frac{2 \times 10^{-3} \text{ N/m}^2 \times (40 \times 10^{-2} \text{ m})}{2 \times \pi \times 14 \times 10^4 \text{ N/m}^2} = \frac{200}{7\pi} \text{ A}$$

$$= 13.2 \text{ A}$$

### SPEED OF SOUND WAVES :

Velocity of sound waves in a linear solid medium is given by

$$v = \sqrt{\frac{Y}{\rho}} \quad (4.1)$$

where  $Y$  = young's modulus of elasticity and  $\rho$  = density

Velocity of sound wave in a fluid medium (liquid or gas) is given by

$$v = \sqrt{\frac{B}{\rho}} \quad (4.2)$$

where,  $\rho$  = density of the medium and  $B$  = Bulk modulus of the medium given by,

$$B = -V \frac{dP}{dV} \quad (4.3)$$

Newton's formula : Newton assumed propagation of sound through a gaseous medium to be an isothermal process.

$$PV = \text{constant}$$

$$\Rightarrow \frac{dP}{dV} = \frac{-P}{V} \quad \text{and} \quad \text{hence } B = P$$

and thus he obtained for velocity of sound in a gas.

$$v = \sqrt{\frac{P}{\rho}} = \sqrt{\frac{RT}{M}} \quad \text{where } M = \text{molar mass}$$

**Laplace's correction :** Later Laplace established that propagation of sound in a gas is not an isothermal but an adiabatic process and hence  $PV^\gamma = \text{constant}$

$$\Rightarrow \frac{dP}{dV} = -\gamma \frac{P}{V}$$

$$\text{where, } B = -V \frac{dP}{dV} = \gamma P$$

and hence speed of sound in a gas,

$$v = \sqrt{\frac{\gamma P}{\rho}} = \sqrt{\frac{\gamma RT}{M}} \quad (4.4)$$

### FACTORS AFFECTING SPEED OF SOUND IN ATMOSPHERE

**(a) Effect of temperature :** as temperature ( $T$ ) increases velocity ( $v$ ) increases. For small change in temperature above room temperature  $v$  increases linearly by 0.6 m/s for every  $1^\circ\text{C}$  rise in temp.

**(b) Effect of humidity :** With increase in humidity density decreases. This is because the molar mass of water vapour is less than the molar mass of air.

**(c) Effect of pressure :**

$$\text{The speed of sound in a gas is given by } v = \sqrt{\frac{\gamma P}{\rho}} = \sqrt{\frac{\gamma RT}{M}}$$

So at constant temperature, if  $P$  changes then  $\rho$  also changes in such a way that  $P/\rho$  remains constant. Hence pressure does not have any effect on velocity of sound as long as temperature is constant.

**Ex.** The constant  $\gamma$  for oxygen as well as for hydrogen is 1.40. If the speed of sound in oxygen is 450 m/s, what will be the speed of hydrogen at the same temperature and pressure?

$$\text{Sol. } v = \sqrt{\frac{\gamma RT}{M}}$$

since temperature,  $T$  is constant

$$\therefore \frac{v_{(\text{H}_2)}}{v_{(\text{O}_2)}} = \sqrt{\frac{M_{\text{O}_2}}{M_{\text{H}_2}}} = \sqrt{\frac{32}{2}} = 4 \Rightarrow v_{(\text{H}_2)} = 4 \times 450 = 1800 \text{ m/s}$$

**Aliter :** The speed of sound in a gas is given by  $u = \sqrt{\frac{\gamma P}{\rho}}$ . At STP, 22.4 litres of oxygen has a mass of 32g whereas the same volume of hydrogen has a mass of 2g. Thus, the density of oxygen is 16times the density of hydrogen at the same temperature and pressure. As  $\gamma$  is same for both the gases.

$$\frac{f_{(\text{hydrogen})}}{f_{(\text{oxygen})}} = \sqrt{\frac{\rho_{(\text{oxygen})}}{\rho_{(\text{hydrogen})}}}$$

$$\text{or } f_{(\text{hydrogen})} = 4f_{(\text{oxygen})} = 4 \times 450 \text{ m/s} = 1800 \text{ m/s}$$

## INTENSITY OF PERIODIC SOUND WAVES

Like any other progressive wave, sound waves also carry energy from one point of space to the other. This energy can be used to work, for example, forcing the eardrums to vibrate or in the extreme case of a sonic boom created by supersonic jet, can even cause glass panes of windows to crack.

The amount of energy carried per unit by a wave is called its power and power per unit area held perpendicular a sound wave travelling along positive x-axis described by the equation.

We define the intensity  $I$  of a wave, or the power per unit area, to be the rate at which the energy being transported by the wave transfers through a unit area  $A$  perpendicular to the direction of travel of the wave :

$$I = \frac{P}{A}$$

In the present case, therefore, the intensity is

$$I = \frac{P}{A} = \frac{1}{2} \rho A v (\omega s_{\max})^2$$

Thus, we see that the intensity of a periodic sound wave is proportional to the square of the displacement amplitude and to the square of the angular frequency (as in the case of a periodic string wave). this can also be written in terms of the pressure amplitude  $\Delta P_{\max}$ ; in this case, we use Equation to obtain

$$I = \frac{\Delta P_{\max}^2}{2\rho v}$$

**Ex.** The pressure amplitude in a sound wave from a radio receiver is  $4.0 \times 10^{-3} \text{ N/m}^2$  and the intensity at a point is  $10^{-6} \text{ W/m}^2$ . If by turning the “Volume” knob the pressure amplitude is increased to  $6 \times 10^{-3} \text{ N/m}^2$ , evaluate the intensity.

**Sol.** The intensity is proportional to the square of the pressure amplitude.

Thus, 
$$\frac{I'}{I} = \left( \frac{P'_0}{P_0} \right)^2$$

or 
$$I' = \left( \frac{P'_0}{P_0} \right)^2 I = \left( \frac{6}{4} \right)^2 \times 10^{-6} \text{ W/m}^2 = 2.25 \times 10^{-6} \text{ W/m}^2.$$

## APPEARANCE OF SOUND TO HUMAN EAR

### Pitch and Frequency

Frequency as we have discussed till now is an objective property measured its units of Hz and which can be assigned a unique value. However a person's perception of frequency is subjective. The brain interprets frequency primarily in terms of a subjective quality called Pitch. A pure note of high frequency is interpreted as high-pitched sound and a pure note of low frequency as low-pitched sound.

Pitch of a sound is that sensation by which we differentiate a buffalo voice, a male voice is of low pitch, a male voice has higher pitch and a female voice has (generally) still higher pitch. This sensation primarily depends on the dominant frequency present in the sound. Higher the frequency, higher will be the pitch and vice versa. The dominant frequency of a buffalo voice is smaller than that of a male voice which in turn is smaller than that of a female voice.

### Loudness and Intensity

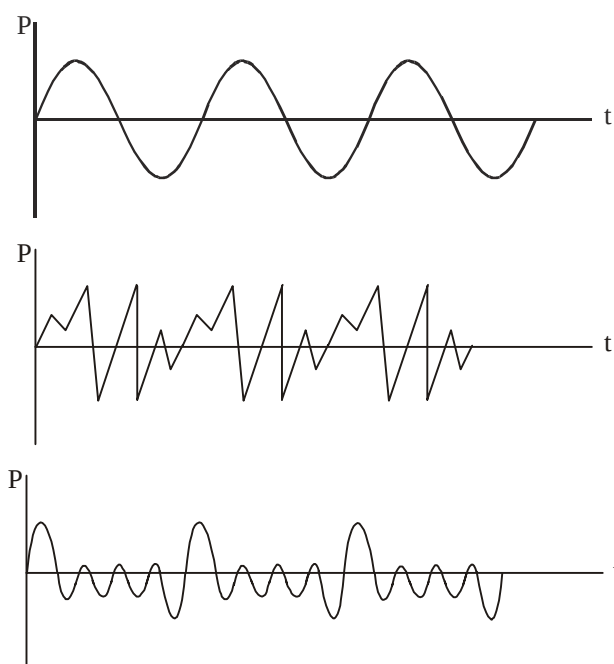
The loudness that we sense is related to the intensity of sound through it is not directly proportional to it. Our perception of loudness is better correlated with the sound level measured in decibels (abbreviated and dB) and defined as follows.

$$\beta = 10 \log_{10} \left( \frac{I}{I_0} \right),$$

where  $I$  is the intensity of the sound and  $I_0$  is a constant reference intensity  $10^{-12} \text{ W/m}^2$ . The reference intensity represents roughly the minimum intensity that is just audible at intermediate frequencies. For  $I = I_0$ , the sound level  $\beta = 0$ . Table shows the approximate sound levels of some of the sounds commonly encountered.

### Quality and Waveform

A sound generated by a source may contain a number of frequency components in it. Different frequency components have different amplitudes and superposition of them results in the actual waveform. The appearance of sound depends on this waveform apart from the dominant frequency and intensity. Figure shows waveforms for a tuning fork, a clarinet and a cornet playing the same note (fundamental frequency = 440 Hz) with equal loudness.





We differentiate between the sound from a table and that from a mridang by saying that they have different quality. A musical sound has certain well-defined frequencies which have considerable amplitude. These frequencies are generally harmonics of a fundamental frequency. Such a sound is particularly pleasant to the ear. On the other hand, a noise has frequencies that do not bear well-defined relationship among themselves.

**Ex.** A bird is singing on a tree. A person approaches the tree and perceives that the intensity has increased by 10 dB. Find the ratio of initial and final separation between the man and the bird.

**Sol.**  $\beta_1 = 10 \log \frac{I_1}{I_0}$

$$\beta_2 = 10 \log \frac{I_2}{I_0} \Rightarrow \beta_2 - \beta_1 = 10 \log \frac{I_2}{I_1} \quad \text{or } 10 = 10 \log_{10} \left( \frac{I_2}{I_1} \right)$$

$$\Rightarrow \frac{I_2}{I_1} = 10^1 = 10$$

for point source  $I \propto \frac{1}{r^2}$

$$\Rightarrow \frac{r_1}{r_2} = \sqrt{\frac{I_2}{I_1}} = \sqrt{10}$$

### INTERFERENCE OF SOUND WAVES :

If  $p_1 = p_{m1} \sin (\omega t - kx_1 + \theta_1)$

and  $p_2 = p_{m2} \sin (\omega t - kx_2 + \theta_1)$

resultant excess pressure at point O is

$$p = p_1 + p_2$$

$$\Rightarrow p = p_0 \sin (\omega t - kx + \theta)$$

where,  $p_0 = \sqrt{p_{m1}^2 + p_{m2}^2 + 2p_{m1}p_{m2} \cos \phi}$ ,  $\phi = |k(x_1 - x_2) + (\theta_1 - \theta_2)| \dots$  (6.1)

(i) for constructive interference

$$\phi = 2n\pi \Rightarrow p_0 = p_{m1} + p_{m2}$$

(ii) for destructive interference

$$\phi = (2n + 1)\pi \Rightarrow p_0 = |p_{m1} - p_{m2}|$$

If  $\phi$  is only due to path difference, then  $\phi = \frac{2\pi}{\lambda} \Delta x$ , and

Condition for constructive interference :  $\Delta x = n\lambda$ ,  $n = 0, \pm 1, \pm 2$

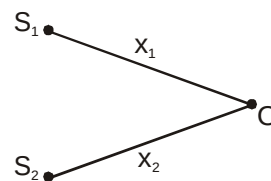
Condition for destructive interference :  $\Delta x = (2n + 1) \frac{\lambda}{2}$ ,  $n = 0, \pm 1, \pm 2$

from equation (6.1)

$$P_0^2 = P_{m1}^2 + P_{m2}^2 + 2P_{m1}P_{m2} \cos \phi$$

Since intensity,  $I \propto (\text{Pressure amplitude})^2$ ,

we have, for resultant intensity,  $I = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos \phi \dots$  (6.2)



$$\text{If } I_1 = I_2 = I_0$$

$$I = 2I_0 (1 + \cos \phi)$$

$$\Rightarrow I = 4I_0 \cos^2 \frac{\phi}{2} \quad \dots (6.3)$$

Hence in this case,

for constructive interference :  $\phi = 0, 2\pi, 4\pi, \dots$  and

and for destructive interference :  $\phi = \pi, 3\pi, \dots$  and

$$I_{\max} = 4I_0$$

$$I_{\min} = 0$$

**Coherence :** Two sources are said to be coherent if the phase difference between them does not change with time. In this case their resultant intensity at any point in space remains constant with time. Two independent sources of sound are generally incoherent in nature, i.e. phase difference between them changes with time and hence the resultant intensity due to them at any point in space changes with time.

- Ex.** Figure shows a tube structure in which a sound signal is sent from one end and is received at the other end. The semicircular part has a radius of 10.0 cm. The frequency of the sound source can be varied from 1 to 10 kHz. Find the frequencies at which the ear perceives maximum intensity. The speed of sound in air = 342 m/s.



- Sol.** The sound wave bifurcates at the junction of the straight and the semicircular parts. The wave through the straight part travels a distance  $s_1 = 2 \times 10$  cm and the wave through the curved part travels a distance  $s_2 = \pi \times 10$  cm = 31.4 cm before they meet again and travel to the receiver. The path difference between the two waves received is, therefore,

$$\Delta s = s_2 - s_1 = 31.4 \text{ cm} - 20 \text{ cm} = 11.4 \text{ cm}$$

The wavelength of either wave is  $\frac{v}{\nu} = \frac{330 \text{ m/s}}{\nu}$ . For constructive interference,  $\Delta p = n\lambda$ , where  $n$  is an integer.

$$\text{or, } \Delta p = n \cdot \frac{v}{\nu} \Rightarrow \nu = \frac{n \cdot v}{\Delta p} \Rightarrow \frac{n \cdot 342}{(0.114)} = 3000 n$$

Thus, the frequencies within the specified range which cause maximum of intensity are

|                 |                 |     |                            |
|-----------------|-----------------|-----|----------------------------|
| $3000 \times 1$ | $3000 \times 2$ | and | $3000 \times 3 \text{ Hz}$ |
|-----------------|-----------------|-----|----------------------------|

### LONGITUDINAL STANDING WAVES :

Two longitudinal waves of same frequency and amplitude travelling in opposite directions interfere to produce a standing wave.

If the two interfering wave are given by

$$p_1 = p_0 \sin (\omega t - kx)$$

$$\text{and } p_2 = p_0 \sin (\omega t + kx + \phi)$$

then the equation of the resultant standing wave would be given by

$$p + p_1 + p_2 = 2p_0 \cos \left(kx + \frac{\phi}{2}\right) \sin \left(\omega t + \frac{\phi}{2}\right)$$

$$\Rightarrow p = p_0' \sin \left(\omega t + \frac{\phi}{2}\right) \quad \dots (8.1)$$

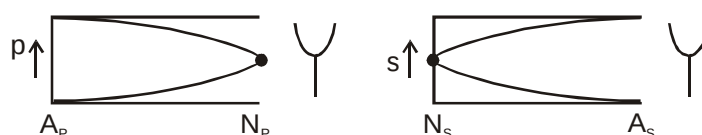
# VIBRATION OF AIR COLUMNS

Standing waves can be set up in air-columns trapped inside cylindrical tubes if frequency of the tuning fork sounding the air column matches one of the natural frequency of air columns. In such a case the sound of the tuning fork becomes markedly louder, and we say there is resonance between the tuning fork and air column. To determine the natural frequency of the air column, notice that there is a displacement node (pressure antinode) at each closed end of the tube as air molecules there are not free to move, and a displacement antinode (pressure-node) at each open end of the air-column.

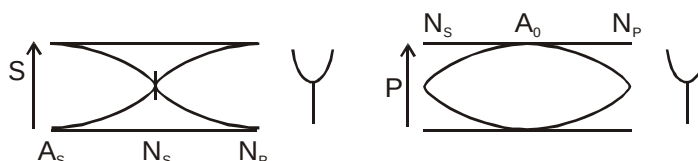
In reality antinodes do not occurs exactly at the open end but a little distance outside. However if diameter of tube is small compared to its length, this end correction can be neglected.

## Closed organ pipe

(In the diagram,  $A_p$  = Pressure antinode,  $A_s$  = displacement antinode,  $N_p$  = pressure node,  $N_s$  = displacement node.)

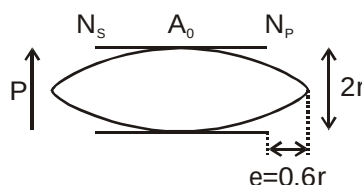


## Open organ pipe :



**End correction :** As mentioned earlier the displacement antinode at an open end of an organ pipe lies slightly outside the open lend. The distance of the antinode from the open end is called end correction and its value is given by

$$e = 0.6 r$$



where  $r$  = radius of the organ pipe.

with end correction, the fundamental frequency of a closed pipe ( $f_c$ ) and an open organ pipe ( $f_0$ ) will be given by

$$f_c = \frac{v}{4(\ell + 0.6r)} \text{ and}$$

$$f_0 = \frac{v}{2(\ell + 1.2r)} \quad \dots\dots\dots (9.5)$$

**Ex.** A tuning fork is vibrating at frequency 200 Hz. When another tuning fork is sounded simultaneously, 6 beats per second are heard. When some mass is added to the tuning fork of 200 Hz, beat frequency decreases. Find the frequency of the other tuning fork.

**Sol.**  $|f - 200| = 6$

$\Rightarrow f = 194 \text{ or } 206$

when 1st tuning fork is loaded its frequency decreases and so does beat frequency

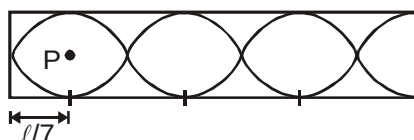
$\Rightarrow 200 > f$

$\Rightarrow f = 194 \text{ Hz}$

**Ex.** A closed organ pipe has length ' $\ell$ '. The air in it is vibrating in 3<sup>rd</sup> overtone with maximum amplitude ' $a$ '. Find the amplitude at a distance of  $\ell/7$  from closed end of the pipe.

**Sol.** The figure shows variation of displacement of particles in a closed organ pipe for 3<sup>rd</sup> overtone.

For third overtone  $\ell = \frac{7\lambda}{4}$  or  $\lambda = \frac{4\ell}{7}$  or  $\frac{\lambda}{4} = \frac{\ell}{7}$



Hence the amplitude at P at a distance  $\frac{\ell}{7}$  from closed end is ' $a$ ' because there is an antinode at that point.

## BEATS

When two sound waves of same amplitude and different frequency superimpose, then intensity at any point in space varies periodically with time. This effect is called beats.

If the equation of the two interfering sound waves emitted by  $s_1$  and  $s_2$  at point O are,

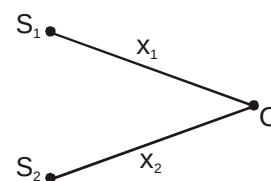
$$p_1 = p_0 \sin(2\pi f_1 t - kx_1 + \theta_1)$$

$$p_2 = p_0 \sin(2\pi f_2 t - kx_2 + \theta_2)$$

By principle of superposition

$$p = p_1 + p_2$$

$$= 2p_0 \cos \left\{ \pi(f_1 - f_2)t + \frac{\theta_1 + \theta_2}{2} \right\} \sin \left\{ \pi(f_1 + f_2)t + \frac{\theta_1 + \theta_2}{2} \right\}$$



i.e., the resultant sound at point O has frequency  $\left( \frac{f_1 + f_2}{2} \right)$  while pressure amplitude  $p'_0(t)$  varies with time as

$$p_0(t) = 2p_0 \cos \left\{ \pi(f_1 - f_2)t + \frac{\phi_1 + \phi_2}{2} \right\}$$

Hence pressure amplitude at point O varies with time with a frequency of  $\left( \frac{f_1 - f_2}{2} \right)$ .

Hence sound intensity will vary with a frequency  $f_1 - f_2$ .

This frequency is called beat frequency ( $f_B$ ) and the time interval between two successive intensity maxima (or minima) is called beat time period ( $T_B$ )

$$f_B = f_1 - f_2$$

$$T_B = \frac{1}{f_1 - f_2}$$

### IMPORTANT POINTS :

- (i) The frequency  $|f_1 - f_2|$  should be less than 16 Hz, for it to be audible.
- (ii) Beat phenomenon can be used for determining an unknown frequency by sounding it together with a sound of known frequency.

**Ex.** Two strings X and Y of a sitar produces a beat of frequency 4Hz. When the tension of string Y is slightly increased, the beat frequency is found to be 2Hz. If the frequency of X is 300 Hz, then the original frequency of Y was.

**Ans.** 296 Hz

### DOPPLER EFFECT

We can express the general relationship for the observed frequency when a source is moving and an observer is at rest as equation , with the same sign convention applied to  $v_s$  as was applied to  $v_0$  : a positive value is substituted for  $v_s$  when the source moves toward the observer and a negative value is substituted when the source moves away from the observer.

Finally, we find the following general relationship for the observed frequency :

$$f' = \left( \frac{v + v_0}{v - v_s} \right) f$$

In this expression, the signs for the values substituted for  $v_0$  and  $v_s$  depend on the direction of the velocity. A positive value is used for motion of the observer or the source toward the other, and a negative sign for motion of one away from the other.

A convenient rule concerning signs for you to remember when working with all Doppler effect problems is as follows :

The word toward is associated with an increase in observed frequency. The words away from are associated with a decrease in observed frequency.

**Ex.** A submarine (sub A) travels through water at a speed of 8.00 m/s, emitting a sonar wave at a frequency of 1400 Hz. The speed of sound in the water is 1533 m/s. A second submarine (sub B) is located such that both submarines are traveling directly toward one another. The second submarine is moving at 9.00 m/s.

- (A) What frequency is detected by an observer riding on sub B as the subs approach each other?
- (B) The subs barely miss each other and pass. What frequency is detected by an observer riding on sub B as the subs recede from each other?

**Sol.** (A) We use equation to find the Doppler - shifted frequency. As the two submarines approach each other, the observer in sub B hears the frequency

$$f' = \left( \frac{v + v_0}{v - v_s} \right) f = \left( \frac{1533 \text{ m/s} + (+9.00 \text{ m/s})}{1533 \text{ m/s} - (+8.00 \text{ m/s})} \right) (1400 \text{ Hz}) = 1416 \text{ Hz}$$

(B) As the two submarines recede from each other, the observer in sub B hears the frequency

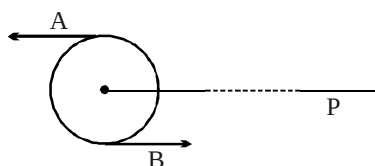
$$f' = \left( \frac{v + v_0}{v - v_s} \right) f$$

$$= \left( \frac{1533 \text{ m/s} + (-9.00 \text{ m/s})}{1533 \text{ m/s} - (-8.00 \text{ m/s})} \right) (1400 \text{ Hz}) = 1385 \text{ Hz}$$

**Ex.** A whistle of frequency 540 Hz is moving in a circle of radius 2m at a constant angular speed of 15rad/s. What are the lowest and height frequencies heard by a listener standing at rest, a long distance away from the centre of the circle ? (velocity of sound in air is 330 m/s ft/sec.)

**Sol.** The whistle is moving along a circular path with constant angular velocity  $\omega$ . The linear velocity of the whistle is given by

$$v_s = \omega R$$



where,  $R$  is radius of the circle

At points A and B, the velocity  $v_s$  of whistle is parallel to line OP, i.e. with respect to observer at P, whistle has maximum velocity  $v_s$  away from P at point A, and towards P at point B. (Since distance OP is large compared to radius  $R$ , whistle may be assumed to be moving along line OP) Observer, therefore, listens maximum frequency when source is at B moving towards observer.

$$f_{\max} = f \frac{v}{v - v_s} = 540 \times \frac{330}{330 - 30} = 540 \times \frac{330}{300} = 594 \text{ Hz}$$

where,  $v$  is speed of sound in air. Similarly, observer listens minimum frequency when source is at A, moving away from observer.

$$f_{\min} = \frac{f v}{v + v_s} = 540 \times \frac{330}{360} = 495 \text{ Hz}$$

### 1. Vibrating air columns :

(i) In a pipe of length  $L$  closed at one end, the fundamental note has a frequency  $f_1 = \frac{v}{4L}$ , where  $v$  is the velocity of sound in air.

(ii) The first overtone  $f_2 = \frac{v}{L} = 2f_1$

### 2. Propagation of sound in solids :

(i) The velocity of propagation of a longitudinal wave in a rod of Young's modulus  $Y$  and

density  $\rho$  is given by  $v = \sqrt{\frac{Y}{\rho}}$

(ii) In a sonometer wire of length  $L$  and mass per unit length  $m$  under tension  $T$  vibrating in  $n$

loops  $f_n = \frac{n}{2L} \sqrt{\frac{T}{m}}$

(iii) Propagation of sound in gases

Laplace formula  $v = \sqrt{\frac{\gamma P}{\rho}}$  where  $\gamma$  is the ratio of specific heats,  $P$  is the pressure and  $\rho$  is the

density.  $\frac{v_t}{v_0} = \sqrt{\frac{T}{T_0}} = \sqrt{\frac{273+t}{273}}$

### 3. Doppler Effect :

(i) When a source of sound moves with a velocity  $v_s$  in a certain direction, the wavelength decreases in front of the source and increases behind the source.

$$\lambda' \text{ (in front)} = \frac{v - v_s}{f_s}; f' = \frac{v}{\lambda'} = \frac{v}{v - v_s} f_s$$

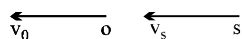
$$\lambda'' \text{ (behind)} = \frac{v + v_s}{f_s}; f'' = \frac{v}{\lambda''} = \frac{v}{v + v_s} f_s$$

Here  $v$  is the velocity of sound in air.

(ii) The apparent frequency =  $\frac{v - v_0}{v} f_s$

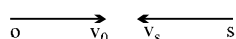
(a) When the source is moving towards the observer and the observer is moving away from the source, the apparent frequency

$$f_a = \frac{v - v_0}{v - v_s} f_s$$



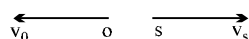
(b) When the source and the observer are moving towards each other.

$$f_a = \frac{v + v_0}{v - v_s} f_s$$



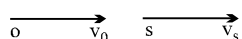
(c) When the source and observer are moving away from each other,

$$f_a = \frac{v - v_0}{v + v_s} f_s$$



(d) When the source is moving away from the observer and the observer is moving towards the source

$$f_a = \frac{v + v_0}{v + v_s} f_s$$



Here all velocities are relation to the medium.

**4. Loudness of sound :**

The loudness level  $B$  of sound is expressed in decibels,  $B = 10 \log \frac{I}{I_0}$  where  $I$  is the intensity,  $I_0$  is a reference intensity.

**5. Beats :**

When two tuning forks of close but different frequencies  $f_1$  and  $f_2$  are vibrating simultaneously at nearby places, a listener observes a fluctuation in the intensity of sound, called beats. The number of beats heard per second is  $f_1 - f_2$ .

**WORKED OUT EXAMPLES**

**Ex.1** It is found that an increase in pressure of 100 kPa causes a certain volume of water to decrease by  $5 \times 10^{-3}$  percent of its original volume. Then the speed of sound in the water is about (density of water  $10^3 \text{ kg/m}^3$ )

- (A) 330 m/s (B) 1414 m/s (C) 1732 m/s (D) 2500 m/s

**Ans. (B)**

**Sol.** Bulk modulus  $\beta = -\frac{\Delta P}{\frac{\Delta V}{V}} = \frac{100 \times 10^3}{\frac{5 \times 10^{-3}}{100}} = 2 \times 10^9$

$$\text{speed } v = \sqrt{\frac{\beta}{\rho}} = \sqrt{\frac{2 \times 10^9}{10^3}} \approx 1414 \text{ m/s}$$

**Ex.2** A sound wave is travelling in a uniform pipe with gas of adiabatic exponent  $\gamma$ . If  $u$  is the particle velocity at any point in medium and  $c$  is the wave velocity, then relative change in pressure  $\frac{dP}{P}$  while wave passes through this point is :-

- (A)  $\frac{u}{\gamma c}$  (B)  $\gamma \sqrt{\frac{u}{c}}$  (C)  $\gamma \frac{u}{c}$  (D)  $\frac{u^2}{\gamma c^2}$

**Ans. (C)**

**Sol.**  $\Delta P = -B \left( \frac{dv}{v} \right)$

$$\Delta P = -B \left( \frac{\delta y}{\delta x} \right)$$

$$\Delta P = -\gamma P \frac{\delta y}{\delta x}$$

$$\frac{\Delta P}{P} = \gamma \left( \frac{u}{c} \right)$$



**Ex.3** A point source emits sound equally in all directions in a non-absorbing medium. Two points P and Q are at a distance of 9 meters and 25 meters respectively from the source. The ratio of the amplitudes of the waves at P and Q is :-

- (A) 5 : 3 (B) 3 : 5 (C) 25 : 9 (D) 625 : 81

**Ans.** (C)

**Sol.**  $I \propto A^2$ ,  $I \propto \frac{1}{r^2}$ ,  $\frac{A_1}{A_2} = \frac{r_2}{r_1}$

**Ex.4** In a resonance tube experiment, an 80 cm air column is in resonance with a tuning fork in first overtone. Which equation can represent correct pressure variation in the air column ( $x = 0$  is the top point of the tube, neglect end correction, speed of sound = 320 m/sec) :-

- (A)  $A \sin \frac{15\pi}{8} x \cos 600\pi t$  (B)  $A \cos \frac{15\pi}{8} x \sin 600 \pi t$   
(C)  $A \cos \frac{15\pi}{8} x \sin 300 \pi t$  (D)  $A \sin \frac{15\pi}{8} x \sin 300 \pi t$

**Ans.** (A)

**Sol.**  $\frac{\omega}{k} = 320$

**Ex.5** The displacement of the medium in a sound wave is given by the equation ;  $y_1 = A \cos (ax + bt)$  where A, a & b are positive constants. The wave is reflected by an obstacle situated at  $x = 0$ . The intensity of the reflected wave is 0.64 times that of the incident wave.

- (a) what are the wavelength & frequency of the incident wave.  
(b) write the equation for the reflected wave.  
(c) in the resultant wave formed after reflection, find the maximum & minimum values of the particle speeds in the medium.

**Ans.** (a)  $2\pi/a$ ,  $b/2\pi$ , (b)  $y_2 = \pm 0.8 A \cos (ax - bt)$ , (c) max. =  $1.8 b A$ , min. = 0,

**Sol.** (a)  $\omega = b$  &  $k = a$

$$f = \frac{2\pi}{\omega} \text{ \& \; } \lambda = \frac{2\pi}{k}$$

(b)  $I \propto A^2$

So  $A_r = 0.8 A$

(c)  $(A_{\text{net}})_{\text{max}} = A + 0.8 A = 1.8 A$

**Ex.6** An observer moves towards a stationary source of sound, with a velocity one-fifth of the velocity of sound. what is the percentage increase in the apparent frequency ?[AIEEE - 2005]

- (1) zero (2) 0.5% (3) 5% (4) 20%

**Ans.** (4)

**Sol.**  $\frac{f_{\text{app}}}{f} = \left( \frac{v + \frac{v}{5}}{v} \right) = \frac{6}{5}$

**Ex.7** The equation of a longitudinal standing wave due to superposition of the progressive waves produced by two sources of sound is  $s = -20 \sin 10 \pi x \sin 100 \pi t$  where  $s$  is the displacement from mean position measured in mm ;  $x$  is in meters and  $t$  in seconds. The specific gravity of the medium is  $10^{-3}$ . Find

- (a) wavelength, frequency and velocity of the progressive waves.  
 (b) Bulk modulus of the medium and the pressure amplitude of the progressive waves.  
 (c) minimum distance between pressure antinode and the displacement antinode.

**Ans.** (a)  $1/5$  m, 50 Hz, 10 m/s; (b) 100 Pa,  $10\pi$  Pa, (c)  $1/20$  m

**Sol.**  $k = 10\pi$

$$\omega = 100 \pi$$

$$\therefore \lambda = \frac{1}{5} \text{ \& } f = 50 \text{ Hz}$$

$$v = \lambda f = 10 \text{ m/s}$$

$$B = \rho v^2$$

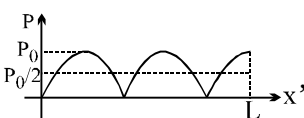
$$P_0 = BKs$$

$$\text{Minimum distance between pressure antinode and displacement antinode} = \frac{\lambda}{4}$$

**Ex.8** The air column in a pipe closed at one end and open to atmosphere at the other end is made to vibrate in its fifth harmonic by a tuning fork of frequency 470 Hz. The length of air column is

$\frac{15}{16}$  m. Neglect end correction. Let  $p_0$  denote the maximum gauge pressure at the closed end

- (a) Find the speed of sound in air.  
 (b) Draw the graph of pressure amplitude vs distance from the open end of the tube.  
 (c) Find the points where the maximum gauge pressure is  $\frac{p_0}{2}$ .

**Ans.** (a) 352.5 m/s, (b) , (c)  $\frac{l}{15}, \frac{5l}{15}, \frac{7l}{15}, \frac{11l}{15}, \frac{13l}{15}$

**Sol.**  $\frac{5\lambda}{4} = \frac{15}{16}$

$$\lambda = 0.75 \text{ m}$$

$$f = 470$$

$$v = \lambda f = 352.5 \text{ m/s}$$

$$P = P_0 \sin\left(\frac{2\pi}{\lambda} x\right)$$

$$\text{where } \lambda = \frac{4\ell}{5}$$

$$P = P_0 \sin\left(\frac{5\pi}{2\ell} x\right)$$

**Ex.9** A metal rod of length  $l = 100$  cm is clamped at two points. Distance of each clamp from nearer end is  $a = 30$  cm. If density and Young's modulus of elasticity of rod material are  $\rho = 9000 \text{ kg m}^{-3}$  and  $Y = 144 \text{ GPa}$  respectively, calculate minimum and next higher frequency of natural longitudinal oscillations of the rod.

**Ans.** 10kHz, 30kHz

**Sol.**  $v = \sqrt{\frac{Y}{\rho}} = 4 \times 10^3 \text{ m/s}$

$$\lambda = 0.4 \text{ m}$$

$$f_0 = \frac{v}{\lambda} = 10 \text{ kHz}$$

**Ex.10** When two tuning forks (fork 1 and fork 2) are sounded simultaneously, 4 beats per second are heard. Now, some tape is attached on the prong of the fork 2. When the tuning forks are sounded again, 6 beats per second are heard. If the frequency of fork 1 is 200 Hz, then what was the original frequency of fork 2 ?

[AIEEE - 2005]

- (1) 200 Hz                      (2) 202 Hz                      (3) 196 Hz                      (4) 204 Hz

**Ans.** (3)

**Sol.**  $f_1 - f_2 = 4$  or  $f_2 - f_1 = 4$   
But according to question  
 $f_1 - f_2 = 4$   
So  $f_2 = 196$

**Ex.11** A whistling train approaches a junction. An observer standing at junction observes the frequency to be 2.2 KHz and 1.8 KHz of the approaching and the receding train. Find the speed of the train (speed of sound = 300 m/s).

[JEE 2005]

**Ans.**  $V_s = 30 \text{ m/s}$

**Sol.**  $\left( \frac{v}{v - v_s} \right) f_0 = 2.2 \times 10^3$

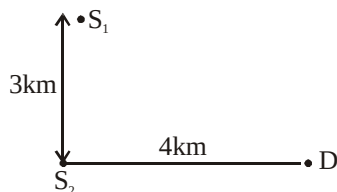
$$\left( \frac{v}{v + v_s} \right) f_0 = 1.8 \times 10^3$$

Dividing both we get

$$\frac{v + v_s}{v - v_s} = \frac{11}{9}$$

$$v_s = 30$$

**Ex.12** Two point sound source  $S_1$  and  $S_2$  are both have the same power and send out sound waves in the same phase. The wavelength of both the waves is  $\frac{48}{5}$  m. The intensity due to  $S_2$  alone at D is  $25 \text{ W/m}^2$ . The resultant intensity at D is :



- (A)  $59 \text{ W/m}^2$       (B)  $61 \text{ W/m}^2$       (C)  $65 \text{ W/m}^2$       (D) None of these

**Ans. (B)**

**Sol.**  $I \propto \frac{1}{r^2}$

Intensity due to  $S_1$  alone at D =  $16 \text{ W/m}^2$

$$\text{Phase difference} = \frac{2\pi}{\left(\frac{48}{5}\right)} \times 1000 = \frac{625}{3} \pi$$

$$\begin{aligned} I_{\text{res}} &= I_1 + I_2 + 2\sqrt{I_1}\sqrt{I_2} \cos\phi \\ &= 25 + 16 + 2(5)(4) \left(\frac{1}{2}\right) \end{aligned}$$

**Ex.13** Spherical sound waves are emitted uniformly in all directions from a point source. The variation in sound level SL as a function of distance  $r(> 0)$  from the source can be written as :- (where a and b are positive constant)

- (A)  $SL = -b \log r^a$       (B)  $SL = a - b \log r$       (C)  $SL = a - b(\log r)^2$       (D)  $SL = a - \frac{b}{r^2}$

**Ans. (B)**

**Sol.**  $I \propto \frac{1}{r^2}$

$$SL = 20 \log_{10} \frac{I}{I_0}$$

## EXERCISE (S-1)

### Sound basics

1. Find the intensity of sound wave whose frequency is 250 Hz. The displacement amplitude of particles of the medium at this position is  $1 \times 10^{-8}$  m. The density of the medium is  $1 \text{ kg/m}^3$ , bulk modulus of elasticity of the medium is  $400 \text{ N/m}^2$ .

SW0001

2. In a mixture of gases, the average number of degrees of freedom per molecule is 6. The rms speed of the molecules of the gas is  $c$ . Find the velocity of sound in the gas.

SW0002

3. The loudness level at a distance  $R$  from a long linear source of sound is found to be 40dB. At this point, the amplitude of oscillations of air molecules is 0.01 cm. Then find the loudness level & amplitude at a point located at a distance '10R' from the source.

SW0003

### Superposition of sound

4. The first overtone of a pipe closed at one end resonates with the third harmonic of a string fixed at its ends. The ratio of the speed of sound to the speed of transverse wave travelling on the string is 2 : 1. Find the ratio of the length of pipe to the length of string.

SW0004

5. In a resonance-column experiment, a long tube, open at the top, is clamped vertically. By a separate device, water level inside the tube can be moved up or down. The section of the tube from the open end to the water level act as a closed organ pipe. A vibrating tuning fork is held above the open end, first and the second resonances occur when the water level is 24.1 cm and 74.1 cm respectively below the open end. Find the diameter of the tube.

SW0005

6. A tuning fork of frequency 480 Hz resonates with a tube closed at one end of length, 16 cm and diameter 5 cm in fundamental mode. Calculate velocity of sound in air.

SW0006

7. An open organ pipe filled with air has a fundamental frequency 500Hz. The first harmonic of another organ pipe closed at one end and filled with carbon dioxide has the same frequency as that of the first harmonic of the open organ pipe. Calculate the length of each pipe. Assume that the velocity of sound in air and in carbondioxide to be 330 and 264 m/s respectively.

SW0007

8. A steel rod having a length of 1 m is fastened at its middle. Assuming young's modulus to be  $2 \times 10^{11} \text{ Pa}$ , and density to be  $8 \text{ gm/cm}^3$  find the fundamental frequency of the longitudinal vibration and frequency of first overtone.

SW0008

9. Two narrow cylindrical pipes A and B have the same length. Pipe A is open at both ends and is filled with a monoatomic gas of molar mass  $M_A$ . Pipe B is open at one end and closed at the other end, and is filled with a diatomic gas of molar mass  $M_B$ . Both gases are at the same temperature.

[JEE 2002]

SW0009

(a) If the frequency of the second harmonic of the fundamental mode in pipe A is equal to the frequency of the third harmonic of the fundamental mode in pipe B, determine the value of  $M_A/M_B$ .

(b) Now the open end of pipe B is also closed (so that the pipe is closed at both ends). Find the ratio of the fundamental frequency in pipe A to that in pipe B.

10. A tube of a certain diameter and of length 48 cm is open at both ends. Its fundamental frequency of resonance is found to be 320 Hz. The velocity of sound in air is 320 m/sec. Estimate the diameter of the tube.

[IIT-1980]

SW0010

**Beats**

11. A stretched uniform wire of a sonometer between two fixed knife edges, when vibrates in its second harmonic gives 1 beat per second with a vibrating tuning fork of frequency 200 Hz. Find the percentage change in the tension of the wire to be in unison with the tuning fork.

SW0011

12. A, B and C are three tuning forks. Frequency of A is 350 Hz. Beats produced by A and B are 5 per second and by B and C are 4 per second. When a wax is put on A beat frequency between A and B is 2 Hz and between A and C is 6 Hz. Then, find the frequency of B and C respectively.

SW0012

13. A source of sound of frequency 256 Hz is moving rapidly towards wall with a velocity of 5 m/sec. How many beats per second will be heard if sound travels at a speed of 330 m/sec?

[IIT-1981]

SW0013

14. Two tuning forks with natural frequencies of 340 Hz each move relative to a stationary observer. One fork moves away from the observer, while the other moves towards him at the same speed. The observer hears beats of frequency 3 Hz. Find the speed of the tuning fork (assume  $v_{\text{sound}} = 340$  m/s)

[IIT-1986]

SW0014

**Doppler effect**

15. Two tuning forks A and B lying on opposite sides of observer 'O' and of natural frequency 85 Hz move with velocity 10 m/s relative to stationary observer O. Fork A moves away from the observer while the fork B moves towards him. A wind with a speed 10 m/s is blowing in the direction of motion of fork A. Find the beat frequency measured by the observer in Hz. [Take speed of sound in air as 340 m/s]

SW0015

16. A car is moving towards a huge wall with a speed  $= c/10$ , where  $c$  = speed of sound in still air. A wind is also blowing parallel to the velocity of the car in the same direction and with the same speed. If the car sounds a horn of frequency  $f$ , then what is the frequency of the reflected sound of the horn heard by driver of the car?

SW0016

17. A plane sound wave of frequency  $f_0$  and wavelength  $\lambda_0$  travels horizontally toward the right. It strikes and is reflected from a large, rigid, vertical plane surface, perpendicular to the direction of propagation of the wave and moving towards the left with a speed  $v$ .
- How many positive wave crests strike the surface in a time interval  $t$ ?
  - At the end of this time interval, how far to the left of the surface is the wave that was reflected at the beginning of the time interval?
  - What is the wavelength of the reflected waves, in terms of  $\lambda_0$ ?
  - What is the frequency, in terms of  $f_0$ ?
  - A listener is at rest at the left of the moving surface. Describe the sensation of sound that he hears as a result of the combined effect of the incident and reflected wave trains.

SW0017

18. A bus is moving towards a huge wall with a velocity of  $5 \text{ ms}^{-1}$ . The driver sounds a horn of frequency 200 Hz. The frequency of the beats heard by a passenger of the bus will be..... Hz (speed of sound in air =  $342 \text{ ms}^{-1}$ )

[IIT-1994]

SW0018

## EXERCISE (S-2)

1. A boat is travelling in a river with a speed of 10 m/s along the stream flowing with a speed 2 m/s. From this boat, a sound transmitter is lowered into the river through a rigid support. The wavelength of the sound emitted from the transmitter inside the water is 14.45 mm. Assume that attenuation of sound in water and air is negligible. [JEE 2001]

- (a) What will be the frequency detected by a receiver kept inside the river downstream ?  
 (b) The transmitter and the receiver are now pulled up into air. The air is blowing with a speed 5 m/sec in the direction opposite the river stream. Determine the frequency of the sound detected by the receiver.

(Temperature of the air and water = 20°C ; Density of river water =  $10^3 \text{ Kg/m}^3$ ; Bulk modulus of the water =  $2.088 \times 10^9 \text{ Pa}$ ; Gas constant  $R = 8.31 \text{ J/mol-K}$ ; Mean molecular mass of air =  $28.8 \times 10^{-3} \text{ kg/mol}$ ;  $C_p/C_v$  for air = 1.4)

SW0019

**Note:** Boat velocity is with respect to ground & receiver is stationary w.r.t. ground

2. The air column in a pipe closed at one end is made to vibrate in its second overtone by a tuning fork of frequency 440 Hz. The speed of sound in air is  $330 \text{ ms}^{-1}$ . End corrections may be neglected. Let  $P_0$  denote the mean pressure at any point in the pipe &  $\Delta P_0$  the maximum amplitude of pressure variation. [JEE '98]

- (i) Find the length  $L$  of the air column.  
 (ii) What is the amplitude of pressure variation at the middle of the column ?  
 (iii) What are the maximum & minimum pressures at the open end of the pipe .  
 (iv) What are the maximum & minimum pressures at the closed end of the pipe?

SW0020

3. A train of length  $l$  is moving with a constant speed  $v$  along a circular track of radius  $R$ , The engine of the train emits a whistle of frequency  $f$ . Find the frequency heard by a guard at the rear end of the train. Make suitable assumption.

SW0021

4. A string 25 cm long and having a mass of 2.5 gm is under tension. A pipe closed at one end is 40 cm long. When the string is set vibrating in its first overtone and the air in the pipe in its fundamental frequency, 8 beats per second are heard. It is observed that decreasing the tension in the string decreases beat frequency. If the speed of sound in air is 320 m/s, find the tension in the string.

[IIT-1982]

SW0022



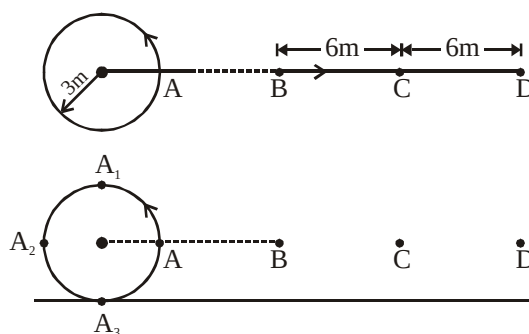
5. A train approaching a hill at speed of 40 km/hr sound a whistle of frequency 580 Hz when it is at a distance of 1 km from a hill. A wind a speed of 40 km/hr blowing the direction of motion of the train. Find
- the frequency of the whistle as heard by an observer on the hill.
  - the distance from the hill at which the echo from the hill is heard by the driver and its frequency.
- (Velocity of sound in air = 1,200 km/hr)

[IIT-1988]

SW0023

6. A source of sound is moving along a circular orbit of radius 3 meters with an angular velocity of 10 rad/s. A sound detector located far away from the source is executing linear simple harmonic motion along the line BD with an amplitude BC = CD = 6 metres. The frequency of oscillation of the detector is  $\frac{5}{\pi}$  per second. The source is at the point A when the detector is at the point B. If the source is at the point A when the detector is at the point B. If the source emits a continuous sound wave of frequency 340 Hz, find the maximum and the minimum frequencies recorded by the detector.

[IIT-1990]



SW0024

7. Two radio stations broadcast their programmes at the same amplitude A and at slight different frequencies  $\omega_1$  and  $\omega_2$  respectively, where  $\omega_1 - \omega_2 = 10^3$  Hz A detector receives the signals from the two stations simultaneously. It can only detect signals of intensity  $\geq 2 A^2$ .
- Find the time interval between successive maxima of the intensity of the signal received by the detector.
  - Find the time for which the detector remains idle in each cycle of the intensity of the signal.

[IIT-1993]

SW0025

8. A sonometer wire under tension of 64 Newtons vibrating in its fundamental mode is in resonance with a vibrating tuning fork. The vibrating portion of the sonometer wire has a length of 10 cm and a mass of 1 gm. The vibrating tuning fork is now moved away of 1 gm. The vibrating wire with a constant speed and an observer standing near the sonometer hears oen beat per second. Calculate the speed with which the tuning fork is moved if the speed of sound in air is 300 m/s.

[IIT-1983]

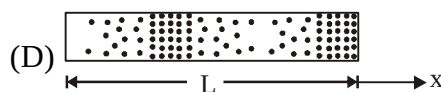
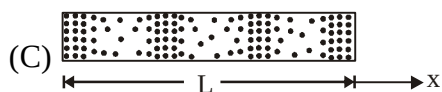
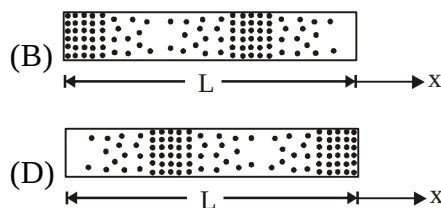
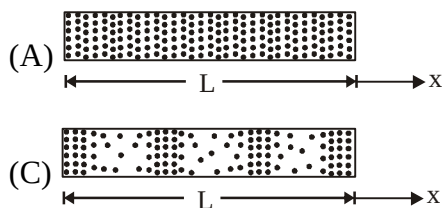
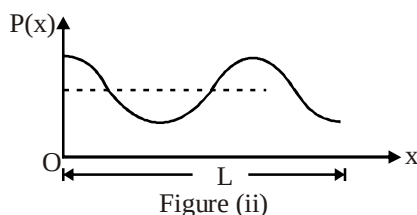
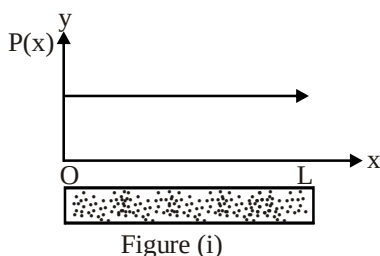
SW0026

9. The displacement of the medium in a sound wave is given by the equation  $y_1 = A \cos(ax + bt)$  where  $A$ ,  $a$  and  $b$  are positive constants. The wave is reflected by an obstacle situated at  $x = 0$ . The intensity of the reflected wave is 0.64 times that of the incident wave. [IIT-1991]
- (a) What are the wavelength and frequency of incident wave?  
(b) Write the equation for the reflected wave.  
(c) In the resultant wave formed after reflection, find the maximum and minimum values of the particle speeds in the medium.  
(d) Express the resultant wave as a superpositions of standing wave and a travelling wave. What are the positions of the antinodes of the standing wave? What is the directions of propagation of travelling wave?
- SW0027
10. The air column in a pipe closed at one end is made to vibrate in its second overtone by a tuning fork of frequency 440 Hz. The speed of sound in air is  $330 \text{ ms}^{-1}$ . End corrections may be neglected. Let  $P_0$  denote the mean pressure at any point in the pipe, and  $\Delta P_0$  the maximum amplitude of pressure variation.
- (a) Find the length  $L$  of the air column.  
(b) What is the amplitude of pressure variation at the middle of the column?  
(c) What are the maximum and minimum pressure at the open end of the pipe?  
(d) What are the maximum and minimum pressure at the closed end of the pipe? [IIT-1998]
- SW0028
11. A 3.6m long vertical pipe resonates with a source of frequency 212.5 Hz when water level is at certain height in the pipe. Find the height of water level (from the bottom of the pipe) at which resonance occurs. Neglect end correction. Now, the pipe is filled to a height  $H$  ( $\approx 3.6 \text{ m}$ ). A small hole is drilled very close to its bottom and water is allowed to leak. Obtain an expression for the rate of fall of water level in the pipe as a function of  $H$ . If the radii of the pipe and the hole are  $2 \times 10^{-2} \text{ m}$  and  $1 \times 10^{-3} \text{ m}$  respectively calculate the time interval between the occurrence of first two resonances. Speed of sound in air is  $340 \text{ m/s}$  and  $g = 10 \text{ m/s}^2$ . [IIT-2000]
- SW0029

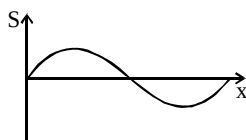
## EXERCISE (O-1)

### Sound basics

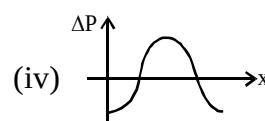
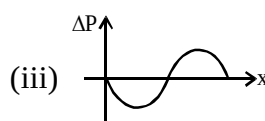
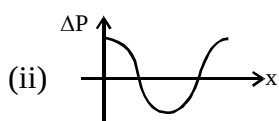
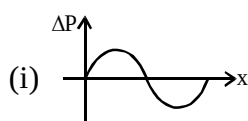
- A sound wave has a wavelength of 3.0 m. The distance from a compression center to the adjacent rarefaction center is :-  
(A) 0.75 m (B) 1.5 m (C) 3.0 m (D) need to know wave speed  
**SW0030**
- You are listening to an "A" note played on a violin string. Let the subscript "s" refer to the violin string and "a" refer to the air. Then :-  
(A)  $f_s = f_a$  but  $\lambda_s \neq \lambda_a$  (B)  $f_s = f_a$  and  $\lambda_s = \lambda_a$  (C)  $\lambda_s = \lambda_a$  but  $f_s \neq f_a$  (D)  $\lambda_s \neq \lambda_a$  and  $f_s \neq f_a$   
**SW0031**
- The fig.(i) shows the graphical representation of the air molecules in a tube of air (length = L) at atmospheric pressure on the absolute pressure  $P(x)$  graph. Which one of the following pictures corresponds to the absolute pressure  $P(x)$  graph of fig. (ii).



- If a sound wave is travelling and snap shot at  $t = 0$  is as shown in figure.



Choose snapshot of pressure variation.

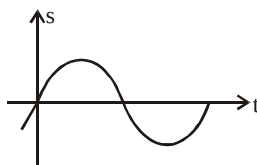


**SW0032**

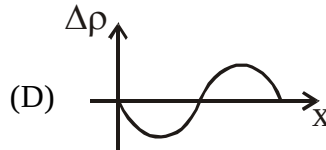
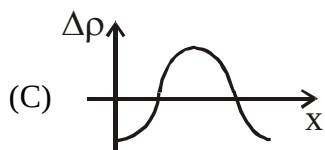
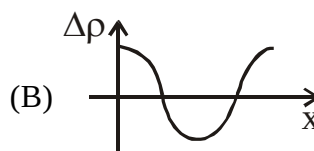
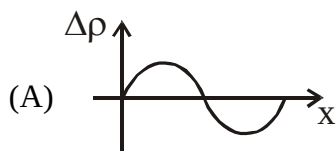
- (A) For wave travelling towards right or left (i) is correct.  
 (B) For wave travelling towards right graph (iv) and for wave travelling towards left graph (iv) is correct.  
 (C) For wave travelling towards right graph (i) and for wave travelling towards left graph (iii) is correct  
 (D) For wave travelling towards right or left (ii) is correct.

SW0033

5. A sound waves is travelling towards right and its s-t graph is as shown for  $x = 0$ .



What will be the variation in density vs  $x$  graph at  $t = T/4$  :-



SW0034

6. A point source of sound is located somewhere along the  $x$ -axis. Experiments show that the same wave front simultaneously reaches listeners at  $x = -8$  m and  $x = +2.0$  m. A third listener is positioned along the positive  $y$ -axis. What is her  $y$ -coordinate (in m) if the same wave front reaches her at the same instant as it does the first two listeners ?  
 (A) 4 (B) 3 (C) 2 (D) 5

SW0035

7. Two monatomic ideal gases 1 and 2 of molecular masses  $m_1$  and  $m_2$  respectively are enclosed in separate container kept at the same temperature. The ratio of the speed of sound in gas 1 to that in gas 2 is given by [JEE 2000 (Scr)]

(A)  $\sqrt{\frac{m_1}{m_2}}$

(B)  $\sqrt{\frac{m_2}{m_1}}$

(C)  $\frac{m_1}{m_2}$

(D)  $\frac{m_2}{m_1}$

SW0036

8. A firecracker exploding on the surface of a lake is heard as two sounds a time interval  $t$  apart by a man on a boat close to water surface. Sound travels with a speed  $u$  in water and a speed  $v$  in air. The distance from the exploding firecracker to the boat is

(A)  $\frac{uvt}{u+v}$

(B)  $\frac{t(u+v)}{uv}$

(C)  $\frac{t(u-v)}{uv}$

(D)  $\frac{uvt}{u-v}$

SW0037

9. The speed of longitudinal wave is 100 times the speed of transverse wave in a taut brass wire. If the Young's modulus of brass is  $1.0 \times 10^{11} \text{ N/m}^2$ , the stress in wire is :-  
 (A)  $1.0 \times 10^7 \text{ N/m}^2$  (B)  $1.0 \times 10^6 \text{ N/m}^2$  (C)  $1.0 \times 10^5 \text{ N/m}^2$  (D)  $1.0 \times 10^8 \text{ N/m}^2$

SW0038

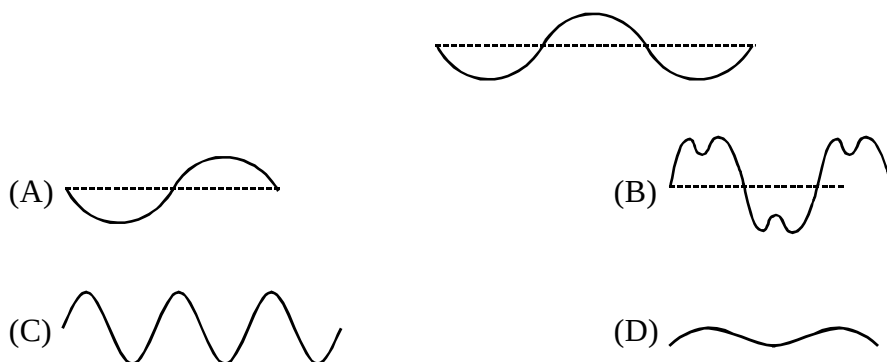
10. The equations of S.H.M. of medium particle due to sound waves propagating in a medium are given by  $s_1 = 2 \sin(200\pi t)$  and  $s_2 = 5 \sin(150\pi t)$ . The ratio of average intensities of sound at these points is:  
 (A) 4 : 25 (B) 9 : 100 (C) 8 : 15 (D) 64 : 225

SW0039

11. A is singing a note and at the same time B is also singing a note with  $1/8$ th the frequency of A. The energies of the two sounds are equal. The displacement amplitude of the note of B is:  
 (A) same as that of A (B) twice that of A  
 (C) four times that of A (D) eight times that of A

SW0040

12. A microphone is connected to an oscilloscope. The diagram shows the trace on the screen when the microphone receives a pure note. Which trace can be obtained when a musical instrument produces a note of the same pitch but of a different quality ?



SW0041

13. Choose **correct** statement?  
 (A) Two different acoustic musical instruments can not have same loudness  
 (B) Two different acoustic musical instruments can not have same pitch  
 (C) Two different acoustic musical instruments can not have same quality  
 (D) Two different acoustic musical instruments can have more than two characteristics same

SW0042

14. A plane transverse wave is propagating in a direction making an angle of  $30^\circ$  with positive x-axis in the x-y plane. Find phase difference between points (0, 0, 0) and (1, 1, 1). Wavelength of the wave is 1m :-

- (A)  $2\pi \text{ rad}$  (B)  $(\sqrt{3} + 1)\pi \text{ rad}$  (C)  $(\sqrt{2} + 1)\pi \text{ rad}$  (D) None

SW0043

15. Which of the following is the equation of a spherical wave :-

- (A)  $S = S_0 \sin (Kx - \omega t)$  (B)  $S = S_0 \cos (Kx - \omega t)$   
 (C)  $S = (S_0/x) \sin (\omega t - Kx)$  (D)  $S = (S_0/x^2) \sin(\omega t - Kx)$

SW0044

16. A note is produced when you blow air across the top of a test tube. Two students were asked about the effect of blowing harder.

**Student-A :** The pitch of sound would increase.

**Student-B :** The intensity of sound would increase

- (A) A is correct, B is wrong (B) B is correct, A is wrong  
 (C) both are correct (D) both are wrong

SW0045

17. A sound absorber attenuates the sound level by 20 dB. The intensity decreases by a factor of-

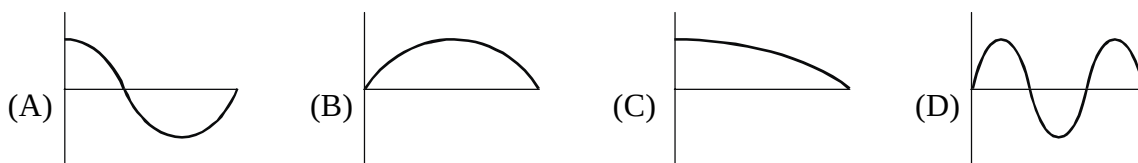
[AIEEE - 2007]

- (A) 1000 (B) 10000 (C) 10 (D) 100

SW0046

### Standing waves

18. Which of the figures, shows the pressure difference from regular atmospheric pressure for an organ pipe of length L closed at one end, corresponds to the 1st overtone for the pipe?



SW0047

19. In an organ pipe whose one end is at  $x = 0$ , the pressure is expressed by

$$p = p_0 \cos \frac{3\pi x}{2} \sin 300\pi t \quad \text{where } x \text{ is in meter and } t \text{ in sec. The organ pipe can be}$$

- (A) closed at one end, open at another with length = 0.5m  
 (B) open at both ends, length = 1m  
 (C) closed at both ends, length = 2m  
 (D) closed at one end, open at another with length =  $\frac{2}{3}$  m

SW0048

20. If  $l_1$  and  $l_2$  are the lengths of air column for the first and second resonance when a tuning fork of frequency  $n$  is sounded on a resonance tube, then the distance of the displacement antinode from the top end of the resonance tube is:

- (A)  $2(l_2 - l_1)$  (B)  $\frac{1}{2}(2l_1 - l_2)$  (C)  $\frac{l_2 - 3l_1}{2}$  (D)  $\frac{l_2 - l_1}{2}$

SW0049

21. A student is experimenting with resonance tube apparatus in Physics lab to find the speed of sound at room temperature. He got resonating lengths of air column as 17 cm and 51 cm, using tuning fork of frequency 512 Hz. Find speed of sound at room temperature and specify, whether the side water reservoir was moved upward or downward to obtain the second resonance (51 cm)?
- (A) 348 m/s, downwards (B) 348 m/s, upwards  
(C) 332 m/s, downwards (D) 332 m/s, upwards

SW0050

### Interference

22. The ratio of maximum to minimum intensity due to superposition of two waves is  $\frac{49}{9}$ . Then the ratio of the intensity of component waves is :-

- (A)  $\frac{25}{4}$  (B)  $\frac{16}{25}$  (C)  $\frac{4}{49}$  (D)  $\frac{9}{49}$

SW0051

23. Two waves of sound having intensities  $I$  and  $4I$  interfere to produce interference pattern. The phase difference between the waves is  $\pi/2$  at point A and  $\pi$  at point B. Then the difference between the resultant intensities at A and B is
- (A)  $2I$  (B)  $4I$  (C)  $5I$  (D)  $7I$

SW0052

24. Three coherent waves of equal frequencies having amplitude  $10 \mu\text{m}$ ,  $4 \mu\text{m}$  and  $7 \mu\text{m}$  respectively, arrive at a given point with successive phase difference of  $\pi/2$ . The amplitude of the resulting wave in  $\mu\text{m}$  is given by
- (A) 5 (B) 6 (C) 3 (D) 4

SW0053

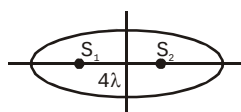
25. The ratio of intensities between two coherent sound sources is  $4 : 1$ . The difference of loudness in dB between maximum and minimum intensities when they interfere in space is:
- (A)  $10 \log 2$  (B)  $20 \log 3$  (C)  $10 \log 3$  (D)  $20 \log 2$

SW0054

26. In Quincke's tube a detector detects minimum intensity. Now one of the tube is displaced by 5 cm. During displacement detector detects maximum intensity 10 times, then finally a minimum intensity (when displacement is complete). The wavelength of sound is:
- (A)  $10/9 \text{ cm}$  (B)  $1 \text{ cm}$  (C)  $1/2 \text{ cm}$  (D)  $5/9 \text{ cm}$

SW0055

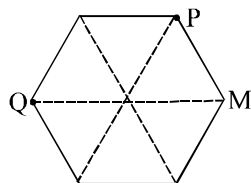
27.  $S_1, S_2$  are two coherent sources of sound located along x-axis separated by  $4\lambda$  where  $\lambda$  is wavelength of sound emitted by them. Number of maxima located on the elliptical boundary around it will be :



- (A) 16 (B) 12 (C) 8 (D) 4

SW0056

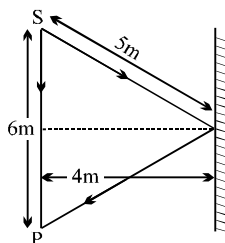
28. Two sound source emitting sound of wavelength 1 m are located at points P and Q as shown in figure. All sides of the polygon are equal and of length 1m. The intensity of sound at M due to both the individual sources is  $I_0$ . What will be the intensity of sound at point M when both the sources are on.



- (A)  $4I_0$  (B)  $2I_0$  (C)  $I_0$  (D)  $(1/2)I_0$

SW0057

29. A person standing at a distance of 6 m from a source of sound receives sound wave in two ways, one directly from the source and other after reflection from a rigid boundary as shown in the figure. The maximum wavelength for which, the person will receive maximum sound intensity, is



- (A) 4 m (B)  $\frac{16}{3}$  m (C) 2 m (D)  $\frac{8}{3}$  m

SW0058

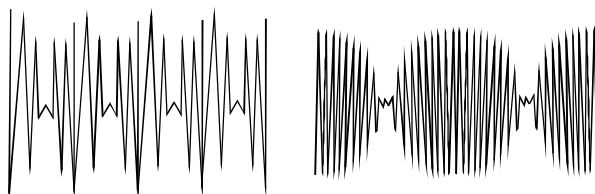
### Beats

30. Beats are heard when the A strings of two violins are played. The beat frequency decreases as the tension in the A string of violin 1 is slowly increased. Which of the following statement is correct ?
- (A) the fundamental frequency of the A string in violin 1 is less than that for violin 2  
 (B) the fundamental frequency of the A string in violin 1 is greater than that for violin 2  
 (C) the fundamental frequency of the A string in violin 1 may be greater or less than that for violin 2 depending on the linear mass densities of the two strings.  
 (D) None of these

SW0059



31. Two waves with similar frequencies are added. The resulting waveform oscillates with the average frequency and with an oscillating amplitude that changes with a frequency equal to the difference between the original frequencies. These oscillations in the amplitude are known as beats. The traces show the resulting waveforms that occur when two different pairs of waves are added. Graph is for the same time interval in both cases, which of the following statements is TRUE?



- (A) On average, the waves on the left had higher frequencies, but the difference between frequencies less  
 (B) On average, the waves on the right had higher frequencies, but the difference between frequencies less  
 (C) On average, the waves on the left had higher frequencies, but the difference between frequencies more  
 (D) On average, the waves on the right had higher frequencies, but the difference between frequencies more

SW0060

### Doppler effect

32. A source when at rest in a medium produces waves with a velocity  $v$  and a wavelength of  $\lambda$ . If the source is set in motion with a velocity  $v_s$  what would be the wavelengths produced directly in front of the source?

(A)  $\lambda \left(1 - \frac{v_s}{v}\right)$       (B)  $\lambda \left(1 + \frac{v_s}{v}\right)$       (C)  $\lambda \left(1 + \frac{v}{v_s}\right)$       (D)  $\frac{\lambda v}{v + v_s}$

SW0061

33. A source of sound  $S$  having frequency  $f$ . Wind is blowing from source to observer  $O$  with velocity  $u$ . If speed of sound with respect to air is  $C$ , the wavelength of sound detected by  $O$  is:

(A)  $\frac{C + u}{f}$       (B)  $\frac{C - u}{f}$       (C)  $\frac{C(C + u)}{(C - u)f}$       (D)  $\frac{C}{f}$

SW0062

34. A train moving towards a hill at a speed of 72 km / hr sounds a whistle of frequency 500 Hz. A wind is blowing from the hill at a speed of 36 km / hr. If the speed of sound in air is 340 m / s, the frequency heard by a man on the hill is

(A) 532.25 Hz.      (B) 565.0 Hz.      (C) 516.1 Hz.      (D) none of the above.

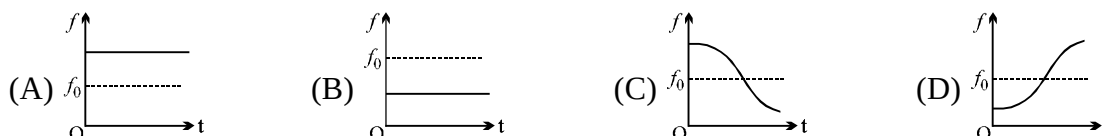
SW0063

35. A source is moving with constant speed  $v_s = 20$  m/sec towards a stationary observer due east of source. Wind is blowing at the speed of 20 m/sec due to  $60^\circ$  north of east. The source is generating of frequency 500 Hz. Then frequency registered by observer is: [Speed of sound in still air = 330 m/sec.]

(A) 500 Hz (B) 532 Hz (C) 531 Hz (D) 530 Hz

SW0064

36. Source and observer both start moving simultaneously from origin, one along x-axis and the other along y-axis with speed of source = twice the speed of observer. The graph between the apparent frequency observed by observer  $f$  and time  $t$  would approximately be :



SW0065

37. A siren placed at a railway platform is emitting sound of frequency 5 kHz. A passenger sitting in a moving train A records a frequency of 5.5 kHz while the train approaches the siren. During his return journey in a different train B he records a frequency of 6.0 kHz while approaching the same siren. The ratio of the velocity of train B to that of train A is :-

(A) 242/252 (B) 2 (C) 5/6 (D) 11/6

SW0066

38. A police van moving with velocity 22 m/s and emitting sound of frequency 176 Hz, follows a motor cycle in turn is moving towards a stationary car and away from the police van. The stationary car is emitting frequency 165 Hz. If motorcyclist does not hear any beats then his velocity is ( $v_s = 330$  m/s)

[JEE 2003 (Scr)]

(A) 22 m/s (B) 24 m/s (C) 20 m/s (D) 18 m/s

SW0067

## EXERCISE (O-2)

1. Two tuning forks of frequency 250 Hz and 256 Hz produce beats. If a maximum of intensity is observed just now, after how much time the minimum is observed at the same place ?

(A)  $\frac{1}{18}$  sec                      (B)  $\frac{1}{4}$  sec.                      (C)  $\frac{1}{3}$  sec.                      (D)  $\frac{1}{12}$  sec.

SW0068

2. The particle displacement of a travelling longitudinal wave is represented by  $S = S(x, t)$ . The midpoints of a compression zone and an adjacent rarefaction zone are represented by the letter 'C' and 'R'. Which of the following is true?

(A)  $|\partial S / \partial x|_C = |\partial S / \partial x|_R$   
 (B)  $|\partial S / \partial t|_C = |\partial S / \partial t|_R = 0$   
 (C)  $(\text{pressure})_C - (\text{pressure})_R = 2 |\partial S / \partial x|_C \times \text{Bulk modulus of air.}$   
 (D) Particles of air are stationary mid-way between 'C' and 'R'.

SW0069

3. Which of the following statements are wrong about the velocity of sound in air:  
 (A) decreases with increases in temperature      (B) increases with decrease in temperature  
 (C) decreases as humidity increases                      (D) independent of density of air.

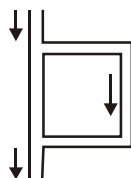
SW0070

4. A car moves towards a hill with speed  $v_c$ . It blows a horn of frequency  $f$  which is heard by an observer following the car with speed  $v_o$ . The speed of sound in air is  $v$ .

(A) the wavelength of sound reaching the hill is  $\frac{v}{f}$   
 (B) the wavelength of sound reaching the hill is  $\frac{v - v_c}{f}$   
 (C) the beat frequency observed by the observer is  $\left( \frac{v + v_o}{v - v_c} \right) f$   
 (D) the beat frequency observed by the observer is  $\frac{2v_c(v + v_o)f}{v^2 - v_c^2}$

SW0071

5. Three coherent source kept along the same line produce intensity  $I_0$  each at point P on this line. When  $S_1$  &  $S_2$  are switched on simultaneously, intensity at point P is  $2I_0$ . When  $S_2$  and  $S_3$  are switched on simultaneously, intensity at point P is  $2I_0$ . Then
- (A) When  $S_1$  and  $S_3$  are switched on simultaneously, intensity at point P can be  $2I_0$   
 (B) When  $S_1$  and  $S_3$  are switched on simultaneously, intensity at point P can be 0  
 (C) When all 3 sources are switched on simultaneously, intensity at point P can be  $I_0$   
 (D) When all 3 sources are switched on simultaneously, intensity at point P can be  $3I_0$
6. A sound consists of four frequencies  $\rightarrow$  300 Hz, 900 Hz, 2400 Hz and 4500 Hz. A sound 'filter' is made by passing this sound through a bifurcated pipe as shown. The sound waves have to travel a distance of 50 cm more in the right branch-pipe than in the straight pipe. The speed of sound in air is 300 m/s. Then, which of the following frequencies will be almost completely muffled or "silenced" at the outlet ?



- (A) 300 Hz                      (B) 900 Hz                      (C) 2400 Hz                      (D) 4500 Hz

SW0073

### Paragraph for Question No. 7 to 9

A metallic rod of length 1m has one end free and other end rigidly clamped. Longitudinal stationary waves are set up in the rod in such a way that there are total six antinodes present along the rod. The amplitude of an antinode is  $4 \times 10^{-6}$  m. Young's modulus and density of the rod are  $6.4 \times 10^{10}$  N/m<sup>2</sup> and  $4 \times 10^3$  Kg/m<sup>3</sup> respectively. Consider the free end to be at origin and at  $t=0$  particles at free end are at positive extreme.

7. The equation describing displacements of particles about their mean positions is
- (A)  $s = 4 \times 10^{-6} \cos\left(\frac{11\pi}{2}x\right) \cos(22\pi \times 10^3 t)$       (B)  $s = 4 \times 10^{-6} \cos\left(\frac{11\pi}{2}x\right) \sin(22\pi \times 10^3 t)$   
 (C)  $s = 4 \times 10^{-6} \cos(5\pi x) \cos(20\pi \times 10^3 t)$       (D)  $s = 4 \times 10^{-6} \cos(5\pi x) \sin(20\pi \times 10^3 t)$

SW0074

8. The equation describing stress developed in the rod is

(A)  $140.8\pi \times 10^4 \cos\left(\frac{11}{2}\pi x + \pi\right) \cos(22\pi \times 10^3 t)$

(B)  $140.8\pi \times 10^4 \sin\left(\frac{11}{2}\pi x + \pi\right) \cos(22\pi \times 10^3 t)$

(C)  $128\pi \times 10^4 \cos(5\pi x + \pi) \cos(20\pi \times 10^3 t)$

(D)  $128\pi \times 10^4 \sin(5\pi x + \pi) \cos(20\pi \times 10^3 t)$

SW0074

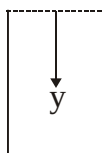
9. The magnitude of strain at midpoint of the rod at  $t = 1$  sec is

(A)  $11\sqrt{3}\pi \times 10^{-6}$  (B)  $11\sqrt{2}\pi \times 10^{-6}$  (C)  $10\sqrt{3}\pi \times 10^{-6}$  (D)  $10\sqrt{2}\pi \times 10^{-6}$

SW0074

**Paragraph for Question No. 10 to 12**

In an organ pipe (may be closed or open) of length 1m standing wave is setup, whose equation for longitudinal displacement is given by  $\xi = (0.1 \text{ mm}) \cos \frac{2\pi}{0.8}(y) \cos(400)t$  where  $y$  is measured from the top of the tube in meters and  $t$  in second.



10. The upper end and the lower ends of the tube are respectively:

(A) open – closed (B) closed – open (C) open – open (D) closed – closed

SW0075

11. The air column is vibrating in

(A) First overtone (B) Second overtone (C) Third harmonic (D) Fundamental mode

SW0075

12. Equation of the standing wave in terms of excess pressure is (Bulk modulus of air  $B = 5 \times 10^5 \text{ N/m}^2$ )

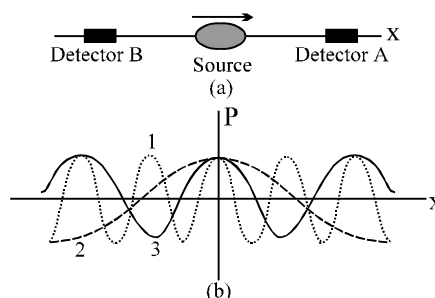
(A)  $P_{\text{ex}} = (125 \pi \text{ N/m}^2) \sin \frac{2\pi}{0.8}(y) \cos(400t)$  (B)  $P_{\text{ex}} = (125 \pi \text{ N/m}^2) \cos \frac{2\pi}{0.8}(y) \sin(400t)$

(C)  $P_{\text{ex}} = (225 \pi \text{ N/m}^2) \sin \frac{2\pi}{0.8}(y) \cos(200t)$  (D)  $P_{\text{ex}} = (225 \pi \text{ N/m}^2) \cos \frac{2\pi}{0.8}(y) \sin(200t)$

SW0075

**Paragraph for Question No. 13 to 16**

A source emitting a sound wave at a certain frequency moves with constant speed along an x-axis figure (a). The source moves directly towards a stationary detector A and directly away from another stationary detector B. The superimposed three plots of figure (b) indicate the pressure function  $P(x)$  of the sound wave as measured by detector A, by detector B, and by someone (c) in the rest frame of the source.



13. Which of the following plot corresponds to the measurement done by detector A?  
 (A) 1 (B) 2 (C) 3 (D) These plots are not possible
14. The plot corresponding to the measurement done by detector B is  
 (A) 1 (B) 2 (C) 3 (D) These plots are not possible
15. The plot corresponding to the measurement done by the detector C is  
 (A) 1 (B) 2 (C) 3 (D) These plots are not possible
16. Now the source stops and begins to move along y-axis with same speed, the plot which corresponds to the measurement of B now is  
 (A) 1 (B) 2 (C) 3 (D) none of these

SW0076

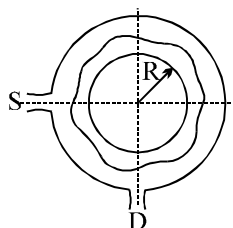
SW0076

SW0076

SW0076

**Paragraph for Question No. 17 to 21**

A narrow tube is bent in the form of a circle of radius  $R$ , as shown in the figure. Two small holes S and D are made in the tube at the positions right angle to each other. A source placed at S generated a wave of intensity  $I_0$  which is equally divided into two parts : One part travels along the longer path, while the other travels along the shorter path. Both the part waves meet at the point D where a detector is placed



17. If a maxima is formed at the detector then, the magnitude of wavelength  $\lambda$  of the wave produced is given by :-

(A)  $\pi R$  (B)  $\frac{\pi R}{2}$  (C)  $\frac{\pi R}{4}$  (D)  $\frac{2\pi R}{3}$

SW0077

18. If the minima is formed at the detector then, the magnitude of wavelength  $\lambda$  of the wave produced is given by :-

(A)  $2\pi R$  (B)  $\frac{3\pi R}{2}$  (C)  $\frac{2\pi R}{3}$  (D)  $\frac{2\pi R}{5}$

SW0077

19. The maximum intensity produced at D is given by :-

(A)  $4I_0$  (B)  $2I_0$  (C)  $I_0$  (D)  $3I_0$

SW0077

20. The maximum value of  $\lambda$  to produce a maxima at D is given by :-

(A)  $\pi R$  (B)  $2\pi R$  (C)  $\frac{\pi R}{2}$  (D)  $\frac{3\pi R}{2}$

SW0077

21. The maximum value of  $\lambda$  to produce a minima at D is given by :-

(A)  $\pi R$  (B)  $2\pi R$  (C)  $\frac{\pi R}{2}$  (D)  $\frac{3\pi R}{2}$

SW0077

**Paragraph for Question Nos. 22 to 24**

Two waves  $y_1 = A \cos (0.5 \pi x - 100 \pi t)$  and  $y_2 = A \cos (0.46 \pi x - 92 \pi t)$  are travelling in a pipe placed along x-axis.

[JEE 2006]

22. Find the number of times intensity is maximum in time interval of 1 sec.

(A) 4 (B) 6 (C) 8 (D) 10

SW0078

23. Find wave velocity of louder sound

(A) 100 m/s (B) 192 m/s (C) 200 m/s (D) 96 m/s

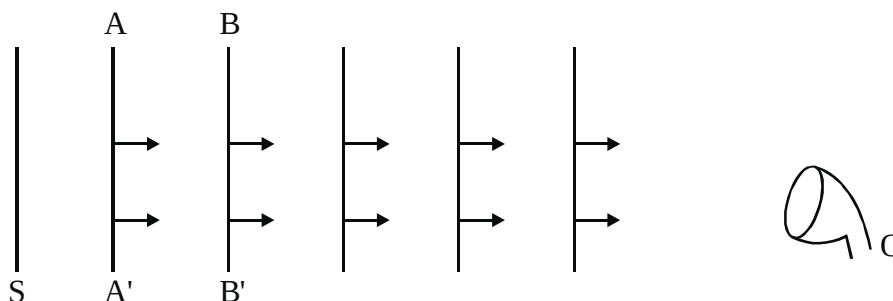
SW0078

24. At  $x = 0$  how many times the value of  $y_1 + y_2$  is zero in one second?

(A) 100 (B) 46 (C) 192 (D) 96

SW0078

25. Consider a large plane diaphragm 'S' emitting sound and a detector 'O'. The diagram shows plane wavefronts for the sound wave travelling in air towards right when source, observer and medium are at rest. AA' and BB' are fixed imaginary planes. Column-I describes about the motion of source, observer or medium and column-II describes various effects. Match them correctly.

**Column-I**

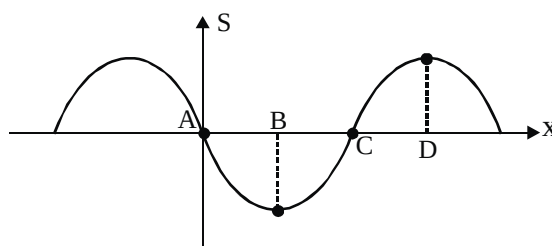
- (A) Source starts moving towards right  
(B) Air starts moving towards right  
(C) Observer and source both move towards left with same speed.  
(D) Source and medium (air) both move towards right with same speed.

**Column-II**

- (P) Distance between any two wavefronts will increase.  
(Q) Distance between any two wavefronts will decrease.  
(R) The time needed by sound to move from plane AA' to BB' will increase.  
(S) The time needed by sound to move from plane AA' to BB' will decrease.  
(T) Frequency received by observer increases.

**SW0079**

26. Figure shows a graph of particle displacement function of  $x$  at  $t = 0$  for a longitudinal wave travelling in positive  $x$ -direction in a gas. A, B, C, D denote position of particles in space.

**Column-I**

- (A) point A  
(B) point B  
(C) point C  
(D) point D

**Column II**

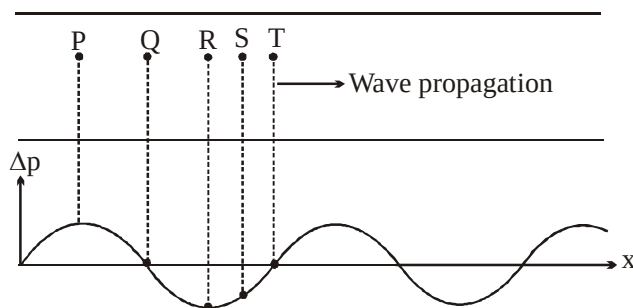
- (P) Particle velocity is in direction of wave propagation  
(Q) Maximum magnitude of strain  
(R) Excess pressure is zero  
(S) Maximum density  
(T) Maximum magnitude of excess pressure

**SW0080**



27. Sound is travelling in a long tube towards right and the graph of excess pressure variation versus position (at some instant) is given below.

Match velocities in column-I with column-II. P,Q,R,S,T are medium particles inside the tube.



**Column-I**

- (A) velocity is towards right
- (B) velocity is towards left
- (C) velocity is zero
- (D) Speed is maximum

**Column-II**

- (P) P
- (Q) Q
- (R) R
- (S) S
- (T) T

SW0081

## EXERCISE - JM

1. **Statement-1:** Two longitudinal waves given by equations :  $y_1(x, t) = 2a \sin(\omega t - kx)$  and  $y_2(x, t) = a \sin(2\omega t - 2kx)$  will have equal intensity.

**Statement-1:** Intensity of waves of given frequency in same medium is proportional to square of amplitude only. [AIEEE - 2011]

- (1) Statement-1 is false, statement-2 is true  
 (2) Statement-1 is true, statement-2 is false  
 (3) Statement-1 is true, statement-2 true; statement-2 is the correct explanation of statement-1  
 (4) Statement-1 is true, statement-2 is true; statement-2 is not correct explanation of statement-1.

SW0082

2. A pipe of length 85 cm is closed from one end. Find the number of possible natural oscillations of air column in the pipe whose frequencies lie below 1250 Hz. The velocity of sound in air is 340 m/s. [JEE Main - 2014]

- (1) 6 (2) 4 (3) 12 (4) 8

SW0083

3. A train is moving on a straight track with speed  $20 \text{ ms}^{-1}$ . It is blowing its whistle at the frequency of 1000 Hz. The percentage change in the frequency heard by a person standing near the track as the train passes him is (speed of sound =  $320 \text{ ms}^{-1}$ ) close to :- [JEE Main - 2015]

- (1) 18% (2) 24% (3) 6% (4) 12%

SW0084

4. A pipe open at both ends has a fundamental frequency  $f$  in air. The pipe is dipped vertically in water so that half of it is in water. The fundamental frequency of the air column is now :- [JEE-Main-2016]

- (1)  $f$  (2)  $\frac{f}{2}$  (3)  $\frac{3f}{4}$  (4)  $2f$

SW0085

5. An observer is moving with half the speed of light towards a stationary microwave source emitting waves at frequency 10 GHz. What is the frequency of the microwave measured by the observer? (speed of light =  $3 \times 10^8 \text{ ms}^{-1}$ ) [JEE Main - 2017]

- (1) 17.3 GHz (2) 15.3 GHz (3) 10.1 GHz (4) 12.1 GHz

SW0086

## EXERCISE (JA)

1. When two progressive waves  $y_1 = 4 \sin(2x - 6t)$  and  $y_2 = 3 \sin\left(2x - 6t - \frac{\pi}{2}\right)$  are superimposed, the

amplitude of the resultant wave is

[IIT-JEE 2010]

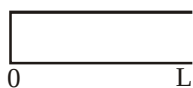
SW0087

2. **Column I** shows four systems, each of the same length  $L$ , for producing standing waves. The lowest possible natural frequency of a system is called its fundamental frequency, whose wavelength is denoted as  $\lambda_f$ . Match each system with statements given in **Column II** describing the nature and wavelength of the standing waves.

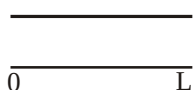
[IIT-JEE 2011]

### Column I

- (A) Pipe closed at one end



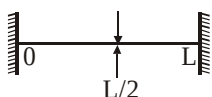
- (B) Pipe open at both ends



- (C) Stretched wire clamped at both ends



- (D) Stretched wire clamped at both ends and at mid-point



### Column II

- (p) Longitudinal waves

- (q) Transverse Waves

- (r)  $\lambda_f = L$

- (s)  $\lambda_f = 2L$

- (t)  $\lambda_f = 4L$

SW0088

3. A police car with a siren of frequency 8 kHz is moving with uniform velocity 36 km/hr towards a tall building which reflects the sound waves. The speed of sound in air is 320 m/s. The frequency of the siren heard by the car driver is [JEE 2011]

(A) 8.50 kHz (B) 8.25 kHz (C) 7.75 kHz (D) 7.50 kHz

SW0089

4. A person blows into open-end of a long pipe. As a result, a high-pressure pulse of air travels down the pipe. When this pulse reaches the other end of the pipe. [JEE 2012]

(A) a high-pressure pulse starts travelling up the pipe, if the other end of the pipe is open  
(B) a low -pressure pulse starts travelling up the pipe, if the other end of the pipe is open  
(C) a low pressure pulse starts travelling up the pipe, if the other end of the pipe is closed  
(D) a high-pressure pulse starts travelling up the pipe, if the other end of the pipe is closed

SW0090

5. A student is performing the experiment of Resonance Column. The diameter of the column tube is 4cm. The frequency of the tuning fork is 512 Hz. The air temperature is 38°C in which the speed of sound is 336 m/s. The zero of the meter scale coincides with the top end of the Resonance column tube. When the first resonance occurs, the reading of the water level in the column is :-

[JEE 2012]

(A) 14.0 cm (B) 15.2 cm (C) 16.4 cm (D) 17.6 cm

SW0091

6. A student is performing an experiment using a resonance column and a tuning fork of frequency  $244 \text{ s}^{-1}$ . He is told that the air in the tube has been replaced by another gas (assume that the column remains filled with the gas). If the minimum height at which resonance occurs is  $(0.350 \pm 0.005) \text{ m}$ , the gas in the tube is

(Useful information :  $\sqrt{167RT} = 640 \text{ J}^{1/2} \text{ mole}^{-1/2}$ ;  $\sqrt{140RT} = 590 \text{ J}^{1/2} \text{ mole}^{-1/2}$ . The molar masses

M in grams are given in the options. Take the values of  $\sqrt{\frac{10}{M}}$  for each gas as given there.)

[JEE Advanced-2014]

(A) Neon  $\left( M = 20, \sqrt{\frac{10}{20}} = \frac{7}{10} \right)$

(B) Nitrogen  $\left( M = 28, \sqrt{\frac{10}{28}} = \frac{3}{5} \right)$

(C) Oxygen  $\left( M = 32, \sqrt{\frac{10}{32}} = \frac{9}{16} \right)$

(D) Argon  $\left( M = 36, \sqrt{\frac{10}{36}} = \frac{17}{32} \right)$

SW0092

7. Four harmonic waves of equal frequencies and equal intensities  $I_0$  have phase angles  $0, \pi/3, 2\pi/3$  and  $\pi$ . When they are superposed, the intensity of the resulting wave is  $nI_0$ . The value of  $n$  is.

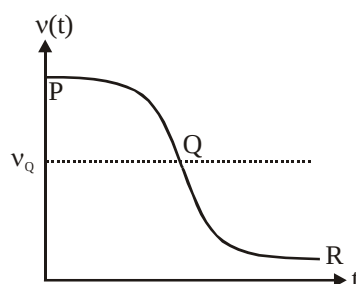
[JEE-Advance-2015]

SW0093

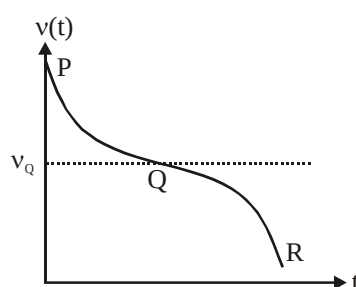
8. Two loudspeakers M and N are located 20m apart and emit sound at frequencies 118 Hz and 121 Hz, respectively. A car is initially at a point P, 1800 m away from the midpoint Q of the line MN and moves towards Q constantly at 60 km/hr along the perpendicular bisector of MN. It crosses Q and eventually reaches a point R, 1800 m away from Q. Let  $v(t)$  represent the beat frequency measured by a person sitting in the car at time  $t$ . Let  $v_P, v_Q$  and  $v_R$  be the beat frequencies measured at locations P, Q and R, respectively. The speed of sound in air is  $330 \text{ ms}^{-1}$ . Which of the following statement(s) is(are) true regarding the sound heard by the person ?

[JEE Advanced 2016]

- (A) The plot below represents schematically the variation of beat frequency with time



- (B) The plot below represents schematically the variations of beat frequency with time



- (C) The rate of change in beat frequency is maximum when the car passes through Q

- (D)  $v_P + v_R = 2v_Q$

SW0094

9. A stationary source emits sound of frequency  $f_0 = 492 \text{ Hz}$ . The sound is reflected by a large car approaching the source with a speed of  $2 \text{ ms}^{-1}$ . The reflected signal is received by the source and superposed with the original. What will be the beat frequency of the resulting signal in Hz ? (Given that the speed of sound in air is  $330 \text{ ms}^{-1}$  and the car reflects the sound at the frequency it has received).

[JEE Advanced 2017]

SW0095

10. Two men are walking along a horizontal straight line in the same direction. The man in front walks at a speed  $1.0 \text{ ms}^{-1}$  and the man behind walks at a speed  $2.0 \text{ ms}^{-1}$ . A third man is standing at a height  $12 \text{ m}$  above the same horizontal line such that all three men are in a vertical plane. The two walking men are blowing identical whistles which emit a sound of frequency  $1430 \text{ Hz}$ . The speed of sound in air is  $330 \text{ ms}^{-1}$ . At the instant, when the moving men are  $10 \text{ m}$  apart, the stationary man is equidistant from them. The frequency of beats in  $\text{Hz}$ , heard by the stationary man at this instant, is \_\_\_\_\_

[JEE Advanced 2018]

SW0096

11. In an experiment to measure the speed of sound by a resonating air column, a tuning fork of frequency  $500 \text{ Hz}$  is used. The length of the air column is varied by changing the level of water in the resonance tube. Two successive resonances are heard at air columns of length  $50.7 \text{ cm}$  and  $83.9 \text{ cm}$ . Which of the following statements is (are) true ?

[JEE Advanced 2018]

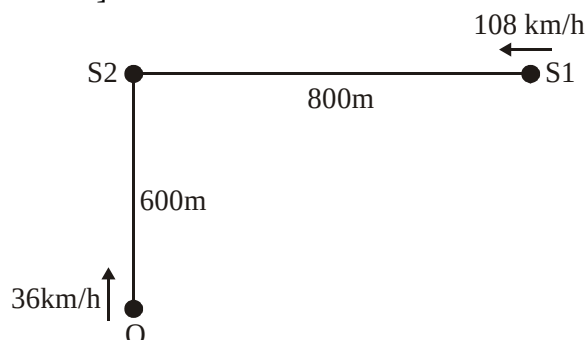
- (A) The speed of sound determined from this experiment is  $332 \text{ ms}^{-1}$   
 (B) The end correction in this experiment is  $0.9 \text{ cm}$   
 (C) The wavelength of the sound wave is  $66.4 \text{ cm}$   
 (D) The resonance at  $50.7 \text{ cm}$  corresponds to the fundamental harmonic

SW0097

12. A train  $S_1$ , moving with a uniform velocity of  $108 \text{ km/h}$ , approaches another train  $S_2$  standing on a platform. An observer  $O$  moves with a uniform velocity of  $36 \text{ km/h}$  towards  $S_2$ , as shown in figure. Both the trains are blowing whistles of same frequency  $120 \text{ Hz}$ . When  $O$  is  $600 \text{ m}$  away from  $S_2$  and distance between  $S_1$  and  $S_2$  is  $800 \text{ m}$ , the number of beats heard by  $O$  is \_\_\_\_\_.

[Speed of the sound =  $330 \text{ m/s}$ ]

[JEE Advanced 2019]



SW0098

# ANSWER KEY

## EXERCISE (S-1)

1. Ans.  $\frac{\pi^2 \times 10^{-9}}{4} \text{ W/m}^2$       2. Ans.  $2c/3$       3. Ans. 30 dB,  $10\sqrt{10} \text{ } \mu\text{m}$   
 4. Ans. 1 : 1      5. Ans. 3 cm      6. Ans. 336 m/s  
 7. Ans. 33 cm and 13.2 cm      8. Ans. 2.5 kHz, 7.5 kHz      9. Ans. (a) 2.116, (b)  $\frac{3}{4}$   
 10. Ans. 3.33 cm, 163 Hz      11. Ans. 1%      12. Ans. 345, 341 or 349 Hz  
 13. Ans. 8      14. Ans. 1.5 m/s      15. Ans. 5  
 16. Ans.  $11f/9$   
 17. Ans. (a)  $\left(\frac{v + \lambda_0 f_0}{\lambda_0}\right)t$  (b)  $(\lambda_0 f_0 - v)t$  (c)  $\lambda_0 \left(\frac{\lambda_0 f_0 - v}{\lambda_0 f_0 + v}\right)$  (d)  $f_0 \left(\frac{\lambda_0 f_0 + v}{\lambda_0 f_0 - v}\right)$  (e)  $\frac{2vf_0}{\lambda_0 f_0 - v}$   
 18. Ans. 6 Hz

## EXERCISE (S-2)

1. Ans. (a) 100696 Hz (b) 103038 Hz  
 2. Ans. (i)  $L = \frac{15}{16} \text{ m}$ , (ii)  $\frac{\Delta P_0}{\sqrt{2}}$ , (iii)  $P_{\max} = P_{\min} = P_0$  (iv)  $P_{\max} = P_0 + \Delta P_0$ ,  $P_{\min} = P_0 - \Delta P_0$   
 3. Ans.  $f$       4. Ans. 27.04 N      5. Ans. (i) 599 Hz, (ii) 0.935 km, 620 Hz  
 6. Ans. 438.7 Hz, 257.3 Hz      7. Ans. (i)  $10^{-3} \text{ sec}$ , (ii)  $2 \times 10^{-3} \text{ sec}$   
 8. Ans. 0.75 m/s  
 9. Ans. (a)  $\frac{2\pi}{a}, \frac{b}{2\pi}$ , (b)  $y = -0.8 A \cos(ax - bt)$  OR  $y = 0.8 A \cos(ax - bt)$ , (c) 1.8 Ab, 0  
 (d)  $y = -1.6 A \sin ax \sin bt + 0.2 A \cos(ax + bt)$ ,  $\left[n + \frac{(-1)^2}{2}\right] \frac{\pi}{a}$ , -X direction OR  
 $y = 0.2 A \cos(ax + bt) + 1.6 A \cos ax \cos bt$ ,  $x = \frac{n\pi}{a}$ , -X direction  
 10. Ans. (a)  $\frac{15}{16} \text{ m}$  (b)  $\frac{\Delta P_0}{\sqrt{2}}$  (c) equal to mean pressure (d)  $P_0 + \Delta P_0$ ,  $P_0 - \Delta P_0$   
 11. Ans.  $\frac{-dH}{dt} = (1.11 \times 10^{-2})\sqrt{H}$ , 43 sec.

## EXERCISE (O-1)

- |              |              |              |              |              |              |
|--------------|--------------|--------------|--------------|--------------|--------------|
| 1. Ans. (B)  | 2. Ans. (A)  | 3. Ans. (B)  | 4. Ans. (B)  | 5. Ans. (A)  | 6. Ans. (A)  |
| 7. Ans. (B)  | 8. Ans. (D)  | 9. Ans. (A)  | 10. Ans. (D) | 11. Ans. (D) | 12. Ans. (B) |
| 13. Ans. (C) | 14. Ans. (B) | 15. Ans. (C) | 16. Ans. (B) | 17. Ans. (D) | 18. Ans. (A) |
| 19. Ans. (C) | 20. Ans. (C) | 21. Ans. (A) | 22. Ans. (A) | 23. Ans. (B) | 24. Ans. (A) |
| 25. Ans. (B) | 26. Ans. (B) | 27. Ans. (A) | 28. Ans. (A) | 29. Ans. (A) | 30. Ans. (A) |
| 31. Ans. (B) | 32. Ans. (A) | 33. Ans. (A) | 34. Ans. (A) | 35. Ans. (C) | 36. Ans. (B) |
| 37. Ans. (B) | 38. Ans. (A) |              |              |              |              |

## EXERCISE (O-2)

- |                  |                   |                   |               |
|------------------|-------------------|-------------------|---------------|
| 1. Ans. (B,D)    | 2. Ans. (A,C,D)   | 3. Ans. (A,B,C,D) | 4. Ans. (B,D) |
| 5. Ans. (B, C)   | 6. Ans. (A, B, D) | 7. Ans. (A)       | 8. Ans. (B)   |
| 9. Ans. (B)      | 10. Ans. (A)      | 11. Ans. (B)      | 12. Ans. (A)  |
| 13. Ans. (A)     | 14. Ans. (B)      | 15. Ans. (C)      | 16. Ans. (D)  |
| 17. Ans. (A,B,C) | 18. Ans. (A,C,D)  | 19. Ans. (B)      | 20. Ans. (A)  |
| 21. Ans. (B)     | 22. Ans. (A)      | 23. Ans. (C)      | 24. Ans. (A)  |
25. Ans. (A) (Q,T) ; (B) (P,S) ; (C) (P) (D) (S,T)
26. Ans. (A) (P,Q,S,T); (B) (R); (C) (Q,T); (D) (R)
27. Ans. (A) (P); (B) (R,S); (C) (Q,T); (D) (P,R)

## EXERCISE (JM)

- |             |             |             |             |             |
|-------------|-------------|-------------|-------------|-------------|
| 1. Ans. (4) | 2. Ans. (1) | 3. Ans. (4) | 4. Ans. (1) | 5. Ans. (1) |
|-------------|-------------|-------------|-------------|-------------|

## EXERCISE (JA)

- |                         |                                         |                         |
|-------------------------|-----------------------------------------|-------------------------|
| 1. Ans. 5               | 2. Ans. (A) p,t (B) p,s (C) q,s (D) q,r | 3. Ans. (A)             |
| 4. Ans. (B,D)           | 5. Ans. (B)                             | 6. Ans. (D)             |
| 8. Ans. (A, C, D)       | 9. Ans. 6                               | 10. Ans. 5 [4.99, 5.01] |
| 12. Ans. (8.12 to 8.13) |                                         | 11. Ans. (A,B,C)        |



# Important Notes

## *Important Notes*

# EXERCISE (S-1)

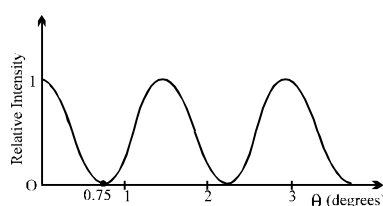
1. In a Young's double slit experiment for interference of light, the slits are 0.2 cm apart and are illuminated by yellow light ( $\lambda = 600 \text{ nm}$ ). What would be the fringe width on a screen placed 1 m from the plane of slits if the whole system is immersed in water of index  $4/3$ ?

WO0001

2. In Young's double slit experiment the slits are 0.5 mm apart and the interference is observed on a screen at a distance of 100 cm from the slit. It is found that the 9th bright fringe is at a distance of 7.5mm from the second dark fringe from the centre of the fringe pattern on same side. Find the wavelength of the light used.

WO0002

3. Light of wavelength 520 nm passing through a double slit, produces interference pattern of relative intensity versus angular position  $\theta$  as shown in the figure. Find the separation  $d$  between the slits.



WO0003

4. In a Young's double slit experiment, two wavelengths of 500 nm and 700 nm were used. What is the minimum distance from the central maximum where their maximas coincide again? (Take  $D/d = 10^3$ . Symbols have their usual meanings.)

[IIT-JEE 2004]

WO0004

5. In a YDSE apparatus,  $d = 1 \text{ mm}$ ,  $\lambda = 600 \text{ nm}$  and  $D = 1 \text{ m}$ . The slits individually produce same intensity on the screen. Find the minimum distance between two points on the screen having 75% intensity of the maximum intensity.

WO0005

6. The distance between two slits in a YDSE apparatus is 3mm. The distance of the screen from the slits is 1m. Microwaves of wavelength 1 mm are incident on the plane of the slits normally. Find the distance of the first maxima on the screen from the central maxima. Also find the total number of maxima on the screen.

WO0006

7. One slit of a double slit experiment is covered by a thin glass plate of refractive index 1.4 and the other by a thin glass plate of refractive index 1.7. The point on the screen, where central bright fringe was formed before the introduction of the glass sheets, is now occupied by the 5<sup>th</sup> bright fringe. Assuming that both the glass plates have same thickness and wavelength of light used is 4800 Å, find their thickness.

WO0007

8. A monochromatic light of  $\lambda = 5000 \text{ Å}$  is incident on two slits separated by a distance of  $5 \times 10^{-4} \text{ m}$ . The interference pattern is seen on a screen placed at a distance of 1 m from the slits. A thin glass plate of thickness  $1.5 \times 10^{-6} \text{ m}$  & refractive index  $\mu = 1.5$  is placed between one of the slits & the screen. Find the intensity at the centre of the screen, if the intensity there is  $I_0$  in the absence of the plate. Also find the lateral shift of the central maximum.

WO0008

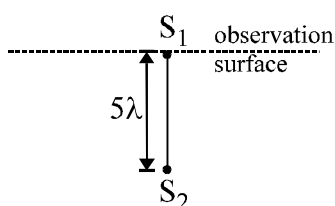
9. In a biprism experiment with sodium light, bands of width 0.0195 cm are observed on screen at 100 cm from slit. On introducing a convex lens 30 cm away from the slit between biprism and screen, two images of the slit are seen 0.7 cm apart on screen. Calculate the wavelength of sodium light.

WO0009

10. A long narrow horizontal slit lies 1 mm above a plane mirror as in Lloyd's mirror. The interference pattern produced by the slit and its image is viewed on a screen distant 1 m from the slit. The wavelength of light is 600 nm. Find the distance of first maximum above the mirror.

WO0010

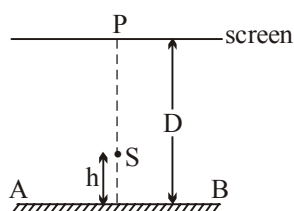
11. Two microwave coherent point sources emitting waves of wavelength  $\lambda$  are placed at  $5\lambda$  distance apart. The interference is being observed on a flat non-reflecting surface along a line passing through one source, in a direction perpendicular to the line joining the two sources (refer figure). Considering  $\lambda$  as 4 mm, calculate the positions of maxima and draw shape of interference pattern. Take initial phase difference between the two sources to be zero.



WO0011

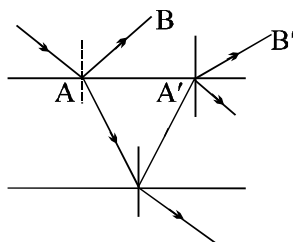
12. A point source  $S$  emitting light of wavelength  $600\text{ nm}$  is placed at a very small height  $h$  above the flat reflecting surface  $AB$  (see figure). The intensity of the reflected light is  $36\%$  of the incident intensity. Interference fringes are observed on a screen placed parallel to the reflecting surface at a very large distance  $D$  from it. [IIT-JEE 2002]

- What is the shape of the interference fringes on the screen?
- Calculate the ratio of the minimum to the maximum intensities in the interference fringes formed near the point  $P$  (shown in the figure).
- If the intensities at point  $P$  corresponds to a maximum, calculate the minimum distance through which the reflecting surface  $AB$  should be shifted so that the intensity at  $P$  again becomes max.



WO0012

13. A ray of light of intensity  $I$  is incident on a parallel glass-slab at a point  $A$  as shown in figure. It undergoes partial reflection and refraction. At each reflection  $20\%$  of incident energy is reflected. The rays  $AB$  and  $A'B'$  undergo interference. Find the ratio  $I_{\max}/I_{\min}$ . [Neglect the absorption of light]



WO0013

## EXERCISE (S-2)

1. A thin glass plate of thickness  $t$  and refractive index  $\mu$  is inserted between screen & one of the slits in a Young's experiment. If the intensity at the centre of the screen is  $I$ , what was the intensity at the same point prior to the introduction of the sheet.

WO0014

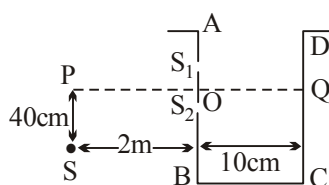
2. In Young's experiment, the source is red light of wavelength  $7 \times 10^{-7} \text{ m}$ . When a thin glass plate of refractive index 1.5 at this wavelength is put in the path of one of the interfering beams, the central bright fringe shifts by  $10^{-3} \text{ m}$  to the position which was previously occupied by the 5<sup>th</sup> bright fringe. Find the thickness of the plate. When the source is now changed to green light of wavelength  $5 \times 10^{-7} \text{ m}$ , the central fringe shifts to a position initially occupied by the 6<sup>th</sup> bright fringe due to red light without the plate. Find the refractive index of glass for the green light. Also estimate the change in fringe width due to the change in wavelength.

[IIT-JEE 1997 C]

WO0015

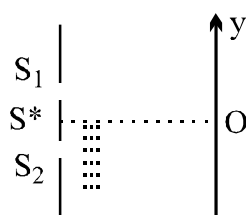
3. A vessel  $ABCD$  of 10 cm width has two small slits  $S_1$  and  $S_2$  sealed with identical glass plates of equal thickness. The distance between the slits is 0.8 mm.  $POQ$  is the line perpendicular to the plane  $AB$  and passing through  $O$ , the middle point of  $S_1$  and  $S_2$ . A monochromatic light source is kept at  $S$ , 40cm below  $P$  and 2 m from the vessel, to illuminate the slits as shown in the figure below. Calculate the position of the central bright fringe on the other wall  $CD$  with respect to the line  $OQ$ . Now, a liquid is poured into the vessel and filled up to  $OQ$ . The central bright fringe is found to be at  $Q$ . Calculate the refractive index of the liquid.

[IIT-JEE 2001]



WO0016

4. The Young's double slit experiment is done in a medium of refractive index  $4/3$ . A light of  $600 \text{ nm}$  wavelength is falling on the slits having  $0.45 \text{ mm}$  separation. The lower slit  $S_2$  is covered by a thin glass sheet of thickness  $10.4 \mu\text{m}$  and refractive index  $1.5$ . The interference pattern is observed on a screen placed  $1.5 \text{ m}$  from the slits as shown [IIT-JEE'99]
- (i) Find the location of the central maximum (bright fringe with zero path difference) on the  $y$ -axis.
- (ii) Find the light intensity at point  $O$  relative to the maximum fringe intensity.
- (iii) Now, if  $600 \text{ nm}$  light is replaced by white light of range  $400$  to  $700 \text{ nm}$ , find the wavelengths of the light that form maxima exactly at point  $O$ . [All wavelengths in this problem are for the given medium of refractive index  $4/3$ . Ignore dispersion]

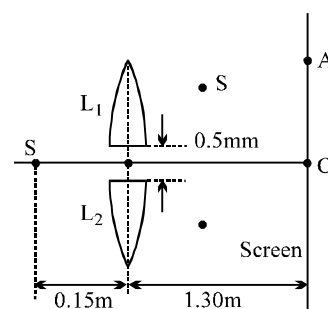


WO0017

5. In a Young's experiment, the upper slit is covered by a thin glass plate of refractive index  $1.4$  while the lower slit is covered by another glass plate having the same thickness as the first one but having refractive index  $1.7$ . Interference pattern is observed using light of wavelength  $5400 \text{ \AA}$ . It is found that the point  $P$  on the screen where the central maximum ( $n = 0$ ) fell before the glass plates were inserted now has  $3/4$  the original intensity. It is further observed that what used to be the 5th maximum earlier, lies below the point  $P$  while the 6th minimum lies above  $P$ . Calculate the thickness of the glass plate. (Absorption of light by glass plate may be neglected). [IIT-JEE 1997]

WO0018

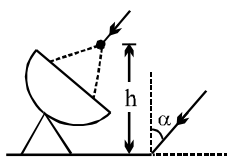
6. In the figure shown  $S$  is a monochromatic point source emitting light of wavelength  $\lambda = 500 \text{ nm}$ . A thin lens of circular shape and focal length  $0.10 \text{ m}$  is cut into two identical halves  $L_1$  and  $L_2$  by a plane passing through a diameter. The two halves are placed symmetrically about the central axis  $SO$  with a gap of  $0.5 \text{ mm}$ . The distance along the axis from  $S$  to  $L_1$  and  $L_2$  is  $0.15 \text{ m}$ , while that from  $L_1$  &  $L_2$  to  $O$  is  $1.30 \text{ m}$ . The screen at  $O$  is normal to  $SO$ . [IIT-JEE 1993]



- (i) If the third intensity maximum occurs at the point  $A$  on the screen, find the distance  $OA$ .
- (ii) If the gap between  $L_1$  &  $L_2$  is reduced from its original value of  $0.5 \text{ mm}$ , will the distance  $OA$  increase, decrease or remain the same ?

WO0019

7. Radio waves coming at angle  $\alpha$  to vertical are received by a radar after reflection from a nearby water surface & directly. What should be height of antenna from water surface so that it records a maximum intensity. (wavelength =  $\lambda$ ).

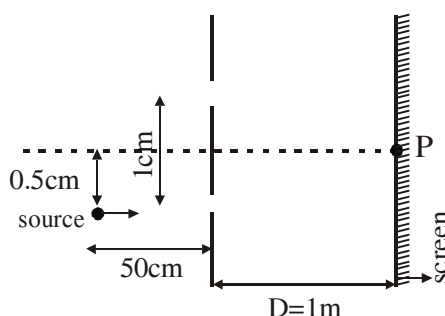


WO0020

8. One radio transmitter  $A$  operating at 60.0 MHz is 10.0 m from another similar transmitter  $B$  that is  $180^\circ$  out of phase with transmitter  $A$ . How far must an observer move from transmitter  $A$  toward transmitter  $B$  along the line connecting  $A$  and  $B$  to reach the nearest point where the two beams are in phase?

WO0021

9. In a typical Young's double slit experiment a point source of monochromatic light is kept as shown in the figure. If the source is given an instantaneous velocity  $v = 1$  mm per second towards the screen, then the instantaneous velocity of central maxima is given as  $\alpha \times 10^{-\beta}$  m/s upward. Find the value of  $\alpha + \beta$ .

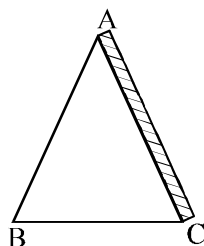


WO0022

10. A prism ( $\mu_p = \sqrt{3}$ ) has an angle of prism  $A = 30^\circ$ . A thin film ( $\mu_f = 2.2$ ) is coated on face  $AC$  as shown in the figure. Light of wavelength 550 nm is incident on the face  $AB$  at  $60^\circ$  angle of incidence. Find

[IIT-JEE 2003]

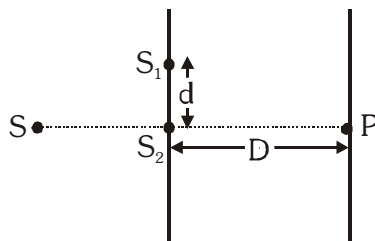
- (i) the angle of its emergence from the face  $AC$  and  
(ii) the minimum thickness (in nm) of the film for which the emerging light is of maximum possible intensity.



WO0023

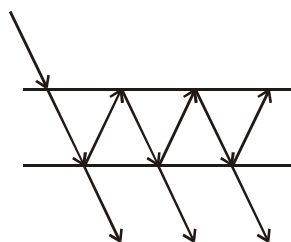


11. In a YDSE experiment two slits  $S_1$  and  $S_2$  have separation of  $d = 2 \text{ mm}$ . The distance of the screen is  $D = 8/5 \text{ m}$ . Source  $S$  starts moving from a very large distance towards  $S_2$  perpendicular to  $S_1S_2$  as shown in figure. The wavelength of monochromatic light is  $500 \text{ nm}$ . The number of maximas observed on the screen at point  $P$  as the source moves towards  $S_2$  is  $3995 + n$ . Find the value of  $n$ .



WO0024

12. A narrow beam of light has entered a large thin glass plate. Each refraction is accompanied by reflection of one third of the beam's energy. What percentage of the light energy is transmitted through the plate?

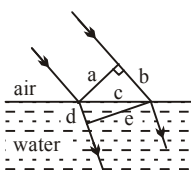


WO0025

## EXERCISE (O-1)

## SINGLE CORRECT TYPE QUESTIONS

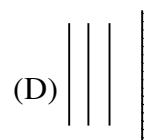
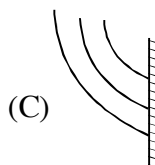
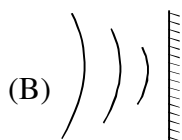
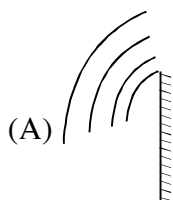
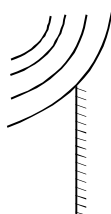
1. Figure shows plane waves refracted from air to water using Huygen's principle (where a, b, c, d, e are lengths on the diagram). The refractive index of water wrt air is the ratio :-



- (A)  $a/e$  (B)  $b/e$  (C)  $b/d$  (D)  $d/b$

WO0026

2. Spherical wavefronts shown in figure, strike a plane mirror. Reflected wavefronts will be as shown in



WO0027

3. Two beams of light having intensities  $I$  and  $4I$  interfere to produce a fringe pattern on a screen. The phase difference between the beams is  $\pi/2$  at point A and  $\pi$  at point B. Then the difference between the resultant intensities at A and B is :

[IIT-JEE (Scr.) 2001]

- (A)  $2I$  (B)  $4I$  (C)  $5I$  (D)  $7I$

WO0028

4. In a young double slit experiment, 12 fringes are observed to be formed in a certain segment of the screen when light of wavelength 600 nm is used. If the wavelength of light is changed to 400 nm, number of fringes observed in the same segment of the screen is given by

[IIT-JEE (Scr.) 2001]

- (A) 12 (B) 18 (C) 24 (D) 30

WO0029

5. Two coherent monochromatic light beams of intensities  $I$  and  $4I$  are superposed. The maximum and minimum possible intensities in the resulting beam are :  
 (A)  $5I$  and  $I$  (B)  $5I$  and  $3I$  (C)  $9I$  and  $I$  (D)  $9I$  and  $3I$

WO0030

6. When light is refracted into a denser medium,  
 (A) its wavelength and frequency both increase  
 (B) its wavelength increases but frequency remains unchanged  
 (C) its wavelength decreases but frequency remains unchanged  
 (D) its wavelength and frequency both decrease.

WO0031

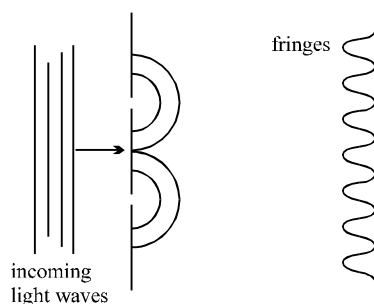
7. In YDSE how many maxima can be obtained on the screen if wavelength of light used is  $200\text{nm}$  and  $d = 700\text{ nm}$ :  
 (A) 12 (B) 7 (C) 18 (D) none of these

WO0032

8. In a YDSE, the central bright fringe can be identified :  
 (A) as it has greater intensity than the other bright fringes.  
 (B) as it is wider than the other bright fringes.  
 (C) as it is narrower than the other bright fringes.  
 (D) by using white light instead of single wavelength light.

WO0033

9. In a Young's double slit experiment, green light is incident on the two slits. The interference pattern is observed on a screen. Which of the following changes would cause the observed fringes to be more closely spaced?



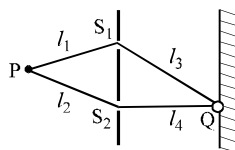
- (A) Reducing the separation between the slits  
 (B) Using blue light instead of green light  
 (C) Used red light instead of green light  
 (D) Moving the light source further away from the slits.

WO0034

10. In Young's double slit experiment, the wavelength of red light is  $7800\text{ \AA}$  and that of blue light is  $5200\text{ \AA}$ . The value of  $n$  for which  $n^{\text{th}}$  bright band due to red light coincides with  $(n + 1)^{\text{th}}$  bright band due to blue light, is :  
 (A) 1 (B) 2 (C) 3 (D) 4

WO0035

11. Two identical narrow slits  $S_1$  and  $S_2$  are illuminated by light of wavelength  $\lambda$  from a point source P. If, as shown in the diagram, the light is then allowed to fall on a screen, and if  $n$  is a positive integer, the condition for destructive interference at  $Q$  is :-



- (A)  $(\ell_1 - \ell_2) = (2n + 1)\lambda/2$  (B)  $(\ell_3 - \ell_4) = (2n + 1)\lambda/2$   
 (C)  $(\ell_1 + \ell_2) - (\ell_3 + \ell_4) = n\lambda$  (D)  $(\ell_1 + \ell_3) - (\ell_2 + \ell_4) = (2n + 1)\lambda/2$

WO0036

12. In Young's double slit experiment, the two slits act as coherent sources of equal amplitude  $A$  and wavelength  $\lambda$ . In another experiment with the same setup the two slits are sources of equal amplitude  $A$  and wavelength  $\lambda$  but are incoherent. The ratio of the average intensity of light at the midpoint of the screen in the first case to that in the second case is :-

- (A) 1 : 1 (B) 2 : 1 (C) 4 : 1 (D) none of these

WO0037

13. In a Young's double slit experiment, a small detector measures an intensity of  $I$  units at the centre of the fringe pattern. If one of the two (identical) slits is now covered, the measured intensity will be :-

- (A)  $2I$  (B)  $I$  (C)  $I/4$  (D)  $I/2$

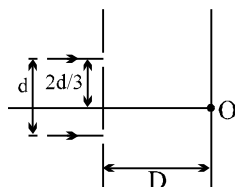
WO0038

14. In a Young's double slit experiment  $D$  equals the distance of screen and  $d$  is the separation between the slit. The distance of the nearest point to the central maximum where the intensity is same as that due to a single slit, is equal to :-

- (A)  $\frac{D\lambda}{d}$  (B)  $\frac{D\lambda}{2d}$  (C)  $\frac{D\lambda}{3d}$  (D)  $\frac{2D\lambda}{d}$

WO0039

15. In the figure shown if a parallel beam of white light is incident on the plane of the slits then the distance of the white spot on the screen from  $O$  is [Assume  $d \ll D$ ,  $\lambda \ll d$ ]



- (A) 0 (B)  $d/2$  (C)  $d/3$  (D)  $d/6$

WO0040

16. In the above question if the light incident is monochromatic and point  $O$  is a maxima, then the wavelength of the light incident cannot be :-

(A)  $d^2/3D$  (B)  $d^2/6D$  (C)  $d^2/12D$  (D)  $d^2/18D$

WO0041

17. In a YDSE bi-chromatic light of wavelengths 400 nm and 560 nm are used. The distance between the slits is 0.1 mm and the distance between the plane of the slits and the screen is 1 m. The minimum distance between two successive regions of complete darkness is :- [IIT-JEE' 2004 (Scr)]

(A) 4 mm (B) 5.6 mm (C) 14 mm (D) 28 mm

WO0042

18. In Young's double slit arrangement, water is filled in the space between screen and slits. Then :

(A) fringe pattern shifts upwards but fringe width remains unchanged.  
 (B) fringe width decreases and central bright fringe shifts upwards.  
 (C) fringe width increases and central bright fringe does not shift.  
 (D) fringe width decreases and central bright fringe does not shift.

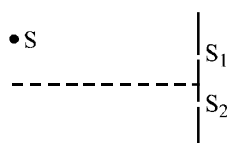
WO0043

19. Light of wavelength  $\lambda$  in air enters a medium of refractive index  $\mu$ . Two points in this medium, lying along the path of this light, are at a distance  $x$  apart. The phase difference between these points is :

(A)  $\frac{2\pi\mu x}{\lambda}$  (B)  $\frac{2\pi x}{\mu\lambda}$  (C)  $\frac{2\pi(\mu-1)x}{\lambda}$  (D)  $\frac{2\pi x}{(\mu-1)\lambda}$

WO0044

20. In YDSE, the source placed symmetrically with respect to the slit is now moved parallel to the plane of the slits so that it is closer to the upper slit, as shown. Then,



(A) the fringe width will increase and fringe pattern will shift down.  
 (B) the fringe width will remain same but fringe pattern will shift up.  
 (C) the fringe width will decrease and fringe pattern will shift down.  
 (D) the fringe width will remain same but fringe pattern will shift down.

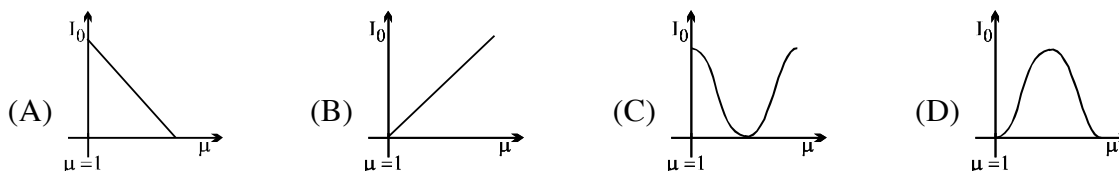
WO0045

21. In the ideal double-slit experiment, when a glass-plate (refractive index 1.5) of thickness  $t$  is introduced in the path of one of the interfering beams (wavelength  $\lambda$ ), the intensity at the position where the central maximum occurred previously remains unchanged. The minimum thickness of the glass-plate is :- [IIT-JEE 2002]

(A)  $2\lambda$  (B)  $\frac{2\lambda}{3}$  (C)  $\frac{\lambda}{3}$  (D)  $\lambda$

WO0046

22. In a YDSE experiment if a slab whose refractive index can be varied is placed in front of one of the slits then the variation of resultant intensity at mid-point of screen with  $\mu$  ( $\mu \geq 1$ ) will be best represented by [Assume slits of equal width and there is no absorption by slab]



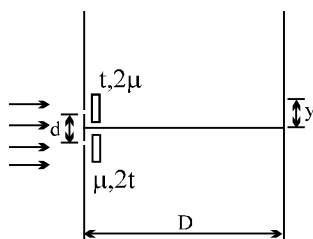
WO0047

23. In a Young's double-slit experiment, let A and B be the two slits. Films of thicknesses  $t_A$  and  $t_B$  and refractive indices  $\mu_A$  and  $\mu_B$ , are placed in front of A and B respectively. If  $\mu_A t_A = \mu_B t_B$ , the central maximum will :

- (A) not shift (B) shift towards A  
(C) shift towards B (D) option (B), if  $t_B > t_A$ ; option (C) if  $t_B < t_A$

WO0048

24. In the YDSE shown the two slits are covered with thin sheets having thickness  $t$  &  $2t$  and refractive index  $2\mu$  and  $\mu$ . Find the position ( $y$ ) of central maxima



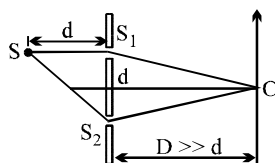
- (A) zero (B)  $\frac{tD}{d}$  (C)  $-\frac{tD}{d}$  (D) None

WO0049

25. In a double slit experiment, when the width of one slit is made twice as wide as the other in compared to normal YDSE having slits of equal width. Then, in the interference pattern [IIT-JEE(Scr.) 2000]
- (A) the intensities of both the maxima and the minima increase.  
(B) the intensity of the maxima increases and the minima has zero intensity.  
(C) the intensity of the maxima decreases and that of the minima increases.  
(D) the intensity of the maxima decreases and the minima has zero intensity.

WO0050

26. To make the central fringe at the centre  $O$ , a mica sheet of refractive index 1.5 is introduced. Choose the correct statements (s).



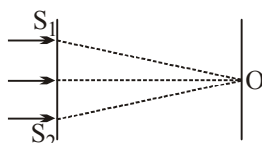
- (A) The thickness of sheet is  $2(\sqrt{2} - 1)d$  in front of  $S_1$ .  
 (B) The thickness of sheet is  $(\sqrt{2} - 1)d$  in front of  $S_2$ .  
 (C) The thickness of sheet is  $2\sqrt{2}d$  in front of  $S_1$ .  
 (D) The thickness of sheet is  $(2\sqrt{2} - 1)d$  in front of  $S_1$ .

WO0051

27. **Statement-1 :** In YDSE, as shown in figure, central bright fringe is formed at  $O$ . If a liquid is filled between plane of slits and screen, the central bright fringe is shifted in upward direction.

and

**Statement-2 :** If path difference at  $O$  increases, y-coordinate of central bright fringe will change.



- (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.  
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.  
 (C) Statement-1 is true, statement-2 is false.  
 (D) Statement-1 is false, statement-2 is true.

WO0052

28. **Statement-1 :** In glass, red light travels faster than blue light.

and

**Statement-2 :** Red light has a wavelength longer than blue.

- (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.  
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.  
 (C) Statement-1 is true, statement-2 is false.  
 (D) Statement-1 is false, statement-2 is true.

WO0053

29. **Statement-1** : In standard YDSE set up with visible light, the position on screen where phase difference is zero appears bright.

and

**Statement-2** : In YDSE set up magnitude of electromagnetic field at central bright fringe is not varying with time.

- (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.  
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.  
(C) Statement-1 is true, statement-2 is false.  
(D) Statement-1 is false, statement-2 is true.

WO0054

30. **Statement-1** : In YDSE, the spacing between any two successive points having intensity half of the maximum intensity is same.

and

**Statement-2** : The intensity on the screen in YDSE varies uniformly with distance from central maximum.

- (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.  
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.  
(C) Statement-1 is true, statement-2 is false.  
(D) Statement-1 is false, statement-2 is true.

WO0055

31. In a double slit experiment, the separation between the slits is  $d = 0.25$  cm and the distance of the screen  $D = 100$  cm from the slits. If the wavelength of light used is  $\lambda = 6000 \text{ \AA}$  and  $I_0$  is the intensity of the central bright fringe, the intensity at a distance  $x = 4 \times 10^{-5}$  m from the central maxima is :-

- (A)  $I_0$                       (B)  $I_0/2$                       (C)  $3I_0/4$                       (D)  $I_0/3$

WO0056

32. Imagine a Young's double slit interference experiment performed with waves associated with fast moving electrons produced from an electron gun. The distance between successive maxima will decrease the most if :-

- (A) the accelerating voltage in the electron gun is decreased  
(B) the accelerating voltage is increased and the distance of the screen from the slits is decreased  
(C) the distance of the screen from the slits is increased.  
(D) the distance between the slits is decreased.

WO0057



33. Two monochromatic and coherent point sources of light are placed at a certain distance from each other in the horizontal plane. The locus of all those points in the horizontal plane which have constructive interference will be
- (A) a hyperbola (B) family of hyperbolas  
(C) family of straight lines (D) family of parabolas

WO0058

### MULTIPLE CORRECT TYPE QUESTIONS

34. To observe a sustained interference pattern formed by two light waves, it is not necessary that they must have :
- (A) the same frequency (B) same amplitude  
(C) a constant phase difference (D) the same intensity

WO0059

35. As a wave propagates, [IIT-JEE 1999]
- (A) the wave intensity remains constant for a plane wave  
(B) the wave intensity decreases with distance from source and is proportional to inverse of the distance, for a spherical wave.  
(C) the wave intensity decreases with distance from source and proportional to inverse square of the distance, for a spherical wave.  
(D) total average power of the spherical wave over any spherical surface centered at the source remains same.

WO0060

36. In a YDSE apparatus, if we use white light then :
- (A) the fringe next to the central will be red (B) the central fringe will be white.  
(C) the fringe next to the central will be violet (D) there will not be a completely dark fringe.

WO0061

37. If the source of light used in a Young's Double Slit Experiment is changed from red to blue, then
- (A) the fringes will become brighter  
(B) consecutive fringes will come closer  
(C) the number of maxima formed on the screen increases  
(D) the central bright fringe will become a dark fringe.

WO0062

38. In a Young's double-slit experiment, let  $A$  and  $B$  be the two slits. A thin film of thickness  $t$  and refractive index  $\mu$  is placed in front of  $A$ . Let  $\beta$  = fringe width. The central maximum will shift :
- (A) towards  $A$  (B) towards  $B$  (C) by  $t(\mu - 1) \frac{\beta}{\lambda}$  (D) by  $\mu t \frac{\beta}{\lambda}$

WO0063

39. If one of the slits of a standard YDSE apparatus is covered by a thin parallel sided glass slab so that it transmit only one half of the light intensity of the other, then :
- (A) the fringe pattern will get shifted towards the covered slit.  
 (B) the fringe pattern will get shifted away from the covered slit.  
 (C) the bright fringes will be less bright and the dark ones will be more bright.  
 (D) the fringe width will remain unchanged.

WO0064

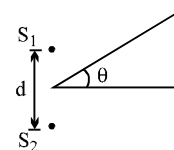
40. In an interference arrangement similar to Young's double-slit experiment, the slits  $S_1$  &  $S_2$  are illuminated with coherent microwave sources, each of frequency  $10^6$  Hz. The sources are synchronized to have zero phase difference. The slits are separated by a distance  $d = 150.0$  m. The intensity  $I(\theta)$  is measured as a function of  $\theta$  at a large distance from  $S_1$  &  $S_2$ , where  $\theta$  is defined as shown. If  $I_0$  is the maximum intensity then  $I(\theta)$  for  $0 \leq \theta \leq 90^\circ$  is given by :

(A)  $I(\theta) = \frac{I_0}{2}$  for  $\theta = 30^\circ$

(B)  $I(\theta) = \frac{I_0}{4}$  for  $\theta = 90^\circ$

(C)  $I(\theta) = I_0$  for  $\theta = 0^\circ$

(D)  $I(\theta)$  is constant for all values of  $\theta$ .



WO0065

41. In a standard YDSE apparatus, a thin film ( $\mu = 1.5$ ,  $t = 2.1 \mu\text{m}$ ) is placed in front of upper slit. How far above or below the centre point of the screen are two nearest maxima located? Take  $D = 1$  m,  $d = 1\text{mm}$ ,  $\lambda = 4500\text{\AA}$ . (Symbols have usual meaning)
- (A) 1.5 mm                      (B) 0.6 mm                      (C) 0.15 mm                      (D) 0.3 mm

WO0066

42. In a YDSE with two identical slits, when the upper slit is covered with a thin, perfectly transparent sheet of mica, the intensity at the centre of screen reduces to 75% of the initial value. Second minima is observed to be above this point and third maxima below it. Which of the following can be a possible value of phase difference caused by the mica sheet

(A)  $\frac{\pi}{3}$

(B)  $\frac{13\pi}{3}$

(C)  $\frac{17\pi}{3}$

(D)  $\frac{11\pi}{3}$

WO0067

43. A beam of 2000 eV electrons are incident normally on the surface of crystal whose inter atomic separation is 0.11 nm. The mass of the electron can be taken as  $9 \times 10^{-31}$  kg. At what angle to the normal can we observe a diffraction maxima.

(A)  $\sin^{-1}\left(\frac{1}{4}\right)$

(B)  $2 \cos^{-1}\left(\frac{1}{4}\right)$

(C)  $\sin^{-1}\left(\frac{1}{2}\right)$

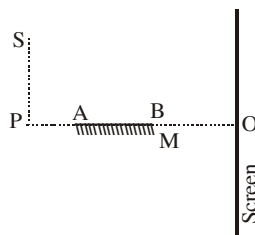
(D)  $2 \cos^{-1}\left(\frac{1}{2}\right)$

WO0068

# COMPREHENSION TYPE QUESTIONS

## Paragraph for Question No. 44 and 45

In an experiment on interference due to single mirror, a light wave emitted directly by the source  $S$  (narrow slit) interferes with the wave reflected from the mirror  $M$  of length 2 mm. Source and screen are separated by distance 90 cm. Source  $S$  is at the height of 3 mm, from the point  $P$  and the middle point of mirror is at distance of 2 mm from point  $P$ . Point  $P$  and mirror are in the same plane of screen is perpendicular to this plane.



44. If fringe width is 0.1 mm then what is the wavelength of light used?  
 (A)  $3.3 \times 10^{-7}$  m      (B)  $6.7 \times 10^{-7}$  m      (C)  $1.0 \times 10^{-7}$  m      (D)  $4 \times 10^{-7}$  m

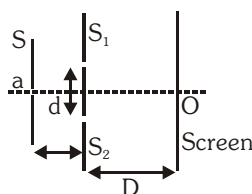
WO0069

45. If the mirror is shifted towards left then how does the fringe pattern on screen change?  
 (A) Fringe width decreases and the region in which interference is formed shifts downward.  
 (B) Fringe width decreases and region in which interference is formed shifts upwards  
 (C) Fringe width does not change and region in which interference is formed shifts upwards  
 (D) Fringe width does not change and region in which interference is formed shifts downwards

WO0069

## Paragraph for Question No. 46 to 48

The figure shows a schematic diagram showing the arrangement of Young's Double Slit Experiment



46. Choose the correct statement(s) related to the wavelength of light used  
 (A) Larger the wavelength of light larger the fringe width  
 (B) The position of central maxima depends on the wavelength of light used  
 (C) If white light is used in YDSE, then the violet colour forms its first maxima closest to the central maxima  
 (D) The central maxima of all the wavelengths coincide

WO0070

47. If the distance  $D$  is varied, then choose the correct statement(s)

- (A) The angular fringe width does not change  
 (B) The fringe width changes in direct proportion  
 (C) The change in fringe width is same for all wavelengths  
 (D) The position of central maxima remains unchanged

WO0070

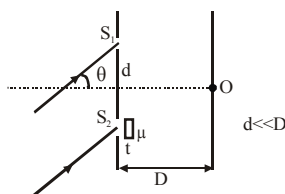
48. If the distance  $d$  is varied, then identify the correct statement(s)

- (A) The angular width does not change  
 (B) The fringe width changes in inverse proportion  
 (C) The positions of all maxima change  
 (D) The positions of all minima change

WO0070

### Paragraph for Question No. 49 to 51

A monochromatic beam of light falls on Young's double slit experiment apparatus as shown in figure. A thin sheet of glass is inserted in front of lower slit  $S_2$ .



49. The central bright fringe can be obtained

- (A) at  $O$  only  
 (B) at  $O$  or below  $O$  only  
 (C) at  $O$  or above  $O$  only  
 (D) Anywhere on the screen

WO0071

50. If central bright fringe is obtained on screen at  $O$  :-

- (A)  $(\mu-1)t = d\sin\theta$     (B)  $(\mu-1)t = d\cos\theta$     (C)  $(\mu-1)t + d\sin\theta = 0$     (D)  $\frac{t}{\mu-1} = \frac{d}{\sin\theta}$

WO0071

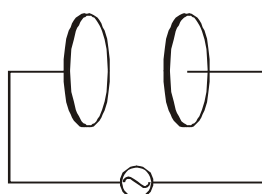
51. The phase difference between the waves interfering at fifth minima is

- (A)  $5\pi$     (B)  $7\pi$     (C)  $9\pi$     (D)  $11\pi$

WO0071

# SUPPLEMENT FOR JEE-MAINS

52. A parallel plate capacitor (fig.) made of circular plates each of radius  $R = 6.0$  cm has a capacitance  $C = 100 \mu\text{F}$ . The capacitor is connected to a 230 V ac supply with a (angular) frequency of  $300 \text{ rad s}^{-1}$ .
- What is the rms value of the conduction current?
  - Is the conduction current equal to the displacement current?
  - Determine the amplitude of  $B$  at a point 3.0 cm from the axis between the plates.



WO0072

53. The amplitude of the magnetic field part of a harmonic electromagnetic wave in vacuum is  $B_0 = 510 \text{ nT}$ . What is the amplitude of the electric field part of the wave?

WO0073

54. In a plane electromagnetic wave, the electric field oscillates sinusoidally at a frequency of  $2.0 \times 10^{10} \text{ Hz}$  and amplitude  $48 \text{ V m}^{-1}$ .
- What is the wavelength of the wave?
  - What is the amplitude of the oscillating magnetic field?
  - Show that the average energy density of the  $E$  field equals the average energy density of the  $B$  field. [ $c = 3 \times 10^8 \text{ m s}^{-1}$ ]

WO0074

55. Suppose that the electric field part of an electromagnetic wave in vacuum is

$$E = \{(3.1 \text{ N/C}) \cos [(1.8 \text{ rad/m}) y + (5.4 \times 10^8 \text{ rad/s}) t]\} \hat{i}$$

- What is the direction of propagation?
- What is the wavelength  $\lambda$ ?
- What is the frequency  $\nu$ ?
- What is the amplitude of the magnetic field part of the wave?
- Write an expression for the magnetic field part of the wave.

WO0075

56. A plane electromagnetic wave traveling in the positive direction of  $x$  axis in vacuum has components  $E_x = E_y = 0$  and  $E_z = (2.0 \text{ V/m}) \cos [(\pi \times 10^{15} \text{ S}^{-1})(t - x/c)]$ . (a) What is the amplitude of the magnetic field component? (b) Parallel to which axis does the magnetic field oscillate? (c) When the electric field component is in the positive direction of the  $z$  axis at a certain point  $P$ , what is the direction of the magnetic field component there?

WO0076

57. An open circuit consists of a  $12\ \mu\text{F}$  parallel plate capacitor charged to  $200\text{V}$  and a  $10\ \Omega$  resistor. At the instant when a switch closes the circuit (with no battery in it) the displacement current between the plates of the capacitor is

WO0077

58. Two coherent waves are described by the following expressions.

$$E_1 = E_0 \sin \left( \frac{2\pi x_1}{\lambda} - 2\pi ft + \frac{\pi}{6} \right); E_2 = E_0 \sin \left( \frac{2\pi x_2}{\lambda} - 2\pi ft + \frac{\pi}{8} \right)$$

Determine the relationship between  $x_1$  and  $x_2$  that produces constructive interference when the two waves are superposed.

WO0078

59. A nickel crystal is used as a diffraction grating for x-rays. Then the same crystal is used to diffract electrons. If the two diffraction patterns are identical, and the energy of each x-ray photon is  $E = 20.0\ \text{keV}$ . What is the kinetic energy of each electron?

WO0079

60. The velocity of electromagnetic waves in a dielectric medium ( $\epsilon_r = 4$ ) is :-  
 (A)  $3 \times 10^8$  metre/second (B)  $1.5 \times 10^8$  metre/second  
 (C)  $6 \times 10^8$  metre/second (D)  $7.5 \times 10^7$  metre/second

WO0080

61. Which of these statements correctly describes the orientation of the electric field ( $\vec{E}$ ), the magnetic field ( $\vec{B}$ ), and the velocity of propagation ( $\vec{v}$ ) of an electromagnetic wave?

- (A)  $\vec{E}$  is perpendicular to  $\vec{B}$ ;  $\vec{v}$  may have any orientation relative to  $\vec{E}$ .  
 (B)  $\vec{E}$  is perpendicular to  $\vec{B}$ ;  $\vec{v}$  may have any orientation perpendicular to  $\vec{E}$   
 (C)  $\vec{E}$  is parallel to  $\vec{B}$ ;  $\vec{v}$  is perpendicular to both  $\vec{B}$  and  $\vec{E}$ .  
 (D) Each of the three vectors is perpendicular to the other two.

WO0081

62. The amplitude of electric field in a parallel light beam of intensity  $4\text{Wm}^{-2}$  is :  
 (A)  $35.5\ \text{NC}^{-1}$  (B)  $45.5\ \text{NC}^{-1}$  (C)  $49.5\ \text{NC}^{-1}$  (D)  $54.8\ \text{NC}^{-1}$

WO0082

63. Instantaneous displacement current of  $1.0\text{A}$  in the space between the parallel plates of  $1\ \mu\text{F}$  capacitor can be established by changing potential difference of:  
 (A)  $10^{-6}\ \text{V/s}$  (B)  $10^6\ \text{V/s}$  (C)  $10^{-8}\ \text{V/s}$  (D)  $10^8\ \text{V/s}$

WO0083

64. A plane electromagnetic wave,  
 $E_z = 100 \cos (6 \times 10^8 t + 4x)\ \text{V/m}$   
 propagates in a medium of dielectric constant:  
 (A) 1.5 (B) 2.0 (C) 2.4 (D) 4.0

WO0084

65. A large parallel plane capacitor, whose plates have an area of  $1 \text{ m}^2$  and are separated from each other by  $1 \text{ mm}$ , is being charged at a rate of  $25.8 \text{ V/s}$ . If the dielectric between the plates has the dielectric constant 10, then the displacement current at this instant is :-  
 (A)  $251 \mu\text{A}$  (B)  $11 \mu\text{A}$  (C)  $2.2 \mu\text{A}$  (D)  $1.1 \mu\text{A}$

WO0085

66. The rms value of the electric field of the light coming from the sun is  $720 \text{ N/C}$ . The average total energy density of the electromagnetic wave is :  
 (A)  $4.58 \times 10^{-6} \text{ J/m}^3$  (B)  $6.37 \times 10^{-9} \text{ J/m}^3$  (C)  $81.35 \times 10^{-12} \text{ J/m}^3$  (D)  $3.3 \times 10^{-3} \text{ J/m}^3$

WO0086

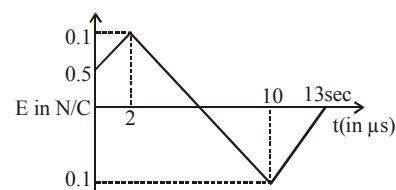
67. A beam of light travelling along x-axis is described by the electric field,  $E_y = (600 \text{ Vm}^{-1}) \sin \omega(t - x/c)$  then maximum magnetic force on a charge  $q = 2e$ , moving along y-axis with a speed of  $3.0 \times 10^7 \text{ ms}^{-1}$  is ( $e = 1.6 \times 10^{-19} \text{ C}$ ) :  
 (A)  $19.2 \times 10^{-17} \text{ N}$  (B)  $1.92 \times 10^{-17} \text{ N}$  (C)  $0.192 \text{ N}$  (D) None of these

WO0087

68. A plane electromagnetic wave travels in free space along x-axis. At a particular point in space, the electric field along y-axis is  $9.3 \text{ Vm}^{-1}$ . The magnetic induction is:  
 (A)  $3.1 \times 10^{-8} \text{ T}$  (B)  $3 \times 10^{-5} \text{ T}$  (C)  $3.1 \times 10^{-6} \text{ T}$  (D)  $9.3 \times 10^{-6} \text{ T}$

WO0088

69. The electric field through an area of  $2 \text{ m}^2$  varies with time as shown in the graph. The greatest displacement current through the area is at :-  
 (A)  $t = 1 \text{ sec}$ .  
 (B)  $t = 4 \text{ sec}$ .  
 (C)  $t = 8 \text{ sec}$ .  
 (D)  $t = 12 \text{ sec}$ .



WO0089

70. A plane electromagnetic wave travelling along the X-direction has a wavelength of  $3 \text{ mm}$ . The variation in the electric field occurs in the Y-direction with an amplitude  $66 \text{ Vm}^{-1}$ . The equation for the electric and magnetic fields as a function of  $x$  and  $t$  are respectively.

- (A)  $E_y = 33 \cos \pi \times 10^{11} \left( t - \frac{x}{c} \right)$ ;  $B_z = 1.1 \times 10^{-7} \cos \pi \times 10^{11} \left( t - \frac{x}{c} \right)$   
 (B)  $E_y = 11 \cos 2\pi \times 10^{11} \left( t - \frac{x}{c} \right)$ ;  $B_y = 1.1 \times 10^{-7} \cos 2\pi \times 10^{11} \left( t - \frac{x}{c} \right)$   
 (C)  $E_x = 33 \cos \pi \times 10^{11} \left( t - \frac{x}{c} \right)$ ;  $B_x = 1.1 \times 10^{-7} \cos \pi \times 10^{11} \left( t - \frac{x}{c} \right)$   
 (D)  $E_y = 66 \cos 2\pi \times 10^{11} \left( t - \frac{x}{c} \right)$ ;  $B_z = 2.2 \times 10^{-7} \cos 2\pi \times 10^{11} \left( t - \frac{x}{c} \right)$

WO0090

71. In a stack of three polarizing sheets the first and third are crossed while the middle one has its axis at  $45^\circ$  to the axes of the other two. The fraction of the intensity of an incident unpolarized beam of light that is transmitted by the stack is:

(A)  $1/2$  (B)  $1/3$  (C)  $1/4$  (D)  $1/8$

WO0091

72. A beam of light strikes a piece of glass at an angle of incidence of  $60^\circ$  and the reflected beam is completely plane polarised. The refractive index of the glass is :-

(A) 1.5 (B)  $\sqrt{3}$  (C)  $\sqrt{2}$  (D)  $(3/2)$

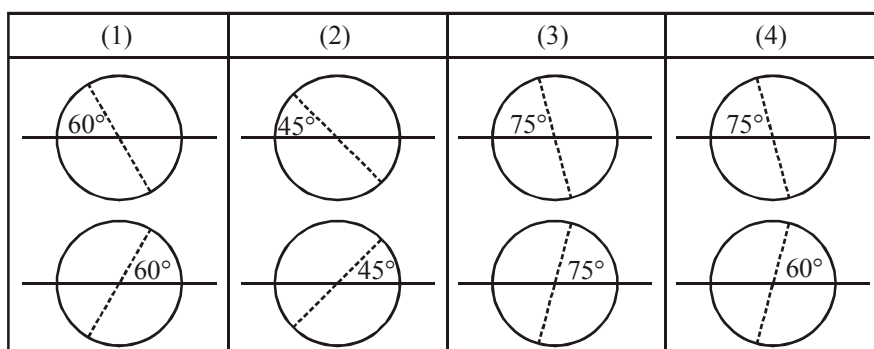
WO0092

73. The Sun is directly overhead and you are facing toward the north. Light coming to your eyes from the sky just above the horizon is :-

(A) Partially polarized north-south (B) Partially polarized east, west  
(C) Partially polarized up-down (D) Randomly polarized

WO0093

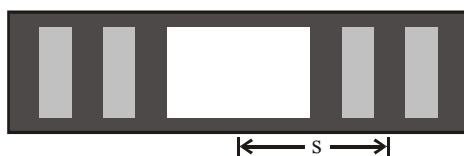
74. The diagrams show four pairs of polarizing sheets, with the polarizing directions indicated by dashed lines. The two 'sheets of each pair are placed one behind the other and the front sheet is illuminated by unpolarized light. The incident intensity is the same for all pairs of sheets. Rank the pairs according to the intensity of the transmitted light, least to greatest.



(A) 1, 2, 3, 4 (B) 4, 2, 1, 3 (C) 2, 4, 3, 1 (D) 2, 1, 4, 3

WO0094

75. Light of wavelength 600 nm is incident upon a single slit with width  $4 \times 10^{-4}$  m. The figure shows the pattern observed on a screen positioned 2 m from the slits. Determine the distance  $s$ .



(A) 0.002 m (B) 0.003 m (C) 0.004 m (D) 0.006 m

WO0095



76. The image of a star (effectively a point source) is made by a convergent lens of focal length 1 m and diameter of aperture 5.0 cm. If the lens is ideal and the effective wavelength in image formation is taken as  $5 \times 10^{-5}$  cm, the diameter of the image formed will be nearest to:  
 (A) zero (B)  $10^{-6}$  cm (C)  $10^{-5}$  cm (D)  $10^{-3}$  cm

WO0096

77. In a single slit diffraction pattern, as the width of the slit is increased,  
 (A) the peak intensity of central maxima increases and its width also increases  
 (B) the peak intensity of central maxima increases and its width decreases.  
 (C) the peak intensity of central maxima decreases and its width increases  
 (D) the peak intensity of central maxima decreases and its width also decreases.

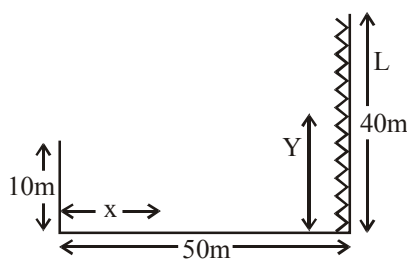
WO0097

78. A beam of electrons with de-broglie wavelength of  $10^{-4}$  m pass through a slit  $10^{-3}$  m wide. Calculate the angular spread introduced because of diffraction by slit.

- (A)  $\frac{9^\circ}{\pi}$  (B)  $\frac{18^\circ}{\pi}$  (C)  $\frac{36^\circ}{\pi}$  (D)  $\frac{4.5^\circ}{\pi}$

WO0098

79. A person lives in a high-rise building on the bank of a river 50 m wide. Across the river is a well lit tower of height 40 m. When the person, who is at a height of 10 m, looks through a polarizer at an appropriate angle at light of the tower reflecting from the river surface, he notes that intensity of light coming from distance X from his building is the least and this corresponds to the light coming from light bulbs at height 'Y' on the tower. The values of X and Y are respectively close to (refractive index of water  $\approx 4/3$ )



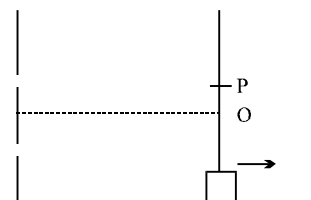
- (A) 13 m, 27 m (B) 22 m, 13 m (C) 25 m, 10 m (D) 17 m, 29 m

WO0099

## EXERCISE (O-2)

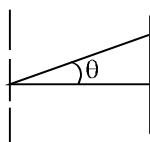
## SINGLE CORRECT TYPE QUESTIONS

1. In a Young's Double slit experiment, first maxima is observed at a fixed point  $P$  on the screen. Now the screen is continuously moved away from the plane of slits. The ratio of intensity at point  $P$  to the intensity at point  $O$  (centre of the screen)
- (A) remains constant  
(B) keeps on decreasing  
(C) first decreases and then increases  
(D) First decreases and then becomes constant



WO0100

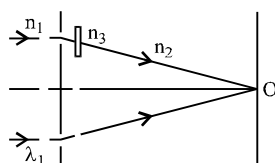
2. Two slits are separated by 0.3 mm. A beam of 500 nm light strikes the slits producing an interference pattern. The number of maxima observed in the angular range  $-30^\circ < \theta < 30^\circ$ .



- (A) 300 (B) 150 (C) 599 (D) 601

WO0101

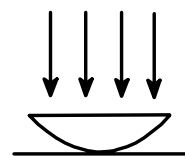
3. In the figure shown in YDSE, a parallel beam of light is incident on the slit from a medium of refractive index  $n_1$ . The wavelength of light in this medium is  $\lambda_1$ . A transparent slab of thickness  $t$  and refractive index  $n_3$  is put in front of one slit. The medium between the screen and the plane of the slits is  $n_2$ . The phase difference between the light waves reaching point ' $O$ ' (symmetrical, relative to the slits) is :



- (A)  $\frac{2\pi}{n_1\lambda_1} (n_3 - n_2) t$  (B)  $\frac{2\pi}{\lambda_1} (n_3 - n_2) t$  (C)  $\frac{2\pi n_1}{n_2 \lambda_1} \left( \frac{n_3}{n_2} - 1 \right) t$  (D)  $\frac{2\pi n_1}{\lambda_1} (n_3 - n_1) t$

WO0102

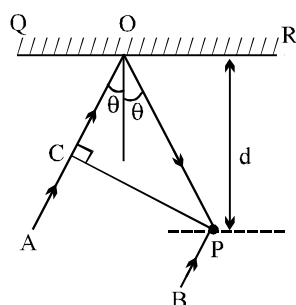
4. A thin slice is cut out of a glass cylinder along a plane parallel to its axis. The slice is placed on a flat glass plate as shown. The observed interference fringes from this combination shall be [IIT-JEE '99]
- (A) straight  
(B) circular  
(C) equally spaced  
(D) having fringe spacing which increases as we go outwards.



WO0103

5. In the adjacent diagram,  $CP$  represents a wavefront and  $AO$  and  $BP$ , the corresponding two rays. Find the condition on  $\theta$  for constructive interference at  $P$  between the ray  $BP$  and reflected ray  $OP$ .

[JEE (Scr.) 2003]



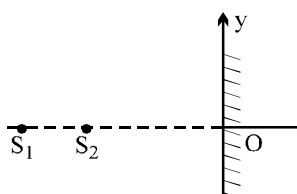
- (A)  $\cos\theta = \frac{3\lambda}{2d}$  (B)  $\cos\theta = \frac{\lambda}{4d}$  (C)  $\sec\theta - \cos\theta = \frac{\lambda}{d}$  (D)  $\sec\theta - \cos\theta = \frac{4\lambda}{d}$

WO0104

6. Two point monochromatic and coherent sources of light of wavelength  $\lambda$  are placed on the dotted line in front of a large screen. The sources emit waves in phase with each other. The distance between  $S_1$  and  $S_2$  is  $d$  while their distance from the screen is much larger.

- (1) If  $d = 7\lambda/2$ ,  $O$  will be a minima  
(2) If  $d = 4.3\lambda$ , there will be a total of 8 minima on  $y$ -axis.  
(3) If  $d = 7\lambda$ ,  $O$  will be a maxima.  
(4) If  $d = \lambda$ , there will be only one maxima on the screen.

Which is the set of correct statement :



- (A) 1, 2 & 3 (B) 2, 3 & 4 (C) 1, 2, 3 & 4 (D) 1, 3 & 4

WO0105

7. A parallel coherent beam of light falls on Fresnel biprism of refractive index  $\mu$  and angle  $\alpha$ . The fringe width on a screen at a distance  $D$  from biprism will be (wavelength =  $\lambda$ )

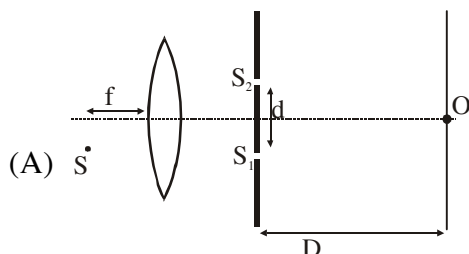
- (A)  $\frac{\lambda}{2(\mu-1)\alpha}$  (B)  $\frac{\lambda D}{2(\mu-1)\alpha}$  (C)  $\frac{D}{2(\mu-1)\alpha}$  (D) none

WO0106

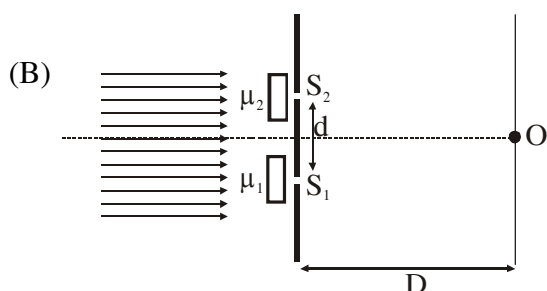
## MATRIX MATCH TYPE QUESTIONS

8. Column-I shows some modifications in a standard YDSE setup. Column-II shows the associated characteristics.

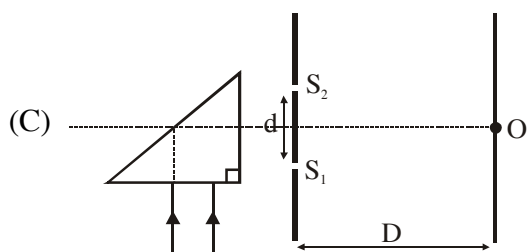
## Column-I



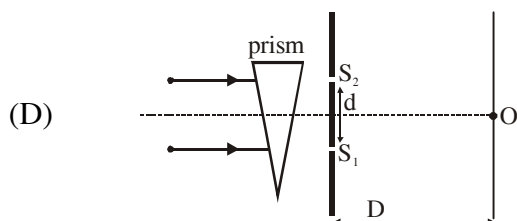
monochromatic point source  
S placed in focal plane.



monochromatic parallel beam incident  
on  $S_1S_2$  through transparent  
slabs of same thickness but  $\mu_1 > \mu_2$



monochromatic parallel beam  
incident on a right angled isosceles  
prism of refractive index 1.50



monochromatic parallel beam  
incident on thin prism

## Column-II

(P) Zero order maxima lies above O.

(Q) If a transparent mica sheet is introduced  
infront of  $S_2$  central bright fringe can be  
obtained at O.

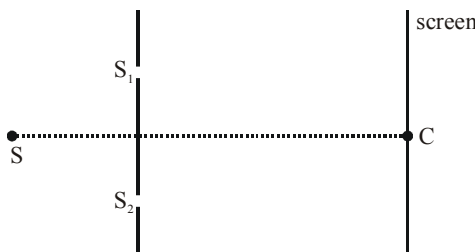
(R) Fringe width  $\beta = \frac{\lambda D}{d}$

(S) Point O can be a minima

(T) Point O can be a least order minima.

WO0107

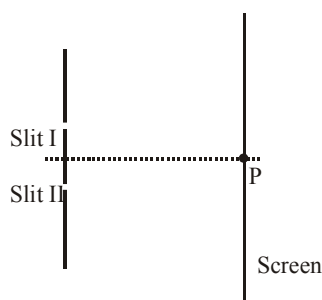
9. In a YDSE setup, light of wavelength  $4000 \text{ \AA}$  is used. Distance of screen from the slits is  $2\text{m}$  and distance between slits is  $1\text{mm}$ . There are three slabs slab 1 (thickness  $2\text{mm}$ ,  $\mu = 2$ ), slab 2 (thickness  $1\text{mm}$ ,  $\mu = 3$ ), slab 3 (thickness  $4\text{mm}$   $\mu = \frac{3}{2}$ )



- (A) If slab 1 is placed in front of slit  $S_1$  (P) Central maxima at C  
 (B) If the slab 2 is placed in front of slit  $S_2$  along with condition (A) (Q) Central maxima above C  
 (C) If slab 3 is placed in front of slab 1 along with condition (B) (R) Fringe width is equal to  $0.8 \text{ mm}$   
 (S) No. of fringes crossing central maxima as a result of slab placing is 5000

WO0108

10. A double slit interference pattern is produced on a screen, as shown in the figure, using monochromatic light of wavelength  $500\text{nm}$ . Point P is the location of the central bright fringe, that is produced when light waves arrive in phase without any path difference. A choice of three strips A, B and C of transparent materials with different thicknesses and refractive indices is available, as shown in the table. These are placed over one or both of the slits, singularly or in conjunction, causing the interference pattern to be shifted across the screen from the original pattern. In the column-I, how the strips have been placed, is mentioned whereas in the column-II, order of the fringe at point P on the screen that will be produced due to the placement of the strips(s), is shown. Correctly match both the column.



| Film                          | A   | B   | C    |
|-------------------------------|-----|-----|------|
| Thickness (in $\mu\text{m}$ ) | 5   | 1.5 | 0.25 |
| Refractive index              | 1.5 | 2.5 | 2    |

**Column I**

- (A) Only strip B is placed over slit-I  
 (B) Strip A is placed over slit-I and strip C is placed over slit-II  
 (C) Strip A is placed over the slit-I and strip B and strip C are placed over the slit-II in conjunction  
 (D) Strip A and strip C are placed over slit-I (in conjunction) and strip B is placed over Slit-II

**Column II**

- (P) First Bright  
 (Q) Fourth Dark  
 (R) Fifth Dark  
 (S) Central Bright

WO0109

## EXERCISE-(J-M)

1. A mixture of light, consisting of wavelength 590 nm and an unknown wavelength, illuminates Young's double slit and gives rise to two overlapping interference patterns on the screen. The central maximum of both lights coincide. Further, it is observed that the third bright fringe of known light coincides with the 4th bright fringe of the unknown light. From this data, the wavelength of the unknown light is :- [AIEEE-2009]

(1) 442.5 nm                      (2) 776.8 nm                      (3) 393.4 nm                      (4) 885.0 nm

WO0110

2. If a source of power 4kW produces  $10^{20}$  photons/second, the radiation belongs to a part of the spectrum called [AIEEE 2010]

(1) microwaves                      (2) y-rays                      (3) X-rays                      (4) ultraviolet rays

WO0111

**Direction : Questions number 3 to 5 are based on the following paragraph.**

An initially parallel cylindrical beam travels in a medium of refractive index  $\mu(I) = \mu_0 + \mu_2 I$ , where  $\mu_0$  and  $\mu_2$  are positive constants and  $I$  is the intensity of the light beam. The intensity of the beam is decreasing with increasing radius.

3. The initial shape of the wavefront of the beam is :- [AIEEE-2010]

(1) planar  
(2) convex  
(3) concave  
(4) convex near the axis and concave near the periphery

WO0112

4. The speed of the light in the medium is :- [AIEEE-2010]

(1) maximum on the axis of the beam                      (2) minimum on the axis of the beam  
(3) the same everywhere in the beam                      (4) directly proportional to the intensity  $I$

WO0112

5. As the beam enters the medium, it will :- [AIEEE-2010]

(1) travel as a cylindrical beam  
(2) diverge  
(3) converge  
(4) diverge near the axis and converge near the periphery

WO0112

6. At two points P and Q on screen in Young's double slit experiment, waves from slits  $S_1$  and  $S_2$  have a path difference of 0 and  $\frac{\lambda}{4}$  respectively. the ratio of intensities at P and Q will be : [AIEEE-2011]

(1) 3 : 2                      (2) 2 : 1                      (3)  $\sqrt{2} : 1$                       (4) 4 : 1

WO0113

7. **Statement-1:** On viewing the clear blue portion of the sky through a Calcite Crystal, the intensity of transmitted light varies as the crystal is rotated.

**Statement-1:** The light coming from the sky is polarized due to scattering of sun light by particles in the atmosphere. The scattering is largest for blue light. [AIEEE-2011]

- (1) Statement-1 is false, statement-2 is true  
 (2) Statement-1 is true, statement-2 is false  
 (3) Statement-1 is true, statement-2 true; statement-2 is the correct explanation of statement-1  
 (4) Statement-1 is true, statement-2 is true; statement -2 is not correct explanation of statement-1.

WO0114

8. In a Young's double slit experiment, the two slits act as coherent sources of waves of equal amplitude  $A$  and wavelength  $\lambda$ . In another experiment with the same arrangement the two slits are made to act as incoherent sources of waves of same amplitude and wavelength. If the intensity at the middle point

of the screen in the first case is  $I_1$  and in the second case  $I_2$ , then the ratio  $\frac{I_1}{I_2}$  is :- [AIEEE-2011]

- (1) 4 (2) 2 (3) 1 (4) 0.5

WO0115

9. Direction :

The question has a paragraph followed by two statement, Statement-1 and statement-2. Of the given four alternatives after the statements, choose the one that describes the statements.

A thin air film is formed by putting the convex surface of a plane-convex lens over a plane glass plate. With monochromatic light, this film gives an interference pattern due to light reflected from the top (convex) surface and the bottom (glass plate) surface of the film [AIEEE-2011]

**Statement-1:**

When light reflects from the air-glass plate interface, the reflected wave suffers a phase change of  $\pi$ .

**Statement-2:** The centre of the interference pattern is dark :-

- (1) Statement-1 is true, Statement-2 is true and Statement-2 is not the correct explanation of Statement-1.  
 (2) Statement-1 is false, Statement-2 is true  
 (3) Statement-1 is true, Statement-2 is false  
 (4) Statement-1 is true, Statement-2 is true and Statement-2 is the correct explanation of statement-1.

WO0116

10. In Young's double slit experiment, one of the slit is wider than other, so that the amplitude of the light from one slit is double of that from other slit. If  $I_m$  be the maximum intensity, the resultant intensity  $I$  when they interfere at phase difference  $\phi$  is given by : [AIEEE-2012]

- (1)  $\frac{I_m}{9} (1 + 8\cos^2 \frac{\phi}{2})$  (2)  $\frac{I_m}{9} (4 + 5\cos\phi)$  (3)  $\frac{I_m}{3} (1 + 2\cos^2 \frac{\phi}{2})$  (4)  $\frac{I_m}{5} (1 + 4\cos^2 \frac{\phi}{2})$

WO0117

11. An electromagnetic wave in vacuum has the electric and magnetic fields  $\vec{E}$  and  $\vec{B}$ , which are always perpendicular to each other. The direction of polarization is given by  $\vec{X}$  and that of wave propagation by  $\vec{k}$ . Then [AIEEE-2012]

- (1)  $\vec{X} \parallel \vec{B}$  and  $\vec{k} \parallel \vec{E} \times \vec{B}$  (2)  $\vec{X} \parallel \vec{E}$  and  $\vec{k} \parallel \vec{B} \times \vec{E}$   
 (3)  $\vec{X} \parallel \vec{B}$  and  $\vec{k} \parallel \vec{B} \times \vec{E}$  (4)  $\vec{X} \parallel \vec{E}$  and  $\vec{k} \parallel \vec{E} \times \vec{B}$

WO0118

12. This question has Statement-1 and statement-2. Of the four choices given after the Statements, choose the one that best describes the two statements. [AIEEE - 2012]

**Statement-1 :** Davisson - Germer experiment established the wave nature of electrons.

**Statement-2:** If electrons have wave nature, they can interfere and show diffraction.

- (1) Statement-1 is true, Statement-2 is true and Statement-2 is the correct explanation for Statement-1.  
 (2) Statement-1 is true, Statement-2 is true and Statement-2 is not the correct explanation of Statement-I  
 (3) Statement-1 is false, Statement-2 is true.  
 (4) Statement-1 is true, Statement-2 is false

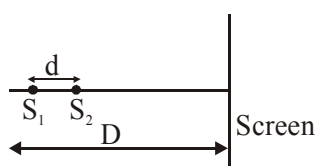
WO0119

13. A beam of unpolarised light of intensity  $I_0$  is passed through a polaroid A and then through another polaroid B which is oriented so that its principal plane makes an angle of  $45^\circ$  relative to that of A. The intensity of the emergent light is :- [JEE-Mains 2013]

- (1)  $I_0$  (2)  $I_0/2$  (3)  $I_0/4$  (4)  $I_0/8$

WO0120

14. Two coherent point sources  $S_1$  and  $S_2$  are separated by a small distance 'd' as shown. The fringes obtained on the screen will be : [JEE-Mains 2013]



- (1) points (2) straight lines (3) semicircles (4) concentric circles

WO0121



15. Match **List-I** (Electromagnetic wave type) with **List-II** (Its association/application) and select the correct option from the choices given below the lists : [JEE-Mains 2014]

| List-I |                  | List-II |                                               |
|--------|------------------|---------|-----------------------------------------------|
| (a)    | Infrared waves   | (i)     | To treat muscular strain                      |
| (b)    | Radio waves      | (ii)    | For broadcasting                              |
| (c)    | X-rays           | (iii)   | To detect fracture of bones                   |
| (d)    | Ultraviolet rays | (iv)    | Absorbed by the ozone layer of the atmosphere |

- |     |       |       |       |       |
|-----|-------|-------|-------|-------|
|     | (a)   | (b)   | (c)   | (d)   |
| (1) | (iii) | (ii)  | (i)   | (iv)  |
| (2) | (i)   | (ii)  | (iii) | (iv)  |
| (3) | (iv)  | (iii) | (ii)  | (i)   |
| (4) | (i)   | (ii)  | (iv)  | (iii) |

WO0122

16. During the propagation of electromagnetic waves in a medium :

[JEE-Mains 2014]

- (1) Electric energy density is equal to the magnetic energy density
- (2) Both electric magnetic energy densities are zero
- (3) Electric energy density is double of the magnetic energy density
- (4) Electric energy density is half of the magnetic energy density.

WO0123

17. Two beams, A and B, of plane polarized light with mutually perpendicular planes of polarization are seen through a polaroid. From the position when the beam A has maximum intensity (and beam B has zero intensity), a rotation of polaroid through  $30^\circ$  makes the two beams appear equally bright. If

the initial intensities of the two beams are  $I_A$  and  $I_B$  respectively, then  $\frac{I_A}{I_B}$  equals : [JEE-Mains 2014]

- |       |                   |       |                   |
|-------|-------------------|-------|-------------------|
| (1) 1 | (2) $\frac{1}{3}$ | (3) 3 | (4) $\frac{3}{2}$ |
|-------|-------------------|-------|-------------------|

WO0124

18. A red LED emits light at 0.1 watt uniformly around it. The amplitude of the electric field of the light at a distance of 1 m from the diode is:- [JEE-Mains 2015]

- |              |              |              |              |
|--------------|--------------|--------------|--------------|
| (1) 5.48 V/m | (2) 7.75 V/m | (3) 1.73 V/m | (4) 2.45 V/m |
|--------------|--------------|--------------|--------------|

WO0125

19. Assuming human pupil to have a radius of 0.25cm and a comfortable viewing distance of 25cm, the minimum separation between two objects that human eye can resolve at 500 nm wavelength is :- [JEE-Mains 2015]

- |                       |                       |                     |                      |
|-----------------------|-----------------------|---------------------|----------------------|
| (1) 100 $\mu\text{m}$ | (2) 300 $\mu\text{m}$ | (3) 1 $\mu\text{m}$ | (4) 30 $\mu\text{m}$ |
|-----------------------|-----------------------|---------------------|----------------------|

WO0126

20. On a hot summer night, the refractive index of air is smallest near the ground and increases with height from the ground. When a light beam is directed horizontally, the Huygens' principle leads us to conclude that as it travels, the light beam : **[JEE-Mains 2015]**

(1) bends downwards (2) bends upwards  
(3) becomes narrower (4) goes horizontally without any deflection

WO0127

21. The box of a pin hole camera, of length  $L$ , has a hole of radius  $a$ . It is assumed that when the hole is illuminated by a parallel beam of light of wavelength  $\lambda$  the spread of the spot (obtained on the opposite wall of the camera) is the sum of its geometrical spread and the spread due to diffraction. The spot would then have its minimum size (say  $b_{\min}$ ) when :- **[JEE-Mains 2016]**

(1)  $a = \frac{\lambda^2}{L}$  and  $b_{\min} = \sqrt{4\lambda L}$  (2)  $a = \frac{\lambda^2}{L}$  and  $b_{\min} = \left(\frac{2\lambda^2}{L}\right)$   
(3)  $a = \sqrt{\lambda L}$  and  $b_{\min} = \left(\frac{2\lambda^2}{L}\right)$  (4)  $a = \sqrt{\lambda L}$  and  $b_{\min} = \sqrt{4\lambda L}$

WO0128

22. Arrange the following electromagnetic radiations per quantum in the order of increasing energy :-

A : Blue light B : Yellow light C : X-ray D : Radiowave

**[JEE-Mains 2016]**

(1) B, A, D, C (2) D, B, A, C (3) A, B, D, C (4) C, A, B, D

WO0129

23. In a Young's double slit experiment, slits are separated by 0.5 mm, and the screen is placed 150 cm away. A beam of light consisting of two wavelengths, 650 nm and 520 nm, is used to obtain interference fringes on the screen. The least distance from the common central maximum to the point where the bright fringes due to both the wavelengths coincide is :- **[JEE-Main 2017]**

(1) 9.75 mm (2) 15.6 mm (3) 1.56 mm (4) 7.8 mm

WO0130

24. The angular width of the central maximum in a single slit diffraction pattern is  $60^\circ$ . The width of the slit is  $1 \mu\text{m}$ . The slit is illuminated by monochromatic plane waves. If another slit of same width is made near it, Young's fringes can be observed on a screen placed at a distance 50 cm from the slits. If the observed fringe width is 1 cm, what is slit separation distance? (i.e. distance between the centres of each slit.) **[JEE-Main 2018]**

(1)  $50 \mu\text{m}$  (2)  $75 \mu\text{m}$  (3)  $100 \mu\text{m}$  (4)  $25 \mu\text{m}$

WO0131

25. An EM wave from air enters a medium. The electric fields are  $\vec{E}_1 = E_{01} \hat{x} \cos \left[ 2\pi v \left( \frac{z}{c} - t \right) \right]$  in air and  $\vec{E}_2 = E_{02} \hat{x} \cos [k(2z - ct)]$  in medium, where the wave number  $k$  and frequency  $v$  refer to their values in air. The medium is non-magnetic. If  $\epsilon_{r1}$  and  $\epsilon_{r2}$  refer to relative permittivity of air and medium respectively, which of the following options is correct ? **[JEE-Main 2018]**

(1)  $\frac{\epsilon_{r1}}{\epsilon_{r2}} = 2$  (2)  $\frac{\epsilon_{r1}}{\epsilon_{r2}} = \frac{1}{4}$  (3)  $\frac{\epsilon_{r1}}{\epsilon_{r2}} = \frac{1}{2}$  (4)  $\frac{\epsilon_{r1}}{\epsilon_{r2}} = 4$

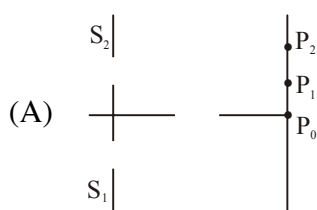
WO0132

# EXERCISE-(J-A)

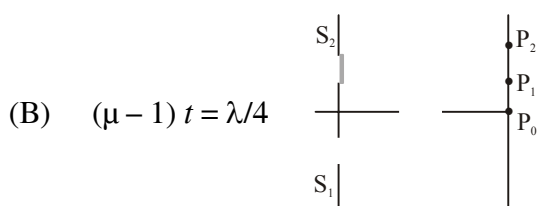
1. **Column I** shows four situations of standard Young's double slit arrangement with the screen placed far away from the slits  $S_1$  and  $S_2$ . In each of these cases  $S_1P_0 = S_2P_0$ ,  $S_1P_1 - S_2P_1 = \lambda/4$  and  $S_1P_2 - S_2P_2 = \lambda/3$ , where  $\lambda$  is the wavelength of the light used. In the cases  $B$ ,  $C$  and  $D$ , a transparent sheet of refractive index  $\mu$  and thickness  $t$  is pasted on slit  $S_2$ . The thicknesses of the sheets are different in different cases. The phase difference between the light waves reaching a point  $P$  on the screen from the two slits is denoted by  $\delta(P)$  and the intensity by  $I(P)$ . Match each situation given in **Column I** with the statement(s) in **Column II** valid for that situation. [IIT-JEE-2009]

## Column-I

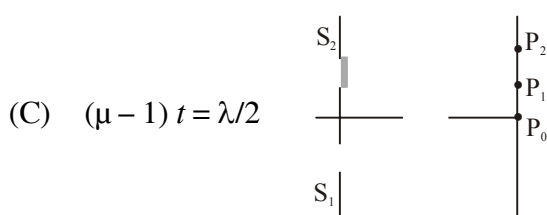
## Column-II



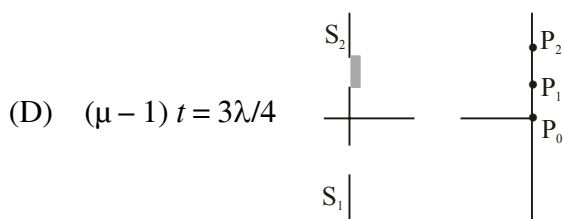
(p)  $\delta(P_0) = 0$



(q)  $\delta(P_1) = 0$



(r)  $I(P_1) = 0$



(s)  $I(P_0) > I(P_1)$

(t)  $I(P_2) > I(P_1)$

WO0133

2. Young's double slit experiment is carried out by using green, red and blue light, one color at a time. The fringe widths recorded are  $\beta_G$ ,  $\beta_R$  and  $\beta_B$ , respectively. Then [IIT-JEE-2012]

- (A)  $\beta_G > \beta_B > \beta_R$  (B)  $\beta_B > \beta_G > \beta_R$   
 (C)  $\beta_R > \beta_B > \beta_G$  (D)  $\beta_R > \beta_G > \beta_B$

WO0134

3. In the Young's double slit experiment using a monochromatic light of wavelength  $\lambda$ , the path difference (in terms of an integer  $n$ ) corresponding to any point having half the peak intensity is :- [JEE Advanced 2013]

- (A)  $(2n+1)\frac{\lambda}{2}$  (B)  $(2n+1)\frac{\lambda}{4}$   
 (C)  $(2n+1)\frac{\lambda}{8}$  (D)  $(2n+1)\frac{\lambda}{16}$

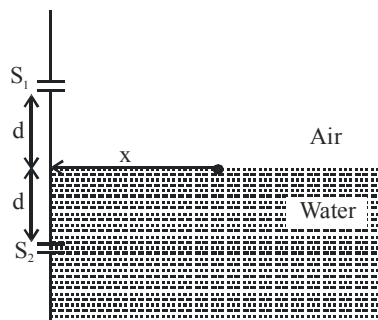
WO0135

4. A light source, which emits two wavelengths  $\lambda_1 = 400$  nm and  $\lambda_2 = 600$  nm, is used in a Young's double slit experiment. If recorded fringe widths for  $\lambda_1$  and  $\lambda_2$  are  $\beta_1$  and  $\beta_2$  and the number of fringes for them within a distance  $y$  on one side of the central maximum are  $m_1$  and  $m_2$ , respectively, then :- [JEE Advanced 2014]

- (A)  $\beta_2 > \beta_1$   
 (B)  $m_1 > m_2$   
 (C) From the central maximum, 3<sup>rd</sup> maximum of  $\lambda_2$  overlaps with 5<sup>th</sup> minimum of  $\lambda_1$   
 (D) The angular separation of fringes of  $\lambda_1$  is greater than  $\lambda_2$

WO0136

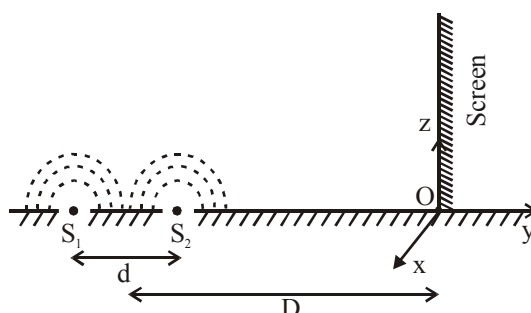
5. A Young's double slit interference arrangement with slits  $S_1$  and  $S_2$  is immersed in water (refractive index =  $4/3$ ) as shown in the figure. The positions of maxima on the surface of water are given by  $x^2 = p^2 m^2 \lambda^2 - d^2$ , where  $\lambda$  is the wavelength of light in air (refractive index = 1),  $2d$  is the separation between the slits and  $m$  is an integer. The value of  $p$  is. [JEE Advanced 2015]



WO0137

6. While conducting the Young's double slit experiment, a student replaced the two slits with a large opaque plate in the  $x$ - $y$  plane containing two small holes that act as two coherent point sources ( $S_1, S_2$ ) emitting light of wavelength 600 nm. The student mistakenly placed the screen parallel to the  $x$ - $z$  plane (for  $z > 0$ ) at a distance  $D = 3\text{m}$  from the mid-point of  $S_1S_2$ , as shown schematically in the figure. The distance between the sources  $d = 0.6003\text{ mm}$ . The origin  $O$  is at the intersection of the screen and the line joining  $S_1S_2$ . Which of the following is (are) true of the intensity pattern on the screen ?

[JEE Advanced 2016]



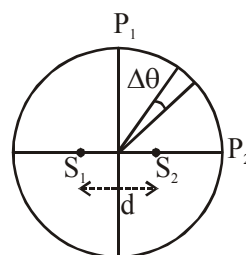
- (A) Hyperbolic bright and dark bands with foci symmetrically placed about  $O$  in the  $x$ -direction  
 (B) Semi circular bright and dark bands centered at point  $O$   
 (C) The region very close to the point  $O$  will be dark  
 (D) Straight bright and dark bands parallel to the  $x$ -axis

WO0138

7. Two coherent monochromatic point sources  $S_1$  and  $S_2$  of wavelength  $\lambda = 600\text{ nm}$  are placed symmetrically on either side of the center of the circle as shown. The sources are separated by a distance  $d = 1.8\text{mm}$ . This arrangement produces interference fringes visible as alternate bright and dark spots on the circumference of the circle. The angular separation between two consecutive bright spots is  $\Delta\theta$ . Which of the following options is/are correct ?

[JEE Advanced 2017]

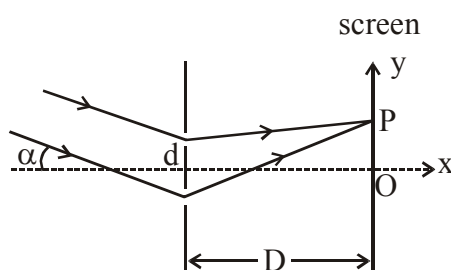
- (A) A dark spot will be formed at the point  $P_2$   
 (B) The angular separation between two consecutive bright spots decreases as we move from  $P_1$  to  $P_2$  along the first quadrant  
 (C) At  $P_2$  the order of the fringe will be maximum  
 (D) The total number of fringes produced between  $P_1$  and  $P_2$  in the first quadrant is close to 3000



WO0139

8. In a Young's double slit experiment, the slit separation  $d$  is 0.3 mm and the screen distance  $D$  is 1m. A parallel beam of light of wavelength 600nm is incident on the slits at angle  $\alpha$  as shown in figure. On the screen, the point O is equidistant from the slits and distance PO is 11.0 mm. Which of the following statement(s) is/are correct ?

[JEE Advanced 2019]



- (1) For  $\alpha = \frac{0.36}{\pi}$  degree, there will be destructive interference at point O.
- (2) Fringe spacing depends on  $\alpha$
- (3) For  $\alpha = \frac{0.36}{\pi}$  degree, there will be destructive interference at point P
- (4) For  $\alpha = 0$ , there will be constructive interference at point P.

WO0140

## GEOMETRICAL OPTICS

### KEY CONCEPTS

#### INTRODUCTION

The branch of Physics called optics deals with the behavior of light and other electromagnetic waves. Under many circumstances, the wavelength of light is negligible compared with the dimensions of the device as in the case of ordinary mirrors and lenses. A light beam can then be treated as a ray whose propagation is governed by simple geometric rules. The part of optics that deals with such phenomenon is known as geometric optics.

#### PROPAGATION OF LIGHT

Light travels along straight line in a medium or in vacuum. The path of light changes only when there is an object in its path or where the medium changes. We call this rectilinear (straight-line) propagation of light. Light that starts from a point A and passes through another point B in the same medium actually passes through all the points on the straight line AB. Such a straight line path of light is called a ray of light. Light rays start from each point of a source and travel along straight lines till they fall on an object or a surface separating two media (mediums). A bundle of light rays is called a beam of light.

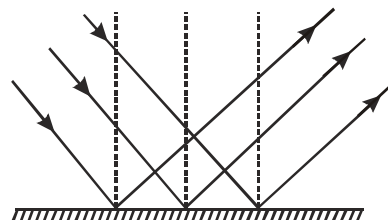
Apart from vacuum and gases, light can travel through some liquids and solids. A medium in which light can travel freely over large distances is called a transparent medium. Water, glycerine, glass and clear plastics are transparent. A medium in which light cannot travel is called opaque. Wood, metals, bricks, etc., are opaque. In materials like oil, light can travel some distance, but its intensity reduces rapidly. Such materials are called translucent.

#### REFLECTION OF LIGHT

When light rays strike the boundary of two media such as air and glass, a part of light is bounced back into the same medium. This is called **Reflection of light**.

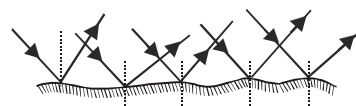
##### (i) Regular / Specular reflection :

When the reflection takes place from a perfect plane surface then after reflection rays remain parallel. It is called **Regular reflection**.



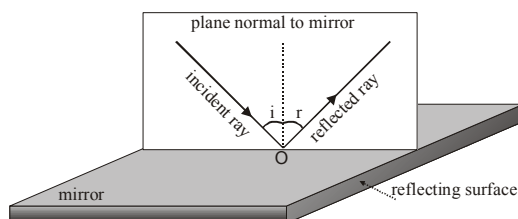
##### (ii) Diffused reflection

When the surface is rough, light is reflected from the surface from bits of its plane surfaces in irregular directions. This is called **diffused reflection**. This process enables us to see an object from any position.

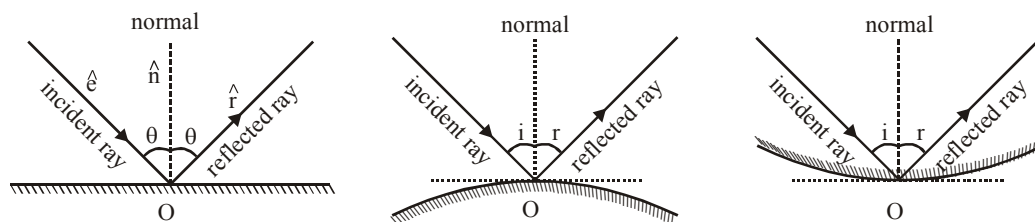


#### LAWS OF REFLECTION

- (1) Incident ray, reflected ray and normal lies in the same plane.



- (2) The angle of reflection is equal to the angle of incident i.e.  $\angle i = \angle r$ .

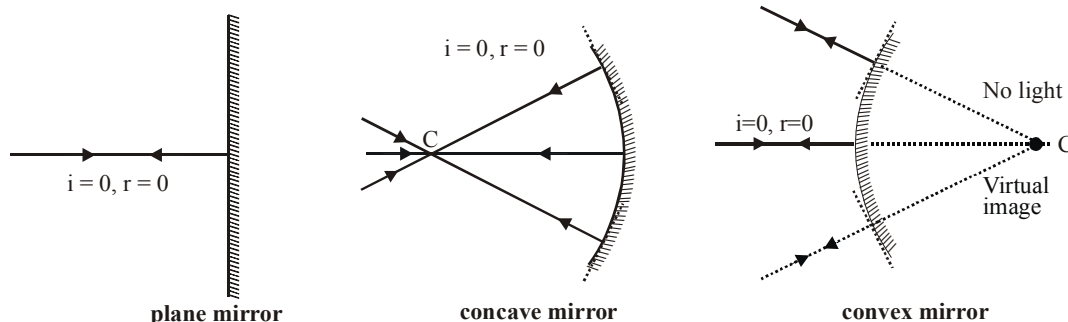


In vector form

$$\hat{r} = \hat{e} - 2(\hat{e} \cdot \hat{n})\hat{n}$$

### KEY POINTS

- **Rectilinear propagation of light** : In a homogeneous transparent medium light travels in straight line.
- When a ray is incident normally on a boundary after reflection it retraces its path.



- The frequency, wavelength and speed does not change on reflection.
- Eye is mostly sensitive for yellow colour and least sensitive for violet and red colour.

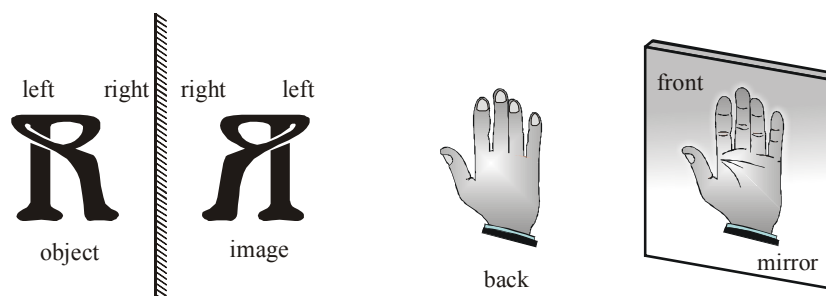
Due to this reason :

- Commercial vehicle's are painted with yellow colour.
- Sodium lamps (yellow colour) are used in road lights.

### REFLECTION FROM PLANE MIRROR

Plane mirror is the perpendicular bisector of the line joining object and image.

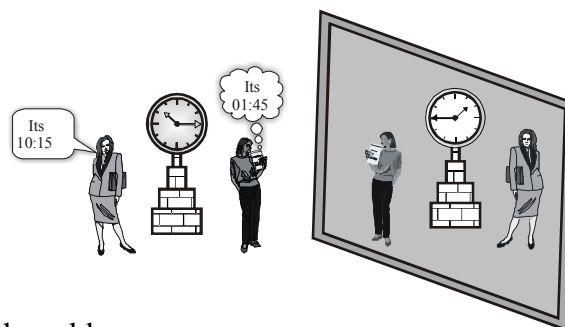
- The image formed by a plane mirror suffers **lateral-inversion**, i.e., in the image formed by a plane mirror left is turned into right and vice-versa with respect to object.



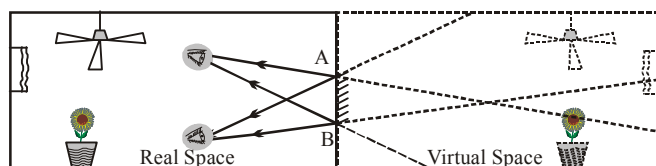
When a watch placed in front of a plane mirror then watch is object and its time is object time and image of watch observed by a person standing in front of mirror then time seen by person.



- (i) Object Time =  $A^H$   
Image Time =  $12 - A^H$
- (ii) Object Time =  $A^H B^M$   
Image Time =  $11 - 60' - A^H B^M$
- (iii) Object Time =  $A^H B^M C^S$   
Image Time =  $11 - 59' - 60'' - A^H B^M C^S$



- A plane mirror behaves like a window to virtual world.



AB  $\rightarrow$  Plane mirror

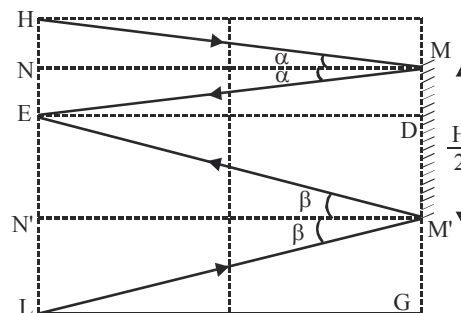
- To see complete image in a plane mirror the minimum length of plane mirror should be half the height of a person.

From figure.  $\triangle HNM$  and  $\triangle ENM$  are congruent

$$\therefore EN = HN \quad \therefore MD = EN = \frac{1}{2} HE$$

Similarly  $\triangle EN'M'$  and  $\triangle LN'M'$  are congruent

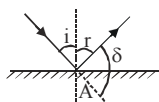
$$\begin{aligned} \text{Length of the mirror } MM' &= MD + M'D = \frac{1}{2} HE + \frac{1}{2} EL \\ &= \frac{1}{2} (HE + EL) = \frac{1}{2} HL \end{aligned}$$



$\therefore$  Minimum of length of mirror is just half of the person.

- This result does not depend on position of eye (height of the eye from ground).
- This result is independent of distance of person in front of mirror.

Deviation for a single mirror



$$\delta = 180 - (i + r); \angle i = \angle r; \delta = 180 - 2i$$

- Total deviation produced by the combination of two plane mirrors which are inclined at an angle  $\theta$  from each other.

$$\delta = \delta_1 + \delta_2 = 180 - 2\alpha + 180 - 2\beta = 360 - 2(\alpha + \beta) \dots (i)$$

$$\text{From } \triangle QAB, \theta + 90 - \alpha + 90 - \beta = 180 \Rightarrow \theta = \alpha + \beta \dots (ii)$$

Putting the value of  $\theta$  in (i) from (ii),  $\delta = 360 - 2\theta$

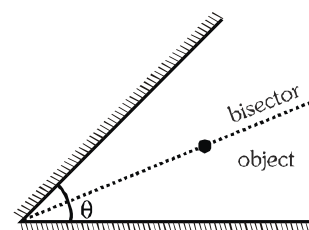
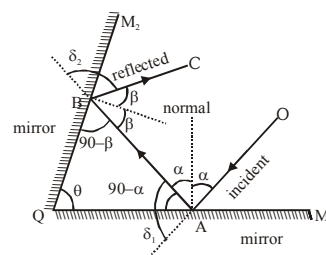
- If there are two plane mirror inclined to each other at an angle  $\theta$  the number of image (n) of a point object formed are determined as follows.

(a) If  $\frac{360^\circ}{\theta} = m$  is even then number of images  $n = m - 1$

(b) If  $\frac{360^\circ}{\theta} = m$  is odd. There will be two case.

(i) When object is not on bisector, then number of images  $n = m$

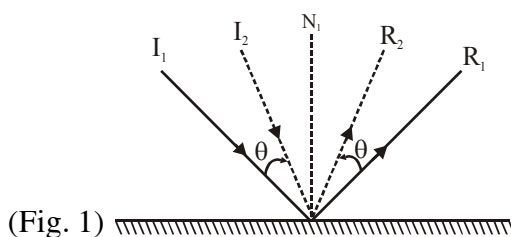
(ii) When object is at bisector, then number of images  $n = m - 1$



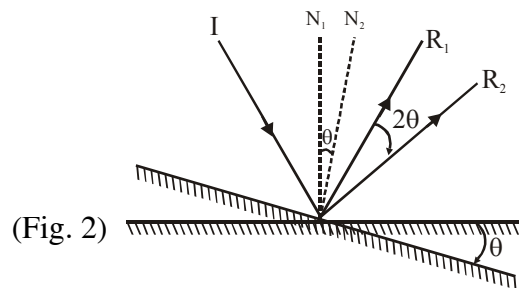
- (c) If  $\frac{360^\circ}{\theta} = m$  is a fraction, and the object is placed symmetrically then no. of images  $n =$  nearest even integer

| S.No. | $\theta$ in degree | $m = \frac{360^\circ}{\theta}$ | Number of images formed if object is placed |               |
|-------|--------------------|--------------------------------|---------------------------------------------|---------------|
|       |                    |                                | asymmetrically                              | symmetrically |
| 1.    | 0                  | $\infty$                       | $\infty$                                    | $\infty$      |
| 2.    | 30                 | 12                             | 11                                          | 11            |
| 3.    | 45                 | 8                              | 7                                           | 7             |
| 4.    | 60                 | 6                              | 5                                           | 5             |
| 5.    | 72                 | 5                              | 5                                           | 4             |
| 6.    | 75                 | 4.8                            | —                                           | 4             |
| 7.    | 90                 | 4                              | 3                                           | 3             |
| 8.    | 112.5              | 3.2                            | —                                           | 4             |
| 9.    | 120                | 3                              | 3                                           | 2             |

- If the object is placed between two plane mirrors then images are formed due to multiple reflections. At each reflection, a part of light energy is absorbed. Therefore, distant images get fainter.
- Keeping the mirror fixed if the incident ray is rotated by some angle, the reflected ray is also rotated by the same angle but in opposite sense. (See Fig. 1)

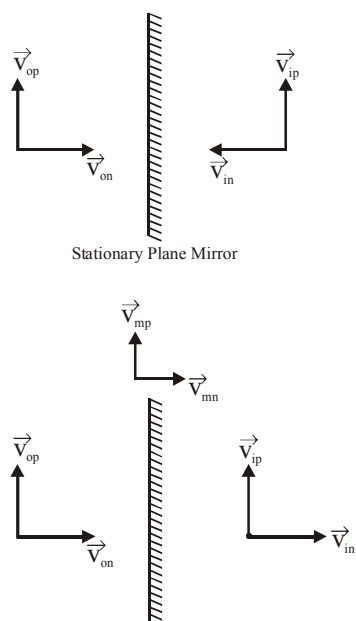


(Fig. 1)



(Fig. 2)

- Keeping the incident ray fixed, if the mirror is rotated by some angle, then the reflected ray rotates by double the angle in the same sense. (See Fig. 2)



Stationary Plane Mirror

$$\vec{v}_{on} = -\vec{v}_{in}, \quad \vec{v}_{op} = \vec{v}_{ip}$$

though speed of object and image are the same

$v_{op}$  = component of velocity of object along parallel to mirror.

$v_{on}$  = component of velocity of object along normal to mirror.

$v_{ip}$  = component of velocity of image along parallel to mirror.

$v_{in}$  = component of velocity of image along normal to mirror.

If mirror is moving  $\vec{v}_{ip} = \vec{v}_{op}$  and  $(\vec{v}_{im})_n = -(\vec{v}_{om})_n$

$$\Rightarrow \vec{v}_{in} - \vec{v}_{mn} = -(\vec{v}_{on} - \vec{v}_{mn}) \Rightarrow \vec{v}_{in} = 2\vec{v}_{mn} - \vec{v}_{on}$$

$\vec{v}_{mn}$  = component of velocity of mirror along normal.

$\vec{v}_{op}$  = component of velocity of object along mirror.

$\vec{v}_{on}$  = component of velocity of object along normal

$\vec{v}_{ip}$  = component of velocity of image along mirror.

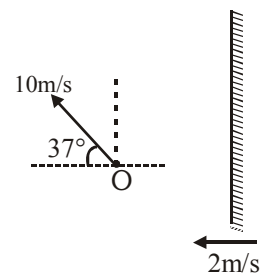
$\vec{v}_{in}$  = component of velocity of image along normal.

**Ex.** Find the velocity of the image.

**Sol.**  $\vec{v}_{ox} = (-10 \cos 37^\circ) \hat{i} = -8\hat{i}$  and

$$\vec{v}_{oy} = (10 \sin 37^\circ) \hat{j} = 6\hat{j}$$

$$\vec{v}_{ix} = 2\vec{v}_{mx} - \vec{v}_{ox} = 2(-2\hat{i}) - (-8\hat{i}) = 4\hat{i} \text{ and } \vec{v}_{iy} = \vec{v}_{oy} = 6\hat{j}$$

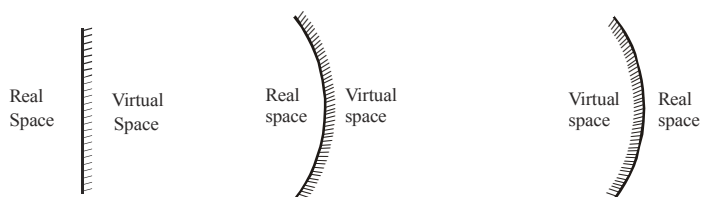


## REAL AND VIRTUAL SPACES

A mirror, plane or spherical divides the space into two ;

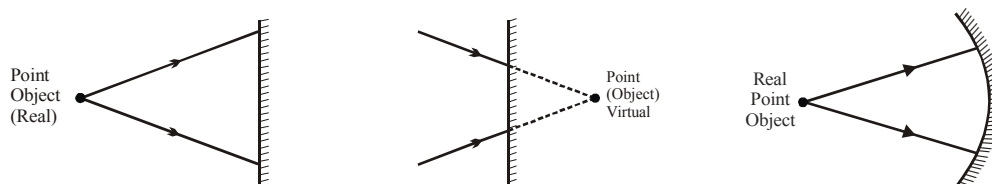
(a) Real space, a side where the reflected rays exist.

(b) Virtual space is on the other side where the reflected rays do not exist.



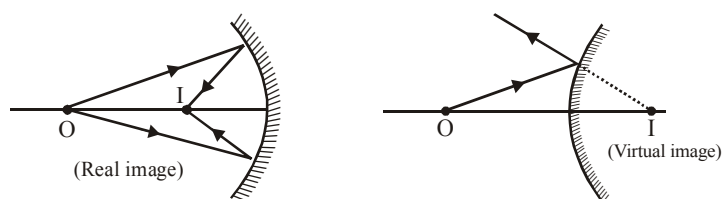
## OBJECT

Object is decided by incident rays only. The point object is that point from which the incident rays actually diverge (Real object) or towards which the incident rays appear to converge (virtual object).



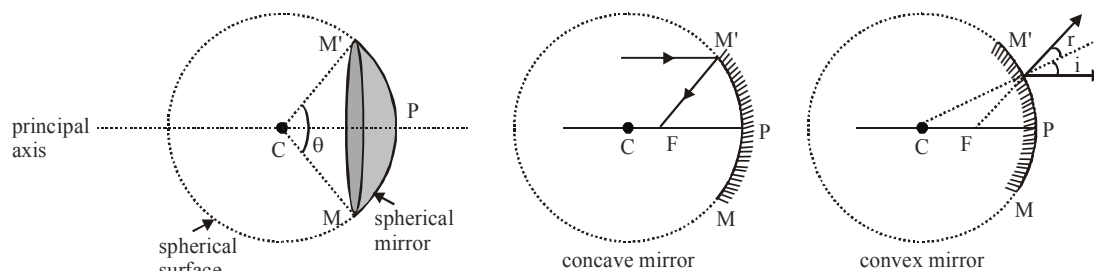
## IMAGE

Image is decided by reflected or refracted rays only. The point image is that point at which the refracted / reflected rays reflected from the mirror, actually converge (real image) or from which the refracted /reflected rays appear to diverge (virtual image).



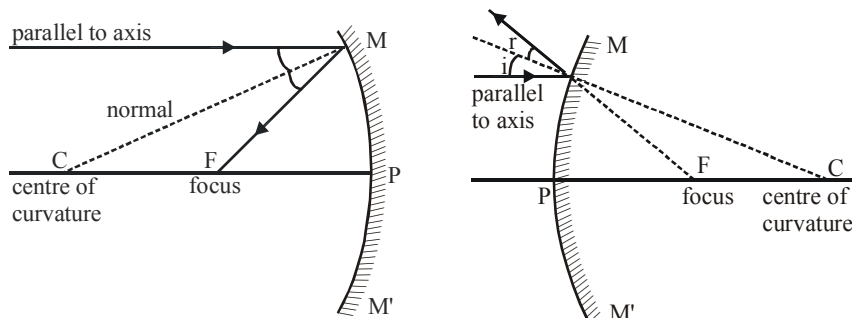
## SPHERICAL (CURVED) MIRROR

Curved mirror is part of a hollow sphere. If reflection takes place from the inner surface then The mirror is called concave and if its outer surface acts as reflector it is convex.

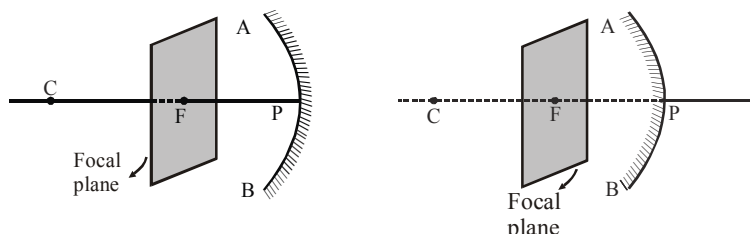


## DEFINITIONS FOR THIN SPHERICAL MIRRORS

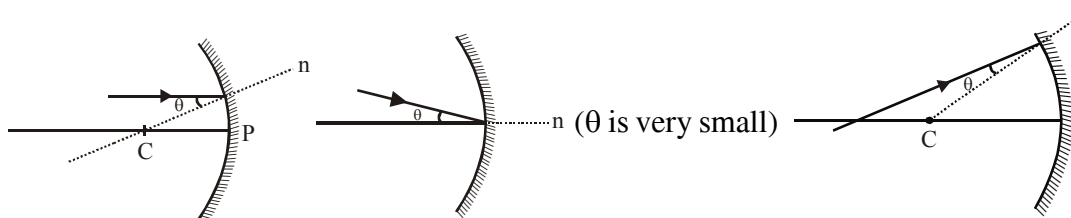
- (i) **Pole** is any point on the reflecting surface of the mirror. For convenience we take it as the midpoint P of the mirror (as shown).
- (ii) **Principal-section** is any section of the mirror such as MM' passing through the pole is called principal-section.
- (iii) **Centre of curvature** is the centre C of the sphere of which the mirror is a part.
- (iv) **Radius of curvature** is the radius R of the sphere of which the mirror is a part.
- (v) **Principal-axis** is the line CP, joining the pole and centre of curvature of the mirror.
- (vi) **Principal-focus** is an image point F on principal axis for which object is at infinity.



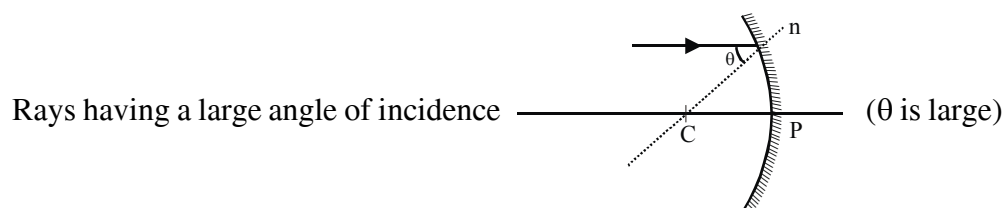
- (vii) **Focal-length** is the distance PF between pole P and focus F along principal axis.
- (viii) **Aperture**, in reference to a mirror, means the effective diameter of the light reflecting area of the mirror.
- (ix) **Focal Plane** is the plane passing through focus and perpendicular to principal axis.



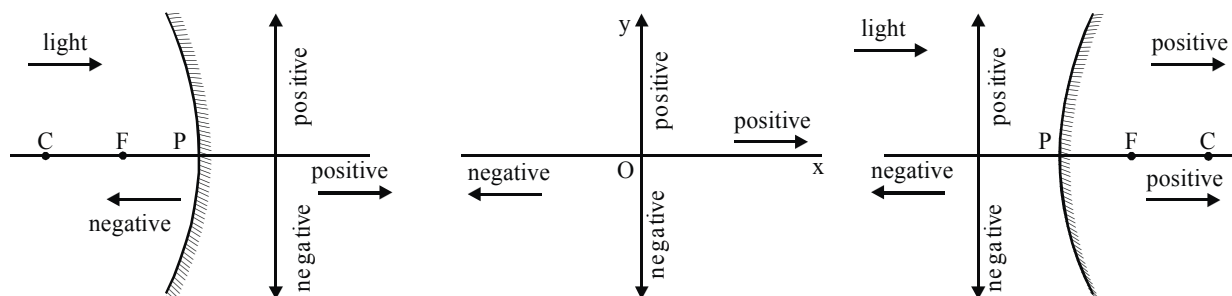
- (x) **Paraxial Rays** Those rays which make small angle with normal at point of incidence and hence are close to principal axis.



- (xi) **Marginal rays** :



## SIGN- CONVENTION

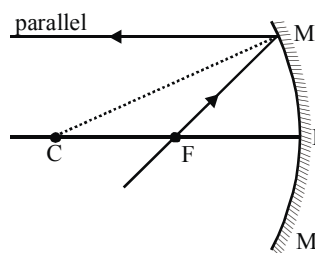
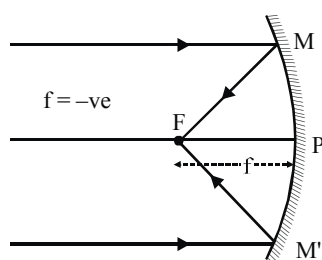


- Along principal axis, distances are measured from the pole (pole is taken as the origin).
- Distance in the direction of light are taken to be positive while opposite to be negative.
- The distances above principal axis are taken to be positive while below it negative.
- Whenever and wherever possible the ray of light is taken to travel from left to right.

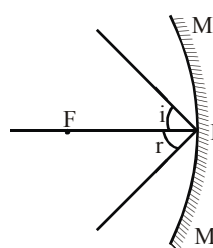
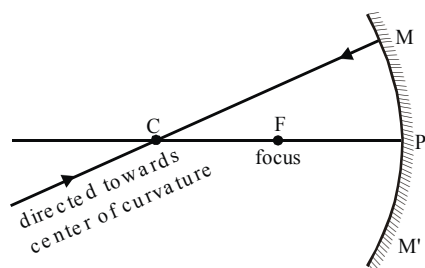
## RULES FOR IMAGE FORMATION (FOR PARAXIAL RAYS ONLY)

(These rules are based on the laws of reflection  $\angle i = \angle r$ )

- A ray parallel to principal axis after reflection from the mirror passes or appears to pass through its focus (by definition of focus).
- A ray passing through or directed towards focus, after reflection from the mirror, becomes parallel to the principal axis.



- A ray passing through or directed towards centre of curvature, after reflection from the mirror, retraces its path (as for it  $\angle i = 0$  and so  $\angle r = 0$ ).
- Incident and reflected rays at the pole of a mirror are symmetrical about the principal axis  $\angle i = \angle r$ .



## RELATIONS FOR SPHERICAL MIRRORS

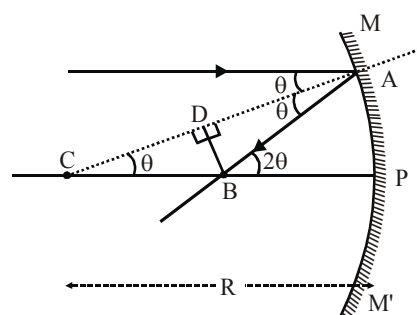
Relation between  $f$  and  $R$  for the spherical mirror

- For Marginal rays

In  $\triangle ABC$ ,  $AB = BC$

$$AC = CD + DA = 2BC \cos \theta \Rightarrow R = 2BC \cos \theta$$

$$\Rightarrow BC = \frac{R}{2 \cos \theta} \text{ and } BP = PC - BC = R - \frac{R}{2 \cos \theta}$$



**Note :** B is not the focus ; it is just a point where a marginal ray after reflection meets.

- **For paraxial rays (parallel to principal axis)**

( $\theta$  small so  $\sin\theta \approx \theta$ ,  $\cos\theta \approx 1$ ,  $\tan\theta \approx \theta$ ) Hence  $BC = \frac{R}{2}$  and  $BP = \frac{R}{2}$

Thus, point B is the midpoint of PC (i.e. radius of curvature) and is defined as FOCUS so  $BP = f = \frac{R}{2}$

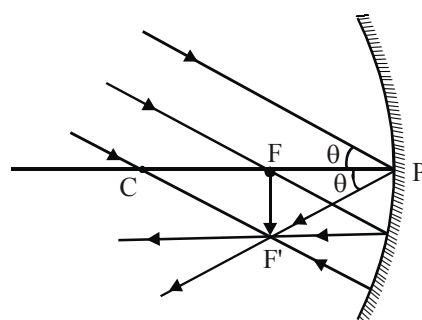
**(Definition :** Paraxial rays parallel to the principal axis after reflection from the mirror meet the principal axis at focus)

- BP is the focal length (f)  $\boxed{f = \frac{R}{2}}$

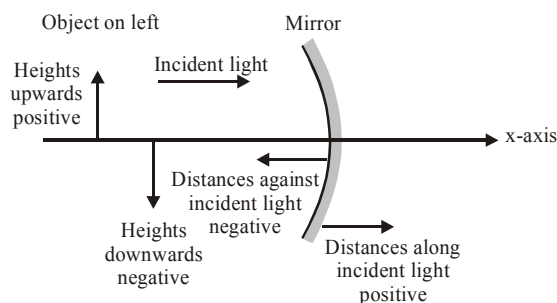
- **Paraxial rays (not parallel to principal axis)**

Such rays after reflection meet at a point in the focal plane ( $F'$ ), such that

$$\frac{FF'}{FP} = \tan\theta \approx \theta \Rightarrow \frac{FF'}{f} = \theta \Rightarrow FF' = f\theta$$



### Sign convention



### Relation between u, v and f for curved mirror

An object is placed at a distance  $u$  from the pole of a mirror and its image is formed at a distance  $v$  (from the pole)

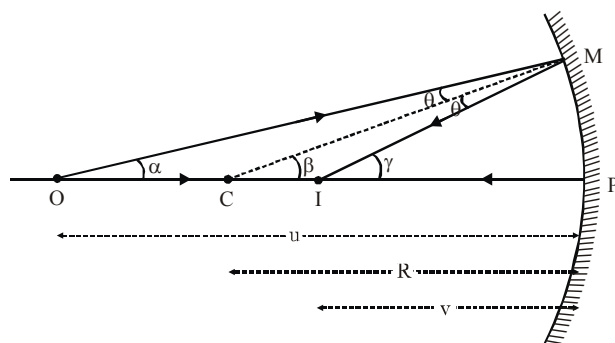
If angle is very small :  $\alpha = \frac{MP}{u}$ ,  $\beta = \frac{MP}{R}$ ,  $\gamma = \frac{MP}{v}$

from  $\triangle CMO$ ,  $\beta = \alpha + \theta \Rightarrow \theta = \beta - \alpha$

from  $\triangle CMI$ ,  $\gamma = \beta + \theta \Rightarrow \theta = \gamma - \beta$

so we can write  $\beta - \alpha = \gamma - \beta \Rightarrow 2\beta = \gamma + \alpha$

$$\therefore \frac{2}{R} = \frac{1}{v} + \frac{1}{u} \Rightarrow \frac{1}{f} = \frac{1}{u} + \frac{1}{v}$$



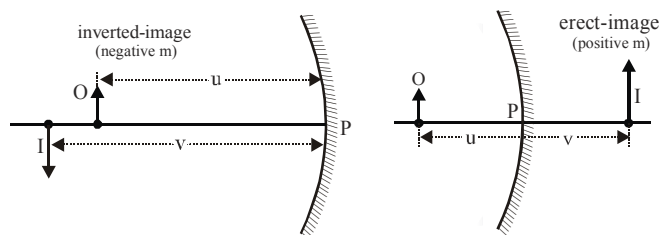
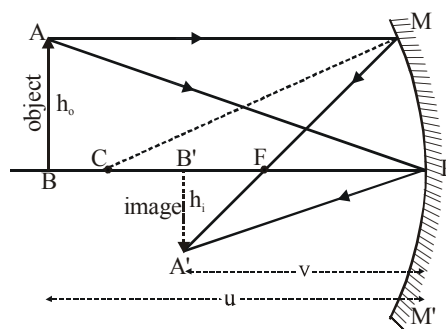
# MAGNIFICATION

## Transverse or lateral magnification

Linear magnification  $m = \frac{\text{height of image}}{\text{height of object}} = \frac{h_i}{h_o}$

$\Delta ABP$  and  $\Delta A'B'P$  are similar so  $\frac{-h_i}{h_o} = \frac{-v}{-u} \Rightarrow \frac{h_i}{h_o} = -\frac{v}{u}$

Magnification  $m = -\frac{v}{u}$ ;  $m = -\frac{v}{u} = \frac{f}{f-u} = \frac{f-v}{f} = \frac{h_i}{h_o}$



If one dimensional object is placed perpendicular to the principal axis then linear magnification is

called transverse or lateral magnification.  $m = \frac{h_i}{h_o} = -\frac{v}{u}$

## Magnification

$|m| > 1$

$m < 0$

## Image

enlarged

inverted

## Magnification

$|m| < 1$

$m > 0$

## Image

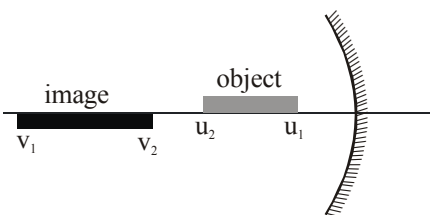
diminished

erect

## Longitudinal magnification

If one dimensional object is placed with its length along the principal axis then linear magnification is called longitudinal magnification.

Longitudinal magnification  $m_L = \frac{\text{length of image}}{\text{length of object}} = \left| \frac{v_2 - v_1}{u_2 - u_1} \right|$



For small objects only :  $m_L = -\frac{dv}{du}$

differentiation of  $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$  gives us  $-\frac{dv}{v^2} - \frac{du}{u^2} = 0 \Rightarrow -\frac{dv}{du} = \left[ \frac{v}{u} \right]^2$  so  $m_L = -\frac{dv}{du} = \left[ \frac{v}{u} \right]^2 = m^2$

## Superficial magnification

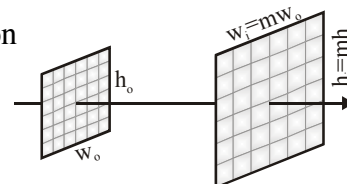
If two dimensional object placed with its plane perpendicular to principal axis its magnification is known as superficial magnification

Linear magnification  $m = \frac{h_i}{h_o} = \frac{w_i}{w_o}$

$h_i = mh_o$ ,  $w_i = mw_o$  and  $A_{obj} = h_o \times w_o$

Area of image :  $A_{image} = h_i \times w_i = mh_o \times mw_o = m^2 A_{obj}$

superficial magnification  $m_s = \frac{\text{area of image}}{\text{area of object}} = \frac{(ma) \times (mb)}{(a \times b)} = m^2$

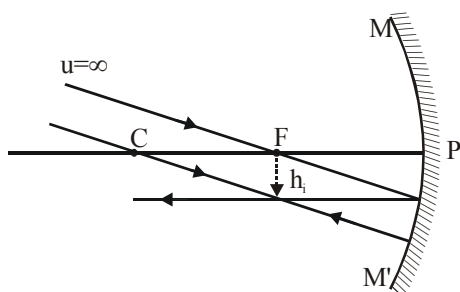


## IMAGE FORMATION BY SPHERICAL MIRRORS

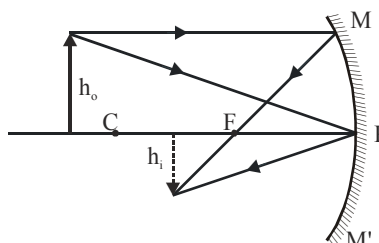
## Concave mirror

(i) **Object** : Placed at infinity**Image** : real, inverted, diminished at F

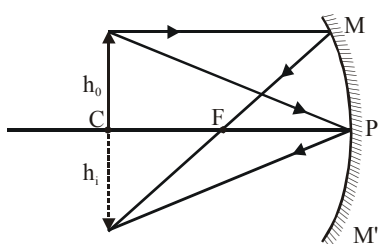
$|m| \ll 1 \text{ \& } m < 0$

(ii) **Object** : Placed in between infinity and C**Image** : real, inverted, diminished in between C and F

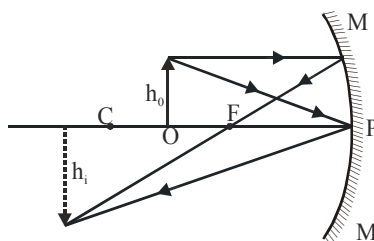
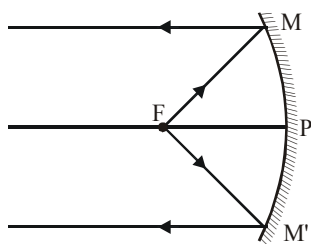
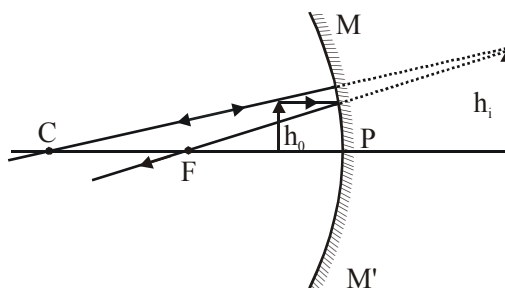
$|m| < 1 \text{ \& } m < 0$

(iii) **Object** : Placed at C**Image** : real, inverted, equal at C

$(m = -1)$

(iv) **Object** : Placed in between F and C**Image** : real, inverted, enlarged beyond C

$|m| > 1 \text{ \& } m < 0$

(v) **Object** : Placed at F**Image** : real, inverted, very large(assumed) at infinity ( $m \ll -1$ )(vi) **Object** : Placed between F and P**Image** : virtual, erect, enlarged andbehind the mirror ( $m > +1$ )

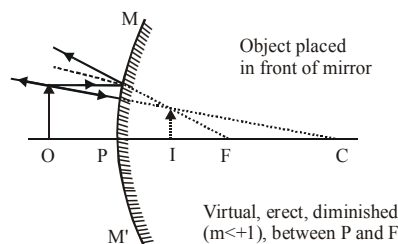
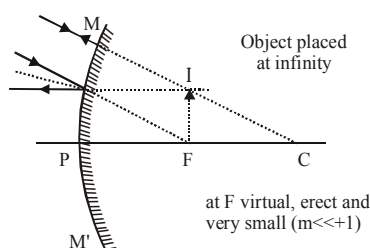
## For concave mirror

| Object                  | Image         | Magnification                 |
|-------------------------|---------------|-------------------------------|
| $-\infty$               | F             | $ m  \ll 1 \text{ \& } m < 0$ |
| $-\infty - C$           | C - F         | $ m  < 1 \text{ \& } m < 0$   |
| C                       | C             | $m = -1$                      |
| C - F                   | $-\infty - C$ | $ m  > 1 \text{ \& } m < 0$   |
| Just before F towards C | $-\infty$     | $m \ll -1$                    |
| Just after F towards P  | $+\infty$     | $m \gg 1$                     |



# Convex mirror

Image is always virtual and erect, whatever be the position of the object and  $m$  is always positive.



## KEY POINTS

### Differences in real & virtual image for spherical mirror

#### Real Image

- (i) Inverted w.r.t. object
- (ii) Can be obtained on screen
- (iii) Its magnification is negative
- (iv) Forms in front of mirror

#### Virtual Image

- (i) Erect w.r.t. object
- (ii) Can not be obtained on screen
- (iii) Its magnification is positive
- (iv) Forms behind the mirror

- For real extended object, if the image formed by a single mirror is erect it is always virtual (i.e.,  $m$  is +ve) and in this situation if the size of image is :

| Smaller than object the mirror is convex | Equal to object the mirror is plane | Larger than object the mirror is concave |
|------------------------------------------|-------------------------------------|------------------------------------------|
| $m < +1$<br>                             | $m = +1$<br>                        | $m > +1$<br>                             |

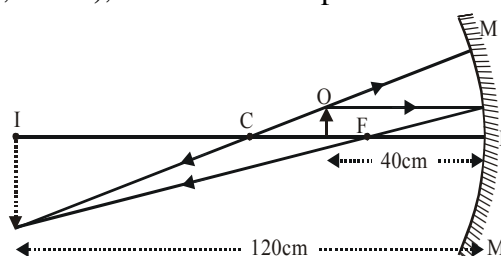
**Ex.** The focal length of a concave mirror is 30cm. Find the position of the object in front of the mirror, so that the image is three times the size of the object.

**Sol.** As the object is in front of the mirror it is real and for real object the magnified image formed by concave mirror can be inverted (i.e., real) or erect (i.e., virtual), so there are two possibilities.

(a) If the image is inverted (i.e., real)

$$m = \frac{f}{f - u} \Rightarrow -3 = \frac{-30}{-30 - u} \Rightarrow u = -40 \text{ cm}$$

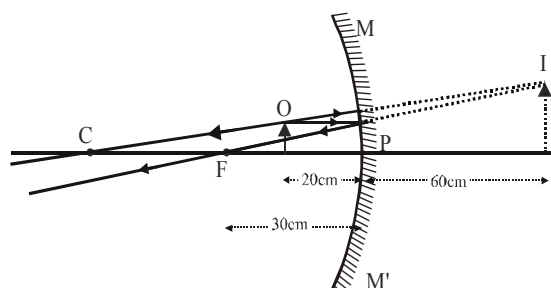
Object must be at a distance of 40 cm in front of the mirror (in between C and F).



(b) If the image is erect (i.e., virtual)

$$m = \frac{f}{f - u} \Rightarrow 3 = \frac{-30}{-30 - u} \Rightarrow u = -20 \text{ cm}$$

Object must be at a distance of 20 cm in front of the mirror (in between F and P).



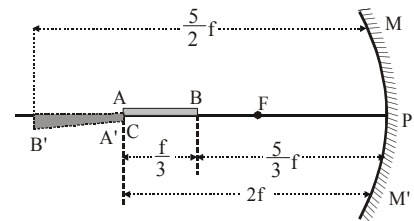
**Ex.** A thin rod of length  $\frac{f}{3}$  is placed along the principal axis of a concave mirror of focal length  $f$  such that its image which is real and elongated, just touches the rod. What is magnification?

**Sol.** Image is real and enlarged, the object must be between C and F. One end A' of the image coincides with the end A of rod itself, so

$$v_A = u_A, \frac{1}{v_A} + \frac{1}{u_A} = \frac{1}{-f} \text{ i.e., } v_A = u_A = -2f$$

so it clear that the end A is at C.  $\therefore$  the length of rod is  $\frac{f}{3}$

$\therefore$  distance of the other end B from P is  $u_B = 2f - \frac{f}{3} = \frac{5f}{3}$



if the distance of image of end B from P is  $v_B$  then  $\frac{1}{v_B} + \frac{1}{\frac{5f}{3}} = \frac{1}{-f} \Rightarrow v_B = -\frac{5f}{2}$

$\therefore$  the length of the image  $|v_B| - |v_A| = \frac{5f}{2} - 2f = \frac{1}{2}f$  and magnification  $m = \frac{|v_B| - |v_A|}{|u_B| - |u_A|} = \frac{\frac{1}{2}f}{-\frac{1}{3}f} = -\frac{3}{2}$

Negative sign implies that image is inverted with respect to object and so it is real.

**Ex** A concave mirror of focal length 10 cm and convex mirror of focal length 15 cm are placed facing each other 40 cm apart. A point object is placed between the mirror on their common axis and 15 cm from the concave mirror. Find the position of image produced by the reflection first at concave mirror and then at convex mirror.

**Sol.** For  $M_1$  mirror O act as a object, let its image is  $I_1$  then,

$$u = -15 \text{ cm}, f = -10 \text{ cm} \Rightarrow \frac{1}{v} + \frac{1}{-15} = \frac{1}{-10} \Rightarrow v = -30 \text{ cm}$$

Image  $I_1$  will act as a object for mirror  $M_2$  its distance from mirror  $M_2$ .

$$u_1 = -(40 - 30) \text{ cm} = -10 \text{ cm}$$

$$\text{so } \frac{1}{v_1} + \frac{1}{u_1} = \frac{1}{f} \Rightarrow \frac{1}{v_1} - \frac{1}{10} = \frac{1}{15} \Rightarrow v_1 = +6 \text{ cm}$$

So final image  $I_2$  is formed at a distance 6 cm behind the convex mirror and is virtual.

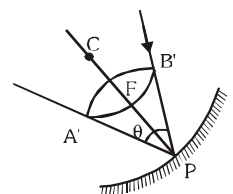
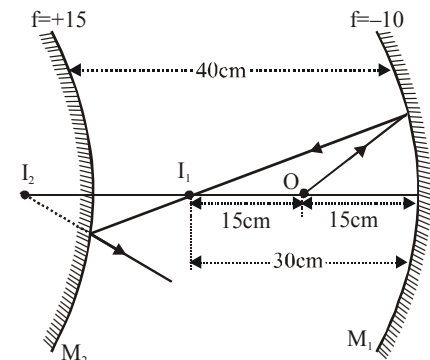
**Ex.** The sun subtends an angle  $\theta$  radians at the pole of a concave mirror of focal length  $f$ . What is the diameter of the image of the sun formed by the mirror.

**Sol.** Since the sun is at large distance very distant,  $u$  is very large and so  $\frac{1}{u} \approx 0$

$$\therefore \frac{1}{v} + \frac{1}{u} = \frac{1}{f} \Rightarrow \frac{1}{v} = -\frac{1}{f} \Rightarrow v = -f$$

The image of sun will be formed at the focus and will be real, inverted and diminished

$$A'B' = \text{height of image and } \theta = \frac{\text{Arc}}{\text{Radius}} = \frac{A'B'}{FP} \Rightarrow \theta = \frac{d}{f} \Rightarrow d = f\theta$$



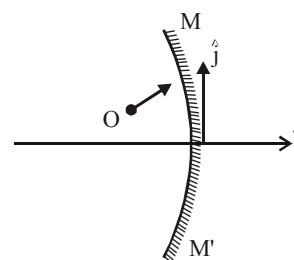
## VELOCITY OF IMAGE OF MOVING OBJECT (SPHERICAL MIRROR)

### (a) Velocity component along axis (Longitudinal velocity)

When an object is coming from infinite towards the focus of concave mirror

$$\therefore -\frac{1}{v} + \frac{1}{u} = \frac{1}{f} \Rightarrow -\frac{1}{v^2} \frac{dv}{dt} - \frac{1}{u^2} \frac{du}{dt} = 0 \Rightarrow \vec{v}_{ix} = -\frac{v^2}{u^2} \vec{v}_{ox} = -m^2 \vec{v}_{ox}$$

$$\left[ \vec{v}_{ix} = \frac{dv}{dt} = \text{velocity of image along principal-axis; } \vec{v}_{ox} = \frac{du}{dt} = \text{velocity of object along principal-axis} \right]$$



### (b) Velocity component perpendicular to axis (Transverse velocity)

$$m = \frac{h_i}{h_o} = -\frac{v}{u} = \frac{f}{f-u} \Rightarrow h_i = \left( \frac{f}{f-u} \right) h_o$$

$$\frac{dh_i}{dt} = \left( \frac{f}{f-u} \right) \frac{dh_o}{dt} + \frac{f h_o}{(f-u)^2} \frac{du}{dt}; \vec{v}_{iy} = \left[ m \vec{v}_{oy} + \frac{m^2 h_o}{f} \vec{v}_{ox} \right] \hat{j}$$

$$\left[ \begin{aligned} \frac{dh_i}{dt} &= \text{velocity of image } \perp^r \text{ to principal-axis} \\ \frac{dh_o}{dt} &= \text{velocity of object } \perp^r \text{ to principal-axis} \end{aligned} \right]$$

**Note :** Here principal axis has been taken to be along x-axis.

## POWER OF A MIRROR

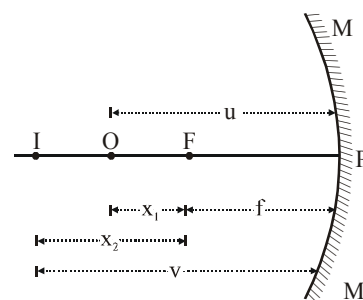
The power of a mirror is defined as  $P = -\frac{1}{f(m)} = -\frac{100}{f(\text{cm})}$

## NEWTON'S FORMULA

In case if spherical mirrors if object distance ( $x_1$ ) and image distance ( $x_2$ ) are measured from focus instead of pole,  
 $u = -(f + x_1)$  and  $v = -(f + x_2)$ ,

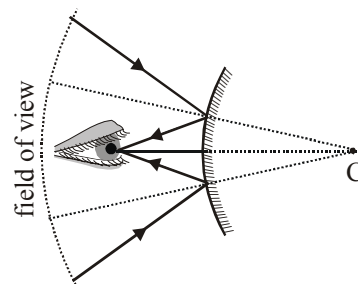
$$\text{by } \frac{1}{v} + \frac{1}{u} = \frac{1}{f} \Rightarrow -\frac{1}{(f+x_2)} - \frac{1}{(f+x_1)} = -\frac{1}{f}$$

on solving  $x_1 x_2 = f^2$  This is Newton's formula.



## POINTS

- Convex mirrors** gives erect, virtual and diminished image. In convex mirror the field of view is increased as compared to plane mirror. It is used as rear-view mirror in vehicles.
- Concave mirrors** give enlarged, erect and virtual image, so these are used by dentists for examining teeth. Due to their converging property concave mirrors are also used as reflectors in automobile head lights and search lights
- As focal length of a spherical mirror  $f = R/2$  depends only on the radius of mirror and is independent of wavelength of light and refractive index of medium so the focal length of a spherical mirror in air or water and for red or blue light is same.



## REFRACTION

Refraction is the phenomenon in which direction of propagation of light changes at the boundary when it passes from one medium to the other. In case of refraction frequency does not change.

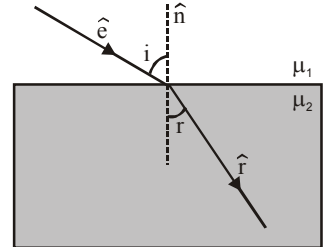
⊙ **Laws of Refraction**

- (i) Incident ray, refracted ray and normal always lie in the same plane.

In vector form  $(\hat{e} \times \hat{n}) \cdot \hat{r} = 0$

- (ii) The product of refractive index and sine of angle of incidence at a point in a medium is constant.  $\mu_1 \sin i = \mu_2 \sin r$  (Snell's law)

In vector form  $\mu_1 |\hat{e} \times \hat{n}| = \mu_2 |\hat{r} \times \hat{n}|$



**Absolute refractive index**

It is defined as the ratio of speed of light in free space 'c' to that in a given medium v.  $\mu$  or  $n = \frac{c}{v}$

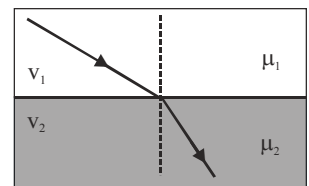
Denser is the medium, lesser will be the speed of light and so greater will be the refractive index,

$$\therefore v_{\text{glass}} < v_{\text{water}}, \quad \therefore \mu_G > \mu_W$$

**Relative refractive index**

When light passes from one medium to the other, the refractive index of medium 2 relative to 1 is written as  ${}_1\mu_2$  and is defined as

$${}_1\mu_2 = \frac{\mu_2}{\mu_1} = \frac{(c/v_2)}{(c/v_1)} = \frac{v_1}{v_2}$$



⊙ **Bending of light ray**

According to Snell's law,  $\mu_1 \sin i = \mu_2 \sin r$

- (i) If light passes from rarer to denser medium  $\mu_1 = \mu_R$  and  $\mu_2 = \mu_D$

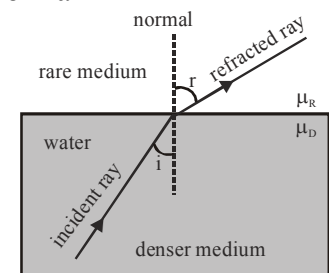
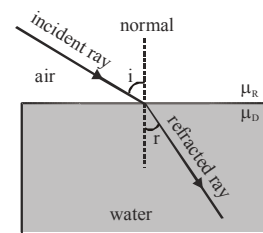
so that  $\frac{\sin i}{\sin r} = \frac{\mu_D}{\mu_R} > 1 \Rightarrow \angle i > \angle r$

In passing from rarer to denser medium, the ray bends towards the normal.

- (ii) If light passes from denser to rarer medium  $\mu_1 = \mu_D$  and  $\mu_2 = \mu_R$

$$\frac{\sin i}{\sin r} = \frac{\mu_R}{\mu_D} < 1 \Rightarrow \angle i < \angle r$$

In passing from denser to rarer medium, the ray bends away from the normal



## APPARENT DEPTH AND NORMAL SHIFT

If a point object in denser medium is observed from rarer medium and boundary is plane, then from Snell's law we have  $\mu_D \sin i = \mu_R \sin r$ ...(i)

If the rays OA and OB are close enough to reach the eye.

$$\sin i \approx \tan i = \frac{p}{d_{ac}} \text{ and } \sin r \approx \tan r = \frac{p}{d_{ap}}$$

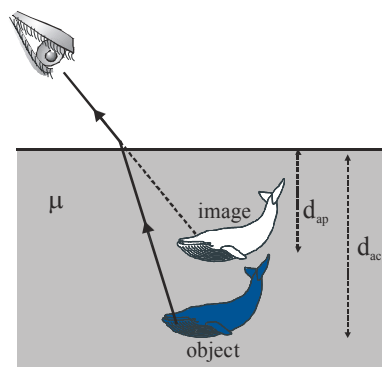
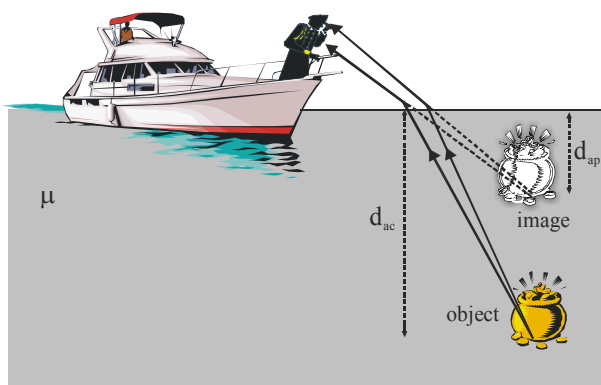
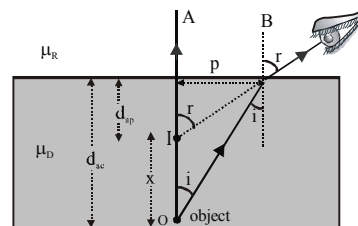
here  $d_{ac}$  = actual depth,  $d_{ap}$  = apparent depth

$$\text{So that equation (i) becomes } \mu_D = \frac{p}{d_{ac}} = \mu_R \frac{p}{d_{ap}} \Rightarrow \frac{d_{ac}}{d_{ap}} = \frac{\mu_D}{\mu_R} = \frac{\mu_1}{\mu_2}$$

$$(\text{If } \mu_R = 1, \mu_D = \mu) \text{ then } d_{ap} = \frac{d_{ac}}{\mu} \text{ so } d_{ap} < d_{ac} \quad \dots(\text{ii})$$

The distance between object and its image, called normal shift (x)

$$x = d_{ac} - d_{ap} \left[ \because d_{ap} = \frac{d_{ac}}{\mu} \right]; x = d_{ac} - \frac{d_{ac}}{\mu} = d_{ac} \left[ 1 - \frac{1}{\mu} \right] \dots(\text{iii}) \quad \text{If } d_{ac} = d \text{ then } x = d \left[ 1 - \frac{1}{\mu} \right]$$



## Object in a rarer medium is seen from a denser medium

$$\frac{d_{ac}}{d_{ap}} = \frac{\mu_1}{\mu_2} = \frac{\mu_R}{\mu_D} = \frac{1}{\mu} (< 1)$$

$$d_{ap} = \mu d_{ac} \text{ i.e., } d_{ap} > d_{ac}$$

A high flying object appears to be higher than in reality.

$$x = d_{ap} - d_{ac} \Rightarrow x = [\mu - 1] d_{ac}$$

## LATERAL SHIFT

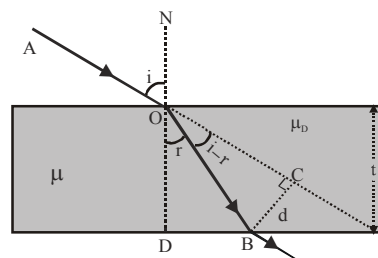
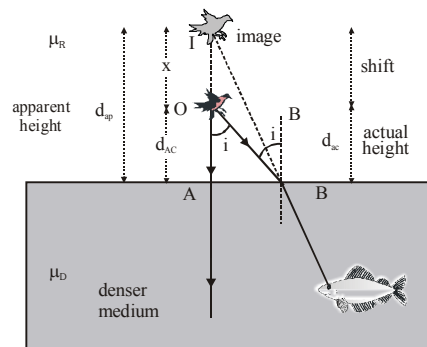
The perpendicular distance between incident and emergent ray is known as lateral shift.

Lateral shift  $d = BC$  and  $t$  = thickness of slab

$$\text{In } \Delta BOC \sin(i-r) \frac{BC}{OB} = \frac{d}{OB} \Rightarrow d = OB \sin(i-r) \dots(\text{i})$$

$$\text{In } \Delta OBD \cos r = \frac{OD}{OB} = \frac{t}{OB} \Rightarrow OB = \frac{t}{\cos r} \dots(\text{ii})$$

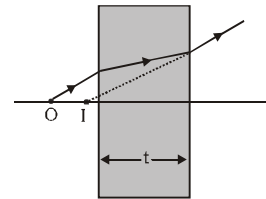
$$\text{From (i) and (ii) } d = \frac{t}{\cos r} \sin(i-r)$$



**TRANSPARENT GLASS SLAB (Normal shift)**

When an object is placed in front of a glass slab, it shifts the object in the direction of incident light and forms an image at a distance  $x$ .

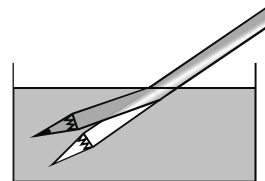
$$x = t \left[ 1 - \frac{1}{\mu} \right]$$

**SOME ILLUSTRATIONS OF REFRACTION**

- Bending of an object**

When a point object in a denser medium is seen

from a rarer medium it appears to bend by  $\frac{d}{\mu}$



- Twinkling of stars**

Due to fluctuations in refractive index of atmosphere the refraction becomes irregular and the light sometimes reaches the eye and sometimes it does not. This gives rise to twinkling of stars.

**KEY POINTS**

- $\mu$  is a scalar and has no units and dimensions.
- If  $\epsilon_0$  and  $\mu_0$  are electric permittivity and magnetic permeability respectively of free space while  $\epsilon$  and  $\mu$  those of a given medium, then according to electromagnetic theory.

$$c = \frac{1}{\sqrt{\epsilon_0 \mu_0}} \text{ and } v_m = \frac{1}{\sqrt{\epsilon \mu}} \Rightarrow n_m = \frac{c}{v} = \sqrt{\frac{\epsilon \mu}{\epsilon_0 \mu_0}} = \sqrt{\epsilon_r \mu_r}$$

- As in vacuum or free space, speed of light of all wavelengths is maximum and equal to  $c$  so for all wavelengths the refractive index of free space is minimum and is  $\mu = \frac{c}{v_m} = \frac{c}{c} = 1$

**Ex.** A ray of light is incident on a transparent glass slab of refractive index 1.62. If the reflected and refracted rays are mutually perpendicular, what is the angle of incidence? [ $\tan^{-1}(1.62) = 58.3^\circ$ ]

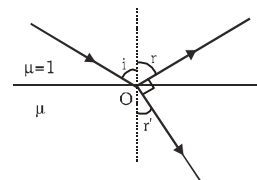
**Sol.** According to given problem:  $r + 90^\circ + r' = 180^\circ$

i.e.,  $r' = 90^\circ - r \Rightarrow r' = (90^\circ - i)$  [ $\because \angle i = \angle r$ ]

And as according to Snell's law:  $1 \sin i = \mu \sin r'$

$\sin i = \mu \sin (90 - i) \Rightarrow \sin i = \mu \cos i$  [ $\because \sin (90 - i) = \cos i$ ]

$\Rightarrow \tan i = \mu \Rightarrow i = \tan^{-1} \mu = \tan^{-1}(1.62) = 58.3^\circ$



**Ex.** A 20 cm thick glass slab of refractive index 1.5 is kept in front of a plane mirror. An object is kept in air at a distance 40 cm from the mirror. Find the position of image w.r.t an observer near the object. What is effect of separation between glass slab and the mirror on image.

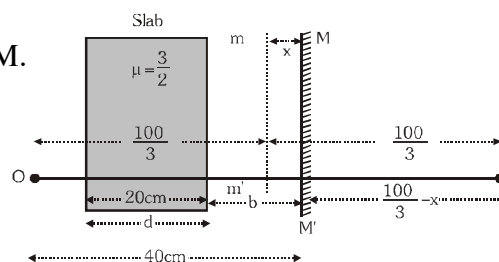
**Sol.** Shifting in object due to glass slab  $x = d \left( 1 - \frac{1}{\mu} \right) = 20 \left[ 1 - \frac{1}{1.5} \right] = \frac{20}{3}$  cm

Distance of object from mirror (as seen by mirror)  $= 40 - \frac{20}{3} = \frac{100}{3}$  cm

Image will be formed at a distance  $\frac{100}{3}$  cm from mirror M.

Shifting in image due to glass slab  $= \frac{20}{3}$  cm

So distance of image from mirror  $= \frac{100}{3} - \frac{20}{3} = \frac{80}{3}$  cm



Distance of image from the actual plane mirror is independent of separation  $b$  between glass slab and the mirror. If the distance is more then brightness of image will be less.

**Ex.** If one face of a prism angle  $30^\circ$  and  $\mu = \sqrt{2}$  is silvered, the incident ray retraces its initial path. What is the angle of incidence ?

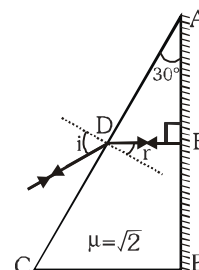
**Sol.** As incident ray retraces its path the ray is incident normally on the silvered face of the prism as shown in figure.

Further, as in  $\triangle AED$   $30^\circ + 90^\circ + \angle D = 180^\circ \Rightarrow \angle D = 60^\circ$

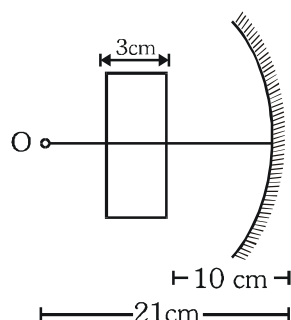
Now as by construction,  $\angle D + \angle r = 90^\circ \Rightarrow \angle r = 90^\circ - 60^\circ = 30^\circ$

$\therefore$  from Snell's law at surface AC,  $1 \sin i = \sqrt{2} \sin 30^\circ = \sqrt{2} \times \frac{1}{2} = \frac{1}{\sqrt{2}}$

$\therefore \sin i = \frac{1}{\sqrt{2}}$  so  $i = 45^\circ$



**Ex.** An object is placed 21 cm in front of a concave mirror of radius of curvature 20 cm. A glass slab of thickness 3 cm and refractive index 1.5 is placed closed to the mirror in space between the object and the mirror. Find the position of final image formed if distance of nearer surface of the slab from the mirror is 10 cm.

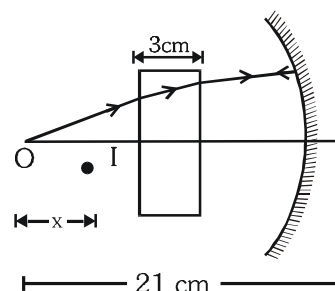


**Sol.** shift by slab  $x = d \left( 1 - \frac{1}{\mu} \right) = 3 \left( 1 - \frac{1}{1.5} \right) = 1 \text{ cm}$

for image formed by mirror  $u = -(21 - 1) \text{ cm} = -20 \text{ cm}$ .

$$\frac{1}{u} + \frac{1}{v} = \frac{1}{f} \Rightarrow \frac{1}{-20} + \frac{1}{v} = \frac{1}{-10} \Rightarrow v = -20 \text{ cm}$$

shift in the direction of light  $v = -(20 + 1) = -21 \text{ cm}$ .

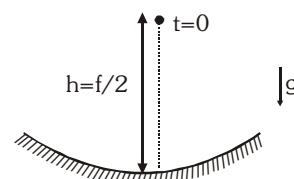


**Ex.** A particle is dropped along the axis from a height  $\frac{f}{2}$  on a concave mirror of focal length  $f$  as shown in figure. Find the maximum speed of image.

**Sol.**  $v_{IM} = -m^2 v_{OM} = -m^2 (gt)$  where

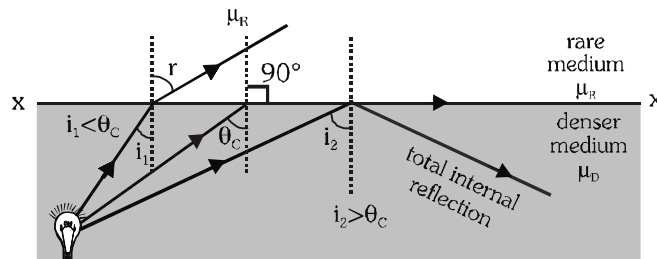
$$m = \frac{f}{f - u} = \frac{-f}{-f + \left( \frac{f}{2} - \frac{gt^2}{2} \right)} = \frac{2f}{f + gt^2} \Rightarrow v_1 = - \left( \frac{2f}{f + gt^2} \right)^2 (gt) = \frac{-4f^2 gt}{(f + gt^2)^2}$$

For maximum speed  $\frac{dv_1}{dt} = 0 \Rightarrow t = \sqrt{\frac{f}{3g}} \Rightarrow v_{\text{max}} = \frac{3}{4} \sqrt{3fg}$



## TOTAL INTERNAL REFLECTION

When light ray travel from denser to rarer medium it bend away from the normal if the angle of incident is increased, angle of refraction will also increased. At a particular value of angle the refracted ray subtend  $90^\circ$  angle with the normal, this angle of incident is known as critical angle ( $\theta_c$ ). If angle of incident further increase the ray come back in the same medium this phenomenon is known as total internal reflection.



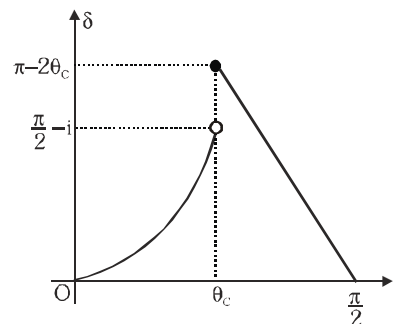
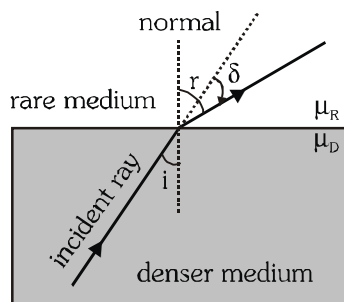
## CONDITIONS

- Angle of incident  $>$  critical angle [ $i > \theta_c$ ]
- Light should travel from denser to rare medium  $\Rightarrow$  Glass to air, water to air, Glass to water

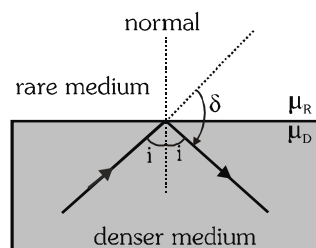
Snell's Law at boundary  $xx'$ ,  $\mu_D \sin \theta_c = \mu_R \sin 90^\circ \Rightarrow \sin \theta_c = \frac{\mu_R}{\mu_D}$

**Graph between angle of deviation ( $\delta$ ) and angle of incidence ( $i$ ) as rays goes from denser to rare medium**

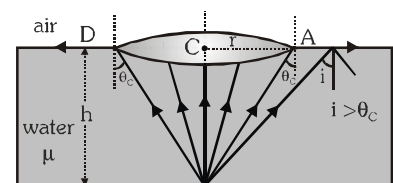
- If  $i < \theta_c$ ,  $\mu_D \sin i = \mu_R \sin r$ ;  $r = \sin^{-1} \left( \frac{\mu_D}{\mu_R} \sin i \right)$  so  $\delta = r - i = \sin^{-1} \left( \frac{\mu_D}{\mu_R} \sin i \right) - i$



- If  $i > \theta_c$ ;  $\delta = \pi - 2i$



- A point object is situated at the bottom of tank filled with a liquid of refractive index  $\mu$  upto height  $h$ . It is found light from the source come out of liquid surface through a circular portion above the object

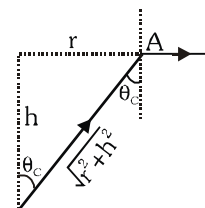




$$\sin \theta_c = \frac{r}{\sqrt{r^2 + h^2}} \quad \& \quad \sin \theta_c = \frac{1}{\mu} \Rightarrow \frac{1}{\mu} = \frac{r}{\sqrt{r^2 + h^2}} \Rightarrow \frac{1}{\mu^2} = \frac{r^2}{r^2 + h^2}$$

$$\Rightarrow \mu^2 r^2 = r^2 + h^2 \Rightarrow (\mu^2 - 1)r^2 = h^2 \Rightarrow \text{radius of circular portion}$$

$$r = \frac{h}{\sqrt{\mu^2 - 1}} \quad \text{and} \quad \text{area} = \pi r^2$$



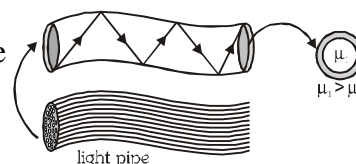
## SOME ILLUSTRATIONS OF TOTAL INTERNAL REFLECTION

### • Sparkling of diamond

The sparkling of diamond is due to total internal reflection inside it. As refractive index for diamond is 2.5 so  $\theta_c = 24^\circ$ . Now the cutting of diamond are such that  $i > \theta_c$ . So TIR will take place again and again inside it. The light which beams out from a few places in some specific directions makes it sparkle.

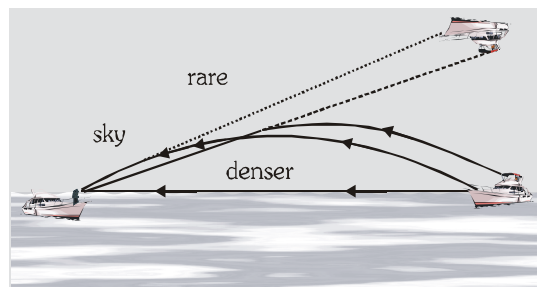
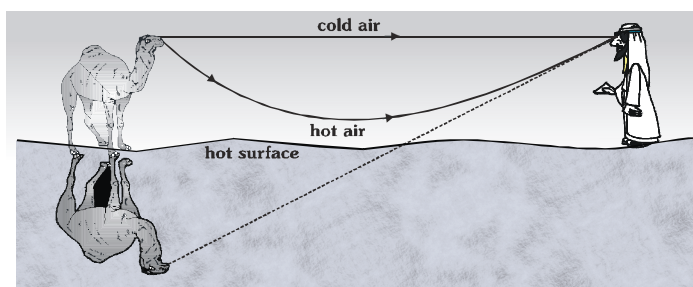
### • Optical Fibre

In it light through multiple total internal reflections is propagated along the axis of a glass fibre of radius of few microns in which index of refraction of core is greater than that of surroundings.



### • Mirage and looming

Mirage is caused by total internal reflection in deserts where due to heating of the earth, refractive index of air near the surface of earth becomes lesser than above it. Light from distant objects reaches the surface of earth with  $i > \theta_c$  so that TIR will take place and we see the image of an object along with the object as shown in figure.



Similar to 'mirage' in deserts, in polar regions 'looming' takes place due to TIR. Here  $\mu$  decreases with height and so the image of an object is formed in air if ( $i > \theta_c$ ) as shown in figure.

## KEY POINTS

- A diver in water at a depth  $d$  sees the world outside through a horizontal circle of radius.  $r = d \tan \theta_c$ .
- In case of total internal reflection, as all (i.e. 100%) incident light is reflected back into the same medium there is no loss of intensity while in case of reflection from mirror or refraction from lenses there is some loss of intensity as all light can never be reflected or refracted. This is why images formed by TIR are much brighter than formed by mirrors or lenses.

**Ex.** A rectangular block of glass is placed on a printed page laying on a horizontal surface. Find the minimum value of the refractive index of glass for which the letters on the page are not visible from any of the vertical faces of the block.

**Sol.** The situation is depicted in figure. Light will not emerge out from the vertical face BC if at it

$$i > \theta_c \text{ or } \sin i > \sin \theta_c \Rightarrow \sin i > \frac{1}{\mu} \left[ \text{as } \sin \theta_c = \frac{1}{\mu} \right] \dots (i)$$

But from Snell's law at O  $1 \times \sin \theta = \mu \sin r$

$$\text{And in } \Delta OPR, r + 90 + i = 180 \Rightarrow r + i = 90^\circ \Rightarrow r = 90 - i$$

$$\text{So } \sin \theta = \mu \sin (90 - i) = \mu \cos i \Rightarrow \cos i = \frac{\sin \theta}{\mu}$$

$$\text{so } \sin i = \sqrt{1 - \cos^2 i} = \sqrt{1 - \left[ \frac{\sin \theta}{\mu} \right]^2} \dots (ii)$$

so substituting the value of  $\sin i$  from equation (ii) in (i),

$$\sqrt{1 - \frac{\sin^2 \theta}{\mu^2}} > \frac{1}{\mu} \text{ i.e., } \mu^2 > 1 + \sin^2 \theta \because (\sin^2 \theta)_{\max} = 1 \therefore \mu^2 > 2 \Rightarrow \mu > \sqrt{2} \therefore$$

$$\mu_{\min} = \sqrt{2}$$

### REFRACTION AT TRANSPARENT CURVED SURFACE

$\mu_1$  = refractive index of the medium in which actual incident ray lies.

$\mu_2$  = refractive index of the medium in which actual refractive ray lies.

O = Object

P = pole

C = centre of curvature

R = PC = radius of curvature

#### Refraction from curved surface

$$\mu_1 \sin \theta_1 = \mu_2 \sin \theta_2$$

if angle is very small :  $\mu_1 \theta_1 = \mu_2 \theta_2 \dots (i)$

$$\text{But } \theta_1 = \alpha + \beta \dots (ii)$$

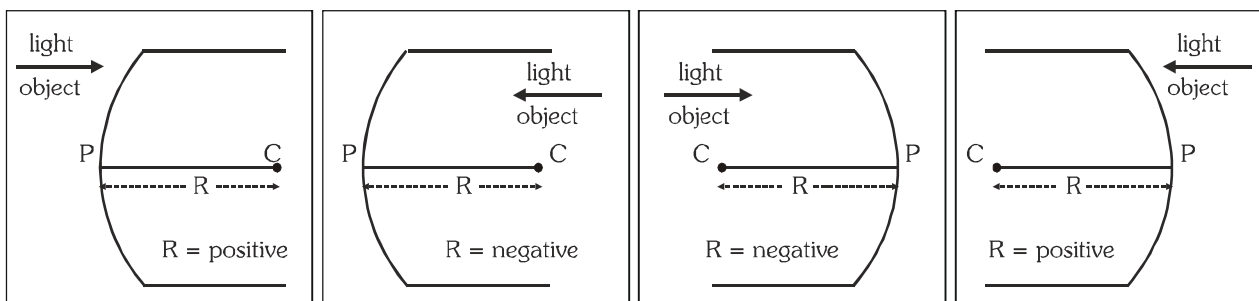
$$\beta = \theta_2 + \gamma \dots (iii)$$

from (i), (ii) and (iii)  $\mu_1 (\alpha + \beta) = \mu_2 (\beta - \gamma)$

$$\Rightarrow \mu_1 \alpha + \mu_1 \beta = \mu_2 \beta - \mu_2 \gamma \Rightarrow \mu_1 \alpha + \mu_2 \gamma = (\mu_2 - \mu_1) \beta$$

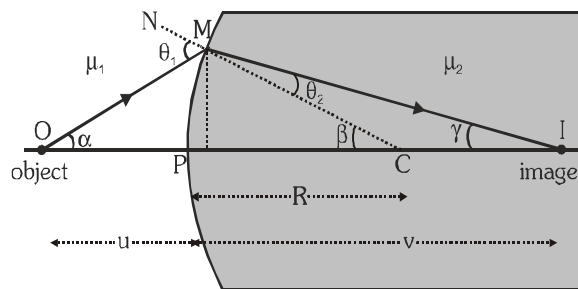
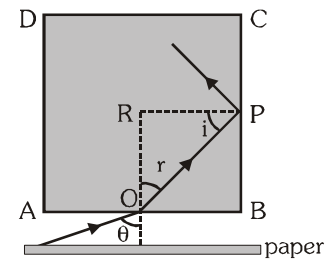
$$\Rightarrow \frac{\mu_1 PM}{-u} + \frac{\mu_2 PM}{v} = \frac{(\mu_2 - \mu_1) PM}{R} \Rightarrow \frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$$

#### SIGN CONVENTION FOR RADIUS OF CURVATURE



These are valid for all single refraction surfaces – convex, concave or plane. In case of plane refracting

$$\text{surface } R \rightarrow \infty, \frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R} \Rightarrow \frac{\mu_2}{v} - \frac{\mu_1}{u} = 0 \text{ i.e. } \frac{u}{v} = \frac{\mu_1}{\mu_2} \text{ or } \frac{d_{Ac}}{d_{Ap}} = \frac{\mu_1}{\mu_2}$$



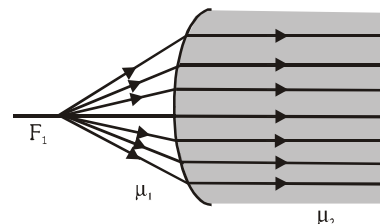
## FOCAL LENGTH OF A SINGLE SPHERICAL SURFACE

A single spherical surface has two principal focus points which are as follows—

- (i) **First focus:** The first principal focus is the point on the axis where when an object is placed, the image is formed at infinity. That is when

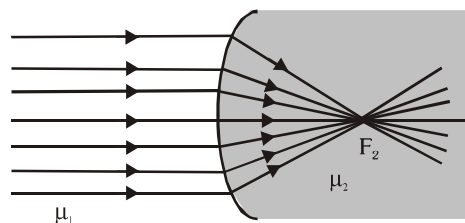
$$u = f_1, v = \infty, \text{ then from } -\frac{\mu_1}{u} + \frac{\mu_2}{v} = \left( \frac{\mu_2 - \mu_1}{R} \right)$$

$$\text{We get } -\frac{\mu_1}{f_1} = \frac{\mu_2 - \mu_1}{R} \Rightarrow f_1 = \frac{-\mu_1 R}{(\mu_2 - \mu_1)}$$



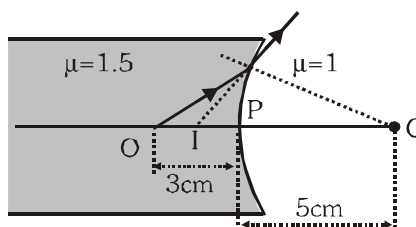
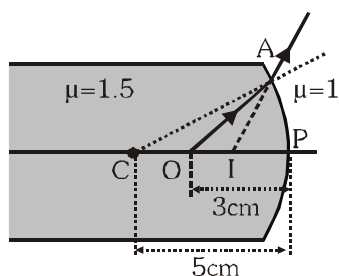
- (ii) **Second focus:** Similarly, the second principal focus is the point where parallel rays focus. That is  $u_1 = -\infty, v_1 = f_2$ , then

$$\frac{\mu_2}{f_2} = \frac{\mu_2 - \mu_1}{R}; f_2 = \frac{\mu_2 R}{(\mu_2 - \mu_1)}$$



- (iii) **Ratio of Focal length:**  $\frac{f_1}{f_2} = -\frac{\mu_1}{\mu_2}$

**Ex** An air bubble in glass ( $\mu = 1.5$ ) is situated at a distance 3 cm from a spherical surface of diameter 10 cm as shown in Figure. At what distance from the surface will the bubble appear if the surface is (a) convex (b) concave.



**Sol.** In case of refraction from curved surface  $\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{(\mu_2 - \mu_1)}{R}$

$$(a) \mu_1 = 1.5, \mu_2 = 1, R = -5 \text{ cm and } u = -3 \text{ cm} \Rightarrow \frac{1}{v} - \frac{(1.5)}{(-3)} = \frac{1-1.5}{(-5)} \Rightarrow v = -2.5 \text{ cm}$$

the bubble will appear at a distance 2.5 cm from the convex curved surface inside the glass.

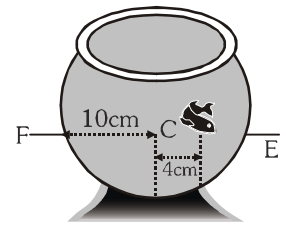
$$(b) \mu_1 = 1.5, \mu_2 = 1, R = 5 \text{ cm and } u = -3 \text{ cm} \Rightarrow \frac{1}{v} - \frac{(1.5)}{(-3)} = \frac{1-1.5}{(5)} \Rightarrow v = -1.66 \text{ cm}$$

the bubble will appear at a distance 1.66 cm from the concave curved surface inside the glass.

**Note :** If the surface is plane then  $R \rightarrow \infty$

$$\text{case (a) or (b) would yield } \frac{1}{v} - \frac{(1.5)}{(-3)} = \frac{(1-1.5)}{\infty} \Rightarrow v = -2 \text{ cm}$$

**Ex.** In a thin spherical fish bowl of radius 10 cm filled with water of refractive index ( $4/3$ ), there is a small fish at a distance 4 cm from the centre C as shown in Figure. Where will the fish appear to be, if seen from (a) E and (b) F (neglect the thickness of glass) ?

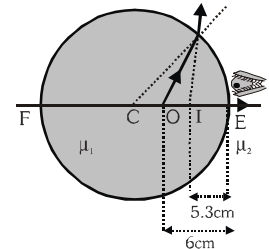


**Sol.** In the case of refraction from curved surface  $\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{(\mu_2 - \mu_1)}{R}$

(a) Seen from E  $\mu_1 = \frac{4}{3}$ ,  $\mu_2 = 1$ ,  $R = -10$  cm &  $u = -(10 - 4) = -6$  cm

$$\Rightarrow \frac{1}{v} - \frac{\frac{4}{3}}{-6} = \frac{1 - \frac{4}{3}}{-10} \Rightarrow v = \frac{90}{17} = -5.3 \text{ cm}$$

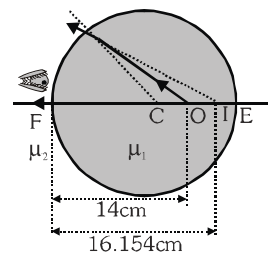
i.e., fish will appear at a distance 5.3 cm from E towards F (lesser than actual distance, i.e., 6 cm)



(b) Seen from F  $\mu_1 = \frac{4}{3}$ ,  $\mu_2 = 1$ ,  $R = -10$  cm and  $u = -(10 + 4) = -14$  cm

$$\Rightarrow \frac{1}{v} - \frac{\frac{4}{3}}{-14} = \frac{1 - \frac{4}{3}}{-10} \Rightarrow v = \frac{-210}{13} = -16.154 \text{ cm}$$

so fish will appear at a distance 16.154 cm from F toward E (more than actual distance, i.e., 14 cm)

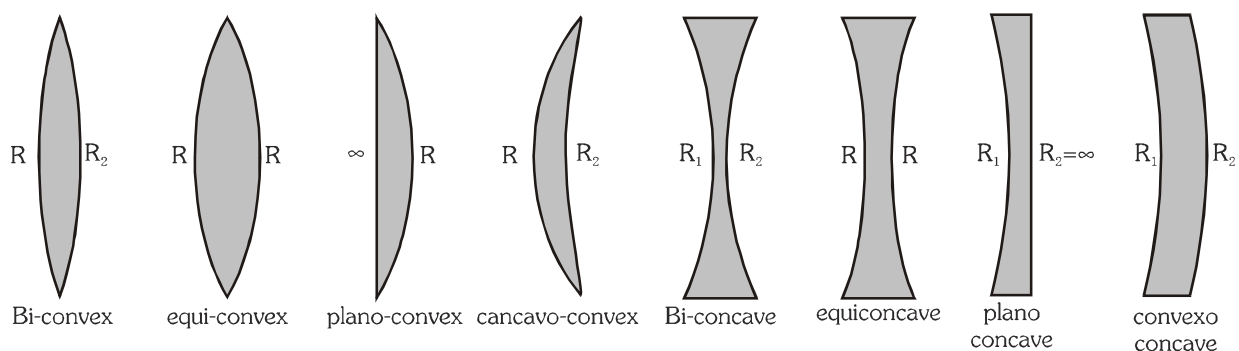


## LENS

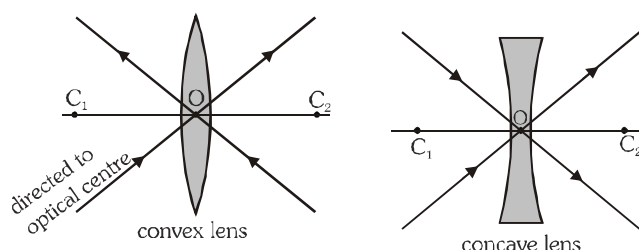
A lens is a piece of transparent material with two refracting surfaces such that at least one is curved and refractive index of its material is different from that of the surroundings.

A thin spherical lens with refractive index greater than that of surroundings behaves as a convergent or convex lens, i.e., converges parallel rays if its central (i.e. paraxial) portion is thicker than marginal one.

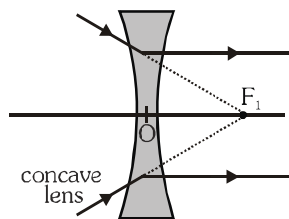
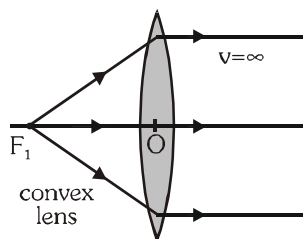
However if the central portion of a lens is thinner than marginal, it diverges parallel rays and behaves as divergent or concave lens. This is how we can identify convergent and divergent lenses.



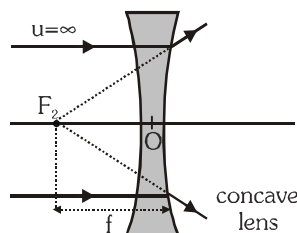
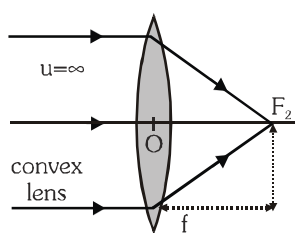
- **Optical Centre :** O is a point for a given lens through which any ray passes undeviated



- **Principal Axis :**  $C_1 C_2$  is a line passing through optical centre and perpendicular to the lens.
- **Principal Focus :** A lens has two surfaces and hence two focal points. First focal point is an object point on the principal axis for which image is formed at infinity.



While second focal point is an image point on the principal axis for which object lies at infinity



- **Focal Length  $f$**  is defined as the distance between optical centre of a lens and the point where the parallel beam of light converges or appears to converge.
- **Aperture :** In reference to a lens, aperture means the effective diameter. Intensity of image formed by a lens which depends on the light passing through the lens will depend on the square of aperture, i.e., Intensity  $\propto$  (Aperture)<sup>2</sup>

### LENS-MAKER'S FORMULA

#### In case of image formation by a lens

Image formed by first surface acts as object for the second.

So, from the formula of refraction at curved surface.

$$\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$$

For first surface A  $\frac{\mu_L}{v_1} - \frac{\mu_M}{u} = \frac{\mu_L - \mu_M}{R_1} \dots (i) \quad [\because \mu_2 = \mu_L, \mu_1 = \mu_M]$

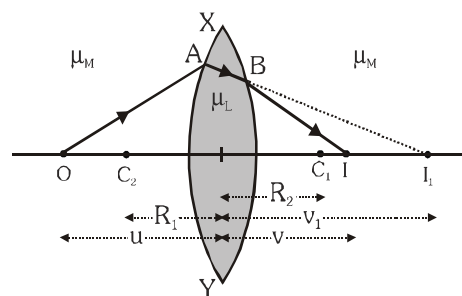
For second surface B  $\frac{\mu_M}{v} - \frac{\mu_L}{v_1} = \frac{\mu_M - \mu_L}{R_2} = -\frac{\mu_L - \mu_M}{R_2} \dots (ii) \quad [\because \mu_2 = \mu_M, \mu_1 = \mu_L, \mu_1 = \mu_2, u \rightarrow v_1]$

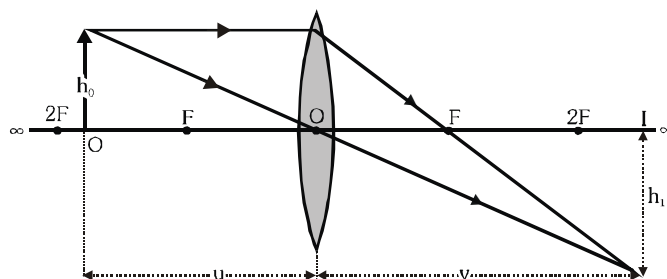
By adding (i) and (ii)

$$\mu_M \left[ \frac{1}{v} - \frac{1}{u} \right] = (\mu_L - \mu_M) \left[ \frac{1}{R_1} - \frac{1}{R_2} \right] \Rightarrow \frac{1}{v} - \frac{1}{u} = \frac{\mu_L - \mu_M}{\mu_M} \left[ \frac{1}{R_1} - \frac{1}{R_2} \right] = (\mu - 1) \left[ \frac{1}{R_1} - \frac{1}{R_2} \right] \dots (iii) \quad \left( \because \mu = \frac{\mu_L}{\mu_M} \right)$$

Now if object is at infinity, Image will be formed at the focus,  $u = -\infty, v = f$

So  $\frac{1}{f} = (\mu - 1) \left[ \frac{1}{R_1} - \frac{1}{R_2} \right] \dots (iv)$





This is known as lens makers formula by equating (iii) and (iv)  $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$

this is known as lens formula

**Magnification** :  $m = \frac{\text{height of image}}{\text{height of object}} = \frac{h_1}{h_0} = \frac{v}{u} = \frac{f}{f+u} = \frac{f-v}{f}$

### RULES FOR IMAGE FORMATION

- A ray passing through optical centre proceeds undeviated through the lens
- A ray passing through first focus or directed towards it, after refraction from the lens, becomes parallel to the principal axis.
- A ray passing parallel to the principal axis after refraction through the lens passes or appears to pass through  $F_2$

#### For Convergent or convex lens

| Object                  | Image            | Magnification         |
|-------------------------|------------------|-----------------------|
| $-\infty$               | F                | $ m  \ll 1$ & $m < 0$ |
| $-\infty - 2F$          | $F-2F$           | $ m  < 1$ & $m < 0$   |
| $2F$                    | $2F$             | $m = -1$              |
| $F - 2F$                | $\infty-2F$      | $ m  > 1$ & $m < 0$   |
| Just before F towards C | $+\infty$        | $m \ll -1$            |
| Just before F towards P | $-\infty$        | $m \gg 1$             |
| $F - O$                 | In front of lens | $m > 1$               |

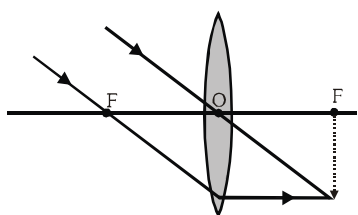
### IMAGE FORMATION FOR CONVEX LENS (CONVERGENT LENS)

- (i) Object is placed at infinity

**Image :**

at F real inverted very small in size

$$|m| \ll 1 \text{ \& } m < 0$$

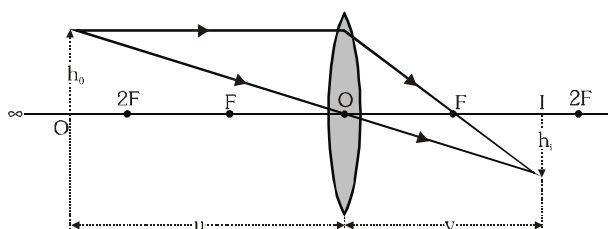


- (ii) Object is placed in between  $\infty - 2F$

**Image :**

real ( $F - 2F$ ) inverted small in size (diminished)

$$|m| < 1 \text{ \& } m < 0$$

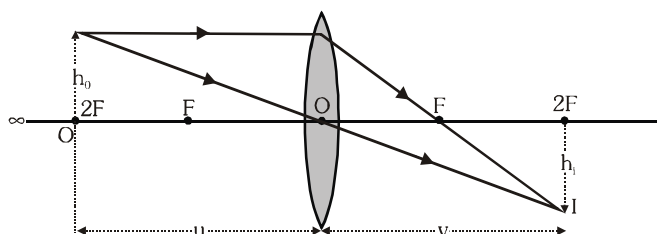


(iii) Object is placed at  $2F$

**Image :**

real (at  $2F$ ) inverted equal (of same size)

( $m = -1$ )

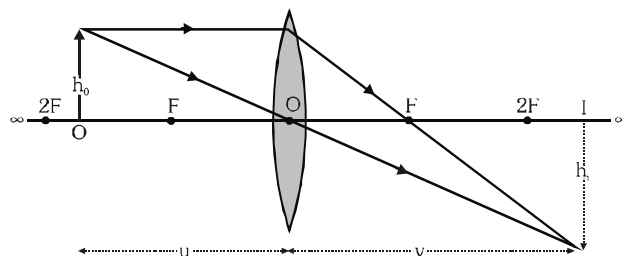


(iv) Object is placed in between  $2F - F$

**Image :**

real ( $2F - \infty$ ) inverted enlarged

$|m| > 1$  &  $m < 0$

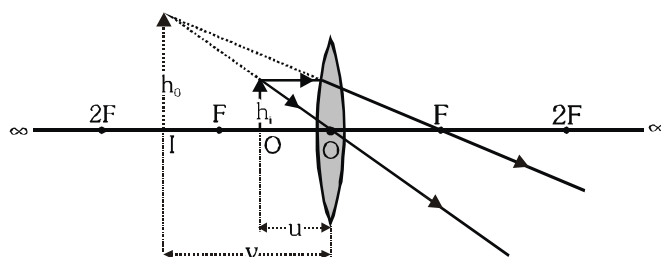


(v) Object is placed in between  $F - O$

**Image :**

virtual (in front of lens) erected enlarge

( $m > +1$ )



### IMAGE FORMATION FOR CONCAVE LENS (DIVERGENT LENS)

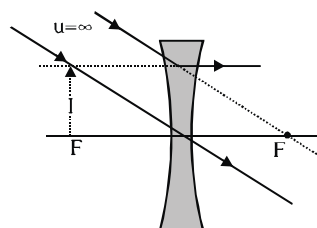
Image is virtual, diminished, erect, towards the object,  $m = +ve$

(i) Object is placed at infinity

**Image :**

At  $F$  virtual erected

diminished ( $m \ll +1$ )

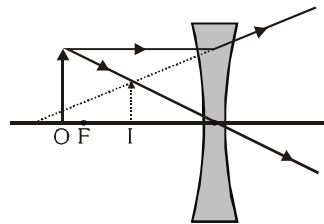


(ii) Object is placed in front of lens

**Image :**

between  $F$  and optical centre

virtual erected diminished ( $m < +1$ )



### Sign convention for object/image for lens

|                |          |
|----------------|----------|
| Real object    | $u - ve$ |
| Real image     | $v + ve$ |
| Virtual object | $u + ve$ |
| Virtual image  | $v - ve$ |

### POWER OF LENS

Reciprocal of focal length in meter is known as power of lens.

**SI UNIT :** dioptre (D)

**Power of lens :**  $P = \frac{1}{f(m)} = \frac{100}{f(cm)}$  dioptre [in air]

## COMBINATION OF LENSES

**Two thin lens are placed in contact with each other**

power of combination.  $P = P_1 + P_2 \Rightarrow \frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2}$

Use sign convention when solve numericals

**Two thin lens are placed in at a small distance d**  
(provided incident rays are parallel to principal axis).

$$\frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2} - \frac{d}{f_1 f_2} \Rightarrow P = P_1 + P_2 - d P_1 P_2$$

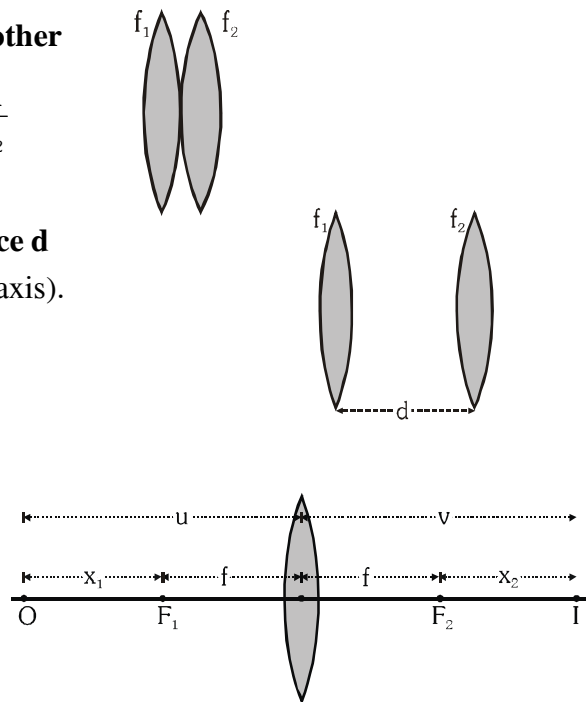
Use sign convention when solving numericals

• **Newton's Formula**

$$f = \sqrt{x_1 x_2}$$

$x_1$  = distance of object from focus.

$x_2$  = distance of image from focus.



## SOME SPECIAL CASES

(i) The focal length of equiconvex lens placed in air

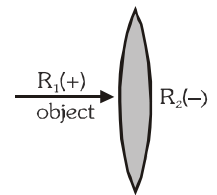
refractive index of lens  $\mu_L = \mu$

$$R_1 = +R,$$

refractive index of medium  $\mu_M = 1$

$$R_2 = -R$$

$$\frac{1}{f} = (\mu - 1) \left[ \frac{1}{R} - \left( -\frac{1}{R} \right) \right] \Rightarrow \text{Focal length } f = \frac{R}{2(\mu - 1)}$$



(ii) Focal length of planoconvex lens placed in air

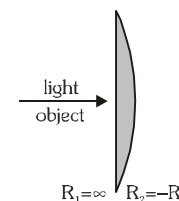
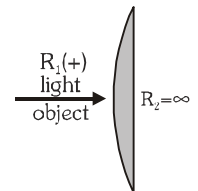
$$\frac{1}{f} = (\mu - 1) \left[ \frac{1}{R} - \frac{1}{\infty} \right]$$

$$\Rightarrow \text{Focal length } f = \frac{R}{(\mu - 1)}$$

If object is placed towards plane surface

$$\frac{1}{f} = (\mu - 1) \left[ \frac{1}{\infty} - \left( -\frac{1}{R} \right) \right]$$

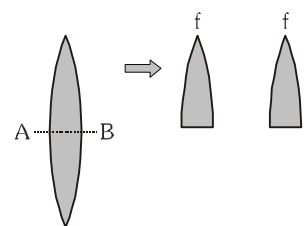
$$\Rightarrow \text{Focal length } f = \frac{R}{(\mu - 1)}$$



(iii) If an equiconvex lens of focal length  $f$  is cut into equal parts by a horizontal plane AB then the focal length of each part will be equal to that of initial lens. Because  $\mu$ ,  $R_1$  and  $R_2$  will remain unchanged. Only intensity will be reduced.

$\therefore$  intensity  $I \propto (\text{apertures})^2$

$\therefore$  intensity through a single part will be reduced





- (iv) If the same lens is cut into equal parts by a vertical plane CD the focal length of each part will be double of initial value but intensity will remain unchanged.

For equiconvex lens  $\frac{1}{f} = \frac{(\mu - 1)2}{R}$  For plano convex lens  $\frac{1}{f_1} = \frac{\mu - 1}{R}$

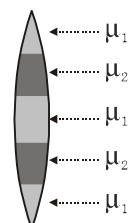
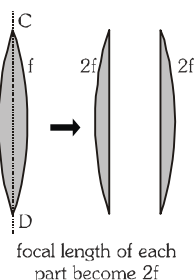
So  $\frac{1}{f} = \frac{2}{f_1} \Rightarrow f_1 = 2f \Rightarrow$  Focal length of each part = 2

(focal length of original lens)

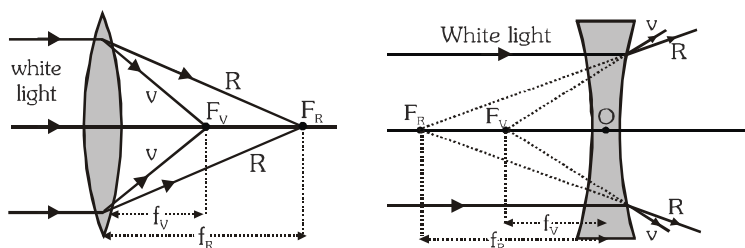
- (v) If a lens is made of number of layers of different refractive index for a given wavelength then no. of images is equal to number of refractive index, as

$\frac{1}{f} \propto (\mu - 1)$

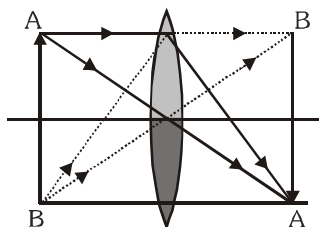
In figure number of images = 2



- (vi) Focal length of lens depends on wavelength.  $\therefore \frac{1}{f} \propto (\mu - 1) \propto \frac{1}{\lambda} [f \propto \lambda]$   $f_R > f_V$



- (vii) If half portion of lens is covered by black paper then intensity of image will be reduced but complete image will be formed.

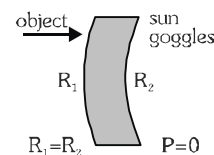


- (viii) Sun-goggles :

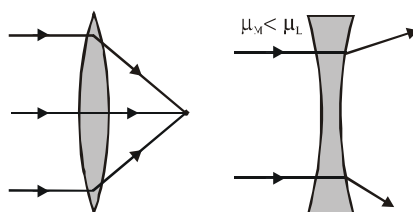
radius of curvature of two surfaces is equal with centre on the same side

$R_1 = R_2 = +R$  so  $\frac{1}{f} = (\mu - 1) \left[ \frac{1}{R} - \frac{1}{R} \right]$

$\Rightarrow \frac{1}{f} = 0 \Rightarrow f = \infty$  and  $P = 0 \Rightarrow$  sun goggles have no power



- (ix) If refractive index of medium < Refractive index of lens



If  $\mu_M < \mu_L$  then  $f = +ve$

Convex lens behave as convex lens. While concave lens behave as concave lens.

- (x) Refractive index of medium = Refractive index of lens ( $\mu_M = \mu_L$ )

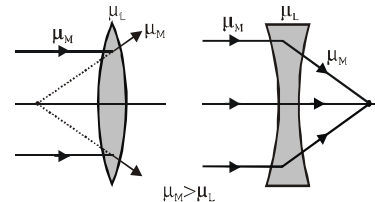
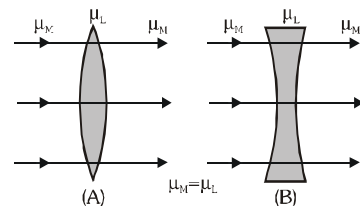
$$\frac{1}{f} = \left( \frac{\mu_L}{\mu_M} - 1 \right) \left( \frac{1}{R_1} - \frac{1}{R_2} \right); \frac{1}{f} = 0 \Rightarrow f = \infty \text{ \& } P = 0$$

Lens will behave as plane glass plate

- (xi) Refractive index of medium > Refractive index of lens

$$\mu_M > \mu_L \Rightarrow \frac{\mu_L}{\mu_M} < 1 \text{ [ f will be negative ]}$$

convex lens will behave as concave lens and concave lens will behave as convex lens. If a air bubble is formed in water it behaves as concave lens.

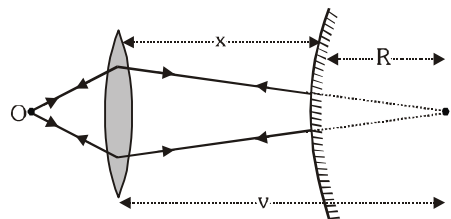
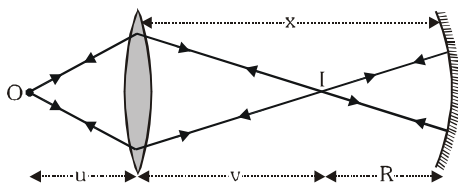


- Ex.** A point source S is placed at distance of 15 cm from a converging lens of focal length 10 cm. Where should a (i) concave mirror (ii) convex mirror of focal length 12 cm be placed so that real image is formed on object itself.

**Sol.**  $u = -15\text{cm}, f = +10\text{cm}; \frac{1}{v} - \frac{1}{u} = \frac{1}{f} \Rightarrow \frac{1}{v} - \frac{1}{(-15)} = \frac{1}{10} \Rightarrow v = 30\text{ cm}$

(i)  $x = v + 2f \Rightarrow 30 + 2 \times 12 = 54\text{ cm}$

(ii)  $x = v - 2f = 30 - 2 \times 12 = 6\text{ cm}$



- Ex.** A convex lens of focal length  $f$  is producing real image which is  $\frac{1}{n}$  times of the size of the object. Find out position of the object.

**Sol.** Image is real so  $m = \frac{v}{u} = -\frac{1}{n} \Rightarrow v = -\frac{u}{n}$

from lens formula  $\frac{1}{v} - \frac{1}{u} = \frac{1}{f} \Rightarrow \frac{1}{-\frac{u}{n}} - \frac{1}{u} = \frac{1}{f} \Rightarrow \frac{-(n+1)}{u} = \frac{1}{f} \Rightarrow u = -f(1+n)$

- Ex.** (a) If  $f = +0.5\text{m}$ , what is the power of the lens ?  
 (b) The radii of curvature of the faces of a double convex lens are 10 cm and 15 cm. Its focal length is 12 cm. What is the refractive index of glass ?  
 (c) A convex lens has 20 cm focal length in air. What is the focal length in water ? (Refractive index of air-water = 1.33, refractive index for air glass is 1.5)

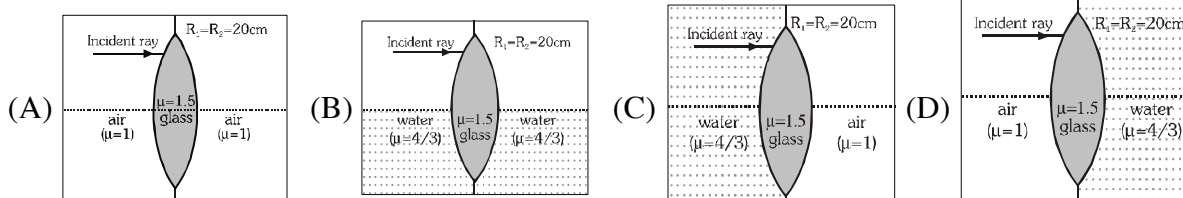
**Sol.** (a)  $P = \frac{1}{f(\text{m})} = \frac{1}{0.5} = +2\text{D}$

(b)  $\frac{1}{f} = (\mu - 1) \left( \frac{1}{R_1} - \frac{1}{R_2} \right) \Rightarrow \frac{1}{12} = (\mu - 1) \left( \frac{1}{10} - \frac{1}{-15} \right) \Rightarrow \mu = 1.5$

$$(c) \frac{1}{f} = (\mu - 1) \left( \frac{1}{R_1} - \frac{1}{R_2} \right) \text{ so } \frac{1}{f} \propto (\mu - 1)$$

$$\frac{f_w}{f_a} = \frac{(\mu_l - 1)}{(\mu_w \mu_l - 1)} \Rightarrow f_w = \frac{(1.5 - 1)}{\left( \frac{1.5}{1.33} - 1 \right)} \times 20 = 78.2 \text{ cm}$$

**Ex. Column I (optical system)**



**Column II (focal length)**

(P) 80 cm

(Q) 40 cm

(R) 30 cm

(S) 20 cm

**Sol. Ans. (A) -S (B) -P (C) -R**

$$\text{For (A): } \frac{1}{f} = (\mu - 1) \left( \frac{1}{p_1} + \frac{1}{p_2} \right) = (1.5 - 1) \left( \frac{1}{20} + \frac{1}{20} \right) = \frac{1}{20} \Rightarrow f = 20 \text{ cm}$$

$$\text{For (B): } \frac{1}{f} = \left( \frac{1.5}{4/3} - 1 \right) \left( \frac{1}{20} + \frac{1}{20} \right) = \frac{1}{80} \Rightarrow f = 80 \text{ cm}$$

$$\text{For (C): } \frac{1.5}{v_1} - \frac{4/3}{\infty} = \left( 1.5 - \frac{4}{3} \right) \left( \frac{1}{20} \right) \text{ \& } \frac{1}{f} - \frac{1.5}{v_1} = (1 - 1.5) \left( \frac{1}{-20} \right) \Rightarrow f = 30 \text{ cm}$$

$$\text{For (D): } \frac{1.5}{v_1} - \frac{1}{\infty} = (1.5 - 1) \left( \frac{1}{20} \right) \text{ \& } \frac{4/3}{f} - \frac{1.5}{v_1} = \left( \frac{4}{3} - 1.5 \right) \left( \frac{1}{-20} \right) \Rightarrow f = 40 \text{ cm}$$

**DISPLACEMENT METHOD**

It is used for determination of focal length of convex lens in laboratory.

A thin convex lens of focal length  $f$  is placed between an object and a screen fixed at a distance  $D$  apart.

If  $D > 4f$  there are two position of lens at which a sharp image of the object is formed on the screen

$$\text{By lens formula } \frac{1}{v} - \frac{1}{u} = \frac{1}{f} \Rightarrow \frac{1}{D-u} - \frac{1}{-u} = \frac{1}{f}$$

$$\Rightarrow u^2 - Du + Df = 0 \Rightarrow u = \frac{D \pm \sqrt{D(D-4f)}}{2}$$

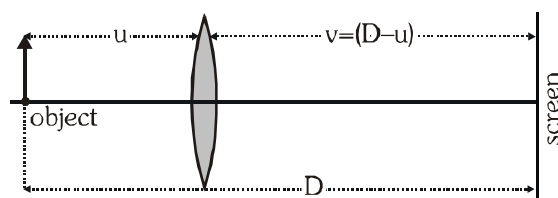
there are three possibilities

(i) for  $D < 4f$   $u$  will be imaginary hence physically no position of lens is possible

(ii) for  $D = 4f$   $u = \frac{D}{2} = 2f$  so only one position of lens is possible

and since  $v = D - u = 4f - 2f = u = 2f$

(iii) for  $D > 4f$   $u_1 = \frac{D - \sqrt{D(D-4f)}}{2}$  and  $u_2 = \frac{D + \sqrt{D(D-4f)}}{2}$



So there are two positions of lens for which real image will be formed on the screen.  
(for two distances  $u_1$  and  $u_2$  of the object from lens)

If the distance between two positions of lens is  $x$

$$\text{then } x = u_2 - u_1 = \frac{D + \sqrt{D(D-4f)}}{2} - \frac{D - \sqrt{D(D-4f)}}{2}$$

$$= \sqrt{D(D-4f)} \Rightarrow x^2 = D^2 - 4Df \Rightarrow f = \frac{D^2 - x^2}{4D}$$

Distance of image corresponds to two positions of the lens :

$$v_1 = D - u_1 = D - \frac{1}{2}[D - \sqrt{D(D-4f)}] = \frac{1}{2}[D + \sqrt{D(D-4f)}] = u_2 \Rightarrow v_1 = u_2$$

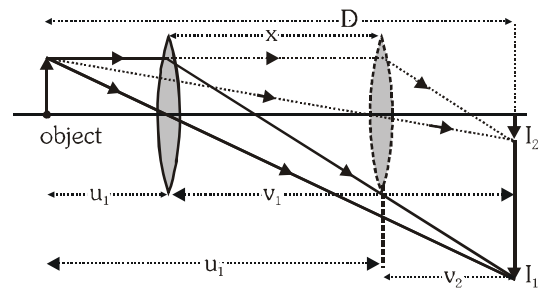
$$v_2 = D - u_2 = D - \frac{1}{2}[D + \sqrt{D(D-4f)}] = \frac{1}{2}[D - \sqrt{D(D-4f)}] = u_1 \Rightarrow v_2 = u_1$$

for two positions of the lens distances of object and image are interchangeable.

$$\text{Now } x = u_2 - u_1 \text{ and } D = v_1 + u_1 = u_2 + u_1 \quad [\because v_1 = u_2]$$

$$\text{so } u_1 (= v_2) = \frac{D-x}{2} \text{ and } v_1 = \frac{D+x}{2} = u_2; m_1 = \frac{I_1}{O} = \frac{v_1}{u_1} = \frac{D+x}{D-x} \text{ and } m_2 = \frac{I_2}{O} = \frac{v_2}{u_2} = \frac{D-x}{D+x}$$

$$\text{Now } m_1 \times m_2 = \frac{D+x}{D-x} \times \frac{D-x}{D+x} \Rightarrow \frac{I_1 I_2}{O^2} = 1 \Rightarrow O = \sqrt{I_1 I_2}$$



**Ex.** A convex lens is placed between an object and a screen which are at a fixed distance apart for one position of the lens. The magnification of the image obtained on the screen is  $m_1$ . When the lens is moved by a distance  $d$  the magnification of the image obtained on the same screen is  $m_2$ , Find the focal length of the lens.

**Sol.** If  $D$  is the distance between the object and the screen,  $d$  the separation of the two position of lens throwing two images on the screen then

$$m_1 = \frac{(D+d)}{(D-d)} \text{ and } m_2 = \frac{(D-d)}{(D+d)} \therefore m_1 - m_2 = \frac{4Dd}{D^2 - d^2} \text{ but } \frac{D^2 - d^2}{4D} = f \text{ so } m_1 - m_2 = \frac{d}{f} \Rightarrow f = \frac{d}{m_1 - m_2}$$

**Ex.** In a displacement method using lens, we obtain two images for separation of the lens  $d$ . One image is magnified as much as the other is diminished. If  $m$  is the magnifications of one image, find the focal length of the lens.

**Sol.** From above question  $f = \frac{d}{m_1 - m_2}$  here if  $m_1$  is taken as  $m$ ,  $m_2 = \frac{1}{m}$ , so  $f$  becomes  $\frac{md}{(m^2 - 1)}$

**Ex.** In the displacement method the distance between the object and the screen is 70 cm and the focal length of the lens is 16 cm, find the separations of the magnified and diminished image position of the lens.

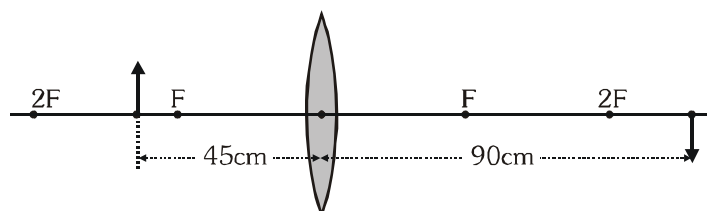
$$\text{Sol. } d = \sqrt{D^2 - 4fd} = \sqrt{(70)^2 - 4 \times 16 \times 70} = \sqrt{420} = 20.5 \text{ cm}$$

**Ex.** An object 25 cm high is placed in front of a convex lens of focal length 30 cm. If the height of image formed is 50 cm, find the distance between the object and the image (real and virtual) ?

**Sol.** As object is in front of the lens, it is real and as  $h_1 = 25$  cm,  $f = 30$  cm,  $h_2 = -50$  cm ;

$$m = \frac{h_2}{h_1} = \frac{-50}{25} = -2$$

$$m = \frac{f}{f+u} \Rightarrow -2 = \frac{30}{30+u} \Rightarrow u = -45 \text{ cm} \Rightarrow m = \frac{v}{u} \Rightarrow -2 = \frac{v}{-45} \Rightarrow v = 90 \text{ cm}$$



As in this situation object and image are on opposite sides of lens, the distance between object and image  $d_1 = u + v = 45 + 90 = 135$  cm. If the image is erect (i.e., virtual)

$$m = \frac{f}{f+u} \Rightarrow 2 = \frac{30}{30+u} \Rightarrow u = -15 \text{ cm} \Rightarrow m = -\frac{v}{u} \Rightarrow 2 = \frac{-v}{-15} \Rightarrow v = 30 \text{ cm}$$

As in the situation both image and object are in front of the lens, the distance between object and image

$$d_2 = v - u = 30 - 15 = 15 \text{ cm.}$$

### COMBINATION OF LENSES AND MIRRORS

When several lenses or mirrors are used, the image formation is considered one after another in steps, The image formed by the lens facing the object serves as an object for the next lens or mirror, the image formed by the second lens acts as an object for the third, and so on, The total magnification in such situations will be given by

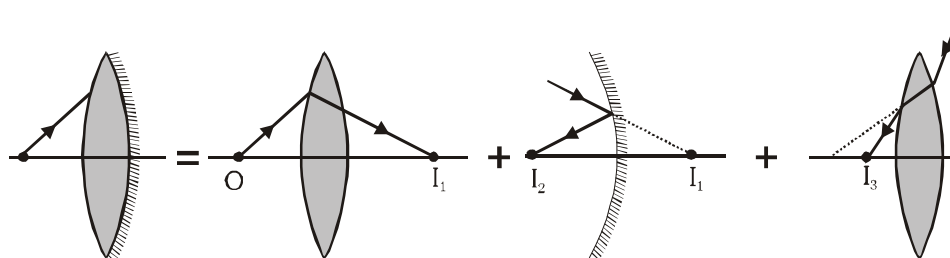
$$m = \frac{I}{O} = \frac{I_1}{O} \times \frac{I_2}{I_1} \times \dots \Rightarrow m = m_1 \times m_2 \times \dots$$

Power of Lens [in air]  $P_L = \frac{1}{f_L}$     Converging lens  $P_L = +ve$     Diverging lens  $P_L = -ve$

Power For mirror  $P_M = -\frac{1}{f_m}$     Convex mirror  $P_M = -ve$     Concave mirror  $P_M = +ve$

### SILVERING OF LENS

Calculate equivalent focal length of a equiconvex lens silvered at one side.

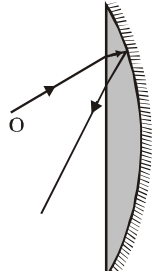


$$P = P_L + P_M + P_L = 2P_L + P_M$$

$$\frac{1}{F} = \frac{1}{f_L} + \frac{1}{f_m} + \frac{1}{f_L} = \frac{2}{f_L} + \frac{1}{f_m} = \frac{2(\mu-1) \times 2}{R} + \frac{2}{R} = \frac{4\mu-4+2}{R} \Rightarrow F = \frac{R}{4\mu-2}$$

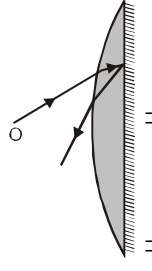
**Ex.** Calculate equivalent focal length of plano convex lens for following case :-

(i) When curved surface is silvered. (ii) When plane surface is silvered.

**Sol.** (i)   $P = 2P_L + P_M$

$$\Rightarrow \frac{1}{F} = \frac{2}{f_L} + \frac{1}{f_m} \Rightarrow \frac{1}{F} = \frac{2(\mu - 1)}{R} + \frac{2}{R}$$

$$\Rightarrow \frac{1}{F} = \frac{2\mu}{R} \Rightarrow F = \frac{R}{2\mu}$$

(ii)   $P = 2P_L + P_M$

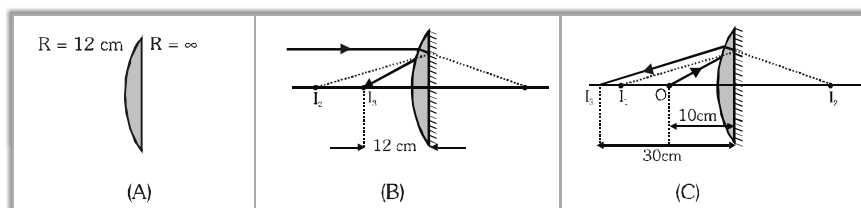
$$\Rightarrow \frac{1}{F} = \frac{2}{f_L} + \frac{1}{f_m} \Rightarrow \frac{1}{F} = \frac{2(\mu - 1)}{R} + \frac{1}{\infty}$$

$$\Rightarrow F = \frac{R}{2(\mu - 1)}$$

**Ex.** The radius of curvature of the convex face of a plano-convex lens is 12 cm and its refractive index is 1.5. (a) Find the focal length of this lens. The plane surface of the lens is now silvered. (b) At what distance from the lens will parallel rays incident on the convex face converge? (c) Sketch the ray diagram to locate the image, when a point object is placed on the axis 20 cm from the lens. (d) Calculate the image distance when the object is placed as in (c).

**Sol.** (a) As for a lens, by lens-maker's formula  $\frac{1}{f} = (\mu - 1) \left[ \frac{1}{R_1} - \frac{1}{R_2} \right]$  Here  $\mu = 1.5$ ;  $R_1 = 12$  cm and  $R_2 = \infty$

So  $\frac{1}{f} = (1.5 - 1) \left[ \frac{1}{12} - \frac{1}{\infty} \right]$  i.e.  $f = 24$  cm i.e., the lens as convergent with focal length 24 cm.



(b) As light after passing through the lens will be incident on the mirror which will reflect it back through

the lens again, so  $P = P_L + P_M + P_L = 2P_L + P_M$  But  $P_L = \frac{1}{f_L} = \frac{1}{0.24}$  and  $P_M = -\frac{1}{\infty} = 0$  [as  $f_M = \frac{R}{2} = \infty$ ]

So  $P = 2 \times \frac{1}{0.24} + 0 = \frac{1}{0.12}$  D. The system is equivalent to a concave mirror of focal length F,

$$P = -\frac{1}{F}$$

i.e.,  $F = -\frac{1}{P} = -0.12$  m = -12 cm i.e., the rays will behave as a concave mirror of focal length 12 cm.

So as for parallel incident rays  $u = -\infty$ , from mirror formula  $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$  we have  $\frac{1}{v} + \frac{1}{-\infty} = \frac{1}{-12}$

$\Rightarrow v = -12$  cm i.e., parallel incident rays will focus will at a distance of 12 cm in front of the lens as shown in Figure (c) and (d) When object is at 20 cm in front of the given silvered lens which behaves

as a concave mirror of focal length 12 cm, from mirror formula  $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$  we have  $\frac{1}{v} + \frac{1}{-20} = \frac{1}{-12}$

$\Rightarrow v = -30$  cm i.e., the silvered lens will form image at a distance of 30 cm in front of it as shown in fig. (C)

**Ex.** A pin is placed 10 cm in front of a convex lens of focal length 20 cm, made of material having refractive index 1.5. The surface of the lens farther away from the pin is silvered and has a radius of curvature 22 cm. Determine the position of the final image. Is the image real or virtual ?

**Sol.** As radius of curvature of silvered surface is 22 cm, so

$$f_M = \frac{R}{2} = \frac{-22}{2} = -11 \text{ cm} = -0.11 \text{ m} \text{ and hence,}$$

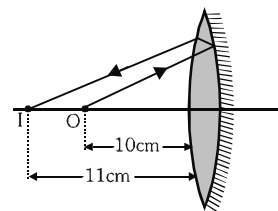
$$P_M = -\frac{1}{f_M} = -\frac{1}{-0.11} = \frac{1}{0.11} \text{ D}$$

Further as the focal length of lens is 20 cm, i.e., 0.20 m its power will be given by :

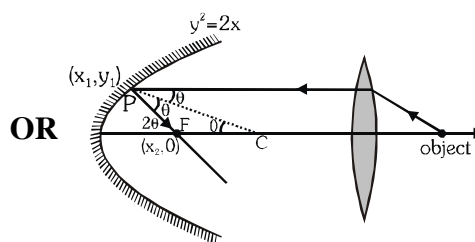
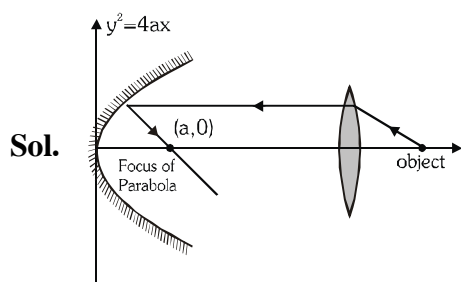
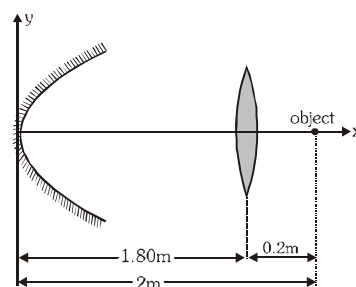
$$P_L = \frac{1}{f_L} = \frac{1}{0.20} \text{ D. Now as in image formation, light after passing through the lens will be reflected back}$$

$$\text{by the curved mirror through the lens again } P = P_L + P_M + P_L = 2P_L + P_M \text{ i.e. } P = \frac{2}{0.20} + \frac{1}{0.11} = \frac{210}{11} \text{ D.}$$

So the focal length of equivalent mirror  $F = -\frac{1}{P} = -\frac{11}{210} \text{ m} = -\frac{110}{21} \text{ cm}$  i.e., the silvered lens behave as a concave mirror of focal length  $(110/21) \text{ cm}$ . So for object at a distance 10 cm in front of it,  $\frac{1}{v} + \frac{1}{-10} = -\frac{21}{110}$  i.e.,  $v = -11 \text{ cm}$  i.e., image will be 11 cm in front of the silvered lens and will be real as shown in Figure.



**Ex.** A point object is kept at a distance of 2m from a parabolic reflecting surface  $y^2 = 2x$ . An equiconvex lens is kept at a distance of 1.80 m from the parabolic surface. The focal length of the lens is 20 cm. Find the position from origin of the image in cm, after reflection from the surface.



$$\text{Comparing with } y^2 = 4ax \Rightarrow a = 0.5$$

$$\Rightarrow \text{final position of image} = 0.5 \text{ m} = 50 \text{ cm}$$

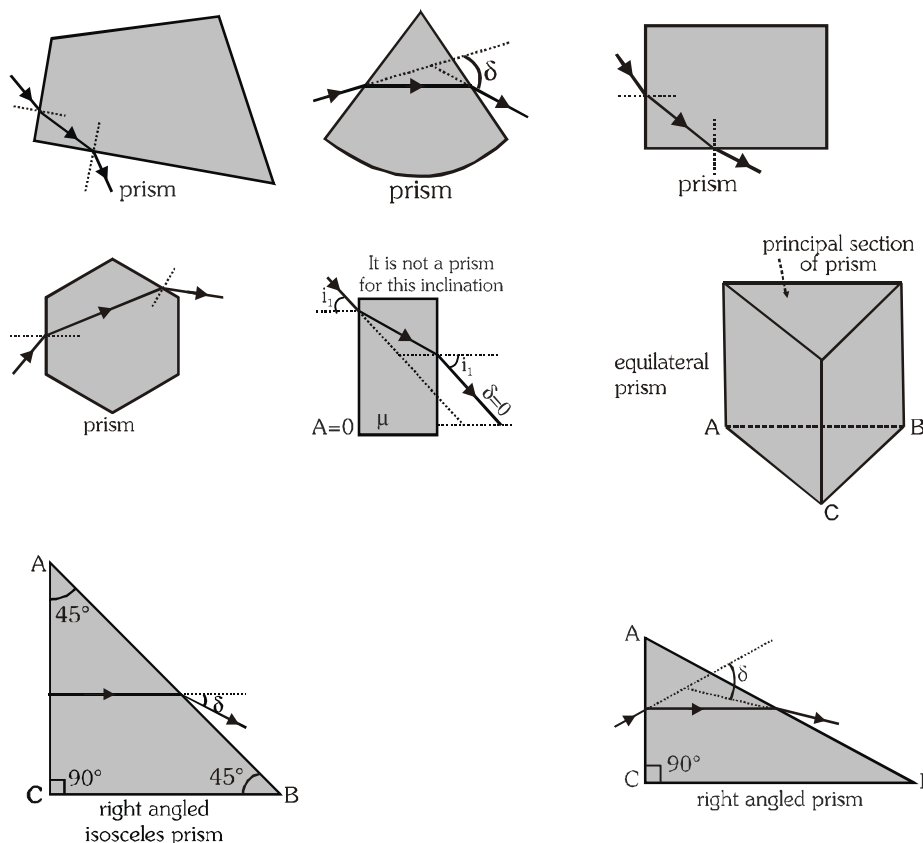
$$\text{PC is a normal so } \tan(\pi - \theta) = \frac{-1}{(dy/dx)_{x_1, y_1}} = -y_1$$

$$\text{But } \tan 2\theta = \frac{y_1 - 0}{x_2 - x_1} \text{ \& } \tan 2\theta = \frac{2 \tan \theta}{1 - \tan^2 \theta}$$

$$\Rightarrow \frac{y_1}{x_2 - x_1} = \frac{2(y_1)}{1 - y_1^2} x_2 = \frac{1}{2} \text{ m}$$

## PRISM

A prism is a homogeneous, transparent medium (such as glass) enclosed by two plane surfaces inclined at an angle. These surfaces are called the 'refracting surfaces' and the angle between them is called the 'refracting angle' or the 'angle of prism'. The section cut by a plane perpendicular to the refracting surfaces is called the 'principal section' of the prism.



### DEVIATION

PQ = incident ray

QR = Refracted ray

RS = emergent ray

A = Prism angle

$i_1$  = incident angle on face AB

$i_2$  = emergent angle on face AC

$r_1$  = refracted angle on face AB

$r_2$  = incident angle on face AC

Angle of deviation on face AB.

Angle of deviation on face AC

Total angle of deviation

$$\delta = \delta_1 + \delta_2 \Rightarrow \delta = (i_1 - r_1) + (i_2 - r_2) = i_1 + i_2 - (r_1 + r_2) \dots (i)$$

$$\text{In } \triangle QOR \quad r_1 + r_2 + \theta = 180^\circ \dots (ii)$$

$$\text{In } \triangle AQR \quad A + \theta = 180^\circ \dots (iii)$$

$$\text{from (ii) and (iii)} \quad r_1 + r_2 = A \dots (iv)$$

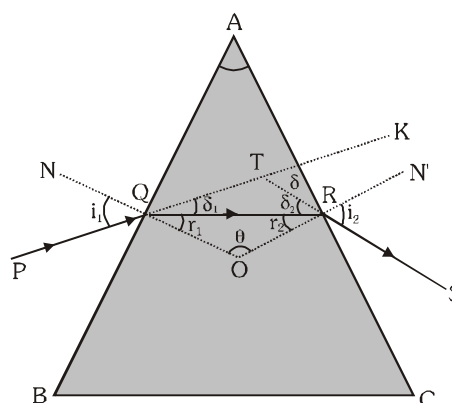
$$\text{from (i) and (iv)} \quad \text{Total angle of deviation } \delta = i_1 + i_2 - A$$

from Snell's law at surface AB

$$\mu_1 \sin i_1 = \mu_2 \sin r_1$$

and at surface AC

$$\mu_2 \sin r_2 = \mu_1 \sin i_2$$





## CONDITION OF MINIMUM DEVIATION

### For minimum deviation

In this condition  $i_1 = i_2 = i \Rightarrow r_1 = r_2 = r$  and since  $r_1 + r_2 = A \therefore r + r = A \Rightarrow 2r = A \Rightarrow r = \frac{A}{2}$

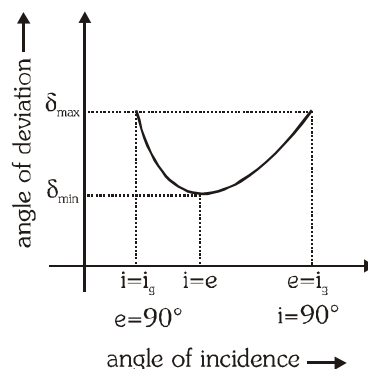
Minimum deviation  $\delta_{\min} = 2i - A$ ;  $i = \frac{A + \delta_{\min}}{2}$ ,  $r = \frac{A}{2}$

if prism is placed in air  $\mu_1 = 1$ ;  $1 \times \sin i = \mu \sin r$

$$\sin \left[ \frac{A + \delta_{\min}}{2} \right] = \mu \sin \frac{A}{2} \Rightarrow \mu = \frac{\sin \left[ \frac{A + \delta_{\min}}{2} \right]}{\sin \frac{A}{2}}$$

if angle of prism is small  $A < 10^\circ$  then  $\sin \theta \approx \theta$

$$\mu = \frac{\frac{A + \delta_{\min}}{2}}{\frac{A}{2}} = \frac{A + \delta_{\min}}{A} \Rightarrow A + \delta_{\min} = \mu A \Rightarrow \delta_{\min} = (\mu - 1)A$$



## CONDITION FOR MAXIMUM DEVIATION/GRAZING EMERGENCE

- **Angle of incidence ( $i_g$ ) for grazing emergence**

For  $i_g$ ,  $e = 90^\circ$

Applying Snell's law at face AC

$$\mu \sin r_2 = 1 \times 1 \Rightarrow \sin r_2 = \frac{1}{\mu}; r_2 = \sin^{-1} \left( \frac{1}{\mu} \right) = \theta_c$$

$$r_1 + r_2 = A \Rightarrow r_1 = A - \theta_c$$

Again, Applying Snell's law at face AB

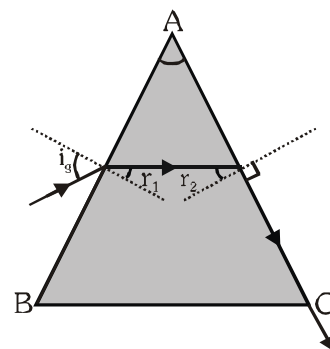
$$1 \times \sin i_g = \mu \sin r_1; 1 \times \sin i_g = \mu \sin (A - \theta_c)$$

$$\sin i_g = \mu [\sin A \cos \theta_c - \cos A \sin \theta_c]$$

$$i_g = \sin^{-1} \left[ \sqrt{\mu^2 - 1} \sin A - \cos A \right] \quad \left[ \text{as } \sin \theta_c = \frac{1}{\mu}, \cos \theta_c = \frac{\sqrt{\mu^2 - 1}}{\mu} \right]$$

If  $i$  increases beyond  $i_g$ ,  $r_1$  increases thus  $r_2$  decreases and becomes less than  $\theta_c$  and ray emerges.

Thus  $i \geq i_g \Rightarrow$  ray emerges, otherwise TIR.  $\delta_{\max} = i_g + 90^\circ - A$



## NO EMERGENCE CONDITION

Let maximum incident angle on the face AB  $i_{\max} = 90^\circ$

$$1 \times \sin 90^\circ = \mu \sin r_1; \sin r_1 = \frac{1}{\mu} = \sin \theta_c; r_1 = \theta_c \dots (i)$$

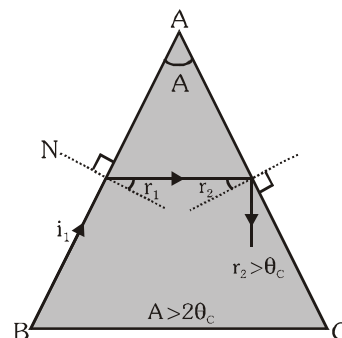
if TIR occur at face AC then

$$r_2 > \theta_c \dots (ii)$$

$$r_1 + r_2 = A \dots (iii)$$

$$\text{from (i) and (ii) } r_1 + r_2 > \theta_c + \theta_c \Rightarrow r_1 + r_2 > 2\theta_c \dots (iv)$$

$$\text{from (iii) and (iv) } A > 2\theta_c \Rightarrow \frac{A}{2} > \theta_c \Rightarrow \sin \frac{A}{2} > \sin \theta_c \Rightarrow \sin \frac{A}{2} > \frac{1}{\mu} \Rightarrow \frac{1}{\sin \frac{A}{2}} < \mu$$



**Ex.** A ray of light passes through an equilateral prism such that angle of incidence is equal of emergence and the later is equal to  $3/4^{\text{th}}$  of the angle of prism. Calculate the angle of deviation. Refractive index of prism is 1.5.

**Sol.**  $A = 60^\circ$ ,  $\mu = 1.5$ ;  $i_1 = i_2 = \frac{3}{4}A = 45^\circ$ ,  $\delta = ?$   
 $\therefore A + \delta = i_1 + i_2$   $\therefore 60^\circ + \delta = 45^\circ + 45^\circ \Rightarrow \delta = 90^\circ - 60^\circ = 30^\circ$

**Ex.** A prism of refractive index 1.53 is placed in water of refractive index 1.33. If the angle of prism is  $60^\circ$ , calculate the angle of minimum deviation in water. ( $\sin 35.1^\circ = 0.575$ )

**Sol.** Here,  ${}^a\mu_g = 1.33$ ,  ${}^a\mu_w = 1.53$ ,  $A = 60^\circ$ ,  $\delta_m = ?$   ${}^w\mu_g = \frac{{}^a\mu_g}{{}^a\mu_w} = \frac{1.53}{1.33} = 1.15$   $\therefore {}^w\mu_g = \frac{\sin \frac{A + \delta_m}{2}}{\sin \frac{A}{2}}$   
 $\therefore \frac{\sin(A + \delta_m)}{2} = {}^w\mu_g \times \sin \frac{A}{2} = 1.15 \sin \frac{60^\circ}{2} = 0.575 \Rightarrow \frac{A + \delta_m}{2} = \sin^{-1}(0.575) = 35.1^\circ$   
 $\therefore \delta_m = 35.1 \times 2 - 60 = 10.2^\circ$

### KEY POINTS

- Angle of prism or refracting angle of prism means the angle between the faces on which light is incident and from which it emerges.
- If the faces of a prism on which light is incident and from which it emerges are parallel then the angle of prism will be zero and as incident ray will emerge parallel to itself, deviation will also be zero, i.e., the prism will act as a transparent plate.
- If  $\mu$  of the material of the prism is equal to that of surroundings, no refraction at its faces will take place and light will pass through it undeviated, i.e.,  $\delta = 0$ .

### DISPERSION OF LIGHT

When white light is incident on a prism then it is splitted into seven colours. This phenomenon is known as dispersion. Prism introduces different refractive index with different wavelength

As  $\delta_{\min} = (\mu - 1) A \therefore \lambda_R > \lambda_V$

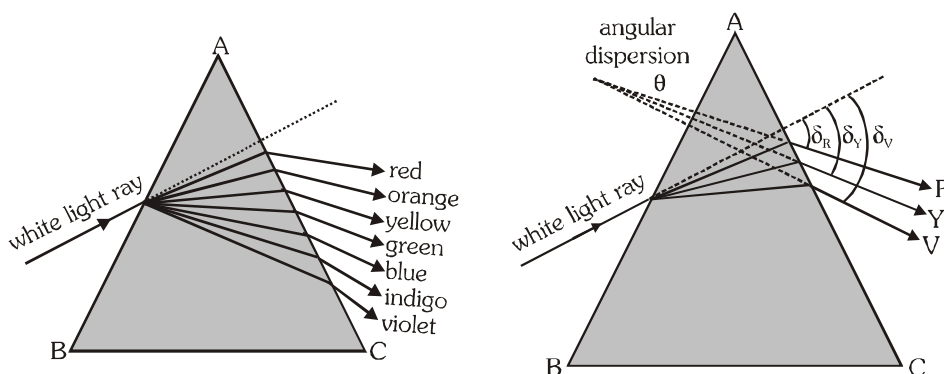
So  $\mu_V > \mu_R \Rightarrow \delta_{m(\text{violet})} > \delta_{m(\text{red})}$

### ANGULAR DISPERSION

It is the difference of angle of deviation for violet colour and red colour

Angular dispersion  $\theta = \delta_V - \delta_R = (\mu_V - 1)A - (\mu_R - 1)A = (\mu_V - \mu_R) A$

It depends on prism material and on the angle of prism  $\theta = (\mu_V - \mu_R) A$



## DISPERSIVE POWER ( $\omega$ )

It is ratio of angular dispersion ( $\theta$ ) to mean colour deviation ( $\delta_y$ )

$$\text{Dispersive power } \omega = \frac{\theta}{\delta_y} \Rightarrow \omega = \frac{(\mu_V - \mu_R)A}{(\mu_y - 1)A} = \frac{\mu_V - \mu_R}{\mu_y - 1} \Rightarrow \omega = \frac{\mu_V - \mu_R}{\mu_y - 1}$$

$$\text{Refractive index of mean colour } \mu_y = \frac{\mu_V + \mu_R}{2}$$

Dispersive power depends only on the material of the prism.

## COMBINATION OF PRISM

### Deviation without dispersion ( $\theta = 0^\circ$ )

Two or more than two thin prism are combined in such a way that deviation occurs i.e. emergent light ray makes angle with incident light ray but dispersion does not occur i.e., light is not splitted into seven colours.

$$\text{Total dispersion } \theta = \theta_1 + \theta_2 = (\mu_V - \mu_R)A + (\mu'_V - \mu'_R)A'$$

$$\text{For no dispersion } \theta = 0; (\mu_V - \mu_R)A + (\mu'_V - \mu'_R)A' = 0$$

$$\text{Therefore, } A' = -\frac{(\mu_V - \mu_R)A}{\mu_V - \mu_R}$$

—ve sign indicates that prism angles are in opposite direction.

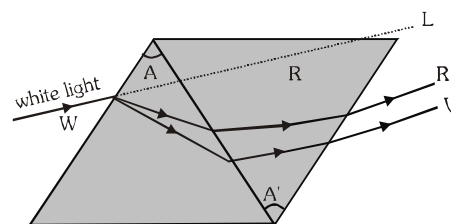
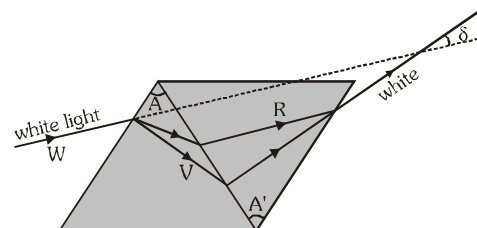
### Dispersion without deviation ( $\delta = 0^\circ$ )

Two or more than two prisms combine in such a way that dispersion occurs i.e., light is splitted into seven colours but deviation do not occur i.e., emergent light ray becomes parallel to incident light ray.

$$\text{Total deviation } \delta = \delta_1 + \delta_2$$

$$\Rightarrow \delta = 0; (\mu - 1)A + (\mu' - 1)A' = 0 \Rightarrow A' = -\frac{(\mu - 1)A}{\mu' - 1}$$

—ve sign indicates that prism angles are in opposite direction.



## KEY POINTS

- Dispersive power like refractive index has no units and dimensions and depends on the material of the prism and is always positive.
- As for a given prism dispersive power is constant, i.e., dispersion of different wavelengths will be different and will be maximum for violet and minimum for red (as deviation is maximum for violet and minimum for red).
- As for a given prism  $\theta \propto \delta$  a single prism produces both deviation and dispersion simultaneously, i.e., a single prism cannot give deviation without dispersion or dispersion without deviation.

**Ex.** White light is passed through a prism of angle  $5^\circ$ . If the refractive indices for red and blue colours are 1.641

and 1.659 respectively, calculate the angle of dispersion between them.

**Sol.** As for small angle of prism  $\delta = (\mu - 1)A$ ,

$$\delta_B = (1.659 - 1) \times 5^\circ = 3.295^\circ \text{ and } \delta_R = (1.641 - 1) \times 5^\circ = 3.205^\circ$$

$$\text{so } \theta = \delta_B - \delta_R = 3.295^\circ - 3.205^\circ = 0.090^\circ$$

**Ex.** Prism angle of a prism is  $10^\circ$ . Their refractive index for red and violet color is 1.51 and 1.52 respectively. Then find the dispersive power.

**Sol.** Dispersive power of prism  $\omega = \left( \frac{\mu_v - \mu_r}{\mu_y - 1} \right)$  but  $\mu_y = \frac{\mu_v + \mu_r}{2} = \frac{1.52 + 1.51}{2} = 1.515$

$$\text{Therefore } \omega = \frac{1.52 - 1.51}{1.515 - 1} = \frac{0.01}{1.515} = 0.019$$

**Ex.** The refractive indices of flint glass for red and violet colours are 1.644 and 1.664. Calculate its dispersive power.

**Sol.** Here,  $\mu_r = 1.644$ ,  $\mu_v = 1.664$ ,  $\omega = ?$

$$\text{Now } \mu_y = \frac{\mu_v + \mu_r}{2} = \frac{1.664 + 1.644}{2} = 1.654 \quad \therefore \omega = \frac{\mu_v - \mu_r}{\mu_y - 1} = \frac{1.664 - 1.644}{1.654 - 1} = 0.0305$$

**Ex.** In a certain spectrum produced by a glass prism of dispersive power 0.031, it was found that  $\mu_r = 1.645$  and  $\mu_v = 1.665$ . What is the refractive index for yellow colour ?

**Sol.** Here,  $\omega = 0.031$ ,  $\mu_r = 1.645$ ,  $\mu_v = 1.665$ ,  $\mu_y = ?$

$$\therefore \omega = \frac{\mu_v - \mu_r}{\mu_y - 1} \quad \therefore \mu_y - 1 = \frac{\mu_v - \mu_r}{\omega} = \frac{1.665 - 1.645}{0.031} = \frac{0.020}{0.31} = 0.0645 \quad \therefore \mu_y = 0.0645 + 1 = 1.0645$$

**Ex.** A combination of two prisms, one of flint and other of crown glass produces dispersion without deviation. The angle of flint glass prism is  $15^\circ$ . Calculate the angle of crown glass prism and angular dispersion of red and violet. ( $\mu$  for crown glass = 1.52,  $\mu$  for flint glass = 1.65,  $\omega$  for crown glass 0.20,  $\omega$  for flint glass = 0.03).

**Sol.** Here,  $A = 15^\circ$ ,  $A' = ?$ ,  $\omega = 0.03$ ,  $\omega' = 0.02$ ,  $\mu = 1.65$ ,  $\mu' = 1.52$ ,

For no deviation,  $\delta + \delta' = 0$

$$(\mu - 1)A + (\mu' - 1)A' = 0 \Rightarrow (1.65 - 1)15^\circ + (1.52 - 1)A' = 0 \Rightarrow A' = \frac{-0.65 \times 15}{0.52} = -18.75^\circ$$

Negative sign indicates that two prisms must be joined in opposition. Net angular dispersion

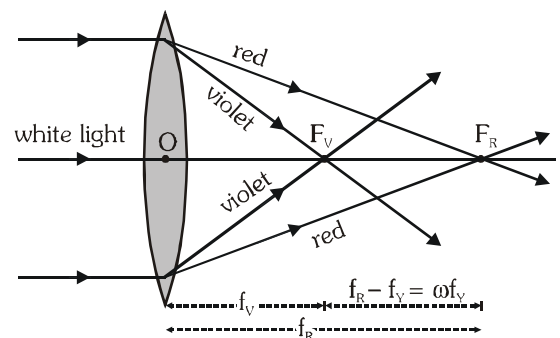
$$(\mu_v - \mu_r)A + (\mu'_v - \mu'_r)A' = \omega(\mu - 1)A + \omega'(\mu' - 1)A' = 0.03(1.65 - 1)15^\circ + 0.02(1.52 - 1)(-18.75^\circ) \\ = 0.2925 - 0.195 = 0.0975^\circ$$

## CHROMATIC ABERRATION

The image of an object in white light formed by a lens is usually colored and blurred. This defect of image is called chromatic aberration and arises due to the fact that focal length of a lens is different for different colors.

For a single lens  $\frac{1}{f} = (\mu - 1) \left[ \frac{1}{R_1} - \frac{1}{R_2} \right]$  and as  $\mu$  of lens is

maximum for violet while minimum for red, violet is focused nearest to the lens while red farthest from it. It is defect of lens.



### Longitudinal or Axial Chromatic Aberration

When a white object O is situated on the axis of a lens, then images of different colors are formed at different points along the axis. The formation of images of different colors at different positions is called 'axial' or longitudinal chromatic aberration. The axial distance between the red and the violet images  $I_R - I_V$  is known as longitudinal aberration. When white light is incident on lens, image is obtained at different point on the axis because focal length of lens depend on wavelength.

$$f \propto \lambda \Rightarrow f_R > f_V$$

$$f_R - f_V = \omega f_y \Rightarrow \text{Axial or longitudinal chromatic aberration}$$

If the object is at infinity, then the longitudinal chromatic aberration is equal to the difference in focal-lengths ( $f_R - f_V$ ) for the red and the violet rays.

### LATERAL CHROMATIC ABERRATION

As the focal-length of the lens varies from color

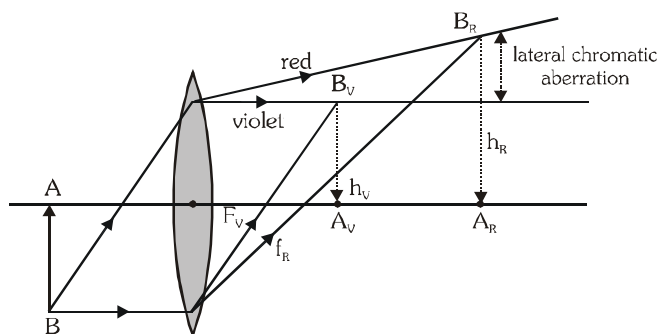
to color, the magnification  $m = \left[ \frac{f}{u+f} \right]$

produced by the lens also varies from color to color.

Therefore, for a finite-size white object AB, the images of different colors formed by the lens are of different sizes.

The formation of images of different colors in different sizes is called lateral chromatic aberration.

The difference in the height of the red image  $B_R A_R$  and the violet image  $B_V A_V$  is known as lateral chromatic aberration.  $LCA = h_R - h_V$



### ACHROMATISM

If two or more lens combined together in such a way that this combination produce image at a same point then this combination is known as achromatic combination of lenses.

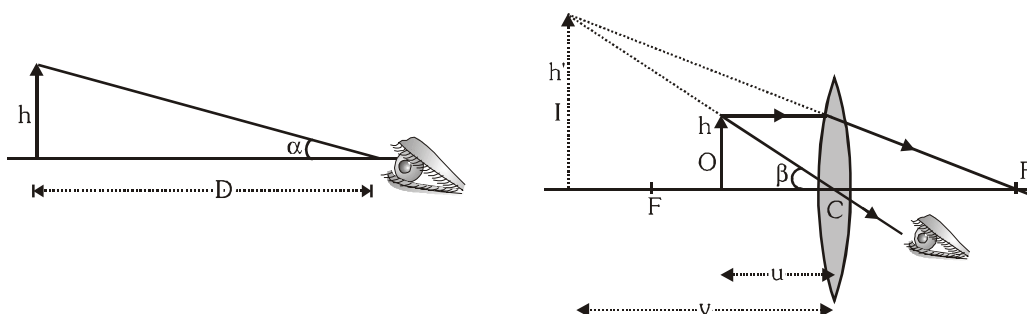
$$\frac{\omega}{f_y} + \frac{\omega'}{f'_y} = 0 \Rightarrow \frac{\omega_1}{f_1} + \frac{\omega_2}{f_2} = 0 \Rightarrow \frac{\omega_1}{\omega_2} = -\frac{f_1}{f_2}$$

For combination of lens.  $\frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2}$  (Apply sign convention in numerical)

### OPTICAL INSTRUMENTS

#### Simple microscope

When object is placed between focus and optical centre a virtual, magnified and erect image is formed



$$\text{Magnifying power (MP)} = \frac{\text{visual angle with instrument } (\beta)}{\text{maximum visual angle for unaided eye } (\alpha)} \Rightarrow \text{MP} = \frac{\frac{h}{-u}}{\frac{h}{-D}} = \frac{D}{u}$$

(i) When the image is formed at infinity :

$$\text{by lens equation } \frac{1}{v} - \frac{1}{u} = \frac{1}{f} \Rightarrow \frac{1}{-\infty} - \frac{1}{-u} = \frac{1}{f} \Rightarrow u = f \quad \text{So } \text{MP} = \frac{D}{u} = \frac{D}{f}$$

(ii) If the image is at minimum distance of clear vision D :

$$\frac{1}{-D} - \frac{1}{-u} = \frac{1}{f} \Rightarrow \frac{1}{u} = \frac{1}{D} + \frac{1}{f} \quad [v = -D \text{ and } u = -ve]$$

$$\text{Multiplying by } D \text{ both the sides } \frac{D}{u} = 1 + \frac{D}{f} \Rightarrow \text{MP} = \frac{D}{u} = 1 + \frac{D}{f}$$

**Ex.** A man with normal near point 25 cm reads a book with small print using a magnifying glass, a thin convex lens of focal length 5 cm.

(a) What is the closest and farthest distance at which he can read the book when viewing through the magnifying glass ?

(b) What is the maximum and minimum MP possible using the above simple microscope ?

**Sol.** (a) As for normal eye far and near point are  $\infty$  and 25 cm respectively, so for magnifier  $v_{\max} = -\infty$  and

$$v_{\min} = -25 \text{ cm. However, for a lens as } \frac{1}{v} - \frac{1}{u} = \frac{1}{f} \Rightarrow u = \frac{f}{(f/v) - 1}$$

$$\text{So } u \text{ will be minimum when } v = \text{minimum} = -25 \text{ cm i.e. } (u)_{\min} = \frac{5}{-(5/25) - 1} = -\frac{25}{6} = -4.17 \text{ cm}$$

$$\text{Ans } u \text{ will be maximum when } v = \text{maximum} = \infty \text{ i.e., } u_{\max} = \frac{5}{\left(\frac{5}{\infty} - 1\right)} = -5 \text{ cm}$$

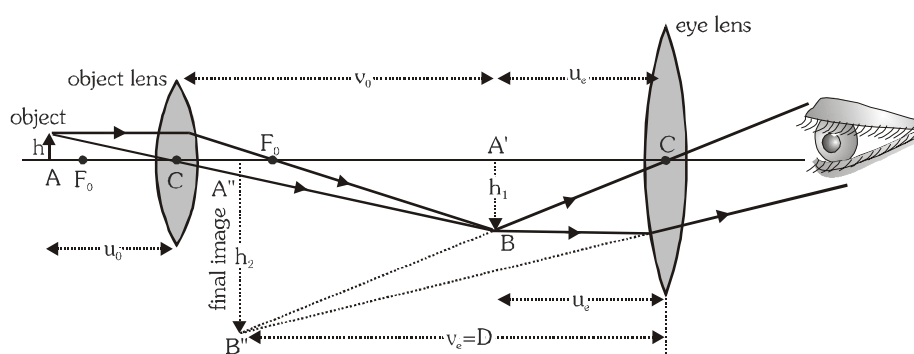
So the closest and farthest distance of the book from the magnifier (or eye) for clear viewing are 4.17 cm and 5 cm respectively.

(b) As in case of simple magnifier  $\text{MP} = (D/u)$ . So MP will be minimum when  $u = \max = 5 \text{ cm}$

$$\Rightarrow (\text{MP})_{\min} = \frac{-25}{-5} = 5 \left[ = \frac{D}{f} \right] \text{ and MP will be maximum when } u = \min = (25/6) \text{ cm}$$

$$\Rightarrow (\text{MP})_{\max} = \frac{-25}{-(25/6)} = 6 \left[ = 1 + \frac{D}{f} \right]$$

## COMPOUND MICROSCOPE



Compound microscope is used to get more magnified image. Object is placed in front of objective lens and image is seen through eye piece. The aperture of objective lens is less as compared to eye piece because object is very near so collection of more light is not required. Generally object is placed between  $F - 2F$  due to this a real inverted and magnified image is formed between  $2F - \infty$ . It is known as intermediate image A'B'. The intermediate image acts as an object for eye piece. Now the distance between both the lenses are adjusted in such a way that intermediate image falls between the optical centre of eye piece and its focus. In this condition, the final image is virtual, inverted and magnified.

Total magnifying power = Linear magnification  $\times$  angular magnification  $MP = m_o m_e = \frac{v_o}{u_o} \frac{D}{u_e}$

(i) When final image is formed at minimum, distance of distinct vision.

$$MP = \frac{v_o}{u_o} \left[ 1 + \frac{D}{f_e} \right] = \frac{f_o}{f_o + u_o} \left[ 1 + \frac{D}{f_e} \right] = \frac{f_o - v_o}{f_o} \left[ 1 + \frac{D}{f_e} \right] = \frac{h_2}{h_1} \left( 1 + \frac{D}{f_e} \right)$$

$$\text{Length of the tube} = v_o + |u_e|$$

(ii) When final image is formed at infinity  $\frac{1}{v_e} - \frac{1}{u_e} = \frac{1}{f_e} \Rightarrow \frac{1}{\infty} + \frac{1}{u_e} = \frac{1}{f_e} \Rightarrow u_e = f_e$

$$MP = \frac{v_o}{u_o} \left[ \frac{D}{f_e} \right] = \frac{f_o}{f_o + u_o} \left[ \frac{D}{f_e} \right] = \frac{f_o - v_o}{f_o} \left[ \frac{D}{f_e} \right] = \frac{h_2}{h_1} \left[ \frac{D}{f_e} \right]. \text{ Length of the tube } L = v_o + f_e$$

Sign convention for solving numerical  $u_o = -ve, v_o = +ve, f_o = +ve,$   
 $u_e = -ve, v_e = -ve, f_e = +ve, m_o = -ve, m_e = +ve, M = -ve$

**Ex.** A thin convex lens of focal length 5 cm is used as a simple microscope by a person with normal near point (25 cm). What is the magnifying power of the microscope?

**Sol.** Here,  $f = 5$  cm;  $D = 25$  cm,  $M = ?$   $MP = 1 + \frac{D}{f} = 1 + \frac{25}{5} = 6$

**Ex.** A compound microscope consists of an objective lens of focal length 2.0 cm and an eye piece of focal length 6.25 cm, separated by a distance of 15 cm. How far from the objective should an object be placed in order to obtain the final image at (a) the least distance of distinct vision (25 cm) (b) infinity?

**Sol.** Here,  $f_o = 2.0$  cm;  $f_e = 6.25$  cm,  $u_o = ?$

$$(a) v_e = -25 \text{ cm} \therefore \frac{1}{v_e} - \frac{1}{u_e} = \frac{1}{f_e} \therefore \frac{1}{u_e} = \frac{1}{v_e} - \frac{1}{f_e} = \frac{1}{-25} - \frac{1}{6.25} = \frac{-1-4}{25} = \frac{-5}{25} \Rightarrow u_e = -5 \text{ cm}$$

As distance between objective and eye piece = 15 cm;  $v_o = 15 - 5 = 10$  cm

$$\therefore \frac{1}{v_o} - \frac{1}{u_o} = \frac{1}{f_o} \therefore \frac{1}{u_o} = \frac{1}{v_o} - \frac{1}{f_o} = \frac{1}{10} - \frac{1}{2} = \frac{1-5}{10} \Rightarrow u_o = \frac{-10}{4} = -2.5 \text{ cm}$$

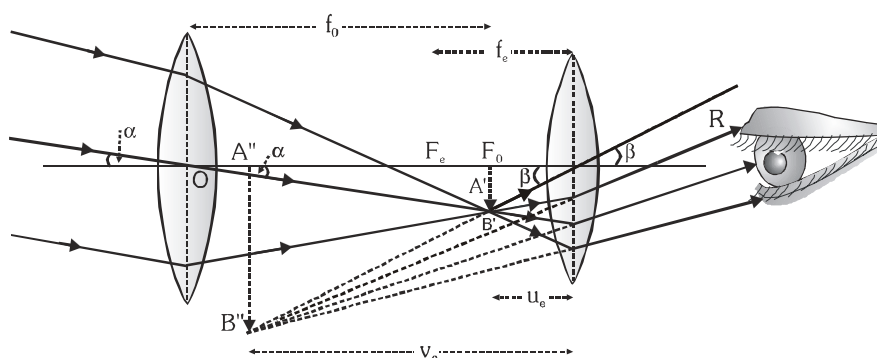
$$\text{Magnifying power} = \frac{v_o}{|u_o|} \left[ 1 + \frac{D}{f_e} \right] = \frac{10}{2.5} \left[ 1 + \frac{25}{6.25} \right] = 20$$

(b)  $\therefore v_e = \infty, u_e = f_e = 6.25$  cm  $\therefore v_o = 15 - 6.25 = 8.75$  cm.

$$\therefore \frac{1}{v_o} - \frac{1}{u_o} = \frac{1}{f_o} \Rightarrow \frac{1}{u_o} = \frac{1}{v_o} - \frac{1}{f_o} = \frac{1}{8.75} - \frac{1}{2} = \frac{2-8.75}{17.5} \Rightarrow u_o = \frac{-17.5}{6.75} = -2.59 \text{ cm}$$

$$\text{Magnifying power} = \frac{v_o}{|u_o|} \times \left[ 1 + \frac{D}{f_e} \right] = \frac{v_o}{|u_o|} \times \frac{D}{|u_e|} = \frac{8.75}{2.59} \times \frac{25}{6.25} = 13.51$$

## ASTRONOMICAL TELESCOPE



A telescope is used to see distant object, objective lens forms the image A'B' at its focus. This image A'B' acts as a object for eyepiece and it forms final image A''B''.

$$MP = \frac{\text{visual angle with instrument } (\beta)}{\text{visual angle for unaided eye } (\alpha)} \Rightarrow MP = \frac{\frac{h'}{f_0}}{\frac{h}{-u_e}} = -\frac{f_0}{u_e} [A'B' = h']$$

(i) If the final image is at infinity  $v_e = -\infty$ ,  $u_e = -ve$

$$\frac{1}{-\infty} - \frac{1}{-u_e} = \frac{1}{f_e} \Rightarrow u_e = f_e. \text{ So } MP = -\frac{f_0}{f_e} \text{ and length of the tube } L = f_0 + f_e$$

(ii) If the final image is at D :  $v_e = -D$   $u_e = -ve$

$$\frac{1}{-D} - \frac{1}{-u_e} = \frac{1}{f_e} \Rightarrow \frac{1}{u_e} = \frac{1}{f_e} + \frac{1}{D} = \frac{1}{f_e} \left[ 1 + \frac{f_e}{D} \right] \text{ So } MP = -\frac{f_0}{u_e} = -\frac{f_0}{f_e} \left[ 1 + \frac{f_e}{D} \right]$$

Length of the tube is  $L = f_0 + u_e$

| S. No. | Compound - Microscope                                                                                                          | S. No. | Astronomical - Telescope                                                                                                              |
|--------|--------------------------------------------------------------------------------------------------------------------------------|--------|---------------------------------------------------------------------------------------------------------------------------------------|
| 1.     | It is used to increase visual angle of near tiny object.                                                                       | 1.     | It is used to increase visual angle of distant large objects.                                                                         |
| 2.     | In it field and eye lens both are convergent, of short focal length and aperture.                                              | 2.     | In it objective lens is of large focal length and aperture while eye lens of short focal length and aperture and both are convergent. |
| 3.     | Final image is inverted, virtual and enlarged and at a distance D to $\infty$ from the eye.                                    | 3.     | Final image is inverted, virtual and enlarged at a distance D to $\infty$ from the eye.                                               |
| 4.     | MP does not change appreciably if objective and eye lens are interchanged as $[MP \sim (LD / f_0 f_e)]$                        | 4.     | MP becomes $(1/m^2)$ times of its initial value if objective and eye-lenses are interchanged as $MP \sim [f_0 / f_e]$                 |
| 5.     | MP is increased by decreasing the focal length of both the lenses.                                                             | 5.     | MP is increased by increasing the focal length of objective lens and by decreasing the focal length of eyepiece                       |
| 6.     | RP is increased by decreasing the wavelength of light used.<br>$\left( \because RP = \frac{2\mu \sin \theta}{\lambda} \right)$ | 6.     | RP is increased by increasing the aperture of objective.<br>$\left( \because RP = \frac{D}{1.22\lambda} \right)$                      |



**Ex.** A small telescope has an objective lens of focal length 144 cm and an eyepiece of focal length 6.0 cm. What is the magnifying power of the telescope ? What is the separation between the objective and the eyepiece? When final image is formed at infinity.

**Sol.** Here,  $f_o = 144$  cm;  $f_e = 6.0$  cm,  $MP = ?$ ,  $L = ?$

$$MP = \frac{-f_o}{f_e} = \frac{-144}{6.0} = -24 \text{ and } L = f_o + f_e = 144 + 6.0 = 150.0 \text{ cm}$$

**Ex.** Diameter of the moon is  $3.5 \times 10^3$  km and its distance from earth is  $3.8 \times 10^5$  km. It is seen by a telescope whose objective and eyepiece have focal lengths 4m and 10cm respectively. What will the angular diameter of the image of the moon.

**Sol.**  $MP = -\frac{f_o}{f_e} = -\frac{400}{10} = -40$ . Angle subtended by the moon at the objective  $= \frac{3.5 \times 10^3}{3.8 \times 10^5} = 0.009$  radian.

Thus angular diameter of the image  $= MP \times \text{visual angle} = 40 \times 0.009 = 0.36$  radian  $= \frac{0.36 \times 180}{3.14} \approx 21^\circ$

**Ex.** A telescope consisting of an objective of focal length 60 cm and a single-lens eyepiece of focal length 5 cm is focussed at a distant object in such a way that parallel rays emerge from the eye piece. If the object subtends an angle of  $2^\circ$  at the objective, then find the angular width of the image.

**Sol.**  $MP = \frac{f_o}{f_e} = \frac{\beta}{\alpha} \Rightarrow \beta = \alpha \frac{f_o}{f_e} = 2^\circ \times \frac{60}{5} = 24^\circ$

**Ex.** The focal lengths of the objective and the eye piece of an astronomical telescope are 60 cm and 5 cm respectively. Calculate the magnifying power and the length of the telescope when the final image is formed at (i) infinity, (ii) least distance of distinct vision (25 cm)

**Sol.** (i) When the final image is at infinity, then :

$$MP = -\frac{f_o}{f_e} = -\frac{60}{5} = -12 \text{ and length of the telescope is } L = f_o + f_e = 60 + 5 = 65 \text{ cm}$$

(ii) For least distance of distinct vision, the magnifying power is :

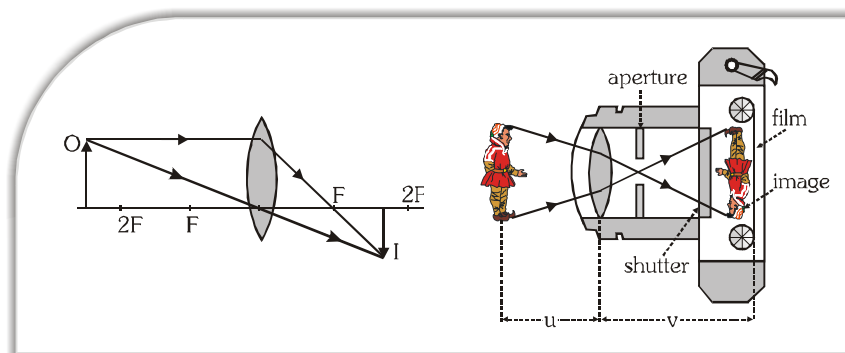
$$MP = -\frac{f_o}{f_e} \left( 1 + \frac{f_e}{D} \right) = -\frac{60}{5} \left( 1 + \frac{5}{25} \right) = -\frac{12 \times 6}{5} = -14.4$$

$$\text{Now } \frac{1}{f_e} = \frac{1}{v_e} - \frac{1}{u_e} \Rightarrow \frac{1}{5} = -\frac{1}{25} - \frac{1}{u_e} \Rightarrow \frac{-1}{u_e} = \frac{1}{25} + \frac{1}{5} \Rightarrow u_e = -4.17 \text{ cm} \Rightarrow |u_e| = 4.17 \text{ cm}$$

The length of telescope in this position is  $L = f_o + |u_e| = 60 + 4.17 = 64.17 \text{ cm}$

## LENS – CAMERA

There is a convex lens whose aperture and distance from the film can be adjusted. Object is real and placed between  $\infty$  and  $2F$ , so the image is real, inverted diminished and between  $F$  and  $2F$ .



If  $I$  is the intensity of light,  $S$  is the light transmitting area of lens and  $t$  is the exposure time,

then for proper exposure,  $I \times S \times t = \text{constant}$

light transmitting area of a lens is proportional to the square of its aperture  $D$ ;  $I \times D^2 \times t = \text{constant}$

If aperture is kept fixed, for proper exposure,  $I \times t = \text{constant}$ , i.e.,  $I_1 t_1 = I_2 t_2$

If intensity is kept fixed, for proper exposure,  $D^2 \times t = \text{constant}$

$$\text{Time of exposure} \propto \frac{1}{(\text{aperture})^2} \dots (i)$$

The ratio of focal length to the aperture of lens is called  $f$ -number of the camera,

$$f\text{-number} = \frac{\text{focal length}}{\text{aperture}} \Rightarrow \text{Aperture} \propto \frac{1}{f\text{-number}} \dots (ii)$$

From equation (i) and (ii)  $\Rightarrow \text{Time of exposure} \propto (f\text{-number})^2$

**Ex.** With diaphragm of the camera lens set at  $\frac{f}{2}$ , the correct exposure time is  $\frac{1}{100}$ , then with diaphragm set at  $\frac{f}{4}$ . Calculate the correct exposure time.

**Sol.** As exposure time  $\propto \frac{1}{(\text{aperture})^2} \Rightarrow t_1 \propto \frac{1}{[f/2]^2}$  and  $t_2 \propto \frac{1}{[f/4]^2}$

$$\text{here } t_1 = \frac{1}{100} \text{ s then } \frac{t_2}{t_1} = \frac{16}{4} = 4 \Rightarrow t_2 = 4t_1 = \frac{4}{100} \text{ s}$$

**Ex.** A good photographic print is obtained by an exposure of two seconds at a distance of 20 cm from the lamp. Calculate the time of exposure required to get an equally good result at a distance of 40 cm.

**Sol.** We know that the intensity of light varies inversely as the  $(\text{distance})^2$ . When distance is doubled, the intensity becomes one-fourth. So, the time of exposure should be four times. Hence, time of exposure  $= 2 \times 4 = 8 \text{ s}$

**Ex.** Photograph of the ground are taken from an aircraft, flying at an altitude of 2000 m, by a camera with a lens of focal length 50 cm. The size of the film in the camera is  $18 \text{ cm} \times 18 \text{ cm}$ . What area of the ground can be photographed by this camera at any one time.

**Sol.** As here  $u = -2000 \text{ m}$ ,  $f = 0.50 \text{ m}$ , so from lens formula  $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$ ,

$$\text{we have } \frac{1}{v} - \frac{1}{(-2000)} = \frac{1}{0.5} \Rightarrow \frac{1}{v} = \frac{1}{0.5} - \frac{1}{2000} \approx \frac{1}{0.5} \left[ \text{as } \frac{1}{0.5} \gg \frac{1}{2000} \right] \Rightarrow v = 0.5 \text{ m} = 50 \text{ cm} = f$$

$$\text{Now as in case of a lens, } m = \frac{v}{u} = \frac{0.5}{-2000} = -\frac{1}{4} \times 10^{-3} \text{ So } I_1 = (ma)(mb) = m^2 A [\because A = ab]$$

$$A = \frac{I_1}{m^2} = \frac{18 \text{ cm} \times 18 \text{ cm}}{\left[\left(\frac{1}{4}\right) \times 10^{-3}\right]^2} = (720 \text{ m} \times 720 \text{ m})$$

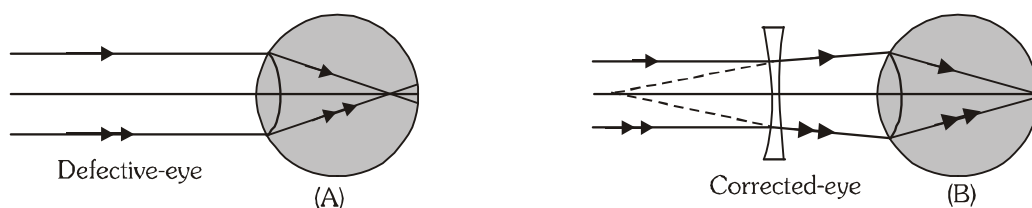
**Ex.** The proper exposure time for a photographic print is 20 s at a distance of 0.6 m from a 40 candle power lamp. How long will you expose the same print at a distance of 1.2 m from a 20 candle power lamp ?

**Sol.** In case of camera, for proper exposure  $I_1 D_1^2 t_1 = I_2 D_2^2 t_2$

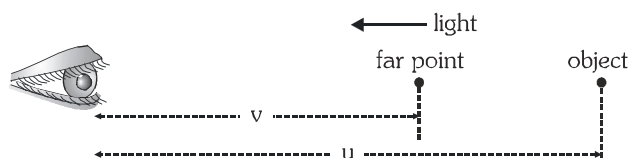
As here D is constant and  $I = (L/r^2)$ ;  $\frac{L_1}{r_1^2} \times t_1 = \frac{L_2}{r_2^2} \times t_2$  So  $\frac{40}{(0.6)^2} \times 20 = \frac{20}{(1.2)^2} t \Rightarrow t = 160 \text{ s}$

## DEFECTS OF EYES

### MYOPIA [or Short-sightedness or Near-sightedness]



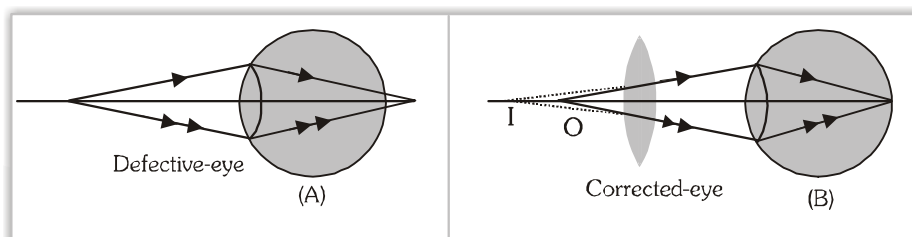
- (i) Distant objects are not clearly visible, but near objects are clearly visible because the image is formed before the retina.
- (ii) To remove the defect, a concave lens is used.
- ⊙ The maximum distance which a person can see without the help of spectacles is known as the far point.
- ⊙ If the reference of the object is not given, then it is taken as infinity.
- ⊙ In this case, the image of the object is formed at the far point of the person.



$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f} = P \Rightarrow \frac{1}{\text{distance of far point (in m)}} - \frac{1}{\text{distance of object (in m)}} = \frac{1}{f} = P$$

$$\frac{1}{100} - \frac{1}{\infty} = \frac{1}{f} = P$$

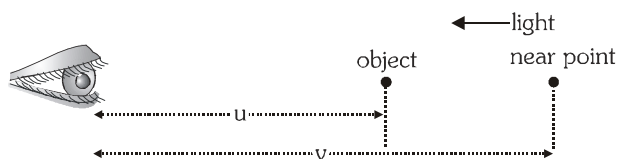
### LONG-SIGHTEDNESS OR HYPERMETROPIA



- (i) Near objects are not clearly visible but far objects are clearly visible.
- (ii) The image of a near object is formed behind the retina.
- (iii) To remove this defect, a convex lens is used.

**Near Point :-**

The minimum distance which a person can see without help of spectacles.



- In this case image of the object is formed at the near point.
- If reference of object is not given it is taken as 25 cm.

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f} = P \Rightarrow \frac{1}{\text{distance of near point (in m)}} - \frac{1}{\text{distance of object (in m)}} = \frac{1}{f} = P$$

distance of near point = -ve, distance of object = -ve,  $P = +ve$

**PRESBYOPIA**

In this case both near and far object are not clearly visible. To remove this defect two separate spectacles one for myopia and other for hypermetropia are used or bifocal lenses are used.

**ASTIGMATISM**

In this defect eye can not see object in two orthogonal direction clearly. It can be removed by using cylindrical lens in particular direction.

**Ex.** A person can not see clearly an object kept at a distance beyond of 100 cm. Find the nature and the power of lens to be used for seeing clearly the object at infinity.

**Sol.** For lens  $u = -\infty$  and  $v = -100$  cm

$$\therefore \frac{1}{v} - \frac{1}{u} = \frac{1}{f} = \frac{1}{v} = \frac{1}{f} \Rightarrow f = v = -100 \text{ cm (concave)} \therefore \text{Power of lens } P = \frac{1}{f} = -\frac{1}{1} = -1\text{D}$$

**Ex.** A far sighted person has a near point of 60 cm. What power lens should be used for eye glasses such that the person can read this book at a distance of 25 cm.

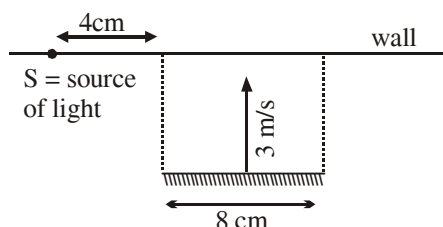
**Sol.** Here  $v = -60$  cm,  $u = -25$  cm

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u} = -\frac{1}{60} + \frac{1}{25} \Rightarrow f = \frac{300}{7} \text{ cm} \therefore \text{Power} = \frac{1}{f(\text{in m})} = \frac{1}{(3/7)} = +2.33\text{D}$$

## EXERCISE (S-1)

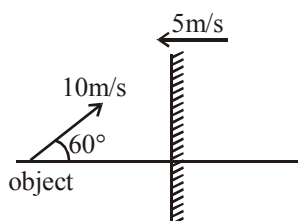
### Plane mirror :

1. A fluorescent lamp of length 1 m is placed horizontally at a depth of 1.2 m below a ceiling. A plane mirror of length 0.6 m is placed below the lamp parallel to and symmetric to the lamp at a distance 2.4m from it. The length of the reflected patch of length on the ceiling \_\_\_\_\_ .  
GO0001
2. A plane mirror of length 8 cm is moving with speed 3 m/s towards a wall in situation as shown in figure. Size of spot formed on the wall is  $32/x$  cm. Find the value of x.



GO0002

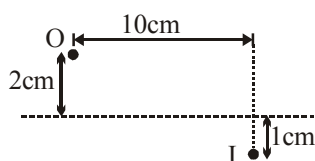
3. The angle between the velocity vector of object and image is  $\frac{\pi}{\alpha}$ . Velocity are shown in the figure. Then the value of  $\alpha$  is.



GO0003

### Spherical mirror :

4. The principal axis of a spherical mirror is shown by dotted line. O is the point object whose real image is I. Find the distance of the pole and centre of curvature of the mirror from object measured along principal axis by drawing ray diagram.



GO0004

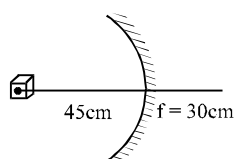
5. An experimentalist devises a method for finding the radius of curvature of a convex mirror. He uses a plane mirror strip between the object and the convex mirror and adjusts it till the two virtual images formed by reflection at both the mirrors coincide without parallax. In his observations, the object distance from the convex mirror is 0.5 m while it is 0.30 m in front of the plane mirror. Find the radius of curvature (in cm) of the convex mirror.

GO0005

6. A thin rod of length  $d/3$  is placed along the principal axis of a concave mirror of focal length  $= d$  such that its image, which is real and elongated, just touches the rod. Find the length of the image?

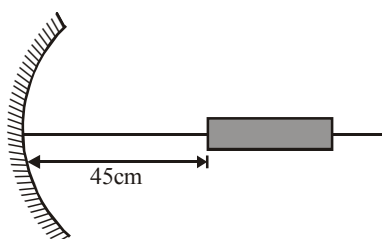
GO0006

7. A cube of side length 1 mm is placed on the axis of a concave mirror at a distance of 45 cm from the pole as shown in the figure. One edge of the cube is parallel to the axis. The focal length of the mirror is 30 cm. Find approximate volume of the image.



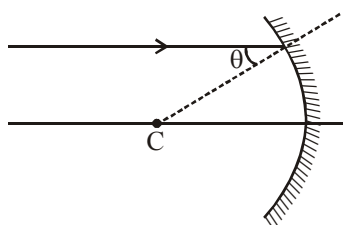
GO0007

8. An object of length 30 cm is placed on principal axis of a concave mirror of focal length 30 cm. Its one end at a distance of 45 cm as shown. If length of image is  $10x$  (in cm) find the value of  $x$ .



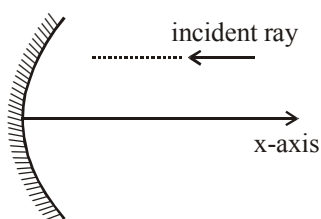
GO0008

9. A ray incident parallel to principal axis on a concave mirror at an angle  $\theta$  after two reflection, reflected ray becomes parallel to incident ray. If  $\theta = \frac{\pi}{N}$  (in radian) then find the value of  $N$ .



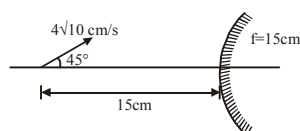
GO0009

10. A parabolic reflecting surface given by  $x = \frac{y^2}{2}$ , is placed at origin, as shown. An incident ray moving along  $y = 1$  is incident on it. The ray gets reflected by the surface twice. The deviation suffered by the ray is  $\frac{\pi}{n}$  radians. Find  $n$ .



GO0010

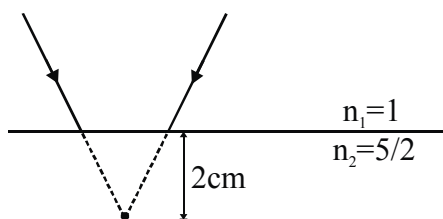
11. In the figure shown, the speed cm/s of image with respect to mirror is :



GO0011

### Refraction on plane surface :

12. In the given figure rays incident on an interface would converge 2 cm below the interface if they continued to move in straight lines without bending. But due to refraction, the rays will bend and meet somewhere else. Find the distance of meeting point of refracted rays below the interface (in cm). (Assuming the rays to be making small angles with the normal to the interface)



GO0012

13. A point object is placed 33 cm from a convex mirror of curvature radius = 40 cm. A glass plate of thickness 6 cm and index 2.0 is placed between the object and mirror, close to the mirror. Find the distance of final image from the object?

GO0013

14. An opaque cylindrical tank with an open top has a diameter of 3.00 m and is completely filled with water. When the setting sun reaches an angle of  $37^\circ$  above the horizon, sunlight ceases to illuminate any part of the bottom of the tank. How deep is tank in meter?

GO0014

15. A cylindrical bucket of depth 60 cm is partly filled with a liquid of refractive index 1.5 and with oil (on top of liquid) of refractive index 2. It appears that the volume of air, volume of liquid and volume of oil are equal, to an observer who views from top of the bucket. The apparent depth of the bucket as

seen by the observer is given as  $\alpha$  cm. Fill  $\left(\frac{\alpha}{5}\right)$  in OMR sheet.

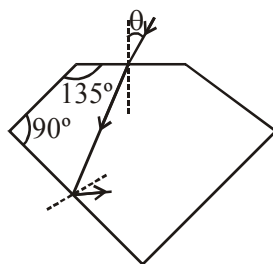
GO0015

### Total internal reflection :

16. A room contains air in which the speed of sound is 340 m/s. The walls of the room are made of concrete, in which the speed of sound is 1700 m/s. (a) Find the critical angle for total internal reflection of sound at the concrete-air boundary. (b) In which medium must the sound be traveling to undergo total internal reflection? (Assume sound waves follow snell's law)

GO0016

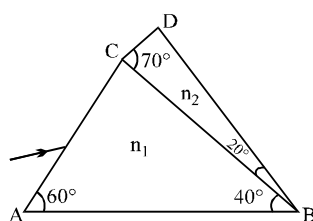
17. A ray of light enters a diamond ( $n = 2$ ) from air and is being internally reflected near the bottom as shown in the figure. Find maximum value of angle  $\theta$  possible ?



GO0017

18. A prism of refractive index  $n_1$  & another prism of refractive index  $n_2$  are stuck together without a gap as shown in the figure. The angles of the prisms are as shown.  $n_1$  &  $n_2$  depend on  $\lambda$ , the wavelength

of light according to  $n_1 = 1.20 + \frac{10.8 \times 10^4}{\lambda^2}$  &  $n_2 = 1.45 + \frac{1.80 \times 10^4}{\lambda^2}$  where  $\lambda$  is in nm. [JEE 1998]



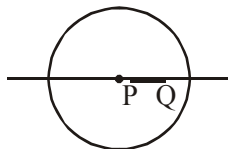
- (i) Calculate the wavelength  $\lambda_0$  for which rays incident at any angle on the interface BC pass through without bending at that interface .  
 (ii) For light of wavelength  $\lambda_0$ , find the angle of incidence  $i$  on the face AC such that the deviation produced by the combination of prisms is minimum.

GO0018



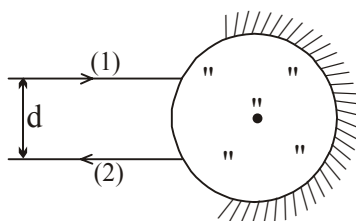
**Refraction at curved surface :**

19. A small object of length 1 mm lies along the principal axis of a spherical glass of radius  $R = 10$  cm and refractive index is  $3/2$ . The object is seen from air along the principal axis from left. The distance of object from the centre is 5 cm. Find the size of the image. Is real, inverted ?



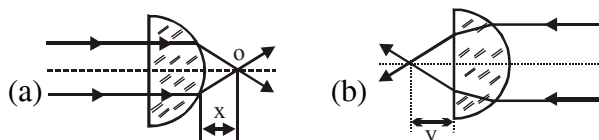
GO0019

20. A transparent sphere of radius  $R = 2.0$  m has a mirrored surface on its right half as shown in figure. A light ray travelling in air is incident on the left side of the sphere. The incident light ray (1) & exiting light ray (2) are parallel & separated by distance  $d = 2.0$  m. Then find the refractive index of the material. (take :  $\sin 15^\circ = 0.25$ )



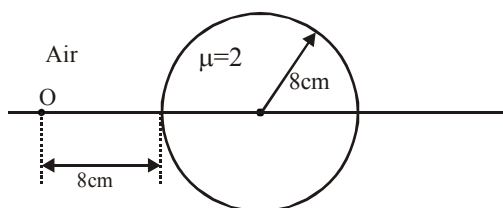
GO0020

21. A narrow beam of light passing through the hemisphere of material with refractive index  $n$ , intersects at point O. Where does the beam converge (i.e.  $y$  in cm) if beam were to travel in the opposite direction as shown in figure (b)? The value of  $x$  is given to be 10 cm. Radius of the hemisphere is also 10 cm :-



GO0021

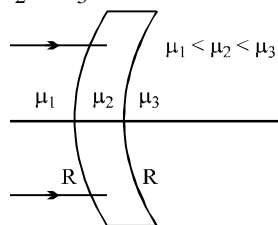
22. On the axis of a transparent sphere of refractive index ( $n=2$ ) & radius 8 cm, an object is kept at the distance 8 cm from the surface of the sphere. Find the minimum distance of the image after all possible refraction from the surface of the sphere in cm.



GO0022

**Lens :**

23. A meniscus lens is made of a material of refractive index  $\mu_2$ . Both its surfaces have radii of curvature  $R$ . It has two different media of refractive indices  $\mu_1$  and  $\mu_3$  respectively, on its two sides (see figure). Calculate its focal length for  $\mu_1 < \mu_2 < \mu_3$ , when light is incident on it as shown. [IIT-JEE 2003]

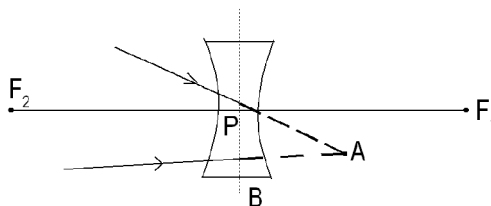


GO0023

24. A converging beam of rays is incident on a diverging lens. Having passed through the lens the rays intersect at a point 15 cm from the lens. If the lens is removed, the point where the rays meet will move 5 cm closer to the mounting that holds. Find the focal length (in cm) of the lens without sign.

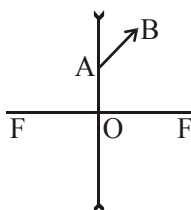
GO0024

25. The rays of a converging beam meet at a point A. A diverging lens is placed in their path in the plane B. Plot the position of the point where the rays meet after passing through the lens. The position of the principal foci FF is known.



GO0025

26. Figure shows ray AB that has passed through a divergent lens. Construct the path of the ray up to the lens if the position of its foci F is known.



GO0026

27. A plano convex lens ( $\mu = 1.5$ ) has a maximum thickness of 1 mm. If diameter of its aperture is 4 cm. Find (i) Radius of curvature of curved surface; (ii) its focal length in air.

GO0027

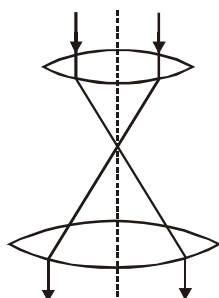
28. A point source of light is kept at a distance of 150 cm from a converging lens, on its optical axis. The focal length of the lens is 100 cm and its diameter is 3 cm. A screen is placed on the other side of the lens, perpendicular to the axis of lens, at a distance 200 cm from it. Then find the area of the illuminated part of the screen? (Assume all rays incident on lens as paraxial)

GO0028

29. There are two thin symmetrical lenses, one is converging, with refractive index  $n_1 = 1.70$ , and the other is diverging with refractive index  $n_2 = 1.50$ . Both lenses have the same curvature radius of their surface equal to  $R = 10$  cm. The lenses are put close together and submerged into water. The focal length of lens system is found to be  $\frac{100}{x}$  cm in water. What is the value of  $x$ . (R.I. of water =  $4/3$ )

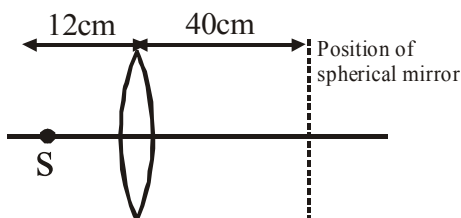
GO0029

30. Consider a 'beam expander' which consists of two converging lenses of focal lengths 40 cm and 100 cm having a common optical axis. A laser beam of diameter 2 mm is incident on the 40 cm focal length lens. Then what is the diameter (in mm) of the final beam (see figure)?



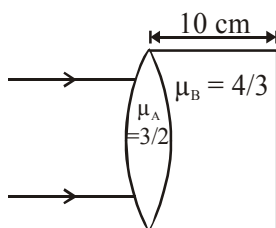
GO0030

31. A converging lens and a spherical mirror are placed with their principal axis coinciding. Their separation equals 40 cm. A point source  $S$  is placed on the principal axis at a distance of 12 cm from the lens as shown in the figure. It is found that the final beam comes out parallel to the principal axis. Focal length of the lens equals 15 cm. Find the focal length of the mirror.



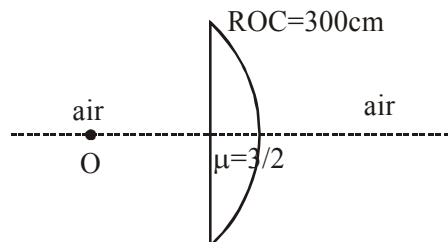
GO0031

32. A parallel paraxial beam of light is incident on the arrangement as shown  $\mu_A = 3/2$ ,  $\mu_B = 4/3$ , the two spherical surfaces are very close and each has radius of curvature 10 cm. Find the point where the rays are focussed. (w.r.t. point of entry)



GO0032

33. Plane surface of a thin planoconvex lens reflects 50% of light, while the curved surface is completely transparent, if final image of 'O' after refraction through thin lens coincides with the image formed due to partial reflection from plane surface. If distance between O and lens is  $x$  (in m) then find the value of  $x/4$ .

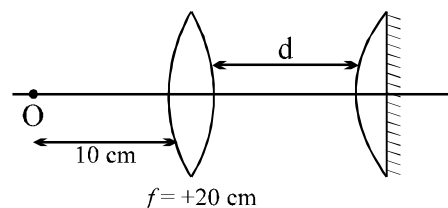


GO0033

34. A lens is placed at origin, with  $x$ -axis as its principal axis. A ray of light is incident on it from the -ve side of  $x$ -axis along the line  $y = \frac{x}{400} + 0.1$ , where  $x, y$  are in cm. Focal length of lens is 30 cm. Find the equation of the ray after passing through the lens.

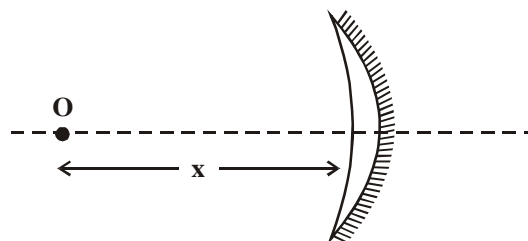
GO0034

35. A convex lens of focal length 20 cm and another plano convex lens of focal length 40 cm are placed co-axially (see fig.). The plano convex lens is silvered on plane surface. What should be the  $d$  (in m) so that final image of the object 'O' is formed on O itself.



GO0035

36. Radii of curvature of a concavo-convex lens (refractive index = 1.5) are 40 cm and 20 cm as shown. The convex side is silvered. The distance  $x$  (in cm) on the principal axis where an object is placed so that its image is created on the object itself, is given as  $4\beta$ . Find the value of  $\beta$ .



GO0036

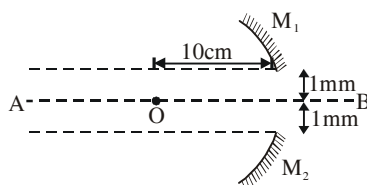
### Dispersion :

37. A flint glass convex lens of focal length 16 cm is placed in contact with crown glass lens. The refractive indices of crown glass for violet and red colour are  $\mu_v = 1.525$ ,  $\mu_r = 1.515$  and for flint glass  $\mu_v = 1.655$  and  $\mu_r = 1.645$ .
- Find the nature and focal length of the crown glass lens which will form an achromatic combination with the flint glass lens.
  - What is the focal length of the combination?

GO0037

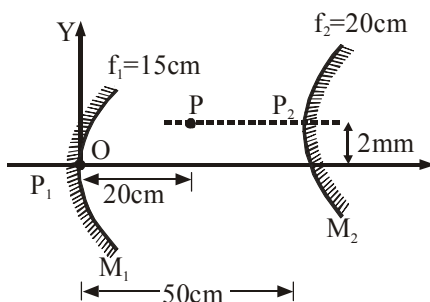
## EXERCISE (S-2)

1. A concave mirror of focal length 20 cm is cut into two parts from the middle and the two parts are moved perpendicularly by a distance 1 mm from the previous principal axis AB. If an object is placed at a distance of 10 cm from the mirror on the line AB then, find the distance between the images formed by the two parts?



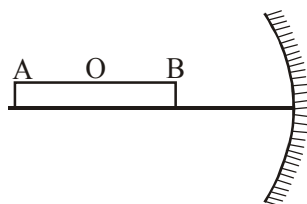
GO0038

2. Find the co-ordinates of image of point object P formed after two successive reflection in situation as shown in figure considering first reflection at concave mirror and then at convex.



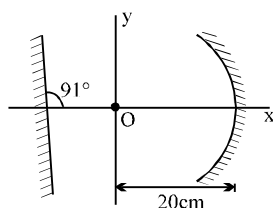
GO0039

3. A rod of length  $\ell$  is moving with constant speed 1 m/s towards the pole of a concave mirror of focal length  $\ell$ . Find the rate of change in length of the image of the rod (in m/s) at the moment centre of the rod coincides with the centre of curvature of the mirror.



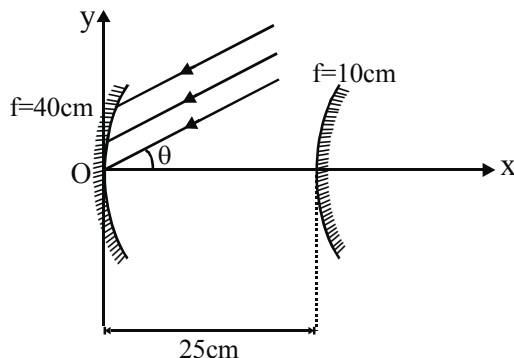
GO0040

4. A point object is placed at the centre of curvature of a concave mirror (taken as origin). A plane mirror is also placed at a distance of 10 cm from the object as shown. Consider two reflection first at plane mirror and then at concave mirror ( $x_0, y_0$ ). Find  $\frac{\pi x_0}{y_0}$  the coordinates of the image thus formed are.



GO0041

5. Consider a concave mirror kept at origin with focal length 40 cm. Parallel rays which subtend an angle  $\theta = \frac{1}{40}$  radian are incident on it. A convex mirror is kept at a distance 25 cm from the first mirror as shown. Find the y-coordinate of the image (in cm) formed by the system of mirrors after two reflections.



GO0042

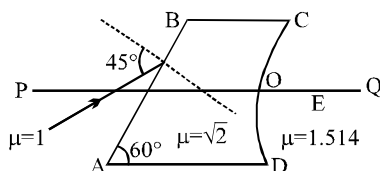
6. The x-y plane is the boundary between two transparent media. Medium-1 with  $z > 0$  has refractive index  $\sqrt{2}$  and medium -2 with  $z < 0$  has a refractive index  $\sqrt{3}$ . A ray of light in medium -1 given by the vector  $A = 6\sqrt{3}\hat{i} + 8\sqrt{3}\hat{j} - 10\hat{k}$  is incident on the plane of separation. Find the unit vector in the direction of refracted ray in medium -2.

[1999, 10M]

GO0043

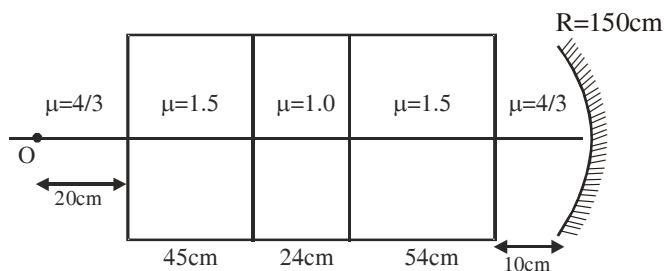
7. Figure shows an irregular block of material of refractive index  $\sqrt{2}$ . A ray of light strikes the face AB as shown in the figure. After refraction it is incident on a spherical surface CD of radius of curvature 0.4 m and enters a medium of refractive index 1.514 to meet PQ at E. Find the distance OE upto two places of decimal.

[IIT-JEE 2004]



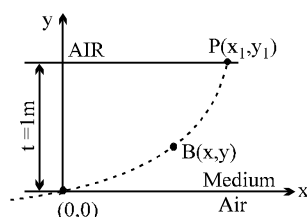
GO0044

8. A composite slab consisting of different media is placed in front of a concave mirror of radius of curvature 150 cm. The whole arrangement is placed in water. An object O is placed at a distance 20cm from the slab. The R.I. of different media are given in the diagram. Find the position of the final image formed by the system.



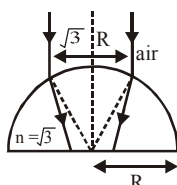
GO0045

9. A ray of light travelling in air is incident at grazing angle (incident angle =  $90^\circ$ ) on a long rectangular slab of a transparent medium of thickness  $t = 1.0$  (see figure). The point of incidence is the origin A (O, O). The medium has a variable index of refraction  $n(y)$  given by:  $n(y) = [ky^{3/2} + 1]^{1/2}$ , where  $k = 1.0 \text{ m}^{-3/2}$ . The refractive index of air is 1.0.
- Obtain a relation between the slope ( $dy/dx$ ) of the trajectory of the ray at a point B (x, y) in the medium and the incident angle ( $i$ ) at that point. [JEE-1995]
  - Find the value of  $n \sin i$ .
  - Obtain an equation for the trajectory  $y(x)$  of the ray in the medium.
  - Determine the coordinates ( $x_1, y_1$ ) of the point P, where the ray intersects the upper surface of the slab-air boundary.
  - Indicate the path of the ray subsequently.



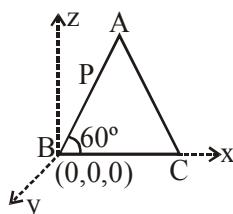
GO0046

10. A light beam of diameter  $\sqrt{3} R$  is incident symmetrically on a glass hemisphere of radius  $R$  and of refractive index  $n = \sqrt{3}$ . Find the radius of the beam at the base of hemisphere.



GO0047

11. An equilateral prism ABC is placed in air with its base side C lying horizontally along X-axis as shown in the figure. A ray given by  $\sqrt{3}z + x = 10$  is incident at a point P on face AB of prism
- Find the value of  $\mu$  for which the ray grazes the faces AC.
  - Find direction of the finally refracted ray if  $\mu = 3/2$ .
  - Find the equation of ray coming out of prism if bottom BC is silvered ?

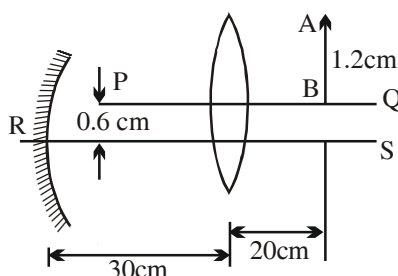


GO0048

12. An object is approaching a thin convex lens of focal length 0.3 m with a speed of 0.01 m/s. Find the magnitudes of the rates of change of position and lateral magnification of image when the object is at a distance of 0.4 m from the lens. [IIT-JEE 2004]

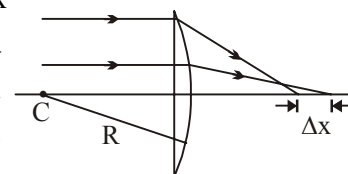
GO0049

13. A convex lens of focal length 15 cm and a concave mirror of focal length 30 cm are kept with their optic axes  $PQ$  and  $RS$  parallel but separated in vertical direction by 0.6 cm as shown. The distance between the lens and mirror is 30 cm. An upright object  $AB$  of height 1.2 cm is placed on the optic axis  $PQ$  of the lens at a distance of 20 cm from the lens. If  $A'B'$  is the image after refraction from the lens and reflection from the mirror, find the distance of  $A'B'$  from the pole of the mirror and obtain its magnification. Also locate positions of  $A'$  and  $B'$  with respect to the optic axis  $RS$ . [IIT-JEE 2000]



GO0050

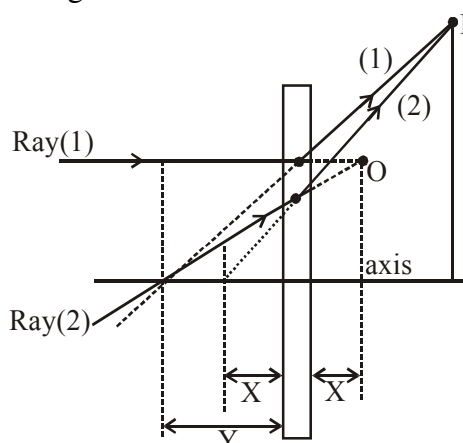
14. Two rays travelling parallel to the principal axis strike a large plano-convex lens having a refractive index of 1.60. If the convex face is spherical, a ray near the edge does not pass through the focal point (spherical aberration occurs). If this face has a radius of curvature of magnitude 20.0 cm and the two rays are  $h_1 = 0.500$  cm and  $h_2 = 12.0$  cm from the principal axis, find the difference in the positions where they cross the principal axis.



GO0051

15. The rectangular box shown is the place of lens. By looking at the ray diagram, answer the following questions :

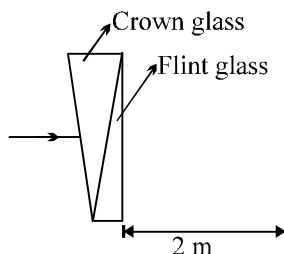
- (i) If  $X$  is 5 cm then what is the focal length of the lens ?  
 (ii) If the point  $O$  is 1 cm above the axis then what is the position of the image ? Consider the optical center of the lens to be the origin.



GO0052



16. The refractive indices of the crown glass for violet and red lights are 1.51 and 1.49 respectively and those of the flint glass are 1.77 and 1.73 respectively. A prism of angle  $6^\circ$  is made of crown glass. A beam of white light is incident at a small angle on this prism. The other thin flint glass prism is combined with the crown glass prism such that the net mean deviation is  $1.5^\circ$  anticlockwise.



- Determine the angle of the flint glass prism.
- A screen is placed normal to the emerging beam at a distance of 2m from the prism combination. Find the distance between red and violet spot on the screen. Which is the topmost colour on screen.

GO0053

17. A large temple has a depression in one wall. On the floor plan it appears as a indentation having spherical shape of radius 2.50 m. A worshiper stands on the center line of the depression, 2.00 m out from its deepest point, and whispers a prayer. Where is the sound concentrated after reflection from the back wall of the depression?

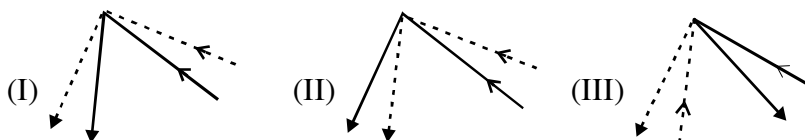
GO0054

## EXERCISE (O-1)

## SINGLE CORRECT TYPE QUESTIONS

*Plane mirror :*

1. Each of these diagrams is supposed to show two different rays being reflected from the same point on the same mirror. Which option is correct :-



- (A) Only I                      (B) Only II                      (C) Only III                      (D) All

GO0055

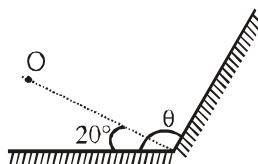
2. If two mirrors are kept at  $60^\circ$  to each other, then the number of images formed by them is :-

[AIEEE- 2002]

- (A) 5                      (B) 6                      (C) 7                      (D) 8

GO0056

3. If  $\theta = 110^\circ$  then total number of images formed by the mirror system will be :-



- (A) 2                      (B) 3                      (C) 4                      (D) 5

GO0057

4. A point object approaches a plane mirror with a speed of  $10 \text{ ms}^{-1}$ , while the image recedes away from the mirror with a speed of  $6 \text{ ms}^{-1}$  (as seen by stationary observer). The direction and magnitude of the velocity of mirror is :-

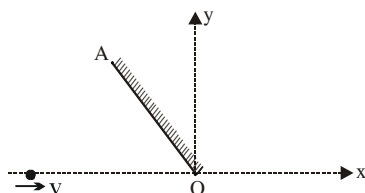
- (A) towards the object,  $8 \text{ ms}^{-1}$   
 (B) towards the image,  $6 \text{ ms}^{-1}$   
 (C) away from the object,  $8 \text{ ms}^{-1}$   
 (D) away from the object,  $2 \text{ ms}^{-1}$

GO0058

5. A boy of TOA3 batch is 1.8m tall and can see his image in a plane mirror fixed on a wall. His eyes are 1.6m from the floor level. The minimum length of the mirror to see his full image is :-  
 (A) 0.9 m (B) 0.85 m  
 (C) 0.8 m (D) Can't be determined

GO0059

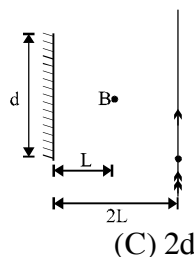
6. A plane mirror OA of length 2 m is kept along the line  $y = -x$  as shown in the figure. An insect having velocity of  $4\hat{i}$  cm/s is moving along the x-axis from far away. The time span for which the insect can see its image will be :-



- (A) 50 sec (B) 25 sec (C)  $25\sqrt{2}$  sec (D)  $50\sqrt{2}$  sec

GO0060

7. A point source of light B is placed at a distance L in front of the centre of a mirror of width d hung vertically on a wall. A man walks in front of the mirror along a line parallel to the mirror at a distance 2L from it as shown. The greatest distance over which he can see the image of the light source in the mirror is :-  
 [IIT-JEE '2000 (Scr)]

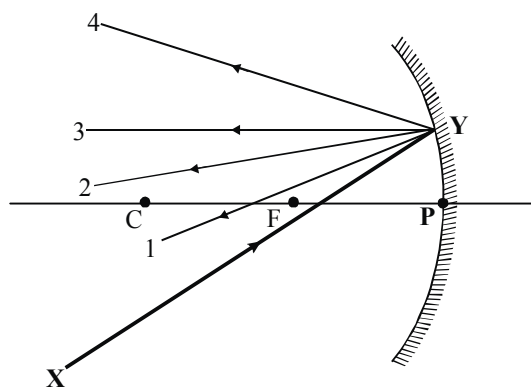


- (A)  $d/2$  (B) d (C) 2d (D) 3d

GO0061

### Spherical mirror :

8. Figure shows a small concave mirror with CP as its principal axis. A ray XY is incident on the mirror. Which of the four rays can be the reflected ray.



- (A) 1 (B) 2 (C) 3 (D) 4

GO0062

9. The distance of an object from a spherical mirror is equal to the focal length of the mirror. Then the image:
- (A) must be at infinity (B) may be at infinity  
(C) may be at the focus (D) none

GO0063

10. An object is placed in front of a spherical mirror whose 2 times magnified image is formed on screen. Then choose **CORRECT** option :-
- (A) Mirror is concave  $m = +2$  (B) Mirror is concave  $m = -2$   
(C) Mirror is convex  $m = +2$  (D) Mirror is convex  $m = -2$

GO0064

11. The table below lists object and image positions for four objects placed in front of mirrors, using Cartesian sign convention with pole of the mirror as origin and direction of incident rays as positive. In the following cases, the case in which images is formed by a convex spherical mirror, is :-

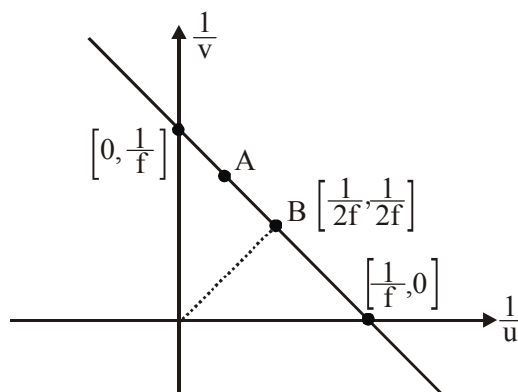
| Object position | Image position |
|-----------------|----------------|
| (A) $-25.0$ cm  | $-16.7$ cm     |
| (B) $-5.0$ cm   | $10.0$ cm      |
| (C) $-20.0$ cm  | $5.71$ cm      |
| (D) $-40.0$ cm  | $80.0$ cm      |

GO0065

12. A particle approaches from very large distance towards concave mirror along the principal axis. By the time the particle reaches the mirror the distance between the particle and its image
- (A) first decreases then increases  
(B) first increases then decreases  
(C) first increases then decreases and then again increases  
(D) first decreases then increases and then again decreases

GO0066

13. Figure shows  $\frac{1}{v}$  vs  $\frac{1}{u}$  curve for convex mirror. Nature of image at point A is :-



- (A) Real, erect, magnified (B) virtual, inverted, magnified  
(C) virtual, inverted, diminished (D) virtual, erect, diminished

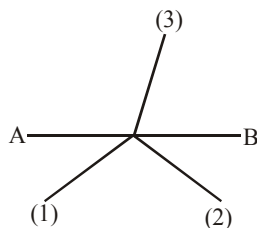
GO0067

14. A boy of height 1 m stands in front of a convex mirror. His distance from the mirror is equal to its focal length. The height of his image is :- (Assume paraxial ray approximation holds)
- (A) 0.25 m                      (B) 0.33 m                      (C) 0.5 m                      (D) 0.67 m

GO0068

**Refraction at plane surface :**

15. AB is a boundary separating two media of different refractive indices. A ray is incident on the boundary is partially reflected and partially transmitted. Choose the **CORRECT** statement.



- (A) 3 is incident ray and 1 is refracted ray  
 (B) 2 is incident ray and 1 is partially reflected ray  
 (C) 1 is incident ray and 3 is refracted ray  
 (D) 3 is incident ray and 2 is partially reflected ray

GO0069

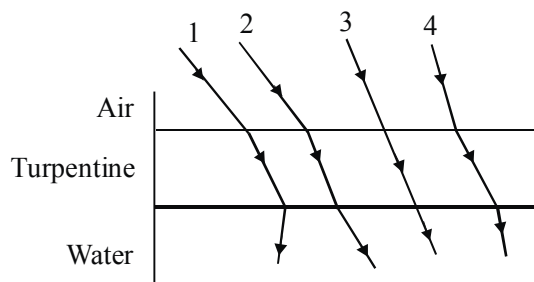
16. **Statement-1:** You see a geostationary satellite above the horizon. You desire to communicate with the satellite by sending a beam of laser light. You should aim your laser slightly higher than the line of sight of the satellite.

**Statement-2:** Light bends away from the normal while moving from denser to rarer medium.

- (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.  
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.  
 (C) Statement-1 is true, statement-2 is false.  
 (D) Statement-1 is false, statement-2 is true.

GO0070

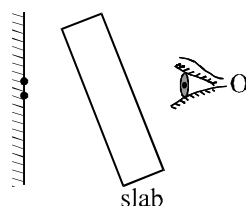
17. The optical density of turpentine is higher than that of water while its mass density is lower. Figure shows a layer of turpentine floating over water in a container. For which one of the four rays incident on turpentine in figure, the path shown is **CORRECT** ?



- (A) 1                      (B) 2                      (C) 3                      (D) 4

GO0071

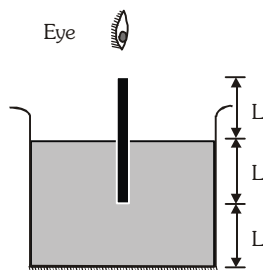
18. When a ray of light of frequency  $6 \times 10^{14}$  Hz travels from water of refractive index  $4/3$  to the glass of refractive index  $8/5$ , its :-  
 (A) frequency decreases to  $5/6$  of its initial value  
 (B) speed decreases to  $5/6$  of its initial value  
 (C) wavelength decreases to  $6/5$  of its initial value  
 (D) speed increases to  $6/5$  of its initial value
- GO0072
19. The observer at O views two closely spaced spots on a vertical wall through an angled glass slab as shown. As seen by observer, the spots appear.



- (A) shifted upward  
 (B) shifted downward  
 (C) spaced farther apart  
 (D) spaced closer together

GO0073

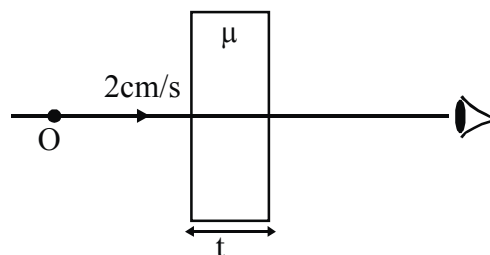
20. What is the length of the image of the rod in mirror, according to the observer in air? (refractive index of the liquid is  $\mu$ )



- (A)  $\mu L + L$       (B)  $L + \frac{L}{\mu}$       (C)  $L\mu + \frac{L}{\mu}$       (D) None of these

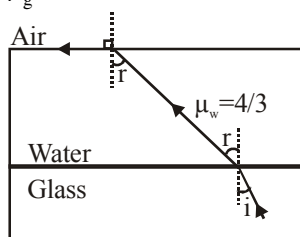
GO0074

21. A glass slab of width ' $t$ ', refractive index ' $\mu$ ' is placed as shown in the figure. If the point object, moves with a speed 2 cm/s towards the slab the speed observed will be  
 (A) 2 cm/s  
 (B) less than 2 cm/s  
 (C) greater than 2 cm/s  
 (D) dependent on the refractive index of surrounding medium



GO0075

22. A ray of light is incident at the glass–water interface at an angle  $i$ , it emerges finally parallel to the surface of water, then the value of  $\mu_g$  would be :— [IIT-JEE 2003]



- (A)  $(4/3) \sin i$  (B)  $1/\sin i$  (C)  $4/3$  (D) 1

GO0076

**Total internal reflection :**

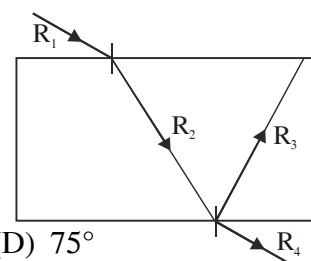
23. **Statement-1:** A point source of light is placed inside water. A light detector present out side, in air can detect light only in a conical region, with the apex at the source and circumscribing the circle of illuminance.

**Statement-2:** Ray incident from denser to rarer medium undergoes total internal reflection when their angle of incidence become more than the critical angle, this situation create a circular region from which light escapes and is called circle of illuminance.

- (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.  
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.  
 (C) Statement-1 is true, statement-2 is false.  
 (D) Statement-1 is false, statement-2 is true.

GO0077

24. A ray  $R_1$  is incident on the plane surface of the glass slab (kept in air) of refractive index  $\sqrt{2}$  at an angle of incidence equal to the critical angle for this air glass system. The refracted ray  $R_2$  undergoes partial reflection and refraction at the other surface. The angle between reflected ray  $R_3$  and the refracted ray  $R_4$  at that surface is :

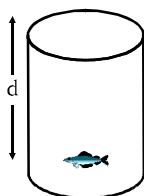


- (A)  $45^\circ$  (B)  $135^\circ$  (C)  $105^\circ$  (D)  $75^\circ$

GO0078

25. A fish floats in liquid with its eye at the centre of an opaque walled full tank of liquid of circular cross section. When the fish look upwards, it can see a fish-eye view of the surrounding scene i.e. it is able to view the entire space above the liquid surface. The diameter of the tank is 30 cm, and the critical angle for liquid is  $37^\circ$ . At what maximum depth below the surface of the liquid,  $d$ , must the fish be

floating?  $\left( \sin 37^\circ = \frac{3}{5} \right)$



- (A) 16 cm (B) 20 cm (C) 11.25 cm (D) 25 cm

GO0079

26. **Statement-1:** When light falls on a sphere made of diamond total internal reflection takes place which makes it shine more than a similar sphere made of common glass.

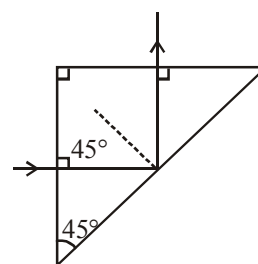
**Statement-2:** Refractive index for diamond is more than refractive index of cheap glass.

- (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.  
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.  
 (C) Statement-1 is true, statement-2 is false.  
 (D) Statement-1 is false, statement-2 is true.

GO0080

27. A light ray is incident perpendicular to one face of a  $90^\circ$  prism and is totally internally reflected at the glass-air interface. If the angle of reflection is  $45^\circ$ , we conclude that the refractive index  $n$  : [AIEEE-2004]

- (A)  $n < \frac{1}{\sqrt{2}}$  (B)  $n > \sqrt{2}$   
 (C)  $n > \frac{1}{\sqrt{2}}$  (D)  $n < \sqrt{2}$



GO0081

### Prism :

28. A ray of light is incident at  $60^\circ$  on a prism of refracting angle  $30^\circ$ . The emerging ray is at an angle  $30^\circ$  with the incident ray. The value of refractive index of the prism is :-

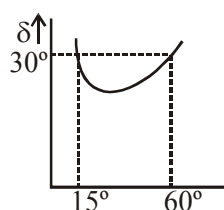
- (A)  $\frac{\sqrt{3}}{4}$  (B)  $\frac{\sqrt{3}}{2}$  (C)  $\sqrt{3}$  (D)  $\frac{2}{\sqrt{3}}$

GO0082

29. The refracting angle of the prism is  $60^\circ$  and minimum deviation of  $30^\circ$ , then the angle of incidence is:-  
 (A)  $30^\circ$  (B)  $45^\circ$  (C)  $25^\circ$  (D)  $60^\circ$

GO0083

30. Figure shows graph of deviation  $\delta$  versus angle of incidence for a light ray striking a prism. Angle of prism is :-



- (A)  $30^\circ$  (B)  $45^\circ$  (C)  $60^\circ$  (D)  $75^\circ$

GO0084



31. There is a prism with refractive index equal to  $\sqrt{2}$  and the refracting angle equal to  $30^\circ$ . One of the refracting surface of the prism is polished. A beam of monochromatic light will retrace its path if its angle of incidence over the first refracting surface of the prism is :-

(A)  $0^\circ$  (B)  $30^\circ$  (C)  $45^\circ$  (D)  $60^\circ$

GO0085

32. The refractive index for the material of a  $60^\circ$  prism is 1.50. Then the angle of incidence for minimum deviation is nearly. ( $\sin 42^\circ \approx \frac{2}{3}$  and  $\sin 49^\circ \approx \frac{3}{4}$ )

(A)  $30^\circ$  (B)  $49^\circ$  (C)  $38^\circ$  (D)  $28^\circ$

GO0086

33. A ray of light is incident on an equilateral glass prism placed on a horizontal table. For minimum deviation which of the following is true ?

(A) PQ is horizontal  
(B) QR is horizontal  
(C) RS is horizontal  
(D) Either PQ or RS is horizontal



GO0087

34. An isosceles glass prism having refractive index  $\mu$  has one of its faces coated with silver. A ray of light is incident normally on the other face (which is equal to the silvered face). The ray of light is reflected twice on the same sized faces and then emerges through the base of the prism perpendicularly. The angles of prism are :-

(A)  $40^\circ, 70^\circ, 70^\circ$  (B)  $50^\circ, 65^\circ, 65^\circ$   
(C)  $36^\circ, 72^\circ, 72^\circ$  (D) data insufficient

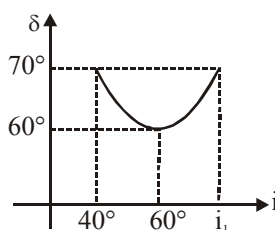
GO0088

35. A given ray of light suffers minimum deviation in an equilateral prism P. If refractive index increases slightly then the ray will now suffer :-

(A) greater deviation (B) no deviation  
(C) same deviation as before (D) Lesser deviation

GO0089

36. The curve of angle of incidence versus angle of deviation shown has been plotted for prism. The value of refractive index of the prism used is :



(A)  $\sqrt{3}$  (B)  $\sqrt{2}$  (C)  $\frac{\sqrt{3}}{\sqrt{2}}$  (D)  $\frac{2}{\sqrt{3}}$

GO0090

37. A beam of monochromatic light is incident at  $i = 50^\circ$  on one face of an equilateral prism, the angle of emergence is  $40^\circ$ , then the angle of minimum deviation is :  
 (A)  $30^\circ$  (B)  $< 30^\circ$  (C)  $\leq 30^\circ$  (D)  $\geq 30^\circ$

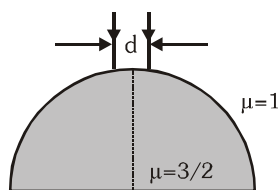
GO0091

38. The angle of a prism is  $6^\circ$  and its refractive index for green light is 1.5. If a green ray passes through it, the deviation will be :-  
 (A)  $30^\circ$  (B)  $15^\circ$  (C)  $9^\circ$  (D)  $3^\circ$

GO0092

### Refraction at curved surface :

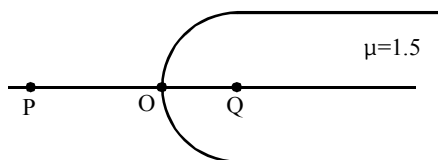
39. A beam of diameter 'd' is incident on a glass hemisphere as shown. If the radius of curvature of the hemisphere is very large in comparison to d, then the diameter of the beam at the base of the hemisphere will be :-



- (A)  $\frac{3}{4}d$  (B) d (C)  $\frac{d}{3}$  (D)  $\frac{2}{3}d$

GO0093

40. One end of a glass rod of refractive index  $n = 1.5$  is a spherical surface of radius of curvature R. The centre of the spherical surface lies inside the glass. A point object placed in air on the axis of the rod at the point P has its real image inside glass at the point Q (see fig.). A line joining the points P and Q cuts the surface at O such that  $OP = 2OQ$ . The distance PO is :-



- (A) 8 R (B) 7 R (C) 2 R (D) None of these

GO0094

41. An air bubble is inside water. The refractive index of water is  $4/3$ . At what distance from the air bubble should a point object be placed so as to form a real image at the same distance from the bubble:  
 (A) 2R (B) 3R  
 (C) 4R (D) The air bubble cannot form a real image

GO0095

42. A point object is placed at the centre of a glass sphere of radius 6 cm and refractive index 1.5. The distance of the virtual image from the surface of the sphere is :-  
 (A) 2 cm (B) 4 cm (C) 6 cm (D) 12 cm

[IIT-JEE 2004 (Scr)]

GO0096

**Lens :**

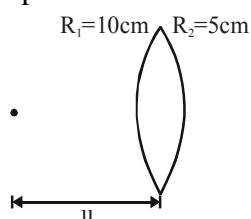
43. A concave lens of glass, refractive index 1.5, has both surfaces of same radius of curvature  $R$ . On immersion in a medium of refractive index 1.75, it will behave as a :-  
 (A) convergent lens of focal length  $3.5R$   
 (B) convergent lens of focal length  $3.0R$   
 (C) divergent lens of focal length  $3.5R$   
 (D) divergent lens of focal length  $3.0R$

GO0097

44. A convergent (biconvex) lens is placed inside a jar filled with a liquid. The lens has focal length  $20\text{cm}$ , when in air and its material has a refractive index of 1.5. If the liquid has a refractive index of 1.6, the focal length of the lens while in the jar, is :-  
 (A)  $-110\text{ cm}$  (B)  $-130\text{ cm}$  (C)  $-160\text{ cm}$  (D)  $-180\text{ cm}$

GO0098

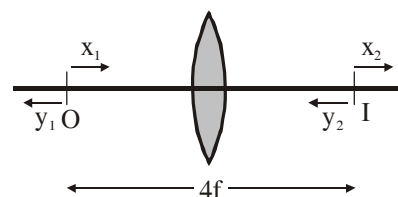
45. A thin convex lens is made of a material of refractive index 1.6. An object is kept at a distance of  $u$  from the lens on the principal axis as shown in the figure. The radius of curvature of the surfaces are  $10\text{ cm}$  and  $5\text{ cm}$ . Now, the lens is reversed such that the face having radius of curvature  $5\text{ cm}$  lies close to the object. The difference in image position as obtained for both the cases is equal to :-



- (A)  $0.4u$  (B)  $0.6u$  (C)  $0.8u$  (D)  $0$

GO0099

46. In a converging lens of focal length  $f$  and the distance between real object and its real image is  $4f$ . If the object moves  $x_1$  distance towards lens its image moves  $x_2$  distance away from the lens and when object moves  $y_1$  distance away from the lens its image moves  $y_2$  distance towards the lens, then choose the correct option:-



- (A)  $x_1 > x_2$  and  $y_1 > y_2$  (B)  $x_1 < x_2$  and  $y_1 < y_2$   
 (C)  $x_1 < x_2$  and  $y_1 > y_2$  (D)  $x_1 > x_2$  and  $y_2 > y_1$

GO0100

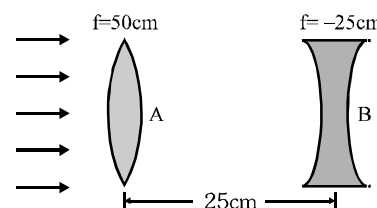
47. A point object is placed on the principal axis of a converging lens and its image ( $I_1$ ) is formed on its principal axis. If the lens is rotated by a small angle  $\theta$  about its optical centre such that its principal axis also rotates by the same amount then the image ( $I_2$ ) of the same object is formed at point P. Choose the correct option.

- (A) Point P lies on the new principal axis.  
 (B) Point P lies on the old principal axis.  
 (C) Point P is anywhere between the two principal axes  
 (D) None of these

GO0101

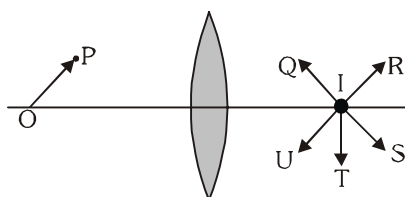
48. The two lenses shown are illuminated by a beam of parallel light from the left. Lens B is then moved slowly toward lens A. The beam emerging from lens B is :

- (A) Initially parallel and then diverging  
(B) Always diverging  
(C) Initially converging and finally parallel  
(D) Always parallel



GO0102

49. A point object O moves from the principal axis of a converging lens in a direction OP. The image of O, will move initially in the direction :



- (A) IQ (B) IR (C) IS (D) IU

GO0103

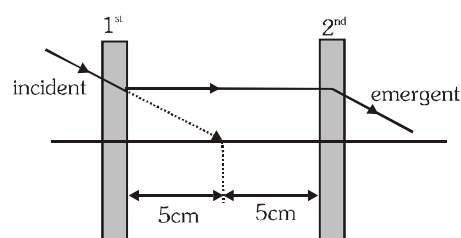
50. When the object is at distances  $u_1$  and  $u_2$  the images formed by the same lens are real and virtual respectively and of the same size. Then focal length of the lens is :

- (A)  $\frac{1}{2}\sqrt{u_1 u_2}$  (B)  $\frac{u_1 + u_2}{2}$  (C)  $\sqrt{u_1 u_2}$  (D)  $\sqrt{(u_1 + u_2)}$

GO0104

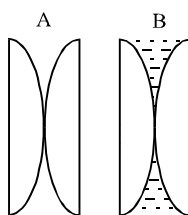
51. Look at the ray diagram shown, what will be the focal length of the 1<sup>st</sup> and the 2<sup>nd</sup> lens, if the incident light ray passes without any deviation ?

- (A) -5 cm and +10 cm  
(B) +5 cm and +10 cm  
(C) -5 cm and +5 cm  
(D) +5 cm and -5 cm



GO0105

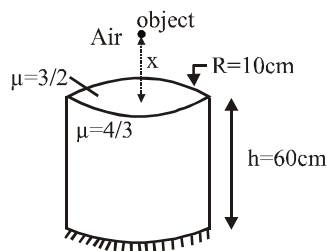
52. Figure A shows two identical plano-convex lenses in contact as shown. The combination has focal length 24 cm. Figure B shows the same with a liquid introduced between them. If refractive index of glass of the lenses is 1.50 and that of the liquid is 1.60, the focal length of the system in figure B will be :-



- (A) -120 cm (B) 120 cm (C) -24 cm (D) 24 cm

GO0106

53. In the arrangement shown given that a biconvex lens of radius of curvature equal to 10 cm and concave mirror has focal length equal to 20cm. Then find the distance 'x' such that the final image formed by the system coincides with the object.



- (A) 15 cm (B) 22.5 cm (C) 20 cm (D) 17.5 cm

GO0107

54. A convex lens of focal length 30 cm forms an image of height 2 cm for an object situated at infinity. If a concave lens of focal length 20 cm is placed coaxially at a distance of 26 cm in front of convex lens then size image would be :-

[IIT-JEE 2003 (Scr)]

- (A) 2.5 cm (B) 5.0 (C) 1.25 (D) None

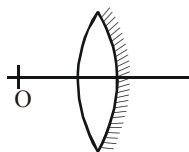
GO0108

55. A plano-convex lens of refractive index 1.5 and radius of curvature 30 cm is silvered at the curved surface. Now, this lens has been used to form the image of an object. At what distance from this lens, an object be placed in order to have a real image of the size of the object? [AIEEE-2004]

- (A) 20 cm (B) 30 cm (C) 60 cm (D) 80 cm

GO0109

56. An equiconvex lens of refractive index  $\mu$  and radius of curvature  $R$  has its one surface silvered. A point source  $O$  is placed before the silvered lens so that its image is coincident with it, the distance of the object from the lens is :-

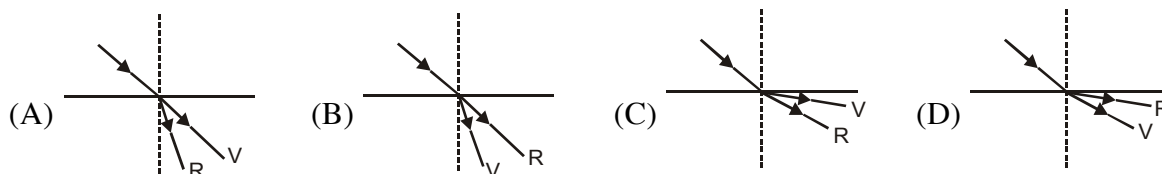


- (A)  $\frac{R}{(\mu-1)}$  (B)  $\frac{2R}{(\mu-1)}$  (C)  $\frac{R}{(2\mu-1)}$  (D)  $\frac{2R}{(2\mu-1)}$

GO0110

### Dispersion :

57. A ray of light is coming from air to water. Which of the following figure shows correct dispersion of light ?



GO0111

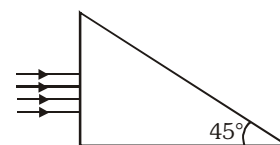
58. When white light passes through a glass prism, one gets spectrum on the other side of the prism. In the emergent beam, the ray which is deviating least is or deviation by a prism is lowest for  
(A) Violet ray (B) Green ray (C) Red ray (D) Yellow ray

GO0112

59. The angle of prism is  $5^\circ$  and its refractive indices for red and violet colours are 1.5 and 1.6 respectively. The angular dispersion produced by the prism is :-  
(A)  $7.75^\circ$  (B)  $5^\circ$  (C)  $0.5^\circ$  (D)  $0.17^\circ$

GO0113

60. A beam of light consisting of red, green and blue and is incident on a right angled prism. The refractive index of the material of the prism for the above red, green and blue wavelengths are 1.39, 1.44 and 1.47 respectively. The prism will  
(A) separate part of the red color from the green and blue colors  
(B) separate part of the blue color from the red and green colors  
(C) separate all the three colors from the other two colors  
(D) not separate even partially, any colors from the other two colors



GO0114

61. Two thin prisms of flint glass, with refracting angles of  $6^\circ$  and  $8^\circ$  respectively, possess dispersive powers in the ratio :  
(A) 4 : 3 (B) 3 : 4 (C) 1 : 1 (D) 9 : 16

GO0115

62. Chromatic aberration will be absent if for two thin lenses in contact :-  
(A)  $(\omega_1/F_1) + (\omega_2/F_2) = 0$  (B)  $(\omega_1/F_2) + (\omega_2/F_1) = 0$   
(C)  $(F_1/\omega_2) + (F_2/\omega_1) = 0$  (D)  $(\omega_1/\omega_2) + (F_1 + F_2) = 0$

GO0116

63. The dispersive powers of the materials of the two lenses are in the ratio 4 : 3. If the achromatic combination of these two lenses in contact is a convex lens of focal length 60 cm then the focal lengths of the component lenses are  
(A) - 20 cm and 25 cm (B) 20 cm and - 25 cm  
(C) - 15 cm and 40 cm (D) 15 cm and - 20 cm

GO0117

64. An achromatic convergent doublet of two lenses in contact has a power of + 2D. The convex lens has a power + 5D. What is the ratio of the dispersive powers of the convergent and divergent lenses  
(A) 2 : 5 (B) 3 : 5 (C) 5 : 2 (D) 5 : 3

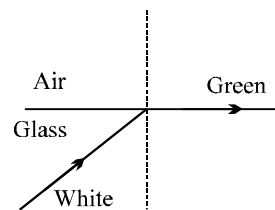
GO0118

65. It is desired to make an achromatic combination of two lenses ( $L_1$  &  $L_2$ ) made of materials having dispersive powers  $\omega_1$  and  $\omega_2$  ( $<\omega_1$ ). If the combination of lenses is converging then  
(A)  $L_1$  is converging  
(B)  $L_2$  is converging  
(C) Power of  $L_1$  is greater than the power of  $L_2$   
(D) None of these

GO0119

66. White light is incident on the interface of glass and air as shown in the figure. If green light is just totally internally reflected then the emerging ray in air contains [IIT-JEE 2004 (Scr)]

- (A) yellow, orange, red  
(B) violet, indigo, blue  
(C) all colours  
(D) all colours except green



GO0120

### MULTIPLE CORRECT TYPE QUESTIONS

#### Spherical mirror :

67. The nature of object and image given with each of the optical condition is shown. Choose the **correct** option(s) :-

- (A) Virtual image I Virtual object
- (B) Virtual image I Real object
- (C) Real object O Virtual image
- (D) Real object O Virtual image

GO0121

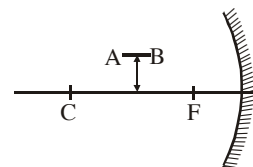
68. In which of the following diagrams the image formed is virtual and inverted ?

- (A)
- (B)
- (C)
- (D)

GO0122

69. An object AB is placed parallel and close to the optical axis between focus F and centre of curvature C of a converging mirror of focal length  $f$  as shown in figure.

- (A) Image of A will be closer than that of B from the mirror.  
 (B) Image of AB will be parallel to the optical axis.  
 (C) Length of image is equal to AB.  
 (D) Length of image is more than AB.



GO0123

70. Mark the **CORRECT** statement :-

- (A) Virtual image of virtual object can be formed by a single optical element  
 (B) Virtual image of virtual object formed by a single optical element can be inverted  
 (C) Virtual image of real object can be formed by a single optical element  
 (D) Virtual image of real object formed by a single optical element can be inverted

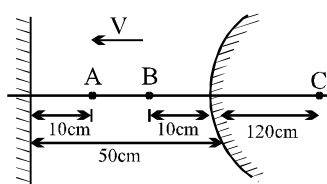
GO0124

71. A concave mirror forms an image of the sun at a distance of 12 cm from it.

- (A) the radius of curvature of this mirror is 6 cm.  
 (B) to use it as a shaving mirror, it can be held at a distance of 8-10 cm from the face  
 (C) if an object is kept at a distance of 24 cm from it, the image formed will be of the same size as the object  
 (D) all the above alternatives are correct.

GO0125

72. In the figure shown consider the first reflection at the plane mirror and second at the convex mirror. AB is object.



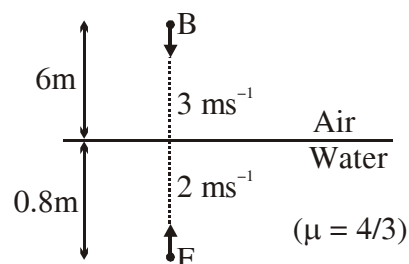
- (A) the second image is real, inverted of  $1/5^{\text{th}}$  magnification w.r.t AB  
 (B) the second image is virtual and erect with magnification  $1/5$  w.r.t AB  
 (C) the second image moves towards the convex mirror  
 (D) the second image moves away from the convex mirror.

GO0126



**Refraction at plane surface :**

73. A fish, F in the pond, is at a depth of 0.8 m from water surface and is moving vertically upwards with velocity  $2 \text{ ms}^{-1}$ . At the same instant, a bird B is at a height of 6 m from water surface and is moving downwards with velocity  $3 \text{ ms}^{-1}$ . At this instant both are on the same vertical lines as shown in the figure. Which of the following statement(s) is(are) correct?



- (A) Height of B, observed by F (from itself) is equal to 8.00 m.  
 (B) Depth of F, observed by B (from itself) is equal to 6.60 m.  
 (C) Velocity of B, observed by F (relative to itself) is equal to  $5.00 \text{ ms}^{-1}$ .  
 (D) Velocity of F, observed by B (relative to itself) is equal to  $4.50 \text{ ms}^{-1}$ .

GO0127

**Total internal reflection :**

74. In the figure (a) light is incident at an angle  $\theta$  which is slightly greater than the critical angle. Now, keeping the incident fixed a slab of refractive index  $n_3$  is placed on surface AB in fig. (b). Which of the following statements are correct:

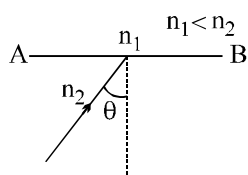


Fig. (a)

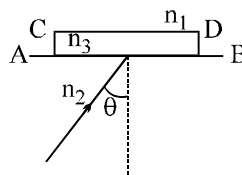


Fig. (b)

- (A) total internal reflection occurs at AB for  $n_3 < n_1$   
 (B) total internal reflection occurs at AB for  $n_3 > n_1$   
 (C) the ray will return back to the same medium for all values of  $n_3$   
 (D) total internal reflection occurs at CD for  $n_3 < n_1$

GO0128

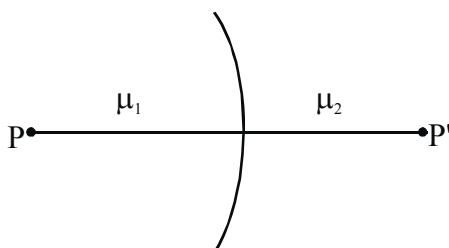
75. For the refraction of light through a prism

- (A) For every angle of deviation there are two angles of incidence.  
 (B) The light travelling inside an equilateral prism is necessarily parallel to the base when prism is set for minimum deviation.  
 (C) There are two angles of incidence for maximum deviation.  
 (D) Angle of minimum deviation will increase if refractive index of prism is increased keeping the outside medium unchanged if  $\mu_p > \mu_s$ .

GO0129

**Refraction at curved surface :**

76. Two refracting media are separated by a spherical interface as shown in the figure.  $PP'$  is the principal axis,  $\mu_1$  and  $\mu_2$  are the refractive indices of medium of incidence and medium of refraction respectively, then



- (A) if  $\mu_2 > \mu_1$ , then there cannot be a real image of real object.  
 (B) if  $\mu_2 > \mu_1$ , then there cannot be a real image of virtual object.  
 (C) if  $\mu_1 > \mu_2$ , then there cannot be a virtual image of virtual object.  
 (D) if  $\mu_1 > \mu_2$ , then there cannot be a real image of real object.

GO0130

**Lens :**

77. Optical axis of a thin equi-convex lens is the X-axis. The co-ordinate of a point object and its image are  $(-20 \text{ cm}, 1 \text{ cm})$  and  $(25 \text{ cm}, -2 \text{ cm})$  respectively  
 (A) the lens is located at  $x = 5 \text{ cm}$   
 (B) the lens is located at  $x = -5 \text{ cm}$   
 (C) the focal length of the lens is  $10 \text{ cm}$   
 (D) the focal length of the lens is  $15 \text{ cm}$

GO0131

78. The principal axis of an optical device is along  $y = -1$ , image of a small body placed at  $(-30, 3)$  is formed at a point  $(60, -3)$ . Then the optical device is  
 (A) A convex lens of focal length  $20 \text{ cm}$   
 (B) A concave mirror of focal length  $60 \text{ cm}$   
 (C) A concave lens of focal length  $20 \text{ cm}$   
 (D) A convex mirror of focal length  $60 \text{ cm}$

GO0132

79. In displacement method, the distance between object and screen is  $96 \text{ cm}$ . The ratio of length of two images formed by a convex lens placed between them is  $4.84$  :-  
 (A) Ratio of the length of object to the length of shorter image is  $11/5$ .  
 (B) Distance between the two positions of the lens is  $36 \text{ cm}$ .  
 (C) Focal length of the lens is  $20.625 \text{ cm}$ .  
 (D) Distance of the lens from the shorter image is  $30 \text{ cm}$ .

GO0133

80. A lens is formed by a material having refracting index  $\frac{3}{2}$  & radii of curvature 20 cm & 10 cm. Then

choose the correct option(s) :-

- (A) If lens is concavo-convex & light is falling on the surface having radius of curvature 20 cm, then focal length is 40 cm  
 (B) If lens is concavo-convex & light is falling on the surface having radius of curvature 10 cm, then focal length is -40 cm  
 (C) If lens is convexo-concave & light is falling on the surface having radius of curvature 20 cm, then focal length is 40 cm  
 (D) If lens is convexo-concave & light is falling on the surface having radius of curvature 10 cm, then focal length is -40 cm

GO0134

81. Which of the following silvered lenses kept in air may form real image of a real object.



GO0135

82. A man wanted to get a picture of a Zebra. He photographed a white donkey after fitting a glass with black streaks onto the objective of his camera.

- (A) the image will look like a white donkey on the photograph.  
 (B) the image will look like a Zebra on the photograph.  
 (C) the image will be more intense compared to the case in which no such glass is used.  
 (D) the image will be less intense compared to the case in which no such glass is used.

GO0136

### COMPREHENSION TYPE QUESTIONS

#### Paragraph for Question No. 83 to 85

A real object is placed perpendicular to the principal axis of a concave mirror at a position (1) such that the image formed is real with a magnification 2. The object is now shifted to another position (2) at a distance 15 cm from the position (1) and a real image is obtained with a magnification 8. Shifting the position of object from (2) to a third position (3) gives an image with magnification  $1/2$ .

83. Position of the image when the object is at position (3) is :-  
 (A) between centre of curvature and  $\infty$  (B) between focus and centre of curvature  
 (C) at focus (D) between pole and focus

GO0137

84. Focal length of the mirror is :-

- (A) - 60 cm (B) -40 cm (C) -30 cm (D) -25 cm

GO0137

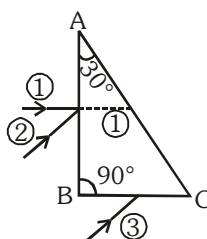
85. Distance of position (1) of the object from pole and that of position (2) of object from pole are, respectively :-  
 (A) 60 cm, 75 cm      (B) 75 cm, 60 cm      (C) 60 cm, 45 cm      (D) 45 cm, 30 cm

GO0137

**Paragraph for Question No. 86 to 88**

ABC is a right-angled prism kept in air. A ray (1) is incident on the face AB along the normal. Refractive index of the material of prism is the minimum value that will be required so that ray (1) undergoes total internal reflection at the face AC.

Another ray (2) is incident on the face AB such that it emerges from face AC along the normal to AC. A third ray (3) falls on the face BC and emerges from face AC such that its angle of emergence is the same as that of incidence. Assuming light (1), (2) and (3) to have the same frequency, answer the following questions.



86. Refractive index of the material of prism is :-  
 (A)  $\sqrt{2}$       (B) 2      (C) 1.5      (D) 2.2
87. Angle of incidence of ray (2) is :-  
 (A)  $0^\circ$       (B)  $45^\circ$       (C)  $60^\circ$       (D)  $90^\circ$
88. Deviation suffered by ray (3) is :-  
 (A)  $60^\circ$       (B)  $30^\circ$       (C)  $90^\circ$       (D)  $120^\circ$

GO0138

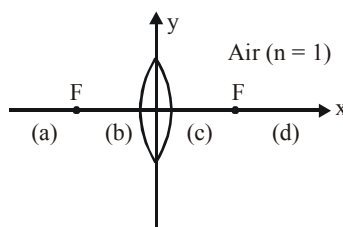
GO0138

GO0138

**Paragraph for Question no. 89 and 90**

This question concerns a symmetrical lens shown, along with its two focal points. It is made of plastic with ( $n = 1.2$ ), and has a focal length  $f$ . Four different regions are shown :

Here (a)  $-\infty < x < -f$ ; (b)  $-f < x < 0$ ; (c)  $0 < x < f$ ; (d)  $f < x < \infty$



89. If an object is placed somewhere in region (a), in which region does the image appear ?  
 (A) (a)      (B) (b)      (C) (c)      (D) (d)

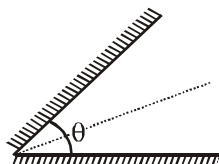
GO0139

90. If incident rays are converging then in which region does the image appear ?  
 (A) (a) (B) (b) (C) (c) (D) (d)

GO0139

**MATRIX MATCH TYPE QUESTIONS**

91. Angle between two mirrors ' $\theta$ ' and location of object is given in column I and some possible number of images are given in column II.



**Column-I**

- (A)  $\theta = 30^\circ$  and object is on bisector  
 (B)  $\theta = 36^\circ$  and object is on bisector  
 (C)  $\theta = 22.5^\circ$  and object is on bisector  
 (D)  $\theta = 50^\circ$  and object is on bisector

**Column-II**

- (P) 9  
 (Q) 11  
 (R) 8  
 (S) 15

GO0140

92. An object O is kept perpendicular to the principal axis of a spherical mirror. In column I, coordinate of, object, u with sign in cm, type of mirror and its focal length in cm is given

**Column I**

- (A)  $u = -18$ , concave mirror, 12  
 (B)  $u = -12$ , concave mirror, 18  
 (C)  $u = -8$ , convex mirror, 10  
 (D)  $u = -10$ , convex mirror, 8

**Column II**

- (P) Image is real  
 (Q) Image is virtual  
 (R) Image diminished  
 (S) Image is enlarged  
 (T) Image is inverted

GO0141

93. **Column-I**

- (A) An object is placed at a distance equal to focal length from pole before convex mirror  
 (B) An object is placed at focus before a concave mirror  
 (C) An object is placed at the centre of curvature before a concave mirror.  
 (D) An object is placed at a distance equal to radius of curvature before a convex mirror.

**Column-II**

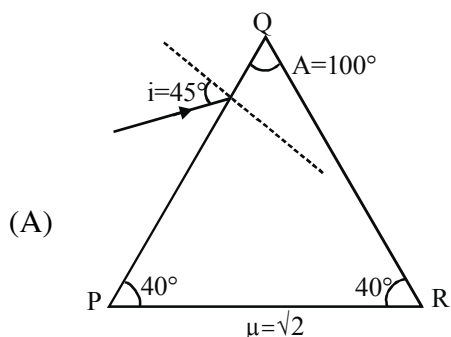
- (P) Magnification is ( $\infty$ )  
 (Q) Magnification is (0.5)  
 (R) Magnification is ( $1/3$ )  
 (S) Magnification is ( $-1$ )

GO0142

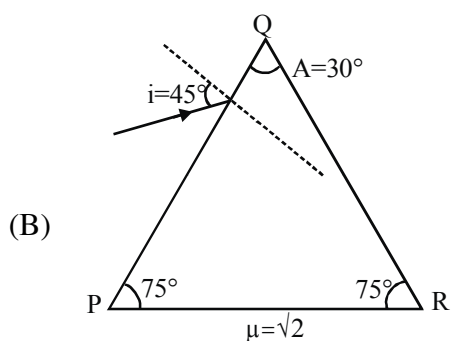
94. Light is incident at surface PQ of prism as shown in column-I then match the column-I with column-II (surrounding medium is air in all cases)

## Column-I

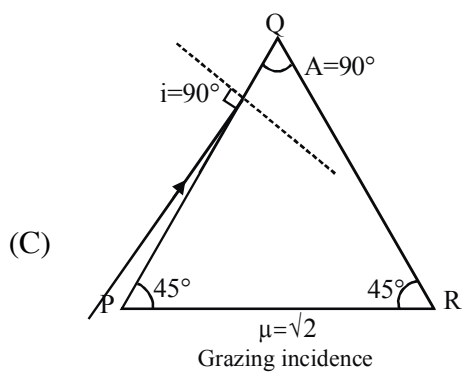
## Column-II



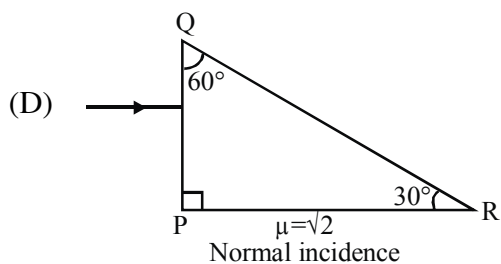
(P) Total internal reflection takes place at surface QR.



(Q) Light emerges normally from the surface QR



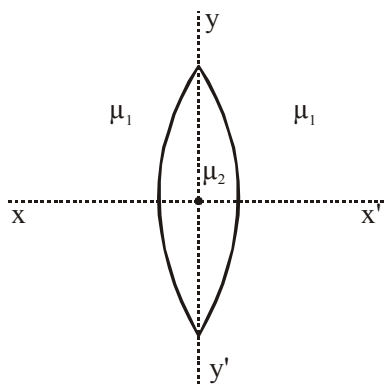
(R) Light emerges parallel to surface QR



(S) The light ray emerges from face PR perpendicularly

(T) When light ray passes through the prism it is parallel to the base PR.

95. An equi-convex lens of refractive index  $\mu_2$  and focal length  $f$  (in air) is kept in medium of refractive index  $\mu_1$ .



**Column A**

- (P) If lens is cut in two equal parts by a plane  $yy'$ . ( $\mu_1 = 1$ )  
 (Q) If lens is cut in two equal parts by a plane  $xx'$ . ( $\mu_1 = 1$ )  
 (R) If  $\mu_1 = \mu_2$ .  
 (S) If  $\mu_1 > \mu_2$ .

**Column B**

- (1) Lens will act as a glass slab  
 (2) Lens will be converging and focal length will change.  
 (3) Lens will be converging and focal length will remain same  
 (4) Lens will be diverging and focal length will change.

**Codes :**

|     | P | Q | R | S |
|-----|---|---|---|---|
| (A) | 2 | 4 | 1 | 3 |
| (B) | 2 | 3 | 1 | 4 |
| (C) | 3 | 4 | 2 | 1 |
| (D) | 1 | 2 | 3 | 4 |

GO0144

96. Column-II shows the optical phenomenon that can be associated with optical components given in column-I. Note that column-I may have more than one matching options in column-II.

**Column-I**

- (A) Convex mirror  
 (B) Converging lens  
 (C) Thin prism  
 (D) Glass slab

**Column-II**

- (P) Dispersion  
 (Q) Deviation  
 (R) Real image of real object  
 (S) Virtual images of real object

GO0145

## SUPPLEMENT FOR MAINS

**Optical Instruments :**

97. Myopia can be removed by using a lenses of  
(A) concave lens (B) convex lens  
(C) cylindrical lens (D) by surgical removal  
GO0146
98. Loss of the ability of eye to focus on near and far objects with advancing age is called  
(A) Presbyopia (B) Astigmatism (C) Hypermetropia (D) Myopia  
GO0147
99. A long sighted person has a minimum distance of distinct vision of 50 cm. He wants to reduce it to 25cm. He should use a  
(A) Concave lens of focal length 50 cm (B) Convex lens of focal length 25 cm  
(C) Convex lens of focal length 50 cm (D) Concave lens of focal length 25 cm  
GO0148
100. A person's near point is 50 cm and his far point is 3m. Powers of the lenses he required for (i) for reading (ii) for viewing distinct stars  
(A) - 2D and 0.33 D (B) 2D and - 0.33 D (C) - 2D and 3D (D) 2D and - 3 D  
GO0149
101. In a simple microscope, if the final image is located at infinity then its magnifying power  
(A)  $25/F$  (B)  $25/D$  (C)  $F/25$  (D)  $(1+25/F)$   
GO0150
102. In a compound microscope, the intermediate image, in normal use is  
(A) Virtual, erect and magnified (B) Real, erect and magnified  
(C) Real, inverted and magnified (D) Virtual, inverted and magnified  
GO0151
103. If tube length of astronomical telescope is 105 cm and magnifying power is 20 for normal setting. Calculate the focal length of objective :—  
(A) 100 cm (B) 10 cm (C) 20 cm (D) 25 cm  
GO0152
104. When length of a compound microscope tube increase, its magnifying power  
(A) decreases (B) increases  
(C) does not change (D) may increase or decrease  
GO0153



105. An astronomical telescope of angular magnification 10, when final image is at infinite has a length of 44 cm. The focal length of the objective is :-  
 (A) 4 cm (B) 40 cm (C) 44 cm (D) 440 cm
106. When the length of an astronomical telescope tube increases its magnifying power  
 (A) Decreases (B) Increases  
 (C) Does not change (D) May increase or decrease
107. The objective of a small telescope has focal length of 120 cm and diameter 5 cm. The focal length of the eye-piece is 2 cm. The magnifying power of telescope for distant object is  
 (A) 12 (B) 24 (C) 60 (D) 300
108. An astronomical telescope has an angular magnification of magnitude 5 for distance objects. The separation between the objective and the eye-piece is 36 cm and the final image is formed at infinity. The focal length  $f_o$  of the objective and  $f_e$  of the eye-piece are  
 (A)  $f_o = 45$  cm and  $f_e = -9$  cm (B)  $f_o = 50$  cm and  $f_e = 10$  cm  
 (C)  $f_o = 7.2$  cm and  $f_e = 5$  cm (D)  $f_o = 30$  cm and  $f_e = 6$  cm
109. The magnifying power of telescope is found to be 9 and separation between lenses is 20cm. what are focal lengths of component lenses  
 (A) 18cm, 2cm (B) 4cm, 16cm (C) 10cm, 10cm (D) 12cm, 8cm

GO0154

GO0155

GO0156

GO0157

GO0158

Paragraph for Question 110 to 113

110. If an astronomical telescope has objective and eye-pieces of focal lengths 200 cm and 4 cm respectively, then the magnifying power of the telescope for the normal vision is  
 (A) 42 (B) 50 (C) 58 (D) 204
111. In the above question the length of the telescope for normal vision, is  
 (A) 204 cm (B) 200 cm (C) 196 cm (D) 203.45 cm
112. In the above question magnifying power of the telescope for distinct vision is  
 (A) 42 (B) 50 (C) 58 (D) 204
113. In the above question the length of the telescope for distinct vision is  
 (A) 204 cm (B) 200 cm (C) 196 cm (D) 203.45 cm

GO0159

GO0159

GO0159

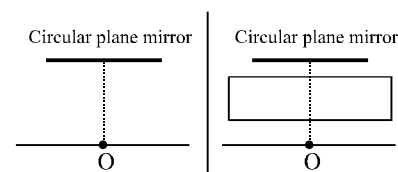
GO0159

## EXERCISE O-2

## SINGLE CORRECT TYPE QUESTIONS

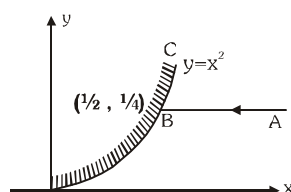
1. In the diagram shown below, a point source O is placed vertically below the center of a circular plane mirror. The light rays starting from the source are reflected from the mirror such that a circular area A on the ground receives light. Now, a glass slab is placed between the mirror and the source O. What will the magnitude of the new area on the ground receiving light?

- (A) A  
(B) Greater than A  
(C) Less than A  
(D) Cannot tell, as the information given is insufficient



GO0160

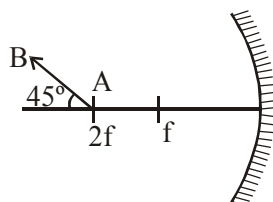
2. In the given figure, incident ray AB is parallel to x-axis. The angle of reflection is :-



- (A)  $30^\circ$  (B)  $60^\circ$  (C)  $45^\circ$  (D) None of these

GO0161

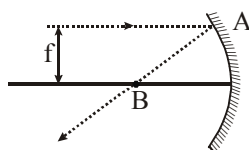
3. The shape of image formed of an object AB due to the concave mirror shown in the figure is best represented by (assume point A is at the centre of curvature of the mirror) :-



- (A) (B) (C) (D)

GO0162

4. A ray of light is incident on a concave mirror. It is parallel to the principal axis and its height from principal axis is equal to the focal length of the mirror. The ratio of the distance of point B to the distance of the focus from the centre of curvature is (AB is the reflected ray) :-



- (A)  $\frac{2}{\sqrt{3}}$  (B)  $\frac{\sqrt{3}}{2}$  (C)  $\frac{2}{3}$  (D)  $\frac{1}{2}$

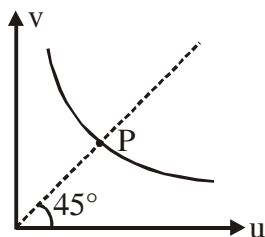
GO0163

5. An infinitely long rod lies along the axis of a concave mirror of focal length  $f$ . The near end of the rod is at distance  $u > f$  from the mirror. Its image will have a length—

(A)  $\frac{uf}{u-f}$       (B)  $\frac{uf}{u+f}$       (C)  $\frac{f^2}{u+f}$       (D)  $\frac{f^2}{u-f}$

GO0164

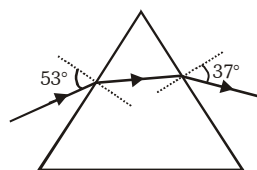
6. The graph shows the variation of  $v$  with change in  $u$  for a mirror. Points plotted above the point P on the curve are for values of  $v$



- (A) smaller than  $f$       (B) smaller than  $2f$       (C) larger than  $2f$       (D) larger than  $f$

GO0165

7. A ray incident at an angle  $53^\circ$  on a prism emerges at an angle  $37^\circ$  as shown. If the angle of incidence is made  $50^\circ$ , which of the following is a possible value of the angle of emergence?

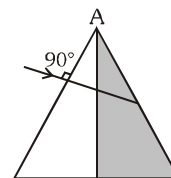


- (A)  $35^\circ$       (B)  $42^\circ$       (C)  $40^\circ$       (D)  $38^\circ$

GO0166

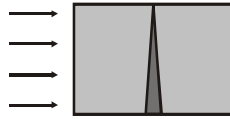
8. A ray of light is incident normally on the first refracting face of the prism of refracting angle  $A$ . The ray of light comes out at grazing emergence. If one half of the prism (shaded position) is knocked off, the same ray will :—

- (A) Emerge at an angle of emergence  $\sin^{-1}\left(\frac{1}{2}\sec A/2\right)$   
 (B) Not emerge out of the prism  
 (C) Emerge at an angle of emergence  $\sin^{-1}\left(\frac{1}{2}\sec A/4\right)$   
 (D) None of these



GO0167

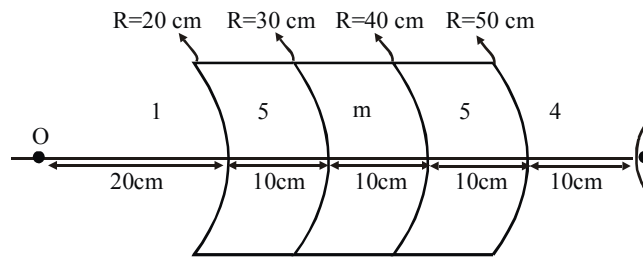
9. A thin isosceles prism with angle  $4^\circ$  and refractive index  $5/4$  is placed inside a transparent tube with liquid (refractive index = 1.5) as shown. The deviation of light due to prism will be :-



- (A)  $0.8^\circ$  upward (B)  $0.8^\circ$  downward (C)  $0.67^\circ$  upward (D)  $0.67^\circ$  downward

GO0168

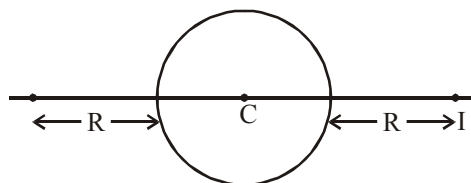
10. In the arrangement shown, separation between observer and object as seen by observer is



- (A) 60 cm (B) more than 60 cm (C) less than 60 cm (D) depends on  $\mu$

GO0169

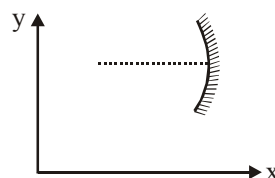
11. In the given situation, a spherical transparent surface of radius  $R$  is placed. The refractive index of sphere is such that image for an object kept at distance  $R$  is formed at the same distance of the opposite side. What is refractive index of sphere?



- (A) 1.5 (B) 1.75 (C) 1.25 (D) 2

GO0170

12. Figure shows a concave mirror with its principal axis parallel to x-axis and focus on the y-axis. The centre of curvature is at  $(-108 \text{ cm}, 54 \text{ cm})$ . The coordinates of the image of a point source A  $(81 \text{ cm}, 48 \text{ cm})$  will be :-



- (A)  $(144 \text{ cm}, 46 \text{ cm})$  (B)  $(144 \text{ cm}, 62 \text{ cm})$   
(C)  $(36 \text{ cm}, 8 \text{ cm})$  (D) None of these

GO0171

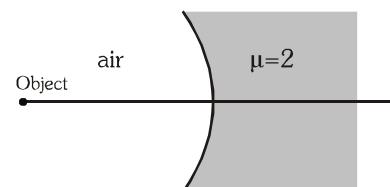
13. A light ray is incident on a transparent sphere of index  $=\sqrt{2}$ , at an angle of incidence  $= 45^\circ$ . What is the deviation of a tiny fraction of the ray, which enters the sphere, undergoes two internal reflections and then refracts out into air ?

(A)  $270^\circ$  (B)  $240^\circ$  (C)  $120^\circ$  (D)  $180^\circ$

GO0172

14. A concave spherical surface with radius  $R$  separates a medium with index of refraction 2 from air. If an object is moved toward the surface from far away along the central axis, its image :

(A) Changes from virtual to real when it is  $R/2$  from the surface  
(B) Changes from virtual to real when it is  $2R$  from the surface  
(C) Changes from real to virtual when it is  $R/2$  from the surface  
(D) Remains virtual



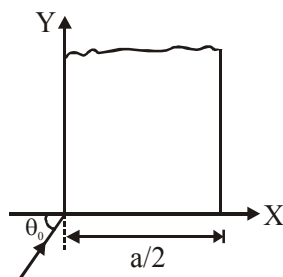
GO0173

### MULTIPLE CORRECT TYPE QUESTIONS

15. A glass slab is of thickness  $d = a/2$  is shown in figure. The refractive index of the slab varies according

to relation  $\mu = \frac{a}{a-x}$  where  $a$  is positive constant. A ray of light enters the slab at origin at an angle

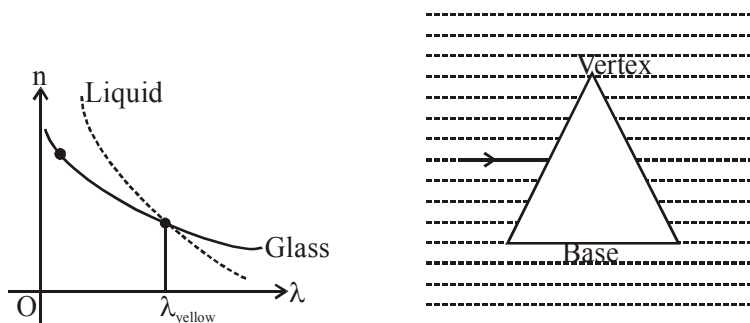
$\theta_0 \rightarrow \pi/2$ , then choose the **CORRECT** statement (s):-



- (A) The path of the light ray inside medium is a circle given by  $(x-a)^2 + y^2 = a^2$   
(B) The path of the light ray inside medium is a parabola given by  $(x-a)^2 = y$   
(C) The deviation of ray from its initial direction is  $\delta = \pi/3$   
(D) Ray emerges grazing at the surface.

GO0174

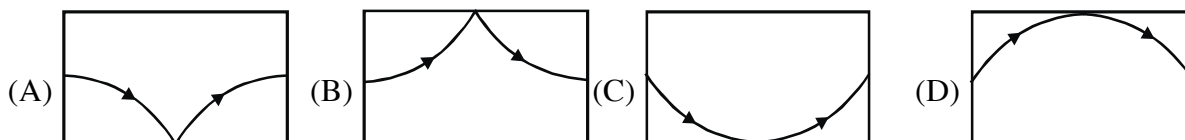
16. A glass prism is immersed in a hypothetical liquid. The curves showing the refractive index  $n$  as a function of wavelength  $\lambda$  for glass and liquid are as shown in the figure. When a ray of white light is incident on the prism parallel to the base :



- (A) yellow ray travels without deviation  
(B) blue ray is deviated towards the vertex  
(C) red ray is deviated towards the base  
(D) there is no dispersion

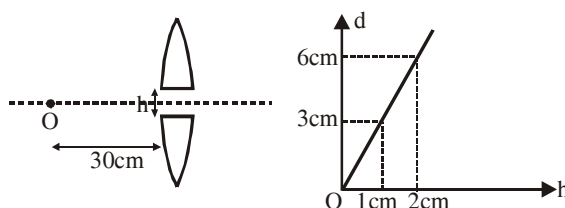
GO0175

17. A cubic container is filled with a liquid whose refractive index increases linearly from top to bottom. Which of the following figures may represent the path of a ray of light inside the liquid?



GO0176

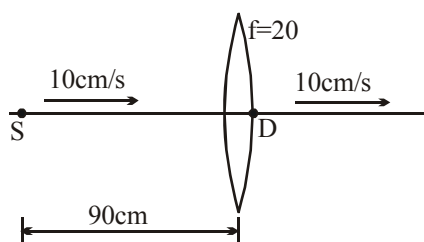
18. Figure shows a convex lens cut symmetrically into two equal halves and separated laterally by a distance  $h$ . A point object  $O$  placed symmetrically at a distance 30 cm, from the lens halves, forms two images separated by a distance  $d$ . A plot of  $d$  versus  $h$  is shown in figure. The focal length of the lens is :-



- (A) 22.5 cm  
(B) 40 cm  
(C) 45 cm  
(D) 20 cm

GO0177

19. The figure shows initial positions of a source of light S and a light detector D. Both S & D move with velocity 10 cm/s, as shown. The converging lens of focal length 20 cm is fixed. Time instant(s) when the detector detects maximum intensity of light is /are :-



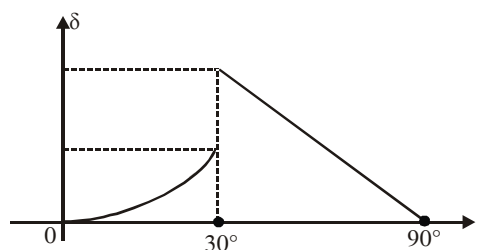
- (A) 3 sec  
(B) 4.5 sec  
(C) 6 sec  
(D) equal intensity at all time

GO0178

20. Figure shows graph of angle of deviation  $\delta$  vs angle of incidence  $i$  for a light ray. Incident ray goes from medium 1 ( $\mu_1$ ) to medium 2 ( $\mu_2$ ). Mark the **correct** option(s).

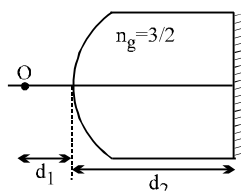
(A)  $\frac{\mu_1}{\mu_2} = \frac{1}{2}$

- (B) Critical angle is  $30^\circ$   
(C)  $\mu_1 > \mu_2$   
(D) Maximum deviation is  $120^\circ$



GO0179

21. In the figure shown a point object O is placed in air on the principal axis. The radius of curvature of the spherical surface is 60 cm.  $I_f$  is the final image formed after all the refractions and reflections.

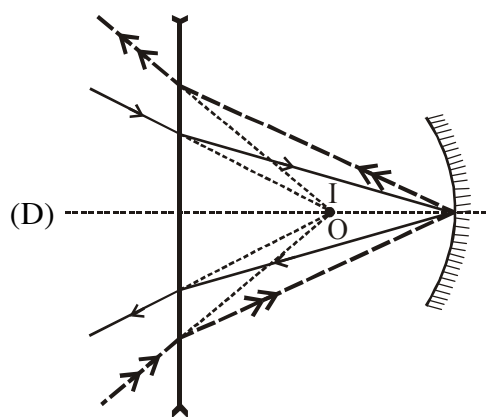
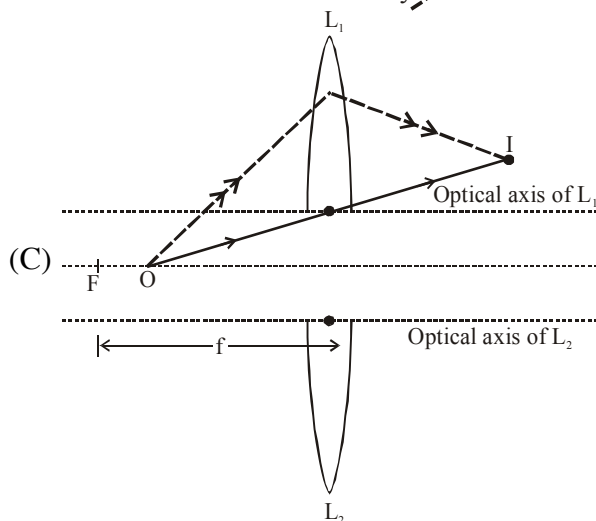
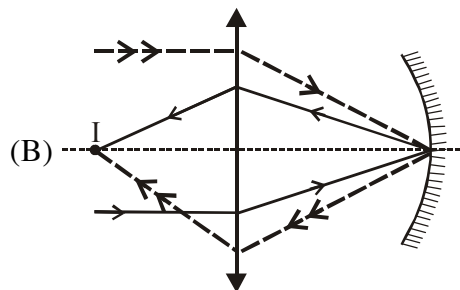
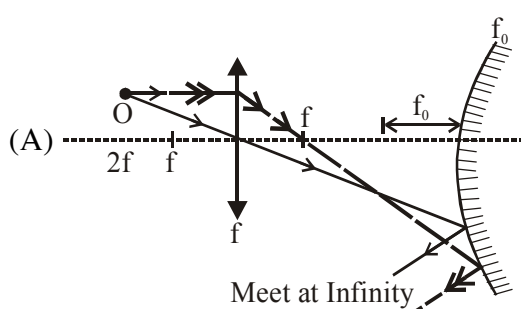


- (A) If  $d_1 = 120$  cm, then the ' $I_f$ ' is formed on 'O' for any value of  $d_2$ .  
(B) If  $d_1 = 240$  cm, then the ' $I_f$ ' is formed on 'O' only if  $d_2 = 360$  cm.  
(C) If  $d_1 = 240$  cm, then the ' $I_f$ ' is formed on 'O' for all values of  $d_2$ .  
(D) If  $d_1 = 240$  cm, then the ' $I_f$ ' cannot be formed on 'O'.

GO0180

22. Choose the incorrect ray diagram(s). All the rays shown are paraxial.

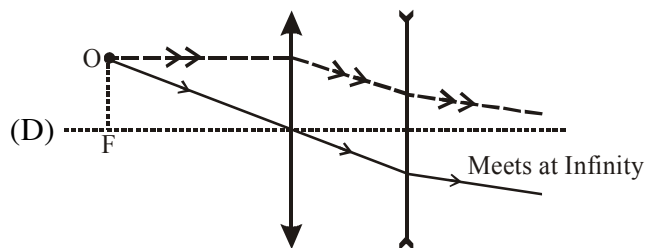
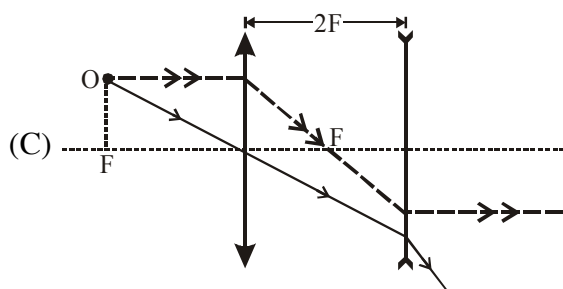
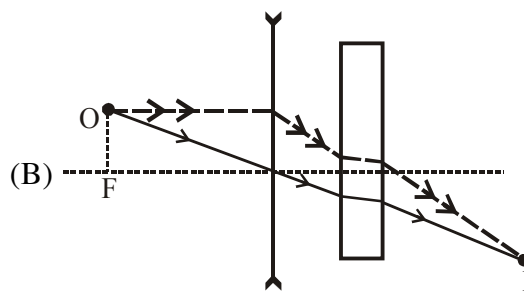
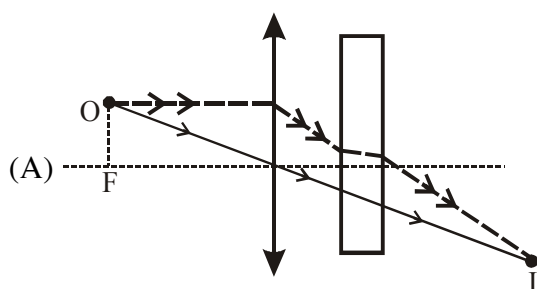
[ $\uparrow$ ] denotes converging lens and [ $\downarrow$ ] denotes diverging lens]



GO0181

23. Choose incorrect ray diagram [ $\uparrow$ ] denotes converging lens and [ $\downarrow$ ] denotes diverging lens]

All symbols have their usual meaning and all the rays shown are paraxial. (focal length of each lens is F)



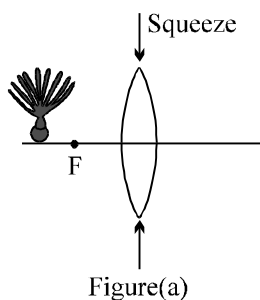
GO0182



COMPREHENSION TYPE QUESTIONS

Paragraph for Question 24 to 26

A turnip sits before a thin converging lens, outside the focal point of the lens. The lens is filled with a transparent gel so that it is flexible; by squeezing its ends toward its center [as indicated in figure(a)], you can change the curvature of its front and rear sides.



24. When you squeeze the lens, the image.

(A) moves towards the lens

(B) moves away from the lens

(C) shifts up

(D) remains as it is

GO0183

25. When you squeeze the lens, the lateral height of image.

(A) increases

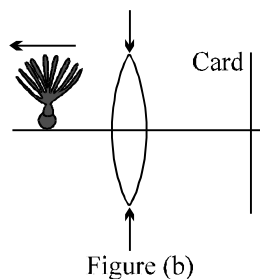
(B) decreases

(C) remains same

(D) data insufficient

GO0183

26. Suppose that a sharp image must be formed on a card which is at a certain distance behind the lens [figure(b)], while you move the turnip away from the lens, then you should



(A) decrease the squeeze of the lens

(B) increase the squeeze of the lens

(C) keep the card and lens as it is.

(D) move the card away from the lens

GO0183

## EXERCISE - (JM)

1. A car is fitted with a convex side-view mirror of focal length 20 cm. A second car 2.8 m behind the first car is overtaking the first car at a relative speed of 15 m/s. The speed of the image of the second car as seen in the mirror of the first one is:- [AIEEE- 2011]

(1) 10 m/s                      (2) 15 m/s                      (3)  $\frac{1}{10}$  m/s                      (4)  $\frac{1}{15}$  m/s

GO0184

2. When monochromatic red light is used instead of blue light in a convex lens, its focal length will :- [AIEEE- 2011]

(1) Does not depend on colour of light                      (2) Increase  
(3) Decrease                      (4) Remain same

GO0185

3. A beaker contains water up to a height  $h_1$  and kerosene of height  $h_2$  above water so that the total height of (water + kerosene) is  $(h_1 + h_2)$ . Refractive index of water is  $\mu_1$  and that of kerosene is  $\mu_2$ . The apparent shift in the position of the bottom of the beaker when viewed from above is :- [AIEEE- 2011]

(1)  $\left(1 - \frac{1}{\mu_1}\right)h_2 + \left(1 - \frac{1}{\mu_2}\right)h_1$                       (2)  $\left(1 + \frac{1}{\mu_1}\right)h_1 - \left(1 + \frac{1}{\mu_2}\right)h_2$   
(3)  $\left(1 - \frac{1}{\mu_1}\right)h_1 + \left(1 - \frac{1}{\mu_2}\right)h_2$                       (4)  $\left(1 + \frac{1}{\mu_1}\right)h_2 - \left(1 + \frac{1}{\mu_2}\right)h_1$

GO0186

4. An object 2.4 m in front of a lens forms a sharp image on a film 12 cm behind the lens. A glass plate 1 cm thick, of refractive index 1.50 is interposed between lens and film with its plane faces parallel to film. At what distance (from lens) should object be shifted to be in sharp focus on film ? [AIEEE- 2012]

(1) 5.6 m                      (2) 7.2 m                      (3) 2.4 m                      (4) 3.2 m

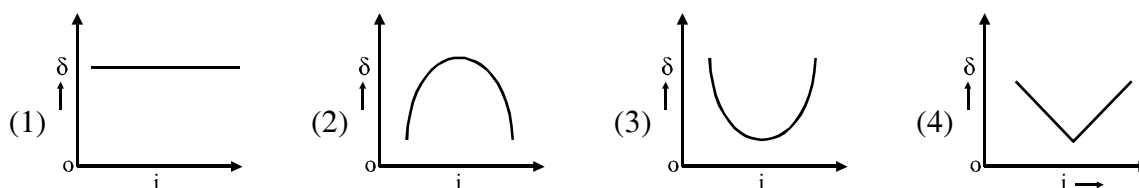
GO0187

5. Diameter of a plano-convex lens is 6cm and thickness at the centre is 3 mm. If speed of light in material of lens is  $2 \times 10^8$  m/s, the focal length of the lens is : [JEE-Main- 2013]

(1) 15 cm                      (2) 20 cm                      (3) 30 cm                      (4) 10 cm

GO0188

6. The graph between angle of deviation ( $\delta$ ) and angle of incidence ( $i$ ) for a triangular prism is represented by :- [JEE-Main- 2013]



GO0189

7. A thin convex lens made from crown glass ( $\mu = \frac{3}{2}$ ) has focal length  $f$ . When it is measured in two different liquids having refractive indices  $\frac{4}{3}$  and  $\frac{5}{3}$ , it has the focal length  $f_1$  and  $f_2$  respectively. The correct relation between the focal lengths is : [JEE-Main- 2014]

- (1)  $f_2 > f$  and  $f_1$  becomes negative  
(2)  $f_1$  and  $f_2$  both become negative  
(3)  $f_1 = f_2 < f$   
(4)  $f_1 > f$  and  $f_2$  become negative

GO0190

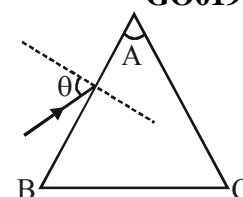
8. A green light is incident from the water to the air - water interface at the critical angle ( $\theta$ ). Select the correct statement. [JEE-Main- 2014]

- (1) The spectrum of visible light whose frequency is more than that of green light will come out to the air medium.  
(2) The entire spectrum of visible light will come out of the water at various angles to the normal  
(3) The entire spectrum of visible light will come out of the water at an angle of  $90^\circ$  to the normal.  
(4) The spectrum of visible light whose frequency is less than that of green light will come out to the air medium.

GO0191

9. Monochromatic light is incident on a glass prism of angle  $A$ . If the refractive index of the material of the prism is  $\mu$ , a ray, incident at an angle  $\theta$ , on the face AB would get transmitted through the face AC of the prism provided :

[JEE-Main- 2015]



- (1)  $\theta > \cos^{-1} \left[ \mu \sin \left( A + \sin^{-1} \left( \frac{1}{\mu} \right) \right) \right]$   
(2)  $\theta < \cos^{-1} \left[ \mu \sin \left( A + \sin^{-1} \left( \frac{1}{\mu} \right) \right) \right]$   
(3)  $\theta > \sin^{-1} \left[ \mu \sin \left( A - \sin^{-1} \left( \frac{1}{\mu} \right) \right) \right]$   
(4)  $\theta < \sin^{-1} \left[ \mu \sin \left( A - \sin^{-1} \left( \frac{1}{\mu} \right) \right) \right]$

GO0192

10. An observer looks at a distant tree of height 10 m with a telescope of magnifying power of 20. To the observer the tree appears : [JEE-Main- 2016]

- (1) 20 times nearer  
(2) 10 times taller  
(3) 10 times nearer  
(4) 20 times taller

GO0193

11. In an experiment for determination of refractive index of glass of a prism by  $i - \delta$ , plot, it was found that a ray incident at angle  $35^\circ$ , suffers a deviation of  $40^\circ$  and that it emerges at angle  $79^\circ$ . In that case which of the following is closest to the maximum possible value of the refractive index?

[JEE-Main- 2016]

- (1) 1.8                      (2) 1.5                      (3) 1.6                      (4) 1.7

GO0194

12. A diverging lens with magnitude of focal length 25 cm is placed at a distance of 15 cm from a converging lens of magnitude of focal length 20 cm. A beam of parallel light falls on the diverging lens. The final image formed is :

[JEE-Main- 2017]

- (1) real and at a distance of 40 cm from the divergent lens  
(2) real and at a distance of 6 cm from the convergent lens  
(3) real and at a distance of 40 cm from convergent lens  
(4) virtual and at a distance of 40 cm from convergent lens.

GO0195

# EXERCISE-(JA)

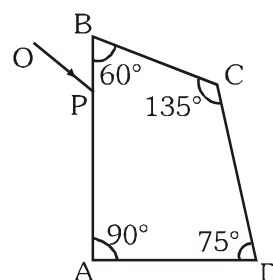
1. A biconvex lens of focal length 15 cm is in front of a plane mirror. The distance between the lens and the mirror is 10 cm. A small object is kept at a distance of 30 cm from the lens. The final image is  
 (A) virtual and at a distance of 16 cm from the mirror  
 (B) real and at a distance of 16 cm from the mirror  
 (C) virtual and at a distance of 20 cm from the mirror  
 (D) real and at a distance of 20 cm from the mirror

[IIT-JEE 2010]

GO0196

2. A ray OP of monochromatic light is incident on the face AB of prism ABCD near vertex B at an incident angle of  $60^\circ$  (see figure). If the refractive index of the material of the prism is  $\sqrt{3}$ , which of the following is (are) correct?  
 [IIT-JEE 2010]

- (A) The ray gets totally internally reflected at face CD
- (B) The ray comes out through face AD
- (C) The angle between the incident ray and the emergent ray is  $90^\circ$
- (D) The angle between the incident ray and the emergent ray is  $120^\circ$



GO0197

3. Two transparent media of refractive indices  $\mu_1$  and  $\mu_3$  have a solid lens shaped transparent material of refractive index  $\mu_2$  between them as shown in figures in **Column II**. A ray traversing these media is also shown in the figures. In **Column I** different relationships between  $\mu_1$ ,  $\mu_2$  and  $\mu_3$  are given. Match them to the ray diagrams shown in **Column II**.  
 [IIT-JEE 2010]

## Column I

- (A)  $\mu_1 < \mu_2$
- (B)  $\mu_1 > \mu_2$
- (C)  $\mu_2 = \mu_3$
- (D)  $\mu_2 > \mu_3$

## Column II

- (p)
- (q)
- (r)
- (s)
- (t)

GO0198

4. The focal length of a thin biconvex lens is 20 cm. When an object is moved from a distance of 25 cm in front of it to 50 cm, the magnification of its image changes from  $m_{25}$  to  $m_{50}$ . The ratio  $\frac{m_{25}}{m_{50}}$  is

[IIT-JEE 2010]

GO0199

5. Image of an object approaching a convex mirror of radius of curvature 20 m along its optical axis is observed to move from  $\frac{25}{3}$  m to  $\frac{50}{7}$  m in 30 seconds. What is the speed of the object in km per hour?

[IIT-JEE 2010]

GO0200

6. A large glass slab  $\left(\mu = \frac{5}{3}\right)$  of thickness 8 cm is placed over a point source of light on a plane surface.

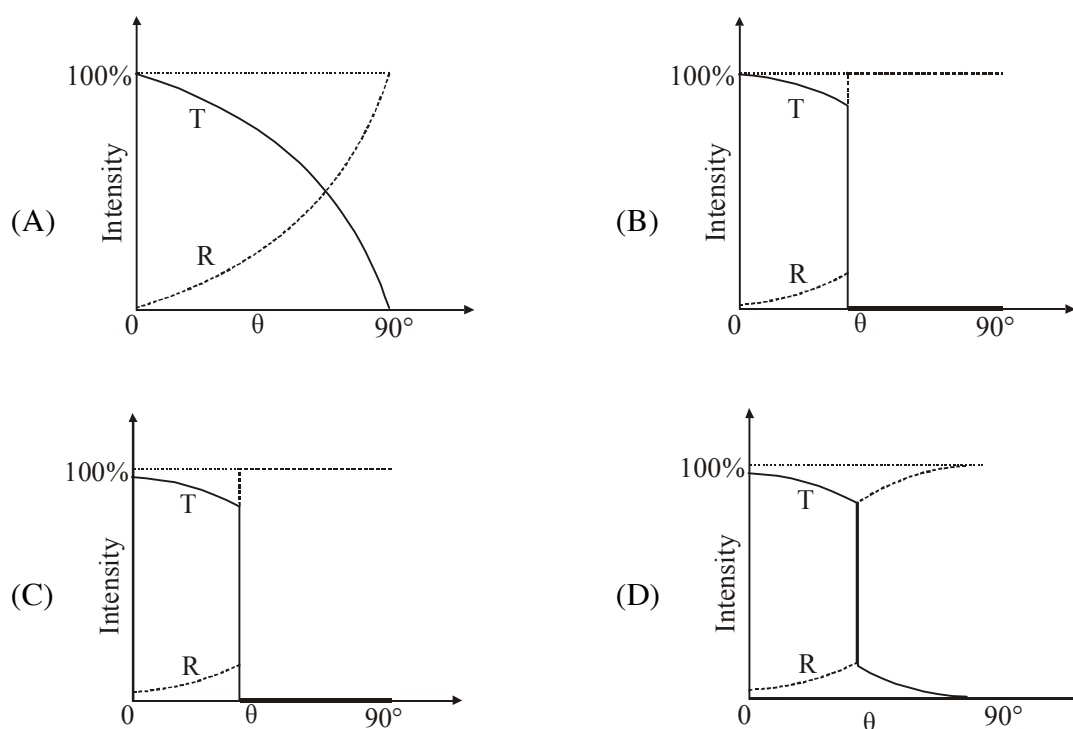
It is seen that light emerges out of the top surface of the slab from a circular area of radius R cm. What is the value of R?

[IIT-JEE 2010]

GO0201

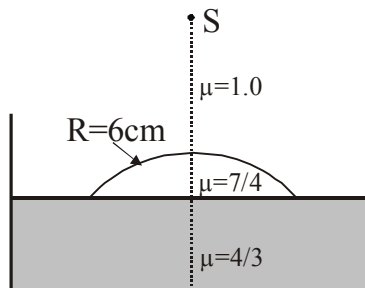
7. A light ray traveling in glass medium is incident on glass-air interface at an angle of incidence  $\theta$ . The reflected (R) and transmitted (T) intensities, both as function of  $\theta$ , are plotted. The correct sketch is

[IIT-JEE 2011]



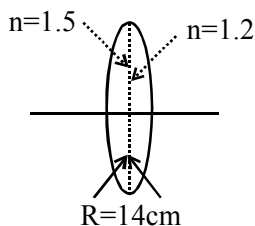
GO0202

8. Water (with refractive index  $= \frac{4}{3}$ ) in a tank is 18 cm deep. Oil of refractive index  $\frac{7}{4}$  lies on water making a convex surface of radius of curvature ' $R = 6$  cm' as shown. Consider oil to act as a thin lens. An object 'S' is placed 24 cm above water surface. The location of its image is at ' $x$ ' cm above the bottom of the tank. Then ' $x$ ' is [IIT-JEE 2011]



GO203

9. A biconvex lens is formed with two thin plano-convex lenses as shown in the figure, Refractive index  $n$  of the first lens is 1.5 and that of the second lens is 1.2. Both the curved surfaces are of the same radius of curvature  $R = 14$  cm. For this biconvex lens, for an object distance of 40 cm, the image distance will be :- [IIT-JEE 2012]



- (A) -280.0 cm (B) 40.0 cm (C) 21.5 cm (D) 13.3 cm

GO204

### Paragraph for Questions 10 and 11

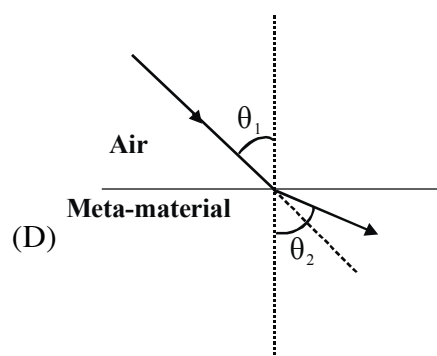
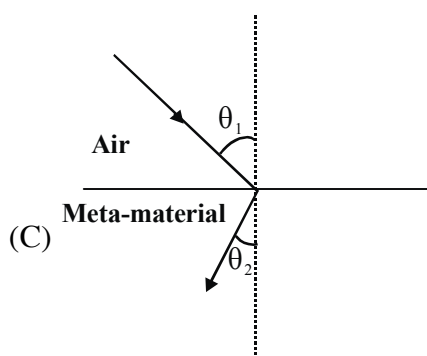
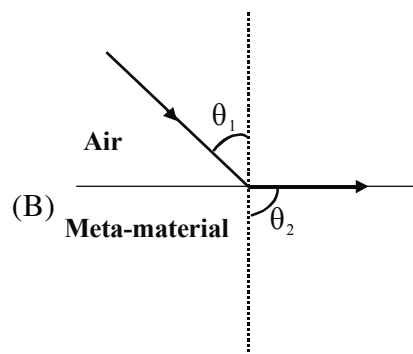
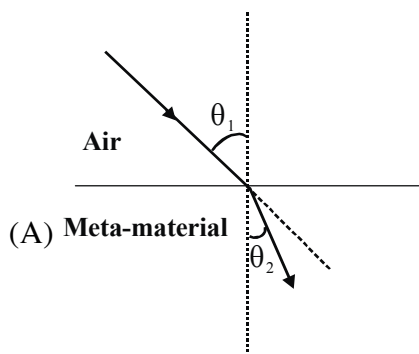
Most materials have the refractive index,  $n > 1$ . So, when a light ray from air enters a naturally occurring material, then by Snell's law,  $\frac{\sin \theta_1}{\sin \theta_2} = \frac{n_2}{n_1}$ , it is understood that the refracted ray bends towards the normal. But it never emerges on the same side of the normal as the incident ray. According to

electromagnetism, the refractive index of the medium is given by the relation,  $n = \left( \frac{c}{v} \right) = \pm \sqrt{\epsilon_r \mu_r}$ ,

where  $c$  is the speed of electromagnetic waves in vacuum,  $v$  its speed in the medium,  $\epsilon_r$  and  $\mu_r$  are the relative permittivity and permeability of the medium respectively.

In normal materials, both  $\epsilon_r$  and  $\mu_r$  are positive, implying positive  $n$  for the medium. When both  $\epsilon_r$  and  $\mu_r$  are negative, one must choose the negative root of  $n$ . Such negative refractive index materials can now be artificially prepared and are called meta-materials. They exhibit significantly different optical behaviour, without violating any physical laws. Since  $n$  is negative, it results in a change in the direction of propagation of the refracted light. However, similar to normal materials, the frequency of light remains unchanged upon refraction even in meta-materials. [IIT-JEE-2012]

10. For light incident from air on a meta-material, the appropriate ray diagram is



11. Choose the correct statement.

(A) The speed of light in the meta-material is  $v = c|n|$

(B) The speed of light in the meta-material is  $v = \frac{c}{|n|}$

(C) The speed of light in the meta-material is  $v = c$ .

(D) The wavelength of the light in the meta-material ( $\lambda_m$ ) is given by  $\lambda_m = \lambda_{\text{air}}|n|$ , where  $\lambda_{\text{air}}$  is the wavelength of the light in air.

GO0205  
[IIT-JEE 2012]

12. A ray of light travelling in the direction  $\frac{1}{2}(\hat{i} + \sqrt{3}\hat{j})$  is incident on a plane mirror. After reflection, it

travels along the direction  $\frac{1}{2}(\hat{i} - \sqrt{3}\hat{j})$ . The angle of incidence is :-

(A)  $30^\circ$

(B)  $45^\circ$

(C)  $60^\circ$

(D)  $75^\circ$

[JEE-Advance-2013]

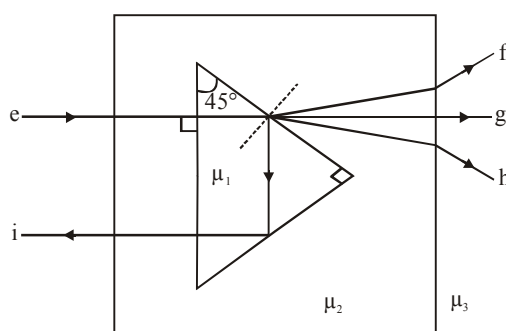
GO0206



13. The image of an object, formed by a plano-convex lens at a distance of 8 m behind the lens, is real and is one-third the size of the object. The wavelength of light inside the lens is  $\frac{2}{3}$  times the wavelength in free space. The radius of the curved surface of the lens is :- [JEE-Advance-2013]  
 (A) 1 m (B) 2 m (C) 3 m (D) 6 m

GO0207

14. A right angled prism of refractive index  $\mu_1$  is placed in a rectangular block of refractive index  $\mu_2$ , which is surrounded by a medium of refractive index  $\mu_3$ , as shown in the figure. A ray of light 'e' enters the rectangular block at normal incidence. Depending upon the relationships between  $\mu_1$ ,  $\mu_2$ , and  $\mu_3$ , it takes one of the four possible paths 'ef', 'eg', 'eh' or 'ei'. [JEE-Advance-2013]



Match the paths in List I with conditions of refractive indices in List II and select the correct answer using the codes given below the lists :

**List I**

- P.  $e \rightarrow f$   
 Q.  $e \rightarrow g$   
 R.  $e \rightarrow h$   
 S.  $e \rightarrow i$

**List II**

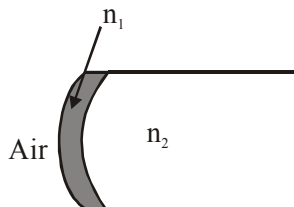
1.  $\mu_1 > \sqrt{2}\mu_2$
2.  $\mu_2 > \mu_1$  and  $\mu_2 > \mu_3$
3.  $\mu_1 = \mu_2$
4.  $\mu_2 < \mu_1 < \sqrt{2}\mu_2$  and  $\mu_2 > \mu_3$

**Codes :**

|     | P | Q | R | S |
|-----|---|---|---|---|
| (A) | 2 | 3 | 1 | 4 |
| (B) | 1 | 2 | 4 | 3 |
| (C) | 4 | 1 | 2 | 3 |
| (D) | 2 | 3 | 4 | 1 |

GO0208

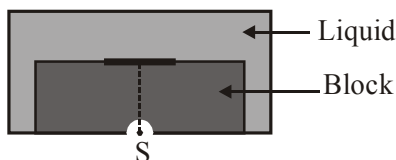
15. A transparent thin film of uniform thickness and refractive index  $n_1 = 1.4$  is coated on the convex spherical surface of radius  $R$  at one end of a long solid glass cylinder of refractive index  $n_2 = 1.5$ , as shown in the figure. Rays of light parallel to the axis of the cylinder traversing through the film from air to glass get focused at distance  $f_1$  from the film, while rays of light traversing from glass to air get focused at distance  $f_2$  from the film. Then [JEE-Advance-2014]



- (A)  $|f_1| = 3R$  (B)  $|f_1| = 2.8R$  (C)  $|f_2| = 2R$  (D)  $|f_2| = 1.4R$

GO0209

16. A point source  $S$  is placed at the bottom of a transparent block of height 10 mm and refractive index 2.72. It is immersed in a lower refractive index liquid as shown in the figure. It is found that the light emerging from the block to the liquid forms a circular bright spot of diameter 11.54 mm on the top of the block. The refractive index of the liquid is :- [JEE-Advance-2014]



- (A) 1.21 (B) 1.30 (C) 1.36 (D) 1.42

GO0210

17. Four combinations of two thin lenses are given in List-I. The radius of curvature of all curved surfaces is  $r$  and the refractive index of all the lenses is 1.5. Match lens combinations in List-I with their focal length in List-II and select the correct answer using the code given below the lists. [JEE-Advance-2014]

## List-I



## List-II

(1)  $2r$

(2)  $\frac{r}{2}$

(3)  $-r$

(4)  $r$

Code :

- (A) P-1, Q-2, R-3, S-4  
(C) P-4, Q-1, R-2, S-3

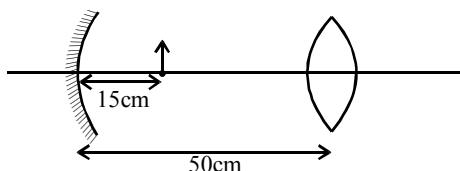
- (B) P-2, Q-4, R-3, S-1  
(D) P-2, Q-1, R-3, S-4

GO0211

18. Consider a concave mirror and a convex lens (refractive index = 1.5) of focal length 10 cm each, separated by a distance of 50 cm in air (refractive index = 1) as shown in the figure. An object is placed at a distance of 15 cm from the mirror. Its erect image formed by this combination has magnification  $M_1$ . When the set-up is kept in a medium of refractive index  $7/6$  the magnification

becomes  $M_2$ . The magnitude  $\left| \frac{M_2}{M_1} \right|$  is.

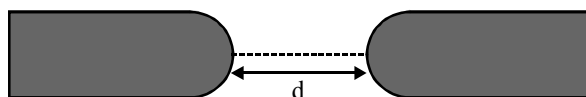
[JEE-Advance-2015]



GO0212

19. Two identical glass rods  $S_1$  and  $S_2$  (refractive index = 1.5) have one convex end of radius of curvature 10 cm. They are placed with the curved surfaces at a distance  $d$  as shown in the figure, with their axes (shown by the dashed line) aligned. When a point source of light  $P$  is placed inside rod  $S_1$  on its axis at a distance of 50 cm from the curved face, the light rays emanating from it are found to be parallel to the axis inside  $S_2$ . The distance  $d$  is :

[JEE-Advance-2015]



(A) 60 cm

(B) 70 cm

(C) 80 cm

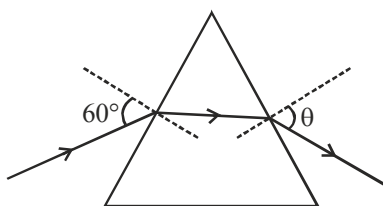
(D) 90 cm

GO0213

20. A monochromatic beam of light is incident at  $60^\circ$  on one face of an equilateral prism of refractive index  $n$  and emerges from the opposite face making an angle  $\theta(n)$  with the normal (see the figure).

For  $n = \sqrt{3}$  the value of  $\theta$  is  $60^\circ$  and  $\frac{d\theta}{dn} = m$ . The value of  $m$  is

[JEE-Advance-2015]

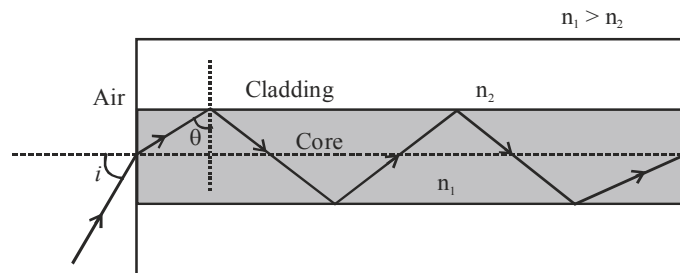


GO0214

## Paragraph for Question No. 21 and 22

Light guidance in an optical fiber can be understood by considering a structure comprising of thin solid glass cylinder of refractive index  $n_1$  surrounded by a medium of lower refractive index  $n_2$ . The light guidance in the structure takes place due to successive total internal reflections at the interface of the media  $n_1$  and  $n_2$  as shown in the figure. All rays with the angle of incidence  $i$  less than a particular value of  $i_m$  are confined in the medium of refractive index  $n_1$ . The numerical aperture (NA) of the structure is defined as  $\sin i_m$ .

[JEE-Advance-2015]



21. For two structures namely  $S_1$  with  $n_1 = \sqrt{45}/4$  and  $n_2 = 3/2$ , and  $S_2$  with  $n_1 = 8/5$  and  $n_2 = 7/5$  and taking the refractive index of water to be  $4/3$  and that of air to be 1, the correct option(s) is (are)

(A) NA of  $S_1$  immersed in water is the same as that of  $S_2$  immersed in liquid of refractive index  $\frac{16}{3\sqrt{15}}$ .

(B) NA of  $S_1$  immersed in liquid of refractive index  $\frac{6}{\sqrt{15}}$  is the same as that of  $S_2$  immersed in water.

(C) NA of  $S_1$  placed in air is the same as that of  $S_2$  immersed in liquid of refractive index  $\frac{4}{\sqrt{15}}$ .

(D) NA of  $S_1$  placed in air is the same as that of  $S_2$  placed in water.

GO0215

22. If two structures of same cross-sectional area, but different numerical apertures  $NA_1$  and  $NA_2$  ( $NA_2 < NA_1$ ) are joined longitudinally, the numerical aperture of the combined structure is

(A)  $\frac{NA_1 NA_2}{NA_1 + NA_2}$

(B)  $NA_1 + NA_2$

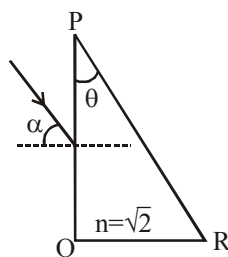
(C)  $NA_1$

(D)  $NA_2$

GO0215

23. A parallel beam of light is incident from air at an angle  $\alpha$  on the side PQ of a right angled triangular prism of refractive index  $n = \sqrt{2}$ . Light undergoes total internal reflection in the prism at the face PR when  $\alpha$  has a minimum value of  $45^\circ$ . The angle  $\theta$  of the prism is :

[JEE-Advance-2016]



(A)  $15^\circ$

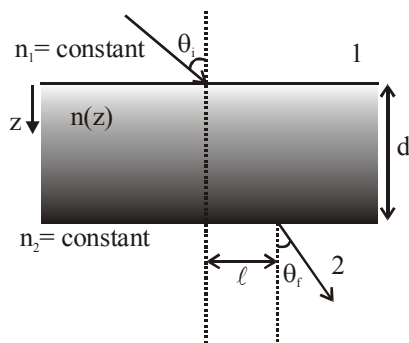
(B)  $22.5^\circ$

(C)  $30^\circ$

(D)  $45^\circ$

GO0216

24. A transparent slab of thickness  $d$  has a refractive index  $n(z)$  that increases with  $z$ . Here  $z$  is the vertical distance inside the slab, measured from the top. The slab is placed between two media with uniform refractive indices  $n_1$  and  $n_2$  ( $> n_1$ ), as shown in the figure. A ray of light is incident with angle  $\theta_i$  from medium 1 and emerges in medium 2 with refraction angle  $\theta_f$  with a lateral displacement  $\ell$ . Which of the following statement(s) is(are) true ? [JEE-Advance-2016]



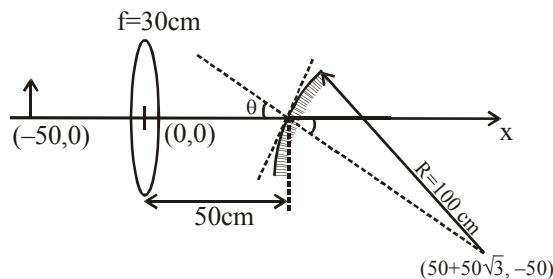
- (A)  $\ell$  is independent of  $n_2$  (B)  $\ell$  is dependent on  $n(z)$   
(C)  $n_1 \sin \theta_i = (n_2 - n_1) \sin \theta_f$  (D)  $n_1 \sin \theta_i = n_2 \sin \theta_f$

GO0217

25. A plano-convex lens is made of a material of refractive index  $n$ . When a small object is placed 30 cm away in front of the curved surface of the lens, an image of double the size of the object is produced. Due to reflection from the convex surface of the lens, another faint image is observed at a distance of 10 cm away from the lens. Which of the following statement(s) is(are) true? [JEE-Advance-2016]
- (A) The refractive index of the lens is 2.5  
(B) The radius of curvature of the convex surface is 45 cm  
(C) The faint image is erect and real  
(D) The focal length of the lens is 20 cm.

GO0218

26. A small object is placed 50 cm to the left of thin convex lens of focal length 30 cm. A convex spherical mirror of radius of curvature 100 cm is placed to the right of the lens at a distance of 50 cm. The mirror is tilted such that the axis of the mirror is at an angle  $\theta = 30^\circ$  to the axis of the lens, as shown in the figure. If the origin of the coordinate system is taken to be at the centre of the lens, the coordinates (in cm) of the point  $(x, y)$  at which the image is formed are : [JEE-Advance-2016]



- (A)  $(25, 25\sqrt{3})$  (B)  $\left(\frac{125}{3}, \frac{25}{\sqrt{3}}\right)$  (C)  $(50 - 25\sqrt{3}, 25)$  (D)  $(0, 0)$

GO0219

27. For an isosceles prism of angle  $A$  and refractive index  $\mu$ , it is found that the angle of minimum deviation  $\delta_m = A$ . Which of the following options is/are correct ? [JEE-Advance-2017]

(A) At minimum deviation, the incident angle  $i_1$  and the refracting angle  $r_1$  at the first refracting surface are related by  $r_1 = (i_1/2)$

(B) For this prism, the refractive index  $\mu$  and the angle of prism  $A$  are related as  $A = \frac{1}{2} \cos^{-1} \left( \frac{\mu}{2} \right)$

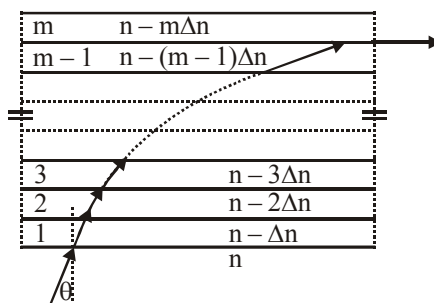
(C) For this prism, the emergent ray at the second surface will be tangential to the surface when the

angle of incidence at the first surface is  $i_1 = \sin^{-1} \left[ \sin A \sqrt{4 \cos^2 \frac{A}{2} - 1} - \cos A \right]$

(D) For the angle of incidence  $i_1 = A$ , the ray inside the prism is parallel to the base of the prism.

GO0220

28. A monochromatic light is travelling in a medium of refractive index  $n = 1.6$ . It enters a stack of glass layers from the bottom side at an angle  $\theta = 30^\circ$ . The interfaces of the glass layers are parallel to each other. The refractive indices of different glass layers are monotonically decreasing as  $n_m = n - m\Delta n$ , where  $n_m$  is the refractive index of the  $m^{\text{th}}$  slab and  $\Delta n = 0.1$  (see the figure). The ray is refracted out parallel to the interface between the  $(m-1)^{\text{th}}$  and  $m^{\text{th}}$  slabs from the right side of the stack. What is the value of  $m$  ? [JEE-Advance-2017]

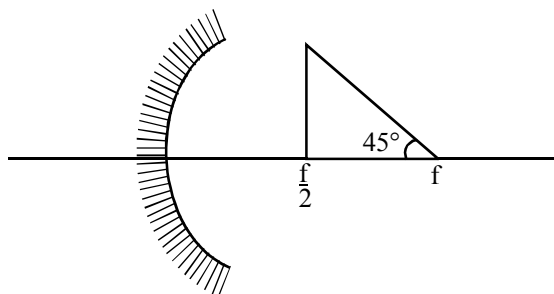


GO0221

29. Sunlight of intensity  $1.3 \text{ kW m}^{-2}$  is incident normally on a thin convex lens of focal length 20 cm. Ignore the energy loss of light due to the lens and assume that the lens aperture size is much smaller than its focal length. The average intensity of light, in  $\text{kW m}^{-2}$ , at a distance 22 cm from the lens on the other side is \_\_\_\_\_. [JEE-Advance-2018]

GO0222

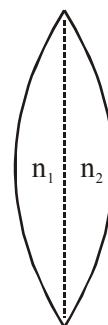
30. A wire is bent in the shape of a right angled triangle and is placed in front of a concave mirror of focal length  $f$ , as shown in the figure. Which of the figures shown in the four options qualitatively represent(s) the shape of the image of the bent wire ? (These figures are not to scale.) ? [JEE-Advance-2018]



- (A)
- (B)
- (C)
- (D)

31. A thin convex lens is made of two materials with refractive indices  $n_1$  and  $n_2$ , as shown in figure. The radius of curvature of the left and right spherical surfaces are equal.  $f$  is the focal length of the lens when  $n_1 = n_2 = n$ . The focal length is  $f + \Delta f$  when  $n_1 = n$  and  $n_2 = n + \Delta n$ . Assuming  $\Delta n \ll (n-1)$  and  $1 < n < 2$ , the correct statement(s) is/are : [JEE-Advance-2019]

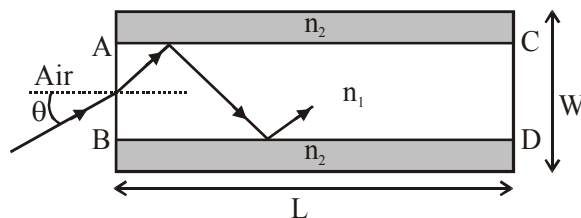
GO0223



- (1) The relation between  $\frac{\Delta f}{f}$  and  $\frac{\Delta n}{n}$  remains unchanged if both the convex surfaces are replaced by concave surfaces of the same radius of curvature.
- (2)  $\left| \frac{\Delta f}{f} \right| < \left| \frac{\Delta n}{n} \right|$
- (3) For  $n = 1.5$ ,  $\Delta n = 10^{-3}$  and  $f = 20$  cm, the value of  $|\Delta f|$  will be 0.02 cm (round off to 2<sup>nd</sup> decimal place)
- (4) If  $\frac{\Delta n}{n} < 0$  then  $\frac{\Delta f}{f} > 0$

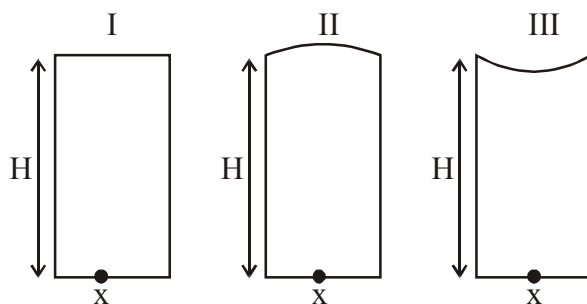
GO0224

32. A planar structure of length  $L$  and width  $W$  is made of two different optical media of refractive indices  $n_1 = 1.5$  and  $n_2 = 1.44$  as shown in figure. If  $L \gg W$ , a ray entering from end AB will emerge from end CD only if the total internal reflection condition is met inside the structure. For  $L = 9.6$  m, if the incident angle  $\theta$  is varied, the maximum time taken by a ray to exit the plane CD is  $t \times 10^{-9}$  s, where  $t$  is \_\_\_\_\_. [Speed of light  $c = 3 \times 10^8$  m/s] **[JEE-Advance-2019]**



GO0225

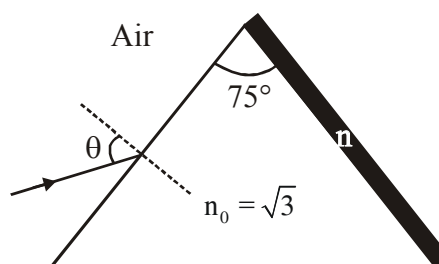
33. Three glass cylinders of equal height  $H = 30$  cm and same refractive index  $n = 1.5$  are placed on a horizontal surfaces shown in figure. Cylinder I has a flat top, cylinder II has a convex top and cylinder III has a concave top. The radii of curvature of the two curved tops are same ( $R = 3$  m). If  $H_1$ ,  $H_2$  and  $H_3$  are the apparent depths of a point X on the bottom of the three cylinders, respectively, the correct statement(s) is/are **[JEE-Advance-2019]**



- (1)  $H_3 > H_1$  (2)  $0.8 \text{ cm} < (H_2 - H_1) < 0.9 \text{ cm}$   
 (3)  $H_2 > H_3$  (4)  $H_2 > H_1$

GO0226

34. A monochromatic light is incident from air on a refracting surface of a prism of angle  $75^\circ$  and refractive index  $n_0 = \sqrt{3}$ . The other refracting surface of a prism is coated by a thin film of material of refractive index  $n$  as shown in figure. The light suffers total internal reflection at the coated prism surface for an incidence angle of  $\theta \leq 60^\circ$ . The value of  $n^2$  is \_\_\_\_\_. **[JEE-Advance-2019]**

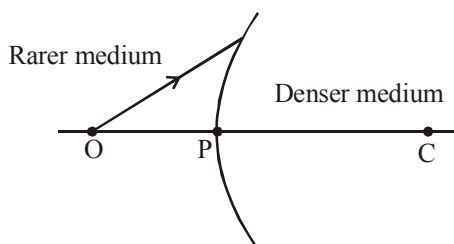


GO0227



## CBSE PREVIOUS YEAR'S QUESTIONS

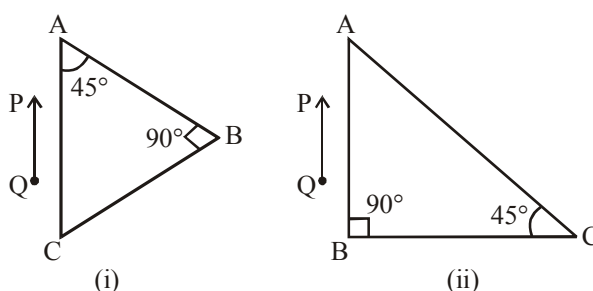
1. Draw a ray diagram of an astronomical telescope in the normal adjustment position. Write down the expression for its magnifying power. [2; CBSE-2004]
2. A spherical surface of radius of curvature  $R$ , separates a rarer and a denser medium as shown in the figure.



Complete the path of the incident ray of light, showing the formation of a real image. Hence derive the relation connecting object distance ' $u$ ', image distance ' $v$ ', radius of curvature  $R$  and the refractive indices  $n_1$  and  $n_2$  of the two media.

Briefly explain, how the focal length of a convex lens changes, with increase in wavelength of incident light. [5; CBSE-2004]

3. A right-angled crown glass prism with critical angle  $41^\circ$  is placed before an object PQ, in two positions as shown in the figures (i) and (ii). Trace the paths of the rays from P and Q the prisms in the two cases. [2; CBSE-2005]



4. (a) Draw a labelled ray diagram to show the formation of an image by a compound microscope. Write the expression for its magnifying power.  
(b) How does the resolving power of a compound microscope change, when (i) refractive index of the medium between the object and the objective lens increases; and (ii) wavelength of the radiation used is increased? [3; CBSE-2005]
5. What is the function of cladding in a typical optical fiber? [1; CBSE-2005]
6. Draw a labelled ray diagram to show the image formation in a refracting type astronomical telescope. Why should the diameter of the objective of a telescope be large? [2; CBSE-2006]

7. A beam of light converges to a point P. A lens is placed in the path of the convergent beam 12 cm from P. At what point does the beam converge if the lens is  
(i) a convex lens of focal length 20 cm,  
(ii) a concave lens of focal length 16 cm ?  
Do the required calculations. [3; CBSE-2006]
8. Define resolving power of a compound microscope. How does the resolving power of a compound microscope change when  
(i) refractive index of the medium between the object and objective lens increases ?  
(ii) wavelength of the radiation used is increased [2; CBSE-2007]
9. A double convex lens of glass of refractive index 1.6 has its both surfaces of equal radii of curvature of 30 cm each. An object of height 5 cm is placed at a distance of 12.5 cm from the lens. Calculate the size of the image formed. [2; CBSE-2007]
10. A glass lens of refractive index 1.5 is placed in a trough of liquid. What must be the refractive index of the liquid in order to make the lens disappear? [1; CBSE-2008]
11. A ray of light passing through an equilateral triangular glass prism from air undergoes minimum deviation when angle of incidence is  $3/4^{\text{th}}$  of the angle of prism. Calculate the speed of light in the prism. [2; CBSE-2008]
12. (a) For a ray of light traveling from a denser medium of refractive index  $n_1$  to a rarer medium of

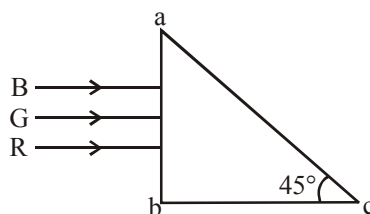
refractive index  $n_2$ , prove that  $\frac{n_2}{n_1} = \sin i_c$  where  $i_c$  is the critical angle of incidence for the media.

(b) Explain with the help of a diagram, how the above principle is used for transmission of video signals using optical fibres. [5; CBSE-2008]

13. You are given following three lenses. Which two lenses will you use as an eyepiece and as an objective to construct an astronomical telescope? [1; CBSE-2009]

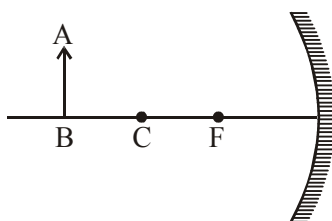
| Lenses | Power (P) | Aperture (A) |
|--------|-----------|--------------|
| L1     | 3D        | 8 cm         |
| L2     | 6D        | 1 cm         |
| L3     | 10D       | 1 cm         |

14. Three light rays red (R), green (G) and blue (B) are incident on a right angled prism 'abc' at face 'ab'. The refractive indices of the material of the prism for red, green and blue wavelengths are 1.39, 1.44 and 1.47 respectively. Out of the three which colour ray will emerge out of face 'ac'? Justify your answer. Trace the path of these rays after passing through face 'ab'. [3; CBSE-2009]

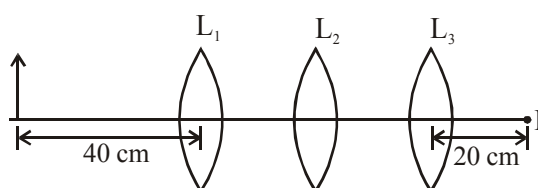


15. Trace the rays of light showing the formation of an image due to a point object placed on the axis of a spherical surface separating the two media of refractive indices  $n_1$  and  $n_2$ . Establish the relation between the distances of the object, the image and the radius of curvature from the central point the spherical surface. Hence derive the expression of lens maker's formula. [5; CBSE-2009]
16. Draw the labeled ray diagram for the formation of image by a compound microscope. Derive the expression for the total magnification of a compound microscope. Explain why both the objective and the eyepieces of a compound microscope must have short focal lengths. [5; CBSE-2009]
17. A converging lens is kept coaxially in contact with a diverging lens- both the lenses being of equal focal lengths. What is the focal length of the combination? [1; CBSE-2010]
18. (i) Draw a net labelled ray diagram of an astronomical telescope in normal adjustment. Explain briefly its working.  
(ii) An astronomical telescope uses two lenses of powers 10 D and 1 D. What is its magnifying power in normal adjustment? [3; CBSE-2010]
19. (i) Draw a neat labelled ray diagram of a compound microscope. Explain briefly its working,  
(ii) Why must both the objective and the eye piece of a compound microscope have short focal lengths? [3; CBSE-2010]
20. An illuminated object and a screen are placed 90 cm apart. Determine the focal length and nature of the lens required to produce a clear image on the screen, twice the size of the object, [3; CBSE-2010]
21. A convex lens made up of glass of refractive index 1.5 is dipped, in turn, in (i) a medium of refractive index 1.65, (ii) a medium of refractive index 1.33.  
(a) Will it behave as a converging or a diverging lens in the two cases ?  
(b) How will its focal length change in the two media ? [3; CBSE-2011]
22. Use the mirror equation to show that (a) an object placed between  $f$  and  $2f$  of a concave mirror produces a real image beyond  $2f$ . (b) a convex mirror always produces a virtual image independent of the location of the object, (c) an object placed between the pole and focus of a concave mirror produces a virtual and enlarged image. [3; CBSE-2011]
23. A compound microscope uses an objective lens of focal length 4 cm and eyepiece lens of focal length 10cm. An object is placed at 6 cm from the objective lens. Calculate the magnifying power of the compound microscope. Also calculate the length of the microscope. [3; CBSE-2011]
24. A giant refracting telescope at an observatory has an objective lens of focal length 1 m. If an eyepiece lens of focal length 1.0 cm is used, find the angular magnification of the telescope. If this telescope is used to view the moon, what is the diameter of the image of the moon formed by the objective lens ? The diameter of the moon is  $3.42 \times 10^6$  m and the radius of the lunar orbit is  $3.8 \times 10^8$  m. [3; CBSE-2011]
25. For the same value of angle of incidence, the angles of refraction in three media A, B and C are  $15^\circ$ ,  $25^\circ$  and  $35^\circ$  respectively. In which medium would the velocity of light be minimum? [1; CBSE-2012]

26. An object AB is kept in front of a concave mirror as shown in the figure.

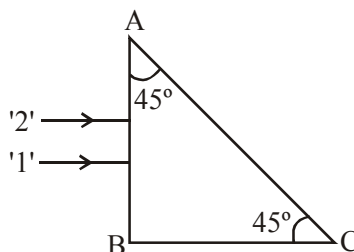


- (i) Complete the ray diagram showing the image formation of the object.  
 (ii) How will the position and intensity of the image be affected if the lower half of the mirror's reflecting surface is painted black? [2; CBSE-2012]
27. Draw a labeled ray diagram of a reflecting telescope. Mention its two advantages over the refracting telescope. [2; CBSE-2012]
28. You are given three lenses  $L_1$ ,  $L_2$  and  $L_3$  each of focal length 20 cm. A object is kept at 40 cm in front of  $L_1$ , as shown. The final real image is formed at the focus "I" of  $L_3$ . Find the separation between  $L_1$ ,  $L_2$  and  $L_3$ . [3; CBSE-2012]

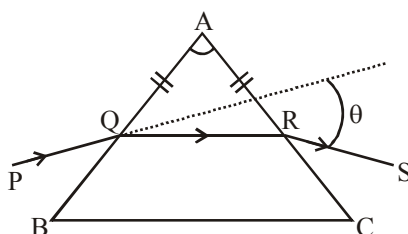


29. A convex lens of focal length  $f_1$  is kept in contact with a concave lens of focal length  $f_2$ . Find the focal length of the combination. [CBSE-2013]
30. Draw a labelled ray diagram of a refracting telescope. Define its magnifying power and write the expression for it. Write two important limitations of a refracting telescope over a reflecting type telescope. [CBSE-2013]
31. One day Chetan's mother developed a severe stomach ache all of a sudden. She was rushed to the doctor who suggested for an immediate endoscopy test and gave an estimate of expenditure for the same. Chetan immediately contacted his class teacher and shared the information with her. The class teacher arranged for the money and rushed to the hospital. On releasing that Chetan belonged to a below average income group family, even the doctor offered concession for the test fee. The test was conducted successfully. [CBSE-2013]
- Answer the following questions based on the above information:
- (a) Which principle in optics is made use of in endoscopy?  
 (b) Briefly explain the values reflected in the action taken by the teacher.  
 (c) In what way do you appreciate the response of the doctor on the given situation?
32. A biconvex lens made of a transparent material of refractive index 1.5 is immersed in water of refractive index 1.33. Will the lens behave as a converging or a diverging lens? Give reason. [CBSE-2014]

33. Two monochromatic rays of light are incident normally on the face AB of an isosceles right-angled prism ABC. The refractive indices of the glass prism for the two rays '1' and '2' are respectively 1.3 and 1.5. Trace the path of these rays after entering through the prism. [CBSE-2014]



34. A Convex lens of focal length 20 cm is placed coaxially with a concave mirror of focal length 10 cm at a distance of 50 cm apart from each other. A beam of light coming parallel to the principal axis is incident on the convex lens. Find the position of the final image formed by this combination. Draw the ray diagram showing the formation of the image. [CBSE-2014]
35. When light travels from an optically denser medium to a rarer medium, why does the critical angle of incidence depend on the colour of light ? [1; CBSE-2015]
36. A ray PQ incident on the refracting face BA is refracted in the prism BAC as shown in the figure and emerges from the other refracting face AC as RS such that  $AQ = AR$ . If the angle of prism  $A = 60^\circ$  and refractive index of material of prism is  $\sqrt{3}$ , calculate angle  $\theta$ . [CBSE-2016]

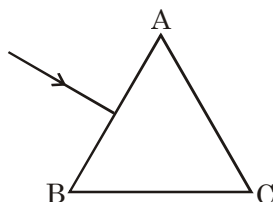


37. (i) A screen is placed at a distance of 100 cm from an object. The image of the object is formed on the screen by a convex lens for two different locations of the lens separated by 20cm. Calculate the focal length of the lens used.  
(ii) A converging lens is kept coaxially in contact with a diverging lens-both the lenses being of equal focal length. What is the focal length of the combination ? [CBSE-2016]
38. How does the angle of minimum deviation of a glass prism vary, if the incident violet light is replaced by red light ? Give reason. [CBSE-2017]
39. (a) Monochromatic light of wavelength 589 nm is incident from air on a water surface. If  $\mu$  for water is 1.33, find the wavelength, frequency and speed of the refracted light. [CBSE-2017]  
(b) A double convex lens is made of a glass of refractive index 1.55, with both faces of the same radius of curvature. Find the radius of curvature required, if the focal length is 20 cm.

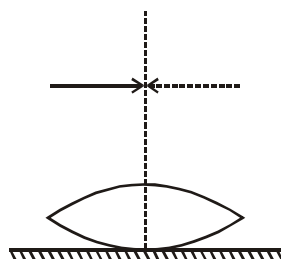
40. (a) Draw a ray diagram depicting the formation of the image by an astronomical telescope in normal adjustment. [CBSE-2017]  
 (b) You are given the following three lenses. Which two lenses will you use as an eyepiece and as an objective to construct an astronomical telescope? Give reason.

| Lenses | Power (D) | Aperture (cm) |
|--------|-----------|---------------|
| $L_1$  | 3         | 8             |
| $L_2$  | 6         | 1             |
| $L_3$  | 10        | 1             |

41. (a) Draw a ray diagram to show the image formation by a combination of two thin convex lenses in contact. Obtain the expression for the power of this combination in terms of the focal lengths of the lenses.  
 (b) A ray of light passing from air through an equilateral glass prism undergoes minimum deviation when the angle of incidence is  $\frac{3}{4}$ th of the angle of prism. Calculate the speed of light in the prism. [CBSE-2017]
42. (b) The figure shows a ray of light falling normally on the face AB of an equilateral glass prism having refractive index  $\frac{3}{2}$ , placed in water of refractive index  $\frac{4}{3}$ . Will this ray suffer total internal reflection on striking the face AC? Justify your answer. [CBSE-2018]



43. A symmetric biconvex lens of radius of curvature  $R$  and made of glass of refractive index 1.5, is placed on a layer of liquid placed on top of a plane mirror as shown in the figure. An optical needle with its tip on the principal axis of the lens is moved along the axis until its real, inverted image coincides with the needle itself. The distance of the needle from the lens is measured to be  $x$ . On removing the liquid layer and repeating the experiment, the distance is found to be  $y$ . Obtain the expression for the refractive index of the liquid in terms of  $x$  and  $y$ . [CBSE-2018]



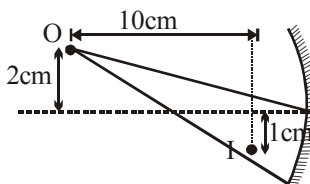
44. (a) Draw a ray diagram to show image formation when the concave mirror produces a real, inverted and magnified image of the object.  
 (b) Obtain the mirror formula and write the expression for the linear magnification.  
 (c) Explain two advantages of a reflecting telescope over a refracting telescope. [CBSE-2018]

# ANSWER KEY

## EXERCISE (S-1)

1. Ans. 3m    2. Ans. 2    3. Ans. 2

4. Ans. 20 cm from object, 20/3 cm from object



5. Ans. 25

6. Ans.  $d/2$     7. Ans.  $16 \text{ mm}^3$     8. Ans. 4    9. Ans. 4    10. Ans. 1    11. Ans. 5

12. Ans. 5    13. Ans. 42 cm    14. Ans. 4    15. Ans. 8    16. Ans. (a)  $\sin^{-1}\left(\frac{1}{5}\right)$  (b) air

17. Ans.  $\theta < \sin^{-1}(2 \sin 15^\circ)$     18. Ans. (i)  $\lambda_0 = 600 \text{ nm}$ ,  $n = 1.5$  (ii)  $i = \sin^{-1}(0.75) = 48.59^\circ$

19. Ans.  $\left(\frac{8}{3} \text{ mm}\right)$

20. Ans. 2

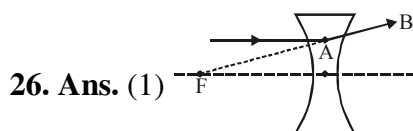
21. Ans. 5cm

22. Ans. 8

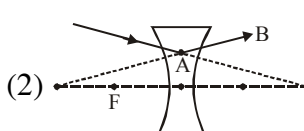
23. Ans.  $f = v = \frac{\mu_3 R}{\mu_3 - \mu_1}$

24. Ans. 30

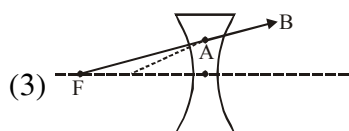
25. Ans. Real, below principal axis, anywhere b/w P &  $\infty$



26. Ans. (1)



- (2)



- (3)

27. Ans. (i) 0.2 m, (ii) 0.4 m    28. Ans.  $(\pi/4) \text{ cm}^2$     29. Ans. 3    30. Ans. 5

31. Ans. (-20 cm)    32. Ans. 17.5 cm    33. Ans. 3    34. Ans.  $y = -\frac{x}{1200} + 0.1$

35. Ans. 1    36. Ans. 4    37. Ans. (a)  $f = -20 \text{ cm}$ , (b) + 80 cm, convergent achromatic lens]

## EXERCISE (S-2)

1. Ans. 2 mm    2. Ans. (30 cm, -14 mm)    3. Ans.  $\frac{32}{9}$     4. Ans. 0180    5. Ans. 2

6. Ans.  $\vec{r} = \frac{3}{5\sqrt{2}}\hat{i} + \frac{2\sqrt{2}}{5}\hat{j} - \frac{1}{\sqrt{2}}\hat{k}$  (angle of incidence =  $60^\circ$ ;  $r = 45^\circ$ )

7. Ans.  $\frac{1.514 \times 0.4}{0.1} = 6.06 \text{ m}$  correct upto two places of decimal.    8. Ans. On the object itself

9. Ans. (i)  $\tan \theta = \frac{dy}{dx} = \cot i$ ; (ii) 1; (iii)  $y = k^2(x/4)^4$ ; (iv) 4.0, 1; (v) It will become parallel to x-axis

10. Ans.  $\frac{R}{\sqrt{3}}$     11. Ans. (a)  $\frac{2}{\sqrt{3}}$  (b) Normal to surface (c) Retrace the path,  $\sqrt{3}z + x = 10$

12. Ans. 0.09 m/s; Magnitude of the rate of change of lateral magnification is  $0.3 \text{ s}^{-1}$ .

13. Ans. 15 cm right of the mirror, magnification = 1.5; (-15 cm, -1.5 cm; -15 cm, 0.3 cm)

14. Ans.  $\frac{64}{3} \text{ cm}$     15. Ans. (i) -10 cm (ii) (10, 2)    16. Ans. (i)  $2^\circ$ , (ii)  $\frac{4\pi}{9} \text{ mm}$     17. Ans.  $\left(\frac{10}{3} \text{ m}\right)$

## EXERCISE (O-1)

1. Ans. (A) 2. Ans. (A) 3. Ans. (B) 4. Ans. (C) 5. Ans. (A) 6. Ans. (D)  
 7. Ans. (D) 8. Ans. (D) 9. Ans. (B) 10. Ans. (B) 11. Ans. (C) 12. Ans. (D)  
 13. Ans. (C) 14. Ans. (C) 15. Ans. (C) 16. Ans. (D) 17. Ans. (B) 18. Ans. (B)  
 19. Ans. (A) 20. Ans. (B) 21. Ans. (A) 22. Ans. (B) 23. Ans. (D) 24. Ans. (C)  
 25. Ans. (B) 26. Ans. (D) 27. Ans. (B) 28. Ans. (C) 29. Ans. (B) 30. Ans. (B)  
 31. Ans. (C) 32. Ans. (B) 33. Ans. (B) 34. Ans. (C) 35. Ans. (A) 36. Ans. (A)  
 37. Ans. (B) 38. Ans. (D) 39. Ans. (D) 40. Ans. (A) 41. Ans. (D) 42. Ans. (C)  
 43. Ans. (A) 44. Ans. (C) 45. Ans. (D) 46. Ans. (C) 47. Ans. (B) 48. Ans. (A)  
 49. Ans. (C) 50. Ans. (B) 51. Ans. (C) 52. Ans. (A) 53. Ans. (B) 54. Ans. (A)  
 55. Ans. (A) 56. Ans. (C) 57. Ans. (B) 58. Ans. (C) 59. Ans. (C) 60. Ans. (A)  
 61. Ans. (C) 62. Ans. (A) 63. Ans. (D) 64. Ans. (B) 65. Ans. (B) 66. Ans. (A)  
 67. Ans. (A,C,D) 68. Ans. (B, C) 69. Ans. (A,D) 70. Ans. (A,B,C)  
 71. Ans. (B,C) 72. Ans. (B,C) 73. Ans. (B,D) 74. Ans. (A,C)  
 75. Ans. (B,C,D) 76. Ans. (A,C) 77. Ans. (B,C) 78. Ans. (A,B)  
 79. Ans. (A,B,C,D) 80. Ans. (A,D) 81. Ans. (A, C) 82. Ans. (A, D)  
 83. Ans. (B) 84. Ans. (B) 85. Ans. (C) 86. Ans. (B)  
 87. Ans. (D) 88. Ans. (D) 89. Ans. (D) 90. Ans. (C)  
 91. Ans. (A)  $\rightarrow$  (Q), (B)  $\rightarrow$  (P), (C)  $\rightarrow$  (S), (D)  $\rightarrow$  (R)  
 92. Ans. (A)  $\rightarrow$  (P,S,T); (B)  $\rightarrow$  (Q,S); (C)  $\rightarrow$  (Q, R) (D)  $\rightarrow$  (Q, R)  
 93. Ans. (A)-Q; (B)-P; (C)-S; (D)-R 94. Ans. (A)-P; (B)-Q; (C)-RT; (D)-PT  
 95. Ans. (B) 96. Ans. (A)-Q,S; (B)-P,Q,R, S; (C)-P,Q,S (D)-S  
 97. Ans. (A) 98. Ans. (A) 99. Ans. (C) 100. Ans. (B) 101. Ans. (A)  
 102. Ans. (C) 103. Ans. (A) 104. Ans. (A) 105. Ans. (B) 106. Ans. (A)  
 107. Ans. (C) 108. Ans. (D) 109. Ans. (A) 110. Ans. (B) 111. Ans. (A)  
 112. Ans. (C) 113. Ans. (D)

## EXERCISE O-2

1. Ans. (A) 2. Ans. (C) 3. Ans. (D) 4. Ans. (A) 5. Ans. (D) 6. Ans. (C)  
 7. Ans. (D) 8. Ans. (A) 9. Ans. (C) 10. Ans. (A) 11. Ans. (D) 12. Ans. (A)  
 13. Ans. (A) 14. Ans. (D) 15. Ans. (A,D) 16. Ans. (A,B,C) 17. Ans. (A,D)  
 18. Ans. (B,D) 19. Ans. (A,C) 20. Ans. (B,C,D) 21. Ans. (A, B)  
 22. Ans. (A,B,C) 23. Ans. (A,B,C,D) 24. Ans. (A) 25. Ans. (B) 26. Ans. (A)

## EXERCISE - (JM)

1. Ans. (4) 2. Ans. (2) 3. Ans. (3) 4. Ans. (1) 5. Ans. (3) 6. Ans. (3)  
 7. Ans. (4) 8. Ans. (4) 9. Ans. (3) 10. Ans. (1) 11. Ans. (2) 12. Ans. (3)

## EXERCISE-(JA)

1. Ans. (B) 2. Ans. (A,B,C) 3. Ans. (A)-pr, (B) -qst, (C) -prt, (D) -qs 4. Ans. 6  
 5. Ans. 3 6. Ans. 6 7. Ans. (C) 8. Ans. 2 9. Ans. (B) 10. Ans. (C)  
 11. Ans. (B) 12. Ans. (A) 13. Ans. (C) 14. Ans. (D) 15. Ans. (A,C) 16. Ans. (C)  
 17. Ans. (B) 18. Ans. 7 19. Ans. (B) 20. Ans. 2 21. Ans. (A,C) 22. Ans. (D)  
 23. Ans. (A) 24. Ans. (A,B,D) 25. Ans. (A,D) 26. Ans. (A) 27. Ans. (A,C,D)  
 28. Ans. 8 29. Ans. 130 [129.95, 130.05] 30. Ans. (D) 31. Ans. (1,3,4)  
 32. Ans. (50.00) 33. Ans. (3,4) 34. Ans. (1.50)



## Important Notes

# Chapter Contents

## MODERN PHYSICS-1

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## MODERN PHYSICS-1

## KEY CONCEPT

## PHOTO ELECTRIC EFFECT

## PHOTOELECTRIC EFFECT

It was discovered by Hertz in 1887. He found that when the negative plate of an electric discharge tube was illuminated with ultraviolet light, the electric discharge took place more readily. Further experiments carried out by Hallwachs confirmed that certain negatively charged particles are emitted, when a Zn plate is illuminated with ultraviolet light. These particles were identified as electrons.

The phenomenon of emission of electrons from the surface of certain substances, when suitable radiations of certain frequency or wavelength are incident upon it is called photoelectric effect.

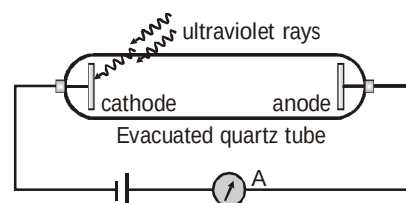
## EXPLANATION OF PHOTOELECTRIC EFFECT

- **On the basis of wave theory** : According to wave theory, light is an electromagnetic radiation consisting of oscillating electric field vectors and magnetic field vectors. When electromagnetic radiations are incident on a metal surface, the free electrons [free electrons means the electrons which are loosely bound and free to move inside the metal] absorb energy from the radiation. This occurs by the oscillations of electron under the action of electric field vector of electromagnetic radiation. When an electron acquires sufficiently high energy so that it can overcome its binding energy, it comes out from the metal.
- **On the basis of photon theory**: According to photon theory of light, light consists of particles (called photons). Each particle carries a certain amount of energy with it. This energy is given by  $E = h\nu$ , where  $h$  is the Planck's constant and  $\nu$  is the frequency. When the photons are incident on a metal surface, they collide with electrons. In some of the collisions, a photon is absorbed by an electron. Thus an electron gets energy  $h\nu$ . If this energy is greater than the binding energy of the electron, it comes out of the metal surface. The extra energy given to the electron becomes its kinetic energy.

## EXPERIMENTS

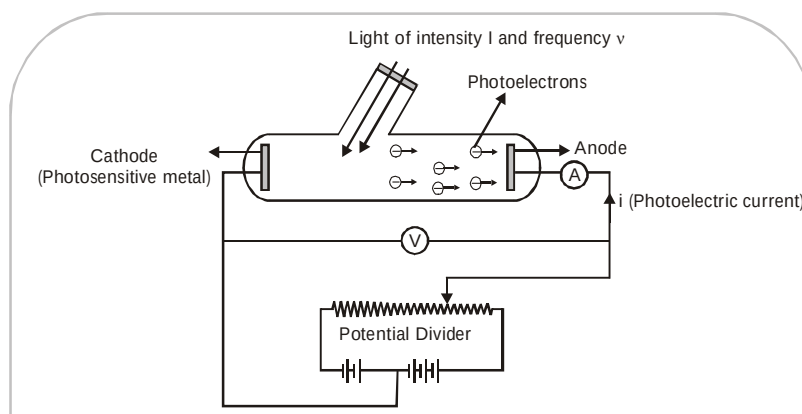
• **Hertz Experiment**

Hertz observed that when ultraviolet rays are incident on negative plate of electric discharge tube then conduction takes place easily in the tube.



- **Hallwachs experiment** : Hallwachs observed that if negatively charged Zn plate is illuminated by U.V. light, its negative charge decreases and it becomes neutral and after some time it gains positive charge. It means in the effect of light, some negative charged particles are emitted from the metal.
- **Lenard Explanation** : He told that when ultraviolet rays are incident on cathode, electrons are ejected. These electrons are attracted by anode and circuit is completed due to flow of electrons and current flows. When U.V. rays incident on anode, electrons are ejected but current does not flow. For the photo electric effect the light of short wavelength (or high frequency) is more effective than the light of long wavelength (or low frequency)

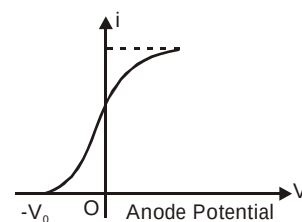
- **Experimental study of photoelectric Effect :** When light of frequency  $\nu$  and intensity  $I$  falls on the cathode, electrons are emitted from it. The electrons are collected by the anode and a current flows in the circuit. This current is called photoelectric current. This experiment is used to study the variation of photoelectric current with different factors like intensity, frequency and the potential difference between the anode and cathode.



(i) **Variation of photoelectric current with potential difference**

With the help of the above experimental setup, a graph is obtained between current and potential difference. The potential difference is varied with the help of a potential divider. The graph obtained is shown below.

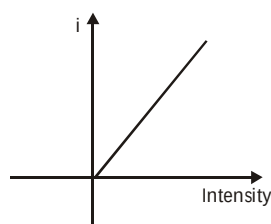
The main points of observation are :



- At zero anode potential, a current exists. It means that electrons are emitted from cathode with some kinetic energy.
- As anode potential is increased, current increases. This implies that different electrons are emitted with different kinetic energies.
- After a certain anode potential, current acquires a constant value called saturation current. Current acquires a saturation value because the number of electrons emitted per second from the cathode are fixed.
- At a certain negative potential, the photoelectric current becomes zero. This is called stopping potential ( $V_0$ ). Stopping potential is a measure of maximum kinetic energy of the emitted electrons. Let  $KE_{\max}$  be the maximum kinetic energy of an emitted electron, then  $KE_{\max} = eV_0$ .

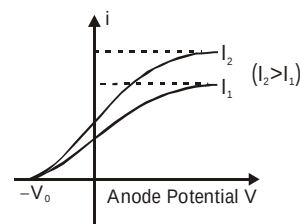
(ii) **Variation of current with intensity**

The photoelectric current is found to be directly proportional to intensity of incident radiation.



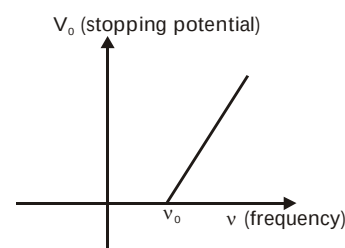
**(iii) Effect of intensity on saturation current and stopping potential**

- Saturation current increases with increase in intensity.
- Stopping potential (and therefore maximum kinetic energy) is independent of intensity.



**(iv) Effect of frequency**

- Stopping potential is found to vary with frequency of incident light linearly. Greater the frequency of incident light, greater the stopping potential.
- There exists a certain minimum frequency  $\nu_0$  below which no stopping potential is required as no emission of electrons takes place. This frequency is called threshold frequency. For photoelectric emission to take place,  $\nu > \nu_0$ .



**IMPORTANT POINTS**

- Photo electric effect is an instantaneous process, as soon as light is incident on the metal, photo electrons are emitted.
- Stopping potential does not depend on the distance between cathode and anode.
- The work function represented the energy needed to remove the least tightly bounded electrons from the surface. It depends only on nature of the metal and independent of any other factors.
- Failure of wave theory of light**
  - According to wave theory when light incident on a surface, energy is distributed continuously over the surface. So that electron has to wait to gain sufficient energy to come out. But in experiment there is no time lag. Emission of electrons takes place in less than  $10^{-9}$  s. This means, electron does not absorb energy. They get all the energy once.
  - When intensity is increased, more energetic electrons should be emitted. So that stopping potential should be intensity dependent. But it is not observed.
  - According to wave theory, if intensity is sufficient then, at each frequency, electron emission is possible. It means there should not be existence of threshold frequency.

**Einstein's Explanation of Photoelectric Effect**

Einstein explained photoelectric effect on the basis of photon–electron interaction. The energy transfer takes place due to collisions between an electrons and a photon. The electrons within the target material are held there by electric force. The electron needs a certain minimum energy to escape from this pull. This minimum energy is the property of target material and it is called the work function. When a photon of energy  $E = h\nu$  collides with and transfers its energy to an electron, and this energy is greater than the work function, the electron can escape through the surface.

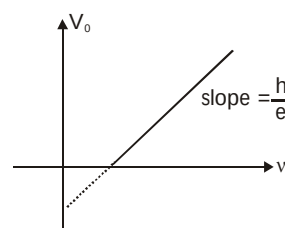
**Einstein's Photoelectric Equation**  $h\nu = \phi + KE_{\max}$

Here  $h\nu$  is the energy transferred to the electron. Out of this,  $\phi$  is the energy needed to escape.

The remaining energy appears as kinetic energy of the electron.

Now  $KE_{\max} = eV_0$  (where  $V_0$  is stopping potential)

$$\therefore h\nu = \phi + eV_0 \Rightarrow V_0 = \left(\frac{h}{e}\right)\nu - \frac{\phi}{e}$$



Thus, the stopping potential varies linearly with the frequency of incident radiation.

Slope of the graph obtained is  $\frac{h}{e}$ . This graph helps in determination of Planck's constant.

### IMPORTANT POINTS

- Einstein's Photo Electric equation is based on conservation of energy.
- Einstein explained P.E.E. on the basis of quantum theory, for which he was awarded noble prize.
- According to Einstein one photon can eject one  $e^-$  only. But here the energy of incident photon should greater or equal to work function.
- In photoelectric effect all photoelectrons do not have same kinetic energy. Their KE range from zero to  $E_{\max}$  which depends on frequency of incident radiation and nature of cathode.
- The photo electric effect takes place only when photons strike bound electrons because for free electrons energy and momentum conservations do not hold together.

**Ex.** Calculate the possible velocity of a photoelectron if the work function of the target material is 1.24 eV and wavelength of light is  $4.36 \times 10^{-7}$  m. What retarding potential is necessary to stop the emission of electrons?

**Sol.** As  $KE_{\max} = h\nu - \phi \Rightarrow \frac{1}{2}mv_{\max}^2 = h\nu - \phi = \frac{hc}{\lambda} - \phi$

$$v_{\max} = \sqrt{\frac{2\left(\frac{hc}{\lambda} - \phi\right)}{m}} = \sqrt{\frac{2\left(\frac{6.63 \times 10^{-34} \times 3 \times 10^8}{4.36 \times 10^{-7}} - 1.24 \times 1.6 \times 10^{-19}\right)}{9.11 \times 10^{-31}}} = 7.523 \times 10^5 \text{ m/s}$$

$\therefore$  The speed of a photoelectron can be any value between 0 and  $7.43 \times 10^5$  m/s

If  $V_0$  is the stopping potential, then  $eV_0 = \frac{1}{2}mv_{\max}^2$

$$\Rightarrow V_0 = \frac{1}{2} \frac{mv_{\max}^2}{e} = \frac{hc}{e\lambda} - \frac{\phi}{e} = \frac{12400}{4360} - 1.24 = 1.60 \text{ V} \quad \left[ \because \frac{hc}{e} = 12400 \times 10^{-10} \text{ V-m} \right]$$

**Ex.** The surface of a metal of work function  $\phi$  is illuminated by light whose electric field component varies with time as  $E = E_0 [1 + \cos \omega t] \sin \omega_0 t$ . Find the maximum kinetic energy of photoelectrons emitted from the surface.

**Sol.** The given electric field component is  $E = E_0 \sin \omega_0 t + E_0 \sin \omega_0 t \cos \omega t = E_0 \sin \omega_0 t + \frac{E_0}{2} [\sin(\omega_0 + \omega)t + \sin(\omega_0 - \omega)t]$

$\therefore$  The given light comprises three different frequencies viz.  $\omega$ ,  $\omega_0 + \omega$ ,  $\omega_0 - \omega$

The maximum kinetic energy will be due to most energetic photon.

$$\therefore KE_{\max} = h\nu - \phi = \frac{h(\omega_0 + \omega)}{2\pi} - \phi \quad \left( \because \omega = 2\pi\nu \text{ or } \nu = \frac{\omega}{2\pi} \right)$$

**Ex.** When light of wavelength  $\lambda$  is incident on a metal surface, stopping potential is found to be  $x$ . When

light of wavelength  $n\lambda$  is incident on the same metal surface, stopping potential is found to be  $\frac{x}{n+1}$ .

Find the threshold wavelength of the metal.

**Sol.** Let  $\lambda_0$  is the threshold wavelength. The work function is  $\phi = \frac{hc}{\lambda_0}$ .

Now, by photoelectric equation  $ex = \frac{hc}{\lambda} - \frac{hc}{\lambda_0} \dots (i) \quad \frac{ex}{n+1} = \frac{hc}{n\lambda} - \frac{hc}{\lambda_0} \dots (ii)$

From (i) and (ii)  $\frac{hc}{\lambda} - \frac{hc}{\lambda_0} = (n+1) \frac{hc}{n\lambda} - (n+1) \frac{hc}{\lambda_0} \Rightarrow \frac{nhc}{\lambda_0} = \frac{hc}{n\lambda} \Rightarrow \lambda_0 = n^2\lambda$

### PHOTON THEORY OF LIGHT

- A photon is a particle of light moving with speed 299792458 m/s in vacuum.
- The speed of a photon is independent of frame of reference. This is the basic postulate of theory of relativity.
- The rest mass of a photon is zero. i.e. photons do not exist at rest.

Effective mass of photon  $m = \frac{E}{c^2} = \frac{hc}{c^2\lambda} = \frac{h}{c\lambda}$  i.e.  $m \propto \frac{1}{\lambda}$

So mass of violet light photon is greater than the mass of red light photon. ( $\because \lambda_R > \lambda_V$ )

- According to Planck the energy of a photon is directly proportional to the frequency of the radiation.

$E \propto \nu$  or  $E = h\nu$

$E = \frac{hc}{\lambda}$  joule ( $\because c = \nu\lambda$ ) or  $E = \frac{hc}{\lambda e} = \frac{12400}{\lambda} \text{ eV} - \text{\AA}$

$\left[ \because \frac{hc}{e} = 12400(\text{\AA} - \text{eV}) \right]$

Here  $E$  = energy of photon,  $c$  = speed of light,  $h$  = Planck's constant,  $e$  = charge of electron

$h = 6.62 \times 10^{-34} \text{ J-s}$ ,  $\nu$  = frequency of photon,  $\lambda$  = wavelength of photon

- Linear momentum of photon  $p = \frac{E}{c} = \frac{h\nu}{c} = \frac{h}{\lambda}$
- A photon can collide with material particles like electron. During these collisions, the total energy and total momentum remain constant.
- Energy of light passing through per unit area per unit time is known as intensity of light.

Intensity of light  $I = \frac{E}{At} = \frac{P}{A} \dots (i)$

Here  $P$  = power of source,

$A$  = Area,  $t$  = time taken

$E$  = energy incident in  $t$  time =  $Nh\nu$ ,

$N$  = number of photon incident in  $t$  time

Intensity  $I = \frac{N(h\nu)}{At} = \frac{n(h\nu)}{A} \dots (ii)$

$\left[ \because n = \frac{N}{t} = \text{no. of photon per sec.} \right]$

From equation (i) and (ii),  $\frac{P}{A} = \frac{n(h\nu)}{A} \Rightarrow n = \frac{P}{h\nu} = \frac{P\lambda}{hc} = 5 \times 10^{24} \text{ J}^{-1} \text{ m}^{-1} \times P \times \lambda$

- When photons fall on a surface, they exert a force and pressure on the surface. This pressure is called radiation pressure.

- Force exerted on perfectly reflecting surface**

Let 'N' photons are there in time  $t$ ,

$$\text{Momentum before striking the surface } (p_1) = \frac{Nh}{\lambda}$$

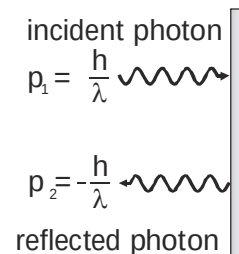
$$\text{Momentum after striking the surface } (p_2) = -\frac{Nh}{\lambda}$$

$$\text{Change in momentum of photons} = p_2 - p_1 = \frac{-2Nh}{\lambda}$$

$$\text{But change in momentum of surface} = \Delta p = \frac{2Nh}{\lambda}$$

$$\text{So that force on surface } F = \frac{2Nh}{t\lambda} = n \left[ \frac{2h}{\lambda} \right] \text{ but } n = \frac{P\lambda}{hc}$$

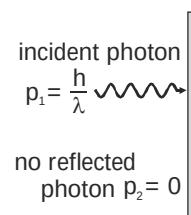
$$\therefore F = \frac{2h}{\lambda} \times \frac{P\lambda}{hc} = \frac{2P}{c} \text{ and Pressure} = \frac{F}{A} = \frac{2P}{cA} = \frac{2I}{c} \left[ \because I = \frac{P}{A} \right]$$



- Force exerted on perfectly absorbing surface**

$$F = \frac{p_1 - p_2}{t} = \frac{\frac{Nh}{\lambda} - 0}{t} = \frac{Nh}{t\lambda} = n \frac{h}{\lambda}$$

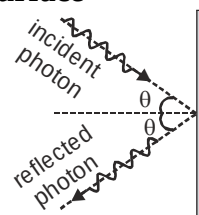
$$F = \frac{P}{c} \left( \because n = \frac{P\lambda}{hc} \right) \text{ and Pressure} = \frac{F}{A} = \frac{P}{Ac} = \frac{I}{c}$$



- When a beam of light is incident at angle  $\theta$  on perfectly reflector surface**

$$F = \frac{2P}{c} \cos \theta = n \left[ \frac{2h}{\lambda} \right] \cos \theta = \frac{2IA \cos \theta}{c}$$

$$\text{Pressure} = \frac{F}{A} = \frac{2I \cos^2 \theta}{c}$$



**Ex.** The intensity of sunlight on the surface of earth is  $1400 \text{ W/m}^2$ . Assuming the mean wavelength of sunlight to be  $6000 \text{ \AA}$ , calculate:-

- The photon flux arriving at  $1 \text{ m}^2$  area on earth perpendicular to light radiations and
- The number of photons emitted from the sun per second (Assuming the average radius of Earth's orbit to be  $1.49 \times 10^{11} \text{ m}$ )

**Sol.** (a) Energy of a photon  $E = \frac{hc}{\lambda} = \frac{12400}{6000} = 2.06 \text{ eV} = 3.3 \times 10^{-19} \text{ J}$

$$\text{Photon flux} = \frac{IA}{E} = \frac{1400 \times 1}{3.3 \times 10^{-19}} = 4.22 \times 10^{21} \text{ photons/sec.}$$

$$(b) \text{ Number of photons emitted per second } n = \frac{P}{E} = \frac{IA}{E} = \frac{1400 \times 4\pi \times (1.49 \times 10^{11})^2}{3.3 \times 10^{-19}} = 1.18 \times 10^{45}$$



**Ex.** In a photoelectric setup, a point source of light of power  $3.2 \times 10^{-3} \text{ W}$  emits monochromatic photons of energy  $5.0 \text{ eV}$ . The source is located at a distance  $0.8 \text{ m}$  from the centre of a stationary metallic sphere of work function  $3.0 \text{ eV}$  and radius  $8 \times 10^{-3} \text{ m}$ . The efficiency of photoelectron emission is one for every  $10^6$  incident photons. Assuming that the sphere is isolated and initially neutral and that photoelectrons are instantly swept away after emission, Find (i) the number of photoelectrons emitted per second. (ii) the time  $t$  after light source is switched on, at which photoelectron emission stops.

**Sol.** Energy of a single photon  $E = 5.0 \text{ eV} = 8 \times 10^{-19} \text{ J}$

Power of source  $P = 3.2 \times 10^{-3} \text{ W}$

$$\therefore \text{number of photons emitted per second } n = \frac{P}{E} = \frac{3.2 \times 10^{-3}}{8 \times 10^{-19}} = 4 \times 10^{15} / \text{s}$$

The number of photons incident per second on metal surface is  $n_0 = \frac{n}{4\pi R^2} \times \pi r^2$

$$n_0 = \frac{4 \times 10^{15}}{4\pi (0.8)^2} \times \pi (8 \times 10^{-3})^2 = 1.0 \times 10^{11} \text{ photon/s}$$

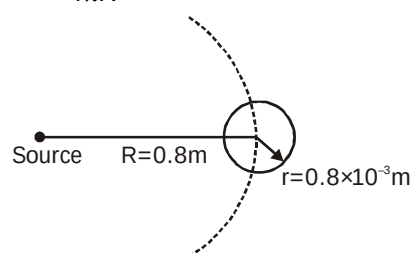
$$\text{Number of electrons emitted} = \frac{1.0 \times 10^{11}}{10^6} = 10^5 / \text{s}$$

$$KE_{\max} = h\nu - \phi = 5.0 - 3.0 = 2.0 \text{ eV}$$

The photoelectron emission stops, when the metallic sphere acquires stopping potential.

$$\text{As } KE_{\max} = 2.0 \text{ eV} \Rightarrow \text{Stopping potential } V_0 = 2\text{V} \Rightarrow 2 = \frac{q}{4\pi\epsilon_0 r} \Rightarrow q = 1.78 \times 10^{-12} \text{ C}$$

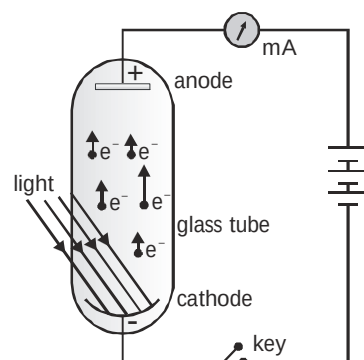
$$\text{Now charge } q = (\text{number of electrons/second}) \times t \times e \Rightarrow t = \frac{1.78 \times 10^{-12}}{10^5 \times 1.6 \times 10^{-19}} = 111 \text{ s}$$



## PHOTO CELL

A photo cell is a practical application of the phenomenon of photo electric effect, with the help of photo cell light energy is converted into electrical energy.

- **Construction :** A photo cell consists of an evacuated sealed glass tube containing anode and a concave cathode of suitable emitting material such as Cesium (Cs).
- **Working:** When light of frequency greater than the threshold frequency of cathode material falls on the cathode, photoelectrons emitted are collected by the anode and an electric current starts flowing in the external circuit. The current increase with the increase in the intensity of light. The current would stop, if the light does not fall on the cathode.
- **Application**
  - In television camera.
  - In automatic door
  - Burglar's alarm
  - Automatic switching of street light and traffic signals.



## MATTER WAVES THEORY

### DUAL NATURE OF LIGHT

Experimental phenomena of light reflection, refraction, interference, diffraction are explained only on the basis of wave theory of light. These phenomena verify the wave nature of light. Experimental phenomena of light photoelectric effect and Compton effect, pair production and positron inhalational can be explained only on the basis of the particle nature of light. These phenomena verify the particle nature of light.

It is inferred that light does not have any definite nature, rather its nature depends on its experimental phenomenon. This is known as the dual nature of light. The wave nature and particle nature both can not be possible simultaneously.

### De-Broglie HYPOTHESIS

De Broglie imagined that as light possess both wave and particle nature, similarly matter must also possess both nature, particle as well as wave. De Broglie imagined that despite particle nature of matter, waves must also be associated with material particles. Wave associated with material particles, are defined as matter waves.

- De Broglie wavelength associated with moving particles**

If a particle of mass  $m$  moving with velocity  $v$

Kinetic energy of the particle  $E = \frac{1}{2}mv^2 = \frac{p^2}{2m}$  momentum of particle  $p = mv = \sqrt{2mE}$  the wave

length associated with the particles is  $\lambda = \frac{h}{p} = \frac{h}{mv} = \frac{h}{\sqrt{2mE}}$   $\lambda \propto \frac{1}{p} \Rightarrow \lambda \propto \frac{1}{v} \Rightarrow \lambda \propto \frac{1}{\sqrt{E}}$

The order of magnitude of wave lengths associated with macroscopic particles is  $10^{-24}$  Å.

The smallest wavelength whose measurement is possible is that of  $\gamma$  - rays ( $\lambda \approx 10^{-5}$  Å). This is the reason why the wave nature of macroscopic particles is not observable.

The wavelength of matter waves associated with the microscopic particles like electron, proton, neutron,  $\alpha$  - particle, atom, molecule etc. is of the order of  $10^{-10}$  m, it is equal to the wavelength of X-rays, which is within the limit of measurement. Hence the wave nature of these particles is observable.

- De Broglie wavelength associated with the charged particles**

Let a charged particle having charge  $q$  is accelerated by potential difference  $V$ .

Kinetic energy of this particle  $E = \frac{1}{2}mv^2 = qV$  Momentum of particle  $p = mv = \sqrt{2mE} = \sqrt{2mqV}$

The De Broglie wavelength associated with charged particle  $\lambda = \frac{h}{p} = \frac{h}{\sqrt{2mE}} = \frac{h}{\sqrt{2mqV}}$

- For an Electron**  $m_e = 9.1 \times 10^{-31}$  kg,  $q = 1.6 \times 10^{-19}$  C,  $h = 6.62 \times 10^{-34}$  J-s

De Broglie wavelength associated with electron  $\lambda = \frac{6.62 \times 10^{-34}}{\sqrt{2 \times 9.1 \times 10^{-31} \times 1.6 \times 10^{-19} V}}$

$$\lambda = \frac{12.27 \times 10^{-10}}{\sqrt{V}} \text{ meter} = \frac{12.27}{\sqrt{V}} \text{ Å} \quad \text{so} \quad \lambda \propto \frac{1}{\sqrt{V}}$$

Potential difference required to stop an electron of wavelength  $\lambda$  is  $V = \frac{150.6}{\lambda^2}$  volt ( $\text{\AA}$ )<sup>2</sup>

- **For Proton**  $m_p = 1.67 \times 10^{-27}$  kg  
De Broglie wavelength associated with proton

$$\lambda_p = \frac{6.62 \times 10^{-34}}{\sqrt{2 \times 1.67 \times 10^{-27} \times 1.6 \times 10^{-19} V}}; \lambda_p = \frac{0.286 \times 10^{-10}}{\sqrt{V}} \text{ meter} = \frac{0.286}{\sqrt{V}} \text{ \AA}$$

- **For Deuteron**  $m_d = 2 \times 1.67 \times 10^{-27}$  kg,  $q_d = 1.6 \times 10^{-19}$  C

$$\lambda_d = \frac{6.62 \times 10^{-34}}{\sqrt{2 \times 2 \times 1.67 \times 10^{-27} \times 1.6 \times 10^{-19} V}} = \frac{0.202}{\sqrt{V}} \text{ \AA}$$

- **For  $\alpha$  Particles**  $q_\alpha = 2 \times 1.6 \times 10^{-19}$  C,  $m_\alpha = 4 \times 1.67 \times 10^{-27}$  kg

$$\therefore \lambda_\alpha = \frac{6.62 \times 10^{-34}}{\sqrt{2 \times 4 \times 1.67 \times 10^{-27} \times 2 \times 1.6 \times 10^{-19} V}} = \frac{0.101}{\sqrt{V}} \text{ \AA}$$

### DE BROGLIE WAVELENGTH ASSOCIATED WITH UNCHARGED PARTICLES

- **Kinetic energy of uncharged particle**  $E = \frac{1}{2}mv^2 = \frac{p^2}{2m}$   
 $m$  = mass of particle,  $v$  = velocity of particle,  $p$  = momentum of particle.
- **Velocity of uncharged particle**  $v = \sqrt{\frac{2E}{m}}$
- **Momentum of particle**  $p = mv = \sqrt{2mE}$

wavelength associated with the particle  $\lambda = \frac{h}{p} = \frac{h}{mv} = \frac{h}{\sqrt{2mE}}$

**Kinetic energy of the particle in terms of its wavelength**  $E = \frac{h^2}{2m\lambda^2} = \frac{h^2}{2m\lambda^2 \times 1.6 \times 10^{-19}} \text{ eV}$

**For a neutron**  $m_n = 1.67 \times 10^{-27}$  kg

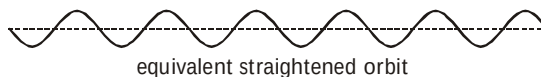
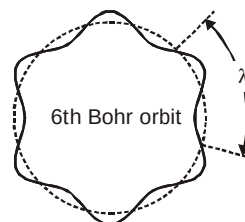
$$\therefore \lambda = \frac{6.62 \times 10^{-34}}{\sqrt{2 \times 1.67 \times 10^{-27} \times E}} = \frac{0.286 \times 10^{-10}}{\sqrt{E}} \text{ meter} \sqrt{\text{eV}} = \frac{0.286}{\sqrt{E}} \text{ \AA} \sqrt{\text{eV}}$$

### EXPLANATION OF BOHR QUANTIZATION CONDITION

According to De Broglie electron revolves round the nucleus in the form of stationary waves (i. e. wave packet) in the similar fashion as stationary waves in a vibrating string. Electron can stay in those circular orbits whose circumference is an integral multiple of De-Broglie wavelength associated with the electron,  $2\pi r = n\lambda$

$$\therefore \lambda = \frac{h}{mv} \text{ and } 2\pi r = n\lambda \therefore mvr = \frac{nh}{2\pi}$$

This is the Bohr quantizations condition.



**Ex.** Find the initial momentum of electron if the momentum of electron is changed by  $p_m$  and the De Broglie wavelength associated with it changes by 0.50 %

**Sol.**  $\frac{d\lambda}{\lambda} \times 100 = 0.5 \Rightarrow \frac{d\lambda}{\lambda} = \frac{0.5}{100} = \frac{1}{200}$  and  $\Delta p = p_m$

$$\because p = \frac{h}{\lambda}, \text{ differentiating } \frac{dp}{d\lambda} = -\frac{h}{\lambda^2} = -\frac{h}{\lambda} \times \frac{1}{\lambda} = -\frac{p}{\lambda} \Rightarrow \left| \frac{dp}{p} \right| = \frac{d\lambda}{\lambda} \therefore \frac{p_m}{p} = \frac{1}{200} \Rightarrow p = 200 p_m$$

**Ex.** An  $\alpha$ -particle moves in circular path of radius 0.83 cm in the presence of a magnetic field of 0.25 Wb/m<sup>2</sup>. Find the De Broglie wavelength associated with the particle.

**Sol.**  $\lambda = \frac{h}{p} = \frac{h}{qBr} = \frac{6.62 \times 10^{-34}}{2 \times 1.6 \times 10^{-19} \times 0.25 \times 83 \times 10^{-4}} \text{ meter} = 0.01 \text{ \AA} \quad \left[ \because \frac{mv^2}{r} = qvB \right]$

**Ex.** A proton and an  $\alpha$ -particle are accelerated through same potential difference. Find the ratio of their de- Broglie wavelength.

**Sol.**  $\lambda = \frac{h}{mv} = \frac{h}{\sqrt{2mE}} = \frac{h}{\sqrt{2mqV}} \quad [\because E = qV]$  For proton  $m_p = m, q = e$

For  $\alpha$ -particle  $m_\alpha = 4m, q = 2e,$   $\frac{\lambda_\alpha}{\lambda_p} = \sqrt{\frac{m_p q_p}{m_\alpha q_\alpha}} = \frac{1}{2\sqrt{2}}$

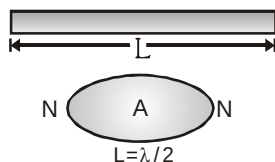
**Ex.** A particle of mass  $m$  is confined to a narrow tube of length  $L$ .

(a) Find the wavelengths of the de-Broglie wave which will resonate in the tube.

(b) Calculate the corresponding particle momenta, and

(c) Calculate the corresponding energies.

**Sol.** (a) The de-Broglie waves will resonate with a node at each end of the tube.



Few of the possible resonance forms are as follows :  $\lambda_n = \frac{2L}{n}, n=1,2,3,\dots$

(b) Since de-Broglie wavelengths are  $\lambda_n = \frac{h}{p_n}$

$$p_n = \frac{h}{\lambda_n} = \frac{nh}{2L}, n = 1, 2, 3, \dots$$



(c) The kinetic energy of the particles are  $K_n = \frac{p_n^2}{2m} = \frac{n^2 h^2}{8L^2 m}, n = 1, 2, 3, \dots$

# ATOMIC STRUCTURE

## VARIOUS MODELS FOR STRUCTURE OF ATOM

### • Dalton's Theory

Every material is composed of minute particles known as atom. Atom is indivisible i.e. it cannot be subdivided. It can neither be created nor be destroyed.

All atoms of same element are identical physically as well as chemically, whereas atoms of different elements are different in properties.

The atoms of different elements are made up of hydrogen atoms. (The radius of the heaviest atom is about 10 times that of hydrogen atom and its mass is about 250 times that of hydrogen).

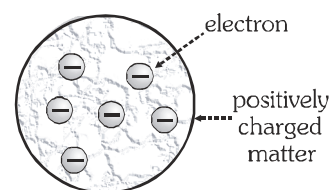
The atom is stable and electrically neutral.

### • Thomson's Atom Model

The atom as a whole is electrically neutral because the positive charge present on the atom (sphere) is equal to the negative charge of electrons present in the sphere.

Atom is a positively charged sphere of radius  $10^{-10}$  m in which electrons are embedded in between.

The positive charge and the whole mass of the atom is uniformly distributed throughout the sphere.



### • Shortcomings of Thomson's model

(i) The spectrum of atoms cannot be explained with the help of this model

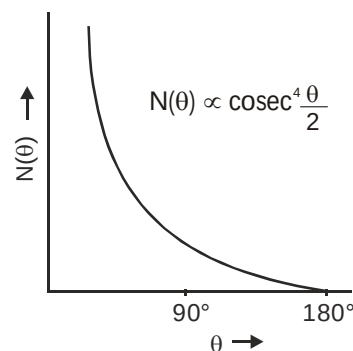
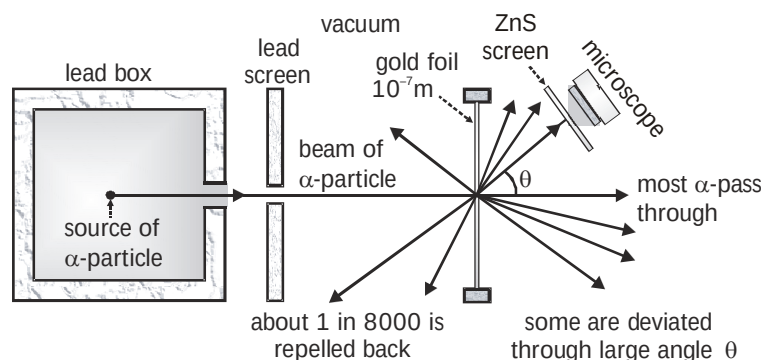
(ii) Scattering of  $\alpha$ -particles cannot be explained with the help of this model

(iii) Angular frequency of electron in  $n^{\text{th}}$  orbit  $\omega_n = \frac{8\pi^2 k^2 Z^2 e^4 m}{n^3 h^3} \Rightarrow \omega_n \propto \frac{Z^2 m}{n^3}$

## RUTHERFORD ATOM MODEL

### • Rutherford experiments on scattering of $\alpha$ – particles by thin gold foil

The experimental arrangement is shown in figure.  $\alpha$ -particles are emitted by some radioactive material (polonium), kept inside a thick lead box. A very fine beam of  $\alpha$ -particles passes through a small hole in the lead screen. This well collimated beam is then allowed to fall on a thin gold foil. While passing through the gold foil,  $\alpha$ -particles are scattered through different angles. A zinc sulphide screen was placed out the other side of the gold foil. This screen was movable, so as to receive the  $\alpha$ -particles, scattered from the gold foil at angles varying from  $0$  to  $180^\circ$ . When an  $\alpha$ -particle strikes the screen, it produces a flash of light and it is observed by the microscope. It was found that :



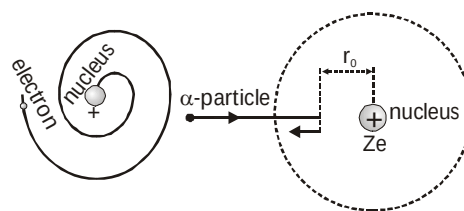
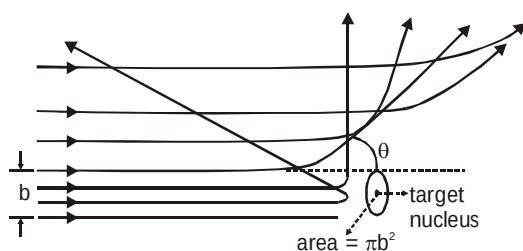
- Most of the  $\alpha$  – particles went straight through the gold foil and produced flashes on the screen as if there were nothing inside gold foil. Thus the atom is hollow.
- Few particles collided with the atoms of the foil which have scattered or deflected through considerable large angles. Few particles even turned back towards source itself.
- The entire positive charge and almost whole mass of the atom is concentrated in small centre called a nucleus.
- The electrons could not deflected the path of a  $\alpha$  – particles i.e. electrons are very light.
- Electrons revolve round the nucleus in circular orbits. So, Rutherford 1911, proposed a new type of model of the atom. According to this model, the positive charge of the atom, instead of being uniformly distributed throughout a sphere of atomic dimension is concentrated in a very small volume (Less than  $10^{-13}\text{m}$  is diameter) at its centre. This central core, now called nucleus, is surrounded by clouds of electron makes.

The entire atom is electrically neutral. According to Rutherford scattering formula, the number of  $\alpha$  – particle scattered at an angle  $\theta$  by a target are given by

$$N_{\theta} = \frac{N_0 n t (2Ze)^2}{16(4\pi\epsilon_0)^2 r^2 (mv_0^2)^2} \times \frac{1}{\sin^4 \frac{\theta}{2}}$$

Where  $N_0$  = number of  $\alpha$  – particles that strike the unit area of the scatter

- $n$  = Number of target atom per  $\text{m}^3$   
 $t$  = Thickness of target  
 $Ze$  = Charge on the target nucleus  
 $2e$  = Charge on  $\alpha$  – particle  
 $r$  = Distance of the screen from target  
 $v_0$  = Velocity of  $\alpha$  – particles at nearer distance of approach the size of a nucleus or the distance of nearer approach is given by.



$$r_0 = \frac{1}{4\pi\epsilon_0} \times \left[ \frac{(2Ze)^2}{\frac{1}{2}mv_0^2} \right] = \frac{1}{4\pi\epsilon_0} \frac{(2Ze)^2}{E_k} \quad \text{where } E_k = \text{K.E. of } \alpha\text{-particle}$$

### Bohr's Atomic Model

In 1913 Neils Bohr, a Danish Physicist, introduced a revolutionary concept i.e., the quantum concept to explain the stability of an atom. He made a simple but bold statement that "The old classical laws which are applicable to bigger bodies cannot be directly applied to the sub-atomic particles such as electrons or protons.

Bohr incorporated the following new ideas now regarded as postulates of Bohr's theory.

1. The centripetal force required for an encircling electron is provided by the electrostatic attraction

between the nucleus and the electron i.e.  $\frac{1}{4\pi\epsilon_0} \frac{(Ze)e}{r^2} = \frac{mv^2}{r} \dots(i)$

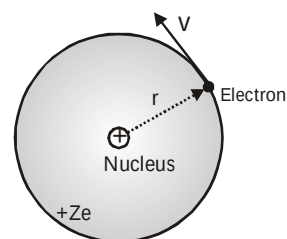
$\epsilon_0$  = Absolute permittivity of free space =  $8.85 \times 10^{-12} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}$

$m$  = Mass of electron

$v$  = Velocity (linear) of electron

$r$  = Radius of the orbit in which electron is revolving.

$Z$  = Atomic number of hydrogen like atom.



2. Electrons can revolve only in those orbits in which angular momentum of electron about nucleus is an

integral multiple of  $\frac{h}{2\pi}$  . i.e.,  $mvr = \frac{nh}{2\pi} \dots(ii)$

$n$  = Principal quantum number of the orbit in which electron is revolving.

3. Electrons in an atom can revolve only in discrete circular orbits called stationary energy levels (shells). An electron in a shell is characterised by a definite energy, angular momentum and orbit number. While in any of these orbits, an electron does not radiate energy although it is accelerated.

4. Electrons in outer orbits have greater energy than those in inner orbits.

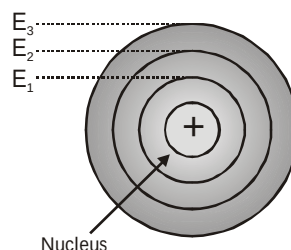
The orbiting electron emits energy when it jumps from an outer orbit (higher energy states) to an inner orbit (lower energy states) and also absorbs energy when it jumps from an inner orbit to an outer orbit.  $E_n$

$$- E_m = h\nu_{n,m}$$

where,  $E_n$  = Outer energy state

$E_m$  = Inner energy state

$\nu_{n,m}$  = Frequency of radiation



5. The energy absorbed or released is always in the form of electromagnetic radiations.

### MATHEMATICAL ANALYSIS OF BOHR'S THEORY

From above equation (i) and (ii) i.e.,  $\frac{1}{4\pi\epsilon_0} \frac{(Ze)e}{r^2} = \frac{mv^2}{r}$  and  $mvr = \frac{nh}{2\pi} \dots(ii)$

We get the following results.

1. **Velocity of electron in nth orbit :** By putting the value of  $mvr$  in equation (i) from (ii) we get

$$\frac{1}{4\pi\epsilon_0} Ze^2 = \left(\frac{nh}{2\pi}\right) \times v \Rightarrow v = \frac{Z}{n} \left[ \frac{e^2}{2\epsilon_0 h} \right] = \frac{Z}{n} v_0 \dots(iii)$$

$$\text{Where, } v_0 = \frac{(1.6 \times 10^{-19})^2}{2 \times 8.85 \times 10^{-12} \times 6.625 \times 10^{-34}} = 2.189 \times 10^6 \text{ ms}^{-1} = \frac{c}{137} = 2.2 \times 10^6 \text{ m/s}$$

where  $c = 3 \times 10^8 \text{ m/s}$  = speed of light in vacuum

2. **Radius of the nth orbit :**

From equation (iii), putting the value of  $v$  in equation (ii), we get

$$m \left( \frac{Z}{n} \times \frac{e^2}{2\epsilon_0 h} \right) r = \frac{nh}{2\pi} \Rightarrow r = \frac{n^2}{Z} \left[ \frac{\epsilon_0 h^2}{\pi m e^2} \right] = \frac{n^2}{Z} r_0 \dots(iv)$$

$$\text{where } r_0 = \frac{8.85 \times 10^{-12} \times (6.625 \times 10^{-34})^2}{3.14 \times 9.11 \times 10^{-31} \times (1.6 \times 10^{-19})^2} = 0.529 \times 10^{-10} \text{ m} \approx 0.53 \text{ \AA}$$

### 3. Total energy of electron in nth orbit :

$$\text{From equation (i) K.E.} = \frac{1}{2}mv^2 = \frac{Ze^2}{8\pi\epsilon_0 r} \text{ and PE} = \frac{1}{4\pi\epsilon_0} \frac{(Ze)(-e)}{r} = -2\text{K.E.} \therefore |\text{PE}| = 2 \text{ KE}$$

$$\text{Total energy of the system } E = \text{KE} + \text{PE} = -2\text{KE} + \text{KE} = -\text{KE} = \frac{-Ze^2}{8\pi\epsilon_0 r}$$

$$\text{By putting the value of } r \text{ from the equation (iv), we get } E = \frac{Z^2}{n^2} \left( -\frac{me^4}{8\epsilon_0^2 h^2} \right) = \frac{Z^2}{n^2} \cdot E_0 \dots (v)$$

$$\text{where } E_0 = \frac{-(9.11 \times 10^{-31})(1.6 \times 10^{-19})^4}{8 \times (8.85 \times 10^{-12})^2 \times (6.625 \times 10^{-34})^2} = -13.6 \text{ eV}$$

### 4. Time period of revolution of electron in nth orbit : $T = \frac{2\pi r}{v}$

$$\text{By putting the values of } r \text{ and } v, \text{ from (iii) and (iv) } T = \frac{n^3}{Z^2} \times \left( \frac{4\epsilon_0^2 h^3}{me^4} \right) = \frac{n^3}{Z^2} \cdot T_0$$

$$\text{where, } T_0 = \frac{4 \times (8.85 \times 10^{-12})^2 \times (6.625 \times 10^{-34})^3}{9.11 \times 10^{-31} \times (1.6 \times 10^{-19})^4} = 1.51 \times 10^{-16} \text{ second}$$

### 5. Frequency of revolution in nth orbit :

$$f = \frac{1}{T} = \frac{Z^2}{n^3} \times \frac{me^4}{4\epsilon_0^2 h^3} = \frac{Z^2}{n^3} \cdot f_0 \text{ where, } f_0 = \frac{9.11 \times 10^{-31} \times (1.6 \times 10^{-19})^4}{4 \times (8.85 \times 10^{-12})^2 \times (6.625 \times 10^{-34})^3} = 6.6 \times 10^{15} \text{ Hz}$$

### 6. Wavelength of photon

$$\Delta E = E_{n_2} - E_{n_1} = \frac{me^4}{8\epsilon_0^2 h^2} \left[ \frac{1}{n_1^2} - \frac{1}{n_2^2} \right] Z^2 = 13.6 \left[ \frac{1}{n_1^2} - \frac{1}{n_2^2} \right] Z^2 \Rightarrow \Delta E = \frac{hc}{\lambda} \Rightarrow \frac{1}{\lambda} = \bar{\nu} = \frac{me^4}{8\epsilon_0^2 h^3 c} \left[ \frac{1}{n_1^2} - \frac{1}{n_2^2} \right] Z^2$$

$$= R_\infty \left[ \frac{1}{n_1^2} - \frac{1}{n_2^2} \right] Z^2 \text{ where, } \bar{\nu} \text{ is called wave number. } R_\infty = R_H = \text{Rydberg constant}$$

$$= \frac{9.11 \times 10^{-31} \times (1.6 \times 10^{-19})^4}{8 \times (8.85 \times 10^{-12})^2 \times (6.625 \times 10^{-34})^3 \times 3 \times 10^8} = 1.097 \times 10^7 \text{ m}^{-1} = 1.097 \times 10^{-3} \text{ \AA}^{-1} \text{ (for stationary nucleus).}$$

If nucleus is not stationary (i.e., mass of nucleus is not much greater than the mass of the revolving particle like electron), then  $R = \frac{R_\infty}{1 + m/M}$  where,  $m$  = mass of revolving particle and  $M$  = mass of nucleus



## SPECTRAL SERIES OF HYDROGEN ATOM

It has been shown that the energy of the outer orbit is greater than the energy of the inner ones. When the Hydrogen atom is subjected to external energy, the electron jumps from lower energy state i.e. the hydrogen atom is excited. The excited state is unstable hence the electron return to its ground state in about  $10^{-8}$  sec. The excess of energy is now radiated in the form of radiations of different wavelength. The different wavelength constitute spectral series. Which are characteristic of atom emitting, then the wave length of different members of series can be found from the following relations

$$\bar{\nu} = \frac{1}{\lambda} = R \left[ \frac{1}{n_1^2} - \frac{1}{n_2^2} \right]$$

This relation explains the complete spectrum of hydrogen. A detailed account of the important radiations are listed below.

- **Lyman Series** : The series consist of wavelength which are emitted when electron jumps from an outer orbits to the first orbit i. e., the electronic jumps to K orbit give rise to lyman series. Here  $n_1 = 1$  &  $n_2 = 2, 3, 4, \dots, \infty$

The wavelengths of different members of Lyman series are :

- **First member** : In this case  $n_1 = 1$  and  $n_2 = 2$

$$\text{hence } \frac{1}{\lambda} = R \left[ \frac{1}{1^2} - \frac{1}{2^2} \right] = \frac{3R}{4} \Rightarrow \lambda = \frac{4}{3R} \Rightarrow \lambda = \frac{4}{3 \times 10.97 \times 10^6} = 1216 \times 10^{-10} \text{ m} = 1216 \text{ \AA}$$

- **Second member** : In this case  $n_1 = 1$  and  $n_2 = 3$  hence

$$\frac{1}{\lambda} = R \left[ \frac{1}{1^2} - \frac{1}{3^2} \right] = \frac{8R}{9} \Rightarrow \lambda = \frac{9}{8R} \Rightarrow \lambda = \frac{9}{8 \times 10.97 \times 10^6} = 1026 \times 10^{-10} \text{ m} = 1026 \text{ \AA}$$

Similarly the wavelength of the other members can be calculated.

- **Limiting members** : In this case  $n_1 = 1$  and  $n_2 = \infty$ , hence

$$\frac{1}{\lambda} = R \left[ \frac{1}{1^2} - \frac{1}{\infty^2} \right] = R \Rightarrow \lambda = \frac{1}{R} \Rightarrow \lambda = \frac{1}{10.97 \times 10^6} = 912 \times 10^{-10} \text{ m} = 912 \text{ \AA}$$

This series lies in ultraviolet region.

- **Balmer Series** : This series is consist of all wave lengths which are emitted when an electron jumps from an outer orbit to the second orbit i. e., the electron jumps to L orbit give rise to Balmer series. Here  $n_1 = 2$  and  $n_2 = 3, 4, 5, \dots, \infty$  The wavelength of different members of Balmer series.
- **First member** : In this case  $n_1 = 2$  and  $n_2 = 3$ , hence

$$\frac{1}{\lambda} = R \left[ \frac{1}{2^2} - \frac{1}{3^2} \right] = \frac{5R}{36} \Rightarrow \lambda = \frac{36}{5R} \Rightarrow \lambda = \frac{36}{5 \times 10.97 \times 10^6} = 6563 \times 10^{-10} \text{ m} = 6563 \text{ \AA}$$

- **Second member** : In this case  $n_1 = 2$  and  $n_2 = 4$ , hence.

$$\frac{1}{\lambda} = R \left[ \frac{1}{2^2} - \frac{1}{4^2} \right] = \frac{3R}{16} \Rightarrow \lambda = \frac{16}{3R} \Rightarrow \lambda = \frac{16}{3 \times 10.97 \times 10^6} = 4861 \times 10^{-10} \text{ m} = 4861 \text{ \AA}$$

- **Limiting members**: In this case  $n_1 = 2$  and  $n_2 = \infty$ , hence  $\frac{1}{\lambda} = R \left[ \frac{1}{2^2} - \frac{1}{\infty} \right] = \frac{R}{4} \Rightarrow \lambda = \frac{4}{R} = 3646 \text{ \AA}$

This series lies in visible and near ultraviolet region.

- **Paschen Series :** This series consist of all wavelength are emitted when an electron jumps from an outer orbit to the third orbit i. e., the electron jumps to M orbit give rise to paschen series. Here  $n_1=3$  &  $n_2=4, 5, 6, \dots, \infty$ . The different wavelengths of this series can be obtained from the formula  $\frac{1}{\lambda} = R \left[ \frac{1}{3^2} - \frac{1}{n_2^2} \right]$

where  $n_2 = 4, 5, 6, \dots, \infty$

For the first member, the wavelength is  $18750\text{\AA}$ . This series lies in infra-red region.

- **Brackett Series :** This series is consist of all wavelengths which are emitted when an electron jumps from an outer orbits to the fourth orbit i. e., the electron jumps to N orbit give rise to Brackett series. Here  $n_1 = 4$  &  $n_2 = 5, 6, 7, \dots, \infty$ .

The different wavelengths of this series can be obtained from the formula  $\frac{1}{\lambda} = R \left[ \frac{1}{4^2} - \frac{1}{n_2^2} \right]$

where  $n_2 = 5, 6, 7, \dots, \infty$

This series lies in infra-red region of spectrum.

- **Pfund series :** The series consist of all wavelengths which are emitted when an electron jumps from an outer orbit to the fifth orbit i. e., the electron jumps to O orbit give right to Pfund series. Here  $n_1 = 5$  and  $n_2 = 6, 7, 8, \dots, \infty$ .

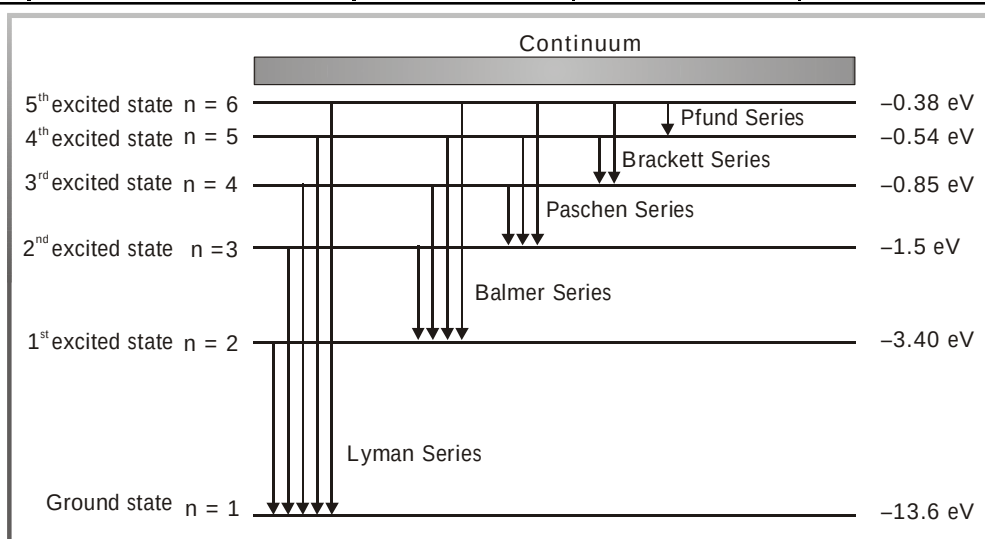
The different wavelengths of this series can be obtained from the formula  $\frac{1}{\lambda} = R \left[ \frac{1}{5^2} - \frac{1}{n_2^2} \right]$

where  $n_2 = 6, 7, 8, \dots, \infty$

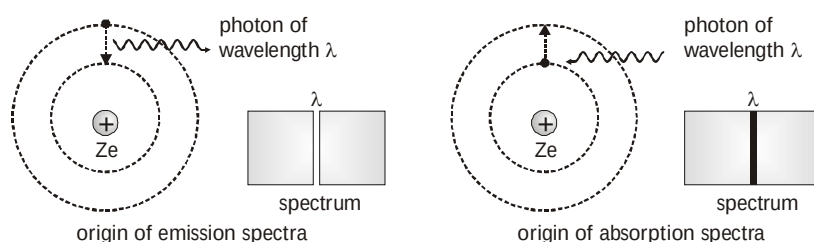
This series lies in infra-red region of spectrum.

The result are tabulated below

| S. No. | Series                | Value of $n_1$ | Value of $n_2$    | Position in the Spectrum |
|--------|-----------------------|----------------|-------------------|--------------------------|
| 1.     | Observed Lyman Series | 1              | 2,3,4... $\infty$ | Ultra Violet             |
| 2.     | Balmer Series         | 2              | 3,4,5... $\infty$ | Visible                  |
| 3.     | Paschen Series        | 3              | 4,5,6... $\infty$ | Infra-red                |
| 4.     | Brackett Series       | 4              | 5,6,7... $\infty$ | Infra-red                |
| 5.     | Pfund Series          | 5              | 6,7,8... $\infty$ | Infra-red                |



Consider the case of simplest atom i. e., hydrogen atom, this has one electron in the innermost orbit i.e., ( $n = 1$ ) and is said to be in the unexcited or normal state. If by some means, sufficient energy is supplied to the electron. It moves to higher energy states. When the atom is in a state of a high energy it is said to be excited. The process of raising or transferring the electron from lower energy state is called excitation. When by the process of excitation, the electron is completely removed from the atom. The atom is said to be ionized. Now the atom has left with a positive charge. Thus the process of raising the atom from the normal state to the ionized state is called ionisation. The process of excitation and ionisation both are absorption phenomena. The excited state is not stationary state and lasts in a very short interval of time ( $10^{-8}$  sec) because the electron under the attractive force of the nucleus jumps to the lower permitted orbit. This is accompanied by the emission of radiation according to BOHR'S frequency condition.



The energy necessary to excite an atom can be supplied in a number of ways. The most commonly kinetic energy (Wholly or partly) of the electrons is transferred to the atom. The atom is now in an excited state. The minimum potential  $V$  required to accelerate the bombarding electrons to cause excitation from the ground state is called the resonance potential. The various values of potential to cause excitation of higher state called **excitation potential**. The potential necessary to accelerate the bombarding electrons to cause ionisation is called the **ionization potential**. The term critical potential is used to include the resonance, excitation and ionisation potential. We have seen that the energy required to excite the electron from first to second state is  $13.6 - 3.4 = 10.2$  eV. from first to third state is  $13.6 - 1.5 = 12.1$  eV., and so on. The energy required to ionise hydrogen atom is  $0 - (-13.6) = 13.6$  eV. Hence ionization potential of hydrogen atom is 13.6 volt.

## SUCCESSSES AND LIMITATIONS

Bohr showed that Planck's quantum ideas were a necessary element of the atomic theory. He introduced the idea of quantized energy levels and explained the emission or absorption of radiations as being due to the transition of an electron from one level to another. As a model for even multielectron atoms, the Bohr picture is still useful. It leads to a good, simple, rational ordering of the electrons in larger atoms and quantitatively helps to predict a greater deal about chemical behavior and spectral detail.

### Bohr's theory is unable to explain the following facts :

- The spectral lines of hydrogen atom are not single lines but each one is a collection of several closely spaced lines.
- The structure of multielectron atoms is not explained.
- No explanation for using the principles of quantization of angular momentum.
- No explanation for Zeeman effect. If a substance which gives a line emission spectrum is placed in a

magnetic field, the lines of the spectrum get splitted up into a number of closely spaced lines. This phenomenon is known as Zeeman effect.

**Ex.** A hydrogen like atom of atomic number  $Z$  is in an excited state of quantum number  $2n$ . It can emit a maximum energy photon of 204 eV. If it makes a transition to quantum state  $n$ , a photon of energy 40.8 eV is emitted. Find  $n$ ,  $Z$  and the ground state energy (in eV) for this atom. Also, calculate the minimum energy (in eV) that can be emitted by this atom during de-excitation. Ground state energy of hydrogen atom is  $-13.6$  eV.

**Sol.** The energy released during de-excitation in hydrogen like atoms is given by :

$$E_{n_2} - E_{n_1} = \frac{me^4}{8\epsilon_0^2 h^2} \left[ \frac{1}{n_1^2} - \frac{1}{n_2^2} \right] Z^2$$

Energy released in de-excitation will be maximum if transition takes place from  $n$ th energy level to ground state i.e.,

$$E_{2n} - E_1 = 13.6 \left[ \frac{1}{1^2} - \frac{1}{(2n)^2} \right] Z^2 = 204 \text{ eV} \dots (i) \quad \& \quad \text{also } E_{2n} - E_n = 13.6 \left[ \frac{1}{n^2} - \frac{1}{(2n)^2} \right] Z^2 = 40.8 \text{ eV} \dots (ii)$$

Taking ratio of (i) to (ii), we will get  $\frac{4n^2 - 1}{3} = 5 \Rightarrow n^2 = 4 \Rightarrow n = 2$

Putting  $n=2$  in equation (i) we get

$$Z^2 = \frac{204 \times 16}{13.6 \times 15} \Rightarrow Z = 4 \therefore E_n = -13.6 \frac{Z^2}{n^2} \Rightarrow E_1 = -13.6 \times \frac{4^2}{1^2} = -217.6 \text{ eV} = \text{ground state energy}$$

$\Delta E$  is minimum if transition will be from  $2n$  to  $2n-1$  i.e. between last two adjacent energy levels.

$$\therefore \Delta E_{\min} = E_{2n} - E_{2n-1} = 13.6 \left[ \frac{1}{3^2} - \frac{1}{4^2} \right] 4^2 = 10.57 \text{ eV}$$

is the minimum amount of energy released during de-excitation.

**Ex.** A single electron orbits around a stationary nucleus of charge  $+Ze$  where  $Z$  is a constant and  $e$  is the magnitude of electronic charge. It requires 47.2 eV to excite the electron from the second orbit to third orbit. Find

(i) The value of  $Z$

(ii) The energy required to excite the electron from the third to the fourth Bohr orbit

(iii) The wavelength of electronic radiation required to remove the electron from first Bohr orbit to infinity

(iv) Find the K.E., P.E. and angular momentum of electron in the 1st Bohr orbit.

[ The ionization energy of hydrogen atom = 13.6 eV, Bohr radius =  $5.3 \times 10^{-11}$  m,

Velocity of light =  $3 \times 10^8$  m/s. Planck's constant =  $6.6 \times 10^{-34}$  J-s ]

**Sol.** The energy required to excite the electron from  $n_1$  to  $n_2$  orbit revolving around the nucleus with

$$\text{charge } +Ze \text{ is given by } E_{n_2} - E_{n_1} = \frac{me^4}{8\epsilon_0^2 h^2} \left[ \frac{1}{n_1^2} - \frac{1}{n_2^2} \right] Z^2 \Rightarrow E_{n_2} - E_{n_1} = Z^2 \times (13.6) \left[ \frac{1}{n_1^2} - \frac{1}{n_2^2} \right]$$

(i) Since 47.2 eV energy is required to excite the electron from  $n_1 = 2$  to  $n_2 = 3$  orbit

$$47.2 = Z^2 \times 13.6 \left[ \frac{1}{2^2} - \frac{1}{3^2} \right] \Rightarrow Z^2 = \frac{47.2 \times 36}{13.6 \times 5} = 24.988 \approx 25 \Rightarrow Z=5$$

(ii) The energy required to excite the electron from  $n_1=3$  to  $n_2=4$  is given by

$$E_4 - E_3 = 13.6 Z^2 \left[ \frac{1}{3^2} - \frac{1}{4^2} \right] = \frac{25 \times 13.6 \times 7}{144} = 16.53 \text{ eV}$$

(iii) The energy required to remove the electron from the first Bohr orbit to infinity ( $\infty$ ) is given by

$$E_\infty - E_3 = 13.6 \times Z^2 \left[ \frac{1}{1^2} - \frac{1}{\infty^2} \right] = 13.6 \times 25 \text{ eV} = 340 \text{ eV}$$

In order to calculate the wavelength of radiation, we use Bohr's frequency relation

$$hf = \frac{hc}{\lambda} = 13.6 \times 25 \times (1.6 \times 10^{-19}) \text{ J} \Rightarrow \lambda = \frac{(6.6 \times 10^{-34}) \times 10^8 \times 3}{13.6 \times 25 \times (1.6 \times 10^{-19})} = 36.397 \text{ \AA}$$

$$(iv) \text{K.E.} = \frac{1}{2} mv_1^2 = \frac{1}{2} \times \frac{Ze^2}{4\pi\epsilon_0 r_1} = 543.4 \times 10^{-19} \text{ J}$$

$$\text{P.E.} = -2 \times \text{K.E.} = -1086.8 \times 10^{-19} \text{ J}$$

$$\text{Angular momentum} = mv_1 r_1 = \frac{h}{2\pi} = 1.05 \times 10^{-34} \text{ Js}$$

The radius  $r_1$  of the first Bohr orbit is given by

$$r_1 = \frac{\epsilon_0 h^2}{\pi m e^2} \cdot \frac{1}{Z} = \frac{0.53 \times 10^{-10}}{5} \left( \because \frac{\epsilon_0 h^2}{\pi m e^2} = 0.53 \times 10^{-10} \text{ m} \right) = 1.106 \times 10^{-10} \text{ m} = 0.106 \text{ \AA}$$

**Ex.** An isolated hydrogen atom emits a photon of 10.2 eV.

(i) Determine the momentum of photon emitted (ii) Calculate the recoil momentum of the atom

(iii) Find the kinetic energy of the recoil atom [Mass of proton =  $m_p = 1.67 \times 10^{-27} \text{ kg}$ ]

**Sol.** (i) Momentum of the photon is  $p_1 = \frac{E}{c} = \frac{10.2 \times 1.6 \times 10^{-19}}{3 \times 10^8} = 5.44 \times 10^{-27} \text{ kg-m/s}$

(ii) Applying the momentum conservation



$$p_2 = p_1 = 5.44 \times 10^{-27} \text{ kg-m/s}$$

(iii)  $K = \frac{1}{2} mv^2$  ( $v$  = recoil speed of atom,  $m$  = mass of hydrogen atom)  $K = \frac{1}{2} m \left( \frac{p}{m} \right)^2 = \frac{p^2}{2m}$

Substituting the value of the momentum of atom, we get  $K = \frac{(5.44 \times 10^{-27})^2}{2 \times 1.67 \times 10^{-27}} = 8.86 \times 10^{-27} \text{ J}$

**Physical quantity**

**Formula**

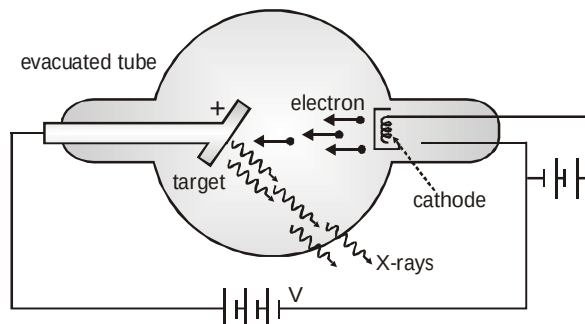
**Ratio Formulae of hydrogen atom**

|                                                              |                                                                                        |                                                                                                      |
|--------------------------------------------------------------|----------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| Radius of Bohr orbit ( $r_n$ )                               | $r_n = \frac{n^2 h^2}{4\pi^2 m k Z e^2}$ ; $r_n = 0.53 \frac{n^2}{Z} \text{ \AA}$      | $r_1 : r_2 : r_3 \dots r_n = 1 : 4 : 9 \dots n^2$                                                    |
| Velocity of electron in $n^{\text{th}}$ Bohr orbit ( $v_n$ ) | $v_n = \frac{2\pi k Z e^2}{nh}$ ; $v_n = 2.2 \times 10^6 \frac{Z}{n}$                  | $v_1 : v_2 : v_3 \dots v_n = 1 : \frac{1}{2} : \frac{1}{3} \dots \frac{1}{n}$                        |
| Momentum of electron ( $p_n$ )                               | $p_n = \frac{2\pi m k Z e^2}{nh}$ ; $p_n \propto \frac{Z}{n}$                          | $p_1 : p_2 : p_3 \dots p_n = 1 : \frac{1}{2} : \frac{1}{3} \dots \frac{1}{n}$                        |
| Angular velocity of electron ( $\omega_n$ )                  | $\omega_n = \frac{8\pi^3 k^2 Z^2 m c^4}{n^3 h^3}$ ; $\omega_n \propto \frac{Z^2}{n^3}$ | $\omega_1 : \omega_2 : \omega_3 \dots \omega_n = 1 : \frac{1}{8} : \frac{1}{27} \dots \frac{1}{n^3}$ |
| Time Period of electron ( $T_n$ )                            | $T_n = \frac{n^3 h^3}{4\pi^2 k^2 Z^2 m e^4}$ ; $T_n \propto \frac{n^3}{Z^2}$           | $T_1 : T_2 : T_3 \dots T_n = 1 : 8 : 27 : \dots : n^3$                                               |
| Frequency ( $f_n$ )                                          | $f_n = \frac{4\pi^2 k^2 Z^2 e^4 m}{n^3 h^3}$ ; $f_n \propto \frac{Z^2}{n^3}$           | $f_1 : f_2 : f_3 \dots f_n = 1 : \frac{1}{8} : \frac{1}{27} \dots \frac{1}{n^3}$                     |
| Orbital current ( $I_n$ )                                    | $I_n = \frac{4\pi^2 k^2 Z^2 m e^5}{n^3 h^3}$ ; $I_n \propto \frac{Z^2}{n^3}$           | $I_1 : I_2 : I_3 \dots I_n = 1 : \frac{1}{8} : \frac{1}{27} \dots \frac{1}{n^3}$                     |
| Angular momentum ( $J_n$ )                                   | $J_n = \frac{nh}{2\pi}$ ; $J_n \propto n$                                              | $J_1 : J_2 : J_3 \dots J_n = 1 : 2 : 3 \dots n$                                                      |
| Centripetal acceleration ( $a_n$ )                           | $a_n = \frac{16\pi^4 k^3 Z^3 m e^6}{n^4 h^4}$ ; $a_n \propto \frac{Z^4}{n^4}$          | $a_1 : a_2 : a_3 \dots a_n = 1 : \frac{1}{16} : \frac{1}{81} \dots \frac{1}{n^4}$                    |
| Kinetic energy ( $E_{K_n}$ )                                 | $E_{K_n} = \frac{RchZ^2}{n^2}$ ; $E_{K_n} \propto \frac{Z^2}{n^2}$                     | $E_{K_1} : E_{K_2} \dots E_{K_n} = 1 : \frac{1}{4} : \frac{1}{9} \dots \frac{1}{n^2}$                |
| Potential energy ( $U_n$ )                                   | $U_n = \frac{-2RchZ^2}{n^2}$ ; $U_n \propto \frac{Z^2}{n^2}$                           | $U_1 : U_2 : U_3 \dots U_n = 1 : \frac{1}{4} : \frac{1}{9} \dots \frac{1}{n^2}$                      |
| Total energy ( $E_n$ )                                       | $E_n = \frac{-RchZ^2}{n^2}$ ; $E_n \propto \frac{Z^2}{n^2}$                            | $E_1 : E_2 : E_3 \dots E_n = 1 : \frac{1}{4} : \frac{1}{9} \dots \frac{1}{n^2}$                      |

# X-RAYS

## ROENTGEN EXPERIMENT

Roentgen discovered X-ray. While performing experiment on electric discharge tube Roentgen observed that when pressure inside the tube is  $10^{-3}$  mm of Hg and applied potential is kept 25kV then some unknown radiation are emitted by anode. These are known as X-ray. X-rays are produced by bombarding high speed electrons on a target of high atomic weight and high melting point.

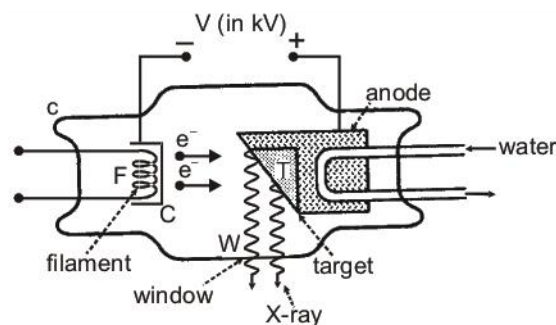


### To Produce X-ray Three Things are Required

- Source of electron
- Means of accelerating these electron to high speed
- Target on which these high speed electron strike

## COOLIDGE METHOD

Coolidge developed thermoionic vacuum X-ray tube in which electron are produced by thermoionic emission method. Due to high potential difference electrons (emitted due to thermoionic method) move towards the target and strike from the atom of target due to which X-ray are produced. Experimentally it is observed that only 1% or 2% kinetic energy of electron beam is used to produce X-ray. Rest of energy is wasted in form of heat.



### Characteristics of target

- Must have high atomic number to produce hard X-rays.
  - High melting point to withstand high temperature produced.
  - High thermal conductivity to remove the heat produced
  - Tantalum, platinum, molybdenum and tungsten serve as target materials
- Control of intensity :** The intensity of X-ray depends on number of electrons striking the target and number of electron depend on temperature of filament which can be controlled by filament current. Thus intensity of X-ray depends on current flowing through filament.
  - Control of Penetrating Power:** The Penetrating power of X-ray depends on the energy of incident electron. The energy of electron can be controlled by applied potential difference. Thus penetrating power of X-ray depend on applied potential difference. Thus the intensity of X-ray depends on current flowing through filament while penetrating power depends on applied potential difference

|                   | Soft X-ray                   | Hard X-ray                   |
|-------------------|------------------------------|------------------------------|
| Wavelength        | 10 Å to 100 Å                | 0.1 Å – 10 Å                 |
| Energy            | $\frac{12400}{\lambda}$ eV-Å | $\frac{12400}{\lambda}$ eV-Å |
| Penetrating power | Less                         | More                         |
| Use               | Radio photography            | Radio therapy                |

- Continuous spectrum of X-ray :** When high speed electron collides from the atom of target and passes close to the nucleus. There is coulomb attractive force due to this electron is deaccelerated i.e. energy is decreased. The loss of energy during deacceleration is emitted in the form of X-rays. X-ray produced in this way are called Braking or Bremstrahlung radiation and form continuous spectrum. In continuous spectrum of X-ray all the wavelength of X-ray are present but below a minimum value of wavelength there is no X-ray. It is called cut off or threshold or minimum wavelength of X-ray. The minimum wavelength depends on applied potential.
- Loss in Kinetic Energy**

$$\frac{1}{2}mv_1^2 - \frac{1}{2}mv_2^2 = h\nu + \text{heat energy if } v_2 = 0, v_1 = v$$

(In first collision, heat = 0)

$$\frac{1}{2}mv^2 = h\nu_{\max} \quad \dots(i)$$

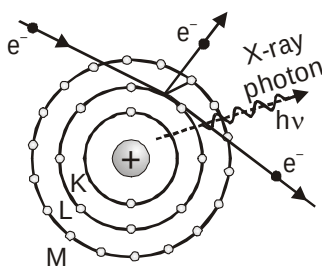
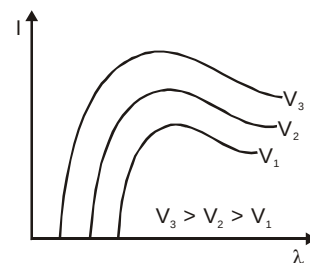
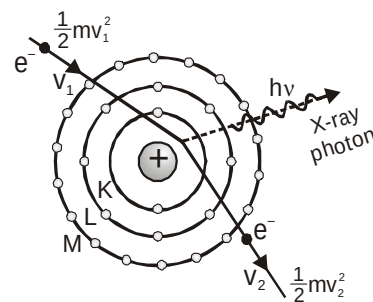
$$\frac{1}{2}mv^2 = eV \quad \dots(ii) \text{ [here } V \text{ is applied potential]}$$

$$\text{from (i) and (ii) } h\nu_{\max} = eV \Rightarrow \frac{hc}{\lambda_{\min}} = eV \Rightarrow \lambda_{\min} = \frac{12400}{V} \times \text{volt} = \frac{12400}{V} \times 10^{-10} \text{ m} \times \text{volt}$$

Continuous X-rays also known as white X-ray. Minimum wavelength of these spectrum only depends on applied potential and doesn't depend on atomic number.

- Characteristic Spectrum of X-ray**

When the target of X-ray tube is collide by energetic electron it emits two type of X-ray radiation. One of them has a continuous spectrum whose wavelength depend on applied potential while other consists of spectral lines whose wavelength depend on nature of target. The radiation forming the line spectrum is called characteristic X-rays. When highly accelerated electron strikes with the atom of target then it knockout the electron of orbit, due to this a vacancy is created. To fill this vacancy electron jumps from higher energy level and electromagnetic radiation are emitted which form characteristic spectrum of X-ray. Whose wavelength depends on nature of target and not on applied potential.





### From Bohr Model

$n_1 = 1,$        $n_2 = 2, 3, 4, \dots$  K series

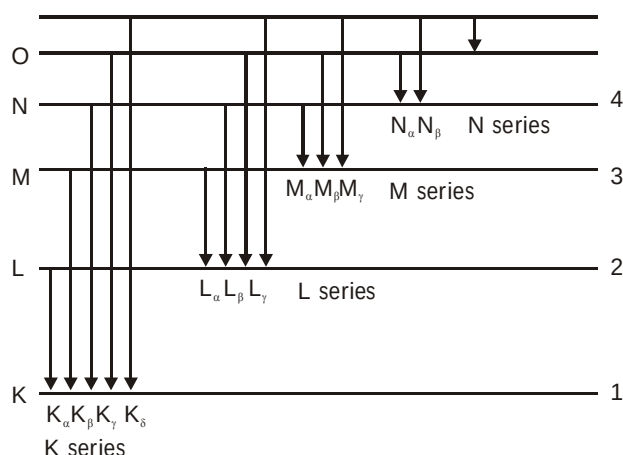
$n_1 = 2,$        $n_2 = 3, 4, 5, \dots$  L series

$n_1 = 3,$        $n_2 = 4, 5, 6, \dots$  M series

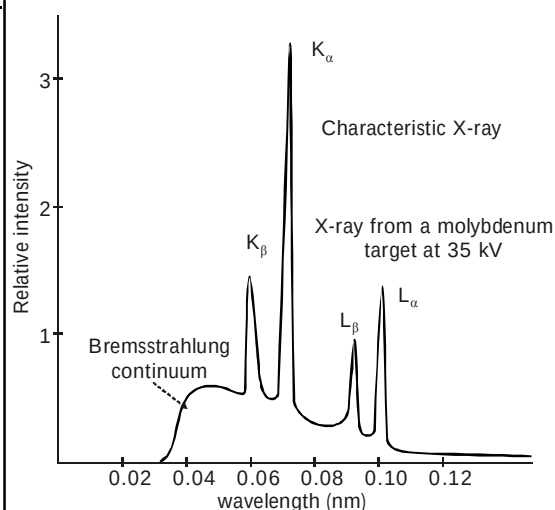
First line of series =  $\alpha$

Second line of series =  $\beta$

Third line of series =  $\gamma$



| Transition                               | Wave-length         | Energy           | Energy difference                    | Wavelength                                                                                         |
|------------------------------------------|---------------------|------------------|--------------------------------------|----------------------------------------------------------------------------------------------------|
| L $\rightarrow$ K<br>(2 $\rightarrow$ 1) | $\lambda_{K\alpha}$ | $h\nu_{K\alpha}$ | $-(E_K - E_L)$<br>$= h\nu_{K\alpha}$ | $\lambda_{K\alpha} = \frac{hc}{(E_K - E_L)}$<br>$= \frac{12400}{(E_K - E_L)} \text{ eV}\text{\AA}$ |
| M $\rightarrow$ K<br>(3 $\rightarrow$ 1) | $\lambda_{K\beta}$  | $h\nu_{K\beta}$  | $-(E_K - E_M)$<br>$= h\nu_{K\beta}$  | $\lambda_{K\beta} = \frac{hc}{(E_K - E_M)}$<br>$= \frac{12400}{(E_K - E_M)} \text{ eV}\text{\AA}$  |
| M $\rightarrow$ L<br>(3 $\rightarrow$ 2) | $\lambda_{L\alpha}$ | $h\nu_{L\alpha}$ | $-(E_L - E_M)$<br>$= h\nu_{L\alpha}$ | $\lambda_{L\alpha} = \frac{hc}{(E_L - E_M)}$<br>$= \frac{12400}{(E_L - E_M)} \text{ eV}\text{\AA}$ |



### MOSELEY'S LAW

Moseley studied the characteristic spectrum of number of many elements and observed that the square root of the frequency of a K- line is closely proportional to atomic number of the element. This is

called Moseley's law.

$$\sqrt{\nu} \propto (Z - b) \Rightarrow \nu \propto (Z - b)^2 \Rightarrow \nu = a(Z - b)^2 \dots (i)$$

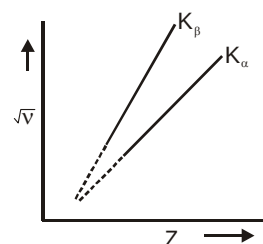
$Z$  = atomic number of target

$\nu$  = frequency of characteristic spectrum

$b$  = screening constant (for K- series  $b=1$ , L series  $b=7.4$ )

$a$  = proportionality constant

From Bohr Model 
$$\nu = RcZ^2 \left[ \frac{1}{n_1^2} - \frac{1}{n_2^2} \right] \dots (ii)$$



Comparing (i) and (ii)  $a = Rc \left[ \frac{1}{n_1^2} - \frac{1}{n_2^2} \right]$

- Thus proportionality constant 'a' does not depend on the nature of target but depend on transition.

| Bohr model |                                                                                  | Moseley's correction |                                                                                       |
|------------|----------------------------------------------------------------------------------|----------------------|---------------------------------------------------------------------------------------|
| 1.         | For single electron species                                                      | 1.                   | For many electron species                                                             |
| 2.         | $\Delta E = 13.6Z^2 \left[ \frac{1}{n_1^2} - \frac{1}{n_2^2} \right] \text{ eV}$ | 2.                   | $\Delta E = 13.6 (Z-1)^2 \left[ \frac{1}{n_1^2} - \frac{1}{n_2^2} \right] \text{ eV}$ |
| 3.         | $\nu = RcZ^2 \left[ \frac{1}{n_1^2} - \frac{1}{n_2^2} \right]$                   | 3.                   | $\nu = Rc(Z-1)^2 \left[ \frac{1}{n_1^2} - \frac{1}{n_2^2} \right]$                    |
| 4.         | $\frac{1}{\lambda} = RZ^2 \left[ \frac{1}{n_1^2} - \frac{1}{n_2^2} \right]$      | 4.                   | $\frac{1}{\lambda} = R(Z-1)^2 \left[ \frac{1}{n_1^2} - \frac{1}{n_2^2} \right]$       |

- For X-ray production, Moseley formulae are used because heavy metal are used.

When target is same  $\lambda \propto \frac{1}{\frac{1}{n_1^2} - \frac{1}{n_2^2}}$

When transition is same  $\lambda \propto \frac{1}{(Z-b)^2}$

## ABSORPTION OF X-RAY

When X-ray passes through x thickness then its intensity  $I = I_0 e^{-\mu x}$

$I_0$  = Intensity of incident X-ray

$I$  = Intensity of X-ray after passing through x distance

$\mu$  = absorption coefficient of material

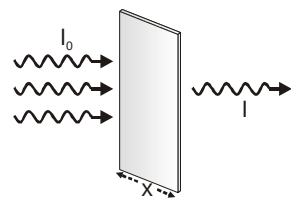
- Intensity of X-ray decrease exponentially.
- Maximum absorption of X-ray  $\rightarrow$  Lead
- Minimum absorption of X-ray  $\rightarrow$  Air

### Half thickness ( $x_{1/2}$ )

The distance travelled by X-ray when intensity become half the original value  $x_{1/2} = \frac{\ln 2}{\mu}$

- Ex.** When X-rays of wavelength  $0.5\text{\AA}$  pass through 10 mm thick Al sheet then their intensity is reduced to one sixth. Find the absorption coefficient for Aluminium .

**Sol.**  $\mu = \frac{2.303}{x} \log \left( \frac{I_0}{I} \right) = \frac{2.303}{10} \log_{10} 6 = \frac{2.303 \times 0.7781}{10} = 0.1752 / \text{mm}$



## DIFFRACTION OF X-RAY

Diffraction of X-ray is possible by crystals because the interatomic spacing in a crystal lattice is order of wavelength of X-rays it was first verified by Laue.

Diffraction of X-ray take place according to Bragg's law  $2d \sin \theta = n\lambda$

$d$  = spacing of crystal plane or lattice constant or distance  
between adjacent atomic plane

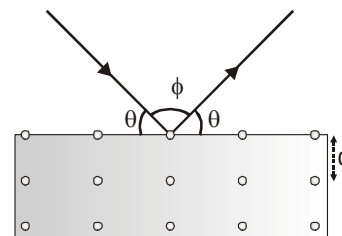
$\theta$  = Bragg's angle or glancing angle

$\phi$  = Diffracting angle  $n = 1, 2, 3, \dots$

### For Maximum Wavelength

$$\sin \theta = 1, n = 1 \Rightarrow \lambda_{\max} = 2d$$

so if  $\lambda > 2d$  diffraction is not possible i.e. solution of Bragg's equation is not possible.



## PROPERTIES OF X-RAY

- X-ray always travel with the velocity of light in straight line because X-rays are em waves
- X-ray is electromagnetic radiation it show particle and wave both nature
- In reflection, diffraction, interference, refraction X-ray shows wave nature while in photoelectric effect it shows particle nature.
- There is no charge on X-ray thus these are not deflected by electric field and magnetic field.
- X-ray are invisible.
- X-ray affects the photographic plate
- When X-ray incidents on the surface of substance it exerts force and pressure and transfer energy and momentum
- Characteristic X-ray can not obtained from hydrogen because the difference of energy level in hydrogen is very small.

**Ex.** Show that the frequency of  $K_{\beta}$  X-ray of a material is equal to the sum of frequencies of  $K_{\alpha}$  and  $L_{\alpha}$  X-rays of the same material.

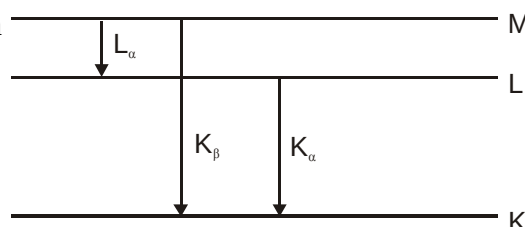
**Sol.** The energy level diagram of an atom with one electron knocked out is shown above.

Energy of  $K_{\alpha}$  X-ray is  $E_{K\alpha} = E_L - E_K$

and of  $K_{\beta}$  X-ray is  $E_{K\beta} = E_M - E_K$

and of  $L_{\alpha}$  X-rays is  $E_{L\alpha} = E_M - E_L$

thus  $E_{K\beta} = E_{K\alpha} + E_{L\alpha}$  or  $\nu_{K\beta} = \nu_{K\alpha} + \nu_{L\alpha}$



## EXERCISE (S-1)

**Photoelectric Effect :**

- In an experiment on photoelectric effect, the slope of the cut-off voltage versus frequency of incident light is found to be  $4.12 \times 10^{-15}$  V s. Calculate the value of Planck's constant. **NCERT**  
**MP0001**
- A 100W sodium lamp radiates energy uniformly in all directions. The lamp is located at the centre of a large sphere that absorbs all the sodium light which is incident on it. The wavelength of the sodium light is 589 nm. (a) What is the energy per photon associated with the sodium light ? (b) At what rate are the photons delivered to the sphere? **NCERT**  
**MP0002**
- In a photoelectric experiment set up, photons of energy 5 eV fall on the cathode having work function 3 eV.  
(a) If the saturation current is  $i_A = 4\mu\text{A}$  for intensity  $10^{-5}$  W/m<sup>2</sup>, then plot a graph between anode potential and current.  
(b) Also draw a graph for intensity of incident radiation of  $2 \times 10^{-5}$  W/m<sup>2</sup>. **[JEE' 2003]**  
**MP0003**
- Monochromatic radiation of wavelength 640.2 nm ( $1\text{nm} = 10^{-9}$  m) from a neon lamp irradiates photosensitive material made of cesium. The stopping voltage is measured to be 0.54 V. The source is replaced by an iron source and its 427.2 nm line irradiates the same photo-cell. Predict the new stopping voltage. **NCERT**  
**MP0004**
- Monochromatic light of wavelength 632.8 nm is produced by a helium-neon laser. The power emitted is 9.42 mW. **NCERT**  
(a) Find the energy and momentum of each photon in the light beam,  
(b) How many photons per second, on the average, arrive at a target irradiated by this beam? (Assume the beam to have uniform cross-section which is less than the target area), and  
(c) How fast does a hydrogen atom have to travel in order to have the same momentum as that of the photon ?  
**MP0005**

**Wave Nature of Matter**

- Calculate the **NCERT**  
(a) momentum, and  
(b) de-Broglie wavelength of the electrons accelerated through a potential difference of 56 V.  
**MP0006**

7. The wavelength of light from the spectral emission line of sodium is 589 nm. Find the kinetic energy at which  
 (a) an electron, and  
 (b) a neutron, would have the same de-Broglie wavelength.  
**NCERT**  
**MP0007**
8. (a) For what kinetic energy of a neutron will the associated de-Broglie wavelength be  $1.40 \times 10^{-10}$  m?  
 (b) Also find the de-Broglie wavelength of a neutron, in thermal equilibrium with matter, having an average kinetic energy of  $(3/2) k T$  at 300 K.  
**NCERT**  
**MP0008**
9. The potential energy of a particle varies as  $U(x) = E_0$  for  $0 \leq x \leq 1$  and  $U(x) = 0$  for  $x > 1$ . For  $0 \leq x \leq 1$ , de-Broglie wavelength is  $\lambda_1$  and for  $x > 1$  the de-Broglie wavelength is  $\lambda_2$ .  
 Total energy of the particle is  $2E_0$ . Find  $\frac{\lambda_1}{\lambda_2}$ .  
**[JEE 2005]**  
**MP0009**

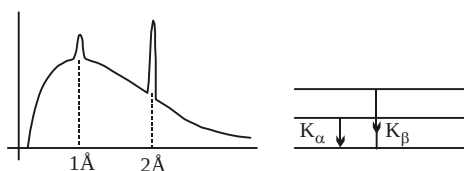
### Bohr's Theory

10. Determine the number of lines in Paschen series which have a wavelength greater than 1000 nm.  
**MP0010**
11. In a photoelectric setup, the radiations from the Balmer series of hydrogen atom are incident on a metal surface of work function 2 eV. The wavelength of incident radiations lies between 450 nm to 700 nm. Find the maximum kinetic energy of photoelectron emitted. (Given  $hc/e = 1242$  eV-nm).  
**[JEE-2004]**  
**MP0011**
12. An electron and a proton are separated by a large distance and the electron approaches the proton with a kinetic energy of 4.11 eV. If the electron is captured by the proton to form hydrogen atom in the ground state, the wavelength of photon given off is  $\alpha \times 10^2$  Å? Fill the value of  $\alpha$  in your OMR sheet.  
**MP0012**
13. A neutron moving through container filled with stationary deuterons. The neutron successively collides elastically and head on with stationary deuterons one at a time. The mass of the neutron is equal to half that of the deuteron. How many such collision would be required to slow the neutron down from 81 eV to 1 eV. [Neglect the relativistic effect]  
**MP0013**
14. What is the shortest wavelength present in the Paschen series of spectral lines?  
**NCERT**  
**MP0014**
15. A 12.5 eV electron beam is used to bombard gaseous hydrogen at room temperature. What series of wavelengths will be emitted?  
**NCERT**  
**MP0015**

16. The total energy of an electron in the first excited state of the hydrogen atom is about  $-3.4$  eV.  
 (a) What is the kinetic energy of the electron in this state? **NCERT**  
 (b) What is the potential energy of the electron in this state?  
 (c) Which of the answers above would change if the choice of the zero of potential energy is changed?  
**MP0016**
17. Obtain the first Bohr's radius and the ground state energy of a muonic hydrogen atom [i.e., an atom in which a negatively charged muon ( $\mu^-$ ) of mass about  $207m_e$  orbits around a proton]. **NCERT**  
**MP0017**
18. A potential difference of  $V$  volts is applied on two parallel electrodes separated by a distance of  $4.0 \times 10^{-2}$  m. The electrons of very low energy are injected into the region between the electrodes which contains argon at low pressure. The average distance the electrons travel between collisions with argon atoms is  $8 \times 10^{-5}$  m. The ionization energy of argon atom is 16 eV. Estimate the minimum value of  $V$  (in kV) such that the electrons will cause ionization in argon atoms by collision.  
**MP0018**
19. In hydrogen-like atom ( $Z = 11$ ),  $n^{\text{th}}$  line of Lyman series has wavelength  $\lambda$  equal to the de-Broglie's wavelength of electron in the level from which it originated. What is the value of  $n$ ?  
 [Take: Bohr radius ( $r_0$ ) =  $0.53 \text{ \AA}$  and Rydberg constant ( $R$ ) =  $1.1 \times 10^7 \text{ m}^{-1}$ ] **[JEE 2006]**  
**MP0019**

### X-Rays

20. The wavelength of characteristic  $K_\alpha$ -line emitted by a hydrogen like element is  $2.5 \text{ \AA}$ . Find the wavelength of the  $K_\gamma$ -line emitted by the same element (in  $\text{\AA}$ ). [Assume the shielding effect to be same as of  $K_\alpha$ ] **MP0020**
21. In an accelerator experiment on high-energy collisions of electrons with positrons, a certain event is interpreted as annihilation of an electron-positron pair of total energy 10.2 BeV into two  $\gamma$ -rays of equal energy. What is the wavelength associated with each  $\gamma$ -ray? ( $1 \text{ BeV} = 10^9 \text{ eV}$ ) **NCERT**  
**MP0021**
22. Figure shows  $K_\alpha$  &  $K_\beta$  X-rays along with continuous X-ray. Find the energy of  $L_\alpha$  X-ray. (Use  $hc = 12420 \text{ eV\AA}$ ).

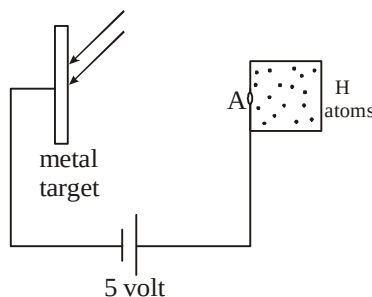


- MP0022**
23. A graph of  $\sqrt{\nu}$  (where  $\nu$  is the frequency of  $K_\alpha$  line of the characteristic X-ray spectrum) is plotted against the atomic number  $Z$  of the elements emitting the characteristic X-ray. The intercept of the graph on the  $Z$ -axis is 1 and the slope of the graph is  $0.5 \times 10^8$  S.I. units. The frequency of the  $K_\alpha$  line for an element of atomic number 41 is given as  $\alpha \times 10^{16} \text{ Hz}$ . Find the value of  $\alpha$ . **MP0023**

## EXERCISE (S-2)

1. A cooling object was emitting radiations of time varying wavelength  $\lambda = 3000 + 40t$ , where  $\lambda$  is in Å and  $t$  is in second is incident on a metal sheet (of work function 2eV) such that the power incident on sheet is constant at 100 watt. This signal is switched on and off for time intervals of 2 minutes and 1 minute respectively. Each time the signal is switched on,  $\lambda$  again starts from fresh value of 3000 Å. If the metal plate is grounded so that it always remains neutral and electron clouding is negligible then find the maximum photocurrent(mA). The photoemission efficiency is 0.01% and remains constant.(Take  $hc=12400\text{eV}\cdot\text{\AA}$ ) **MP0024**
2. A light of wavelength 3540 Å falls on a metal having work function of 2.5 eV. If ejected electron collides with another target metal inelastically and its total kinetic energy is utilized to raise the temperature of target metal. The mass of target metal is  $10^{-3}$  kg and its specific heat is 160 J/kg/°C. If  $10^{18}$  electrons are ejected per second, then find the rate of raise of temperature (in °C/s) of the metal [Assume there is no loss of energy of ejected electron by any other process, all the electron are reaching the target metal with max kinetic energy and take  $hc=12400\text{eV}\cdot\text{\AA}$ ] **MP0025**
3. A beam of light has three wavelengths 4144Å, 4972Å & 6216 Å with a total intensity of  $3.6\times 10^{-3}\text{ W}\cdot\text{m}^{-2}$  equally distributed amongst the three wavelengths. The beam falls normally on an area  $1.0\text{ cm}^2$  of a clean metallic surface of work function 2.3 eV. Assume that there is no loss of light by reflection and that each energetically capable photon ejects one electron. Calculate the number of photoelectrons liberated in two seconds. **MP0026**
4. In a photo electric effect set-up, a point source of light of power  $3.2 \times 10^{-3}\text{ W}$  emits mono energetic photons of energy 5.0 eV. The source is located at a distance of 0.8 m from the centre of a stationary metallic sphere of work function 3.0 eV & of radius  $8.0 \times 10^{-3}\text{ m}$ . The efficiency of photo electrons emission is one for every  $10^6$  incident photons. Assume that the sphere is isolated and initially neutral, and that photo electrons are instantly swept away after emission.
  - (a) Calculate the number of photo electrons emitted per second.
  - (b) Find the ratio of the wavelength of incident light to the De - Broglie wave length of the fastest photo electrons emitted.
  - (c) It is observed that the photo electron emission stops at a certain time  $t$  after the light source is switched on. Why ?
  - (d) Evaluate the time  $t$ .**MP0027**
5. Two identical nonrelativistic particles move at right angles to each other, possessing De Broglie wavelengths,  $\lambda_1$  &  $\lambda_2$ . Find the De Broglie wavelength of each particle in the frame of their centre of mass. **MP0028**
6. Assume that the de-Broglie wave associated with an electron can form a standing wave between the atoms arranged in a one dimensional array with nodes at each of the atomic sites. It is found that one such standing wave is formed if the distance 'd' between the atoms of the array is 2 Å. A similar standing wave is again formed if 'd' is increased to 2.5 Å but not for any intermediate value of d. Find the energy of the electrons in electron volts and the least value of d for which the standing wave of the type described above can form. **MP0029**

7. A gas of identical hydrogen like atoms has some atoms in the lowest (ground) energy level A & some atoms in a particular upper (excited) energy level B & there are no atoms in any other energy level. The atoms of the gas make transition to a higher energy level by the absorbing monochromatic light of photon energy  $2.55\text{ eV}$ . Subsequently, the atoms emit radiation of only six different photon energies. Some of the emitted photons have energy  $2.55\text{ eV}$ . Some have energy more and some have less than  $2.55\text{ eV}$ .
- Find the principal quantum number of the initially excited level B.
  - Find the ionisation energy for the gas atoms.
  - Find the maximum and the minimum energies of the emitted photons.
- MP0030**
8. A monochromatic light source of frequency  $\nu$  illuminates a metallic surface and ejects photoelectrons. The photoelectrons having maximum energy are just able to ionize the hydrogen atoms in ground state. When the whole experiment is repeated with an incident radiation of frequency  $(5/6)\nu$ , the photoelectrons so emitted are able to excite the hydrogen atom beam which then emits a radiation of wavelength of  $1215\text{ \AA}$ . Find the work function of the metal and the frequency  $\nu$ . **MP0031**
9. A neutron of kinetic energy  $65\text{ eV}$  collides inelastically with a singly ionized helium atom at rest. It is scattered at an angle of  $90^\circ$  with respect of its original direction.  
(Given : Mass of he atom =  $4 \times (\text{mass of neutron})$ , ionization energy of H atom =  $13.6\text{ eV}$ )
- Find the allowed values of the energy of the neutron & that of the atom after collision.
  - If the atom gets de-excited subsequently by emitting radiation, find the frequencies of the emitted radiation.
- MP0032**
10. A beam of ultraviolet light of wavelength  $100\text{ nm} - 200\text{ nm}$  is passed through a box filled with hydrogen gas in ground state. The light coming out of the box is split into two beams 'A' and 'B'. A contains unabsorbed light from the incident light and B contains the emitted light by hydrogen atoms. The beam A is incident on the emitter in a photoelectric tube. The stopping potential in this case is 5 volts. Find the work function of the emitter. In the second case the beam B is incident on the same emitter. Find the stopping potential in this case. You can assume that the transition to higher energy states are not permitted from the excited states. Use  $hc = 12400\text{ eV\AA}$ . **MP0033**
11. Electromagnetic waves of wavelength  $1242\text{ \AA}$  are incident on a metal of work function  $2\text{ eV}$ . The target metal is connected to a 5 volt cell, as shown. The electrons pass through hole A into a gas of hydrogen atoms in their ground state. Find the number of spectral lines emitted when hydrogen atoms come back to their ground states after having been excited by the electrons. Assume all excitations in H-atoms from ground state only. ( $hc = 12420\text{ eV\AA}$ )



12. A  $\text{He}^+$  ion in ground state is fired towards a Hydrogen atom in ground state and at rest. What should be the minimum kinetic energy (in eV) of  $\text{He}^+$  ion so that both single electron species may get excited.

**MP0035**



13. Consider a universe in which the  $\pi$ -meson orbits around the nucleus instead of electron. Assuming a Bohr model for a  $\pi$ -meson of mass  $m_\pi$  and of the same charge as the electron is in a circular orbit of radius  $r$  about the nucleus with an orbital angular momentum  $\frac{h}{2\pi}$ . If the radius of a nucleus of atomic number  $Z$  is given by  $R = 1.6 \times 10^{-15} Z^{1/3}$  m. The total number of elements in this universe that can exist is given as 'N'. Fill  $\left[ \frac{N-1}{12} \right]$  in OMR sheet. [ Given  $\frac{\epsilon_0 h^2}{\pi m_e e^2} = 0.53 \text{ \AA}$ ;  $\frac{m_\pi}{m_e} = 265$ ; neglect any shielding effect for the heavier atoms and assume non relativistic physics to be applicable and take  $5^{1/4} \approx 1.5$  ] **MP0036**
14. The peak emission from a black body at a certain temperature occurs at a wavelength of  $6000 \text{ \AA}$ . On increasing its temperature, the total radiation emitted is increased 16 times. These radiations are allowed to fall on a metal surface. Photoelectrons emitted by the peak radiation at higher temperature can be brought to rest by applying a potential equivalent to the excitation potential corresponding to the transition for the level  $n = 4$  to  $n = 2$  in the Bohr's hydrogen atom. The work function of the metal is given by  $\frac{\alpha}{100} \text{ eV}$  where  $\alpha$  is the numerical constant. Find the value of  $\alpha$ . [Take :  $hc = 12420 \text{ eV-\AA}$  ] **MP0037**
15. A neutron beam, in which each neutron has same kinetic energy, is passed through a sample of hydrogen like gas (but not hydrogen) in ground state. Due to collision of neutrons with the ions of the gas, ions are excited and then they emit photons. Six spectral lines are obtained in which one of the lines is of wavelength  $(6200/51) \text{ nm}$ . What is the minimum possible value of kinetic energy of the neutrons for this to be possible. The mass of neutron and proton can be assumed to be nearly same. Find the answer in the form  $25\alpha \times 10^{-2} \text{ eV}$  and fill value of  $\alpha$ . **MP0038**
16. Electrons in a hydrogen like atom ( $Z = 3$ ) make transitions from the fourth excited state to the third excited state and from the third excited state to the second excited state. The resulting radiations are incident on a metal plate and eject photoelectrons. The stopping potential for photoelectrons ejected by shorter wavelength is  $3.95 \text{ eV}$ . Find the work function (in eV) of the metal plate. **MP0039**
17. A hydrogen atom at rest is in ground state. It is struck by a  $\text{He}^+$  ion in first excited state. Assuming the collision to be head on and the mass of  $\text{He}^+$  to be four times that of hydrogen atom, find the least value of kinetic energy of incoming particle which can excite both the particles to second excited state. **MP0040**
18. In an X-ray tube the accelerating voltage is  $20 \text{ KV}$ . Two targets A and B are used one by one. For 'A' the wavelength of the  $K_\alpha$  line is  $62 \text{ pm}$ . For 'B' the wavelength of the  $L_\alpha$  line is  $124 \text{ pm}$ . The energy of the 'B' ion with vacancy in 'L' shell is  $15.5 \text{ KeV}$  higher than the atom of B. [Take  $hc = 12400 \text{ eV-\AA}$ ]  
(i) Find  $\lambda_{\min}$  in  $\text{\AA}$ .  
(ii) Can  $K_\alpha$  - photon be emitted by 'A'? Explain with reason.  
(iii) Can  $L$  - photons be emitted by 'B'? What is the minimum wavelength (in  $\text{\AA}$ ) of the characteristic X-ray that will be emitted by 'B'. **MP0041**
19. An X-rays tube is working at a potential difference of  $38.08 \text{ kV}$ . The potential difference is decreased to half its initial value. It is found that difference of the wavelength of  $K_\alpha$  X-ray and the most energetic continuous X-rays becomes  $1/4$  times of the difference prior to the change of voltage. Assuming  $K_\alpha$  line is present in both cases, find the atomic number of the target element. [Take  $R_{\text{ch}} = 13.6 \text{ eV}$  ] **MP0042**

## EXERCISE (O-1)

## SINGLE CORRECT TYPE QUESTIONS

*Photoelectric Effect*

- Statement 1 :** Photoelectric effect establishes quantum nature of light.  
**and**  
**Statement 2 :** There is negligible time lag between photon collisions with the material and photoelectron emission irrespective of intensity of incident light. (Assume incident light is of frequency greater than threshold frequency of the material).

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.  
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.  
 (C) Statement-1 is true, statement-2 is false.  
 (D) Statement-1 is false, statement-2 is true. **MP0043**
- Statement-1 :** Work function of aluminum is 4.2 eV. If two photons each of energy 2.5 eV strikes on a piece of aluminum, the photo electric emission does not occur  
**Statement-2 :** In photo electric effect a single photon interacts with a single electron and electron is emitted only if energy of each incident photon is greater then the work function.

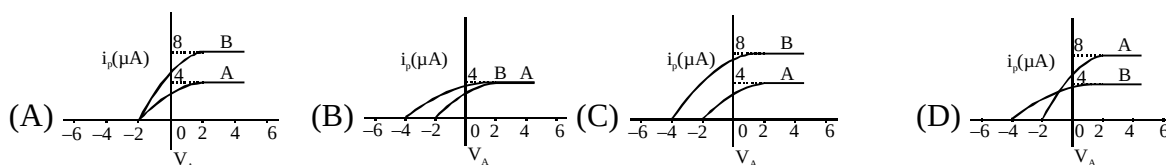
(A) Statement-1 is True, Statement-2 is True, Statement-2 is a correct explanation for statement-1  
 (B) Statement-1 is True, Statement-2 is True, Statement-2 is NOT a correct explanation for Statement-1  
 (C) Statement-1 is True, Statement-2 is False  
 (D) Statement-1 is False, Statement-2 is True **MP0044**
- A photocell is illuminated by a small bright source placed 1 m away. When the same source of light is place  $\frac{1}{2}$  m away, the number of electrons emitted by photocathode would- **[AIEEE - 2005]**

(A) decrease by a factor of 4  
 (B) increase by a factor of 4  
 (C) decrease by a factor of 2  
 (D) increase by a factor of 2 **MP0045**
- Two monochromatic light sources, A and B, emit the same number of photons per second. The wavelength of A is  $\lambda_A = 400$  nm, and that of B is  $\lambda_B = 600$  nm. The power radiated by source B is

(A) equal to that of source A  
 (B) less than that of source A  
 (C) greater than that of source A  
 (D) cannot be compared to that from source A using the available data. **MP0046**
- The energy flux of sunlight reaching the surface of the earth is  $1.388 \times 10^3 \text{ W/m}^2$ . How many photons (nearly) per square metre are incident on the Earth per second? Assume that the photons in the sunlight have an average wavelength of 550 nm. **NCERT**

(A)  $8 \times 10^{21}$  (B)  $4 \times 10^{21}$  (C)  $4 \times 10^{38}$  (D)  $8 \times 10^{38}$  **MP0047**

6. The threshold frequency for a certain metal is  $3.3 \times 10^{14}$  Hz. If light of frequency  $8.2 \times 10^{14}$  Hz is incident on the metal, the cutoff voltage for the photoelectric emission is. **NCERT**  
 (A) 8 V (B) 6 V (C) 2 V (D) 4 **MP0048**
7. Light of frequency  $7.21 \times 10^{14}$  Hz is incident on a metal surface. Electrons with a maximum speed of  $6.0 \times 10^5$  m/s are ejected from the surface. The threshold frequency for photoemission of electrons is **NCERT**  
 (A)  $4.73 \times 10^{14}$  Hz (B)  $4.73 \times 10^{10}$  Hz (C)  $2.08 \times 10^{10}$  Hz (D) None of these **MP0049**
8. In a photoelectric effect experiment, photons of energy 5 eV are incident on the photo-cathode of work function 3 eV. For photon intensity  $I_A = 10^{15} \text{ m}^{-2}\text{s}^{-1}$ , saturation current of  $4.0 \mu\text{A}$  is obtained. Sketch of the variation of photocurrent  $i_p$  against the anode voltage  $V_a$  for photon intensity  $I_A$  (curve A) and  $I_B = 2 \times 10^{15} \text{ m}^{-2}\text{s}^{-1}$  (curve B) will be :

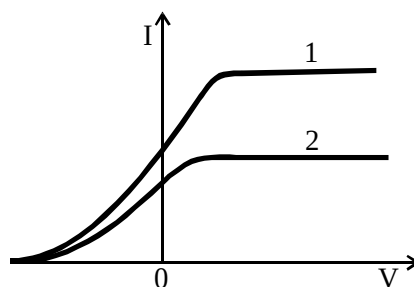


**MP0050**

9. **Statement-1** : When ultraviolet light is incident on a photocell, its stopping potential is  $V_0$  and the maximum kinetic energy of the photoelectrons is  $K_{\max}$ . When the ultraviolet light is replaced by infrared light, both  $V_0$  and  $K_{\max}$  increase.  
**Statement-2** : Photoelectrons are emitted with speeds ranging from zero to a maximum value.  
 (A) Statement-1 is true, Statement-2 is false  
 (B) Statement-1 is true, Statement-2 is true; Statement-2 is the correct explanation of Statement-1  
 (C) Statement-1 is true, Statement-2 is true; Statement-2 is not the correct explanation of Statement-1  
 (D) Statement-1 is false, Statement-2 is true

**MP0051**

10. A photocathode can be illuminated by the light from two sources, each of which emits monochromatic radiation. The sources are positioned at equal distances from the photocathode. The dependence of the photocurrent on the voltage between the cathode and the anode is depicted by curve 1 for one source and by curve 2 for the other. In what respect do these **sources** differ ?



- (A) Highest frequency photon (B) Number of photons emitted per second  
 (C) Number of photoelectrons emitted per second  
 (D) None of these

**MP0052**

11. A radiation of energy  $E$  falls normally on a perfectly reflecting surface. The momentum transferred to the surface is- [AIEEE - 2004]

(A)  $E/c$  (B)  $2E/c$  (C)  $Ec$  (D)  $E/c^2$

MP0053

12. The threshold frequency for a metallic surface corresponds to an energy of 6.2 eV and the stopping potential for a radiation incident on this surface is 5V. The incident radiation lies in- [AIEEE - 2006]

(A) ultra-violet region (B) infra-red region (C) visible region (D) X-ray region

MP0054

13. The time taken by a photoelectron to come out after the photon strikes is approximately-

[AIEEE - 2006]

(A)  $10^{-4}$  s (B)  $10^{-10}$  s (C)  $10^{-16}$  (D)  $10^{-1}$  s

MP0055

14. The anode voltage of a photocell is kept fixed. The wavelength  $\lambda$  of the light falling on the cathode is gradually changed. The plate current  $I$  of the photocell varies as follows : [AIEEE - 2006]



MP0056

15. Photon of frequency  $\nu$  has a momentum associated with it. If  $c$  is the velocity of light, the momentum is- [AIEEE - 2007]

(A)  $\nu/c$  (B)  $h\nu c$  (C)  $h\nu/c^2$  (D)  $h\nu/c$

MP0057

### Wave Nature of Matter

16. **Statement-1** : If an electron has the same wavelength as a photon, they have the same energy.

**Statement-2** : by de Broglie hypothesis,  $p = h/\lambda$  for both the electron and the photon.

- (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.  
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.  
 (C) Statement-1 is true, statement-2 is false.  
 (D) Statement-1 is false, statement-2 is true.

MP0058

17. A particle of mass  $4m$  at rest decays into two particles of masses  $m$  and  $3m$  having non-zero velocities. The ratio of the de-Broglie wavelengths of the particles 1 and 2 is

(A)  $\frac{1}{2}$  (B)  $\frac{1}{4}$  (C) 2 (D) 1

MP0059

18. A free particle with initial kinetic energy  $E$  and de-Broglie wavelength  $\lambda$  enters a region in which it has potential energy  $U$ . What is the particle's new de-Broglie wavelength?

(A)  $\lambda(1-U/E)^{-1/2}$  (B)  $\lambda(1-U/E)$  (C)  $\lambda(1-E/U)^{-1}$  (D)  $\lambda(1+U/E)^{1/2}$

MP0060

19. Proton, deuteron and  $\alpha$  particles are accelerated through the same potential difference. Then the ratio of their de-Broglie wavelength as

(A)  $1:\sqrt{2}:1$  (B)  $1:1:1$  (C)  $1:2:2\sqrt{2}$  (D)  $2\sqrt{2}:2:1$

MP0061

20. **Statement-1** : An electron and a proton are accelerated through the same potential difference. The de-Broglie wavelength associated with the electron is longer.

**Statement-2** : De-Broglie wavelength associated with a moving particle is  $\lambda = \frac{h}{p}$  where,  $p$  is the linear momentum and both have same KE.

- (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.  
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.  
(C) Statement-1 is true, statement-2 is false.  
(D) Statement-1 is false, statement-2 is true.

MP0062

21. An  $\alpha$ -particle of energy 5 MeV is scattered through  $180^\circ$  by a fixed uranium nucleus. The distance of the closest approach is of the order of-

[AIEEE-2004]

(A) 1 Å (B)  $10^{-10}$  cm (C)  $10^{-12}$  cm (D)  $10^{-15}$  cm

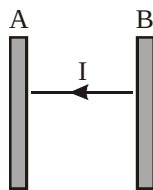
MP0063

22. After absorbing a slowly moving neutron of mass  $m_N$  (momentum  $\sim 0$ ) a nucleus of mass  $M$  breaks into two nuclei of masses  $m_1$  and  $3m_1$  ( $4m_1 = M + m_N$ ), respectively. If the de Broglie wavelength of the nucleus with mass  $m_1$  is  $\lambda$ , then de-Broglie wavelength of the other nucleus will be:-

(A)  $9\lambda$  (B)  $3\lambda$  (C)  $\frac{\lambda}{3}$  (D)  $\lambda$

MP0064

23. A parallel beam of light of intensity  $I$  is incident normally on a plane surface A which absorbs 50% of the incident light. The reflected light falls on B which is perfect reflector, the light reflected by B is again partly reflected and partly absorbed and this process continues. For all absorption by A, absorption coefficient is 0.5. The pressure experienced by A due to light is :-



(A)  $\frac{1.5I}{c}$  (B)  $\frac{I}{c}$  (C)  $\frac{3I}{2c}$  (D)  $\frac{3I}{c}$

MP0065

24. **Statement-1** : If the accelerating potential in an X-ray tube is increased, the wavelengths of the characteristic X-rays do not change. [JEE 2007]

**because**

**Statement-2** : When an electron beam strikes the target in an X-ray tube, part of the kinetic energy is converted into X-ray energy

- (A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1  
 (B) Statement-1 is True, Statement-2 is True; Statement-2 is **NOT** a correct explanation for Statement-1  
 (C) Statement-1 is True, Statement-2 is False  
 (D) Statement-1 is False, Statement-2 is True

MP0066

25. Electrons with de-Broglie wavelength  $\lambda$  fall on the target in an X-ray tube. The cut-off wavelength of the emitted X-rays is [JEE 2007]

(A)  $\lambda_0 = \frac{2mc\lambda^2}{h}$       (B)  $\lambda_0 = \frac{2h}{mc}$       (C)  $\lambda_0 = \frac{2m^2c^2\lambda^3}{h^2}$       (D)  $\lambda_0 = \lambda$

MP0067

### Bohr's Theory

26. The de-Broglie wavelength of an electron in the first Bohr orbit is

- (A) equal to the circumference of first orbit      (B) equal to  $\frac{1}{2} \times$  (circumference of first orbit)  
 (C) equal to  $\frac{1}{4} \times$  (circumference of first orbit)      (D) equal to  $\frac{3}{4} \times$  (circumference of first orbit)

MP0068

27. **Statement-1** : When light is passed through a sample of hydrogen atoms in ground state, then wavelengths of absorption lines are same as wavelengths of lines of Lyman series in emission spectrum.

**and**

**Statement-2** : In ground state hydrogen atom will absorb only those radiation which will excite to higher energy level.

- (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.  
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.  
 (C) Statement-1 is true, statement-2 is false.  
 (D) Statement-1 is false, statement-2 is true.

MP0069

28. **Statement-1** : In a laboratory experiment, on emission from atomic hydrogen in a discharge tube, only a small number of lines are observed whereas a large number of lines are present in the hydrogen spectrum of a star.

**Statement-2** : The temperature of discharge tube is much smaller than that of the star.

- (A) Statement-1 is True, Statement-2 is True, Statement-2 is a correct explanation for statement-1  
 (B) Statement-1 is True, Statement-2 is True, Statement-2 is NOT a correct explanation for

Statement-1

(C) Statement-1 is True, Statement-2 is False

(D) Statement-1 is False, Statement-2 is True

MP0070

29. If the binding energy of the electron in a hydrogen atom is 13.6 eV, the energy required to remove the electron from the first excited state of  $\text{Li}^{2+}$  is- [AIEEE - 2003]

(A) 30.6 eV (B) 13.6 eV (C) 3.4 eV (D) 122.4 eV

MP0071

30. A neutron having kinetic energy 13.6 eV collides with a hydrogen atom in ground state at rest. Assume that the mass of neutron and hydrogen atoms are same and the neutron does not leave its line of motion. Then which of the following is the possible KE of the neutron after the collision?

(A) zero (B) 3.4 eV (C) 1.5 eV (D) 6.8 eV.

MP0072

31. In the spectrum of single ionised helium, the wavelength of a line observed is almost the same as the first line of Balmer series of hydrogen. It is due to transition of electron from

(A)  $n_1 = 6$  to  $n_2 = 4$  (B)  $n_1 = 5$  to  $n_2 = 3$  (C)  $n_1 = 4$  to  $n_2 = 2$  (D)  $n_1 = 3$  to  $n_2 = 2$

MP0073

32. Light coming from a discharge tube filled with hydrogen falls on the cathode of the photoelectric cell. The work function of the surface of cathode is 4eV. Which one of the following values of the anode voltage (in Volts) with respect to the cathode will likely to make the photo current zero.

(A) - 4 (B) - 6 (C) - 8 (D) - 10

MP0074

33. According to Bohr model, magnetic field at the centre (at the nucleus) of a hydrogen atom due to the motion of the electron in  $n^{\text{th}}$  orbit is proportional to

(A)  $1/n^3$  (B)  $1/n^5$  (C)  $n^5$  (D)  $n^3$

MP0075

34. A photon of 10.2 eV energy collides with a hydrogen atom in ground state inelastically. After few microseconds one more photon of energy 15 eV collides with the same hydrogen atom. Then what can be detected by a suitable detector. [JEE' 2005 (Scr)]

(A) one photon of 10.2 eV and an electron of energy 1.4 eV  
(B) 2 photons of energy 10.2 eV  
(C) 2 photons of energy 3.4 eV  
(D) 1 photon of 3.4 eV and one electron of 1.4 eV

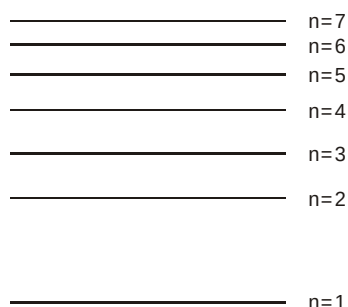
MP0076

35. Two hydrogen atoms are in excited state with electrons residing in  $n = 2$ . First one is moving towards left and emits a photon of energy  $E_1$  towards right. Second one is moving towards left with same speed and emits a photon of energy  $E_2$  towards left. Taking recoil of nucleus into account during emission process

(A)  $E_1 > E_2$  (B)  $E_1 < E_2$   
(C)  $E_1 = E_2$  (D) information insufficient

MP0077

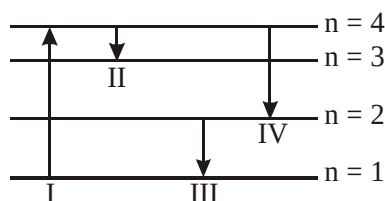
36. The maximum number of emission lines for atomic hydrogen that you would expect to see with naked eye if the only electronic levels involved are those shown in the figure, is



- (A) 6 (B) 5 (C) 21 (D)  $\infty$

MP0078

37. The diagram shows the energy levels for an electron in a certain atom. Which transition shown represents the emission of a photon with maximum wavelength?



- (A) III (B) IV (C) I (D) II

MP0079

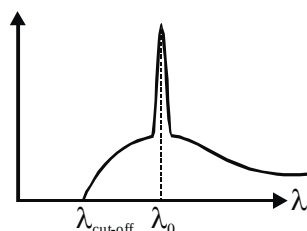
38. The largest wavelength in the ultraviolet region of the hydrogen spectrum is 122 nm. The smallest wavelength in the infrared region of the hydrogen spectrum (to the nearest integer) is [JEE 2007]

- (A) 802 nm (B) 823 nm (C) 1882 nm (D) 1648 nm

MP0080

### X-Rays

39. **Assertion (A):** Variation of intensity of X-rays is plotted against  $\lambda$ . On increasing the accelerating potential,  $(\lambda_0 - \lambda_{\text{cut-off}})$  increases.



**Reasoning (R):**  $\lambda_{\text{cut-off}}$  will decrease but  $\lambda_0$  will be same, as wavelength of characteristic X-rays is independent of the accelerating potential.

**Choose the correct statement from the following.**

- (A) A is correct and R is the correct explanation of A.



- (B) Both A and R are correct but R is not the correct explanation of A.  
 (C) A is correct but R is wrong.  
 (D) Both A and R are wrong.

MP0081

40. An X-ray tube is run at 50 kV. The current flowing in it is 20mA . The power of the tube is :  
 (A) 1000 W (B) 200 W (C) 20000 W (D) 20 W

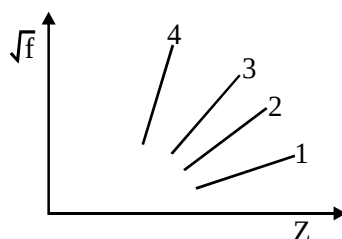
MP0082

41. Choose the **INCORRECT** statement.

- (A) Cut-off wavelength of X-rays is independent of filament voltage.  
 (B) Presence of  $K_{\alpha}$ -line in X-ray spectrum means that L-series will also be present.  
 (C) Increase in filament current increases intensity of X-ray.  
 (D) Presence of L-series in X-ray spectrum means that K-series will also be present.

MP0083

42. The given graph shows the variation of  $\sqrt{f}$  vs Z for characteristics X-rays. Lines 1, 2, 3, 4 shown in the graph corresponds to any one of  $k_{\alpha}$ ,  $k_{\beta}$ ,  $L_{\alpha}$ ,  $L_{\beta}$ . Then  $L_{\beta}$  is represent by :-



- (A) line 1 (B) line 2 (C) line 3 (D) line 4

MP0084

43. The X-ray beam coming from an X-ray tube will be

- (A) monochromatic  
 (B) having all wavelengths smaller than a certain maximum wavelength  
 (C) having all wavelengths larger than a certain minimum wavelength  
 (D) having all wavelengths lying between a minimum and a maximum wavelength

MP0085

44.  $E_1$  is energy of  $k_{\alpha}$  photon of aluminium,  $E_2$  is energy of  $k_{\beta}$  photon of aluminium and  $E_3$  is energy of  $k_{\alpha}$  photon from sodium, then the correct order of energies is given

- (A)  $E_1 > E_2 > E_3$  (B)  $E_3 > E_2 > E_1$  (C)  $E_3 > E_1 > E_2$  (D)  $E_2 > E_1 > E_3$

MP0086

45. The K, L and M energy levels of platinum lie roughly at 78, 12 and 3 keV respectively. The ratio of wavelength of  $K_{\alpha}$  line to that of  $K_{\beta}$  line in X-ray spectrum is-

- (A)  $\frac{22}{3}$  (B)  $\frac{3}{22}$  (C)  $\frac{22}{25}$  (D)  $\frac{25}{22}$

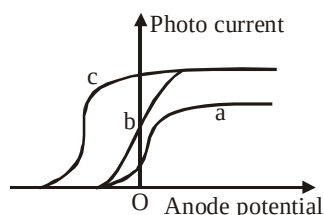
MP0087

46. What is the essential distinction between X-rays and  $\gamma$ -rays  
 (A)  $\gamma$ -rays have shorter wavelength than X-rays  
 (B)  $\gamma$ -rays are extraterrestrial, X-rays are man-made  
 (C)  $\gamma$ -rays have less penetrating power than X-rays  
 (D)  $\gamma$ -rays originate from within an atomic nucleus, X-rays from outside an atomic nucleus.  
**MP0088**
47. Mosley law relates :  
 (A) Frequency of emitted X-ray with applied voltage  
 (B) Wavelength and intensity of X-ray.  
 (C) Frequency of emitted X-ray with atomic number  
 (D) Wavelength and angle of scattering.  
**MP0089**
48. The intensity of gamma radiation from a given source is I. On passing through 36 mm of lead, it is reduced to  $I/8$ . The thickness of lead, which will reduce the intensity to  $I/2$  will be- [AIEEE-2005]  
 (A) 6 mm (B) 9 mm (C) 18 mm (D) 12 mm  
**MP0090**
49. The wavelength of  $K_{\alpha}$  X-ray of an element having atomic number  $Z = 11$  is  $\lambda$ . The wavelength of  $K_{\alpha}$  X-ray of another element of atomic number  $Z'$  is  $4\lambda$ . Then  $Z'$  is [JEE' 2005 (Scr)]  
 (A) 11 (B) 44 (C) 6 (D) 4  
**MP0091**
50. Characteristic X-ray  
 (A) Have only discrete wavelength which are characteristic of the target.  
 (B) Have all the possible wavelength.  
 (C) Are characteristic of speed of projectile electrons.  
 (D) None  
**MP0092**
51. Which of the following transitions in hydrogen atoms emit photons of highest frequency ? [AIEEE - 2007]  
 (A)  $n = 2$  to  $n = 6$  (B)  $n = 6$  to  $n = 2$  (C)  $n = 2$  to  $n = 1$  (D)  $n = 1$  to  $n = 2$   
**MP0093**

### MULTIPLE CORRECT TYPE QUESTIONS

52. Photoelectric effect supports quantum nature of light because  
 (A) there is minimum frequency of light below which no photoelectrons are emitted  
 (B) the maximum kinetic energy of photo-electrons depends only on the frequency of light and not on its intensity  
 (C) even when the metal surface is faintly illuminated, the photoelectrons leave the surface immediately  
 (D) electric charge of photo-electrons is quantized  
**MP0094**

53. The figure shows the variation of photo current with anode potential for a photosensitive surface for three different radiations. Let  $I_a$ ,  $I_b$  and  $I_c$  be the intensities and  $f_a$ ,  $f_b$  and  $f_c$  be the frequencies for the curves a, b and c respectively. Choose correct options



- (A)  $f_a = f_b$                       (B)  $I_a < I_b$                       (C)  $f_c < f_b$                       (D)  $I_c > I_b$

MP0095

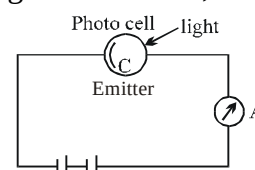
54. In photoelectric effect, stopping potential depends on  
(A) frequency of the incident light  
(B) intensity of the incident light by varies source distance  
(C) emitter's properties  
(D) frequency and intensity of the incident light

MP0096

55. Which of the following phenomena can be explained only on the basis of quantum theory of light?  
(A) Energy spectrum of black body radiation    (B) Atomic spectra  
(C) Photoelectric effect                              (D) Doppler effect

MP0097

56. The figure shows a photo cell circuit. The cathode of the photo cell is illuminated by a monochromatic light. If the intensity is kept constant and the frequency of the incident light is increased, then the  
(A) photo electric current in the circuit increases  
(B) photo electric current in the circuit decreases  
(C) maximum kinetic energy of the photo electrons increases  
(D) photo electric current in the circuit can be reduced to zero, when the polarity of the terminals is reversed



MP0098

57. In a photoelectric effect, electrons are emitted  
(A) at a rate that is proportional to the square of the amplitude of the incident radiation.  
(B) with a maximum velocity proportional to the frequency of the incident radiation.  
(C) at a rate that is independent of the emitter.  
(D) only if the frequency of the incident radiation is above a certain threshold value  
(E) only if the temperature of the emitter is high

MP0099

58. A small plate of area  $1 \text{ cm}^2$  is placed at a distance of  $\frac{1}{\sqrt{\pi}} \text{ m}$  from an isotropic point source emitting light of frequency  $\frac{1}{6.63} \times 10^{14} \text{ Hz}$ , at a power of  $2.00 \text{ mW}$ . Assume the plate to be normal to the incident photons. [where  $h = 6.63 \times 10^{-34} \text{ J-s}$ ]. Select **CORRECT** alternative(s)  
(A) Energy possessed by each photon is  $10^{-20} \text{ J}$   
(B) Photon emission rate is  $2 \times 10^{17} \text{ s}^{-1}$   
(C) The fraction of area of beam intercepted by the plate is  $\frac{1}{4} \times 10^{-4}$   
(D) The rate of photons striking the plate is  $5 \times 10^{12}$  per second.

MP0100

59. A metallic sphere of radius  $r$  remote from all other bodies is irradiated with a radiation of wavelength  $\lambda$  which is capable of causing photoelectric effect.
- (A) the maximum potential gained by the sphere will be independent of its radius  
 (B) the net positive charge appearing on the sphere after a long time will depend on the radius of the sphere  
 (C) the maximum kinetic energy of the electrons emanating from the sphere will keep on declining with time  
 (D) the kinetic energy of the most energetic electrons emanating from the sphere initially will be independent of the radius of the sphere.

MP0101

60. Two electrons are moving with the same speed  $v$ . One electron enters a region of uniform electric field while the other enters a region of uniform magnetic field, then after sometime if the de-Broglie wavelengths of the two are  $\lambda_1$  and  $\lambda_2$ , then select the possible option(s) :-
- (A)  $\lambda_1 = \lambda_2$  (B)  $\lambda_1 > \lambda_2$   
 (C)  $\lambda_1 < \lambda_2$  (D)  $\lambda_1 > \lambda_2$  or  $\lambda_1 < \lambda_2$

MP0102

61. A particle moves in a closed orbit around the origin, due to a central force which is directed towards the origin. The de Broglie wavelength of the particle varies cyclically between two values  $\lambda_1, \lambda_2$  with  $\lambda_1 > \lambda_2$ . Which of the following statements is/are true ?
- (A) The particle could be moving in a circular orbit with origin as centre  
 (B) The particle could be moving in an elliptic orbit with origin as its focus.  
 (C) When the de Broglie wave length is  $\lambda_1$ , the particle is nearer the origin than when its value is  $\lambda_2$ .  
 (D) When the de Broglie wavelength is  $\lambda_2$ , the particle is nearer the origin than when its value is  $\lambda_1$ .

MP0103

62. According to Bohr's theory of hydrogen atom, for the electron in the  $n^{\text{th}}$  permissible orbit,
- (A) linear momentum  $\propto \frac{1}{n}$   
 (B) radius of orbit  $\propto n$   
 (C) kinetic energy  $\propto \frac{1}{n^2}$   
 (D) angular momentum  $\propto n$

MP0104

63. The magnitude of angular momentum, orbit radius and frequency of an electron in hydrogen atom corresponding to the quantum number  $n$  are  $L, r$  and  $f$  respectively, then according to Bohr's theory of hydrogen atom.

- (A)  $frL$  is constant for all orbits (B)  $Lf \propto \frac{1}{n^2}$   
 (C)  $fr \propto \frac{1}{n}$  (D)  $Lr \propto \frac{1}{n^3}$

MP0105

64. A particular hydrogen like atom has its ground state binding "energy 122.4eV. Its is in ground state. Then:
- (A) Its atomic number is 3
  - (B) An electron of 90eV can excite it.
  - (C) An electron of kinetic energy nearly 91.8eV can be brought to almost rest by this atom.
  - (D) An electron of kinetic energy 2.6eV may emerge from the atom when electron of kinetic energy 125eV collides with this atom.

MP0106

65. A beam of ultraviolet light of all wavelengths passes through hydrogen gas at room temperature, in the x-direction. Assume that all photons emitted due to electron transition inside the gas emerge in the y-direction. Let A and B denote the lights emerging from the gas in the x and y directions respectively.
- (A) Some of the incident wavelengths will be absent in A.
  - (B) Only those wavelengths will be present in B which are absent in A.
  - (C) B will contain some visible light.
  - (D) B will contain some infrared light.

MP0107

66. In the hydrogen atom, if the reference level of potential energy is assumed to be zero at the ground state level. Choose the incorrect statement.
- (A) The total energy of the shell increases with increase in the value of n
  - (B) The total energy of the shell decrease with increase in the value of n.
  - (C) The difference in total energy of any two shells remains the same.
  - (D) The total energy at the ground state becomes 13.6 eV.

MP0108

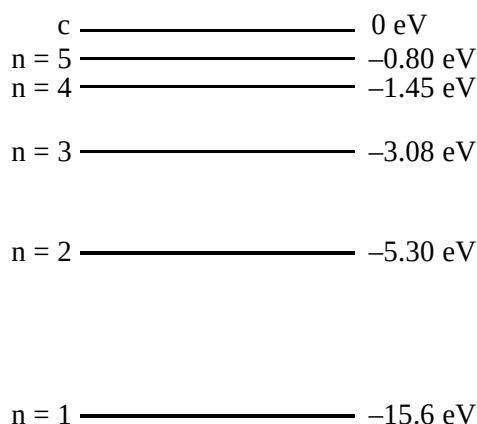
67. A neutron collides head-on with a stationary hydrogen atom in ground state. Which of the following statements are correct (Assume that the hydrogen atom and neutron has same mass) :
- (A) If kinetic energy of the neutron is less than 20.4 eV collision must be elastic.
  - (B) If kinetic energy of the neutron is less than 20.4 eV collision may be inelastic.
  - (C) Inelastic collision may be take place only when initial kinetic energy of neutron is greater than 20.4 eV.
  - (D) Perfectly inelastic collision can not take place.

MP0109

68. A free hydrogen atom in ground state is at rest. A neutron of kinetic energy 'K' collides with the hydrogen atom. After collision hydrogen atom emits two photons in succession one of which has energy 2.55 eV. (Assume that the hydrogen atom and neutron has same mass)
- (A) minimum value of 'K' is 25.5 eV.
  - (B) minimum value of 'K' is 12.75 eV
  - (C) the other photon has energy 10.2 eV if K is minium.
  - (D) the upper energy level is of excitation energy 12.75 eV.

MP0110

69. The energy levels of a hypothetical one electron atom are shown in the figure



- (A) The ionization potential of this atom is 15.6 V  
 (B) The short wavelength limit of the series terminating at  $n = 2$  is 2339 Å  
 (C) The excitation potential for the state  $n = 3$  is 12.52 V  
 (D) Wave number of the photon emitted for the transition  $n = 3$  to  $n = 1$  is  $1.009 \times 10^7 \text{ m}^{-1}$

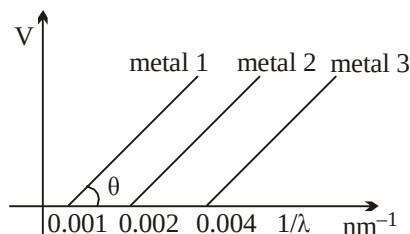
MP0111

70. Suppose frequency of emitted photon is  $f_0$  when electron of a stationary hydrogen atom jumps from a higher state  $m$  to a lower state  $n$ . If the atom is moving with a velocity  $v$  ( $\ll c$ ) and emits a photon of frequency  $f$  during the same transition, then which of the following statements are possible :-  
 (A)  $f$  may be equal to  $f_0$  (B)  $f$  may be greater than  $f_0$   
 (C)  $f$  may be less than  $f_0$  (D)  $f$  cannot be equal to  $f_0$

MP0112

71. The graph between  $1/\lambda$  and stopping potential ( $V$ ) of three metals having work functions  $\phi_1$ ,  $\phi_2$  and  $\phi_3$  in an experiment of photo-electric effect is plotted as shown in the figure. Which of the following statement(s) is/are correct? [Here  $\lambda$  is the wavelength of the incident ray].

[JEE 2006]



- (A) Ratio of work functions  $\phi_1 : \phi_2 : \phi_3 = 1 : 2 : 4$   
 (B) Ratio of work functions  $\phi_1 : \phi_2 : \phi_3 = 4 : 2 : 1$   
 (C)  $\tan \theta$  is directly proportional to  $hc/e$ , where  $h$  is Planck's constant and  $c$  is the speed of light  
 (D) The violet colour light can eject photoelectrons from metals 2 and 3.

MP0113

MATRIX MATCH TYPE QUESTION

72. Some quantities related to the photoelectric effect are mentioned under Column I and Column II. Match each quantity in Column I with the corresponding quantities in Column II on which it depends

**Column-I**

- (A) Saturation current  
(B) Stopping potential  
(C) de-Broglie wavelength of photoelectron  
(D) Force due to radiation falling on the photo-plate.

**Column-II**

- (P) Frequency of light  
(Q) Work function  
(R) Area of photosensitive plate  
(S) Intensity of light (at constant frequency)  
(T) None of these

MP0114

73. When we write expression for energy of electron in  $n^{\text{th}}$  orbit of helium ion ( $\text{He}^+$ ) we take zero potential energy for  $n = \infty$ , but the potential energy depends on reference. If we take total energy of atom for  $n = 1$  orbit as zero then

**Column-I**

- (A) Total energy of electron in  $n = 2$   
(B) Ionization energy from ground state  
(C) Energy required to exit electron from  $n = 1$  to  $n = 2$   
(D) Negative of potential energy of electron in  $n = 1$

**Column-II**

- (P) 54.4 eV  
(Q) 40.8 eV  
(R) depends on reference level  
(S) independent of reference level.  
(T) 70.3 eV

MP0115

74. In each situation of column-I a physical quantity related to orbiting electron in a hydrogen like atom is given. The terms 'Z' and 'n' given in column-II have usual meaning in Bohr's theory. Match the quantities in column-I with the terms they depend on in column-II :-

**Column-I**

- (A) Frequency of orbiting electron  
(B) Angular momentum of orbiting electron  
(C) Magnetic moment of orbiting electron  
(D) The average current due to orbiting of electron

**Column-II**

- (P) Is directly proportional to  $Z^2$   
(Q) Is directly proportional to  $n$   
(R) Is inversely proportional to  $n^3$   
(S) Is independent of  $Z$   
(T) None of these

MP0116

75. Match the entries of column-I with the entries of column-II :-

**Column-I**

- (I) Characteristic X-ray  
(II) Photoelectric effect  
(III) Thermo-ionic emission  
(IV) Continuous X-ray

**Column-II**

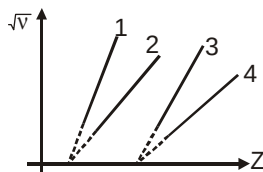
- (P) Inverse process of photoelectric effect  
(Q) Emission of electrons  
(R) Moseley's law  
(S) Emission of radiations

Then choose the correct matching.

- (A) (i)  $\rightarrow$ (RS) ; (ii)  $\rightarrow$  (RS) ; (iii)  $\rightarrow$ (S) ; (iv)  $\rightarrow$ (PS)  
 (B) (i)  $\rightarrow$ (RS) ; (ii)  $\rightarrow$  (Q) ; (iii)  $\rightarrow$ (Q) ; (iv)  $\rightarrow$ (PS)  
 (C) (i)  $\rightarrow$ (RS) ; (ii)  $\rightarrow$  (S) ; (iii)  $\rightarrow$ (S) ; (iv)  $\rightarrow$ (PRS)  
 (D) (i)  $\rightarrow$ (RS) ; (ii)  $\rightarrow$  (Q) ; (iii)  $\rightarrow$ (Q) ; (iv)  $\rightarrow$ (PRS)

MP0117

76.  $\sqrt{v}$  versus  $Z$  graph for characteristic X-rays is as shown in figure. Match the following (assume screening constant for  $K_\alpha$  and  $K_\beta$  is same and for  $L_\alpha$  &  $L_\beta$  is same) :-

**Column-I**

- |     |          |     |
|-----|----------|-----|
| (A) | Line – 1 | (P) |
| (B) | Line – 2 | (Q) |
| (C) | Line – 3 | (R) |
| (D) | Line – 4 | (S) |
|     |          | (T) |

**Column-II**

- $L_\alpha$   
 $L_\beta$   
 $K_\alpha$   
 $K_\beta$   
 Both  $K_\alpha$  and  $L_\beta$

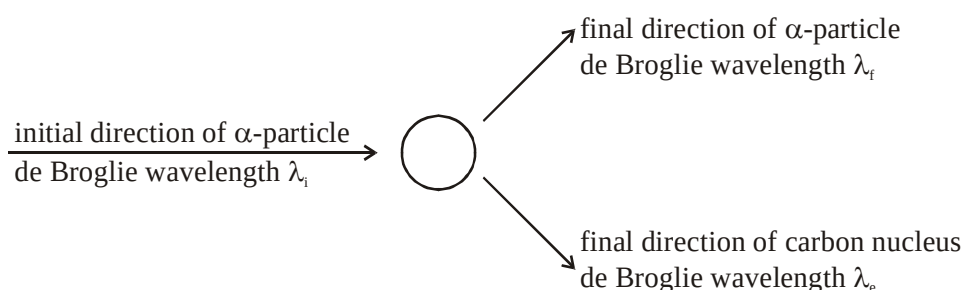
MP0118



## EXERCISE (O-2)

### SINGLE CORRECT TYPE QUESTIONS

- The maximum kinetic energy of photo-electron liberated from the surface of lithium (work function = 2.35 eV) by electromagnetic radiation whose electric component varies with time as  $E = a[1 + \cos(2\pi f_1 t)] \cos(2\pi f_2 t)$  where 'a' is a constant,  $f_1 = 3.6 \times 10^{15}$  Hz and  $f_2 = 1.2 \times 10^{15}$  Hz is [Take :  $h = 6.6 \times 10^{-34}$  J-s]  
 (A) 2.6 eV (B) 7.55 eV (C) 12.5 eV (D) 17.45 eV **MP0119**
- A photocell in the saturation mode is irradiated by light of wavelength  $\lambda = 6600 \text{ \AA}$ . The corresponding spectral sensitivity of the cell is  $s_\lambda = 4.8 \text{ mA/W}$ . Find the yield of photoelectrons, i.e. the number of photoelectrons produced by each incident photon. [Take :  $h = 6.6 \times 10^{-34}$  J-s]  
 (A)  $9 \times 10^{-2}$  (B)  $9 \times 10^{-4}$  (C)  $9 \times 10^{-3}$  (D) 9 **MP0120**
- An  $\alpha$ -particle having a de Broglie wavelength  $\lambda_i$  collides with a stationary carbon nucleus. The  $\alpha$ -particle moves off in a different direction as shown below.



After the collision, the de Broglie wavelength of the  $\alpha$ -particle and the carbon nucleus are  $\lambda_f$  and  $\lambda_e$  respectively. Which of the following relations about de Broglie wavelength is correct ?

- (A)  $\lambda_i < \lambda_f$  (B)  $\lambda_i > \lambda_f$  (C)  $\lambda_f = \lambda_e$  (D)  $\lambda_i = \lambda_e$  **MP0121**
- Choose the correct statement(s) for hydrogen and deuterium atoms (considering motion of nucleus)  
 (A) The radius of first Bohr orbit of deuterium is less than that of hydrogen  
 (B) The speed of electron in the first Bohr orbit of deuterium is more than that of hydrogen.  
 (C) The wavelength of first Balmer line of deuterium is more than that of hydrogen  
 (D) The angular momentum of electron in the first Bohr orbit of deuterium is more than that of hydrogen. **MP0122**

5. Apply Bohr's atomic model to a lithium atom. Assuming that its two K-shell electrons are too close to nucleus such that nucleus and K-shell electron act as a nucleus of effective positive charge equivalent to electron. The ionization energy of its outermost electron is:-

(A) 30.6 eV (B) 3.4 eV (C) 32.4 eV (D) 13.6 eV

**MP0123**

6. The attractive potential for an atom is given by  $v = v_0 \ln(r/r_0)$ ,  $v_0$  and  $r_0$  are constant and  $r$  is the radius of the orbit. The radius  $r$  of the  $n^{\text{th}}$  Bohr's orbit depends upon principal quantum number  $n$  as:

**[JEE' 2003 (Scr)]**

(A)  $r \propto n$  (B)  $r \propto 1/n^2$  (C)  $r \propto n^2$  (D)  $r \propto 1/n$

**MP0124**

7. A force of attraction between the positively charged nucleus and the negatively charged electron in the hydrogen atom is given by  $F = \frac{ke^2}{r^2}$  where  $k$  is the constant. The electron, initially moving in a circle of radius  $R_1$  about the nucleus, jumps suddenly into a circular orbit of radius  $R_2$ . The total energy of the atom decreased in this process is :-

(A)  $ke^2 \left[ \frac{1}{R_1} + \frac{1}{R_2} \right]$  (B)  $\frac{ke^2}{2} \left[ \frac{1}{R_2} - \frac{1}{R_1} \right]$  (C)  $ke^2 \left[ \frac{R_1 R_2}{R_2 - R_1} \right]$  (D)  $ke^2 \left[ \frac{R_1 R_2}{R_1 + R_2} \right]$

**MP0125**

8. A stationary hydrogen atom of mass  $M$  emits a photon corresponding to the longest wavelength of Balmer series. The recoil velocity acquired by the atom is ( $R$  = Rydberg constant and  $h$  = plank's constant)

(A)  $\frac{Rh}{M}$  (B)  $\frac{Rh}{4M}$  (C)  $\frac{3}{4} \frac{Rh}{M}$  (D)  $\frac{5}{36} \frac{Rh}{M}$

**MP0126**

9. Let the potential energy of a hydrogen atom in the ground state be zero. Then its energy in the first excited state will be :

(A) 10.2eV (B) 13.6eV (C) 23.8eV (D) 27.2 eV

**MP0127**

10. Hydrogen atoms in ground state are excited by monochromatic radiation of wavelength 975 Å. The number of lines in the resulting spectrum will be :

(A) 3 (B) 4 (C) 6 (D) 10

**MP0128**

11. The relation between  $\lambda_1$  : wavelength of series limit of lyman  $\lambda_2$  : the wavelength of the series limit of Balmer series and  $\lambda_3$  : the wavelength of first line of lyman series is

(A)  $\lambda_1 = \lambda_2 + \lambda_3$  (B)  $\lambda_3 = \lambda_1 + \lambda_2$  (C)  $\lambda_2 = \lambda_3 - \lambda_1$  (D) none

**MP0129**

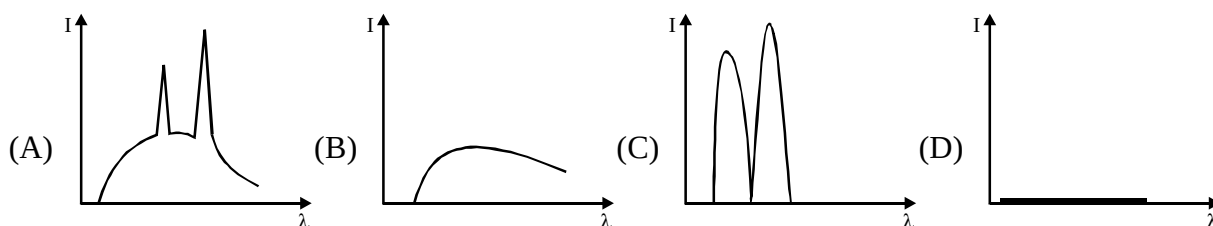
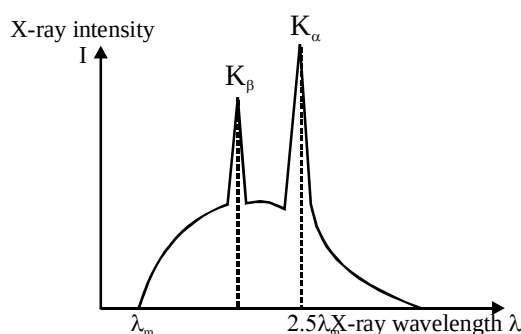
12. A hydrogen like gas atoms absorb radiations of wavelength  $\lambda_0$  and consequently emit radiations of 6 difference wavelengths of which, three wavelengths are shorter than  $\lambda_0$ . Choose the correct alternative(s).
- (A) The final excited state of the atoms is  $n = 3$ .  
 (B) The final excited state of the atoms is  $n = 4$ .  
 (C) The initial state of the atoms is  $n = 1$ .  
 (D) The initial state of the atoms is  $n = 3$ .

MP0130

13. According to the Bohr theory of the hydrogen atom, electron starting in the 4th energy level and eventually ending in the ground state could produce a total of how many lines in the hydrogen spectra?
- (A) 7 (B) 6 (C) 5 (D) 4  
 (E) 3

MP0131

14. When an electron accelerated by potential difference  $U$  is bombarded on a specific metal the emitted X-ray spectrum obtained is shown in adjoining graph. If the potential difference is reduced to  $U/3$ , the correct spectrum is :-



MP0132

15. In an X-ray experiment target is made up of copper ( $Z = 29$ ) having some impurity. The  $K_\alpha$  line of copper have wavelength  $\lambda_0$ . It was observed that another  $K_\alpha$  line due to impurity have wavelength

$$\frac{784}{625} \lambda_0 . \text{ The atomic number of the impurity element is}$$

- (A) 22 (B) 23 (C) 24 (D) 26

MP0133

## MULTIPLE CORRECT TYPE QUESTIONS

16. In a photoelectric effect experiment, if  $f$  is the frequency of radiations incident on the metal surface and  $I$  is the intensity of the incident radiations, then choose the correct statement(s).
- (A) If  $f$  is increased keeping  $I$  and work function constant then maximum kinetic energy of photoelectron increases.
- (B) If distance between cathode and anode is increased stopping potential increases
- (C) If  $I$  is increased keeping  $f$  and work function constant then stopping potential remains same and saturation current increases.
- (D) Work function is decreased keeping  $f$  and  $I$  constant then stopping potential increases

MP0134

17. When a monochromatic point-source of light is at a distance of 0.2 m from a small photoelectric cell, the stopping potential and the saturation current are respectively 0.6 volt and 18.0 mA. If the same source is placed 0.6 m away from the photoelectric cell, then :-
- (A) The stopping potential will be 0.2 volt      (B) The stopping potential will be 1.8 volt
- (C) The saturation current will be 6.0 mA      (D) The saturation current will be 2.0 mA

MP0135

18. The collector of the photocell (in photoelectric experiment) is made of tungsten while the emitter is of Platinum having work function of 10 eV. Monochromatic radiation of wavelength  $124 \text{ \AA}$  & power 100 watt is incident on emitter which emits photo electrons with a quantum efficiency of 1%. The accelerating voltage across the photocell is of 10,000 volts (Use :  $hc = 12400 \text{ eV \AA}$ )
- (A) The power supplied by the accelerating voltage source is 100 watt
- (B) The minimum wavelength of radiation coming from the tungsten target (collector) is  $1.23 \text{ \AA}$
- (C) The power supplied by the accelerating voltage source is 10 watt
- (D) The minimum wavelength of radiation coming from the tungsten target (collector) is  $2.23 \text{ \AA}$

MP0136

19. Light of wavelength  $\lambda_1$  &  $\lambda_2$  are falling on two metal surface A & B. For wavelength  $\lambda_1$  electron ejected from both the surfaces and for wavelength  $\lambda_2$  electron ejected from only surface B. On the basis of these facts which one of the following is a false statement
- (A) more energy is required for ejection of electron from metal 'A'
- (B)  $\lambda_1 > \lambda_2$
- (C) threshold wavelength for A is greater than B
- (D) energy of electron ejected from metal A will be greater than electron ejected from metal B for wavelength  $\lambda$

MP0137

20. The accelerating potential (V) applied between a photocathode and the respective anode is such that the fastest photoelectron can fly only one fourth of the distance between the cathode and the anode. If the distance between photocathode and anode is reduced to  $(1/4)^{\text{th}}$  of the original value while maintaining the accelerating potential constant, then
- (A) The fastest electron will reach the anode
- (B) The fastest electron will reach upto one fourth of the new distance between the cathode and the anode

- (C) Kinetic energy of the fastest photoelectron emitted from photocathode will not change due to change in plate separation (keeping  $V$  constant)  
 (D) Kinetic energy of the fastest photoelectron will increase due to decrease in plate separation (keeping  $V$  constant)

MP0138

21. For the electron in the  $n^{\text{th}}$  orbit of hydrogen atom. Under the assumption of the Bohr's atomic model choose the **CORRECT** option(s) (Here  $n$  is the principal quantum number) :-  
 (A) Frequency of the electron is inversely proportional to  $n^3$   
 (B) The magnitude of potential energy of the electron in an orbit is greater than its kinetic energy  
 (C) Magnetic induction at the nucleus produced due to the motion of electron in the  $n^{\text{th}}$  orbit is proportional to  $n^5$   
 (D) Magnetic moment produced due to the motion of electron in the  $n^{\text{th}}$  orbit is proportional to  $n$

MP0139

22. Energy liberated in the de-excitation of hydrogen atom from  $3^{\text{rd}}$  level to  $1^{\text{st}}$  level falls on a photocathode. Later when the same photocathode is exposed to a spectrum of some unknown hydrogen like gas, excited to  $2^{\text{nd}}$  energy level, it is found that the de-Broglie wavelength of the fastest photoelectrons, now ejected has decreased by a factor of 3. For this new gas, difference of energies of  $2^{\text{nd}}$  Lyman line and  $1^{\text{st}}$  Balmer line is found to be 3 times the ionization potential of the hydrogen atom. Select the correct statement(s) :  
 (A) The gas is lithium  
 (B) The gas is helium  
 (C) The work function of photocathode is 8.5 eV  
 (D) The work function of photocathode is 5.5 eV

MP0140

23. An electron of the kinetic energy 10eV collides with a hydrogen atom in 1st excited state. Assuming loss of kinetic energy in the collision to be quantized, the collision :  
 (A) may be perfectly inelastic  
 (B) may be inelastic  
 (C) may be elastic  
 (D) must be inelastic

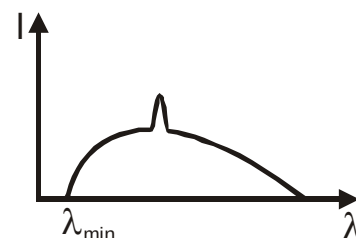
MP0141

24. An X-ray tube has three main controls.  
 (i) the target material (its atomic number  $Z$ )  
 (ii) the filament current ( $I_f$ )  
 (iii) the accelerating voltage ( $V$ )

Figure shows a typical intensity distribution against wavelength.

Which of the following is **CORRECT**?

- (A) The limit  $\lambda_{\text{min}}$  is proportional to  $V^{-1}$   
 (B) The sharp peak shifts to the right as  $Z$  is increased  
 (C) The penetrating power of X-ray increases if  $V$  is increased  
 (D) The intensity everywhere increases if filament current  $I_f$  is increased



MP0142

## COMPREHENSION TYPE QUESTIONS

## Paragraph for Question 25 to 27

A mercury arc lamp provides 0.1 watt of ultra-violet radiation at a wavelength of  $\lambda = 2537 \text{ \AA}$  only. The photo tube (cathode of photo electric device) consists of potassium and has an effective area of  $4 \text{ cm}^2$ . The cathode is located at a distance of 1m from the radiation source. The work function for potassium is  $\phi_0 = 2.22 \text{ eV}$ .

25. According to classical theory, the radiation from arc lamp spreads out uniformly in space as spherical wave. What time of exposure to the radiation should be required for a potassium atom (radius  $2 \text{ \AA}$ ) in the cathode to accumulate sufficient energy to eject a photo-electron ?  
 (A) 352 second (B) 176 second (C) 704 seconds (D) No time lag

MP0143

26. To what saturation current does the flux of photons at the cathode corresponds if the photo conversion efficiency is 5%.  
 (A) 32.5 nA (B) 10.15 nA (C) 65 nA (D) 3.25 nA

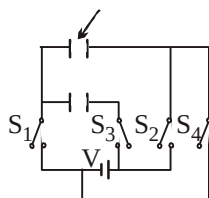
MP0143

27. What is the cut off potential  $V_0$  ?  
 (A) 26.9 V (B) 2.69 V (C) 1.35 V (D) 5.33 V

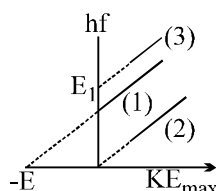
MP0143

## Paragraph for question nos. 28 to 30

The circuit shown is placed in vacuum. Both the capacitors are identical and they have the same capacitance  $C$ . Light is incident on the left plate of the upper capacitor. When all the switches are open then the  $hf$  versus  $KE_{\max}$  is shown by the straight line (A). In all the cases, we are measuring the  $KE_{\max}$  when the electron reaches the opposite plate.



When only  $S_1$  the switches and  $S_2$  are closed, the graph becomes (B).  
 When only  $S_3$  and  $S_4$  are closed then the graph becomes (C).



28. What is the work function of the cathode?  
 (A)  $E$  (B)  $E_1$  (C)  $E + E_1$  (D) none of these

MP0144

29. What is the value of  $eV$ ?

- (A)  $E$  (B)  $E_1$  (C)  $E + E_1$  (D) none of these

MP0144

30. What is the value of  $E_1$ ?

- (A)  $3E$  (B)  $3E/2$  (C)  $E/2$  (D) none of these

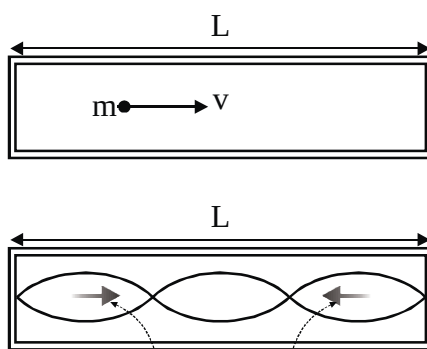
MP0144

### Paragraph for Question. 31 to 33

While conducting his doctoral research in theoretical physics and with no experimental evidence to go on, de Broglie reasoned by analogy with Einstein's equation  $E = hf$  and with some of the ideas of his theory of relativity. The details need not concern us, but they led de Broglie to postulate that if a material particle of momentum  $p = mv$  has a wave-like nature, then its wavelength must be given by

$$\lambda = \frac{h}{p} = \frac{h}{mv} \text{ where } h \text{ is Planck's constant. This is called the de-Broglie wavelength.}$$

de-Broglie considered a matter wave to be a traveling wave. But suppose that a "particle" of matter is confined to a small region of space and cannot travel. How do the wave-like properties manifest themselves? This is the problem of "a particle in a box." Figure shows a particle of mass  $m$  moving in one dimension as it bounces back and forth with speed  $v$  between the ends of a box of length  $L$ . We'll call this a one-dimensional box; its width isn't relevant. A particle in a box creates a standing de Broglie wave as it reflects back and forth.



Matter waves travel in both directions.

31. What should be de-Broglie wavelength of confined particle in the box [here  $n \in \mathbb{N}$ ]

- (A)  $\frac{L}{2n}$  (B)  $\frac{2L}{n}$  (C)  $\frac{L}{n}$  (D)  $nL$

MP0145

32. Confined particle's energy is given by

- (A)  $\frac{n^2 h^2}{2mL^2}$  (B)  $\frac{2n^2 h^2}{mL^2}$  (C)  $\frac{n^2 h^2}{8mL^2}$  (D)  $\frac{n^2 h^2}{4mL^2}$

MP0145

33. Consider an oil drop from Millikan's oil drop experiment having diameter  $1\text{ }\mu\text{m}$  confined between the plates separated by  $10\text{ }\mu\text{m}$ . Density of oil is  $900\text{ kg/m}^3$ . What is minimum energy of such an oil drop? [Given :  $h = 6.63 \times 10^{-34}\text{ Js}$ ]

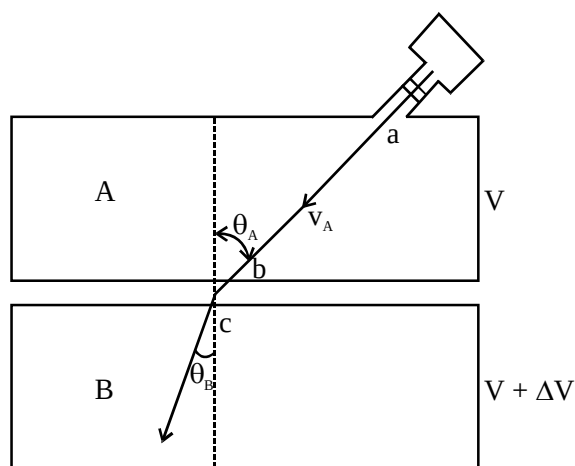
(A)  $2.4 \times 10^{-42}\text{ J}$  (B)  $1.2 \times 10^{-42}\text{ J}$  (C)  $3.6 \times 10^{-42}\text{ J}$  (D)  $4.8 \times 10^{-42}\text{ J}$

MP0145

### Paragraph for Question Nos. 34 to 36

Let a pencil of electrons from a suitable gun G enter through orifice in an enclosed metal box A, which has potential  $V$  relative to filament of the gun. Let these electrons emerge from A through orifice b and enter through c to another box B, which is maintained at a potential  $V + \Delta V$ . The electric field between the two boxes changes the component of velocity of the electrons perpendicular to the adjacent surface, and the electrons enter B with a change in their direction of motion. Let  $v_A$  and  $v_B$  be the velocities of the electrons in A and B, respectively, and  $\theta_A$ ,  $\theta_B$  the angles between these directions and the normal to the box faces at b and c. Since the electric field does not change the horizontal component of velocity,

$$v_A \sin \theta_A = v_B \sin \theta_B \quad \frac{\sin \theta_A}{\sin \theta_B} = \frac{v_B}{v_A}$$



The "refraction" of a pencil of electrons

Now if we dealing with light waves undergoing refraction, or any other kind of wave, the relation would be

$$\frac{\sin \theta_A}{\sin \theta_B} = \mu = \frac{u_A}{u_B}$$



where  $\mu$  is the relative refractive index of the two media and  $u_A, u_B$  are the corresponding velocities of light wave. Comparison of the last two equations gives the result  $\frac{u_A}{u_B} = \frac{v_B}{v_A}$ . We may conclude that if matter waves follow the electron along its path, the wave speed  $u$  is inversely proportional to the speed  $v$  of the electron, or  $u = \frac{b}{v}$ .

34. If we define the frequency of matter waves as  $f = \frac{u}{\lambda}$ , the

- (A) Frequency of matter waves in medium A is more than that in medium B
- (B) Frequency of matter waves in medium A is less than that in medium B
- (C) Frequency of matter waves in medium A is same as that in medium B
- (D) Cannot be predicted

MP0146

35. Suppose  $V_A = 20$  volt and  $V_B = 15$  volt. Choose the **CORRECT** statement :-

- (A) The speed of electrons as well as speed of matter waves inside box B is more
- (B) The speed of electrons as well as speed of matter waves inside box B is less
- (C) The speed of electrons inside the box B is more, but speed of matter waves in box B is less
- (D) The speed of electrons inside the box B is less, but speed of matter waves in box B is more

MP0146

36. The refractive index for matter waves can be defined as  $\frac{c}{u}$  where  $c$  is some constant. So refractive index for matter waves

- (A) is inversely proportional to  $\lambda$
- (B) is independent of  $\lambda$
- (C) is directly proportional of  $\lambda$
- (D) is proportional to  $\sqrt{\lambda}$

MP0146

## EXERCISE (JM)

1. Suppose an electron is attracted towards the origin by a force  $\frac{k}{r}$  where 'k' is a constant and 'r' is the distance of the electron from the origin. By applying Bohr model to this system, the radius of the  $n^{\text{th}}$  orbital of the electron is found to be ' $r_n$ ' and the kinetic energy of the electron to be ' $T_n$ '. Then which of the following is true? [AIEEE - 2008]

(1)  $T_n \propto \frac{1}{n^2}, r_n \propto n^2$

(2)  $T_n$  independent of  $n, r_n \propto n$

(3)  $T_n \propto \frac{1}{n}, r_n \propto n$

(4)  $T_n \propto \frac{1}{n}, r_n \propto n^2$

MP0147

2. The transition from the state  $n = 4$  to  $n = 3$  in a hydrogen like atom results in ultraviolet radiation. Infrared radiation will be obtained in the transition from :- [AIEEE - 2009]

(1)  $4 \rightarrow 2$

(2)  $5 \rightarrow 4$

(3)  $2 \rightarrow 1$

(4)  $3 \rightarrow 2$

MP0148

3. The surface of a metal is illuminated with the light of 400 nm. The kinetic energy of the ejected photoelectrons was found to be 1.68 eV. The work function of the metal is : ( $hc = 1240 \text{ eV-nm}$ )

[AIEEE - 2009]

(1) 1.51 eV

(2) 1.68 eV

(3) 3.09 eV

(4) 1.41 eV

MP0149

**Directions :** Question number 4 contain Statement-1 and Statement-2. Of the four choices given after the statements, choose the one that best describes the two statements.

4. Statement-1 : When ultraviolet light is incident on a photocell, its stopping potential is  $V_0$  and the maximum kinetic energy of the photoelectrons is  $K_{\text{max}}$ . When the ultraviolet light is replaced by X-rays, both  $V_0$  and  $K_{\text{max}}$  increase.

**Statement-2 :** Photoelectrons are emitted with speeds ranging from zero to a maximum value because of the range of frequencies present in the incident light. [AIEEE - 2010]

(1) Statement-1 is true, Statement-2 is false

(2) Statement-1 is true, Statement-2 is true; Statement-2 is the correct explanation of Statement-1

(3) Statement-1 is true, Statement-2 is true; Statement-2 is not the correct explanation of Statement-1

(4) Statement-1 is false, Statement-2 is true

MP0150

5. If a source of power 4kW produces  $10^{20}$  photons/second, the radiation belongs to apart of the spectrum called :- [AIEEE - 2010]

(1)  $\gamma$ -rays

(2) X-rays

(3) ultraviolet rays

(4) microwaves

MP0151

6. Energy required for the electron excitation in  $\text{Li}^{++}$  from the first to the third Bohr orbit is:-

[AIIEEE-2011]

- (1) 108.8 eV                      (2) 122.4 eV                      (3) 12.1 eV                      (4) 36.3 eV

MP0152

7. This question has Statement-1 and Statement-2. Of the four choices given after the statements, choose the one that best describes the two statements.

**Statement-1 :** A metallic surface is irradiated by a monochromatic light of frequency  $\nu > \nu_0$  (the threshold frequency). The maximum kinetic energy and the stopping potential are  $K_{\max}$  and  $V_0$  respectively. If the frequency incident on the surface is doubled, both the  $K_{\max}$  and  $V_0$  are also doubled.

**Statement-2 :** The maximum kinetic energy and the stopping potential of photoelectrons emitted from a surface are linearly dependent on the frequency of incident light. [AIIEEE-2011]

- (1) Statement-1 is true, Statement-2 is true, Statement-2 is not the correct explanation of Statement-1  
(2) Statement-1 is false, Statement-2 is true  
(3) Statement-1 is true, Statement-2 is false  
(4) Statement-1 is true, Statement-2 is true, Statement-2 is the correct explanation of Statement-1

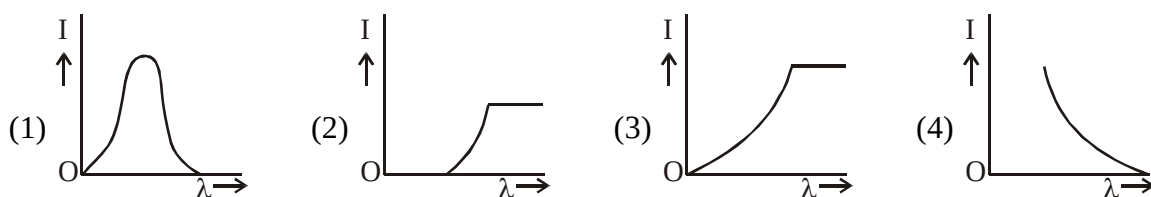
MP0153

8. Hydrogen atom is excited from ground state to another state with principal quantum number equal to 4. Then the number of spectral lines in the emission spectra will be :- [AIIEEE-2012]

- (1) 6                      (2) 2                      (3) 3                      (4) 5

MP0154

9. The anode voltage of photocell is kept fixed. The wavelength  $\lambda$  of the light falling on the cathode is gradually changed. The plate current  $I$  of the photocell varies as follows : [AIIEEE-2013]



MP0155

10. In a hydrogen like atom electron makes transition from an energy level with quantum number  $n$  to another with quantum number  $(n-1)$ . If  $n \gg 1$ , the frequency of radiation emitted is proportional to :

[JEE Main-2013]

- (1)  $\frac{1}{n}$                       (2)  $\frac{1}{n^2}$                       (3)  $\frac{1}{n^{3/2}}$                       (4)  $\frac{1}{n^3}$

MP0156

11. As an electron makes a transition from an excited state to the ground state of a hydrogen - like atom/ion : [JEE Main-2015]

- (1) kinetic energy decreases, potential energy increases but total energy remains same
- (2) kinetic energy and total energy decrease but potential energy increases
- (3) its kinetic energy increases but potential energy and total energy decreases
- (4) kinetic energy, potential energy and total energy decrease

MP0157

12. Match List-I (Fundament Experiment) with List-II (its conclusion) and select the correct option from the choices given below the list : [JEE Main-2015]

|     | List-I                    |       | List-II                        |
|-----|---------------------------|-------|--------------------------------|
| (A) | Franck-Hertz Experiment.  | (i)   | Particle nature of light       |
| (B) | Photo-electric experiment | (iii) | Discrete energy levels of atom |
| (C) | Davison-Germer Experiment | (iii) | Wave nature of electroc        |
|     |                           | (iv)  | Structure of atom              |

- (1) A-ii, B-i, C-iii
- (2) A-iv, B-iii, C-ii
- (3) A-i, B-iv, C-iii
- (4) A-ii, B-iv, C-iii

MP0158

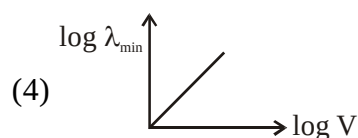
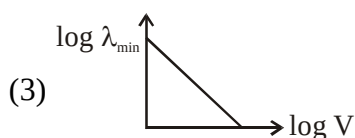
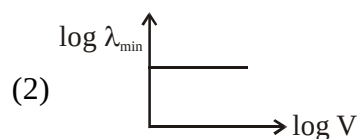
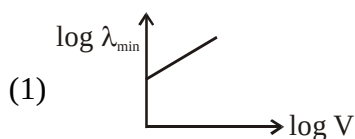
13. Radiation of wavelength  $\lambda$ , is incident on a photocell. The fastest emitted electron has speed  $v$ . If the wavelength of changed to  $\frac{3\lambda}{4}$ , the speed of the fastest emitted electron will be :-[JEE Main-2016]

- (1)  $= v\left(\frac{3}{5}\right)^{1/2}$
- (2)  $> v\left(\frac{4}{3}\right)^{1/2}$
- (3)  $< v\left(\frac{4}{3}\right)^{1/2}$
- (4)  $= v\left(\frac{4}{3}\right)^{1/2}$

MP0159

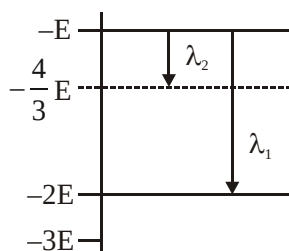
14. An electron beam is accelerated by a potential difference  $V$  to hit a metallic target to produce X-rays. It produces continuous as well as characteristic X-rays. If  $\lambda_{\min}$  is the smallest possible wavelength of X-ray in the spectrum, the variation of  $\log \lambda_{\min}$  with  $\log V$  is correctly represented in :

[JEE Main-2017]



MP0160

15. Some energy levels of a molecule are shown in the figure. The ratio of the wavelengths  $r = \lambda_1/\lambda_2$ , is given by : [JEE Main-2017]



- (1)  $r = \frac{3}{4}$       (2)  $r = \frac{1}{3}$       (3)  $r = \frac{4}{3}$       (4)  $r = \frac{2}{3}$

MP0161

16. A particle A of mass  $m$  and initial velocity  $v$  collides with a particle B of mass  $\frac{m}{2}$  which is at rest. The collision is head on, and elastic. The ratio of the de-Broglie wavelengths  $\lambda_A$  to  $\lambda_B$  after the collision is: [JEE Main-2017]

- (1)  $\frac{\lambda_A}{\lambda_B} = \frac{2}{3}$       (2)  $\frac{\lambda_A}{\lambda_B} = \frac{1}{2}$       (3)  $\frac{\lambda_A}{\lambda_B} = \frac{1}{3}$       (4)  $\frac{\lambda_A}{\lambda_B} = 2$

MP0162

17. If the series limit frequency of the Lyman series is  $\nu_L$ , then the series limit frequency of the Pfund series is : [JEE Main-2018]

- (1)  $16 \nu_L$       (2)  $\nu_L/16$       (3)  $\nu_L/25$       (4)  $25 \nu_L$

MP0163

## EXERCISE (JA)

1. Which one of the following statements is **WRONG** in the context of X-rays generated from a X-ray tube ? [JEE 2008]
- (A) Wavelength of characteristic X-rays decreases when the atomic number of the target increases  
 (B) Cut-off wavelength of the continuous X-rays depends on the atomic number of the target  
 (C) Intensity of the characteristic X-rays depends on the electrical power given to the X-rays tube  
 (D) Cut-off wavelength of the continuous X-rays depends on the energy of the electrons in the X-ray tube

MP0164

## Paragraph for Question Nos. 2 to 4

In a mixture of H – He<sup>+</sup> gas (He<sup>+</sup> is singly ionized He atom), H atoms and He<sup>+</sup> ions are excited to their respective first excited states. Subsequently, H atoms transfer their total excitation energy to He<sup>+</sup> ions (by collisions). Assume that the Bohr model of atom is exactly valid. [JEE 2008]

2. The quantum number  $n$  of the state finally populated in He<sup>+</sup> ions is
- (A) 2 (B) 3 (C) 4 (D) 5
3. The wavelength of light emitted in the visible region by He<sup>+</sup> ions after collisions with H atoms is
- (A)  $6.5 \times 10^{-7}$  m (B)  $5.6 \times 10^{-7}$  m (C)  $4.8 \times 10^{-7}$  m (D)  $4.0 \times 10^{-7}$  m

MP0165

MP0165

4. The ratio of the kinetic energy of the  $n = 2$  electron for the H atom to that of He<sup>+</sup> ion is
- (A)  $\frac{1}{4}$  (B)  $\frac{1}{2}$  (C) 1 (D) 2

MP0165

## Paragraph for Question Nos. 5 to 7

When a particle is restricted to move along x-axis between  $x = 0$  and  $x = a$ , where  $a$  is of nanometer dimension, its energy can take only certain specific values. The allowed energies of the particle moving in such a restricted region, correspond to the formation of standing waves with nodes at its ends  $x = 0$  and  $x = a$ . The wavelength of this standing wave is related to the linear momentum  $p$  of the particle according to the de Broglie relation. The energy of the particle of mass  $m$  is related to its linear

momentum as  $E = \frac{p^2}{2m}$ . Thus, the energy of the particle can be denoted by a quantum number ' $n$ '

taking values 1,2,3, ... ( $n = 1$ , called the ground state) corresponding to the number of loops in the standing wave. Use the model described above to answer the following three questions for a particle moving in the line  $x = 0$  to  $x = a$ . Take  $h = 6.6 \times 10^{-34}$  Js and  $e = 1.6 \times 10^{-19}$  C. [JEE-2009]

5. The allowed energy for the particle for a particular value of  $n$  is proportional to
- (A)  $a^{-2}$  (B)  $a^{-3/2}$  (C)  $a^{-1}$  (D)  $a^2$

MP0166

6. If the mass of the particle is  $m = 1.0 \times 10^{-30}$  kg and  $a = 6.6$  nm, the energy of the particle in its ground state is closest to :
- (A) 0.8 meV                      (B) 8 meV                      (C) 80 meV                      (D) 800 meV

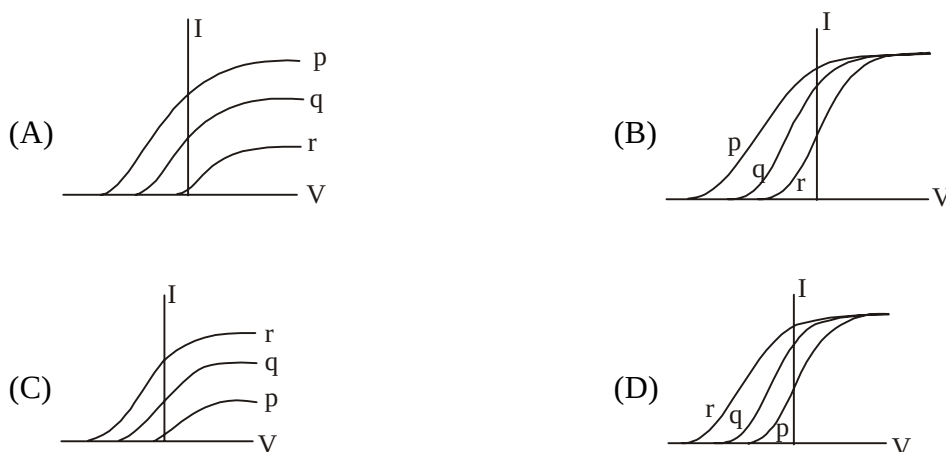
MP0166

7. The speed of the particle, that can take discrete values, is proportional to
- (A)  $n^{-3/2}$                       (B)  $n^{-1}$                       (C)  $n^{1/2}$                       (D)  $n$

MP0166

8. Photoelectric effect experiments are performed using three different metal plates p, q and r having work functions  $\phi_p = 2.0$  eV,  $\phi_q = 2.5$  eV and  $\phi_r = 3.0$  eV, respectively. A light beam containing wavelengths of 550 nm, 450 nm and 350 nm with equal intensities illuminates each of the plates. The correct I-V graph for the experiment is :

[JEE-2009]



MP0167

9. An  $\alpha$ -particle and a proton are accelerated from rest by a potential difference of 100 V. After this, their de Broglie wavelengths are  $\lambda_\alpha$  and  $\lambda_p$  respectively. The ratio  $\frac{\lambda_p}{\lambda_\alpha}$ , to the nearest integer, is

[JEE 2010]

MP0168

### Paragraph for Question Nos. 10 to 12

The key feature of Bohr's theory of spectrum of hydrogen atom is the quantization of angular momentum when an electron is revolving around a proton. We will extend this to a general rotational motion to find quantized rotational energy of a diatomic molecule assuming it to be rigid. The rule to be applied is Bohr's quantization condition.

[JEE 2010]

10. A diatomic molecule has moment of inertia  $I$ . By Bohr's quantization condition its rotational energy in the  $n^{\text{th}}$  level ( $n = 0$  is not allowed) is

(A)  $\frac{1}{n^2} \left( \frac{h^2}{8\pi^2 I} \right)$                       (B)  $\frac{1}{n} \left( \frac{h^2}{8\pi^2 I} \right)$                       (C)  $n \left( \frac{h^2}{8\pi^2 I} \right)$                       (D)  $n^2 \left( \frac{h^2}{8\pi^2 I} \right)$

MP0169

11. It is found that the excitation frequency from ground to the first excited state of rotation for the CO molecule is close to  $\frac{4}{\pi} \times 10^{11}$  Hz. Then the moment of inertia of CO molecule about its center of mass is close to [Take  $h = 2\pi \times 10^{-34}$  Js]
- (A)  $2.76 \times 10^{-46}$  kg m<sup>2</sup> (B)  $1.87 \times 10^{-46}$  kg m<sup>2</sup>  
 (C)  $4.67 \times 10^{-47}$  kg m<sup>2</sup> (D)  $1.17 \times 10^{-47}$  kg m<sup>2</sup>

MP0169

12. In a CO molecule, the distance between C (mass = 12 a.m.u.) and O (mass = 16 a.m.u.), where  $1 \text{ a.m.u.} = \frac{5}{3} \times 10^{-27} \text{ kg}$ , is close to

(A)  $2.4 \times 10^{-10} \text{ m}$  (B)  $1.9 \times 10^{-10} \text{ m}$  (C)  $1.3 \times 10^{-10} \text{ m}$  (D)  $4.4 \times 10^{-11} \text{ m}$

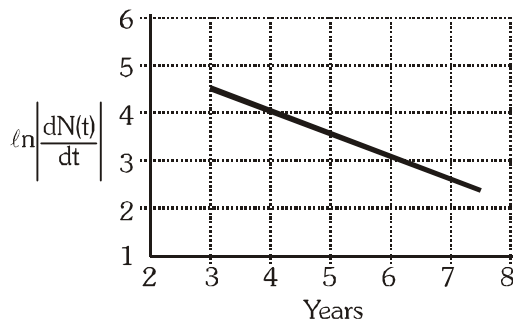
MP0169

13. To determine the half life of a radioactive element, a student plots a graph of  $\ln \left| \frac{dN(t)}{dt} \right|$  versus t. Here

$\frac{dN(t)}{dt}$  is the rate of radioactive decay at time t. If the number of radioactive nuclei of this element

decreases by a factor of p after 4.16 years, the value of p is

[JEE 2010]



MP0170

14. The wavelength of the first spectral line in the Balmer series of hydrogen atom is 6561 Å. The wavelength of the second spectral line in the Balmer series of singly-ionized helium atom is

[JEE 2011]

(A) 1215 Å (B) 1640 Å (C) 2430 Å (D) 4687 Å

MP0171

15. A silver sphere of radius 1 cm and work function 4.7 eV is suspended from an insulating thread in free-space. It is under continuous illumination of 200 nm wavelength light. As photoelectrons are emitted, the sphere gets charged and acquires a potential. The maximum number of photoelectrons emitted from the sphere is  $A \times 10^Z$  (where  $1 < A < 10$ ). The value of 'Z' is

[JEE 2011]

MP0172



16. A proton is fired from very far away towards a nucleus with charge  $Q = 120 e$ , where  $e$  is the electronic charge. It makes a closest approach of  $10 \text{ fm}$  to the nucleus. The de Broglie wavelength (in units of  $\text{fm}$ ) of the proton at its start is (Take : The proton mass,  $m_p = (5/3) \times 10^{-27} \text{ kg}$ ;

$$h/e = 4.2 \times 10^{-15} \text{ J.s/C}; \frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ m/F}; 1 \text{ fm} = 10^{-15} \text{ m})$$

[JEE 2012]

MP0173

17. The work functions of Silver and sodium are  $4.6$  and  $2.3 \text{ eV}$ , repetitively. The ratio of the slope of the stopping potential versus frequency plot for Silver to that of Sodium is. [JEE Advanced-2013]

MP0174

18. The radius of the orbit of an electron in a Hydrogen-like atom is  $4.5 a_0$ , where  $a_0$  is the Bohr radius. Its orbital angular momentum is  $\frac{3h}{2\pi}$ . It is given that  $h$  is Planck constant and  $R$  is Rydberg constant. The possible wavelength (s), when the atom de-excites, is (are) :- [JEE Advanced-2013]

- (A)  $\frac{9}{32R}$  (B)  $\frac{9}{16R}$  (C)  $\frac{9}{5R}$  (D)  $\frac{4}{3R}$

MP0175

19. Consider a hydrogen atom with its electron in the  $n^{\text{th}}$  orbital. An electromagnetic radiation of wavelength  $90 \text{ nm}$  is used to ionize the atom. If the kinetic energy of the ejected electron is  $10.4 \text{ eV}$ , then the value of  $n$  is ( $hc = 1242 \text{ eV nm}$ ). [JEE Advanced-2015]

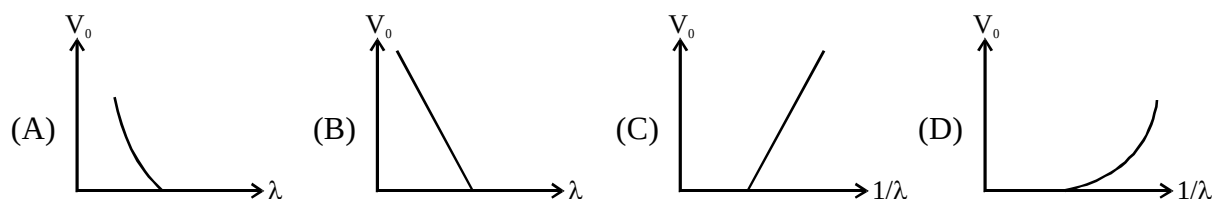
MP0176

20. Planck's constant  $h$ , speed of light  $c$  and gravitational constant  $G$  are used to form a unit of length  $L$  and a unit of mass  $M$ . Then the correct option(s) is(are) :- [JEE Advanced-2015]

- (A)  $M \propto \sqrt{c}$  (B)  $M \propto \sqrt{G}$  (C)  $L \propto \sqrt{h}$  (D)  $L \propto \sqrt{G}$

MP0177

21. For photo-electric effect with incident photon wavelength  $\lambda$ , the stopping potential is  $V_0$ . Identify the correct variation(s) of  $V_0$  with  $\lambda$  and  $1/\lambda$ . [JEE Advanced-2015]



MP0178

22. An electron in an excited state of  $\text{Li}^{2+}$  ion has angular momentum  $3h/2\pi$ . The de Broglie wavelength of the electron in this state is  $p\pi a_0$  (where  $a_0$  is the Bohr radius). The value of  $p$  is [JEE Advanced-2015]

MP0179

23. In a historical experiment to determine Planck's constant, a metal surface was irradiated with light of different wavelengths. The emitted photoelectron energies were measured by applying a stopping potential. The relevant data for the wavelength ( $\lambda$ ) of incident light and the corresponding stopping potential ( $V_0$ ) are given below: [JEE Advanced-2016]

| $\lambda (\mu\text{m})$ | $V_0 (\text{Volt})$ |
|-------------------------|---------------------|
| 0.3                     | 2.0                 |
| 0.4                     | 1.0                 |
| 0.5                     | 0.4                 |

Given that  $c = 3 \times 10^8 \text{ ms}^{-1}$  and  $e = 1.6 \times 10^{-19} \text{ C}$ , Planck's constant (in units of J s) found from such an experiment is :

- (A)  $6.0 \times 10^{-34}$  (B)  $6.4 \times 10^{-34}$  (C)  $6.6 \times 10^{-34}$  (D)  $6.8 \times 10^{-34}$

MP0180

24. Highly excited states for hydrogen like atoms (also called Rydberg states) with nuclear charge  $Ze$  are defined by their principal quantum number  $n$ , where  $n \gg 1$ . Which of the following statement(s) is (are) true? [JEE Advanced-2016]

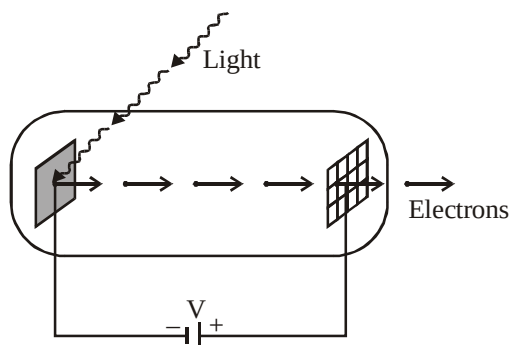
- (A) Relative change in the radii of two consecutive orbitals does not depend on  $Z$   
 (B) Relative change in the radii of two consecutive orbitals varies as  $1/n$   
 (C) Relative change in the energy of two consecutive orbitals varies as  $1/n^3$   
 (D) Relative change in the angular momenta of two consecutive orbitals varies as  $1/n$

MP0181

25. A hydrogen atom in its ground state is irradiated by light of wavelength  $970 \text{ \AA}$ . Taking  $hc/e = 1.237 \times 10^{-6} \text{ eV m}$  and the ground state energy of hydrogen atom as  $-13.6 \text{ eV}$ , the number of lines present in the emission spectrum is [JEE Advanced-2016]

MP0182

26. Light of wavelength  $\lambda_{ph}$  falls on a cathode plate inside a vacuum tube as shown in the figure. The work function of the cathode surface is  $\phi$  and the anode is a wire mesh of conducting material kept at a distance  $d$  from the cathode. A potential difference  $V$  is maintained between the electrodes. If the minimum de Broglie wavelength of the electrons passing through the anode is  $\lambda_e$ , which of the following statement(s) is(are) true ? [JEE Advanced-2016]



- (A) For large potential difference ( $V \gg \phi/e$ ),  $\lambda_e$  is approximately halved if  $V$  is made four times  
 (B)  $\lambda_e$  increases at the same rate as  $\lambda_{ph}$  for  $\lambda_{ph} < hc/\phi$   
 (C)  $\lambda_e$  is approximately halved, if  $d$  is doubled  
 (D)  $\lambda_e$  decreases with increase in  $\phi$  and  $\lambda_{ph}$

**MP0183**

27. An electron in a hydrogen atom undergoes a transition from an orbit with quantum number  $n_i$  to another with quantum number  $n_f$ .  $V_i$  and  $V_f$  are respectively the initial and final potential energies of the electron. If  $\frac{V_i}{V_f} = 6.25$ , then the smallest possible  $n_f$  is.

**[JEE Advanced-2017]**

**MP0184**

28. A photoelectric material having work-function  $\phi_0$  is illuminated with light of wavelength  $\lambda$   $\left( \lambda < \frac{hc}{\phi_0} \right)$ .

The fastest photoelectron has a de-Broglie wavelength  $\lambda_d$ . A change in wavelength of the incident light by  $\Delta\lambda$  results in a change  $\Delta\lambda_d$  in  $\lambda_d$ . Then the ratio  $\Delta\lambda_d/\Delta\lambda$  is proportional to

**[JEE Advanced-2017]**

- (A)  $\lambda_d^3/\lambda^2$  (B)  $\lambda_d^3/\lambda$  (C)  $\lambda_d^2/\lambda^2$  (D)  $\lambda_d/\lambda$

**MP0185**

29. In a photoelectric experiment a parallel beam of monochromatic light with power of 200 W is incident on a perfectly absorbing cathode of work function 6.25 eV. The frequency of light is just above the threshold frequency so that the photoelectrons are emitted with negligible kinetic energy. Assume that the photoelectron emission efficiency is 100%. A potential difference of 500 V is applied between the cathode and the anode. All the emitted electrons are incident normally on the anode and are absorbed. The anode experiences a force  $F = n \times 10^{-4}$  N due to the impact of the electrons. The value of  $n$  is..... Mass of the electron  $m_e = 9 \times 10^{-31}$  kg and  $1.0 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$ .

**[JEE Advanced-2018]**

**MP0186**

30. Consider a hydrogen-like ionized atom with atomic number  $Z$  with a single electron. In the emission spectrum of this atom, the photon emitted in the  $n = 2$  to  $n = 1$  transition has energy 74.8 eV higher than the photon emitted in the  $n = 3$  to  $n = 2$  transition. The ionization energy of the hydrogen atom is 13.6 eV. The value of  $Z$  is.....

**[JEE Advanced-2018]**

**MP0187**

31. A free hydrogen atom after absorbing a photon of wavelength  $\lambda_a$  gets excited from the state  $n = 1$  to the state  $n = 4$ . Immediately after that the electron jumps to  $n = m$  state by emitting a photon of wavelength  $\lambda_e$ . Let the change in momentum of atom due to the absorption and the emission are

$\Delta p_a$  and  $\Delta p_e$ , respectively. If  $\lambda_a/\lambda_e = \frac{1}{5}$ . Which of the option(s) is/are correct ?

[Use  $hc = 1242 \text{ eV nm}$ ;  $1 \text{ nm} = 10^{-9} \text{ m}$ ,  $h$  and  $c$  are Planck's constant and speed of light, respectively]

[JEE Advanced-2019]

(1)  $\lambda_e = 418 \text{ nm}$

(2) The ratio of kinetic energy of the electron in the state  $n = m$  to the state  $n = 1$  is  $\frac{1}{4}$

(3)  $m = 2$

(4)  $\Delta p_a/\Delta p_e = \frac{1}{2}$

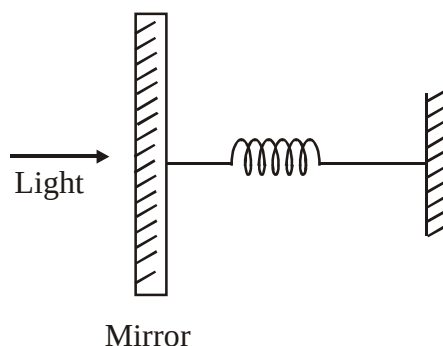
MP0188

32. A perfectly reflecting mirror of mass  $M$  mounted on a spring constitutes a spring-mass system of angular frequency  $\Omega$  such that  $\frac{4\pi M\Omega}{h} = 10^{24} \text{ m}^{-2}$  with  $h$  as Planck's constant.  $N$  photons of wavelength

$\lambda = 8\pi \times 10^{-6} \text{ m}$  strike the mirror simultaneously at normal incidence such that the mirror gets displaced by  $1 \mu\text{m}$ . If the value of  $N$  is  $x \times 10^{12}$ , then the value of  $x$  is \_\_\_\_\_.

[Consider the spring as massless]

[JEE Advanced-2019]



MP0189

# CBSE PREVIOUS YEAR'S QUESTIONS

1. Two metals A and B have work functions 4eV and 10 eV respectively. Which metal has lower threshold wavelength? [1; CBSE-2004]
2. Red light, however bright it is, cannot produce the emission of electrons from a clean zinc surface. But even weak ultraviolet radiation can do so. Why?  
X-rays of wavelength ' $\lambda$ ' fell on photosensitive surface, emitting electrons. Assuming X-rays of wavelength ' $\lambda$ ' fall on a photo sensitive Surface, emitting be neglected, prove that the de Broglie wavelength of electrons emitted will be  $\sqrt{\frac{h\lambda}{2mc}}$
3. Define the terms: 'half-life period' and 'decay constant of a radioactive sample. Derive the relation between these terms. [3; CBSE-2004]
4. When a deuteron of mass 20141 u and negligible kinetic energy is absorbed by a lithium ( ${}^6_3\text{Li}$ ) nucleus of mass 6.0155 u, the compound nucleus disintegrates spontaneously into two alpha particles, each of mass 4.0026 u. Calculate the energy in joules carried by each alpha particle ( $1\text{u} = 1.66 \times 10^{-27}\text{kg}$ ). [3; CBSE-2004]
5. Ultraviolet light is incident on two photosensitive materials having work functions  $W_1$  and  $W_2$  ( $W_1 > W_2$ ) in which case will the kinetic energy of the emitted electrons be greater? Why? [1;CBSE-2005]
6. Mention the significance of Davisson-Germer experiment An  $\alpha$  particle and a proton are accelerated from rest through the same potential difference V. Find the ratio of de-Broglie wavelengths associated with them. [3;CBSE-2005]
7. (a) Draw the energy level diagram showing the emission of  $\beta$ -particles followed by  $\gamma$ -rays by a  ${}^{60}_{27}\text{Co}$  nucleus.  
(b) Plot the distribution of kinetic energy of  $\beta$  particles and state why the energy spectrum is continuous. [3; CBSE-2005]
8. A radioactive sample contains 2.2 mg of pure  ${}^{11}_6\text{C}$  which has half-life period of 1224 seconds. Calculate  
(i) The number of atoms present initially.  
(ii) The activity when 5 $\mu\text{g}$  of the sample will be left. [3; CBSE-2005]
9. De-Broglie wavelength associated with an electron accelerated through a potential difference V is  $\lambda$ . What will be its wavelength when the accelerating potential is increased to 4 V? [1; CBSE-2006]
10. Sketch a graph between frequency of incident radiations and stopping potential for a given photosensitive material. What information can be obtained from the value of the intercept on the potential axis?  
A source of light of frequency greater than the threshold frequency is placed at a distance of 1 m from the cathode of a photo-cell. The stopping potential is found to be V. If the distance of the light source from the cathode is reduced, explain giving reasons, what change will you observe in the  
(i) photoelectric current,  
(ii) stopping potential. [3; CBSE-2006]

11. Define the terms half-life period and decay constant of a radioactive substance. Write their S.I. units. Establish the relationship between the two. [3; CBSE-2006]
12. A neutron is absorbed by a  ${}^6_3\text{Li}$  nucleus with the subsequent emission of an alpha particle.  
(i) Write the corresponding nuclear reaction.  
(ii) Calculate the energy released, in MeV, in this reaction.  
Given mass  ${}^6_3\text{Li} = 6.015126 \text{ u}$ ; mass (neutron) = 1.0086654 u; mass (alpha particle) = 4.0026044 u and mass (triton) = 3.0100000u. Take  $1\text{u} = 931 \text{ MeV}/c^2$ . [3; CBSE-2006]
13. Ultraviolet radiations of different frequencies  $\nu_1$  and  $\nu_2$  are incident on two photosensitive materials having work functions  $W_1$  and  $W_2$  ( $W_1 > W_2$ ) respectively. The kinetic energy of the emitted electrons is same in both the cases. Which one of the two radiations will be of higher frequency ? [1; CBSE-2007]
14. Define the term 'activity' of radionuclide. Write its SI unit. [1; CBSE-2007]
15. Draw a graph showing the variation of potential energy between a pair of nucleons as a function of their separation. Indicate the regions in which the nuclear force is (i) attractive, (ii) repulsive. [2; CBSE-2007]
16. Draw a schematic diagram of the experimental arrangement used by Davisson and Germer to establish the wave nature of electrons. Explain briefly how the de-Broglie relation was experimentally verified in case of electrons. [3; CBSE-2007]
17. Draw the graph to show variation of binding energy per nucleon with mass number of different atomic nuclei. Calculate binding energy/nucleon of  ${}^{40}_{20}\text{Ca}$  nucleus.  
Given:  
mass of  ${}^{40}_{20}\text{Ca} = 39.962589 \text{ u}$   
mass of  ${}^{40}_{20}\text{Ca}$  proton = 1.007825 u  
mass of neutron = 1.008665 u and  $1\text{u} = 931 \text{ MeV}/c^2$  [3; CBSE-2007]
18. An electron and alpha particle have the same de-Broglie wavelength associated with them How are their kinetic energies related to each other? [1; CBSE-2008]
19. A nucleus  ${}^{23}_{10}\text{Ne}$  undergoes decay and becomes  ${}^{23}_{11}\text{Na}$ . Calculate the maximum kinetic energy of electrons emitted assuming that the daughter nucleus and anti-neutrino carry negligible kinetic energy.

$$\left\{ \begin{array}{l} \text{Mass of } {}^{23}_{10}\text{Ne} = 22.994455\text{u} \\ \text{Mass of } {}^{23}_{11}\text{Na} = 22.989770\text{u} \\ 1\text{u} = 931.5\text{MeV} / c^2 \end{array} \right\}$$

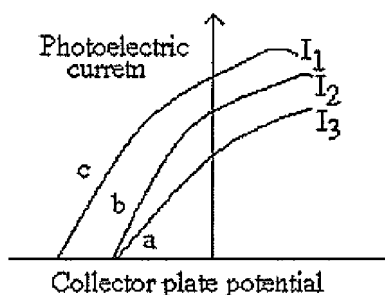
[2; CBSE-2008]

20. An electromagnetic wave of wavelength  $\lambda$  is incident on a photosensitive surface of negligible work function. If the photo-electrons emitted from this surface have the de-Broglie wavelength  $\lambda_1$  prove

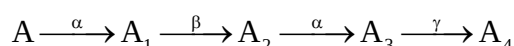
$$\text{that } \lambda = \left( \frac{2mc}{h} \right) \lambda_1^2$$

[3; CBSE-2008]

21. The figure shows a plot of three curves a, b, c showing the variation of photocurrent vs collector plate potential for three different intensities  $I_1$ ,  $I_2$  and  $I_3$  having frequencies  $\nu_1$ ,  $\nu_2$  and  $\nu_3$  respectively incident on a photosensitive surface. Point out the two curves of which the incident radiations have same frequency but different intensities. [1; CBSE-2009]

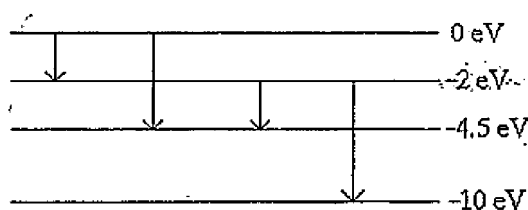


22. Two nuclei have mass numbers in the ratio 1 : 3. What is the ratio of their nuclear densities? [1; CBSE-2009]
23. A radioactive nucleus 'A' Undergoes a series of decays according to the following scheme:



The mass number and atomic number of  $A_4$  are 172 and 69 respectively. What are these numbers for A? [2; CBSE-2009]

24. An electron and a proton are accelerated through the potential. Which one of the two has (i) greater value of de-Broglie wavelength associated with it and (ii) less momentum? Justify your answer. [3; CBSE-2009]
25. (a) The energy levels of an atom are as shown below. Which of them will result in the transition of a photon of wavelength 275 nm?
- (h) Which transition corresponds to emission of radiation of maximum wavelength? [3; CBSE-2009]



26. Define ionisation energy. What is its value for a hydrogen atom? [1; CBSE-2010]
27. An  $\alpha$ -particle and a proton are accelerated from rest by the same potential. Find the ratio of their de-Broglie wavelengths. [2; CBSE-2010]
28. Write Einstein's photoelectric equation. State clearly the three salient features observe in photoelectric effect, which can be explained on the basis of the above equation. [2; CBSE-2010]
29. Draw a plot of potential energy of a pair of nucleons as a function of their separation. Write two important conclusions which you can draw regarding the nature of nuclear forces. [2; CBSE-2010]

30. Draw a plot of the binding energy per nucleon as a function of mass number for a large number of nuclei,  $2 \leq A \leq 240$ . How do you explain the constancy of binding energy per nucleon in the range  $30 < A < 170$  using the property that nuclear force is short-ranged? [2; CBSE-2010]
31. (a) Write symbolically the  $\beta^-$  decay process of  $^{32}_{15}\text{P}$ .  
(b) Derive an expression for the average life of a radionuclide. Give its relationship with the half-life. [3; CBSE-2010]
32. Write any two characteristic properties of nuclear force. [1; CBSE-2011]
33. Define the term 'stopping potential' in relation to photoelectric effect. [1; CBSE-2011]
34. Using the curve for the binding energy per nucleon as a function of mass number  $A$ , state clearly how the release in energy in the processes of nuclear fission and nuclear fusion can be explained. [2; CBSE-2011]
35. Draw a plot showing the variation of photoelectric current with collector plate potential for two different frequencies,  $\nu_1 > \nu_2$ , of incident radiation having the same intensity. In which case will the stopping potential be higher? Justify your answer. [3; CBSE-2011]
36. (a) Using de-Broglie's hypothesis, explain with the help of a suitable diagram, Bohr's second postulate of quantization of energy levels in a hydrogen atom,  
(b) The ground state energy of hydrogen atom is  $-13.6\text{eV}$ . What are the kinetic and potential energies of the electron in this state? [3; CBSE-2011]
37. Define the terms (i) 'cut-off Voltage' and (ii) 'threshold frequency' in relation to the phenomenon of photoelectric effect  
Using Einstein's photoelectric equation shows how the cut-off voltage and threshold frequency for a given photosensitive material can be determined with the help of a suitable plot/graph. [3; CBSE-2012]
38. Draw a plot of potential energy of a pair of nucleons as a function of their separations. Mark the regions where the nuclear is (i) attractive and (ii) repulsive. Write any two characteristic features of nuclear forces. [3; CBSE-2012]
39. In a Geiger-Marsden experiment, calculate the distance of closest approach to the nucleus of  $Z = 80$ , when a  $\alpha$ -particle of  $8\text{MeV}$  energy impinges on it before it comes momentarily to rest and reverses its direction.  
How will the distance of closest approach be affected when the kinetic energy of the  $\alpha$ -particle is doubles?

OR

The ground state energy of hydrogen atom is  $-13.6\text{ eV}$ . If an electron make a transition from the energy level  $-0.85\text{ eV}$  to  $-3.4\text{ eV}$ , calculate spectrum does his wavelength belong? [3; CBSE-2012]

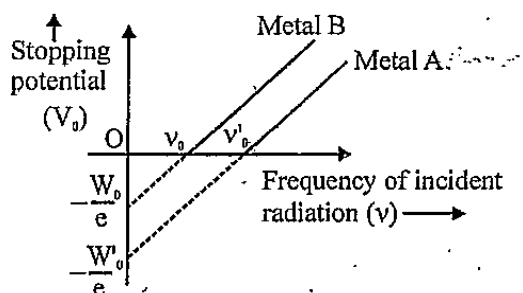
40. Define the activity of a given radioactive substance. Write its S.I. unit. [CBSE-2013]
41. Write the expression for the deBroglie wavelength associated with a charged particle having charge 'q' and mass 'm', when it is accelerated by a potential  $V$ . [CBSE-2013]
42. Write Einstein's photoelectric equation and point out any two characteristic properties of photons on which this equation is based. Briefly explain the three observed features which can be explained' by this equation. [CBSE-2013]



43. Using Bohr's postulates, derive the expression for the frequency of radiation emitted when electron in hydrogen atom undergoes transition from higher energy state (quantum number  $n_i$ ) to the lower state, ( $n_f$ ). When electron in hydrogen atom jumps from energy state  $n_i = 4$  to  $n_f = 3, 2, 1$ , identify the spectral series to which the emission lines belong. [CBSE-2013]

OR

- (a) Draw the plot of binding energy per nucleon ( $BE/A$ ) as a function of mass number  $A$ . Write two important conclusions that can be drawn regarding the nature of nuclear force.  
 (b) Use this graph to explain the release of energy in both the processes of nuclear fusion and fission,  
 (c) Write the basic nuclear process of neutron undergoing  $\beta$ -decay. Why is the detection of neutrinos found very difficult?
44. The graph shows the variation of stopping potential with frequency of incident radiation for two photosensitive metals A and B. Which one of the two has higher value of work-function? Justify your answer. [CBSE-2014]



45. Why is it found experimentally difficult to detect neutrinos in nuclear  $\beta$ -decay? [CBSE-2014]  
 46. Using Rutherford model of the atom, derive the expression for the total energy of the electron in hydrogen atom. What is the significance of total negative energy possessed by the electron? [CBSE-2014]

OR

Using Bohr's postulates of the atomic model, derive the expression for radius of  $n^{\text{th}}$  electron orbit. Hence obtain the expression for Bohr's radius. [CBSE-2014]

47. When the electron orbiting in hydrogen atom in its ground state moves to the third excited state, show how the de Broglie wavelength associated with it would be affected. [CBSE-2015]  
 48. Define the terms 'stopping potential' and 'threshold frequency' in relation to photoelectric effect. How does one determine these physical quantities using Einstein's equation? [CBSE-2015]  
 49. A proton and an  $\alpha$  particle are accelerated through the same potential difference. Which one of the two has (i) greater de-Broglie wavelength, and (ii) less kinetic energy? Justify your answer. [2 ; CBSE-2016]  
 50. When is  $H_\alpha$  line in the emission spectrum of hydrogen atom obtained? Calculate the frequency of the photon emitted during this transition. [2 ; CBSE-2016]

OR

Calculate the wavelength of radiation emitted when electron in a hydrogen atom jumps from  $n = \infty$  to  $n = 1$ . [2 ; CBSE-2016]

51. State two important properties of photon which are used to write Einstein's photoelectric equation. Define (i) stopping potential and (ii) threshold frequency, using Einstein's equation and drawing necessary plot between relevant quantities. [3 ; CBSE-2016]
52. (a) Derive the mathematical expression for law of radioactive decay for a sample of a radioactive nucleus. [3 ; CBSE-2016]  
(b) How is the mean life of a given radioactive nucleus related to the decay constant ?
53. A 12.5 eV electron beam is used to excite a gaseous hydrogen atom at room temperature. Determine the wavelengths and the corresponding series of the lines emitted. [CBSE-2017]
54. Using photon picture of light, show how Einstein's photoelectric equation can be established. Write two features of photoelectric effect which cannot be explained by wave theory. [CBSE-2017]
55. Draw graphs showing variation of photoelectric current with applied voltage for two incident radiations of equal frequency and different intensities. Mark the graph for the radiation of higher intensity. [CBSE-2018]
56. Four nuclei of an element undergo fusion to form a heavier nucleus, with release of energy. Which of the two – the parent or the daughter nucleus – would have higher binding energy per nucleon? [CBSE-2018]
57. If light of wavelength 412.5 nm is incident on each of the metals given below, which ones will show photoelectric emission and why? [CBSE-2018]

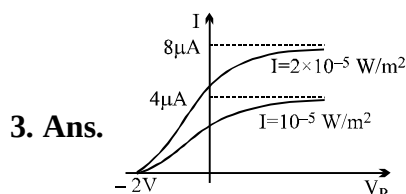
| Metal | Work Function (eV) |
|-------|--------------------|
| Na    | 1.92               |
| K     | 2.15               |
| Ca    | 3.20               |
| Mo    | 4.17               |

58. (a) State Bohr's postulate to define stable orbits in hydrogen atom. How does de-Broglie's hypothesis explain the stability of these orbits?  
(b) A hydrogen atom initially in the ground state absorbs a photon which excites it to the  $n = 4$  level. Estimate the frequency of the photon. [CBSE-2018]
59. (a) Explain the processes of nuclear fission and nuclear fusion by using the plot of binding energy per nucleon (BE/A) versus the mass number A.  
(b) A radioactive isotope has a half-life of 10 years. How long will it take for the activity to reduce to 3.125%? [CBSE-2018]

# ANSWER KEY

## EXERCISE (S-1)

1. Ans.  $6.59 \times 10^{-34}$  Js      2. Ans. (a)  $3.38 \times 10^{-19}$  J = 2.11 eV (b)  $3.0 \times 10^{20}$  photons/s



4. Ans. Use  $eV_0 = h\nu - \phi_0$  for both sources. From the data on the first source,  $\phi_0 = 1.40$  eV. Use the value to obtain for the second source  $V_0 = 1.50$  V.
5. Ans. (a)  $3.14 \times 10^{-19}$  J,  $1.05 \times 10^{-27}$  kg m/s (b)  $3 \times 10^{16}$  photons/s (c) 0.63 m/s
6. Ans. (a)  $4.04 \times 10^{-24}$  kg ms<sup>-1</sup> (b) 0.164 nm
7. Ans. (a)  $6.95 \times 10^{-25}$  J = 4.34 μeV (b)  $3.78 \times 10^{-28}$  J = 0.236 neV
8. Ans. (a)  $6.686 \times 10^{-21}$  J =  $4.174 \times 10^{-2}$  eV (b) 0.145 nm
9. Ans.  $\sqrt{2}$       10. Ans. 4      11. Ans. 0.55 eV      12. Ans. 7
13. Ans. 2      14. Ans. 820 nm.
15. Ans. Lyman series: 103 nm and 122 nm; Balmer series: 656 nm.
16. Ans. (a) +3.4 eV; (b) -6.8 eV; (c) potential energy
17. Ans.  $2.56 \times 10^{-13}$  m; -2.8 keV      18. Ans. 8      19. Ans. n = 24
20. Ans. 2
21. Ans. Use  $\lambda = (hc/E)$  with  $E = 5.1 \times 1.602 \times 10^{-10}$  J to get  $\lambda = 2.43 \times 10^{-16}$  m.
22. Ans. 6210 eV      23. Ans. 400

## EXERCISE (S-2)

1. Ans. 5      2. Ans. 1      3. Ans.  $1.1 \times 10^{12}$
4. Ans. (a)  $10^5$  s<sup>-1</sup>; (b) 286.18; (d) 111 s      5. Ans.  $\lambda = \frac{2\lambda_1\lambda_2}{\lambda_1^2 + \lambda_2^2}$
6. Ans. KE  $\cong$  151 eV,  $d_{\text{least}} = 0.5$  Å      7. Ans. (i) 2; (ii) 13.6 eV; (iii) 12.75 eV, 0.66 eV
8. Ans. 6.8 eV,  $5 \times 10^{15}$  Hz
9. Ans. (i) Allowed values of energy of neutron = 6.36 eV and 0.312 eV; Allowed values of energy of He atom = 17.84 eV and 16.328 eV, (ii)  $18.23 \times 10^{14}$  Hz,  $9.846 \times 10^{15}$  Hz,  $11.6 \times 10^{15}$  Hz
10. Ans. 7.4 eV, 4.7 Volts      11. Ans. 6      12. Ans. 255      13. Ans. 3      14. Ans. 159
15. Ans. 255      16. Ans. 2      17. Ans. 98.25 eV
18. Ans. (i) 0.62 Å, (ii) No, (iii) Yes 0.8 Å      19. Ans. 41

**EXERCISE (O-1)**

1. Ans. (A) 2. Ans. (A) 3. Ans. (B) 4. Ans. (B) 5. Ans. (B) 6. Ans. (C)  
 7. Ans. (A) 8. Ans. (A) 9. Ans. (D) 10. Ans. (B) 11. Ans. (B) 12. Ans. (A)  
 13. Ans. (B) 14. Ans. (B) 15. Ans. (D) 16. Ans. (D) 17. Ans. (D) 18. Ans. (A)  
 19. Ans. (D) 20. Ans. (A) 21. Ans. (C) 22. Ans. (D) 23. Ans. (D) 24. Ans. (B)  
 25. Ans. (A) 26. Ans. (A) 27. Ans. (A) 28. Ans. (A) 29. Ans. (A) 30. Ans. (A)  
 31. Ans. (A) 32. Ans. (D) 33. Ans. (B) 34. Ans. (A) 35. Ans. (B) 36. Ans. (B)  
 37. Ans. (D) 38. Ans. (B) 39. Ans. (A) 40. Ans. (A) 41. Ans. (D) 42. Ans. (B)  
 43. Ans. (C) 44. Ans. (D) 45. Ans. (D) 46. Ans. (D) 47. Ans. (C) 48. Ans. (D)  
 49. Ans. (C) 50. Ans. (A) 51. Ans. (C) 52. Ans. (A,B,C) 53. Ans. (A,B,D)  
 54. Ans. (A,C) 55. Ans. (A,B,C) 56. Ans. (B,C,D) 57. Ans. (A,D)  
 58. Ans. (A,B,C,D) 59. Ans. (A,B,D) 60. Ans. (A,B,C,D) 61. Ans. (B, D)  
 62. Ans. (A,C,D) 63. Ans. (A,B,C) 64. Ans. (A,C,D) 65. Ans. (A,C,D)  
 66. Ans. (B) 67. Ans. (A,C) 68. Ans. (A,C,D) 69. Ans. (A,B,C,D)  
 70. Ans. (A, B, C) 71. Ans. (A, C) 72. Ans. (A) P,R,S (B) P,Q (C) P,Q (D) R,S  
 73. Ans. (A)-QR, (B)-PS, (C)-QS, (D)-PR  
 74. Ans. (A)  $\rightarrow$  (P,R) ; (B)  $\rightarrow$  (Q,S) ; (C)  $\rightarrow$  (Q, S) ; (D)  $\rightarrow$  (P,R) 75. Ans. (B)  
 76. Ans. (A)  $\rightarrow$  (S); (B)  $\rightarrow$  (R); (C)  $\rightarrow$  (Q); (D)  $\rightarrow$  (P)

**EXERCISE (O-2)**

1. Ans. (D) 2. Ans. (C) 3. Ans. (A) 4. Ans. (B,D) 5. Ans. (B) 6. Ans. (A)  
 7. Ans. (B) 8. Ans. (D) 9. Ans. (C) 10. Ans. (C) 11. Ans. (D) 12. Ans. (B)  
 13. Ans. (B) 14. Ans. (B) 15. Ans. (D) 16. Ans. (A,C,D) 17. Ans. (D)  
 18. Ans. (A,B) 19. Ans. (B,C,D) 20. Ans. (B, C) 21. Ans. (A, B, D)  
 22. Ans. (B, C) 23. Ans. (B,C) 24. Ans. (A,C,D) 25. Ans. (A)  
 26. Ans. (A) 27. Ans. (B) 28. Ans. (A) 29. Ans. (A) 30. Ans. (B) 31. Ans. (B)  
 32. Ans. (C) 33. Ans. (B) 34. Ans. (C) 35. Ans. (D) 36. Ans. (A)

**EXERCISE (JM)**

1. Ans. (2) 2. Ans. (2) 3. Ans. (4) 4. Ans. (1) 5. Ans. (2) 6. Ans. (1)  
 7. Ans. (2) 8. Ans. (1) 9. Ans. (4) 10. Ans. (4) 11. Ans. (3) 12. Ans. (1)  
 13. Ans. (2) 14. Ans. (3) 15. Ans. (2) 16. Ans. (4) 17. Ans. (3)

**EXERCISE (JA)**

1. Ans. (B) 2. Ans. (C) 3. Ans. (C) 4. Ans. (A) 5. Ans. (A) 6. Ans. (B)  
 7. Ans. (D) 8. Ans. (A) 9. Ans. 3 10. Ans. (D) 11. Ans. (B) 12. Ans. (C)  
 13. Ans. 8 14. Ans. (A) 15. Ans. 7 16. Ans. 7 17. Ans. 1 18. Ans. (A, C)  
 19. Ans. 2 20. Ans. (A), (C), (D) 21. Ans. (A,C) 22. Ans. 2 23. Ans. (B)  
 24. Ans. (A, B, D) 25. Ans. 6 26. Ans. (A) 27. Ans. 5 28. Ans. (A)  
 29. Ans. 24 [23.60, 24.40] 30. Ans. 3 [3,3] 31. Ans. (2,3) 32. Ans. (1.00)

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## MODERN PHYSICS-2

# KEY CONCEPT

## NUCLEAR PHYSICS

### ATOMIC NUCLEUS

The atomic nucleus consists of two types of elementary particles, viz. protons and neutrons. These particles are called nucleons. The proton (denoted by  $p$ ) has a charge  $+e$  and a mass  $m_p = 1.6726 \times 10^{-27}$  kg, which is approximately 1840 times larger than the electron mass. The proton is the nucleus of the simplest atom with  $Z=1$ , viz the hydrogen atom.

The neutron (denoted by  $n$ ) is an electrically neutral particle (its charge is zero). The neutron mass is  $1.6749 \times 10^{-27}$  kg. The fact that the mass of a neutron exceeds the mass of a proton by about 2.5 times the electronic masses is of essential importance. It follows from this that the neutron in free state (outside the nucleus) is unstable (radioactive). With half life equal to 12 min, the neutron spontaneously transforms into a proton by emitting an electron ( $e^-$ ) and a particle called the antineutrino ( $\bar{\nu}$ ).

This process can be schematically written as follows :  ${}_0n^1 \rightarrow {}_1p^1 + {}_{-1}e^0 + \bar{\nu}$

The most important characteristics of the nucleus are the charge number  $Z$  (coinciding with atomic number of the element) and mass number  $A$ . The charge number  $Z$  is equal to the number of protons in the nucleus, and hence it determines the nuclear charge equal to  $Ze$ . The mass number  $A$  is equal to the number of nucleons in the nucleus (i.e., to the total number of protons and neutrons). Nuclei are symbolically designated as  $X_Z^A$  or  ${}_Z X^A$  where  $X$  stands for the symbol of a chemical element.

For example, the nucleus of the oxygen atom is symbolically written as  $O_8^{16}$  or  ${}_8O^{16}$ .

The shape of nucleus is approximately spherical and its radius is approximately related to the mass number by

$$R = 1.2 A^{1/3} \times 10^{-15} \text{ m} = 1.2 \times 10^{-15} \times A^{1/3} \text{ m}$$

Most of the chemical elements have several types of atoms differing in the number of neutrons in their nuclei. These varieties are called isotopes. For example carbon has three isotopes  ${}_6C^{12}$ ,  ${}_6C^{13}$ ,  ${}_6C^{14}$ . In addition to stable isotopes, there also exist unstable (radioactive) isotopes. Atomic masses are specified in terms of the atomic mass unit or unified mass unit ( $u$ ). The mass of a neutral atom of the carbon  ${}_6C^{12}$  is defined to be exactly 12  $u$ .

$$1u = 1.66056 \times 10^{-27} \text{ kg} = 931.5 \text{ MeV}.$$

### BINDING ENERGY

The rest mass of the nucleus is smaller than the sum of the rest masses of nucleons constituting it. This is due to the fact that when nucleons combine to form a nucleus, some energy (binding energy) is liberated. The binding energy is equal to the work that must be done to split the nucleus into the particles constituting it.

The difference between the total mass of the nucleons and mass of the nucleus is called the mass defect of the nucleus represented by  $\Delta m = [Zm_p + (A-Z)m_n] - m_{\text{nuc}}$

Multiplying the mass defect by the square of the velocity of light, we can find the binding energy of the nucleus.

$$BE = \Delta mc^2 = [(Zm_p + (A-Z)m_n) - m_{\text{nuc}}]c^2$$

If the masses are taken in atomic mass unit, the binding energy is given by

$$BE = [(Zm_p + (A-Z)m_n) - m_{\text{nuc}}] 931.5 \text{ MeV}$$

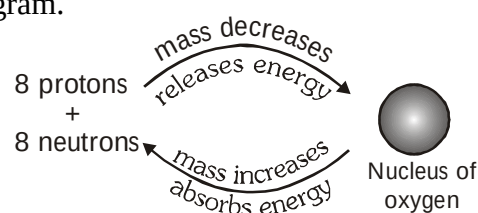
Let us take example of oxygen nucleus. It contains 8 protons and 8 neutrons.

We can discuss concept of binding energy by following diagram.

$$8m_p + 8m_n > \text{mass of nucleus of oxygen}$$

For nucleus we apply mass energy conservation,

$$8m_p + 8m_n = \text{mass of nucleus} + \frac{B.E.}{c^2}$$



For general nucleus  ${}_Z^AX$ , mass defect = difference between total mass of nucleons and mass of the nucleus

$$\Delta m = [Zm_p + (A-Z)m_n] - M$$

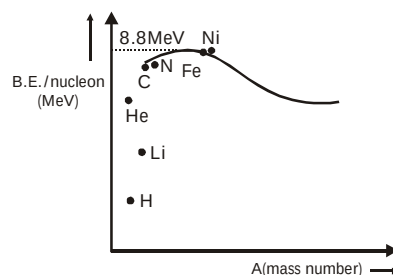
$$B.E. = \Delta mc^2 \text{ (joules)} = (\Delta m)_{\text{in amu}} \times 931.5 \text{ MeV}$$

### Binding Energy per Nucleon

Stability of a nucleus does not depend upon binding energy of a nucleus but it depends upon binding energy per nucleon

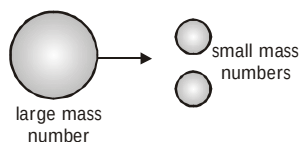
$$B.E./\text{nucleon} = \frac{B.E.}{\text{mass number}}$$

$$\text{Stability} \propto \frac{B.E.}{A}$$

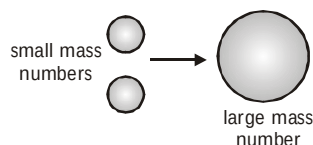


(i)  $B.E./A$  is maximum for  $A = 62$  (Ni), It is  $8.79460 \pm 0.00003 \text{ MeV/nucleon}$ , means most stable nuclei are in the region of  $A = 62$ .

(ii) Heavy nuclei achieve stability by breaking into two smaller nuclei and this reaction is called fission reaction.



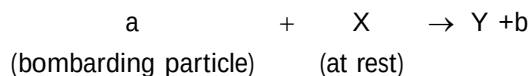
(iii) Nuclei achieve stability by combining and resulting into heavy nucleus and this reaction is called fusion reaction.



(iv) In both reactions products are more stable in comparison to reactants and  $Q$  value is positive.

## NUCLEAR COLLISIONS

We can represent a nuclear collision or reaction by the following notation, which means  $X(a,b)Y$



We can apply :

(i) Conservation of momentum (ii) Conservation of charge (iii) Conservation of mass-energy



By mass energy conservation

$$(i) K_1 + K_2 + (m_a + m_x)c^2 = K_3 + K_4 + (m_Y + m_b)c^2$$

(ii) Energy released in any nuclear reaction or collision is called  $Q$  value of the reaction

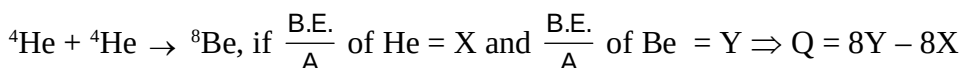
$$(iii) Q = (K_3 + K_4) - (K_1 + K_2) = \Sigma K_p - \Sigma K_R = (\Sigma m_R - \Sigma m_p)c^2$$

(iv) If  $Q$  is positive, energy is released and products are more stable in comparison to reactants.

(v) If  $Q$  is negative, energy is absorbed and products are less stable in comparison to reactants.

$$Q = \Sigma(\text{B.E.})_{\text{product}} - \Sigma(\text{B.E.})_{\text{reactants}}$$

**Ex.** Let us find the  $Q$  value of fusion reaction



$$\text{Q value for } \alpha \text{ decay} \quad {}_Z^AX^A \rightarrow {}_{Z-2}Y^{A-4} + {}_2^4\text{He}^4 \Rightarrow Q = K_\alpha + K_Y \quad \dots(i)$$

$$\text{Momentum conservation, } p_Y = p_\alpha \quad \dots(ii)$$

$$K_\alpha = \frac{p^2}{2 \times m \times 4}$$

$$K_Y = \frac{p^2}{2m(A-4)} = \frac{4K_\alpha}{A-4}$$

$$Q = K_\alpha + \frac{4K_\alpha}{A-4} = \frac{A}{A-4} K_\alpha$$

$$K_\alpha = \frac{A-4}{A} Q$$

For  $\alpha$  decay  $A > 210$  which means maximum part of released energy is associated with K.E. of  $\alpha$ . If  $Q$  is negative, the reaction is endoergic. The minimum amount of energy that a bombarding particle must have in order to initiate an endoergic reaction is called Threshold energy  $E_{th}$ ,

given by  $E_{th} = -Q \left( \frac{m_1}{m_2} + 1 \right)$  where  $m_1$  = mass of the projectile.

$E_{th}$  = minimum kinetic energy of the projectile to initiate the nuclear reaction

$m_2$  = mass of the target

**Ex.** How much energy must a bombarding proton possess to cause the reaction  ${}_3\text{Li}^7 + {}_1\text{H}^1 \rightarrow {}_4\text{Be}^7 + {}_0n^1$  (Mass of  ${}_3\text{Li}^7$  atom is 7.01600, mass of  ${}_1\text{H}^1$  atom is 1.0783, mass of  ${}_4\text{Be}^7$  atom is 7.01693)

**Sol.** Since the mass of an atom includes the masses of the atomic electrons, the appropriate number of electron masses must be subtracted from the given values.

**Reactants :** Total mass =  $(7.01600 - 3m_e) + (1.0783 - 1m_e) = 8.0943 - 4m_e$

**Products :** Total mass =  $(7.01693 - 4m_e) + 1.0087 = 8.02563 - 4m_e$



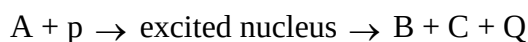
The energy is supplied as kinetic energy of the bombarding proton. The incident proton must have more than this energy because the system must possess some kinetic energy even after the reaction, so that momentum is conserved with momentum conservation taken into account, the minimum kinetic energy that the incident particle must possess can be found with the formula.

where,  $Q = - [(8.02563 - 4m_e) - (8.0943 - 4m_e)] \times 931.5 \text{ MeV} = -63.96 \text{ MeV}$

$$E_{th} = - \left(1 + \frac{m}{M}\right) Q = - \left(1 + \frac{1}{7}\right) (-63.96) = 73.1 \text{ MeV}$$

## NUCLEAR FISSION

In 1938 by Hahn and Strassmann. By attack of a particle splitting of a heavy nucleus ( $A > 230$ ) into two or more lighter nuclei. In this process certain mass disappears which is obtained in the form of energy (enormous amount)

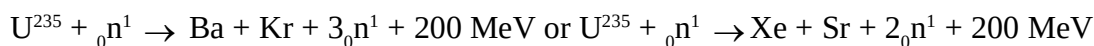


Hahn and Strassmann done the first fission of nucleus of  $U^{235}$ .

When  $U^{235}$  is bombarded by a neutron it splits into two fragments and 2 or 3 secondary neutrons and releases about 190 MeV ( $\approx 200 \text{ MeV}$ ) energy per fission (or from single nucleus)

Fragments are uncertain but each time energy released is almost same.

Possible reactions are



and many other reactions are possible.

- The average number of secondary neutrons is 2.5.
- Nuclear fission can be explained by using "liquid drop model" also.
- The mass defect  $\Delta m$  is about 0.1% of mass of fissioned nucleus
- About 93% of released energy ( $Q$ ) is appear in the form of kinetic energies of products and about 7% part in the form of  $\gamma$  - rays.

## NUCLEAR CHAIN REACTION :

The equation of fission of  $U^{235}$  is  $U^{235} + {}_0n^1 \rightarrow Ba + Kr + 3{}_0n^1 + Q$ .

These three secondary neutrons produced in the reaction may cause of fission of three more  $U^{235}$  and give 9 neutrons, which in turn, may cause of nine more fission of  $U^{235}$  and so on.

Thus a continuous 'Nuclear Chain reaction' would start.

If there is no control on chain reaction then in a short time ( $\approx 10^{-6} \text{ sec.}$ ) a huge amount of energy will be released. (This is the principle of 'Atom bomb'). If chain is controlled then produced energy can be used for peaceful purposes. For example nuclear reactor (Based on fission) are generating electricity.

## NATURAL URANIUM :

It is mixture of  $U^{235}$  (0.7%) and  $U^{238}$  (99.3%).

$U^{235}$  is easily fissionable, by slow neutron (or thermal neutrons) having K.E. of the order of 0.03 eV.

But  $U^{238}$  is fissionable with fast neutrons.

**Note :** Chain reaction in natural uranium can't occur

To improve the quality, percentage of  $U^{235}$  is increased to 3%.

The proposed uranium is called 'Enriched Uranium' (97%  $U^{238}$  and 3%  $U^{235}$ )

## LOSSES OF SECONDARY NEUTRONS :

**Leakage of neutrons from the system :** Due to their maximum K.E. some neutrons escape from the system.

**Absorption of neutrons by  $U^{238}$  :** Which is not fissionable by these secondary neutrons.

## CRITICAL SIZE (OR MASS) :

In order to sustain chain reaction in a sample of enriched uranium, it is required that the number of lost neutrons should be much smaller than the number of neutrons produced in a fission process. For it the size of uranium block should be equal or greater than a certain size called **critical size**.

## REPRODUCTION FACTOR :

$$(K) = \frac{\text{rate of production of neutrons}}{\text{rate of loss of neutrons}}$$

- (i) If size of Uranium used is 'Critical' then  $K = 1$  and the chain reaction will be steady or sustained (As in nuclear reaction)
- (ii) If size of Uranium used is 'Super critical' then  $K > 1$  and chain reaction will accelerate resulting in a explosion (As in atom bomb)
- (iii) If size of Uranium used is 'Sub Critical' then  $K < 1$  and chain reaction will retard and will stop.

## NUCLEAR REACTOR ( $K = 1$ ) : Credit $\rightarrow$ To Enricho Fermi

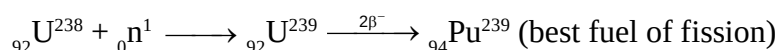
### Construction :

- **Nuclear Fuel :** Commonly used are  $U^{235}$ ,  $Pu^{239}$ .  $Pu^{239}$  is the **best**. Its critical size is less than critical size of  $U^{235}$ .  
But  $Pu^{239}$  is not naturally available and  $U^{235}$  is used in most of the reactors.
- **Moderator :** Its function is to slow down the fast secondary neutrons. Because only slow neutrons can bring the fission of  $U^{235}$ . The moderator should be light and it should not absorb the neutrons. Commonly, Heavy water ( $D_2O$ , molecular weight 20 gm.) Graphite etc. are used. These are rich of protons. Neutrons collide with the protons and interchange their energy. Thus neutrons get slow down.
- **Control rods :** They have the ability to capture the slow neutrons and can control the chain reaction at any stage. Boron and Cadmium are best absorber of neutrons.
- **Coolant :** A substance which absorb the produced heat and transfers it to water for further use. Generally coolant is water at high pressure

## FAST BREADER REACTORS

The atomic reactor in which fresh fissionable fuel ( $Pu^{239}$ ) is produced along with energy. The amount of produced fuel ( $Pu^{239}$ ) is more than consumed fuel ( $U^{235}$ )

- **Fuel :** Natural Uranium.
- **Process:** During fission of  $U^{235}$ , energy and secondary neutrons are produced. These secondary neutrons are absorbed by  $U^{238}$  and  $U^{239}$  is formed. This  $U^{239}$  converts into  $Pu^{239}$  after two beta decay. This  $Pu^{239}$  can be separated, its half life is 2400 years.

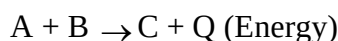


This  $\text{Pu}^{239}$  can be used in nuclear weapons because of its small critical size than  $\text{U}^{235}$ .

- **Moderator** : Are not used in these reactors.
- **Coolant** : Liquid sodium

## NUCLEAR FUSION :

It is the phenomenon of fusing two or more lighter nuclei to form a single heavy nucleus.



The product (C) is more stable than reactants (A and B) and  $m_c < (m_a + m_b)$

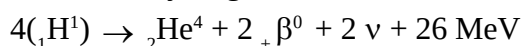
and mass defect  $\Delta m = [(m_a + m_b) - m_c] \text{ amu}$

Energy released is  $E = \Delta m \times 931 \text{ MeV}$

The total binding energy and binding energy per nucleon C both are more than of A and B.

$$\Delta E = E_c - (E_a + E_b)$$

Fusion of four hydrogen nuclei into helium nucleus :



- Energy released per fission  $\gg$  Energy released per fusion
- Energy per nucleon in fission  $\left[ = \frac{200}{235} \approx 0.85 \text{ MeV} \right] \ll$  energy per nucleon in fusion  $\left[ = \frac{24}{4} \approx 6 \text{ MeV} \right]$

## REQUIRED CONDITION FOR NUCLEAR FUSION

- **High temperature** :

Which provide kinetic energy to nuclei to overcome the repulsive electrostatic force between them.

- **High Pressure (or density)** :

Which ensure frequent collision and increases the probability of fusion. The required temperature and pressure at earth (lab) are not possible. These condition exist in the sun and in many other stars. The source of energy in the sun is nuclear fusion, where hydrogen is in plasma state and there protons fuse to form helium nuclei.

## HYDROGEN BOMB

It is based on nuclear fusion and produces more energy than an atom bomb.

### Pair production

A  $\gamma$ -photon of energy more than 1.02 MeV, when interact with a nucleus produces pair of electron ( $e^-$ ) and positron ( $e^+$ ). The energy equivalent to rest mass of  $e^-$  (or  $e^+$ ) = 0.51 MeV. The energy equivalent to rest mass of pair ( $e^- + e^+$ ) = 1.02 MeV.

For pair production Energy of photon  $\geq 1.02 \text{ MeV}$ .

If energy of photon is more than 1.02 MeV, the extra energy  $(E - 1.02) \text{ MeV}$  divides approximately in equal amount to each

$$\text{particle as the kinetic energy or } (KE)_{e^- \text{ or } e^+} = \left[ \frac{E_{\text{ph}} - 1.02}{2} \right] \text{ MeV}$$

If  $E < 1.02 \text{ MeV}$ , pair will not produce.

### Pair Annihilation

When electron and positron combines they annihilates to each other and only energy is released in the form of two gamma photons. If the energy of electron and positron are negligible then energy of each  $\gamma$ -photon is 0.51 MeV

**Ex.** In a nuclear reactor, fission is produced in 1 g for  $^{235}_{92}\text{U}$  (235.0439) in 24 hours by slow neutrons (1.0087 u). Assume that  $^{92}_{35}\text{Kr}$  (91.8973 u) and  $^{141}_{56}\text{Ba}$  (140.9139 amu) are produced in all reactions and no energy is lost.

(i) Write the complete reaction (ii) Calculate the total energy produced in kilowatt hour.

Given  $1\text{u} = 931\text{ MeV}$ .

**Sol.** The nuclear fission reaction is  $^{235}_{92}\text{U} + {}^1_0\text{n} \rightarrow {}^{141}_{56}\text{Ba} + {}^{92}_{36}\text{Kr} + 3{}_^1_0\text{n}$

Mass defect  $\Delta m = [(m_u + m_n) - (m_{\text{Ba}} + m_{\text{Kr}} + 3m_n)] = 256.0526 - 235.8373 = 0.2153\text{ u}$

Energy released  $Q = 0.2153 \times 931 = 200\text{ MeV}$ . Number of atoms in 1 g =  $\frac{6.02 \times 10^{23}}{235} = 2.56 \times 10^{21}$

Energy released in fission of 1 g of  $^{235}\text{U}$  is  $E = 200 \times 2.56 \times 10^{21} = 5.12 \times 10^{23}\text{ MeV}$

$$= 5.12 \times 10^{23} \times 1.6 \times 10^{-13} = 8.2 \times 10^{10}\text{ J}$$

$$= \frac{8.2 \times 10^{10}}{3.6 \times 10^6}\text{ kWh} = 2.28 \times 10^4\text{ kWh}$$

**Ex.** It is proposed to use the nuclear fusion reaction :  ${}_1^2\text{H} + {}_1^2\text{H} \rightarrow {}_2^4\text{He}$  in a nuclear reactor of 200 MW rating. If the energy from above reaction is used with at 25% efficiency in the reactor, how many grams of deuterium will be needed per day. (Mass of  ${}_1^2\text{H}$  is 2.0141 u and mass of  ${}_2^4\text{He}$  is 4.0026 u)

**Sol.** Energy released in the nuclear fusion is  $Q = \Delta mc^2 = \Delta m(931)\text{ MeV}$  (where  $\Delta m$  is in amu)

$$Q = (2 \times 2.0141 - 4.0026) \times 931\text{ MeV} = 23.834\text{ MeV} = 23.834 \times 10^6\text{ eV}$$

Since efficiency of reactor is 25%

$$\text{So effective energy used} = \frac{25}{100} \times 23.834 \times 10^6 \times 1.6 \times 10^{-19}\text{ J} = 9.534 \times 10^{-13}\text{ J}$$

Since the two deuterium nuclei are involved in a fusion reaction,

$$\text{therefore, energy released per deuterium is } \frac{9.534 \times 10^{-13}}{2}.$$

For 200 MW power per day, number of deuterium nuclei required

$$= \frac{200 \times 10^6 \times 86400}{\frac{9.534 \times 10^{-13}}{2}} = 3.624 \times 10^{25}$$

Since 2g of deuterium constitute  $6 \times 10^{23}$  nuclei, therefore amount of deuterium required is

$$= \frac{2 \times 3.624 \times 10^{25}}{6 \times 10^{23}} = 120.83\text{ g/day}$$

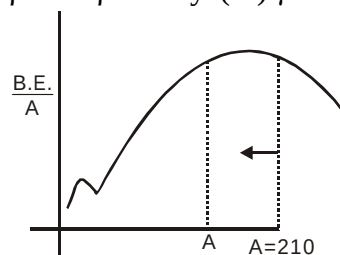
## RADIOACTIVITY

The process of spontaneous disintegration shown by some unstable atomic nuclei is known as natural radioactivity. This property is associated with the emission of certain types of penetrating radiations, called radioactive rays, or Becquerel rays ( $\alpha$ ,  $\beta$ ,  $\gamma$ -rays). The elements or compounds, whose atoms disintegrate and emit radiations are called radioactive elements. Radioactivity is a continuous, irreversible nuclear phenomenon.

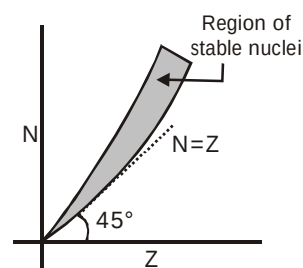
### Radioactive Decays

Generally, there are three types of radioactive decays (i)  $\alpha$  decay (ii)  $\beta^-$  and  $\beta^+$  decay (iii)  $\gamma$  decay

- **$\alpha$  decay :** In  $\alpha$  decay, the unstable nucleus emits an  $\alpha$  particle. By emitting  $\alpha$  particle, the nucleus decreases its mass energy number and move towards stability. Nucleus having  $A > 210$  shows  $\alpha$  decay. By releasing  $\alpha$  particle, it can attain higher stability and  $Q$  value is positive.



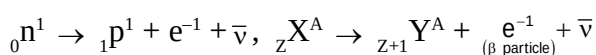
- **$\beta$  decay :** In beta decay ( $N/Z$ ) ratio of nucleus is changed. This decay is shown by unstable nuclei. In beta decay, either a neutron is converted into proton or proton is converted into neutron. For better understanding we discuss  $N/Z$  graph. There are two type of unstable nuclides



- **A type**

For A type nuclides  $(N/Z)_A > (N/Z)_{\text{stable}}$

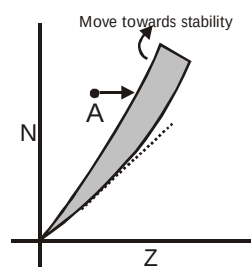
To achieve stability, it increases  $Z$  by conversion of neutron into proton



This decay is called  $\beta^-$  decay.

Kinetic energy available for  $e^{-}$  and  $\bar{\nu}$  is,  $Q = K_{\beta} + K_{\bar{\nu}}$

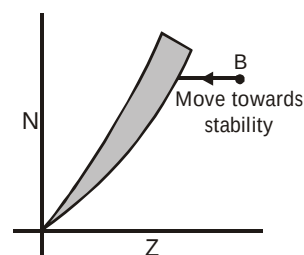
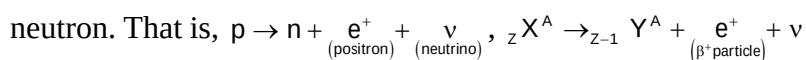
K.E. of  $\beta$  satisfies the condition  $0 < K_{\beta} < Q$



- **B type**

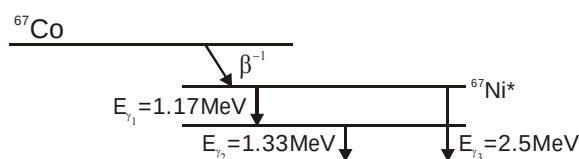
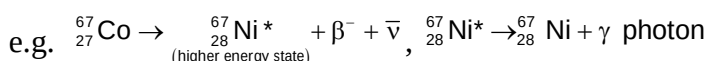
For B type nuclides  $(N/Z)_B < (N/Z)_{\text{stable}}$

To achieve stability it decreases  $Z$  by the conversion of a proton into



- **$\gamma$  decay :** When an  $\alpha$  or  $\beta$  decay takes place, the daughter nucleus is usually in higher energy state, such a nucleus comes to ground state by emitting a photon or photons.

Order of energy of  $\gamma$  photon is 100 KeV



Properties of  $\alpha$ ,  $\beta$  and  $\gamma$  rays

| Features                                        | $\alpha$ -particles                                                                                                                                                                    | $\beta$ -particles                                                                                                                                                                                       | $\gamma$ -rays                                                                                                                                  |
|-------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| Identity                                        | Helium nucleus or doubly ionised helium atom ( ${}_2\text{He}^4$ )                                                                                                                     | Fast moving electrons ( ${}_{-1}\beta^0$ or $\beta^-$ )                                                                                                                                                  | Electromagnetic wave (photons)                                                                                                                  |
| Charge                                          | Twice of proton ( $+2e$ ) $\approx 4m_p$                                                                                                                                               | Electronic ( $-e$ )<br>(rest mass of $\beta$ )<br>$=$ (rest mass of electron)                                                                                                                            | Neutral<br>rest mass $= 0$                                                                                                                      |
| Mass                                            | $m_p$ —mass of proton                                                                                                                                                                  |                                                                                                                                                                                                          |                                                                                                                                                 |
| Speed                                           | $1.4 \times 10^7$ m/s. to $2.2 \times 10^7$ m/s.<br>(Only certain value between this range).<br>Their speed depends on nature of the nucleus.<br>So that it is a characteristic speed. | 1% of $c$ to 99% of $c$<br>(All possible values between this range)<br>$\beta$ -particles come out with different speeds from the same type of nucleus.<br>So that it can not be a characteristic speed. | Only $c = 3 \times 10^8$ m/s<br>$\gamma$ -photons come out with same speed from all types of nucleus.<br>So, can not be a characteristic speed. |
| K.E.                                            | $\approx$ MeV                                                                                                                                                                          | $\approx$ MeV                                                                                                                                                                                            | $\approx$ MeV                                                                                                                                   |
| Energy spectrum                                 | Line and discrete (or linear)                                                                                                                                                          | Continuous (or linear)                                                                                                                                                                                   | Line and discrete                                                                                                                               |
| Ionization power ( $\alpha > \beta > \gamma$ )  | 10,000 times of $\gamma$ -rays                                                                                                                                                         | 100 times of $\gamma$ -rays<br>(or $\frac{1}{100}$ times of $\alpha$ )                                                                                                                                   | 1 (or $\frac{1}{100}$ times of $\beta$ )                                                                                                        |
| Penetration power ( $\gamma > \beta > \alpha$ ) | $\frac{1}{10000}$ times of $\gamma$ -rays                                                                                                                                              | $\frac{1}{100}$ times of $\gamma$ -rays<br>(100 times of $\alpha$ )                                                                                                                                      | 1(100 times of $\beta$ )                                                                                                                        |
| Effect of electric or magnetic field            | Deflection                                                                                                                                                                             | Deflection (More than $\alpha$ )                                                                                                                                                                         | No deflection                                                                                                                                   |
| Explanation of emission                         | By Tunnel effect (or quantum mechanics)                                                                                                                                                | By weak nuclear interactions                                                                                                                                                                             | With the help of energy levels in nucleus                                                                                                       |

Laws of Radioactive Decay

- The radioactive decay is a spontaneous process with the emission of  $\alpha$ ,  $\beta$  and  $\gamma$  rays. It is not influenced by external conditions such as temperature, pressure, electric and magnetic fields.
- The rate of disintegration is directly proportional to the number of radioactive atoms present at that time i.e., rate of decay  $\propto$  number of nuclei.

$$\text{Rate of decay} = \lambda (\text{number of nuclei}) \text{ i.e. } \frac{dN}{dt} = -\lambda N$$

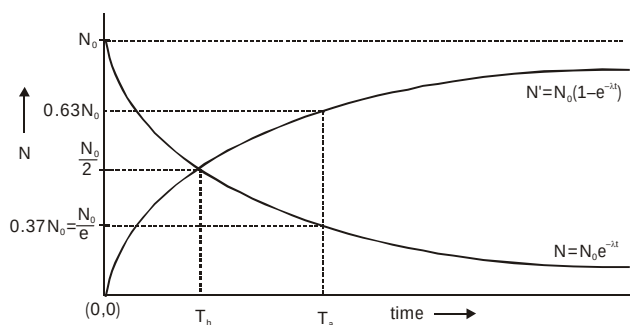
where  $\lambda$  is called the decay constant. This equation may be expressed in the form  $\frac{dN}{N} = -\lambda dt$ .

$$\int_{N_0}^N \frac{dN}{N} = -\lambda \int_0^t dt \Rightarrow \ln\left(\frac{N}{N_0}\right) = -\lambda t$$

where  $N_0$  is the number of parent nuclei at  $t=0$ . The number that survives at time  $t$  is therefore

$$N = N_0 e^{-\lambda t} \text{ and } t = \frac{2.303}{\lambda} \log_{10}\left(\frac{N_0}{N_t}\right) \text{ this function is plotted in figure.}$$

**Graph :** Time versus  $N$  (or  $N'$ )



- **Half life ( $T_h$ ) :** It is the time during which number of active nuclei reduce to half of initial value.

If at  $t = 0$  no. of active nuclei  $N_0$  then at  $t = T_h$  number of active nuclei will be  $\frac{N_0}{2}$

From decay equation  $N = N_0 e^{-\lambda t}$

$$\frac{N_0}{2} = N_0 e^{-\lambda T_h} \Rightarrow T_h = \frac{\ln(2)}{\lambda} = \frac{0.693}{\lambda} \approx \frac{0.7}{\lambda}$$

- **Mean or Average Life ( $T_a$ ) :** It is the average of age of all active nuclei i.e.

$$T_a = \frac{\text{sum of times of existence of all nuclei in a sample}}{\text{initial number of active nuclei in that sample}} = \frac{1}{\lambda}$$

(i) At  $t = 0$ , number of active nuclei  $= N_0$  then number of active nuclei at

$$t = T_a \text{ is } N = N_0 e^{-\lambda T_a} = N_0 e^{-1} = \frac{N_0}{e} = 0.37N_0 = 37\% \text{ of } N_0$$

(ii) Number nuclei which have been disintegrated within duration  $T_a$  is

$$N' = N_0 - N = N_0 - 0.37N_0 = 0.63N_0 = 63\% \text{ of } N_0$$

- $T_a = \frac{1}{\lambda} = \frac{T_h}{\ln(2)} = \frac{T_h}{0.693} = 1.44 T_h$
- Within duration  $T_h \Rightarrow 50\%$  of  $N_0$  decayed and  $50\%$  of  $N_0$  remains active
- Within duration  $T_a \Rightarrow 63\%$  of  $N_0$  decayed and  $37\%$  of  $N_0$  remains active

### ACTIVITY OF A SAMPLE (A OR R) (OR DECAY RATE)

It is the rate of decay of a radioactive sample  $R = -\frac{dN}{dt} = N\lambda$  or  $R = R_0 e^{-\lambda t}$

- Activity of a sample at any instant depends upon number of active nuclei at that instant.  
 $R \propto N$  (or active mass),  $R \propto m$
- $R$  also decreases exponentially w.r.t. time same as the number of active nuclei decreases.

- R is not a constant with N, m and time while  $\lambda$ ,  $T_h$  and  $T_a$  are constant
- At  $t = 0$ ,  $R = R_0$  then at  $t = T_h \Rightarrow R = \frac{R_0}{2}$  and at  $t = T_a \Rightarrow R = \frac{R_0}{e}$  or  $0.37 R_0$
- Similarly active mass of radioactive sample decreases exponentially.  $m = m_0 e^{-\lambda t}$
- Activity of m gm active sample (molecular weight  $M_w$ ) is  $R = \lambda N = \frac{0.693}{T_h} \left[ \frac{N_{AV}}{M_w} \right] m$

here  $N_{AV}$  = Avogadro number =  $6.023 \times 10^{23}$

**SI UNIT of R :** 1 becquerel (1 Bq) = 1 decay/sec

**Other Unit is curie :** 1 Ci =  $3.70 \times 10^{10}$  decays/sec

**1 Rutherford :** (1 Rd) =  $10^6$  decays/s

**Specific activity :** Activity of 1 gm sample of radioactive substance. Its unit is Ci/gm  
e.g. specific activity of radium (226) is 1 Ci/gm.

**Ex.** The half-life of cobalt-60 is 5.25 yrs. After how long does its activity reduce to about one eighth of its original value?

**Sol** The activity is proportional to the number of undecayed atoms: In each half-life, the remaining sample

decays to half of its initial value. Since  $\left(\frac{1}{2}\right) \times \left(\frac{1}{2}\right) \times \left(\frac{1}{2}\right) = \frac{1}{8}$ , therefore, three half-lives or 15.75 years

are required for the sample to decay to 1/8th its original strength.

**Ex.** A count rate meter is used to measure the activity of a given sample. At one instant the meter shows 4750 counts per minute. Five minutes later it shows 2700 counts per minute.

(i) Find the decay constant. (ii) Also, find the half-life of the sample.

**Sol.** Initial activity  $A_i = \left. \frac{-dN}{dt} \right|_{t=0} = \lambda N_0 = 4750$  ... (i) Final activity  $A_f = \left. \frac{-dN}{dt} \right|_{t=5} = \lambda N = 2700$  ... (ii)

Dividing (i) by (ii), we get  $\frac{4750}{2700} = \frac{N_0}{N_t}$

The decay constant is given by  $\lambda = \frac{2.303}{t} \log \frac{N_0}{N_t} = \frac{2.303}{5} \log \frac{4750}{2700} = 0.113 \text{ min}^{-1}$

Half-life of the sample is  $T = \frac{0.693}{\lambda} = \frac{0.693}{0.113} = 6.14 \text{ min}$

### • Parallel radioactive disintegration

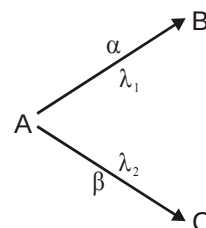
Let initial number of nuclei of A is  $N_0$  then at any time number of nuclei of

A, B & C are given by  $N_0 = N_A + N_B + N_C \Rightarrow \frac{dN_A}{dt} = -\frac{d}{dt}(N_B + N_C)$

A disintegrates into B and C by emitting  $\alpha$ ,  $\beta$  particle.

Now,  $\frac{dN_B}{dt} = -\lambda_1 N_A$  and  $\frac{dN_C}{dt} = -\lambda_2 N_A \Rightarrow \frac{d}{dt}(N_B + N_C) = -(\lambda_1 + \lambda_2) N_A$

$\Rightarrow \frac{dN_A}{dt} = -(\lambda_1 + \lambda_2) N_A \Rightarrow \lambda_{\text{eff}} = \lambda_1 + \lambda_2 \Rightarrow t_{\text{eff}} = \frac{t_1 t_2}{t_1 + t_2}$





**Ex.** The mean lives of a radioactive substances are 1620 and 405 years for  $\alpha$ -emission and  $\beta$ -emission respectively. Find out the time during which three fourth of a sample will decay if it is decaying both by  $\alpha$ -emission and  $\beta$ -emission simultaneously.

**Sol.** When a substance decays by  $\alpha$  and  $\beta$  emission simultaneously, the average rate of disintegration  $\lambda_{av}$  is given by

$\lambda_{av} = \lambda_{\alpha} + \lambda_{\beta}$  when  $\lambda_{\alpha}$  = disintegration constant for  $\alpha$ -emission only  $\lambda_{\beta}$  = disintegration constant for  $\beta$ -emission only

$$\text{Mean life is given by } T_m = \frac{1}{\lambda}, \lambda_{av} = \lambda_{\alpha} + \lambda_{\beta} \Rightarrow \frac{1}{T_m} = \frac{1}{T_{\alpha}} + \frac{1}{T_{\beta}} = \frac{1}{1620} + \frac{1}{405} = \frac{1}{324}$$

$$\lambda_{av} \times t = 2.303 \log \frac{N_0}{N_t}, \frac{1}{324} t = 2.303 \log \frac{100}{25} \Rightarrow t = 2.303 \times 324 \log 4 = 449 \text{ years.}$$

**Ex.** A radioactive decay is given by  $A \xrightarrow{t_{1/2}=8 \text{ yrs}} B$

Only A is present at  $t=0$ . Find the time at which if we are able to pick one atom out of the sample, then probability of getting B is 15 times of getting A.

**Sol.**

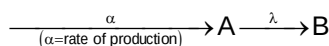
$$\begin{array}{ccc} A & \longrightarrow & B \\ \text{at } t=0 & N_0 & 0 \\ \text{at } t=t & N & N_0 - N \end{array}$$

$$\text{Probability of getting A, } P_A = \frac{N}{N_0}$$

$$\text{Probability of getting B, } P_B = \frac{N_0 - N}{N_0} \Rightarrow P_B = 15 P_A \Rightarrow \frac{N_0 - N}{N_0} = 15 \frac{N}{N_0} \Rightarrow N_0 = 16N \Rightarrow N = \frac{N_0}{16}$$

Remaining nuclei are  $\frac{1}{16}$  th of initial nuclei, hence required time  $t = 4$  half lives = 32 years

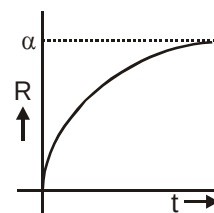
### Radioactive Disintegration with Successive Production



$$\frac{dN_A}{dt} = \alpha - \lambda N_A \dots (i)$$

$$\text{when } N_A \text{ in maximum } \frac{dN_A}{dt} = 0 = \alpha - \lambda N_A = 0, N_A \text{ max} = \frac{\alpha}{\lambda} = \frac{\text{rate of production}}{\lambda}$$

$$\text{By equation (i) } \int_0^t \frac{dN_A}{\alpha - \lambda N_A} = \int_0^t dt, \text{ Number of nuclei is } N_A = \frac{\alpha}{\lambda} (1 - e^{-\lambda t})$$



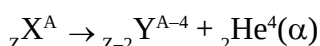
**Ex.**  $\xrightarrow{10^{21} \text{ per sec}} A \xrightarrow{\lambda = \frac{1}{30}} B$

A shows radioactive disintegration and it is continuously produced at the rate of  $10^{21}$  per sec. Find maximum number of nuclei of A.

**Sol.** At maximum,  $r_{\text{production}} = r_{\text{decay}} \Rightarrow 10^{21} = \frac{1}{30} N \Rightarrow N = 30 \times 10^{21}$

### Soddy and Fajan's Group Displacement Laws :

(i)  **$\alpha$ -decay** : The emission of one  $\alpha$ -particle reduces the mass number by 4 units and atomic number by 2 units. If parent and daughter nuclei are represented by symbols X and Y respectively then,

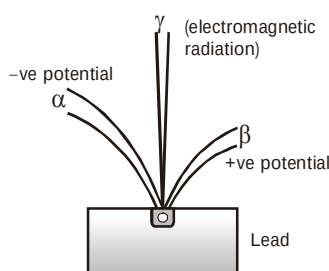


(ii)  **$\beta$ -decay** : Beta particles are said to be fast moving electrons coming from the nucleus of a radioactive substance. Does it mean that a nucleus contains electrons? No, it is an established fact that nucleus does not contain any electrons. When a nucleus emits a beta particle, one of its neutrons breaks into a proton, an electron (i.e.,  $\beta$ -particle) and an antineutrino  $n \rightarrow p + e + \bar{\nu}$

where  $n = \text{neutron}$   $p = \text{proton}$   $e = \beta\text{-particle}$

Thus emission of a beta particle is caused by the decay of a neutron into a proton. The daughter nucleus thus has an atomic number greater than one (due to one new proton in the nucleus) but same mass number as that of parent nucleus. Therefore, representing the parent and daughter nucleus by symbols X and Y respectively, we have  ${}_Z X^A \rightarrow {}_{Z+1} Y^A + \beta + \bar{\nu}$

(iii)  **$\gamma$ -decay** : When parent atoms emit gamma rays, no charge is involved as these are neutral rays. Thus there is no effect on the atomic number and mass number of the parent nucleus. However the emission of  $\gamma$ -rays represents energy. Hence the emission of these rays changes the nucleus from an excited (high energy) state to a less excited (lower energy) state.



## EXERCISE (S-1)

### Nuclear Physics :

- Suppose that the Sun consists entirely of hydrogen atom and releases the energy by the nuclear reaction,  $4 {}^1_1\text{H} \longrightarrow {}^4_2\text{He}$  with 26 MeV of energy released. If the total output power of the Sun is assumed to remain constant at  $3.9 \times 10^{26}$  W, find the time it will take to burn all the hydrogen. Take the mass of the Sun as  $1.7 \times 10^{30}$  kg.

**MP0190**

- When two deuterons ( ${}_1\text{H}^2$ ) fuse to form a helium nucleus  ${}_2\text{He}^4$ , 23.6 MeV energy is released. Find the binding energy of helium if it is 1.1 MeV for each nucleon of deuterium.

**MP0191**

- Highly energetic electrons are bombarded on a target of an element containing 30 neutrons. The ratio of radii of nucleus to that of helium nucleus is  $(14)^{1/3}$ . Find  
(a) atomic number of the nucleus  
(b) the frequency of  $K_\alpha$  line of the X-ray produced. ( $R = 1.1 \times 10^7 \text{ m}^{-1}$  and  $c = 3 \times 10^8 \text{ m/s}$ )

**[JEE 2005]**

**MP0192**

### Radioactivity :

- The kinetic energy of an  $\alpha$  – particle which flies out of the nucleus of a  $\text{Ra}^{226}$  atom in radioactive disintegration is 4.78 MeV. Find the total energy evolved during the escape of the  $\alpha$  – particle.

**MP0193**

- $\text{U}^{238}$  and  $\text{U}^{235}$  occur in nature in an atomic ratio 140 : 1. Assuming that at the time of earth's formation the two isotopes were present in equal amounts. Calculate the age of the earth.  
(Half life of  $\text{U}^{238} = 4.5 \times 10^9$  yrs & that of  $\text{U}^{235} = 7.13 \times 10^8$  yrs)

**MP0194**

- At  $t = 0$ , a sample is placed in a reactor. An unstable nuclide is produced at a constant rate  $R$  in the sample by neutron absorption. This nuclide  $\beta^-$  decays with half life  $\tau$ . Find the time required to produce 80% of the equilibrium quantity of this unstable nuclide.

**MP0195**

## EXERCISE (S-2)

1. An experiment is done to determine the half-life of radioactive substance that emits one  $\beta$ -particle for each decay process. Measurement show that 8.4  $\beta$  are emitted each second by 2.5 mg of the substance. The atomic weight of the substance is 230. Find the half life of the substance.

**MP0196**

2. A wooden piece of great antiquity weighs 50 gm and shows  $C^{14}$  activity of 320 disintegrations per minute. Estimate the length of the time which has elapsed since this wood was part of living tree, assuming that living plants show a  $C^{14}$  activity of 12 disintegrations per minute per gm. The half life of  $C^{14}$  is 5730 yrs.

**MP0197**

3. A body of mass  $m_0$  is placed on a smooth horizontal surface. The mass of the body is decreasing exponentially with disintegration constant  $\lambda$ . Assuming that the mass is ejected backward with a relative velocity  $u$ . Initially the body was at rest. Find the velocity of body after time  $t$ .

**MP0198**

4. A radionuclide with disintegration constant  $\lambda$  is produced in a reactor at a constant rate  $\alpha$  nuclei per sec. During each decay energy  $E_0$  is released. 20% of this energy is utilised in increasing the temperature of water. Find the increase in temperature of  $m$  mass of water in time  $t$ . Specific heat of water is  $S$ . Assume that there is no loss of energy through water surface.

**MP0199**

5. The element Curium  $^{248}_{96}\text{Cm}$  has a mean life of  $10^{13}$  seconds. Its primary decay modes are spontaneous fission and  $\alpha$  decay, the former with a probability of 8% and the latter with a probability of 92%. Each fission releases 200 MeV of energy. The masses involved in  $\alpha$  decay are as follows

$$^{248}_{96}\text{Cm} = 248.072220\text{u}, \quad ^{244}_{94}\text{Pu} = 244.064100\text{u} \quad \& \quad ^4_2\text{He} = 4.002603\text{u}.$$

Calculate the power output from a sample of  $10^{20}$  Cm atoms. ( $1\text{u} = 931\text{ MeV}/c^2$ )

**MP0200**

6. A small quantity of solution containing  $^{24}\text{Na}$  radionuclide (half life 15 hours) of activity 1.0 microcurie is injected into the blood of a person. A sample of the blood of volume  $1\text{ cm}^3$  taken after 5 hours shows an activity of 296 disintegration per minute. Determine the total volume of blood in the body of the person. Assume that the radioactive solution mixes uniformly in the blood of the person. ( $1\text{ Curie} = 3.7 \times 10^{10}$  disintegrations per second)

**MP0201**

7. The positron is a fundamental particle with the same mass as that of the electron and with a charge equal to that of an electron but of opposite sign. When a positron and an electron collide, they may annihilate each other. The energy corresponding to their mass appears in two photons of equal energy. Find the wavelength of the radiation emitted. [Take : mass of electron =  $(0.5/C^2)$  MeV and  $hC = 1.2 \times 10^{-12}$  MeV.m where  $h$  is the Plank's constant and  $C$  is the velocity of light in air]
- MP0202**
8. A  $\pi^+$  meson of negligible initial velocity decays to a  $\mu^+$  (muon) and a neutrino. With what kinetic energy (in eV) does the muon move? (The rest mass of neutrino can be considered zero. The rest mass of the  $\pi^+$  meson is 150 MeV and the rest mass of the muon is 100 MeV.) Take neutrino to behave like a photon. Take  $\sqrt{2} = 1.41$ .
- MP0203**
9. A radioactive decay counter is switched on at  $t = 0$ . A  $\beta$  - active sample is present near the counter. The counter registers the number of  $\beta$ -particles emitted by the sample. The counter registers  $1 \times 10^5$   $\beta$ -particles at  $t = 36$  s and  $1.11 \times 10^5$   $\beta$ -particles at  $t = 108$  s. Find  $T_{1/2}$  of this sample
- MP0204**
10. A radioactive sample emits  $n$   $\beta$ -particles in 2 sec. In next 2 sec it emits  $0.75 n$   $\beta$ -particles, what is the mean life of the sample?
- [JEE 2003]**  
**MP0205**
11. The age of a rock containing lead and uranium is equal to  $1.5 \times 10^9$  yrs. The uranium is decaying into lead with half life equal to  $4.5 \times 10^9$  yrs. Find the ratio of lead to uranium present in the rock, assuming initially no lead was present in the rock. (Given  $2^{1/3} = 1.259$ ).
- [JEE 2004]**  
**MP0206**

## EXERCISE (O-1)

### SINGLE CORRECT TYPE QUESTIONS

#### Nuclear Physics :

- In Rutherford's famous gold foil scattering experiment, he found that most alpha particles would pass through the foil undeflected. Which one of the following nuclear properties can be inferred from this observation?  
 (A) The nucleus must have a positive charge  
 (B) Most of the mass of an atom is in the nucleus  
 (C) The nucleus contains both protons neutrons  
 (D) The diameter of the nucleus is small compared to the diameter of the atom  
 (E) None of these  

**MP0207**
- Let  $u$  be denote one atomic mass unit. One atom of an element of mass number  $A$  has mass exactly equal to  $Au$   
 (A) for any value of  $A$   
 (B) only for  $A = 1$   
 (C) only for  $A = 12$   
 (D) for any value of  $A$  provided the atom is stable  

**MP0208**
- If radius of the  ${}^{27}_{13}\text{Al}$  nucleus is estimated to be 3.6 fermi, then the radius of  ${}^{125}_{52}\text{Te}$  nucleus be nearly-  
 (A) 6 fermi (B) 8 fermi (C) 4 fermi (D) 5 fermi  

**[AIEEE - 2005]**  
**MP0209**
- The surface area of a nucleus varies with mass number  $A$  as  
 (A)  $A^{2/3}$  (B)  $A^{1/3}$  (C)  $A$  (D) None  

**MP0210**
- A nucleus disintegrates into two nuclear parts which have their velocities in the ratio 2 : 1 The ratio of their nuclear sizes will be-  
 (A)  $2^{1/3} : 1$  (B)  $1 : 3^{1/2}$  (C)  $3^{1/2} : 1$  (D)  $1 : 2^{1/3}$   

**[AIEEE - 2004]**  
**MP0211**
- The binding energy per nucleon for  $\text{C}^{12}$  is 7.68 MeV and that for  $\text{C}^{13}$  is 7.5 MeV. The energy required to remove a neutron from  $\text{C}^{13}$  is  
 (A) 5.34 MeV (B) 5.5 MeV (C) 9.5 MeV (D) 9.34 MeV  

**MP0212**

7. The following nuclear reaction is an example of  $^{12}_6\text{C} + ^4_2\text{H} \rightarrow ^{16}_8\text{O} + \text{energy}$   
 (A) fission (B) fusion (C) alpha decay (D) beta decay  
**MP0213**
8. Fast neutrons may most easily be slowed down by which one of the following methods?  
 (A) passing them through a substance rich in hydrogen  
 (B) allowing them to collide elastically with heavy nuclei  
 (C) using lead shielding  
 (D) passing them through an increasing potential gradient space  
**MP0214**
9. A nuclear transformation is denoted by  $X(n, \alpha) \rightarrow ^7_3\text{Li}$ . Which of the following is the nucleus of element X?  
 (A)  $^{12}_6\text{C}$  (B)  $^{10}_5\text{B}$  (C)  $^9_5\text{B}$  (D)  $^{11}_4\text{Be}$   
**[AIEEE - 2005]**  
**MP0215**
10. A certain radioactive nuclide of mass number  $m_x$  disintegrates, with the emission of an electron and  $\gamma$  radiation only, to give second nuclide of mass number  $m_y$ . Which one of the following equation correctly relates  $m_x$  and  $m_y$ ?  
 (A)  $m_y = m_x + 1$  (B)  $m_y = m_x - 2$  (C)  $m_y = m_x - 1$  (D)  $m_y = m_x$   
**MP0216**
11. A nucleus with  $Z = 92$  emits the following in a sequence :  $\alpha, \alpha, \beta^-, \beta^-, \alpha, \alpha, \alpha, \alpha, \beta^-, \beta^-, \alpha, \beta^+, \beta^+, \alpha$ . The  $Z$  of the resulting nucleus is-  
 (A) 76 (B) 78 (C) 82 (D) 74  
**[AIEEE - 2003]**  
**MP0217**
12. In an  $\alpha$ -decay the Kinetic energy of  $\alpha$  particle is 48 MeV and  $Q$ -value of the reaction is 50 MeV. The mass number of the mother nucleus is: (Assume that daughter nucleus is in ground state)  
 (A) 96 (B) 100 (C) 104 (D) none of these  
**MP0218**
13. When  $\text{U}^{238}$  nucleus originally at rest, decays by emitting an alpha particle having a speed  $u$ , the recoil speed of the residual nucleus is-  
 (A)  $\frac{4u}{238}$  (B)  $-\frac{4u}{234}$  (C)  $\frac{4u}{234}$  (D)  $-\frac{4u}{238}$   
**[AIEEE - 2003]**  
**MP0219**
14. If a star converts all of its Helium into oxygen nucleus, find the amount of energy released per nucleus of oxygen.  $\text{O} = 15.9994 \text{ amu}$  and  $\text{He} = 4.0026 \text{ amu}$   
 (A) 7.26 MeV (B) 7 MeV (C) 10.24 MeV (D) 5.12 MeV  
**[JEE' 2005 (Scr)]**  
**MP0220**

15. The nucleus of element X ( $A = 220$ ) undergoes  $\alpha$ -decay. If Q-value of the reaction is 5.5 MeV, then the kinetic energy of  $\alpha$ -particle is :  
 (A) 5.4 MeV (B) 10.8 MeV (C) 2.7 MeV (D) None

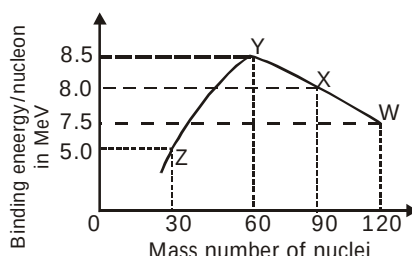
[JEE 2003 (Scr)]

MP0221

16. In the uranium radioactive series the initial nucleus is  ${}_{92}\text{U}^{238}$ , and the final nucleus is  ${}_{82}\text{Pb}^{206}$ . When the uranium nucleus decays to lead, the number of  $\alpha$  - particles emitted is.. and the number of  $\beta$ -particles emitted...  
 (A) 6, 8 (B) 8, 6 (C) 16, 6 (D) 32, 12

MP0222

17. Binding energy per nucleon versus mass number curve for nuclei is shown in figure. W, X, Y and Z are four nuclei indicated on the curve. The process that would release energy is :



- (A)  $Y \rightarrow 2Z$  (B)  $W \rightarrow X + Z$  (C)  $W \rightarrow 2Y$  (D)  $X \rightarrow Y + Z$

MP0223

18. An alpha nucleus of energy  $\frac{1}{2}mv^2$  bombards a heavy nuclear target of charge Ze. Then the distance of closest approach for the alpha nucleus will be proportional to-  
 (A)  $v^2$  (B)  $1/m$  (C)  $1/v^4$  (D)  $1/Ze$

[AIEEE - 2006]

MP0224

19. If the binding energy per nucleon in  ${}^7_3\text{Li}$  and  ${}^4_2\text{He}$  nuclei are 5.60 MeV and 7.06 MeV respectively, then in the reaction :  $p + {}^7_3\text{Li} \rightarrow 2{}^4_2\text{He}$  energy of proton must be-  
 (A) 28.24 MeV (B) 17.28 MeV (C) 1.46 MeV (D) 39.2 MeV

[AIEEE - 2006]

MP0225

20. If  $M_0$  is the mass of an oxygen isotope  ${}^{17}_8\text{O}$ ,  $M_p$  and  $M_n$  are the masses of a proton and a neutron, respectively, the nuclear binding energy of the isotope is-  
 (A)  $(M_0 - 8M_p)c^2$  (B)  $(M_0 - 8M_p - 9M_n)c^2$   
 (C)  $M_0c^2$  (D)  $(M_0 - 17M_n)c^2$

[AIEEE - 2007]

MP0226



21. In the options given below, let  $E$  denote the rest mass energy of a nucleus and  $n$  a neutron. The correct option is :-

$$(A) E\left({}^{236}_{92}\text{U}\right) > E\left({}^{137}_{53}\text{I}\right) + E\left({}^{97}_{39}\text{Y}\right) + 2E(n) \quad (B) E\left({}^{236}_{92}\text{U}\right) < E\left({}^{137}_{53}\text{I}\right) + E\left({}^{97}_{39}\text{Y}\right) + 2E(n)$$

$$(C) E\left({}^{236}_{92}\text{U}\right) < E\left({}^{140}_{56}\text{Ba}\right) + E\left({}^{94}_{36}\text{Kr}\right) + 2E(n) \quad (D) E\left({}^{236}_{92}\text{U}\right) = E\left({}^{140}_{56}\text{Ba}\right) + E\left({}^{94}_{36}\text{Kr}\right) + 2E(n)$$

[JEE 2007]

MP0227

### Radioactivity :

22. The half-life of  ${}^{131}\text{I}$  is 8 days. Given a sample of  ${}^{131}\text{I}$  at time  $t = 0$ , we can assert that :  
 (A) no nucleus will decay before  $t = 4$  days (B) no nucleus will decay before  $t = 8$  days  
 (C) all nuclei will decay before  $t = 16$  days (D) a given nucleus may decay at any time after  $t = 0$

MP0228

23. Activity of a radioactive substance is  $R_1$  at time  $t_1$  and  $R_2$  at time  $t_2$  ( $t_2 > t_1$ ). Then the ratio  $\frac{R_2}{R_1}$  is:

$$(A) \frac{t_2}{t_1} \quad (B) e^{-\lambda(t_1+t_2)} \quad (C) e^{\left(\frac{t_1-t_2}{\lambda}\right)} \quad (D) e^{\lambda(t_1-t_2)}$$

MP0229

24. A particular nucleus in a large population of identical radioactive nuclei did survive 5 half lives of that isotope. Then the probability that this surviving nucleus will survive the next half life :

$$(A) \frac{1}{32} \quad (B) \frac{1}{5} \quad (C) \frac{1}{2} \quad (D) \frac{1}{10}$$

MP0230

25. A certain radio active substance has a half life of 5 years. Thus for a nucleus in a sample of the element, the probability of decay in ten years is

$$(A) 50\% \quad (B) 75\% \quad (C) 100\% \quad (D) 60\%$$

MP0231

26. The activity of a sample reduces from  $A_0$  to  $A_0/\sqrt{3}$  in one hour. The activity after 3 hours more will be :-

$$(A) \frac{A_0}{3\sqrt{3}} \quad (B) \frac{A_0}{9} \quad (C) \frac{A_0}{9\sqrt{3}} \quad (D) \frac{A_0}{27}$$

MP0232

27. Half life of radium is 1620 years. How many radium nuclei decay in 5 hours in 5 gm radium? (Atomic weight of radium = 223)

$$(A) 9.1 \times 10^{12} \quad (B) 3.23 \times 10^{15} \quad (C) 1.72 \times 10^{20} \quad (D) 3.3 \times 10^{17}$$

MP0233

28. The activity of a sample of radioactive material is  $A_1$  at time  $t_1$  and  $A_2$  at time  $t_2$  ( $t_2 > t_1$ ). Its mean life is  $T$ .

(A)  $A_1 t_1 = A_2 t_2$

(B)  $\frac{A_1 - A_2}{t_2 - t_1} = \text{constant}$

(C)  $A_2 = A_1 e^{(t_1 - t_2)/T}$

(D)  $A_2 = A_1 e^{(t_1/Tt_2)}$

MP0234

29. The decay constant of the end product of a radioactive series is

(A) zero

(B) infinite

(C) finite (non zero)

(D) depends on the end product.

MP0235

30. A radioactive substance is dissolved in a liquid and the solution is heated. The activity of the solution

(A) is smaller than that of element

(B) is greater than that of element

(C) is equal to that of element

(D) will be smaller or greater depending upon whether the solution is weak or concentrated.

MP0236

31. In a certain nuclear reactor, a radioactive nucleus is being produced at a constant rate = 1000 /s. The mean life of the radionuclide is 40 minutes. At steady state, the number of radionuclide will be

(A)  $4 \times 10^4$

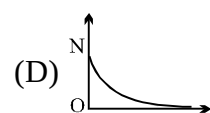
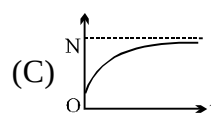
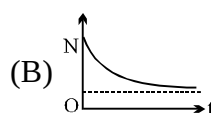
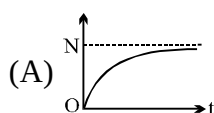
(B)  $24 \times 10^4$

(C)  $24 \times 10^5$

(D)  $24 \times 10^6$

MP0237

32. In the above question, if there were  $20 \times 10^5$  radionuclide at  $t = 0$ , then the graph of  $N$  v/s  $t$  is



MP0238

33. Which of the following cannot be emitted by radioactive substances during their decay ?

(A) Protons

(B) Neutrinos

(C) Helium nuclei

(D) Electrons

[AIEEE - 2003]

MP0239

34. A 280 days old radioactive substance shows an activity of 6000 dps, 140 days later it's activity becomes 3000dps. What was its initial activity.

(A) 20000 dps

(B) 24000 dps

(C) 12000 dps

(D) 6000 dps

[JEE 2004 (Scr)]

MP0240

35. When  ${}_3\text{Li}^7$  nuclei are bombarded by protons and the resultant nuclei are  ${}_4\text{Be}^8$ , the emitted particles will be-

(A) alpha particles

(B) beta particles

(C) gamma photons

(D) neutrons

[AIEEE - 2006]

MP0241

36. The energy spectrum of  $\beta$ -particles [number  $N(E)$  as a function of  $\beta$ -energy  $E$ ] emitted from a radioactive source is- [AIEEE - 2006]



MP0242

37. In gamma ray emission from a nucleus  
 (A) both the neutron number and the proton number change  
 (B) there is no change in the proton number and the neutron number  
 (C) only the neutron number changes  
 (D) only the proton number changes

[AIEEE-2007]

MP0243

38. The half-life period of a radioactive element X is same as the mean life time of another radioactive element Y. Initially they have the same number of atoms. Then-  
 (A) X will decay faster than Y  
 (B) Y will decay faster than X  
 (C) Y and X have same decay rate initially  
 (D) X and Y decay at same rate always

[AIEEE - 2007]

MP0244

39. Given a sample of Radium-226 having half-life of 4 days. Find the probability, a nucleus disintegrates within 2 half lives.  
 (A) 1 (B) 1/2 (C) 3/4 (D) 1/4

[JEE 2006]

MP0245

### MATRIX MATCH TYPE QUESTION

40. Match the following Columns :-

#### Column-I

- (A) Nuclear fusion  
 (B) Nuclear fission  
 (C)  $\beta$ -decay  
 (D) Exothermic nuclear reaction

#### Column-II

- (P) Converts some matter into energy  
 (Q) Generally occurs for nuclei with low atomic number  
 (R) Generally occurs for nuclei with higher atomic number  
 (S) Essentially proceeds by weak nuclear forces

[JEE 2006]

MP0246

41. Some laws/processes are given in **Column I**. Match these with the physical phenomena given in **Column II** and indicate your answer by darkening appropriate bubbles in the  $4 \times 4$  matrix given in the ORS.

**Column-I**

- (A) Transition between two atomic energy levels
- (B) Electron emission from a material
- (C) Mosley's law
- (D) Change of photon energy into kinetic energy of electrons

**Column-II**

- (P) Characteristic X-rays
- (Q) Photoelectric effect
- (R) Hydrogen spectrum
- (S)  $\beta$ -decay

[JEE 2007]

MP0247

## EXERCISE (O-2)

## SINGLE CORRECT TYPE QUESTIONS

- The binding energies of nuclei X and Y are  $E_1$  and  $E_2$  respectively. Two atoms of X fuse to give one atom of Y and an energy Q is released. Then:  
 (A)  $Q = 2E_1 - E_2$  (B)  $Q = E_2 - 2E_1$  (C)  $Q = 2E_1 + E_2$  (D)  $Q = 2E_2 + E_1$   
**MP0248**
- If each fission in a  $U^{235}$  nucleus releases 200 MeV, how many fissions must occur per second to produce a power of 1 KW  
 (A)  $1.325 \times 10^{13}$  (B)  $3.125 \times 10^{13}$  (C)  $1.235 \times 10^{13}$  (D)  $2.135 \times 10^{13}$   
**MP0249**
- The binding energies of the atom of elements A & B are  $E_a$  &  $E_b$  respectively. Three atoms of the element B fuse to give one atom of element A. This fusion process is accompanied by release of energy e. Then  $E_a$ ,  $E_b$  are related to each other as  
 (A)  $E_a + e = 3E_b$  (B)  $E_a = 3E_b$  (C)  $E_a - e = 3E_b$  (D)  $E_a + 3E_b + e = 0$   
**MP0250**
- The rest mass of the deuteron,  ${}^2_1H$ , is equivalent to an energy of 1876 MeV, the rest mass of a proton is equivalent to 939 MeV and that of a neutron to 940 MeV. A deuteron may disintegrate to a proton and a neutron if it :  
 (A) emits a  $\gamma$ -ray photon of energy 2 MeV (B) captures a  $\gamma$ -ray photon of energy 2 MeV  
 (C) emits a  $\gamma$ -ray photon of energy 3 MeV (D) captures a  $\gamma$ -ray photon of energy 3 MeV  
**MP0251**
- The number of  $\alpha$  and  $\beta^-$  emitted during the radioactive decay chain starting from  ${}^{226}_{88}Ra$  and ending at  ${}^{206}_{82}Pb$  is  
 (A)  $3\alpha$  &  $6\beta^-$  (B)  $4\alpha$  &  $5\beta^-$  (C)  $5\alpha$  &  $4\beta^-$  (D)  $6\alpha$  &  $6\beta^-$   
**MP0252**
- In a radioactive element the fraction of initial amount remaining after its mean life time is  
 (A)  $1 - \frac{1}{e}$  (B)  $\frac{1}{e^2}$  (C)  $\frac{1}{e}$  (D)  $1 - \frac{1}{e^2}$   
**MP0253**
- Two radioactive material  $A_1$  and  $A_2$  have decay constants of  $10\lambda_0$  and  $\lambda_0$ . If initially they have same number of nuclei, the ratio of number of their undecayed nuclei will be  $(1/e)$  after a time  
 (A)  $\frac{1}{\lambda_0}$  (B)  $\frac{1}{9\lambda_0}$  (C)  $\frac{1}{10\lambda_0}$  (D) 1  
**MP0254**

8. 90% of a radioactive sample is left undecayed after time  $t$  has elapsed. What percentage of the initial sample will decay in a total time  $2t$  :
- (A) 20% (B) 19% (C) 40% (D) 38%

MP0255

9. A radioactive material of half-life  $T$  was produced in a nuclear reactor at different instants, the quantity produced second time was twice of that produced first time. If now their present activities are  $A_1$  and  $A_2$  respectively then their age difference equals:

(A)  $\frac{T}{\ln 2} \left| \ln \frac{A_1}{A_2} \right|$  (B)  $T \left| \ln \frac{A_1}{A_2} \right|$  (C)  $\frac{T}{\ln 2} \left| \ln \frac{A_2}{2A_1} \right|$  (D)  $T \left| \ln \frac{A_2}{2A_1} \right|$

MP0256

10. The half-life of substance X is 45 years, and it decomposes to substance Y. A sample from a meteorite was taken which contained 2% of X and 14% of Y by quantity of substance. If substance Y is not normally found on a meteorite, what is the approximate age of the meteorite?
- (A) 270 years (B) 135 years (C) 90 years (D) 45 years

MP0257

11. At time  $t = 0$ ,  $N_1$  nuclei of decay constant  $\lambda_1$  &  $N_2$  nuclei of decay constant  $\lambda_2$  are mixed. The decay rate of the mixture is :

(A)  $N_1 N_2 e^{-(\lambda_1 + \lambda_2)t}$  (B)  $\left( \frac{N_1}{N_2} \right) e^{-(\lambda_1 - \lambda_2)t}$

(C)  $\left( N_1 \lambda_1 e^{-\lambda_1 t} + N_2 \lambda_2 e^{-\lambda_2 t} \right)$  (D)  $N_1 \lambda_1 N_2 \lambda_2 e^{-(\lambda_1 + \lambda_2)t}$

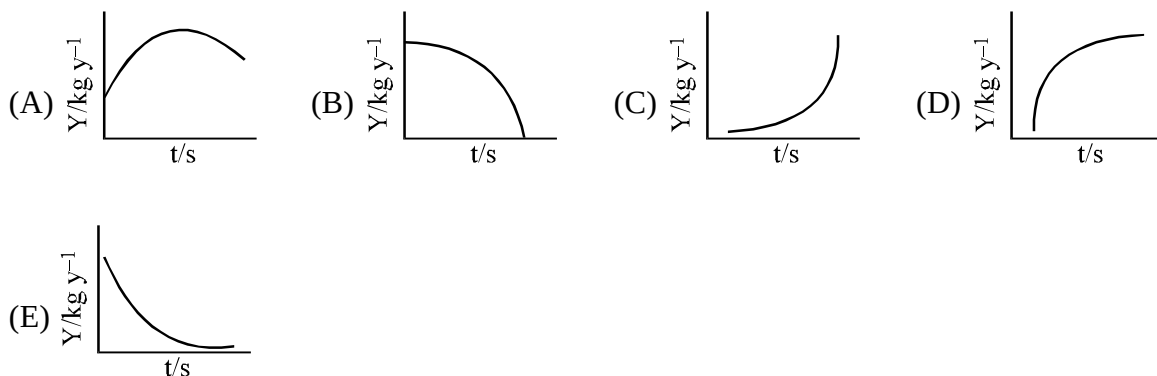
MP0258

12. A radioactive nuclide can decay simultaneously by two different processes which have decay constants  $\lambda_1$  and  $\lambda_2$ . The effective decay constant of the nuclide is  $\lambda$ , then :

(A)  $\lambda = \lambda_1 + \lambda_2$  (B)  $\lambda = 1/2(\lambda_1 + \lambda_2)$  (C)  $\frac{1}{\lambda} = \frac{1}{\lambda_1} + \frac{1}{\lambda_2}$  (D)  $\lambda = \sqrt{\lambda_1 \lambda_2}$

MP0259

13. The radioactive nucleus of an element X decays to a stable nucleus of element Y. A graph of the rate of formation of Y against time would look like



MP0260

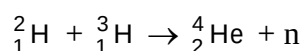
14. The half life of a neutron is 800 sec.  $10^8$  neutrons at a certain instant are projected from one space station towards another space station, situated 3200 km away, with a velocity 2000 m/s. Their velocity remains constant during the journey. How many neutrons reach the other station?  
 (A)  $50 \times 10^6$  (B)  $25 \times 10^6$  (C)  $80 \times 10^5$  (D)  $25 \times 10^5$

MP0261

15. A radioactive source in the form of a metal sphere of diameter  $3.2 \times 10^{-3}$  m emits  $\beta$ -particle at a constant rate of  $6.25 \times 10^{10}$  particle/sec. The source is electrically insulated and all the  $\beta$ -particle are emitted from the surface. The potential of the sphere will rise to 1 V in time  
 (A) 180  $\mu$  sec (B) 90  $\mu$  sec (C) 18  $\mu$  sec (D) 9  $\mu$  sec

MP0262

16. In the nuclear fusion reaction,



given that the repulsive potential energy between the two nuclei is  $7.7 \times 10^{-14}$  J, the temperature at which the gases must be heated to initiate the reaction is nearly [Boltzmann's constant  $k = 1.38 \times 10^{-23}$  J/K]-

- (A)  $10^7$  K (B)  $10^5$  K (C)  $10^3$  K (D)  $10^9$  K

[AIEEE - 2003]

MP0263

17. The binding energy per nucleon of deuteron ( ${}^2_1\text{H}$ ) and helium nucleus ( ${}^4_2\text{He}$ ) is 1.1 MeV and 7 MeV respectively. If two deuteron nuclei react to form a single helium nucleus, then the energy released is-  
 (A) 13.9 MeV (B) 26.9 MeV (C) 23.6 MeV (D) 19.2 MeV

[AIEEE - 2004]

MP0264

### MULTIPLE CORRECT TYPE QUESTIONS

18. When a nucleus with atomic number Z and mass number A undergoes a radioactive decay process:  
 (A) both Z and A will decrease, if the process is  $\alpha$  decay  
 (B) Z will decrease but A will not change, if the process is  $\beta^+$  decay  
 (C) Z will decrease but A will not change, if the process is  $\beta^-$  decay  
 (D) Z and A will remain unchanged, if the process is  $\gamma$  decay.

MP0265

19. When the atomic number A of the nucleus increases  
 (A) initially the neutron-proton ratio is constant = 1  
 (B) initially neutron-proton ratio increases and later decreases  
 (C) initially binding energy per nucleon increases and later decreases  
 (D) the binding energy per nucleon increases when the neutron-proton ratio increases.

MP0266

20. Let  $m_p$  be the mass of a proton,  $m_n$  the mass of a neutron,  $M_1$  the mass of a  ${}^{20}_{10}\text{Ne}$  nucleus and  $M_2$  the mass of a  ${}^{40}_{20}\text{Ca}$  nucleus. Then

- (A)  $M_2 = 2M_1$  (B)  $M_2 > 2M_1$  (C)  $M_2 < 2M_1$  (D)  $M_1 < 10(m_n + m_p)$

MP0267

21. The decay constant of a radio active substance is  $0.173 \text{ (years)}^{-1}$ . Therefore :

- (A) Nearly 63% of the radioactive substance will decay in  $(1/0.173)$  year.
- (B) half life of the radio active substance is  $(1/0.173)$  year.
- (C) one -forth of the radioactive substance will be left after nearly 8 years.
- (D) all the above statements are true.

MP0268

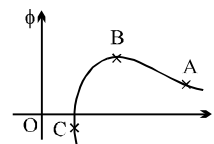
22. Which of the following statement(s) is/are correct?

- (A) The rest mass of a stable nucleus is less than the sum of the rest masses of its separated nucleons.
- (B) The rest mass of a stable nucleus is greater than the sum of the rest masses of its separated nucleons.
- (C) In nuclear fusion, energy is released by fusion of two nuclei of medium mass (approx. 100 amu)
- (D) In nuclear fission, energy is released by fragmentation of a very heavy nucleus.

MP0269

23. The graph shown by the side shows the variation of potential energy  $\phi$  of a proton with its distance 'r' from a fixed sodium nucleus, as it approaches the nucleus, placed at origin O. Then the portion.

- (A) AB indicates nuclear repulsion
- (B) AB indicates electrostatic repulsion
- (C) BC indicates nuclear attraction
- (D) BC represents electrostatic interaction



MP0270

24. The instability of the nucleus can be due to various causes. An unstable nucleus emits radiations if possible to transform into less unstable state. Then the cause and the result can be

- (A) a nucleus of excess nucleons is  $\alpha$  active
- (B) an excited nucleus of excess protons is  $\beta^-$  active
- (C) an excited nucleus of excess protons is  $\beta^+$  active
- (D) an nucleus of excess neutrons is  $\beta^-$  active

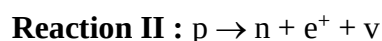
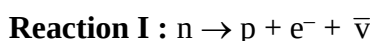
MP0271

25. In  $\beta$ -decay, the Q-value of the process is E. Then

- (A) K.E. of a  $\beta$ -particle cannot exceed E.
- (B) K.E. of anti neutrino emitted lies between Zero and E.
- (C) N/Z ratio of the nucleus is altered.
- (D) Mass number (A) of the nucleus is altered.

MP0272

26. Consider the following nuclear reactions and select the correct statements from the options that follow.



- (A) Free neutron is unstable, therefore reaction I is possible
- (B) Free proton is stable, therefore reaction II is not possible
- (C) Inside a nucleus, both decays (reaction I and II) are possible
- (D) Inside a nucleus, reaction I is not possible but reaction II is possible.

MP0273



27. When the nucleus of an electrically neutral atom undergoes a radioactive decay process, it will remain neutral after the decay if the process is :
- (A)  $\alpha$  decay                      (B)  $\beta^-$  decay                      (C)  $\gamma$  decay                      (D) K-capture

MP0274

**MATRIX MATCH TYPE QUESTION**

28. In the following, column I lists some physical quantities & the column II gives approx. energy values associated with some of them. Choose the appropriate value of energy from column II for each of the physical quantities in column I and write the corresponding letter A, B, C etc. against the number (i), (ii), (iii), etc. of the physical quantity in the answer book. In your answer, the sequence of column I should be maintained.

| Column I                              | Column II    |
|---------------------------------------|--------------|
| (i) Energy of thermal neutrons        | (A) 0.025 eV |
| (ii) Energy of X-rays                 | (B) 0.5 eV   |
| (iii) Binding energy per nucleon      | (C) 3 eV     |
| (iv) Photoelectric threshold of metal | (D) 20 eV    |
|                                       | (E) 10 keV   |
|                                       | (F) 8 MeV    |

MP0275

## EXERCISE - JM

1. This question contains Statement-1 and Statement-2. Out of the four choices given after the statements, choose the one that best describes the two statements.

**Statement-1 :** Energy is released when heavy nuclei undergo fission or light nuclei undergo fusion.

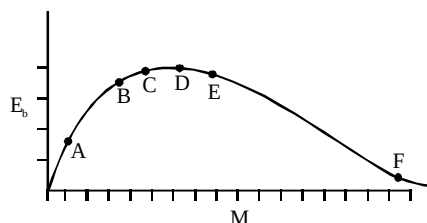
**Statement-2 :** For heavy nuclei, binding energy per nucleon increases with increasing  $Z$  while for light nuclei it decreases with increasing  $Z$ .

- (1) Statement-1 is false, Statement-2 is true.
- (2) Statement-1 is true, Statement-2 is true; Statement-2 is a correct explanation for Statement-1.
- (3) Statement-1 is true, Statement-2 is true; Statement-2 is not a correct explanation for Statement-1.
- (4) Statement-1 is true, Statement-2 is false.

[AIEEE - 2008]

MP0276

2.



The above is a plot of binding energy per nucleon  $E_b$ , against the nuclear mass  $M$ ; A, B, C, D, E, F correspond to different nuclei. Consider four reactions :

- (i)  $A + B \rightarrow C + \varepsilon$
- (ii)  $C \rightarrow A + B + \varepsilon$
- (iii)  $D + E \rightarrow F + \varepsilon$
- (iv)  $F \rightarrow D + E + \varepsilon$

where  $\varepsilon$  is the energy released ? In which reactions is  $\varepsilon$  positive ?

- (i)  $A + B \rightarrow C + \varepsilon$
  - (ii)  $C \rightarrow A + B + \varepsilon$
  - (iii)  $D + E \rightarrow F + \varepsilon$
  - (iv)  $F \rightarrow D + E + \varepsilon$
- (1) (ii) and (iv)
  - (2) (ii) and (iii)
  - (3) (i) and (iv)
  - (4) (i) and (iii)

[AIEEE - 2008]

MP0277

**Directions :** Questions number 3 – 4 are based on the following paragraph.

A nucleus of mass  $M + \Delta m$  is at rest and decays into two daughter nuclei of equal mass  $\frac{M}{2}$  each.

Speed of light is  $c$ .

3. The speed of daughter nuclei is :-

(1)  $c\sqrt{\frac{\Delta m}{M + \Delta m}}$

(2)  $c\frac{\Delta m}{M + \Delta m}$

(3)  $c\sqrt{\frac{2\Delta m}{M}}$

(4)  $c\sqrt{\frac{\Delta m}{M}}$

[AIEEE-2010]

MP0278

4. The binding energy per nucleon for the parent nucleus is  $E_1$  and that for the daughter nuclei is  $E_2$ . Then:-

(1)  $E_1 = 2E_2$

(2)  $E_2 = 2E_1$

(3)  $E_1 > E_2$

(4)  $E_2 > E_1$

[AIEEE - 2010]

MP0279

5. A radioactive nucleus (initial mass number  $A$  and atomic number  $Z$ ) emits 3  $\alpha$ -particles and 2 positrons. The ratio of number of neutrons to that of protons in the final nucleus will be:-

(1)  $\frac{A - Z - 4}{Z - 2}$

(2)  $\frac{A - Z - 8}{Z - 4}$

(3)  $\frac{A - Z - 4}{Z - 8}$

(4)  $\frac{A - Z - 12}{Z - 4}$

[AIEEE - 2010]

MP0280

6. The half life of a radioactive substance is 20 minutes. The approximate time interval ( $t_2 - t_1$ ) between the time  $t_2$  when  $\frac{2}{3}$  of it has decayed and time  $t_1$  when  $\frac{1}{3}$  of it had decayed is :-

(1) 20 min

(2) 28 min

(3) 7 min

(4) 14 min

[AIEEE - 2011]

MP0281

7. After absorbing a slowly moving neutron of mass  $m_N$  (momentum  $\sim 0$ ) a nucleus of mass  $M$  breaks into two nuclei of masses  $m_1$  and  $5m_1$  ( $6m_1 = M + m_N$ ), respectively. If the de Broglie wavelength of the nucleus with mass  $m_1$  is  $\lambda$ , then de Broglie wavelength of the other nucleus will be:-

(1)  $25\lambda$

(2)  $5\lambda$

(3)  $\frac{\lambda}{5}$

(4)  $\lambda$

[AIEEE - 2011]

MP0282

8. **Statement-1:** A nucleus having energy  $E_1$  decays by  $\beta^-$ -emission to daughter nucleus having energy  $E_2$ , but the  $\beta^-$  rays are emitted with a continuous energy spectrum having end point energy  $E_1 - E_2$ .

**Statement-2:** To conserve energy and momentum in  $\beta^-$ -decay at least three particles must take part in the transformation.

(1) Statement-1 is incorrect, statement-2 is correct

(2) Statement-1 is correct, statement-2 is incorrect

(3) Statement-1 is correct, statement-2 correct; statement-2 is the correct explanation of statement-1

(4) Statement-1 is correct, statement-2 is correct; statement -2 is not the correct explanation of statement-1.

[AIEEE - 2011]

MP0283

9. Assume that a neutron breaks into a proton and an electron. The energy released during this process is :

(Mass of neutron =  $1.6725 \times 10^{-27}$  kg

Mass of proton =  $1.6725 \times 10^{-27}$  kg

Mass of electron =  $9 \times 10^{-31}$  kg)

- (1) 5.4 MeV (2) 0.73 MeV (3) 7.10 MeV (4) 6.30 MeV

[AIEEE - 2012]

MP0284

10. Hydrogen ( ${}_1\text{H}^1$ ), Deuterium ( ${}_1\text{H}^2$ ), singly ionised Helium ( ${}_2\text{He}^4$ )<sup>+</sup> and doubly ionised lithium ( ${}_3\text{Li}^6$ )<sup>++</sup> all have one electron around the nucleus. Consider an electron transition from  $n = 2$  to  $n = 1$ . If the wave lengths of emitted radiation are  $\lambda_1$ ,  $\lambda_2$ ,  $\lambda_3$  and  $\lambda_4$  respectively then approximately which one of the following is **correct** ?

(1)  $\lambda_1 = \lambda_2 = 4\lambda_3 = 9\lambda_4$

(2)  $\lambda_1 = 2\lambda_2 = 3\lambda_3 = 4\lambda_4$

(3)  $4\lambda_1 = 2\lambda_2 = 2\lambda_3 = \lambda_4$

(4)  $\lambda_1 = 2\lambda_2 = 2\lambda_3 = \lambda_4$

[JEE-Main-2014]

MP0285

11. The radiation corresponding to  $3 \rightarrow 2$  transition of hydrogen atom falls on a metal surface to produce photoelectrons. These electrons are made to enter a magnetic field of  $3 \times 10^{-4}$  T. If the radius of the largest circular path followed by these electrons is 10.0 mm, the work function of the metal is close to:-

- (1) 0.8 eV (2) 1.6 eV (3) 1.8 eV (4) 1.1 eV

[JEE-Main-2014]

MP0286

12. Half-lives of two radioactive elements A and B are 20 minutes and 40 minutes, respectively. Initially, the samples have equal number of nuclei. After 80 minutes, the ratio of decayed numbers of A and B nuclei will be :-

- (1) 5 : 4 (2) 1 : 16 (3) 4 : 1 (4) 1 : 4

[JEE-Main-2016]

MP0287

13. A radioactive nucleus A with a half life  $T$ , decays into a nucleus B. At  $t = 0$ , there is no nucleus B. At sometime  $t$ , the ratio of the number of B to that of A is 0.3. Then,  $t$  is given by :

- (1)  $t = T \log(1.3)$  (2)  $t = \frac{T}{\log(1.3)}$  (3)  $t = \frac{T \log 2}{2 \log 1.3}$  (4)  $t = T \frac{\log 1.3}{\log 2}$

[JEE-Main-2017]

MP0288

14. An electron from various excited states of hydrogen atom emit radiation to come to the ground state. Let  $\lambda_n$ ,  $\lambda_g$  be the de Broglie wavelength of the electron in the  $n^{\text{th}}$  state and the ground state respectively. Let  $\Lambda_n$  be the wavelength of the emitted photon in the transition from the  $n^{\text{th}}$  state to the ground state. For large  $n$ , (A, B are constants)

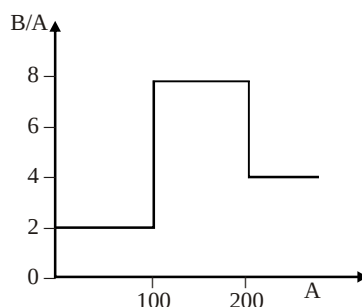
- (1)  $\Lambda_n \approx A + B\lambda_n$  (2)  $\Lambda_n^2 \approx A + B\lambda_n^2$  (3)  $\Lambda_n^2 \approx \lambda$  (4)  $\Lambda_n \approx A + \frac{B}{\lambda_n^2}$

[JEE-Main-2018]

MP0289

## EXERCISE - JA

1. Assume that the nuclear binding energy per nucleon ( $B/A$ ) versus mass number ( $A$ ) is as shown in the figure. Use this plot to choose the correct choice(s) given below : [JEE 2008]



- (A) Fusion of two nuclei with mass numbers lying in the range of  $1 < A < 50$  will release energy  
 (B) Fusion of two nuclei with mass numbers lying in the range of  $51 < A < 100$  will release energy  
 (C) Fission of a nucleus lying in the mass range of  $100 < A < 200$  will release energy when broken into two equal fragments  
 (D) Fission of a nucleus lying in the mass range of  $200 < A < 260$  will release energy when broken into two equal fragments

MP0290

2. A radioactive sample S1 having an activity of  $5\mu\text{Ci}$  has twice the number of nuclei as another sample S2 which has an activity of  $10\mu\text{Ci}$ . The half lives of S1 and S2 can be : [JEE 2008]

- (A) 20 years and 5 years, respectively (B) 20 years and 10 years, respectively  
 (C) 10 years each (D) 5 years each

MP0291

## Paragraph for Question Nos. 3 to 5

Scientists are working hard to develop nuclear fusion reactor. Nuclei of heavy hydrogen,  ${}^2_1\text{H}$ , known as deuteron and denoted by D, can be thought of as a candidate for fusion reactor. The D-D reaction is  ${}^2_1\text{H} + {}^2_1\text{H} \rightarrow {}^3_2\text{He} + n + \text{energy}$ . In the core of fusion reactor, a gas of heavy hydrogen is fully ionized into deuteron nuclei and electrons. This collection of  ${}^2_1\text{H}$  nuclei and electrons is known as plasma. The nuclei move randomly in the reactor core and occasionally come close enough for nuclear fusion to take place. Usually, the temperatures in the reactor core are too high and no material wall can be used to confine the plasma. Special techniques are used which confine the plasma for a time  $t_0$  before the particles fly away from the core. If  $n$  is the density (number/volume) of deuterons, the product  $nt_0$  is called Lawson number. In one of the criteria, a reactor is termed successful if Lawson number is greater than  $5 \times 10^{14} \text{ s/cm}^3$ . It may be helpful to use the following :

$$\text{Boltzmann constant } k = 8.6 \times 10^{-5} \text{ eV/K ; } \frac{e^2}{4\pi\epsilon_0} = 1.44 \times 10^{-9} \text{ eVm.}$$

3. In the core of nuclear fusion reactor, the gas becomes plasma because of [JEE-2009]  
 (A) strong nuclear force acting between the deuterons  
 (B) Coulomb force acting between the deuterons  
 (C) Coulomb force acting between deuteron-electron pairs  
 (D) the high temperature maintained inside the reactor core

MP0292

4. Assume that two deuteron nuclei in the core of fusion reactor at temperature  $T$  are moving towards each other, each with kinetic energy  $1.5 \text{ kT}$ , when the separation between them is large enough to neglect Coulomb potential energy. Also neglect any interaction from other particles in the core. The minimum temperature  $T$  required for them to reach a separation of  $4 \times 10^{-15} \text{ m}$  is in the range  
 (A)  $1.0 \times 10^9 \text{ K} < T < 2.0 \times 10^9 \text{ K}$  (B)  $2.0 \times 10^9 \text{ K} < T < 3.0 \times 10^9 \text{ K}$   
 (C)  $3.0 \times 10^9 \text{ K} < T < 4.0 \times 10^9 \text{ K}$  (D)  $4.0 \times 10^9 \text{ K} < T < 5.0 \times 10^9 \text{ K}$

[JEE-2009]

MP0292

5. Results of calculations for four different designs of a fusion reactor using D-D reaction are given below. Which of these is most promising based on Lawson criterion?  
 (A) deuteron density  $= 2.0 \times 10^{12} \text{ cm}^{-3}$ , confinement time  $= 5.0 \times 10^{-3} \text{ s}$   
 (B) deuteron density  $= 8.0 \times 10^{14} \text{ cm}^{-3}$ , confinement time  $= 9.0 \times 10^{-1} \text{ s}$   
 (C) deuteron density  $= 4.0 \times 10^{23} \text{ cm}^{-3}$ , confinement time  $= 1.0 \times 10^{-11} \text{ s}$   
 (D) deuteron density  $= 1.0 \times 10^{24} \text{ cm}^{-3}$ , confinement time  $= 4.0 \times 10^{-12} \text{ s}$

[JEE-2009]

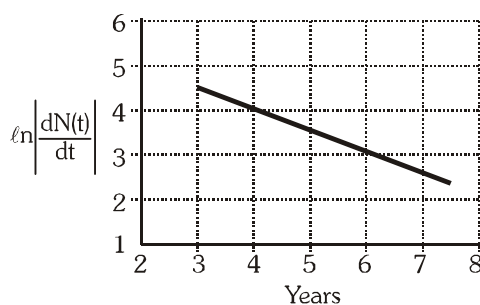
MP0292

6. To determine the half life of a radioactive element, a student plots a graph of  $\ln \left| \frac{dN(t)}{dt} \right|$  versus  $t$ . Here

$\frac{dN(t)}{dt}$  is the rate of radioactive decay at time  $t$ . If the number of radioactive nuclei of this element

decreases by a factor of  $p$  after 4.16 years, the value of  $p$  is

[JEE-2010]



MP0293

7. The activity of a freshly prepared radioactive sample is  $10^{10}$  disintegrations per second, whose mean life is  $10^9$  s. The mass of an atom of this radioisotope is  $10^{-25}$  kg. The mass (in mg) of the radioactive sample is

[JEE-2011]

MP0294

### Paragraph for Questions 8 and 9

The  $\beta$ -decay process, discovered around 1900, is basically the decay of a neutron ( $n$ ). In the laboratory, a proton ( $p$ ) and an electron ( $e^-$ ) are observed as the decay products of the neutron. Therefore, considering the decay of a neutron as a two-body decay process, it was predicted theoretically that the kinetic energy of the electron should be a constant. But experimentally, it was observed that the electron kinetic energy has a continuous spectrum. Considering a three-body decay process, i.e.  $n \rightarrow p + e^- + \bar{\nu}_e$ , around 1930, Pauli explained the observed electron energy spectrum. Assuming the anti-neutrino ( $\bar{\nu}_e$ ) to be massless and possessing negligible energy, and the neutron to be at rest, momentum and energy conservation principles are applied. From this calculation, the maximum kinetic energy of the electron is  $0.8 \times 10^6$  eV. The kinetic energy carried by the proton is only the recoil energy.

8. If the anti-neutrino had a mass of  $3 \text{ eV}/c^2$  (where  $c$  is the speed of light) instead of zero mass, what should be the range of the kinetic energy,  $K$ , of the electron?

(A)  $0 \leq K \leq 0.8 \times 10^6 \text{ eV}$

(B)  $3.0 \text{ eV} \leq K \leq 0.8 \times 10^6 \text{ eV}$

(C)  $3.0 \text{ eV} \leq K < 0.8 \times 10^6 \text{ eV}$

(D)  $0 \leq K < 0.8 \times 10^6 \text{ eV}$

[JEE 2012]

MP0295

9. What is the maximum energy of the anti-neutrino?

(A) zero

(B) much less than  $0.8 \times 10^6$  eV(C) Nearly  $0.8 \times 10^6$  eV(D) Much larger than  $0.8 \times 10^6$  eV

[JEE 2012]

MP0295

10. A freshly prepared sample of a radioisotope of half-life 1386 s has activity  $10^3$  disintegrations per second. Given that  $\ln 2 = 0.693$ , the fraction of the initial number of nuclei (expressed in nearest integer percentage) that will decay in the first 80 s after preparation of the sample is.

[JEE Advance-2013]

MP0296

### Paragraph for Questions 11 and 12

The mass of a nucleus  ${}_Z^AX$  is less than the sum of the masses of  $(A - Z)$  number of neutrons and  $Z$  number of protons in the nucleus. The energy equivalent to the corresponding mass difference is known as the binding energy of the nucleus. A heavy nucleus of mass  $M$  can break into two light nuclei of masses  $m_1$  and  $m_2$  only if  $(m_1 + m_2) < M$ . Also two light nuclei of masses  $m_3$  and  $m_4$  can undergo complete fusion and form a heavy nucleus of mass  $M'$  only if  $(m_3 + m_4) > M'$ . The masses of some neutral atoms are given in the table below :-

[JEE Advance-2013]

|                          |              |                          |              |                          |              |                          |              |
|--------------------------|--------------|--------------------------|--------------|--------------------------|--------------|--------------------------|--------------|
| ${}^1_1\text{H}$         | 1.007825 u   | ${}^2_1\text{H}$         | 2.014102 u   | ${}^3_1\text{H}$         | 3.016050 u   | ${}^4_2\text{He}$        | 4.002603 u   |
| ${}^6_3\text{Li}$        | 6.015123 u   | ${}^7_3\text{Li}$        | 7.016004 u   | ${}^{70}_{30}\text{Zn}$  | 69.925325 u  | ${}^{82}_{34}\text{Se}$  | 81.916709 u  |
| ${}^{152}_{64}\text{Gd}$ | 151.919803 u | ${}^{206}_{82}\text{Pb}$ | 205.974455 u | ${}^{209}_{83}\text{Bi}$ | 208.980388 u | ${}^{210}_{84}\text{Po}$ | 209.982876 u |

$$(1\text{u} = 932 \text{ MeV}/c^2)$$

11. The kinetic energy (in keV) of the alpha particle, when the nucleus  ${}^{210}_{84}\text{Po}$  at rest undergoes alpha decay, is :-

(A) 5319 (B) 5422 (C) 5707 (D) 5818

MP0297

12. The correct statement is :-

(A) The nucleus  ${}^6_3\text{Li}$  can emit an alpha particle  
 (B) The nucleus  ${}^{210}_{84}\text{Po}$  can emit a proton  
 (C) Deuteron and alpha particle can undergo complete fusion  
 (D) The nuclei  ${}^{70}_{30}\text{Zn}$  and  ${}^{82}_{34}\text{Se}$  can undergo complete fusion

MP0297

13. Match List I of the nuclear processes with List II containing parent nucleus and one of the end products of each process and then select the correct answer using the codes given below the lists:

[JEE Advance-2013]

| List I             | List II                                                                |
|--------------------|------------------------------------------------------------------------|
| P. Alpha decay     | 1. ${}^{15}_8\text{O} \rightarrow {}^{15}_7\text{N} + \dots$           |
| Q. $\beta^+$ decay | 2. ${}^{238}_{92}\text{U} \rightarrow {}^{234}_{90}\text{Th} + \dots$  |
| R. Fission         | 3. ${}^{185}_{83}\text{Bi} \rightarrow {}^{184}_{82}\text{Pb} + \dots$ |
| S. Proton emission | 4. ${}^{239}_{94}\text{Pu} \rightarrow {}^{140}_{57}\text{La} + \dots$ |

Codes :

| P     | Q | R | S |
|-------|---|---|---|
| (A) 4 | 2 | 1 | 3 |
| (B) 1 | 3 | 2 | 4 |
| (C) 2 | 1 | 4 | 3 |
| (D) 4 | 3 | 2 | 1 |

MP0298

14. If  $\lambda_{\text{Cu}}$  is the wavelength of  $\text{K}_\alpha$  X-ray line of copper (atomic number 29) and  $\lambda_{\text{Mo}}$  is the wavelength of the  $\text{K}_\alpha$  X-ray line of molybdenum (atomic number 42), then the ratio  $\frac{\lambda_{\text{Cu}}}{\lambda_{\text{Mo}}}$  is close to :-

(A) 1.99 (B) 2.14 (C) 0.50 (D) 0.48

[JEE Advance-2014]

MP0299



15. A metal surface is illuminated by light of two different wavelength 248 nm and 310 nm. The maximum speeds of the photoelectrons corresponding to these wavelengths are  $u_1$  and  $u_2$ , respectively. If the ratio  $u_1 : u_2 = 2 : 1$  and  $hc = 1240 \text{ eV nm}$ , the work function of the metal is nearly :-  
 (A) 3.7 eV (B) 3.2 eV (C) 2.8 eV (D) 2.5 eV

[JEE Advance-2014]

MP0300

16. A nuclear power plant supplying electrical power to a village uses a radioactive material of half life T years as the fuel. The amount of fuel at the beginning is such that the total power requirement of the village is 12.5% of the electrical power available from the plant at that time. If the plant is able to meet the total power needs of the village for a maximum period of  $nT$  years, then the value of  $n$  is.

[JEE Advance-2015]

MP0301

17. Match the nuclear processes given in column-I with the appropriate option(s) in column-II.

[JEE Advance-2015]

**Column-I****Column-II**

- |                                  |                                                                  |
|----------------------------------|------------------------------------------------------------------|
| (A) Nuclear fusion               | (P) Absorption of thermal neutrons by ${}_{92}^{235}\text{U}$    |
| (B) Fission in a nuclear reactor | (Q) ${}_{27}^{60}\text{Co}$ nucleus                              |
| (C) $\beta$ -decay               | (R) Energy production in stars via hydrogen conversion to helium |
| (D) $\gamma$ -ray emission       | (S) Heavy water                                                  |
|                                  | (T) Neutrino emission                                            |

MP0302

18. For a radioactive material, its activity  $A$  and rate of change of its activity  $R$  are defined as  $A = -\frac{dN}{dt}$

and  $R = -\frac{dA}{dt}$ , where  $N(t)$  is the number of nuclei at time  $t$ . Two radioactive sources P (mean life  $\tau$ ) and Q (mean life  $2\tau$ ) have the same activity at  $t = 0$ . Their rates of change of activities at  $t = 2\tau$  are  $R_p$  and  $R_Q$ , respectively. If  $\frac{R_p}{R_Q} = \frac{n}{e}$ , then the value of  $n$  is :

[JEE Advance-2015]

MP0303

19. A fission reaction is given by  ${}_{92}^{236}\text{U} \rightarrow {}_{54}^{140}\text{Xe} + {}_{38}^{94}\text{Sr} + x + y$ , where  $x$  and  $y$  are two particles. Considering  ${}_{92}^{236}\text{U}$  to be at rest, the kinetic energies of the products are denoted by  $K_{\text{Xe}}, K_{\text{Sr}}, K_x$  (2 MeV) and  $K_y$  (2 MeV), respectively. Let the binding energies per nucleon of  ${}_{92}^{236}\text{U}$ ,  ${}_{54}^{140}\text{Xe}$  and  ${}_{38}^{94}\text{Sr}$  be 7.5 MeV, 8.5 MeV and 8.5 MeV, respectively. Considering different conservation laws, the correct option(s) is (are) :-

[JEE Advance-2015]

- (A)  $x = n$ ,  $y = n$ ,  $K_{\text{Sr}} = 129 \text{ MeV}$ ,  $K_{\text{Xe}} = 86 \text{ MeV}$   
 (B)  $x = p$ ,  $y = e^-$ ,  $K_{\text{Sr}} = 129 \text{ MeV}$ ,  $K_{\text{Xe}} = 86 \text{ MeV}$   
 (C)  $x = p$ ,  $y = n$ ,  $K_{\text{Sr}} = 129 \text{ MeV}$ ,  $K_{\text{Xe}} = 86 \text{ MeV}$   
 (D)  $x = n$ ,  $y = n$ ,  $K_{\text{Sr}} = 86 \text{ MeV}$ ,  $K_{\text{Xe}} = 129 \text{ MeV}$

MP0304

20. The isotope  $^{12}_5\text{B}$  having a mass 12.014 u undergoes  $\beta$ -decay to  $^{12}_6\text{C}$ .  $^{12}_6\text{C}$  has an excited state of the nucleus ( $^{12}_6\text{C}^*$ ) at 4.041 MeV above its ground state. If  $^{12}_5\text{B}$  decays to  $^{12}_6\text{C}^*$ , the maximum kinetic energy of the  $\beta$ -particle in units of MeV is ( $1\text{u} = 931.5 \text{ MeV}/c^2$ , where  $c$  is the speed of light in vacuum). **[JEE Advance-2016]**

MP0305

21. The electrostatic energy of  $Z$  protons uniformly distributed throughout a spherical nucleus of radius  $R$  is given by  $E = \frac{3}{5} \frac{Z(Z-1)e^2}{4\pi\epsilon_0 R}$  **[JEE Advance-2016]**

The measured masses of the neutron,  $^1_1\text{H}$ ,  $^{15}_7\text{N}$  and  $^{15}_8\text{O}$  are 1.008665 u, 1.007825 u, 15.000109 u and 15.003065 u respectively. Given that the radii of both the  $^{15}_7\text{N}$  and  $^{15}_8\text{O}$  nuclei are same,  $1\text{u} = 931.5 \text{ MeV}/c^2$

( $c$  is the speed of light) and  $\frac{e^2}{(4\pi\epsilon_0)} = 1.44 \text{ MeV fm}$ . Assuming that the difference between the binding energies of  $^{15}_7\text{N}$  and  $^{15}_8\text{O}$  is purely due to the electrostatic energy, the radius of either of the nuclei is ( $1\text{fm} = 10^{-15}\text{m}$ )

- (A) 2.85 fm (B) 3.03 fm (C) 3.42 fm (D) 3.80 fm

MP0306

22. An accident in a nuclear laboratory resulted in deposition of a certain amount of radioactive material of half-life 18 days inside the laboratory. Tests revealed that the radiation was 64 times more than the permissible level required for safe operation of the laboratory. What is the minimum number of days after which the laboratory can be considered safe for use? **[JEE Advance-2016]**

- (A) 64 (B) 90 (C) 108 (D) 120

MP0307

23.  $^{131}\text{I}$  is an isotope of Iodine that  $\beta$  decays to an isotope of Xenon with a half-life of 8 days. A small amount of a serum labelled with  $^{131}\text{I}$  is injected into the blood of a person. The activity of the amount of  $^{131}\text{I}$  injected was  $2.4 \times 10^5$  Becquerel (Bq). It is known that the injected serum will get distributed uniformly in the blood stream in less than half an hour. After 11.5 hours, 2.5 ml of blood is drawn from the person's body, and gives an activity of 115 Bq. The total volume of blood in the person's body, in liters is approximately (you may use  $e^x \approx 1 + x$  for  $|x| \ll 1$  and  $\ln 2 \approx 0.7$ ).

**[JEE Advance-2017]**

MP0308

24. In a radioactive decay chain,  $^{232}_{90}\text{Th}$  nucleus decays to  $^{212}_{82}\text{Pb}$  nucleus. Let  $N_\alpha$  and  $N_\beta$  be the number of  $\alpha$  and  $\beta^-$  particles, respectively, emitted in this decay process. Which of the following statements is (are) true ?

- (A)  $N_\alpha = 5$  (B)  $N_\alpha = 6$  (C)  $N_\beta = 2$  (D)  $N_\beta = 4$

**[JEE Advance-2018]**

MP0309

25. Suppose a  $^{226}_{88}\text{Ra}$  nucleus at rest and in ground state undergoes  $\alpha$ -decay to a  $^{222}_{86}\text{Rn}$  nucleus in its excited state. The kinetic energy of the emitted  $\alpha$  particle is found to be 4.44 MeV.  $^{222}_{86}\text{Rn}$  nucleus then goes to its ground state by  $\gamma$ -decay. The energy of the emitted  $\gamma$ -photon is \_\_\_\_\_ keV,

[Given: atomic mass of  $^{226}_{88}\text{Ra} = 226.005\text{u}$ , atomic mass of  $^{222}_{86}\text{Rn} = 222.000\text{u}$ , atomic mass of  $\alpha$  particle =  $4.000\text{u}$ ,  $1\text{u} = 931 \text{ MeV}/c^2$ ,  $c$  is speed of the light]

**[JEE Advanced-2019]**

MP0310

## ANSWER KEY

## EXERCISE (S-1)

1. Ans.  $8/3 \times 10^{18}$  sec      2. Ans. 28 MeV      3. Ans.  $v = 1.546 \times 10^{18}$  Hz;  $Z = 26$

4. Ans. 4.87 MeV      5. Ans.  $6.04 \times 10^9$  yrs      6. Ans.  $t = \left( \frac{\ln 5}{\ln 2} \right) \tau$

## EXERCISE (S-2)

1. Ans.  $1.7 \times 10^{10}$  years      2. Ans. 5196 yrs      3. Ans.  $v = u\lambda t$

4. Ans.  $\Delta T = \frac{0.2E_0 \left[ \alpha t - \frac{\alpha}{\lambda} (1 - e^{-\lambda t}) \right]}{mS}$       5. Ans.  $\approx 33.298 \mu\text{W}$

6. Ans. 6 litre      7. Ans.  $2.4 \times 10^{-12}$  m      8. Ans.  $9.00 \times 10^6$

9. Ans. ( $T_{1/2} = 10.8$  sec)      10. Ans.  $1.75n = N_0(1 - e^{-4\lambda})$ , 6.95 sec,  $\frac{2}{\ln\left(\frac{4}{3}\right)}$

11. Ans. 0.259

## EXERCISE (O-1)

## SINGLE CORRECT TYPE QUESTIONS

- |              |              |              |              |              |              |
|--------------|--------------|--------------|--------------|--------------|--------------|
| 1. Ans. (D)  | 2. Ans. (C)  | 3. Ans. (A)  | 4. Ans. (A)  | 5. Ans. (D)  | 6. Ans. (A)  |
| 7. Ans. (B)  | 8. Ans. (A)  | 9. Ans. (B)  | 10. Ans. (D) | 11. Ans. (B) | 12. Ans. (B) |
| 13. Ans. (C) | 14. Ans. (C) | 15. Ans. (A) | 16. Ans. (B) | 17. Ans. (C) | 18. Ans. (B) |
| 19. Ans. (B) | 20. Ans. (B) | 21. Ans. (A) | 22. Ans. (D) | 23. Ans. (D) | 24. Ans. (C) |
| 25. Ans. (B) | 26. Ans. (B) | 27. Ans. (B) | 28. Ans. (C) | 29. Ans. (A) | 30. Ans. (C) |
| 31. Ans. (C) | 32. Ans. (C) | 33. Ans. (A) | 34. Ans. (B) | 35. Ans. (C) | 36. Ans. (C) |
| 37. Ans. (B) | 38. Ans. (B) | 39. Ans. (C) |              |              |              |

## MATRIX MATCH TYPE QUESTION

40. Ans. (A) P, Q; (B) P, R; (C) S, P; (D) P, Q, R

41. Ans. (A) R, P; (B) Q, S; (C) P; (D) Q

## EXERCISE (O-2)

- |                                       |                    |                 |                    |
|---------------------------------------|--------------------|-----------------|--------------------|
| 1. Ans. (B)                           | 2. Ans. (B)        | 3. Ans. (C)     | 4. Ans. (D)        |
| 5. Ans. (C)                           | 6. Ans. (C)        | 7. Ans. (B)     | 8. Ans. (B)        |
| 9. Ans. (C)                           | 10. Ans. (B)       | 11. Ans. (C)    | 12. Ans. (A)       |
| 13. Ans. (E)                          | 14. Ans. (B)       | 15. Ans. (C)    | 16. Ans. (D)       |
| 17. Ans. (C)                          | 18. Ans. (A, B, D) | 19. Ans. (A, C) | 20. Ans. (C, D)    |
| 21. Ans. (A, C)                       | 22. Ans. (A, D)    | 23. Ans. (B, C) | 24. Ans. (A, C, D) |
| 25. Ans. (A, B, C)                    | 26. Ans. (A, B, C) | 27. Ans. (C, D) |                    |
| 28. Ans. (i) A, (ii) E (iii) F (iv) C |                    |                 |                    |

**EXERCISE - JM**

- |                 |              |              |              |
|-----------------|--------------|--------------|--------------|
| 1. Ans. (4)     | 2. Ans. (3)  | 3. Ans. (3)  | 4. Ans. (4)  |
| 5. Ans. (3)     | 6. Ans. (1)  | 7. Ans. (4)  | 8. Ans. (3)  |
| 9. Ans. (Bonus) | 10. Ans. (1) | 11. Ans. (4) | 12. Ans. (1) |
| 13. Ans. (4)    | 14. Ans. (4) |              |              |

**EXERCISE - JA**

- |                                                    |                |                    |              |
|----------------------------------------------------|----------------|--------------------|--------------|
| 1. Ans. (B, D)                                     | 2. Ans. (A)    | 3. Ans. (D)        | 4. Ans. (A)  |
| 5. Ans. (B)                                        | 6. Ans. (8)    | 7. Ans. (1)        | 8. Ans. (D)  |
| 9. Ans. (C)                                        | 10. Ans. 4     | 11. Ans. (A)       | 12. Ans. (C) |
| 13. Ans. (C)                                       | 14. Ans. (B)   | 15. Ans. (A)       | 16. Ans. 3   |
| 17. Ans. (A)-R or R,T; (B)-P & S; (C)-Q & T; (D)-R |                |                    | 18. Ans. 2   |
| 19. Ans. (A)                                       | 20. Ans. 9     | 21. Ans. (C)       | 22. Ans. (C) |
| 23. Ans. 5                                         | 24. Ans. (A,C) | 25. Ans. (135.00 ) |              |

# Important Notes

# Chapter Contents

## SEMICONDUCTOR

|                   |    |
|-------------------|----|
| 01. THEORY        | 1  |
| 02. EXERCISE (S)  | 26 |
| 03. EXERCISE (O)  | 28 |
| 04. EXERCISE (JM) | 37 |
| 05. ANSWER KEY    | 45 |



# KEY CONCEPTS

## ENERGY BANDS IN SOLIDS

Based on Pauli's exclusion principle

In an isolated atom electrons present in energy level but in solid, atoms are not isolated, there is interaction among each other, due to this energy level split into different energy levels. Quantity of these different energy levels depends on the quantity of interacting atoms. Splitting of sharp and closely compact energy levels result into energy band. This is discrete in nature. Order of energy levels in a band is  $10^{23}$  and their energy difference =  $10^{-23}$  eV.

### Energy Band

Range of energy possessed by electron in a solid is known as energy band.

### Valence Band (VB)

Range of energies possessed by valence electron is known as valence band.

- (a) Have bonded electron.
- (b) No flow of current due to such electron.
- (c) Always fulfill by electron.

### Conduction Band (CB)

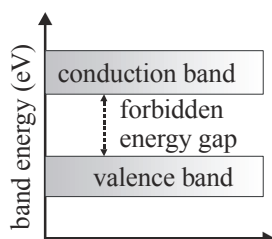
Range of energies possessed by free electron is known as conduction band.

- (a) It has conducting electrons.
- (b) Current flows due to such electrons.
- (c) If conduction band is fully empty then current conduction is not possible.
- (d) Electrons may exist or not in it.

### Forbidden Energy gap (FEG) ( $\Delta E_g$ )

$$\Delta E_g = (C B)_{\min} - (V B)_{\max}$$

Energy gap between conduction band and valence band, where no free electron can exist.



- Width of forbidden energy gap depends upon the nature of substance.
- Width is more, then valence electrons are strongly attached with nucleus
- Width of forbidden energy gap is represented in eV.
- As temperature increases forbidden energy gap decreases (very slightly).

## CLASSIFICATION OF CONDUCTORS, INSULATORS AND SEMICONDUCTOR : -

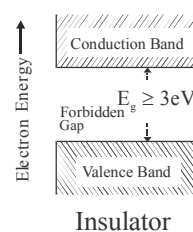
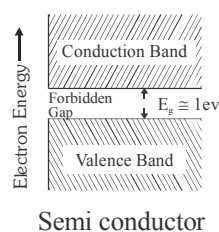
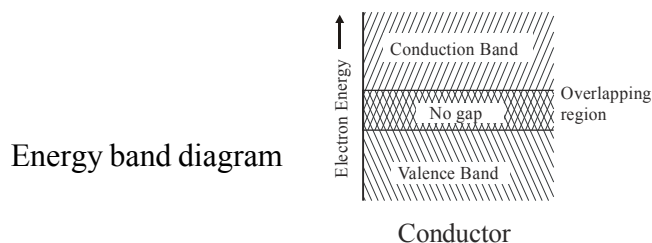
On the basis of the relative values of electrical conductivity and energy bands the solids are broadly classified into three categories

- (i) Conductor
- (ii) Semiconductor
- (iii) Insulator



**Comparison between conductor, semiconductor and insulator :**

| Properties                                   | Conductor                          | Semiconductor                   | Insulator                           |
|----------------------------------------------|------------------------------------|---------------------------------|-------------------------------------|
| Resistivity                                  | $10^{-2} - 10^{-8} \Omega\text{m}$ | $10^{-5} - 10^6 \Omega\text{m}$ | $10^{11} - 10^{19} \Omega\text{m}$  |
| Conductivity                                 | $10^2 - 10^8 \text{ mho/m}$        | $10^5 - 10^{-6} \text{ mho/m}$  | $10^{-11} - 10^{-19} \text{ mho/m}$ |
| Temp. Coefficient of resistance ( $\alpha$ ) | Positive                           | Negative                        | Negative                            |
| Current                                      | Due to free electrons              | Due to electrons and holes      | No current                          |



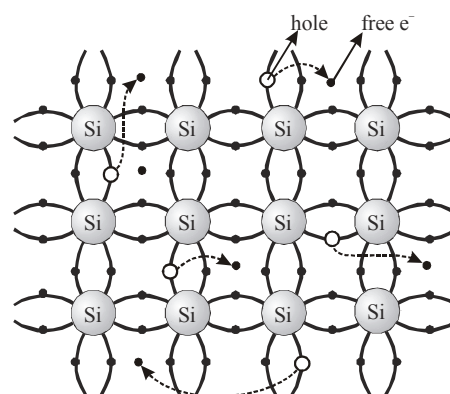
|                      |                    |                                |                              |
|----------------------|--------------------|--------------------------------|------------------------------|
| Forbidden energy gap | $\cong 0\text{eV}$ | $\cong 1\text{eV}$             | $\geq 3\text{eV}$            |
| Example :            | Pt, Al, Cu, Ag     | Ge, Si, GaAs, GaF <sub>2</sub> | Wood, plastic, Diamond, Mica |

**CONCEPT OF "HOLES" IN SEMICONDUCTORS**

Due to external energy (temp. or radiation) when electron goes from valence band to conduction band (i.e. bonded electrons becomes free) a vacancy of free  $e^-$  creates in valence band,

which has same charge as electron but positive. This positively charged vacancy is termed as hole and shown in figure.

- It is deficiency of electron in VB.
- It acts as positive charge carrier.
- It's effective mass is more than electron.
- It's mobility is less than electron.



**Note :** Hole acts as virtual charge carrier, although it has no physical significance.

- Number of electrons reaching from VB to CB at temperature T kelvin

$$n = A T^{3/2} e^{-\frac{E_g}{2kT}} = AT^{3/2} \exp\left[-\frac{E_g}{2kT}\right]$$

where

$k$  = Boltzmann constant =  $1.38 \times 10^{-23} \text{ J/K}$

$A$  = constant

$T$  = absolute temperature

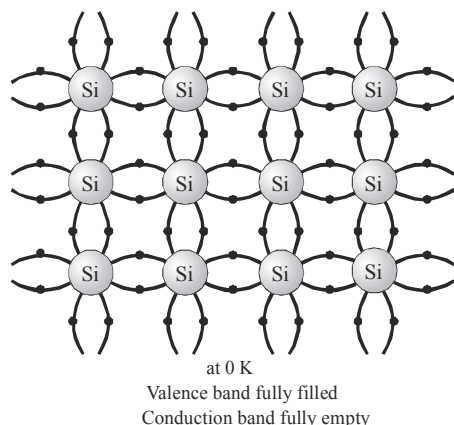
$E_g$  = energy gap between CB and VB

- In silicon at room temperature out of  $10^{12}$  Si atoms only one electron goes from VB to CB.
- In germanium at room temperature out of  $10^9$  Ge atoms only one electron goes from VB to CB.

## EFFECT OF TEMPERATURE ON SEMICONDUCTOR

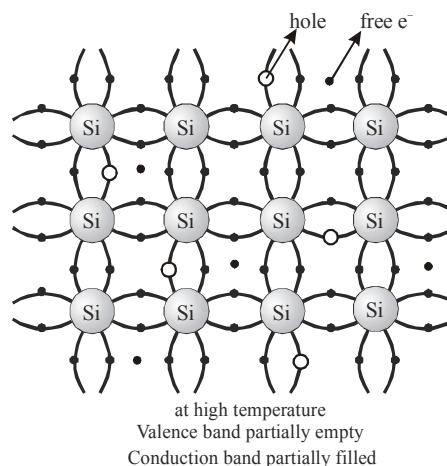
### At absolute zero kelvin temperature

At this temperature covalent bonds are very strong and there are no free electrons and semiconductor behaves as perfect insulator.



### Above absolute temperature

With increase in temperature few valence electrons jump into conduction band and hence it behaves as poor conductor.



## EFFECT OF IMPURITY IN SEMICONDUCTOR

Doping is a method of addition of "desirable" impurity atoms to pure semiconductor to increase conductivity of semiconductor. or

Doping is a process of deliberate addition of a desirable impurity atoms to a pure semiconductor to modify its properties in controlled manner.

Added impurity atoms are called dopants.

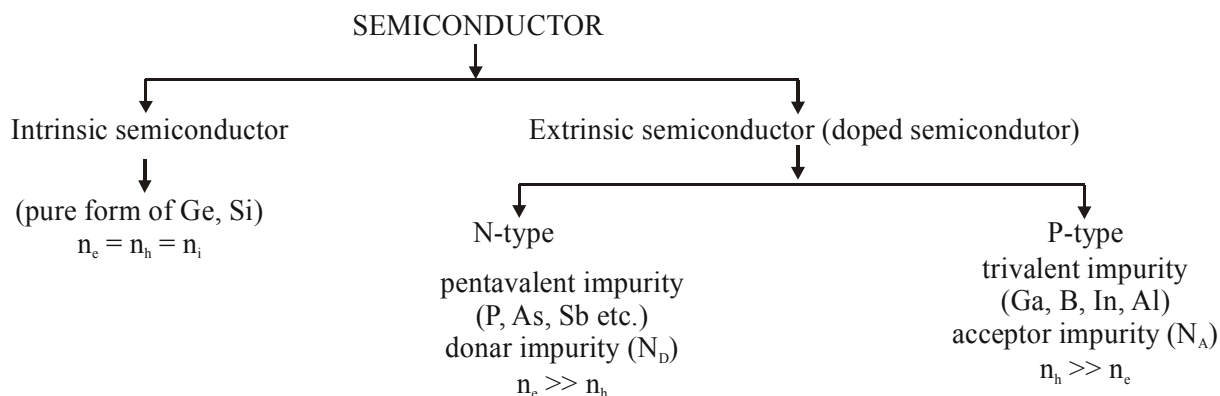
The impurity added may be  $\approx 1$  part per million (ppm).

- The dopant atom should take the position of semiconductor atom in the lattice.
- The presence of the dopant atom should not distort the crystal lattice.
- The size of the dopant atom should be almost the same as that of the crystal atom.
- The concentration of dopant atoms should not be large (not more than 1% of the crystal atom).

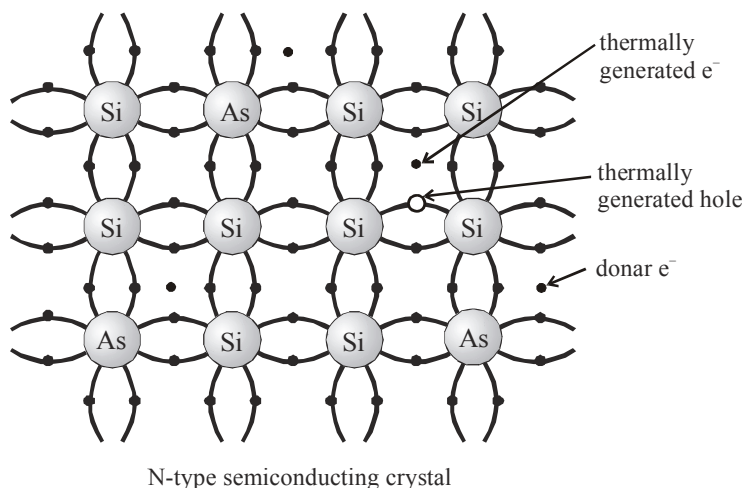
It is to be noted that the doping of a semiconductor increases its electrical conductivity to a great extent.

- The concentration of dopant atoms be very low, doping ratio is vary from impure : pure ::  $1 : 10^6$  to  $1 : 10^{10}$  In general it is  $1 : 10^8$
- There are two main method of doping.  
(i) Alloy method (ii) Diffusion method (The best)
- The size of dopant atom (impurity) should be almost the same as that of crystal atom. So that crystalline structure of solid remain unchanged.

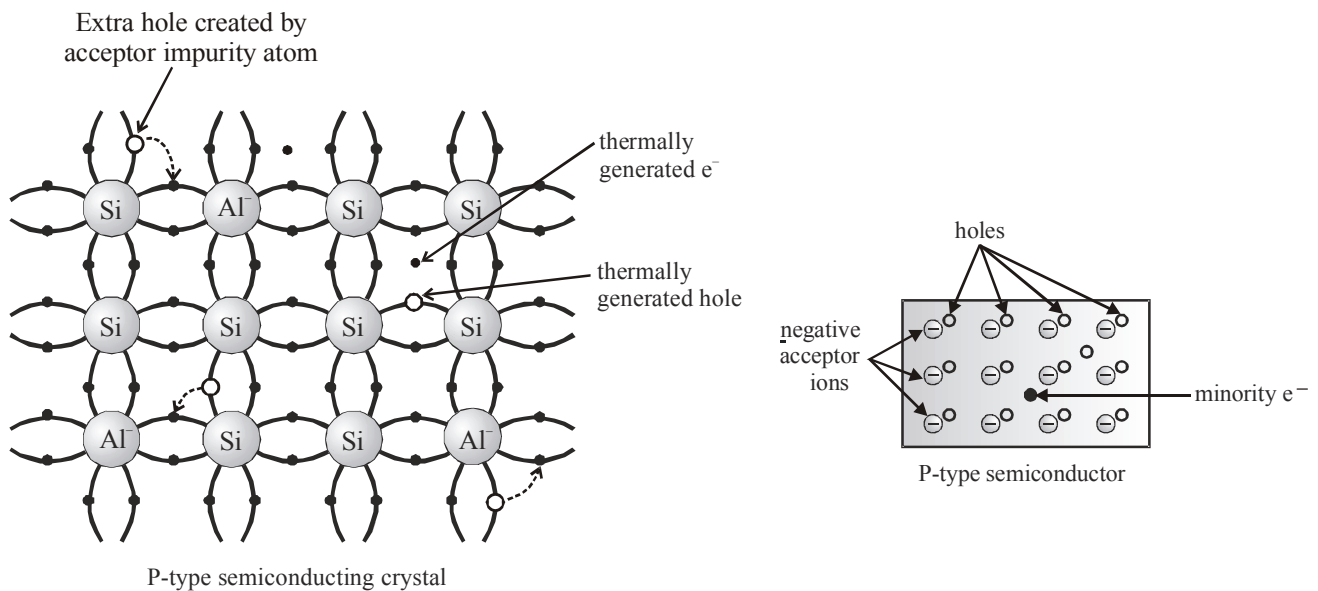
## CLASSIFICATION OF SEMICONDUCTOR

**N type semiconductor**

When a pure semiconductor (Si or Ge) is doped by pentavalent impurity (P, As, Sb, Bi) then four electrons out of the five valence electrons of impurity take part, in covalent bonding, with four silicon atoms surrounding it and the fifth electron is set free. These impurity atoms which donate free  $e^-$  for conduction are called as Donor impurity ( $N_D$ ). Due to donor impurity free  $e^-$  increases very much so it is called as "N" type semiconductor. By donating  $e^-$  impurity atoms get positive charge and hence known as "Immobile Donor positive Ion". In N-type semiconductor free  $e^-$  are called as "majority" charge carriers and "holes" are called as "minority" charge carriers.

**P type semiconductor**

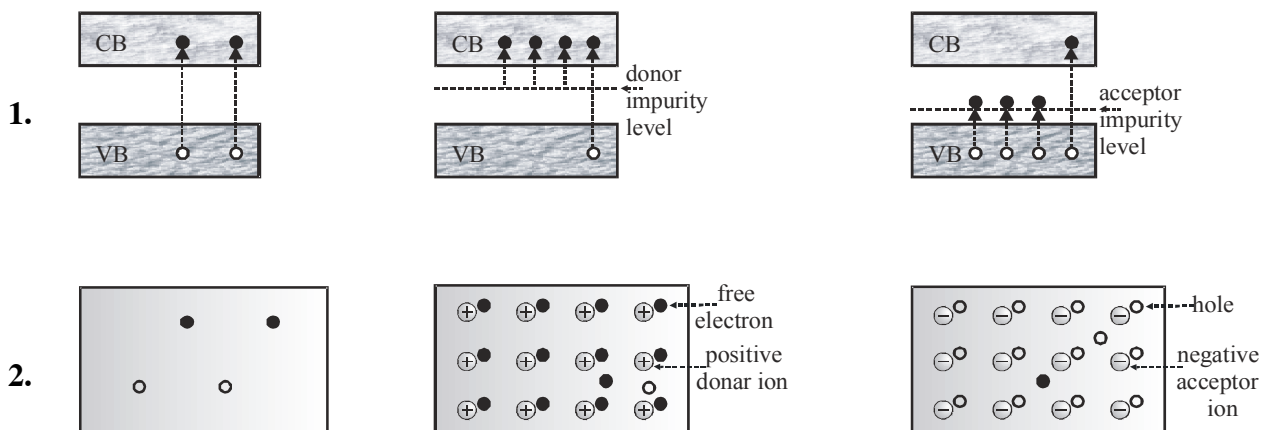
When a pure semiconductor (Si or Ge) is doped by trivalent impurity (B, Al, In, Ga) then outer most three electrons of the valence band of impurity take part, in covalent bonding with four silicon atoms surrounding it and except one electron from semiconductor and make hole in semiconductor. These impurity atoms which accept bonded  $e^-$  from valence band are called as Acceptor impurity ( $N_A$ ). Here holes increases very much so it is called as "P" type semiconductor and impurity ions known as "Immobile Acceptor negative Ion". In P-type semiconductor free  $e^-$  are called as minority charge carriers and holes are called as majority charge carriers.



### Intrinsic Semiconductor

### N-type (Pentavalent impurity)

### P-type (Trivalent impurity)



3. Current due to electron and hole

Mainly due to electrons

Mainly due to holes

4.  $n_e = n_h = n_i$

$n_h \ll n_e$  ( $N_D \approx n_e$ )

$n_h \gg n_e$  ( $N_A \approx n_h$ )

5.  $I = I_e + I_h$

$I \approx I_e$

$I \approx I_h$

6. Entirely neutral

Entirely neutral

Entirely neutral

7. Quantity of electrons and holes are equal

Majority - Electrons  
Minority - Holes

Majority - Holes  
Minority - Electrons

### Mass action Law

In semiconductors due to thermal effect, generation of free  $e^-$  and hole takes place.

Apart from the process of generation, recombination also occurs simultaneously, in which free  $e^-$  further recombine with hole.

At equilibrium rate of generation of charge carries is equal to rate of recombination of charge carrier. The recombination occurs due to  $e^-$  colliding with a hole, larger value of  $n_e$  or  $n_h$ , higher is the probability of their recombination.

Hence for a given semiconductor rate of recombination  $\propto n_e \times n_h$

so rate of recombination =  $R n_e \times n_h$   $R$  = recombination coefficient,

The value of  $R$  remains constant for a solid, according to the law of thermodynamics until crystalline lattice structure remains same.

For intrinsic semiconductor  $n_e = n_h = n_i$

so rate of recombination =  $R n_i^2$

$$R n_e \times n_h = R n_i^2 \Rightarrow n_i^2 = n_e \times n_h$$

Under thermal equilibrium, the product of the concentration ' $n_e$ ' of free electrons and the concentration  $n_h$  of holes is a constant and it is independent of the amount of doping by acceptor and donor impurities.

Thus from mass action law  $n_e \times n_h = n_i^2$

### Electron-hole Recombination :

It is necessarily to complete a band that electron is shared from neighbouring atoms or it may also be received from conduction band. In the second case electron recombines with the hole of valence band. This process is known as electron-hole recombination.

The breaking of bands or generation of electron-hole pairs, and completion of bands due to recombination is taking place continuously.

At equilibrium, the rate of generation becomes equal to the rate of recombination, giving a fixed number of free electrons and holes.

## RESISTIVITY AND CONDUCTIVITY OF SEMICONDUCTOR

### Conduction in conductor

Relation between current ( $I$ ) and drift velocity ( $v_d$ )

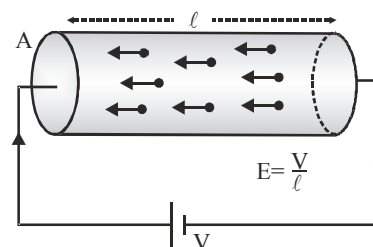
$$I = neAv_d \quad \begin{array}{l} n = \text{number of electron in unit volume} \\ A = \text{cross sectional area} \end{array}$$

$$\text{current density } J = \frac{I}{A} \text{ amp/m}^2 = ne v_d$$

$$J = ne \mu E$$

$$\text{Conductivity } \sigma = ne\mu = 1/\rho$$

$$\text{Mobility } \mu = \frac{v_d}{E}$$



$$\text{drift velocity of electron } v_d = \mu E$$

$$J = \sigma E$$

$$\rho = \text{Resistivity}$$

## Conduction in Semiconductor

### Intrinsic semiconductor

$$n_e = n_h$$

$$J = ne [v_e + v_h]$$

$$\sigma = \frac{1}{\rho} = en [\mu_e + \mu_h]$$

### P - type

$$n_h \gg n_e$$

$$J \cong e n_h v_h$$

$$\sigma = \frac{1}{\rho} \cong e n_h \mu_h$$

### N - type

$$n_e \gg n_h$$

$$J \cong e n_e v_e$$

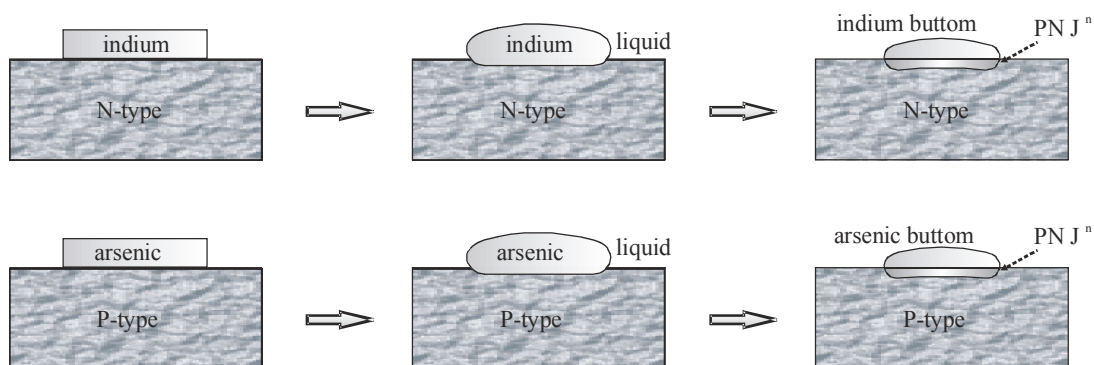
$$\sigma = \frac{1}{\rho} \cong e n_e \mu_e$$

## P - N JUNCTION

### Techniques for making P-N junction

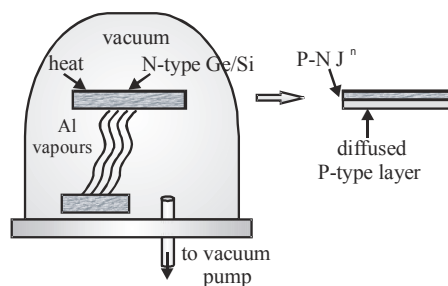
#### (i) Alloy Method or Alloy Junction

Here a small piece of III group impurity like indium is placed over n-Ge or n-Si and melted as shown in figure ultimately P - N junction form.



#### (ii) Diffusion Junction :-

A heated P-type semiconductor is kept in pentavalent impurity vapours which diffuse into P-type semiconductor as shown and make P-N junction.



#### (iii) Vapour deposited junction or epitaxial junction :

If we want to grow a layer of n-Si or p-Si then p-Si wafer is kept in an atmosphere of Silane (a silicon compound which dissociates into Si at high temperatures) plus phosphorous vapours. On cracking of silane at high temperature a fresh layer on n-Si grows on p-Si giving the "P-N junction". Since this junction growth is layer by layer so it is also referred as layer growth or epitaxial junction formation of P-N junction.

### Description of P-N Junction without applied voltage or bias

Given diagram shows a P-N junction immediately after it is formed.

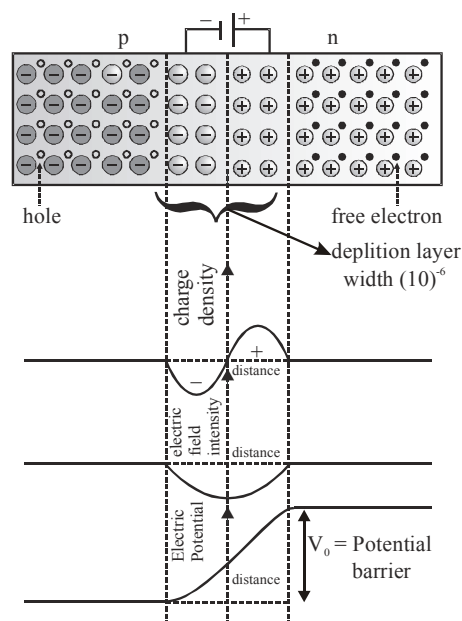
P region has mobile majority holes and immobile negatively charged impurity ions.

N region has mobile majority free electrons and immobile positively charged impurity ions.

Due to concentration difference diffusion of holes starts from P to N side and diffusion of  $e^-$ s starts N to P side.

Due to this a layer of only positive (in N side) and negative (in P-side) started to form which generate an electric field (N to P side) which oppose diffusion process, during diffusion magnitude of electric field increases due to this diffusion it gradually decreased and ultimately stop.

The layer of immobile positive and negative ions, which have no free electrons and holes called as **depletion layer** as shown in diagram.



- Width of depletion layer  $\cong 10^{-6}$  m**

(a) As doping increases depletion layer decreases

(b) As temperature is increased depletion layer also increases.

(c) P-N junction  $\rightarrow$  nonohmic, due to nonlinear relation between I and V.

- Potential Barrier or contact potential**

Ge  $\longrightarrow$  0.3 V

Si  $\longrightarrow$  0.7 V

- Electric field, produce due to potential barrier  $E = \frac{V}{d} = \frac{0.5}{10^{-6}} \Rightarrow E \cong 10^5$  V/m

This field prevents the respective majority carrier from crossing barrier region

### DIFFUSION AND DRIFT CURRENT

(1) Diffusion current – P to N side

(2) Drift current – N to P side

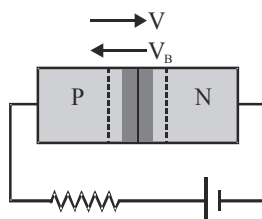
If there is no biasing diffusion current = drift current

So total current is zero

## BEHAVIOUR OF P-N JUNCTION WITH AN EXTERNAL VOLTAGE APPLIED OR BIAS

### Forward Bias

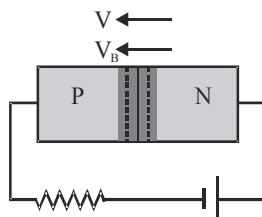
If we apply a voltage " $V$ " such that P-side is positive and N-side is negative as shown in diagram.



The applied voltage is opposite to the junction barrier potential. Due to this effective potential barrier decreases, junction width also decreases, so more majority carriers will be allowed to flow across junction. It means the current flow is principally due to majority charge carriers and it is in the order of mA called as forward Bias.

### Reverse Bias

If we apply a voltage " $V$ " such that P-side is negative and N-side is positive as shown in diagram. The applied voltage is in same direction as the junction barrier potential. Due to this effective potential barrier increases, junction width also increases, so no majority carriers will be allowed to flow across junction.



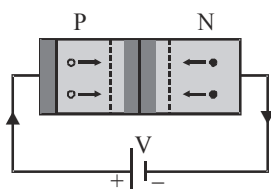
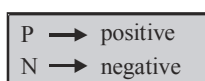
Only minority carriers will drift. It means the current flow is principally due to minority charge carriers and is very small (in the order of  $\mu A$ ). This bias is called as reversed Bias.

- In reverse bias, the current is very small and nearly constant with bias (termed as reverse saturation current). However interesting behaviour results in some special cases if the reverse bias is increased further beyond a certain limit, above particular high voltage breakdown of depletion layer started.
- Breakdown of a diode is of following two types :
  - (i) Zener breakdown
  - (ii) Avalanche breakdown

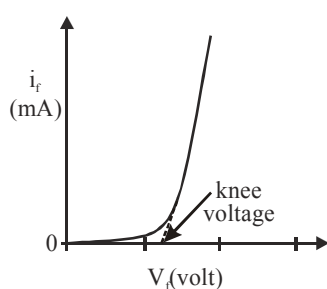


### Comparison between Forward Bias and Reverse Bias

#### Forward Bias



1. Potential Barrier reduces
2. Width of depletion layer decreases
3. P-N jn. provide very small resistance
4. Forward current flows in the circuit
5. Order of forward current is milli ampere.
6. Current flows mainly due to majority carriers.
7. Forward characteristic curves.



8. Forward resistance

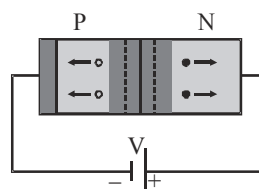
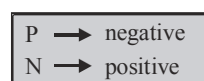
$$R_f = \frac{\Delta V_f}{\Delta I_f} \cong 100 \Omega$$

9. Order of knee or cut in voltage

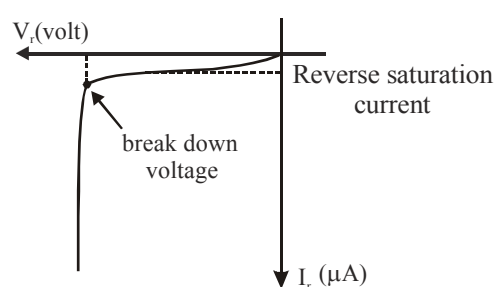
|    |   |       |
|----|---|-------|
| Ge | → | 0.3 V |
| Si | → | 0.7 V |

Special point : Generally  $\frac{R_r}{R_f} = 10^3 : 1$  for Ge

#### Reverse Bias



1. Potential Barrier increases.
2. Width of depletion layer increases.
3. P-N jn. provide high resistance
4. Very small current flows.
5. Order of current is micro ampere for Ge or Nano ampere for Si.
6. Current flows mainly due to minority carriers.
7. Reverse characteristic curve



8. Reverse resistance

$$R_r = \frac{\Delta V_r}{\Delta I_r} \cong 10^6 \Omega$$

9. Breakdown voltage

|    |   |      |
|----|---|------|
| Ge | → | 25 V |
| Si | → | 35 V |

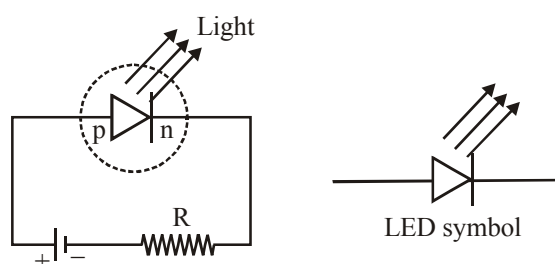
$$\frac{R_r}{R_f} = 10^4 : 1 \text{ for Si}$$

### Light Emitting Diode (LED):

A light emitting diode is simply a forward biased p-n junction which emits spontaneous light radiation. When forward bias is applied, the electron and holes at the junction recombine and energy released is emitted in the form of light. For visible radiation phosphorus doped GaAs is commonly used. The advantages of LEDs are:

- (i) Low operational voltage and less power.
- (ii) Fast action with no warm up time.
- (iii) Emitted light is nearly monochromatic.
- (iv) They have long life.

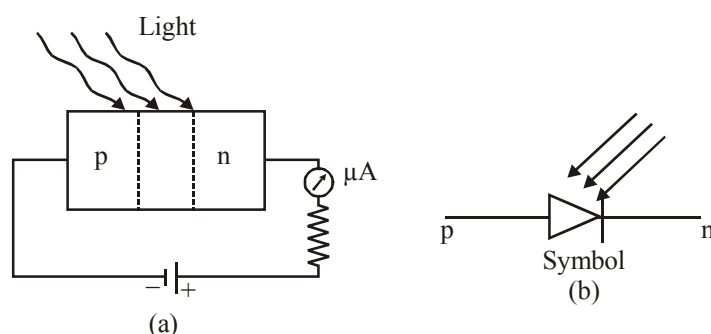
I-V characteristics of LED are similar to that of Si junction diode but the threshold voltages are much higher and slightly different for each colour. The reverse breakdown voltages of LED's are very low, about 5 V.



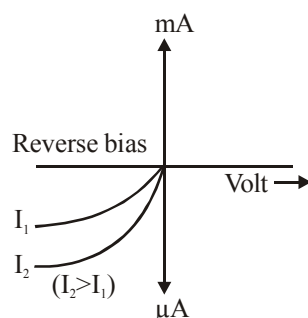
### Photodiode:

**It is a reversed-biased p-n junction, illuminated by radiation.** When p-n junction is reversed biased with no current, a very small reverse saturated current flows across the junction called the dark current. When the junction is illuminated with light, electron-hole pairs are created at the junction, due to which additional current begins to flow across the junction; the current is solely due to minority charge carriers.

- (1) A photodiode is used in reverse bias, although in forward bias current is more than current in reverse bias because in reverse bias it is easier to observe change in current with change in light intensity.
- (2) Photodiode is used to measure light intensity because reverse current increases with increase of intensity of light.

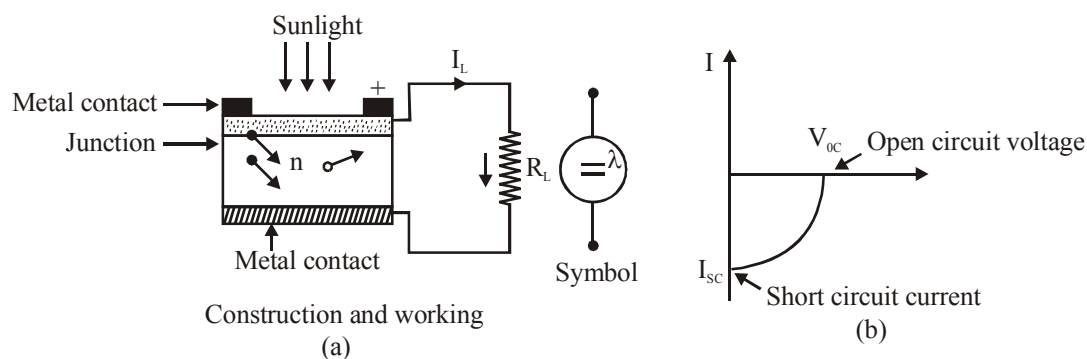


The characteristic curves of a photodiode for two different illuminations  $I_1$  and  $I_2$  ( $I_2 > I_1$ ) are shown in figure.



### Solar Cell

A solar cell is a junction diode which converts light energy into electrical energy. A p-n junction solar cell consists of a large junction with no external biasing.



The surface layer of p-region is made very thin so that the incident photons may easily penetrate to reach the junction which is the active region. In an operation in the photovoltaic mode (i.e., generation of voltage due to bombardment of optical photons); the materials suitable for photocells are silicon (Si), gallium arsenide (GaAs), cadmium sulphide (CdS) and cadmium selenide (CdSe).

#### Working:

When photons of energy greater than band gap energy ( $h\nu > E_g$ ) are made incident on the junction, electron-hole pairs are created which move in opposite directions due to junction field. These are collected at two sides of junction, thus producing photo-voltage; this gives rise to photocurrent. The characteristic curve of solar cell is shown in fig. Solar cells are used in satellites to recharge their batteries.

### 11. REVERSE BREAKDOWN

If the reverse bias voltage is made too high, the current through the PN junction increases rapidly at  $V_z$ . The voltage at which this happens is called **breakdown voltage** or **Zener voltage**.

There two mechanism which causes this breakdown. One is called avalanche breakdown and other is called **Zener breakdown**.

**Zener breakdown:** When reverse bias is increased the electric field at then junction also increases. At some stage the electric field becomes so high that it breaks the covalent bonds creating electron, hole pairs, thus a large number of carriers are generated. This causes a large current to flow. This machanism is know as Zener breakdown.

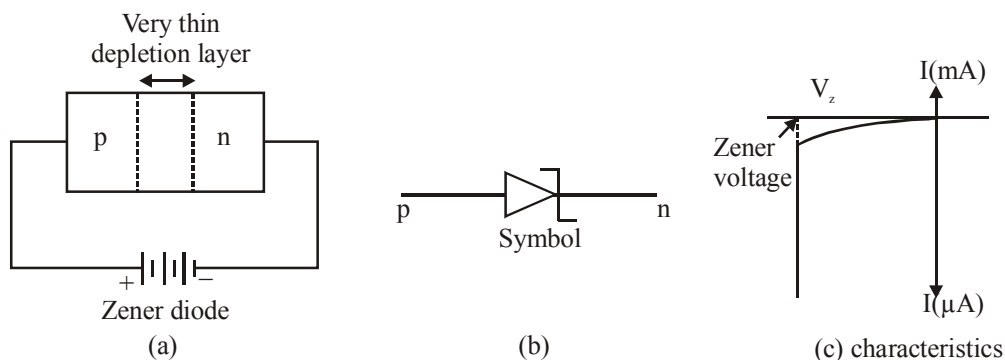
**Avalanche breakdown :** At high reverse voltage, due to high electric field, the rniority charge carriers, while crossing the junction acquires very high velocities. These by collision breaks down the covalent bonds, generating more carriers. A chain reaction is established, giving rise to high current. This mechanism is called **avalanche breakdown..**

### Zener Diode:

A zener diode is a specially designed heavily doped p-n junction, having a very thin depletion layer and having a very sharp breakdown voltage. It is always operated in breakdown region. Its breakdown voltage  $V_z$  is less than 6 V.

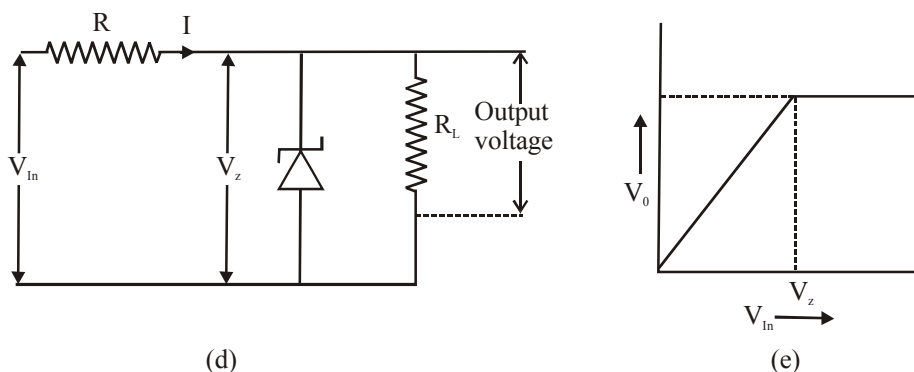
### Zener Diode as a Voltage Regulator:

Zener diode may be used as a voltage regulator. The circuit of zener-diode is shown in figure.



In breakdown region the equation:  $V_0 = V_z = V_{in} - RI$

Clearly, when the input voltage exceeds zener voltage to keep the voltage regularity, the extra input voltage appears across series resistance  $R$ . The voltage regulation curve is shown in figure.



**Zener Break down**

Where covalent bonds of depletion layer, its self break, due to high electric field of very high Reverse bias voltage.  
very-very high reverse bias voltage.

This phenomena predominant

- (i) At lower voltage after "break down"
- (ii) In P – N having "High doping"
- (iii) P – N Jn. having thin depletion layer

Here P – N not damage paramanently

"In D.C voltage stablizer zener phenomenan is used".

**Avalanche Break down**

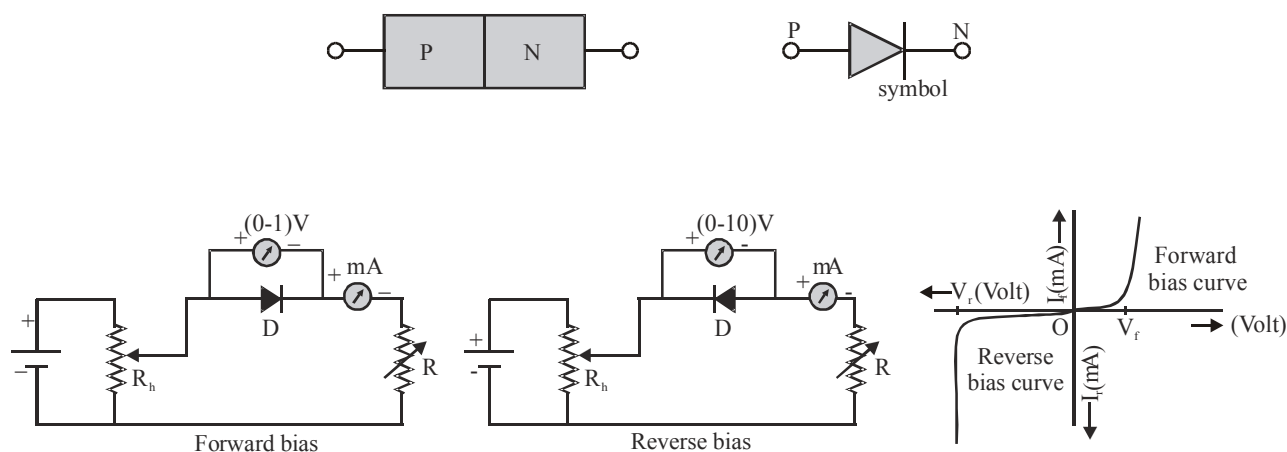
Here covalent bonds of depletion layers are broken by collision of "Minorities" which aquire high kinetic energy from high electric field of

This phenomena predominant

- (i) At high voltage after breakdown
- (ii) In P – N having "Low doping"
- (iii) P – N Jn. having thick depletion layer

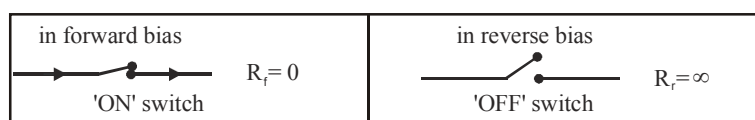
Here P – N damage paramanently due to

"Heating effect" due to abruptly increment of minorities during repeatative collisoins.

**CHARACTERISTIC CURVE OF P-N JUNCTION DIODE**

In forward bias when voltage is increased from 0V is steps and corresponding value of current is measured, the curve comes as OB of figure. We may note that current increase very sharply after a certain voltage knee voltage. At this voltage, barrier potential is completely eliminated and diode offers a low resistance.

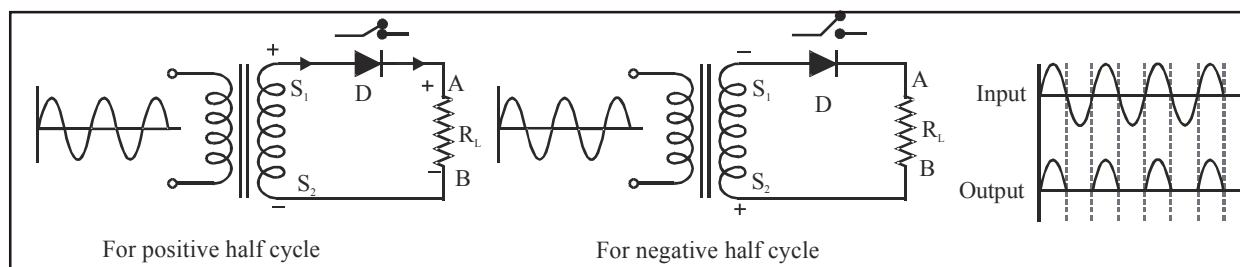
In reverse bias a microammeter has been used as current is very very small. When reverse voltage is increased from 0V and corresponding values of current measured the plot comes as OCD. We may note that reverse current is almost constant hence called reverse saturation current. It implies that diode resistance is very high. As reverse voltage reaches value  $V_B$ , called breakdown voltage, current increases very sharply.

**For Ideal Diode**

## RECTIFIER

It is device which is used for converting alternating current into direct current.

### Half wave rectifier



During the first half (positive) of the input signal, let  $S_1$  is at positive and  $S_2$  is at negative potential. So, the PN junction diode  $D$  is forward biased. The current flows through the load resistance  $R_L$  and output voltage is obtained.

During the second half (negative) of the input signal,  $S_1$  and  $S_2$  would be negative and positive respectively. The PN junction diode will be reversed biased. In this case, practically no current would flow through the load resistance. So, there will be no output voltage.

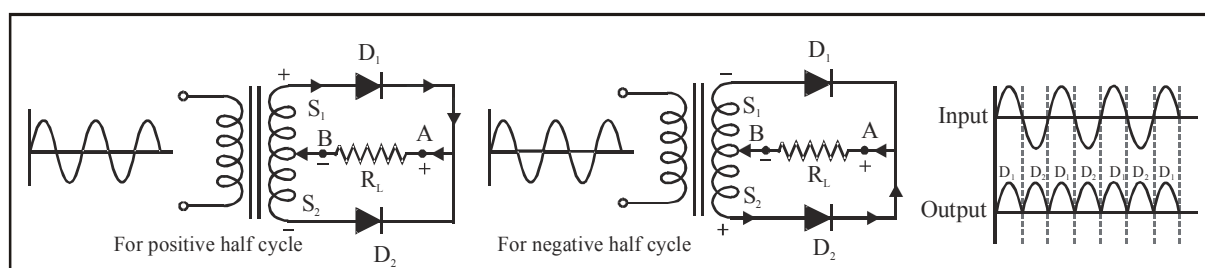
Thus, corresponding to an alternating input signal, we get a unidirectional pulsating output as shown.

### Peak inverse voltage (PIV)

In half wave rectifier  $PIV = \text{maximum voltage across secondary coil of transformer } (V_s)$   
 $= \text{Peak value of output } (V_m)$

### Full wave rectifier

When the diode rectifies the whole of the AC wave, it is called full wave rectifier. Figure shows the experimental arrangement for using diode as full wave rectifier. The alternating signal is fed to the primary a transformer. The output signal appears across the load resistance  $R_L$ .



During the positive half of the input signal :

Let  $S_1$  positive and  $S_2$  negative.

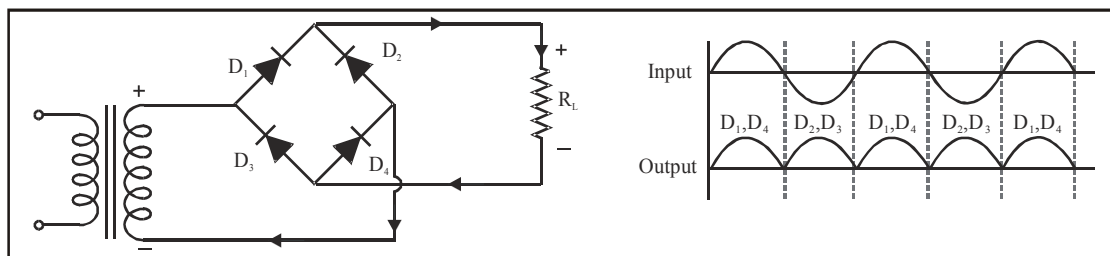
In this case diode  $D_1$  is forward biased and  $D_2$  is reverse biased. So only  $D_1$  conducts and hence the flow of current in the load resistance  $R_L$  is from A to B.

During the negative half of the input signal :

Now  $S_1$  is negative and  $S_2$  is positive. So  $D_1$  is reverse-biased and  $D_2$  is forward biased. So only  $D_2$  conducts and hence the current flows through the load resistance  $R_L$  from A to B.

It is clear that whether the input signal is positive or negative, the current always flows through the load resistance in the same direction and full wave rectification is obtained.

### Bridge Rectifier



## TRANSISTOR

Inventor William Bradford Shockley, John Bardeen and Walter Houser Brattain.

Transistor is a three terminal device which transfers a signal from low resistance circuit to high resistance circuit.

It is formed when a thin layer of one type of extrinsic semiconductor (P or N type) is sandwiched between two thick layers of other type of extrinsic semiconductor.

Each transistor have three terminals which are :-

- (i) Emitter
- (ii) Base
- (iii) Collector

### Emitter

It is the left most part of the transistor. It emits the majority carrier towards base. It is highly doped and medium in size.

### Base

It is the middle part of transistor which is sandwiched by emitter (E) and collector (C). It is lightly doped and very thin in size.

### Collector

It is right part of the transistor which collect the majority carriers emitted by emitter. It has large size and moderate doping.

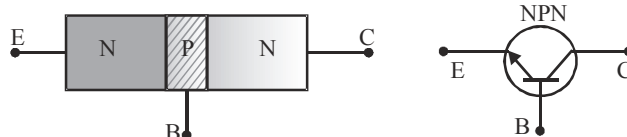
### There are two semiconducting PN-junctions in a transistor

- (i) The junction between emitter and base is known as emitter-base junction ( $J_{EB}$ ).
- (ii) The junction between base and collector is known as base-collector junction ( $J_{CB}$ ).

## TRANSISTOR ARE OF TWO TYPES

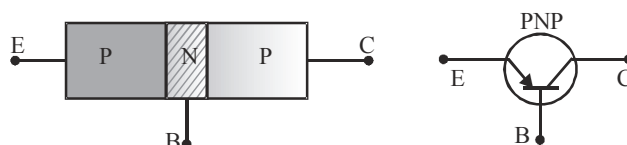
### N-P-N Transistor

If a thin layer of P-type semiconductor is sandwiched between two thick layers of N-type semiconductor is known as NPN transistor.

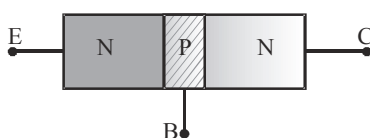


### P-N-P Transistor

If a thin layer of N-type of semiconductor is sandwiched between two thick layer of P-type semiconductor is known as PNP transistor.



- Transistor have two P-N Junction  $J_{EB}$  and  $J_{CB}$ , therefore it can be biased in four following ways as given below:



| Emitter-Base   | Collector-Base | Region of working |
|----------------|----------------|-------------------|
| Forward biased | Reverse biased | Active            |
| Reverse biased | Forward biased | Inverse Active    |
| Reverse biased | Reverse biased | Cut off           |
| Forward biased | Forward biased | Saturation        |

- Comparison between E, B and C

|           |               |               |
|-----------|---------------|---------------|
| Emitter   | Medium size   | High doping   |
| Base      | Smallest size | Low doping    |
| Collector | Largest size  | Medium doping |

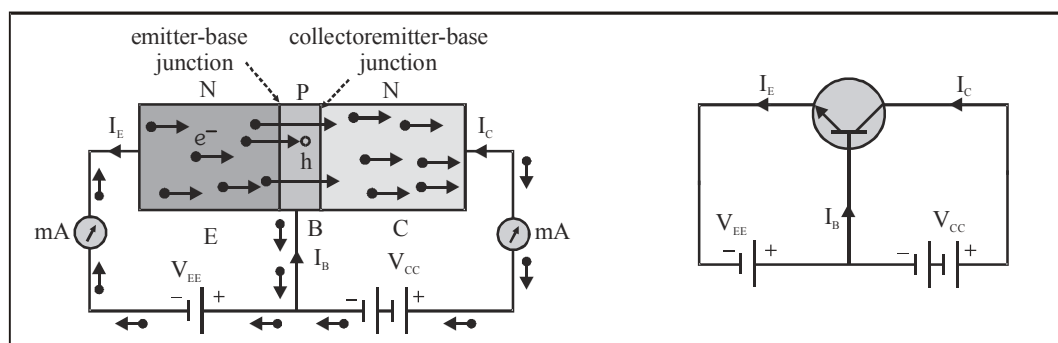
- The collector region is made physically larger than the emitter. Because collector has to dissipate much greater power.
- Transistor all mostly work in active region in electronic devices & transistor work as an amplifier in Active region only.
- Transistor i.e. It is a short form of two words "Transfer resistors". Signal is introduced at low resistance circuit and out put is taken at high resistance circuit.



- Base is lightly doped. Otherwise the most of the charge carrier from the emitter recombine in base region and not reaches at collector.
- Transistor is a current operated device i.e. the action of transistor is controlled by the motion of charge carriers. i.e. current.

### WORKING OF NPN TRANSISTOR

The emitter Base junction is forward bias and collector base junction is reversed biased of n-p-n transistor in circuit (A) and symbolic representation is shown in Figure.



When emitter base junction is forward bias, electrons (majority carriers) in emitter are repelled toward base.

The barrier of emitter base junction is reduced and the electron enter the base, about 5% of these electron recombine with hole in base region result in small current ( $I_b$ ).

The remaining electron ( $\approx 95\%$ ) enter the collector region because they are attracted towards the positive terminal of battery.

For each electron entering the positive terminal of the battery is connected with collector base junction an electron from negative terminal of the battery connected with emitter base junction enters the region.

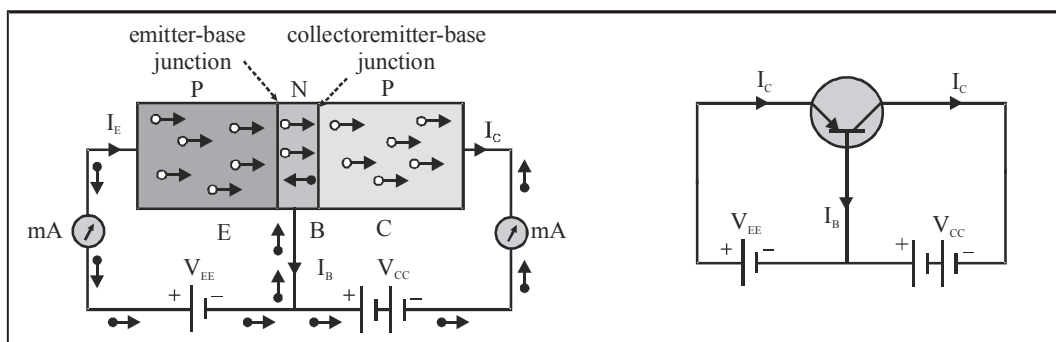
The emitter current ( $I_e$ ) is more than the collector ( $I_c$ ).

The base current is the difference between  $I_e$  and  $I_c$  and proportional to the number of electron hole recombination in the base.

$$I_e = I_b + I_c$$

### WORKING OF PNP TRANSISTOR

When emitter-base junction is forward biased holes (majority carriers) in the emitter are repelled towards the base and diffuse through the emitter base junction. The barrier potential of emitter-base junction decreases and hole enter the n-region (i.e. base). A small number of holes ( $\approx 5\%$ ) combine with electron of base-region resulting small current ( $I_b$ ). The remaining hole ( $\approx 95\%$ ) enter into the collector region because they are attracted towards negative terminal of the battery connected with the collector-base junction. These hole constitute the collector current ( $I_c$ ).



As one hole reaches the collector, it is neutralized by the battery. As soon as one electron and a hole is neutralized in collector a covalent bond is broken in emitter region. The electron hole pair is produced. The released electron enter the positive terminal of battery and hole move towards the collector.

### Basic Transistor Circuit Configurations :-

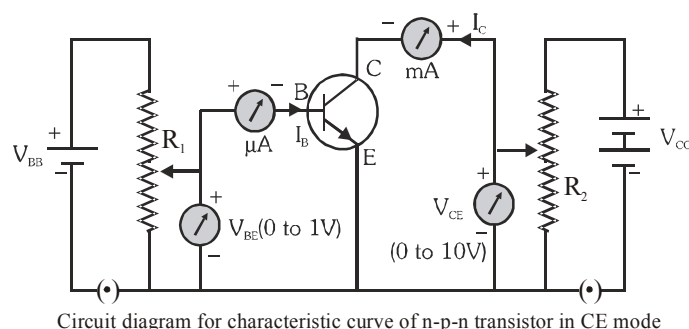
To study about the characteristics of transistor we have to make a circuit [In which four terminals are required]. But the transistor have three terminals, so one of the terminal of transistor is made common in input and output both. Thus, we have three possible configuration of transistor circuit.

- (i) Common base      (ii) Common emitter      (iii) Common collector

In these three common emitter is widely used and common collector is rarely used.

### Common emitter characteristics of a transistor

#### Circuit Diagram :

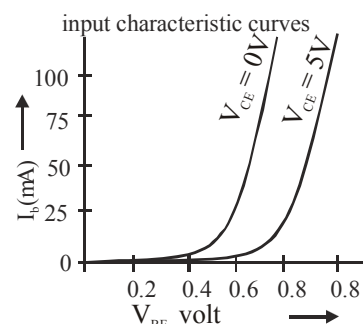


Circuit diagram for characteristic curve of n-p-n transistor in CE mode

### Input characteristics

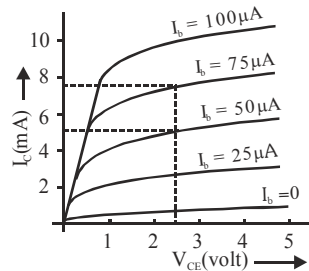
The variation of base current ( $I_b$ ) (input) with base emitter voltage ( $V_{EB}$ ) at constant-emitter voltage ( $V_{CE}$ ) is called input characteristic.

- Keep the collector-emitter voltage ( $V_{CE}$ ) constant (say  $V_{CE} = 1V$ )
- Now change emitter base voltage by  $R_1$  and note the corresponding value of base current ( $I_b$ ).
- Plot the graph between  $V_{EB}$  and  $I_b$ .
- A set of such curves can be plotted at different ( $V_{CE} = 2V$ )



### Output characteristics

The variation of collector current  $I_c$  (output) with collector-emitter voltage ( $V_{CE}$ ) at constant base current ( $I_b$ ) is called output characteristic.



- Keep the base current ( $I_b$ ) constant (say  $I_b = 10 \mu A$ )
- Now change the collector-emitter voltage ( $V_{CE}$ ) using variable resistance  $R_2$  and note the corresponding values of collector current ( $I_c$ ).
- Plot the graph between ( $V_{CE}$  versus  $I_c$ )
- A set of such curves can be plotted at different fixed values of base current (say 0, 20  $\mu A$ , 30  $\mu A$  etc.)

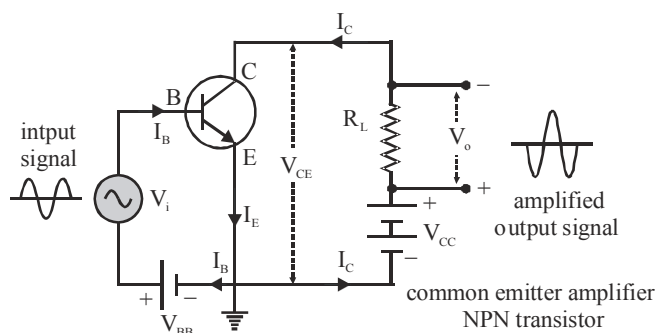
### TRANSISTOR AS AN AMPLIFIER

The process of increasing the amplitude of input signal without distorting its wave shape and without changing its frequency is known as amplification.

A device which increases the amplitude of the input signal is called amplifier.

A transistor can be used as an amplifier in active state.

A basic circuit of a common emitter transistor amplifier is shown.



# LOGIC GATES

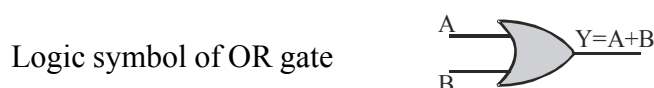
## INTRODUCTION :

- A logic gate is a digital circuit which is based on certain logical relationship between the input and the output voltages of the circuit.
- The logic gates are built using the semiconductor diodes and transistors.
- Each logic gate is represented by its characteristic symbol.
- The operation of a logic gate is indicated in a table, known as truth table. This table contains all possible combinations of inputs and the corresponding outputs.
- A logic gate is also represented by a Boolean algebraic expression. Boolean algebra is a method of writing logical equations showing how an output depends upon the combination of inputs. Boolean algebra was invented by George Boole.

## BASIC LOGIC GATES

There are three basic logic gates. They are (1) OR gate(2) AND gate, and (3) NOT gate

- **The OR gate :-** The output of an OR gate attains the state 1 if one or more inputs attain the state 1.

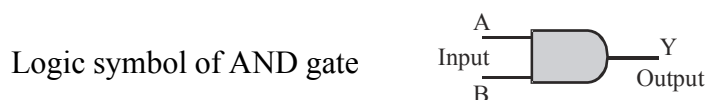


The Boolean expression of OR gate is  $Y = A + B$ , read as Y equals A 'OR' B.

Truth table of a two-input OR gate

| A | B | Y |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 1 |

- **The AND gate :-** The output of an AND gate attains the state 1 if and only if all the inputs are in state 1.



The Boolean expression of AND gate is  $Y = A.B$

It is read as Y equals A 'AND' B

Truth table of a two-input AND gate

| A | B | Y |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 0 |
| 1 | 0 | 0 |
| 1 | 1 | 1 |

- **The NOT gate :** The output of a NOT gate attains the state 1 if and only if the input does not attain the state 1.

Logic symbol of NOT gate 

The Boolean expression is  $Y = \bar{A}$ , read as Y equals NOT A.

Truth table of NOT gate

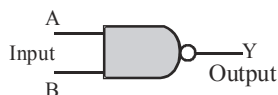
| A | Y |
|---|---|
| 0 | 1 |
| 1 | 0 |

### COMBINATION OF GATES :

The three basis gates (OR, AND and NOT) when connected in various combinations give us logic gates such as NAND, NOR gates, which are the universal building blocks of digital circuits.

- **The NAND gate :**

Logic symbol of NAND gate



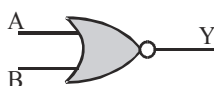
The Boolean expression of NAND gate is  $Y = \overline{A \cdot B}$

Truth table of a NAND gate

| A | B | Y |
|---|---|---|
| 0 | 0 | 1 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 0 |

- **The NOR gate :**

Logic symbol of NOR gate



The Boolean expression of NOR gate is  $Y = \overline{A + B}$

Truth table of a NOR gate

| A | B | Y |
|---|---|---|
| 0 | 0 | 1 |
| 0 | 1 | 0 |
| 1 | 0 | 0 |
| 1 | 1 | 0 |

## UNIVERSAL GATES :

The NAND or NOR gate is the universal building block of all digital circuits. Repeated use of NAND gates (or NOR gates) gives other gates. Therefore, any digital system can be achieved entirely from NAND or NOR gates. We shall show how the repeated use of NAND (and NOR) gates will give us different gates.

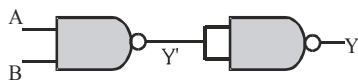
- **The NOT gate from a NAND gate :-** When all the inputs of a NAND gate are connected together, as shown in the figure, we obtain a NOT gate



Truth table of a single input NAND gate

| A | B = (A) | Y |
|---|---------|---|
| 0 | 0       | 1 |
| 1 | 1       | 0 |

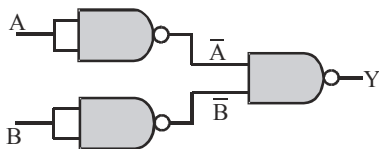
- **The AND gate from a NAND gates :-** If a NAND gate is followed by a NOT gate (i.e., a single input NAND gate), the resulting circuit is an AND gate as shown in figure and truth table given show how an AND gate has been obtained from NAND gates.



Truth table

| A | B | Y' | Y |
|---|---|----|---|
| 0 | 0 | 1  | 0 |
| 0 | 1 | 1  | 0 |
| 1 | 0 | 1  | 0 |
| 1 | 1 | 0  | 1 |

- **The OR gate from NAND gates :-** If we invert the inputs A and B and then apply them to the NAND gate, the resulting circuit is an OR gate.



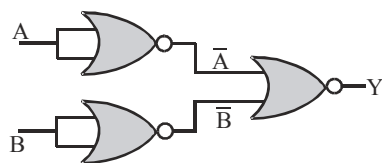
Truth table

| A | B | $\bar{A}$ | $\bar{B}$ | Y |
|---|---|-----------|-----------|---|
| 0 | 0 | 1         | 1         | 0 |
| 0 | 1 | 1         | 0         | 1 |
| 1 | 0 | 0         | 1         | 1 |
| 1 | 1 | 0         | 0         | 1 |

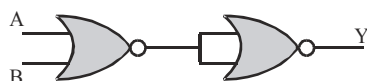
- **The NOT gate from NOR gates :-** When all the inputs of a NOR gate are connected together as shown in the figure, we obtain a NOT gate



- The AND gate from NOR gates :- If we invert the inputs A and B and then apply them to the NOR gate, the resulting circuit is an AND gate.



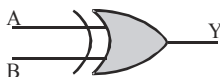
- The OR gate from NOR gate :-** If a NOR gate is followed by a single input NOR gate (NOT gate), the resulting circuit is an OR gate.



### XOR AND XNOR GATES :

- The Exclusive - OR gate (XOR gate):-** The output of a two-input XOR gate attains the state 1 if one and only one input attains the state 1.

Logic symbol of XOR gate



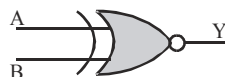
The Boolean expression of XOR gate is  $Y = A.\bar{B} + \bar{A}.B$  or  $Y = A \oplus B$

Truth table of a XOR gate

| A | B | Y |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 0 |

- Exclusive - NOR gate (XNOR gate):-** The output is in state 1 when its both inputs are the same that is, both 0 or both 1.

Logic symbol of XNOR gate



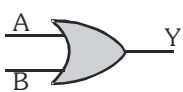
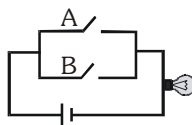
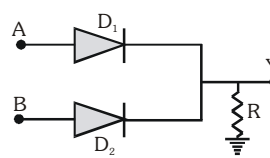
The Boolean expression of XNOR gate is  $Y = A.B + \bar{A}.\bar{B}$  or  $Y = \overline{A \oplus B}$  or  $A \odot B$

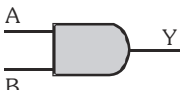
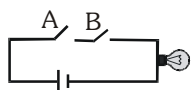
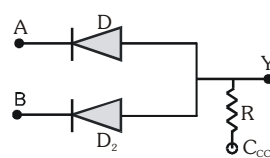
Truth table of a XNOR gate

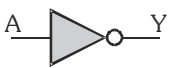
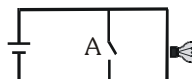
| A | B | Y |
|---|---|---|
| 0 | 0 | 1 |
| 0 | 1 | 0 |
| 1 | 0 | 0 |
| 1 | 1 | 1 |

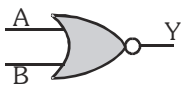
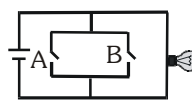
## SUMMARY OF LOGIC GATES

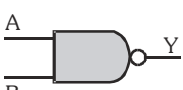
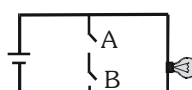
| Names | Symbol | Boolean Expression | Truth table | Electrical analogue | Circuit diagram (Practical Realisation) |
|-------|--------|--------------------|-------------|---------------------|-----------------------------------------|
|-------|--------|--------------------|-------------|---------------------|-----------------------------------------|

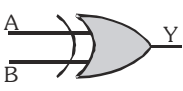
| Names | Symbol                                                                            | Boolean Expression | Truth table                                                                                                                                                                                                                         | Electrical analogue | Circuit diagram (Practical Realisation) |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                    |                                                                                     |
|-------|-----------------------------------------------------------------------------------|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|-----------------------------------------|---|---|---|---|---|---|---|---|---|---|---|---|---|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| OR    |  | $Y = A + B$        | <table border="1"> <tr><th>A</th><th>B</th><th>Y</th></tr> <tr><td>0</td><td>0</td><td>0</td></tr> <tr><td>0</td><td>1</td><td>1</td></tr> <tr><td>1</td><td>0</td><td>1</td></tr> <tr><td>1</td><td>1</td><td>1</td></tr> </table> | A                   | B                                       | Y | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 0 | 1 | 1 | 1 | 1 |  |  |
| A     | B                                                                                 | Y                  |                                                                                                                                                                                                                                     |                     |                                         |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                    |                                                                                     |
| 0     | 0                                                                                 | 0                  |                                                                                                                                                                                                                                     |                     |                                         |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                    |                                                                                     |
| 0     | 1                                                                                 | 1                  |                                                                                                                                                                                                                                     |                     |                                         |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                    |                                                                                     |
| 1     | 0                                                                                 | 1                  |                                                                                                                                                                                                                                     |                     |                                         |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                    |                                                                                     |
| 1     | 1                                                                                 | 1                  |                                                                                                                                                                                                                                     |                     |                                         |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                    |                                                                                     |

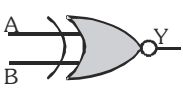
| Names | Symbol                                                                            | Boolean Expression | Truth table                                                                                                                                                                                                                         | Electrical analogue | Circuit diagram (Practical Realisation) |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                    |                                                                                     |
|-------|-----------------------------------------------------------------------------------|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|-----------------------------------------|---|---|---|---|---|---|---|---|---|---|---|---|---|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| AND   |  | $Y = A \cdot B$    | <table border="1"> <tr><th>A</th><th>B</th><th>Y</th></tr> <tr><td>0</td><td>0</td><td>0</td></tr> <tr><td>0</td><td>1</td><td>0</td></tr> <tr><td>1</td><td>0</td><td>0</td></tr> <tr><td>1</td><td>1</td><td>1</td></tr> </table> | A                   | B                                       | Y | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 1 | 1 | 1 |  |  |
| A     | B                                                                                 | Y                  |                                                                                                                                                                                                                                     |                     |                                         |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                    |                                                                                     |
| 0     | 0                                                                                 | 0                  |                                                                                                                                                                                                                                     |                     |                                         |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                    |                                                                                     |
| 0     | 1                                                                                 | 0                  |                                                                                                                                                                                                                                     |                     |                                         |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                    |                                                                                     |
| 1     | 0                                                                                 | 0                  |                                                                                                                                                                                                                                     |                     |                                         |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                    |                                                                                     |
| 1     | 1                                                                                 | 1                  |                                                                                                                                                                                                                                     |                     |                                         |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                    |                                                                                     |

| Names           | Symbol                                                                            | Boolean Expression | Truth table                                                                                                           | Electrical analogue |   |   |   |   |   |                                                                                    |
|-----------------|-----------------------------------------------------------------------------------|--------------------|-----------------------------------------------------------------------------------------------------------------------|---------------------|---|---|---|---|---|------------------------------------------------------------------------------------|
| NOT<br>Inverter |  | $Y = \bar{A}$      | <table border="1"> <tr><th>A</th><th>Y</th></tr> <tr><td>0</td><td>1</td></tr> <tr><td>1</td><td>0</td></tr> </table> | A                   | Y | 0 | 1 | 1 | 0 |  |
| A               | Y                                                                                 |                    |                                                                                                                       |                     |   |   |   |   |   |                                                                                    |
| 0               | 1                                                                                 |                    |                                                                                                                       |                     |   |   |   |   |   |                                                                                    |
| 1               | 0                                                                                 |                    |                                                                                                                       |                     |   |   |   |   |   |                                                                                    |

| Names               | Symbol                                                                              | Boolean Expression     | Truth table                                                                                                                                                                                                                         | Electrical analogue |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                      |
|---------------------|-------------------------------------------------------------------------------------|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|---|---|---|---|---|---|---|---|---|---|---|---|---|---|--------------------------------------------------------------------------------------|
| NOR<br>(OR<br>+NOT) |  | $Y = \overline{A + B}$ | <table border="1"> <tr><th>A</th><th>B</th><th>Y</th></tr> <tr><td>0</td><td>0</td><td>1</td></tr> <tr><td>0</td><td>1</td><td>0</td></tr> <tr><td>1</td><td>0</td><td>0</td></tr> <tr><td>1</td><td>1</td><td>0</td></tr> </table> | A                   | B | Y | 0 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 0 | 1 | 1 | 0 |  |
| A                   | B                                                                                   | Y                      |                                                                                                                                                                                                                                     |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                      |
| 0                   | 0                                                                                   | 1                      |                                                                                                                                                                                                                                     |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                      |
| 0                   | 1                                                                                   | 0                      |                                                                                                                                                                                                                                     |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                      |
| 1                   | 0                                                                                   | 0                      |                                                                                                                                                                                                                                     |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                      |
| 1                   | 1                                                                                   | 0                      |                                                                                                                                                                                                                                     |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                      |

| Names             | Symbol                                                                              | Boolean Expression         | Truth table                                                                                                                                                                                                                         | Electrical analogue |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                      |
|-------------------|-------------------------------------------------------------------------------------|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|---|---|---|---|---|---|---|---|---|---|---|---|---|---|--------------------------------------------------------------------------------------|
| NAND<br>(AND+NOT) |  | $Y = \overline{A \cdot B}$ | <table border="1"> <tr><th>A</th><th>B</th><th>Y</th></tr> <tr><td>0</td><td>0</td><td>1</td></tr> <tr><td>0</td><td>1</td><td>1</td></tr> <tr><td>1</td><td>0</td><td>1</td></tr> <tr><td>1</td><td>1</td><td>0</td></tr> </table> | A                   | B | Y | 0 | 0 | 1 | 0 | 1 | 1 | 1 | 0 | 1 | 1 | 1 | 0 |  |
| A                 | B                                                                                   | Y                          |                                                                                                                                                                                                                                     |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                      |
| 0                 | 0                                                                                   | 1                          |                                                                                                                                                                                                                                     |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                      |
| 0                 | 1                                                                                   | 1                          |                                                                                                                                                                                                                                     |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                      |
| 1                 | 0                                                                                   | 1                          |                                                                                                                                                                                                                                     |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                      |
| 1                 | 1                                                                                   | 0                          |                                                                                                                                                                                                                                     |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                      |

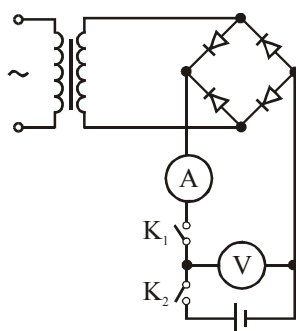
| Names                    | Symbol                                                                              | Boolean Expression                                                | Truth table                                                                                                                                                                                                                         |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|--------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| XOR<br>(Exclusive<br>OR) |  | $Y = A \oplus B$<br>or<br>$Y = \bar{A} \cdot B + A \cdot \bar{B}$ | <table border="1"> <tr><th>A</th><th>B</th><th>Y</th></tr> <tr><td>0</td><td>0</td><td>0</td></tr> <tr><td>0</td><td>1</td><td>1</td></tr> <tr><td>1</td><td>0</td><td>1</td></tr> <tr><td>1</td><td>1</td><td>0</td></tr> </table> | A | B | Y | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 0 | 1 | 1 | 1 | 0 |
| A                        | B                                                                                   | Y                                                                 |                                                                                                                                                                                                                                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 0                        | 0                                                                                   | 0                                                                 |                                                                                                                                                                                                                                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 0                        | 1                                                                                   | 1                                                                 |                                                                                                                                                                                                                                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 1                        | 0                                                                                   | 1                                                                 |                                                                                                                                                                                                                                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 1                        | 1                                                                                   | 0                                                                 |                                                                                                                                                                                                                                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |

| Names                      | Symbol                                                                              | Boolean Expression                                                                                    | Truth table                                                                                                                                                                                                                         |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|----------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| XNOR<br>(Exclusive<br>NOR) |  | $Y = A \odot B$<br>or<br>$Y = A \cdot B + \bar{A} \cdot \bar{B}$<br>or<br>$Y = \overline{A \oplus B}$ | <table border="1"> <tr><th>A</th><th>B</th><th>Y</th></tr> <tr><td>0</td><td>0</td><td>1</td></tr> <tr><td>0</td><td>1</td><td>0</td></tr> <tr><td>1</td><td>0</td><td>0</td></tr> <tr><td>1</td><td>1</td><td>1</td></tr> </table> | A | B | Y | 0 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 0 | 1 | 1 | 1 |
| A                          | B                                                                                   | Y                                                                                                     |                                                                                                                                                                                                                                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 0                          | 0                                                                                   | 1                                                                                                     |                                                                                                                                                                                                                                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 0                          | 1                                                                                   | 0                                                                                                     |                                                                                                                                                                                                                                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 1                          | 0                                                                                   | 0                                                                                                     |                                                                                                                                                                                                                                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 1                          | 1                                                                                   | 1                                                                                                     |                                                                                                                                                                                                                                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |



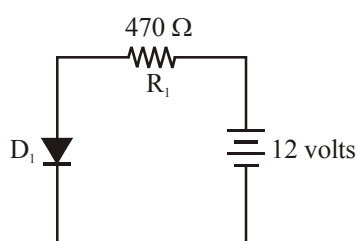
## EXERCISE-S

1. The battery is charged from full wave rectifier fed by a sinusoidal voltage (see figure). Ideal diodes, ammeter and voltmeter show the time-average value. At idle with only key  $K_1$  closed, voltmeter shows 12 V, and current is then absent, ie, reading of ammeter is 0. If only the key  $K_2$  is closed, the voltmeter shows battery voltage at 12.3 V. During charging, when the  $K_2$  and  $K_1$  are closed, the voltmeter shows 12.8 V and ammeter shows 5 A. Find the internal resistance of battery.



SC0001

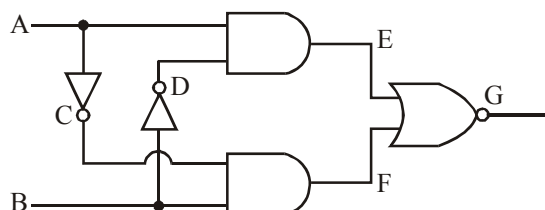
2. Complete the following table of values for this diode circuit, assuming a typical forward voltage drop of 0.65 volts for the diode:



|   | $R_1$        | $D_1$ | Total |
|---|--------------|-------|-------|
| V |              |       | 12 V  |
| I |              |       |       |
| R | 470 $\Omega$ |       |       |
| P |              |       |       |

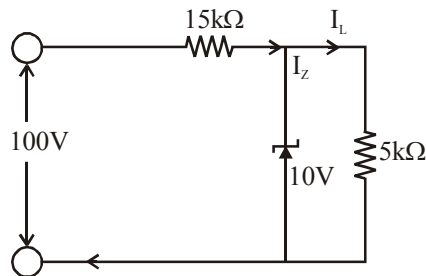
SC0002

3. Write a truth table for the circuit in figure, including the states at C, D, E, F and G.



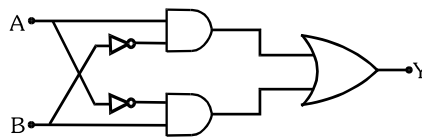
SC0003

4. For the circuit shown in figure, find  
 i) the output voltage  
 ii) the voltage drop across series resistance  
 iii) the current through zener diode.



SC0004

5. Write down the actual logic operation carried by the following circuit. Explain your answer.



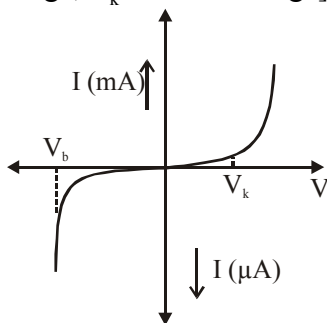
SC0005

## EXERCISE-O

## SINGLE CORRECT TYPE QUESTIONS

1. An electric field is applied to a semi-conductor. Let the number of charge carriers be  $n$  and the average drift speed be  $v$ . If the temperature is increased  
 (A) Both  $n$  and  $v$  will increase (B)  $n$  will increase but  $v$  will decrease  
 (C)  $v$  will increase but  $n$  will decrease (D) Both  $n$  and  $v$  will decrease  
 SC0006
2. The manifestation of band structure in solids is due to - [AIEEE-2004]  
 (A) Bohr's correspondence principle (B) Pauli's exclusion principle  
 (C) Heisenberg's uncertainty principle (D) Boltzmann's law  
 SC0007
3. P-type semiconductor is formed when  
 A. As impurity is mixed in Si B. Al impurity is mixed in Si  
 C. B impurity is mixed in Ge D. P impurity is mixed in Ge  
 (A) A and C (B) A and D (C) B and C (D) B and D  
 SC0008
4. In extrinsic semiconductors -  
 (A) The conduction band and valence band overlap  
 (B) The gap between conduction band and valence band is more than 16 eV  
 (C) The gap between conduction band and valence band is near about 1 eV  
 (D) The gap between conduction band and valence band will be 100 eV and more  
 SC0009
5. A strip of copper and another of germanium are cooled from room temperature to 80 K. The resistance of - [AIEEE-2004]  
 (A) copper strip increases and that of germanium decreases  
 (B) copper strip decreases and that of germanium increases  
 (C) each of these increases  
 (D) each of these decreases  
 SC0010
6. In a P-N junction diode not connected to any circuit -  
 (A) the potential is the same everywhere  
 (B) the P-type side is at a higher potential than the N-type side  
 (C) there is an electric field at the junction directed from the N-type side to the P-type side  
 (D) there is an electric field at the junction directed from the P-type to the N-type side  
 SC0011
7. When p-n junction diode is forward biased, then. [AIEEE-2004]  
 (A) both the depletion region and barrier height are reduced  
 (B) the depletion region is widened and barrier height is reduced  
 (C) the depletion region is reduced and barrier height is increased  
 (D) both the depletion region and barrier height are increased  
 SC0012

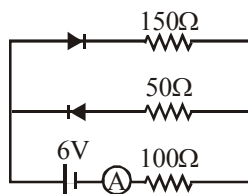
8. The V-I characteristic for a p-n junction diode is plotted as shown in the figure. From the plot we can conclude that [ $V_b \rightarrow$  breakdown voltage,  $V_k \rightarrow$  knee voltage]



- (A) the forward bias resistance of diode is very high; almost infinity for small values of  $V$  and after a certain value it becomes very low  
 (B) the reverse bias resistance of diode is very high in the beginning upto breakdown voltage is not achieved  
 (C) both forward and reverse bias resistances are same for all voltages  
 (D) both (A) and (B) are correct

SC0013

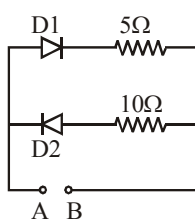
9. In the below given arrangement determine the ammeter reading, if each diodes have a forward resistance of  $50\ \Omega$  and infinite backward resistance -



- (A) zero (B) 0.02 (C) 0.03 (D) 0.036

SC0014

10. A 2V battery is connected across AB as shown in the figure. The value of the current supplied by the battery when in one case battery's positive terminal is connected to A and in other case when positive terminal of battery is connected to B will respectively be :-

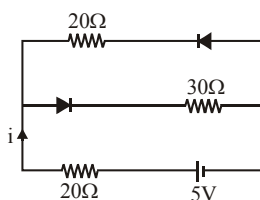


(JEE-Main Online-2015)

- (A) 0.1 A and 0.2 A (B) 0.4 A and 0.2 A (C) 0.2 A and 0.4 A (D) 0.2 A and 0.1 A

SC0015

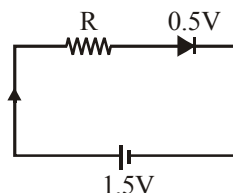
11. Current in the circuit will be -



- (A)  $\frac{5}{40}$  A (B)  $\frac{1}{10}$  A (C)  $\frac{5}{10}$  A (D)  $\frac{5}{20}$  A

SC0016

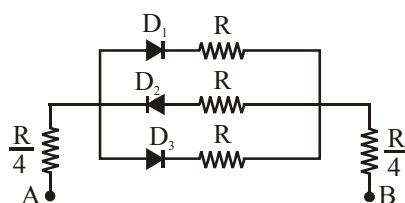
12. The diode used in the circuit shown in the figure has a constant voltage drop of  $0.5\text{ V}$  at all currents and a maximum power rating of  $100\text{ milliwatts}$ . What should be the value of the resistor  $R$  connected in series with the diode for obtaining maximum current -



- (A)  $1.5\ \Omega$  (B)  $5\ \Omega$  (C)  $6.67\ \Omega$  (D)  $200\ \Omega$

SC0017

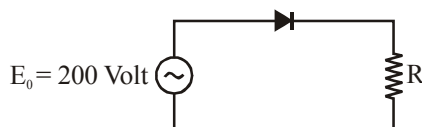
13. In the following circuits PN-junction diodes  $D_1$ ,  $D_2$  and  $D_3$  are ideal for the following potential of A and B, the correct increasing order of resistance between A and B will be -



- (i)  $-10\text{V}, -5\text{V}$ , (ii)  $-5\text{V}, -10\text{V}$  (iii)  $-4\text{V}, -12\text{V}$   
 (A) (i) < (ii) < (iii) (B) (iii) < (ii) < (i) (C) (ii) = (iii) < (i) (D) (i) = (iii) < (ii)

SC0018

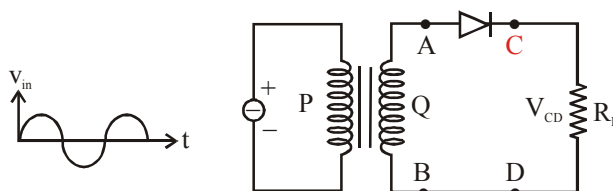
14. A sinusoidal voltage of peak value  $200\text{ volt}$  is connected to a diode and resistor  $R$  in the circuit shown so that half wave rectification occurs. If the forward resistance of the diode is negligible compared to  $R$  then rms voltage (in volt) across  $R$  is approximately -



- (A) 200 (B) 100 (C)  $\frac{200}{\sqrt{2}}$  (D) 280

SC0019

15. In the half-wave rectifier circuit shown. Which one of the following wave forms is true for  $V_{CD}$ , if the input is as shown?



- (A) (B) (C) (D)

SC0020

16. Zener diode is a p-n junction which has -  
 (A) p-end heavily doped, n-end lightly doped  
 (B) n-end heavily doped, p-end lightly doped  
 (C) both p and n-ends heavily doped  
 (D) both p and n-ends lightly doped

SC0021

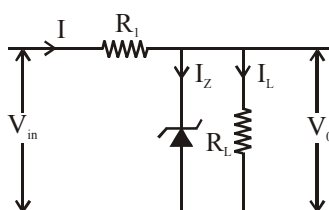
17. Zener diode has both p and n-ends heavily doped so that -  
 (A) it has small thickness of depletion region  
 (B) it has large thickness of depletion region due to large recombination  
 (C) it has large reverse bias voltage  
 (D) it has weak reverse current when reverse biased

SC0022

18. Most important use of zener diode is to have -  
 (A) constant voltage across applied load  
 (B) any desired current at constant voltage  
 (C) a p-n junction working under constant regulated voltage conditions  
 (D) a p-n junction to operate at high voltages

SC0023

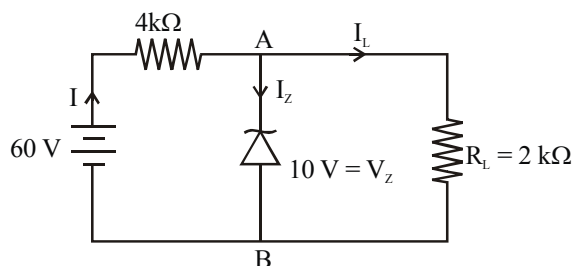
19. In given figure when input voltage increases,



- (A) the current through  $R_s$ ,  $R_L$  and zener increases  
 (B) the current through  $R_s$  increases, zener increases but through  $R_L$  remains constant  
 (C) the current through  $R_s$  increases, through zener decreases,  $R_L$  increases  
 (D) the current through  $R_s$  increases, through zener remains constant but  $R_L$  increases

SC0024

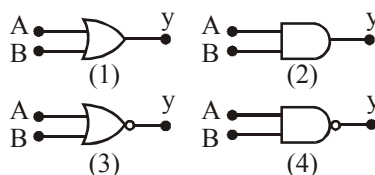
20. A Zener diode is connected to a battery and a load as shown below. The currents  $I$ ,  $I_Z$  and  $I_L$  are respectively.  
 (JEE-Main Online-2014)



- (A) 12.5 mA, 5 mA, 7.5 mA  
 (B) 15 mA, 7.5 mA, 7.5 mA  
 (C) 12.5 mA, 7.5 mA, 5 mA  
 (D) 15 mA, 5 mA, 10 mA

SC0025

21. A transistor is used in common emitter mode as an amplifier, then:  
 (1) the base emitter junction is forward biased  
 (2) the base emitter junction is reverse biased  
 (3) the input signal is connected in series with the voltage applied to bias the base emitter junction  
 (4) the Input signal is connected in series with the voltage applied to bias the base collector junction which is correct-  
 (A) 1, 2, 3 (B) 1, 2, 3, 4 (C) 1, 3 (D) 2, 3, 4
- SC0026**
22. In a n-p-n transistor circuit the collector current is 10 mA. If 90% of the electron emitted reach the collector then -  
 (A) the emitter current will be 9 mA (B) the base current will be 9 mA  
 (C) the emitter current will be 11 mA (D) the base current will be -1 mA
- SC0027**
23. n-p-n transistors are preferred to p-n-p transistors because -  
 (A) they have low cost  
 (B) they have low dissipation energy  
 (C) they are capable of handling large power  
 (D) electrons have high mobility than holes.
- SC0028**
24. When npn transistor is used as an amplifier  
 (A) electrons move from collector to base  
 (B) holes move from emitter to base .  
 (C) electrons move from base to collector  
 (D) holes move from base to emitter
- [AIEEE-2004]**
25. For a transistor,  $\alpha$  is 0.8. The transistor is in CE configuration. The change in the collector current when the base current changes by 6 mA is :-  
 (A) 6 mA (B) 4.8mA (C) 24mA (D) 8mA
- SC0029**
26. Given below are four logic gate symbol (figure). Those for OR, NOR and NAND are respectively



- (A) 1,4,3 (B) 4,1,2 (C) 1,3,4 (D) 4,2,1

27. The following truth table corresponds to the logic gate -

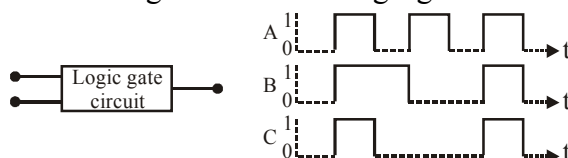
|   |   |   |   |   |
|---|---|---|---|---|
| A | 0 | 0 | 1 | 1 |
| B | 0 | 1 | 0 | 1 |
| X | 0 | 1 | 1 | 1 |

- (A) NAND (B) OR (C) AND (D) XOR

SC0031

SC0032

28. The following figure shows a logic gate circuit with two input A and B output C. The voltage waveforms of A, B and C are as shown in second figure below. The logic gate is :



- (A) OR gate (C) NAND gate (B) AND gate (D) NOR gate

SC0033

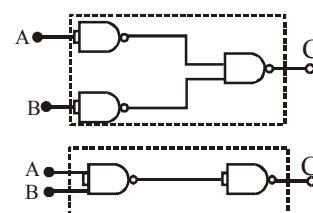
29. Which of the following gates will have an output of 1 -

- (A)  $\begin{matrix} 1 \\ 0 \end{matrix}$  (B)  $\begin{matrix} 0 \\ 1 \end{matrix}$  (C)  $\begin{matrix} 0 \\ 1 \end{matrix}$  (D)  $\begin{matrix} 0 \\ 1 \end{matrix}$

SC0034

30. The combination of 'NAND' gates shown here under (figure) are equivalent to -

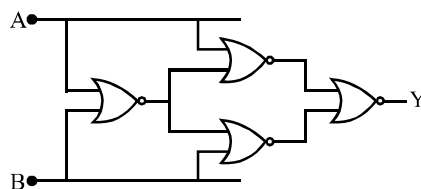
- (A) An OR gate and an AND gate respectively  
(B) An AND gate and a NOT gate respectively  
(C) An AND gate and an OR gate respectively  
(D) An OR gate and a NOT gate respectively



SC0035

31. A system of four gates is set up as shown. The 'truth table' corresponding to this system is :-

(JEE-Main Online-2013)



- (A) 

| A | B | Y |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 0 |
| 1 | 0 | 1 |
| 1 | 1 | 1 |

 (B) 

| A | B | Y |
|---|---|---|
| 0 | 0 | 1 |
| 0 | 1 | 1 |
| 1 | 0 | 0 |
| 1 | 1 | 0 |

 (C) 

| A | B | Y |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 0 |

 (D) 

| A | B | Y |
|---|---|---|
| 0 | 0 | 1 |
| 0 | 1 | 0 |
| 1 | 0 | 0 |
| 1 | 1 | 1 |

SC0036

32. Which of the following circuits correctly represents the following truth table ?

(JEE-Main Online-2013)

| A | B | C |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 0 |
| 1 | 0 | 1 |
| 1 | 1 | 0 |

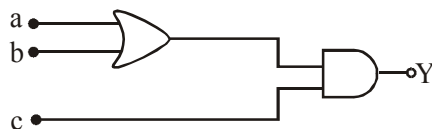
- (A) (B) (C) (D)

SC0037



33. To get an output of 1 from the circuit shown in figure the input must be :-

(JEE-Main Online-2016)



- (A)  $a = 1, b = 0, c = 1$  (B)  $a = 1, b = 0, c = 0$   
(C)  $a = 0, b = 1, c = 0$  (D)  $a = 0, b = 0, c = 1$

SC0038

34. For the given combination of gates, if the logic states of inputs A, B, C are as follows  $A = B = C = 0$  and  $A = B = 1, C = 0$  then the logic states of output D are -



- (A) 0, 0 (B) 0, 1 (C) 1, 0 (D) 1, 1

SC0039

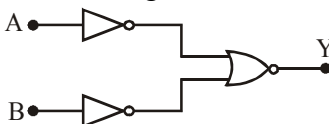
35. This symbol represents



- (A) NOT gate (B) OR (C) AND gate (D) NOR gate

SC0040

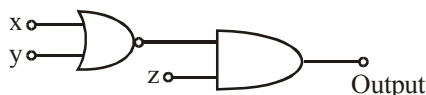
36. Which logic gate is represented by the following combination of logic gates -



- (A) OR (B) NAND (C) AND (D) NOR

SC0041

37. In which of the following cases, we would obtain an output of one.



- (A)  $x = 1, y = 1, z = 1$  (B)  $x = 1, y = 1, z = 0$  (C)  $x = 0, y = 1, z = 1$  (D)  $x = 0, y = 0, z = 1$

SC0042

38. In a full wave rectifier circuit operating from 50 Hz mains frequency, the fundamental frequency in the ripple would be :-

[AIEEE-2005]

- (A) 50 Hz (B) 25 Hz (C) 100 Hz (D) 70.7 Hz

SC0043

39. The electrical conductivity of a semiconductor increases when electromagnetic radiation of wavelength shorter than 2480 nm is incident on it. The band gap (in eV) for the semiconductor is

[AIEEE-2005]

- (A) 1.1 eV (B) 2.5 eV (C) 0.5 eV (D) 0.7 eV

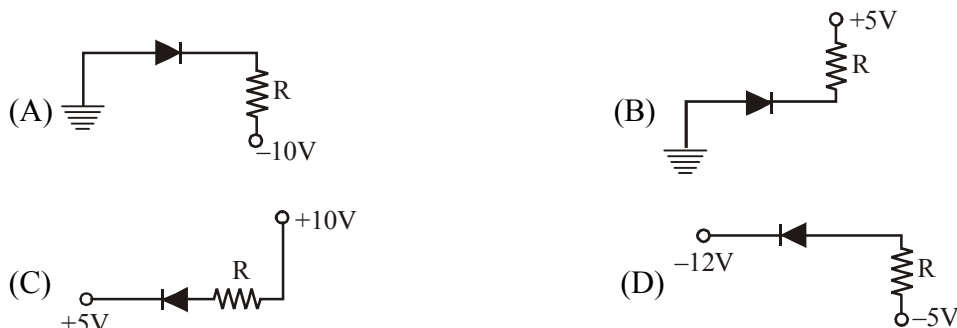
SC0044

40. If the ratio of the concentration of electrons to that of holes in a semiconductor is  $\frac{7}{5}$  and the ratio of currents is  $\frac{7}{4}$  then what is the ratio of their drift velocities - [AIEEE-2006]

(A)  $\frac{5}{4}$  (B)  $\frac{4}{7}$  (C)  $\frac{5}{6}$  (D)  $\frac{4}{5}$

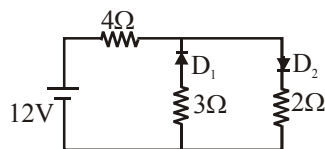
SC0045

41. In the following, which one of the diodes is reverse biased ; [AIEEE-2006]



SC0046

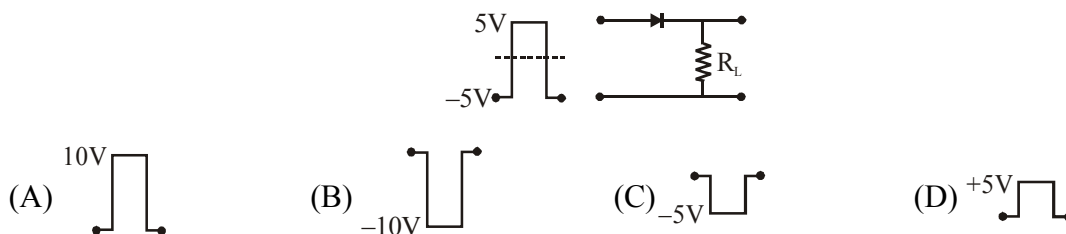
42. The circuit has two oppositely connected ideal diodes in parallel. What is the current flowing in the circuit? [AIEEE-2006]



(A) 2.31 A (B) 1.33 A (C) 1.71 A (D) 2.00 A

SC0047

43. If in a p-n junction diode, a square input signal of 10V is applied as shown. Then the output signal across  $R_L$  will be :- [AIEEE-2007]



SC0048

44. Carbon, silicon and germanium have four valence electrons each. At room temperature which one of the following statements is most appropriate? [AIEEE-2007]

(A) The number of free conduction electrons is significant in C but small in Si and Ge  
 (B) The number of free conduction electrons is negligibly small in all the three  
 (C) The number of free electrons for conduction is significant in all the three  
 (D) The number of free electrons for conduction is significant only in Si and Ge but small in C

SC0049

## MULTIPLE CORRECT TYPE QUESTIONS

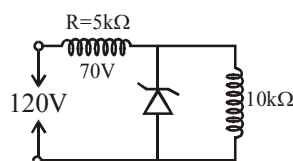
45. Which of the following statements is correct  
 (A) Resistance of semiconductor decreases with increase in temperature  
 (B) In an electric field, displacement of holes is opposite to the displacement of electrons  
 (C) Resistance of a conductor decreases with the increase in temperature  
 (D) n-type semiconductors are neutral

SC0050

46. Pick out the correct statement the reverse current in p-n junction diode  
 (A) can be minimum and constant  
 (B) remains constant even after the breakdown voltage  
 (C) becomes infinity at breakdown  
 (D) reverse current is controlled by external resistance

SC0051

47. For the circuit shown in the figure:



- (A) current through zener diode is 4 mA  
 (B) current through zener diode is 9 mA  
 (C) the output voltage is 50 V  
 (D) the output voltage is 40 V

SC0052

48. Which of the following devices are heavily doped p-n junction  
 (A) Photo diode  
 (B) Light emitting diode  
 (C) Solar cell  
 (D) Zener mode

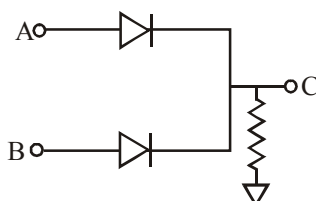
SC0053

49. Which of the following statements is correct for proper working of zener diode ?  
 (A) Reverse bias voltage should be less than or equal to zener breakdown voltage  
 (B) Reverse bias voltage applied must be greater than zener breakdown voltage.  
 (C) Zener is to be reverse biased for zener action  
 (D) For given zener diode there can be different zener breakdown voltages

SC0054

## EXERCISE-JM

1. In the circuit below, A and B represent two inputs and C represents the output. [AIEEE-2008]

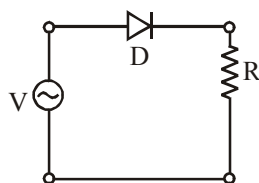


The circuit represents

- (1) AND gate (2) NAND gate (3) OR gate (4) NOR gate

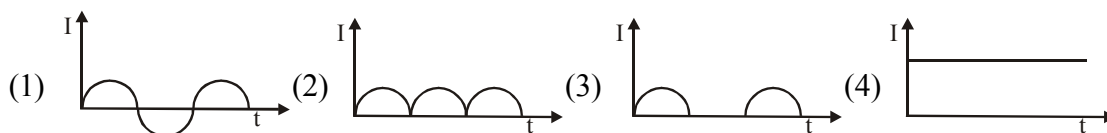
SC0055

2. An p-n junction (D) shown in the figure can act as a rectifier. An alternating current source (V) is connected in the circuit.



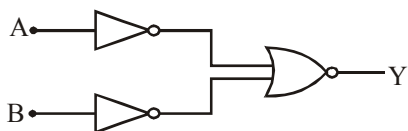
The current (I) in the resistor R can be shown by:

[AIEEE-2009]



SC0056

3. The logic circuit shown below has the input wave forms 'A' and 'B' as shown. Pick out the correct output waveform. [AIEEE-2009]



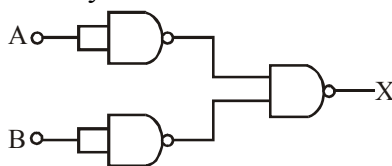
Output is -



SC0057

4. The combination of gates shown below yields.

[AIEEE-2010]

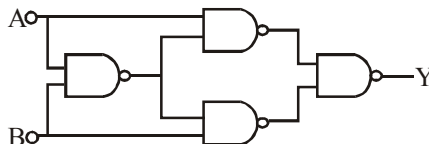


- (1) XOR gate      (2) NAND gate      (3) OR gate      (4) NOT gate

SC0058

5. Truth table for system of four NAND gates as shown in figure is:

[AIEEE-2012]



(1)

| A | B | Y |
|---|---|---|
| 0 | 0 | 1 |
| 0 | 1 | 1 |
| 1 | 0 | 0 |
| 1 | 1 | 0 |

(2)

| A | B | Y |
|---|---|---|
| 0 | 0 | 1 |
| 0 | 1 | 0 |
| 1 | 0 | 0 |
| 1 | 1 | 1 |

(3)

| A | B | Y |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 0 |

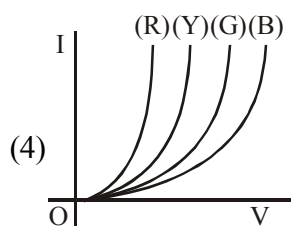
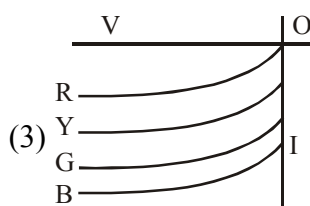
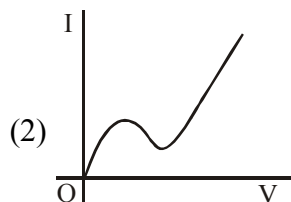
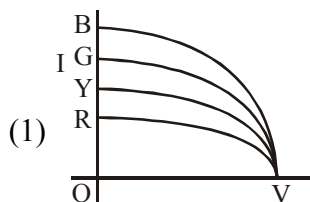
(4)

| A | B | Y |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 0 |
| 1 | 0 | 1 |
| 1 | 1 | 1 |

SC0059

6. The I-V characteristic of an LED is

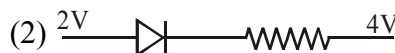
[JEE Main 2013]



SC0060

7. The forward biased diode connection is:

[JEE Main-2014]



SC0061

8. For a common emitter configuration, if  $\alpha$  and  $\beta$  have their usual meanings, the **incorrect** relationship between  $\alpha$  and  $\beta$  is

[JEE Main-2016]

(1)  $\alpha = \frac{\beta^2}{1+\beta^2}$

(2)  $\frac{1}{\alpha} = \frac{1}{\beta} + 1$

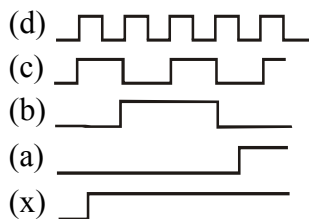
(3)  $\alpha = \frac{\beta}{1-\beta}$

(4)  $\alpha = \frac{\beta}{1+\beta}$

SC0062

9. If a, b, c, d are inputs to a gate and x is its output, then as per the following time graph, the gate is

[JEE Main-2016]



- (1) NAND (2) NOT (3) AND (4) OR

SC0063

10. The temperature dependence of resistances of Cu and undoped Si in the temperature range 300-400K, is best described by :-

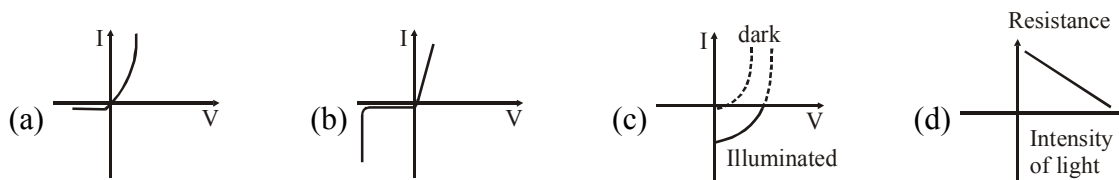
[JEE Main-2016]

- (1) Linear decrease for Cu, linear decrease for Si.  
(2) Linear increase for Cu, linear increase for Si.  
(3) Linear increase for Cu, exponential increase for Si  
(4) Linear increase for Cu, exponential decrease for Si

SC0064

11. Identify the semiconductor devices whose characteristics are given below, in the order (a), (b), (c), (d):-

[JEE Main-2016]



- (1) Zener diode, Solar cell, Simple diode, Light dependent resistance  
(2) Simple diode, Zener diode, Solar cell, Light dependent resistance  
(3) Zener diode, Simple diode, Light dependent resistance, Solar cell  
(4) Solar cell, Light dependent resistance, Zener diode, Simple diode

SC0065

12. In a common emitter amplifier circuit using an n-p-n transistor, the phase difference between the input and the output voltages will be :

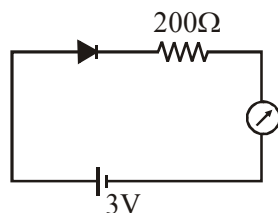
[JEE Main-2017]

- (1)  $135^\circ$  (2)  $180^\circ$  (3)  $45^\circ$  (4)  $90^\circ$

SC0066

13. The reading of the ammeter for a silicon diode in the given circuit is :-

[JEE Main-2018]

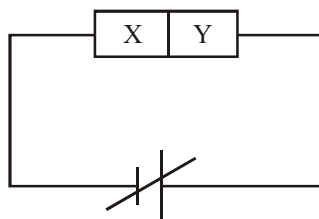


- (1) 15 mA (2) 11.5 mA (3) 13.5 mA (4) 0

SC0067

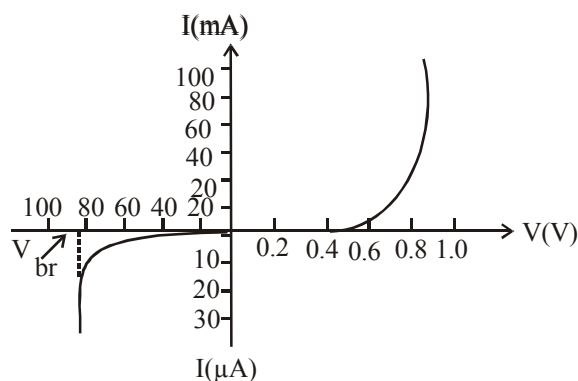
### CBSE Previous Year's Questions

1. Draw the voltage current characteristic of a zener diode. [1; CBSE-2004]
2. With the help of a labelled circuit diagram, explain how an n-p-n transistor can be used as an amplifier in common-emitter configuration. Explain how the input and output voltages are out of phase by  $180^\circ$  for a common-emitter transistor amplifier. [5; CBSE-2004]
3. How does the resistivity of (i) a conductor and (ii) a semiconductor vary with temperature? Give reason for each case. [2; CBSE-2005]
4. On the basis of the energy band diagrams. Distinguish between metals, insulators and semiconductors. [3; CBSE-2005]
5. (a) With the help of a circuit diagram explain the working of transistor as oscillator.  
(b) Draw a circuit diagram for a two input OR gate and explain its working with the help of input, output waveforms. [5; CBSE-2005]
6. (a) Explain briefly with the help of a circuit diagram, how V - I characteristics of a p-n junction diode are obtained in (i) forward bias, and (ii) reverse bias.  
(b) A photodiode is fabricated from a semiconductor with a band gap of 2.8 eV. Can it detect wave length of 6000 nm? Justify. [5; CBSE-2005]
7. Explain (i) forward biasing, (ii) reverse biasing of a P-N junction diode. With the help of a circuit diagram, explain the use of this device as a half - wave rectifier. [3; CBSE-2006]
8. What are energy bands? How are these formed? Distinguish between a conductor, an insulator and a semiconductor on the basis of energy band diagram. [5; CBSE-2006]
9. Explain the function of base region of a transistor. Why is this region made thin and lightly doped? Draw a circuit diagram to study the input and output characteristics of n-p-n transistor in a common emitter (CE) configuration. Show these characteristics graphically. Explain how current amplification factor of the transistor is calculated using output characteristics. [5; CBSE-2006]
10. Two semiconductor materials X and Y shown in the given figure, are made by doping germanium crystal with indium and arsenic respectively. The two are joined end to end and connected to a battery as shown. [2; CBSE-2007]

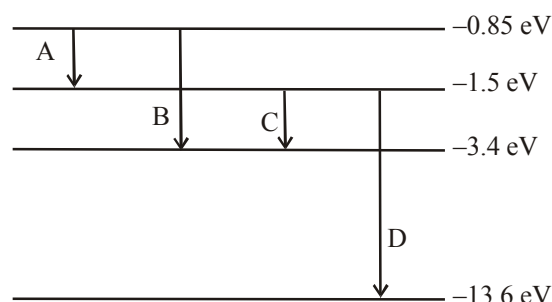


- (i) Will the junction be forward biased or reverse biased ?
  - (ii) Sketch a V-I graph for this arrangement.
11. Draw the circuit diagram of a common emitter amplifier using n-p-n transistor. What is the phase difference between the input signal and output voltage ? State two reasons why a common emitter amplifier is preferred to a common base amplifier. [3; CBSE-2007]

12. Explain the formation of energy band in solids. Draw energy band diagram for (i) a conductor, (ii) an intrinsic semiconductor. [3; CBSE-2007]
13. Write the acronym LASER in expanded form. State any four reasons for preferring diode lasers as light sources for optical communication links. [3; CBSE-2007]
14. Distinguish between an intrinsic semiconductor and P-type semiconductor. Give reason, why, a P-type semiconductor crystal is electrically neutral, although  $n_h \gg n_e$ ? [2; CBSE-2008]
15. The figure below shows the V-I characteristic of a semiconductor diode.



- (i) Identify the semiconductor diode used.
- (ii) Draw the circuit diagram to obtain the given characteristic of this device.
- (iii) Briefly explain how this diode can be used as a voltage regulator. [3; CBSE-2008]
16. The energy level diagram of an element is given below. Identify, by doing necessary calculations, which transition corresponds to emission of a spectral line of wavelength 102.7 nm.

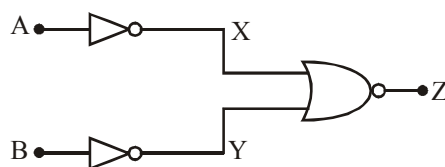


[3; CBSE-2008]

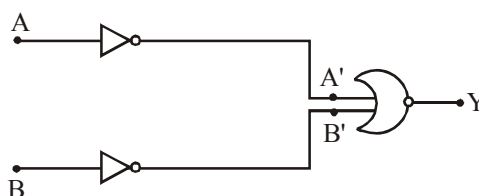
17. (i) Draw a circuit diagram to study the input and output characteristics of an n-p-n transistor in its common emitter configuration. Draw the typical input and output characteristics.  
(ii) Explain, with the help of a circuit diagram, the working of n -p-n transistor as a common emitter amplifier. [5; CBSE-2009]
18. How is a zener diode fabricated so as to make it a special purpose diode? Draw I -V characteristics of zener diode and explain the significance of break down voltage. Explain briefly, with the help of a circuit diagram, how a-p-n junction diode work [5; CBSE-2009]



19. (a) Draw the circuit diagram of a p-n junction diode in (i) forward bias, (ii) reverse bias. How are these circuits used to study the V-I characteristics of a silicon diode? Draw the typical V-I characteristics.
- (b) What is a light emitting diode (LED)? Mention two important advantages of LEDs over conventional lamps. [5; CBSE-2010]
20. (a) Draw the circuit arrangement for studying the input and output characteristics of an n-p-n transistor in CE configuration. With the help of these characteristics define (i) input resistance, (ii) current amplification factor.
- (b) Describe briefly with the help of a circuit diagram how an n-p-n transistor is used to produce self-sustained oscillations. [5; CBSE-2010]
21. What happens to the width of depletion layer of a p-n junction when it is (i) forward biased, (ii) reverse biased? [1; CBSE-2011]
22. Draw a labelled diagram of a full wave rectifier circuit. State its working principle. Show the input-output waveforms. [3; CBSE-2011]
23. You are given a circuit below. Write its truth table, hence, identify the logic operation carried out by this circuit. Draw the logic symbol of the gate it corresponds to. [3; CBSE-2011]



24. Describe briefly with the help of a circuit diagram, how the flow of current carriers in a p-n-p transistor is regulated with emitter-base junction forward and base-collector junction reverse biased. [2; CBSE-2012]
25. (a) Describe briefly, with the help of a diagram, the role of the two important processes involved in the formation of a p-n junction
- (b) Name the device which is used as a voltage regulator. Draw the necessary circuit diagram and explain its working

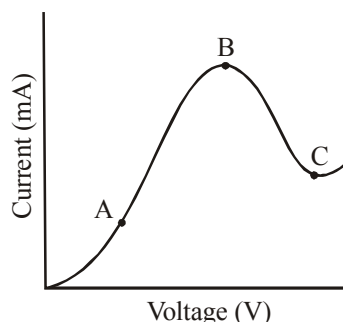


OR

- (a) Explain briefly the principle on which a transistor-amplifier works as an oscillator. Draw the necessary circuit diagram and explain its working.
- (b) Identify the equivalent gate for the following circuit and write its truth table. [5; CBSE-2012]

26. The graph shown in the figure represents a plot of current versus voltage for a given semiconductor. Identify the region, if any, over which the semiconductor has a negative resistance.

[3; CBSE-2013]



27. Draw typical output characteristics of an n-p-n transistor in CE configuration. Show how these characteristics can be used to determine output resistance. [5; CBSE-2013]

OR

28. Draw  $V-I$  characteristics of a p-n junction diode. Answer the following questions, giving reasons:  
(i) Why is the current under reverse bias almost independent of the applied potential upto a critical voltage?  
(ii) Why does the reverse current show a sudden increase at the critical voltage?

Name any semiconductor device which operates under the reverse bias in the breakdown region.

[5; CBSE-2013]

29. Draw a circuit diagram of n-p-n transistor amplifier in CE configuration. Under what condition does the transistor acts as an amplifier? [2; CBSE-2014]

30. Explain, with the help of a circuit diagram, the working of a p-n junction diode as a half-wave rectifier. [2; CBSE-2014]

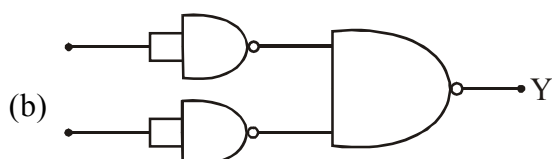
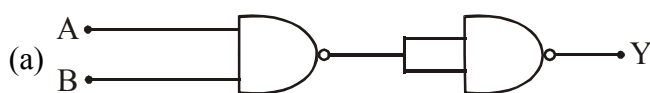
31. Write any two distinguishing features between conductors, semiconductors and insulators on the basis of energy band diagrams. [3; CBSE-2014]

32. With the help of a circuit diagram, explain the working of a junction diode as a full wave rectifier. Draw its input and output waveforms. Which characteristic property makes the junction diode suitable for rectification? [3; CBSE-2015]

33. The outputs of two NOT gates are fed to a NOR gate. Draw the logic circuit of the combination of gates. Write its truth table. Identify the gate equivalent to this circuit. [3; CBSE-2015]

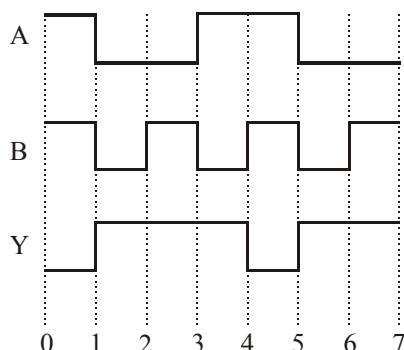
OR

You are given two circuits (a) and (b) as shown in the figures, which consist of NAND gates. Identify the logic operation carried out by the two. Write the truth tables for each. Identify the gates equivalent to the two circuits.

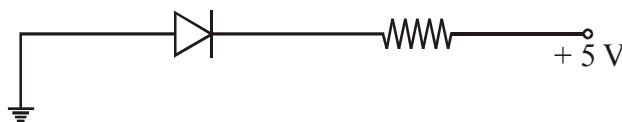


[3; CBSE-2015]

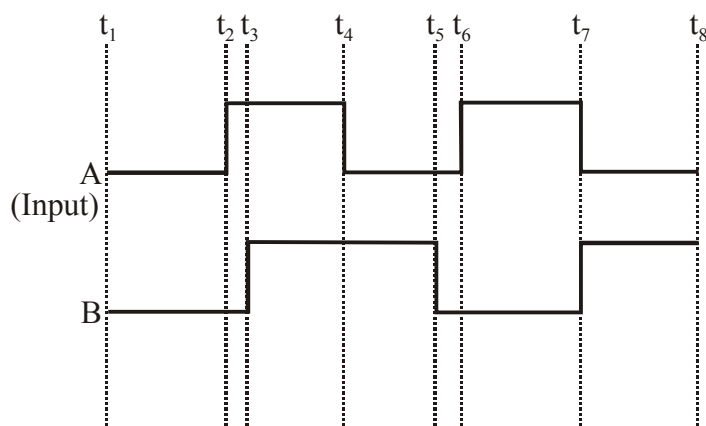
34. (i) Name two important processes that occur during the formation of a pn junction.  
 (ii) Draw the circuit diagram of a full wave rectifier along with the input and output waveforms. Briefly explain how the output voltage/current is unidirectional. [CBSE-2016]
35. (i) Distinguish between a conductor and a semi conductor on the basis of energy band diagram.  
 (ii) The following figure shows the input waveforms (A, B) and the output waveform (Y) of a gate. Identify the gate, write its truth table and draw its logic symbol. [CBSE-2016]



36. (a) In the following diagram, is the junction diode forward biased or reverse biased ? [CBSE-2017]



- (b) Draw the circuit diagram of a full wave rectifier and state how it works.
37. (a) Write the functions of the three segments of a transistor. [CBSE-2017]  
 (b) The figure shows the input waveforms A and B for 'AND' gate. Draw the output waveform and write the truth table for this logic gate.



38. (a) A student wants to use two p-n junction diodes to convert alternating current into direct current. Draw the labelled circuit diagram she would use and explain how it works.  
 (b) Give the truth table and circuit symbol for NAND gate. [CBSE-2018]
39. Draw the typical input and output characteristics of an n-p-n transistor in CE configuration. Show how these characteristics can be used to determine (a) the input resistance ( $r_i$ ) and (b) current amplification factor ( $\beta$ ). [CBSE-2018]

# ANSWER KEY

## EXERCISE-S

1. Ans.  $0.1 \Omega$

2. Ans.

|   | $R_i$        | $D_i$    | Total    |
|---|--------------|----------|----------|
| V | 11.35V       | 0.65V    | 12V      |
| I | 24.15 mA     | 24.15 mA | 24.15 mA |
| R | 470 $\Omega$ |          |          |
| P | 274.1 mW     | 15.7 mW  | 289.8 mW |

3. Ans.

| A | B | C | D | E | F | G |
|---|---|---|---|---|---|---|
| 0 | 0 | 1 | 1 | 0 | 0 | 1 |
| 0 | 1 | 1 | 0 | 0 | 1 | 0 |
| 1 | 0 | 0 | 1 | 1 | 0 | 0 |
| 1 | 1 | 0 | 0 | 0 | 0 | 1 |

4. Ans. (i) 10 V (ii) 90 V (iii) 4 mA

## EXERCISE-O

### SINGLE CORRECT TYPE QUESTIONS

- |              |              |              |              |              |              |
|--------------|--------------|--------------|--------------|--------------|--------------|
| 1. Ans. (B)  | 2. Ans. (B)  | 3. Ans. (C)  | 4. Ans. (C)  | 5. Ans. (B)  | 6. Ans. (C)  |
| 7. Ans. (A)  | 8. Ans. (D)  | 9. Ans. (B)  | 10. Ans. (B) | 11. Ans. (B) | 12. Ans. (B) |
| 13. Ans. (C) | 14. Ans. (B) | 15. Ans. (B) | 16. Ans. (C) | 17. Ans. (A) | 18. Ans. (A) |
| 19. Ans. (B) | 20. Ans. (C) | 21. Ans. (C) | 22. Ans. (C) | 23. Ans. (D) | 24. Ans. (C) |
| 25. Ans. (C) | 26. Ans. (C) | 27. Ans. (B) | 28. Ans. (C) | 29. Ans. (C) | 30. Ans. (A) |
| 31. Ans. (D) | 32. Ans. (B) | 33. Ans. (A) | 34. Ans. (D) | 35. Ans. (A) | 36. Ans. (C) |
| 37. Ans. (D) | 38. Ans. (C) | 39. Ans. (C) | 40. Ans. (A) | 41. Ans. (B) | 42. Ans. (D) |
| 43. Ans. (D) | 44. Ans. (D) |              |              |              |              |

### MULTIPLE CORRECT TYPE QUESTIONS

45. Ans. (A,B,D) 46. Ans. (A,C,D) 47. Ans. (B,C) 48. Ans. (B,D) 49. Ans. (B,C)

## EXERCISE-JM

- |              |                  |             |              |              |              |
|--------------|------------------|-------------|--------------|--------------|--------------|
| 1. Ans. (3)  | 2. Ans. (3)      | 3. Ans. (1) | 4. Ans. (3)  | 5. Ans. (3)  | 6. Ans. (4)  |
| 7. Ans. (4)  | 8. Ans. (1 or 3) | 9. Ans. (4) | 10. Ans. (4) | 11. Ans. (2) | 12. Ans. (2) |
| 13. Ans. (2) |                  |             |              |              |              |

# KEY CONCEPTS

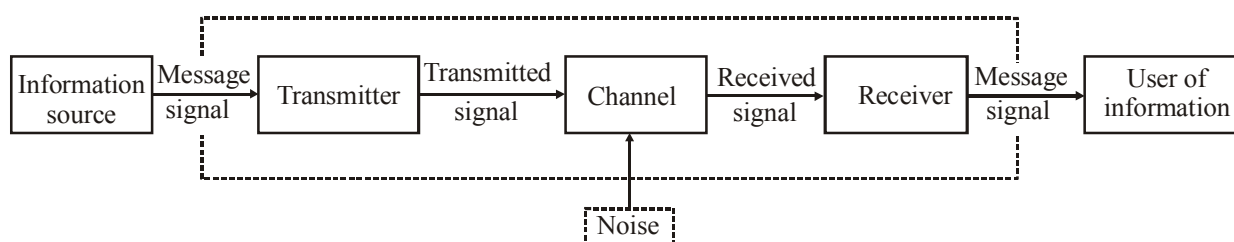
## 1. INTRODUCTION

Communication means transmission of information. Everyone experiences the need to impart or receive information continuously in the surrounding and for this, we speak, listen, send message by a messenger, use coded signalling methods through smoke or flags or beating of drum etc. and these days we are using telephones, TV, radio, satellite communication etc. The aim of this chapter is to introduce the concepts of communication namely the mode of communication, the need for modulation, production and detection of amplitude modulation.

### Elements of a Communication System :

Every communication system has three essential elements-

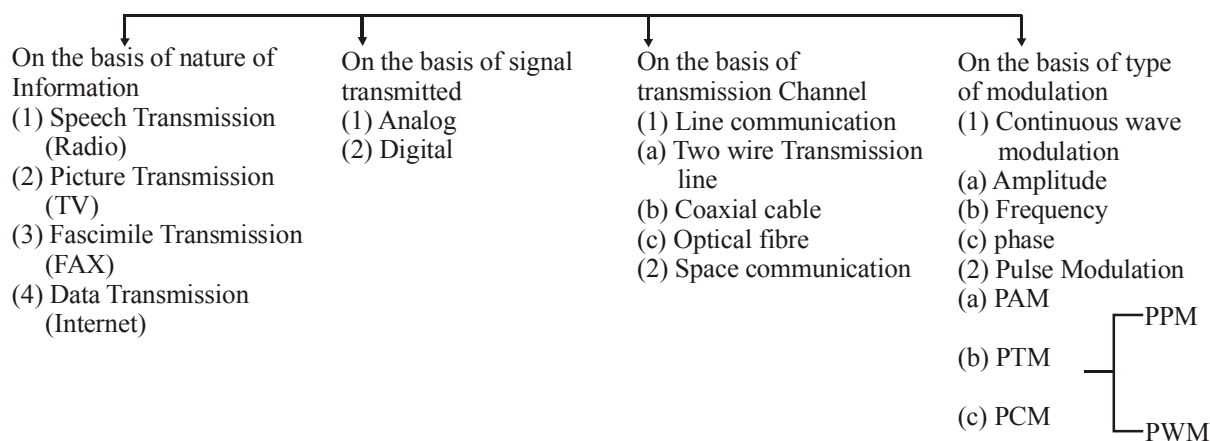
(i) transmitter (ii) medium/channel (iii) receiver



Transmitter converts the message signal into an electric signal and transmits through channel. The receiver receives the transmitted signal and reconstructs the original message signal to the end user. There are two basic modes of communication: (i) point-to-point and (ii) broadcast.

In point-to-point communication mode, communication takes place over a link between a single transmitter and a receiver as in telephony. In the broadcast mode, there are a large number of receivers corresponding to a single transmitter. Radio and television are most common examples of broadcast mode of communication. However the communication system can be classified as follows :

### Types of Communication Systems



**Basic terminology Used in Electronic Communication systems :**

(i) **Transducer.** Transducer is the device that converts one form of energy into another. Microphone, photo detectors and piezoelectric sensors are types of transducer. They convert information into electrical signal.

(ii) **Signal** Signal is the information converted in electrical form. Signals can be analog or digital. Sound and picture signals in TV are analog.

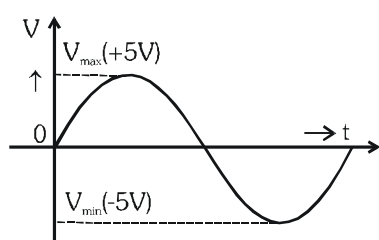
It is defined as a single-valued function of time which has a unique value at every instant of time.

(1) **Analog Signal :-**

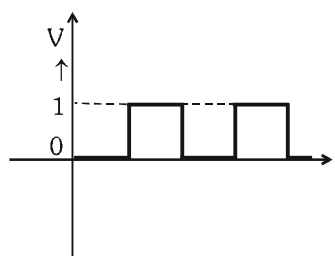
A continuously varying signal (Voltage or Current) is called an analog signal. A decimal number with system base 10 is used to deal with analog signal.

(2) **Digital Signal :-**

A signal that can have only discrete stepwise values is called a digital signal. A binary number system with base 2 is used to deal with digital signals.



**An Analog Signal**



**An Digital Signal**

(iii) **Noise :** There are unwanted signals that tend to disturb the transmission and processing of message signals. The source of noise can be inside or outside the system.

(iv) **Transmitter :** A transmitter processes the incoming message signal to make it suitable for transmission through a channel and subsequent reception.

(v) **Receiver :** A receiver extracts the desired message signals from the received signals at the channel output.

(vi) **Attenuation :** It is the loss of strength of a signals while propagating through a medium. It is like damping of oscillations.

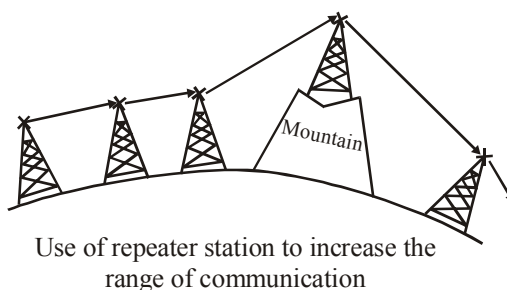
(vii) **Amplification :** It is the process of increasing the amplitude (and therefore the strength) of a signal using an electronic circuit called the amplifier. Amplification is absolutely necessary to compensate for the attenuation of the signal in communication systems.

(viii) **Range :** It is the largest distance between the source and the destination upto which the signal gets received with sufficient strength.

(ix) **Bandwidth :** It is the frequency range over which an equipment operates or the portion of the spectrum occupied by the signal.

(x) **Modulation :** The original low frequency message/information signal cannot be transmitted to long distances. So, at the transmitter end, information contained in the low frequency message signal is superimposed on a high frequency wave, which acts as a carrier of the information. This process is known as modulation.

- (xi) **Demodulation** : The process of retrieval of original information from the carrier wave at the receiver end is termed as demodulation. This process is the reverse of modulation.
- (xii) **Repeater** : A repeater acts as a receiver and a transmitter. A repeater picks up the signal which is coming from the transmitter, amplifies and retransmits it with a change in carrier frequency. Repeaters are necessary to extend the range of a communication system as shown in figure A communication satellite is basically a repeater station in space.



## BANDWIDTH

### Bandwidth of signals :

Different signals used in a communication system such as voice, music, picture, computer data etc. all have different ranges of frequency. The difference of maximum and minimum frequency in the range of each signal is called bandwidth of that signal.

Bandwidth can be of message signal as well as of transmission medium.

#### (i) Bandwidth for analog signals

Bandwidth for some analog signals are listed below :

| Signal  | Frequency range                                                                  | Bandwidth required   |
|---------|----------------------------------------------------------------------------------|----------------------|
| Speech  | 300-3100 Hz                                                                      | 3100-300<br>=2800 Hz |
| Music   | High frequencies produced by musical instrument<br>Audible range =20 Hz - 20 kHz | 20 kHz               |
| Picture | -                                                                                | 4.2 MHz              |
| TV      | Contains both voice and picture                                                  | 6 MHz                |

#### (ii) Bandwidth for digital signal

Basically digital signals are rectangular waves and these can be splitted into a superposition of sinusoidal waves of frequencies  $v_0, 2v_0, 3v_0, 4v_0, nv_0$ , where  $n$  is an integer extending to infinity. This implies that the infinite band width is required to reproduce the rectangular waves. However, for practical purposes, higher harmonics are neglected for limiting the bandwidth

### Band width of Transmission Medium

Different types of transmission media offer different band width in which some of are listed below

| Frequency Bands |                                          |                                                      |                                                                        |
|-----------------|------------------------------------------|------------------------------------------------------|------------------------------------------------------------------------|
|                 | Service                                  | Frequency range                                      | Remarks                                                                |
| 1               | Wire<br>(most common :<br>Coaxial Cable) | 750 MHz<br>(Bandwidth)                               | Normally<br>operated below<br>18 GHz                                   |
| 2               | Free space<br>(radio waves)              | Few hundred kHz to<br>GHz                            |                                                                        |
|                 | (i) Standard AM<br>broadcast             | 540kHz -1600 kHz                                     |                                                                        |
|                 | (ii) FM                                  | 88-108 MHz                                           |                                                                        |
|                 | (iii) Television                         | 54-72 MHz<br>76-88 MHz<br>174-216 MHz<br>420-890 MHz | VHF (Very) high<br>frequencies) TV<br>UHF (Ultra high<br>frequency) TV |
|                 | (iv) Cellular<br>mobile<br>radio         | 896-901 MHz<br>840-935 MHz                           | Mobile to base<br>Station<br>Base station to<br>mobile                 |
|                 | (v) Satellite<br>Communication           | 5.925-6.425 GHz<br>3.7 - 4.2 GHz                     | Uplinking<br>Downlinking                                               |
| 3               | Optical<br>communication<br>using fibres | 1THz-1000 THz<br>(microwaves-<br>ultra violet)       | One single<br>optical fibre<br>offers<br>bandwidth > 100<br>GHz        |

### Propagation of Electromagnetic Waves :

In case of radio waves communication, an antenna at the transmitter radiates the electromagnetic waves (em waves). The em waves travel through the space and reach the receiving antenna at the other end. As the em wave travels away from the transmitter, their strength keeps on decreasing. Many factors influence the propagation of em waves including the path they follow. The composition of the earth's atmosphere also plays a vital role in the propagation of em waves, as summarised below.



**Table 4 Layers of atmosphere and their interaction with the propagating em waves**

| Atmospheric stratum (layer)               | Height over earth's surface (approx)       | Exists during                                | Frequencies most likely affected                                     |
|-------------------------------------------|--------------------------------------------|----------------------------------------------|----------------------------------------------------------------------|
| 1. Troposphere<br>2. Ionosphere           | 10 km                                      | Day and night                                | VHF (upto several GHz)                                               |
| (i) D (part of stratosphere)              | 65-75 km                                   | Day only                                     | Reflects LF, absorbs MF & HF to some degree                          |
| (ii) E (part of stratosphere)             | 100 km                                     | Day only                                     | Helps surface waves, reflects HF                                     |
| (iii) F <sub>1</sub> (Part of Mesosphere) | 170-190 km                                 | Daytime, merges with F <sub>2</sub> at night | Partially absorbs HF waves yet allowing them to reach F <sub>2</sub> |
| (iv) F <sub>2</sub> (Thermosphere)        | 300 km at night, 250-400 km during daytime | Day and night                                | Efficiently reflects HF waves particularly at night                  |

### Ground Wave Propagation :

- The radio waves which travel through atmosphere following the surface of earth are known as ground waves or surface waves and their propagation is called ground wave propagation or surface wave propagation.
- The ground wave transmission becomes weaker with increase in frequency because more absorption of ground waves takes place at higher frequency during propagation through atmosphere.
- The ground wave propagation is suitable for low and medium frequency i.e. upto 2 or 3 MHz only.
- The ground wave propagation is generally used for local band broadcasting and is commonly called medium wave.
- The maximum range of ground or surface wave propagation depends on two factors :
  - The frequency of the radio waves and
  - Power of the transmitter

### Sky Wave Propagation :

- The sky waves are the radio waves of frequency between 2 MHz to 30 MHz.
- The ionospheric layer acts as a reflector for a certain range of frequencies (3 to 30 MHz). Electromagnetic waves of frequencies higher than 30 MHz penetrate the ionosphere and escape.
- The highest frequency of radiowaves which when sent straight (i.e. normally) towards the layer of ionosphere gets reflected from ionosphere and returns to the earth is called critical frequency.. If is given by  $f_c = 9 (N_{\max})^{1/2}$ , where N is the number density of electron/m<sup>3</sup>.

### Space wave propagation :

- The space waves are the radiowaves of very high frequency (i.e. between 30 MHz. to 300 MHz or more).
- the space waves can travel through atmosphere from transmitter antenna to receiver antenna either directly or after reflection from ground in the earth's troposphere region. That is why the space wave propagation is also called as tropospheric propagation or line of sight propagation.
- The range of communication of space wave propagation can be increased by increasing the heights of transmitting and receiving antenna.

- (d) If the transmitting antenna is at a height  $h_T$ , then you can show that the distance to the horizon  $d_T$  is given as  $d_T = \sqrt{2Rh_T}$ , where  $R$  is the radius of the earth (approximately 6400 km).  $d_T$  is also called the radio maximum line-of sight distance  $d_m$  between the two antennas having heights  $h_T$  and  $h_R$  above the earth is given by :

$$d_m = \sqrt{2Rh_T} + \sqrt{2Rh_R}$$

where  $h_R$  is the height of receiving antenna.

## Modulation

- \* It is a process by which any electrical signal called input / baseband or modulating signal, is mounted onto another signal of high frequency which is known as carrier signal.
- \* It is defined as the process by which some characteristic (called parameter) of carrier signal is varied in accordance with the instantaneous value of the baseband signal.
- \* The signal which results from this process is known as modulated signal.

### Need for Modulation :

#### (i) To avoid interference :

If many modulating signals travel directly through the same transmission channel, they will interfere with each other and result in distortion.

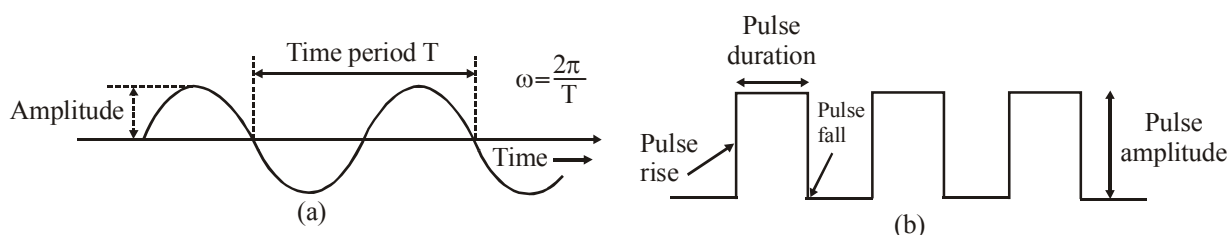
#### (ii) To design antennas of practicable size :

The minimum height of antenna (not of antenna tower) should be  $\lambda/4$  where  $\lambda$  is wavelength of modulating signal. This minimum size becomes impracticable because the frequency of the modulating signal can be upto 5 kHz which corresponds to a wavelength of  $3 \times 10^8 / 5 \times 10^3 = 60$  km. This will require an antenna of the minimum height of  $\lambda/4 = 15$  km. This size of an antenna is not practical.

#### (iii) Effective Power Radiated by an Antenna :

A theoretical study of radiation from a linear antenna (length  $\ell$ ) shows that the power radiated is proportional to (frequency)<sup>2</sup> i.e.  $(\ell/\lambda)^2$ . For a good transmission, we need high powers and hence this also points out to the need of using high frequency transmission.

The above discussion suggests that there is a need for translating the original low frequency baseband message signal into high frequency wave before transmission. In doing so, we take the help of a high frequency signal, which we already know now, is known as the carrier wave, and a process known as modulation which attaches information to it. The carrier wave may be continuous (sinusoidal) or in the form of pulses, as shown in figure

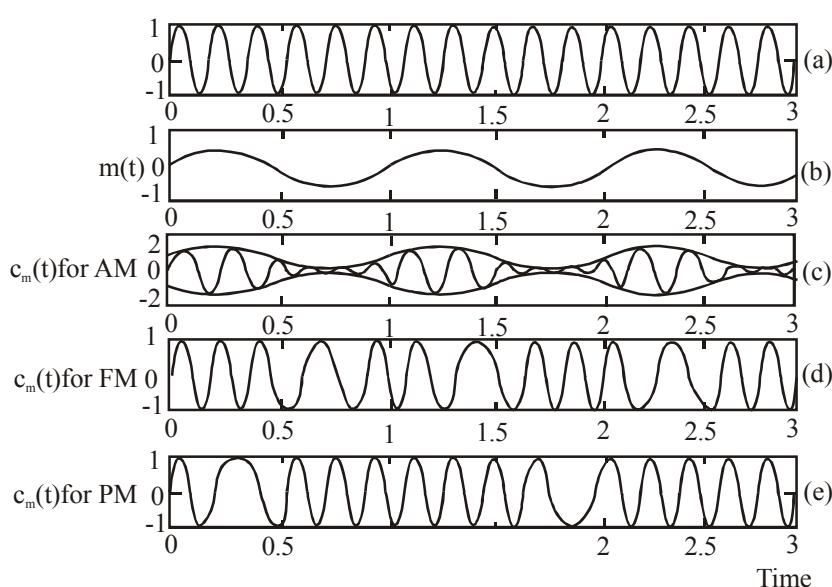


### Carrier wave : Sinusoidal

A sinusoidal carrier wave can be represented as

$$c(t) = A_c \sin(\omega_c t + \phi)$$

where  $c(t)$  is the signal strength (voltage or current),  $A_c$  is the amplitude,  $\omega_c (= 2\pi f_c)$  is the angular frequency and  $\phi$  is the initial phase of the carrier wave. Thus, modulation can be affected by varying, any of three parameters, viz  $A_c$ ,  $\omega_c$  and  $\phi$ , of the carrier wave can as per the parameter of the message or information signal. This results in three types of modulation : (i) Amplitude modulation (AM) (ii) Frequency modulation (FM) and (iii) Phase modulation (PM), as shown in figure.



Modulation of a carrier wave : (a) a sinusoidal carrier wave  
(b) a modulating signal : (c) amplitude modulation :  
(d) frequency modulation : and (e) phase modulation

### Carrier Wave Pulses :

Similarly, the significant characteristics of a pulse are : Pulse Amplitude, Pulse duration or pulse Width, and pulse Position (denoting the time of rise or fall of the pulse amplitude) Hence, different types of pulse modulation are (a) pulse amplitude modulation (PAM), (b) Pulse duration modulation (PDM) or pulse width modulation (PWM), and (c) Pulse position modulation (PPM).

**Ex.1 A separate high freq. wave (i.e. carrier wave) is needed in modulation why ?**

**Ans.** This is because we cannot change any of the characteristics (amplitude, frequency or phase) of the audio signal as this would change the message to be communicated. So keeping the audio signal same, the amplitude of freq. or phase of the high freq. carrier wave is modified in accordance with the modulating (i.e. audio signal) signal.

**Amplitude Modulation :**

In amplitude modulation the amplitude of the carrier is varied in accordance with the information signals. Let  $c(t) = A_c \sin \omega_c t$  represent carrier wave and  $m(t) = A_m \sin \omega_m t$  represent the message or the modulating signal where  $\omega_m = 2\pi f_m$  is the angular frequency of the message signal. The modulated signal  $c_m(t)$  can be written as

$$c_m(t) = (A_c + A_m \sin \omega_m t) \sin \omega_c t$$

$$= A_c \left( 1 + \frac{A_m}{A_c} \sin \omega_m t \right) \sin \omega_c t \quad \dots(1)$$

Note that the modulated signal now contains the message signal & it can be written as :

$$c_m(t) = A_c \sin \omega_c t + \mu A_c \sin \omega_m t \sin \omega_c t \quad \dots(2)$$

Here  $\mu = A_m/A_c$  is the modulation index

Often,  $\mu$  is expressed in percentage and is called the percentage modulation. Importance of modulation index is that it determine the quality of the transmitted signal. When modulation index is small, variation in carrier amplitude will be small. Therefore, audio signal being transmitted will be weak. As the modulation index increases, the audio signal on reception becomes clearer. If the modulation index becomes greater than one, the signal gets lost partially.

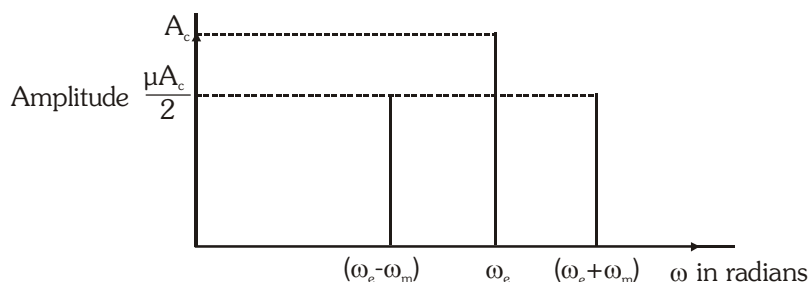
If a carrier wave is modulated by several sine waves the total modulated index  $m_t$  is given by

$$m_t = \sqrt{m_1^2 + m_2^2 + m_3^2 + \dots}$$

Using the trigonometric relation  $\sin A \sin B = \frac{1}{2} (\cos(A - B) - \cos(A + B))$ , we can write  $c_m(t)$  of eq. (15.4) as

$$c_m(t) = A_c \sin \omega_c t + \frac{\mu A_c}{2} \cos(\omega_c - \omega_m) t - \frac{\mu A_c}{2} \cos(\omega_c + \omega_m) t \quad \dots(3)$$

Here  $\omega_c - \omega_m$  and  $\omega_c + \omega_m$  are respectively called the lower side and upper side frequencies. The modulated signal now consists of the carrier wave of frequency  $\omega_c$  plus two sinusoidal waves each with a frequency slightly different from, known as side bands. The frequency spectrum of the amplitude modulated signal is shown in figure :



As long as the broadcast frequencies (carrier waves) are sufficiently spaced out so that sidebands do not overlap, different stations can operate without interfering with each other.

$$f_{SB} = f_c \pm f_m$$

∴ Frequency of lower side band is

$$f_{LSB} = f_c - f_m \quad \dots (4)$$

and frequency of upper side band is

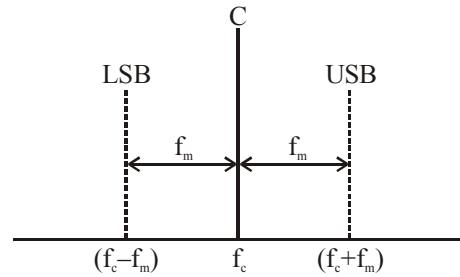
$$f_{USB} = f_c + f_m$$

Band width of amplitude modulated wave is

$$\begin{aligned} &= f_{USB} - f_{LSB} \\ &= (f_c + f_m) - (f_c - f_m) = 2f_m \quad \dots (5) \end{aligned}$$

Band width = twice the frequency of the modulating signal

Phase modulation & frequency modulation are 2 different types of angular modulation.



Power in AM waves : Power dissipated in any circuit  $P = \frac{V_{rms}^2}{R}$

Hence :

(i) Carrier power  $P_c = \frac{\left(\frac{E_c}{\sqrt{2}}\right)^2}{R} = \frac{E_c^2}{2R}$

(ii) Total power of side bands  $P_{sb} = \frac{\left(\frac{m_a E_c}{2\sqrt{2}}\right)^2}{R} \times 2 = \frac{m_a^2 E_c^2}{4R}$

(iii) Total power of AM wave  $P_{Total} = P_c + P_{ab} = \frac{E_c^2}{2R} \left(1 + \frac{m_a^2}{2}\right)$

(iv)  $\frac{P_t}{P_c} = \left(1 + \frac{m_a^2}{2}\right)$  and  $\frac{P_{ab}}{P_t} = \frac{m_a^2 / 2}{\left(1 + \frac{m_a^2}{2}\right)}$

(v) Maximum power in the AM (without distortion) will occur when  $m_a = 1$   
i.e.,  $P_t = 1.5 P = 3P_{ab}$

(vi) If  $I_c$  = Unmodulated current and  $I_t$  = total or modulated current

$$\Rightarrow \frac{P_t}{P_c} = \frac{I_t^2}{I_c^2} \Rightarrow \frac{I_t}{I_c} = \sqrt{\left(1 + \frac{m_a^2}{2}\right)}$$

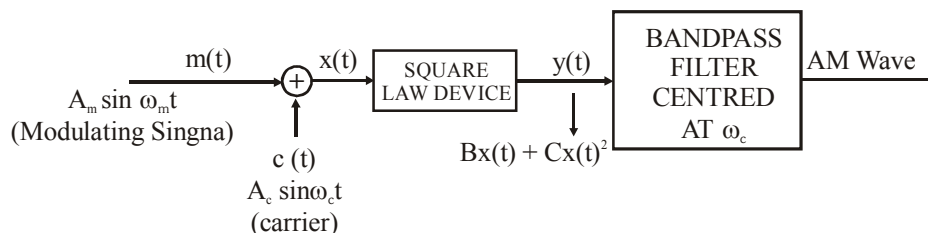
**Ex.2** A message signal of frequency 10 kHz and peak voltage of 10 volts is used to modulate a carrier of frequency 1 Mhz and peak voltage of 20 volts. Determine (a) modulation index, (b) the side bands produced.

**Sol.** (a) Modulation index =  $10/20 = 0.5$

(b) The side bands are at  $(1000 + 10 \text{ kHz}) = 1010 \text{ kHz}$  and  $(1000 - 10 \text{ kHz}) = 990 \text{ kHz}$ .

### Production of Amplitude modulated Wave :

Amplitude modulation can be produced by a variety of methods. A conceptually simple method is shown in the block diagram of figure.



Here the modulating signal  $A_m \sin \omega_m t$  is added to the carrier signal  $A_c \sin \omega_c t$  to produce the signal  $x(t)$ . This signal  $x(t) = A_m \sin \omega_m t + A_c \sin \omega_c t$  is passed through a square law device which is a non-linear device which produces an output

$$y(t) = Bx(t) + Cx^2(t) \quad \dots\dots(4)$$

where  $B$  and  $C$  are constants. Thus,

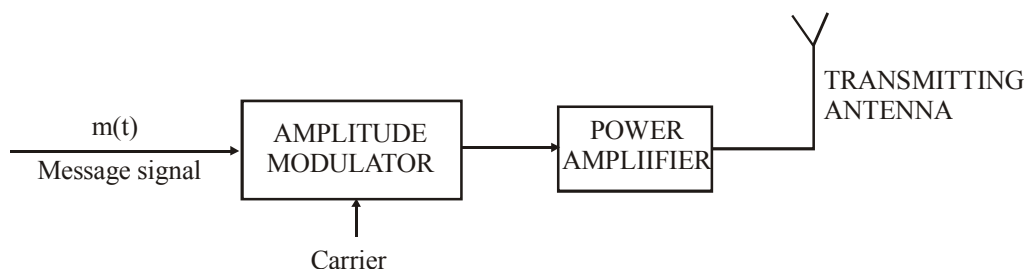
$$y(t) = BA_m \sin \omega_m t + BA_c \sin \omega_c t + C[A_m^2 \sin^2 \omega_m t + A_c^2 \sin^2 \omega_c t + 2A_m A_c \sin \omega_m t \sin \omega_c t] \quad \dots\dots(5)$$

$$= BA_m \sin \omega_m t + BA_c \sin \omega_c t + \frac{CA_m^2}{2} + A_c^2 - \frac{CA_m^2}{2} \cos 2\omega_m t - \frac{CA_c^2}{2} \cos 2\omega_c t + CA_m A_c \cos (\omega_c - \omega_m) t - CA_m A_c \cos (\omega_c + \omega_m) t \quad \dots\dots(6)$$

where the trigonometric relations  $\sin^2 A = (1 - \cos 2A)/2$  and the relation for  $\sin A \sin B$  mentioned earlier are used.

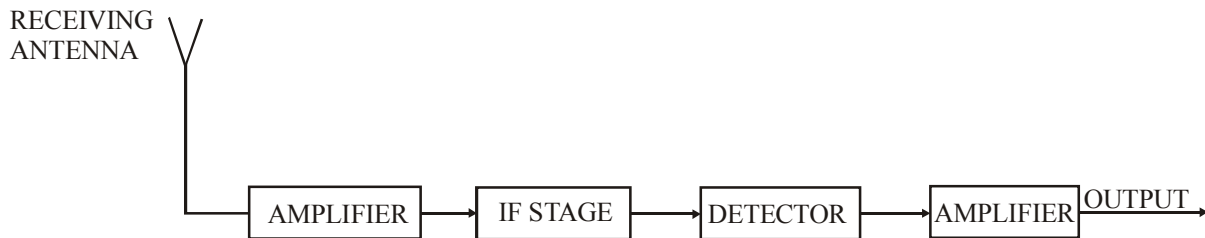
In equation (6), there is a dc term  $C/2 (A_m^2 + A_c^2)$  and sinusoids of frequencies  $\omega_m$ ,  $2\omega_m$ ,  $\omega_c - \omega_m$  and  $\omega_c + \omega_m$ . The output of the band pass filter therefore is of the same form as equation (3) and is therefore an AM wave.

It is to be mentioned that the modulated signal cannot be transmitted as such. The modulator is to be followed by a power amplifier which provides the necessary power and then the modulated signal is fed to an antenna of appropriate size for radiation as shown in figure.

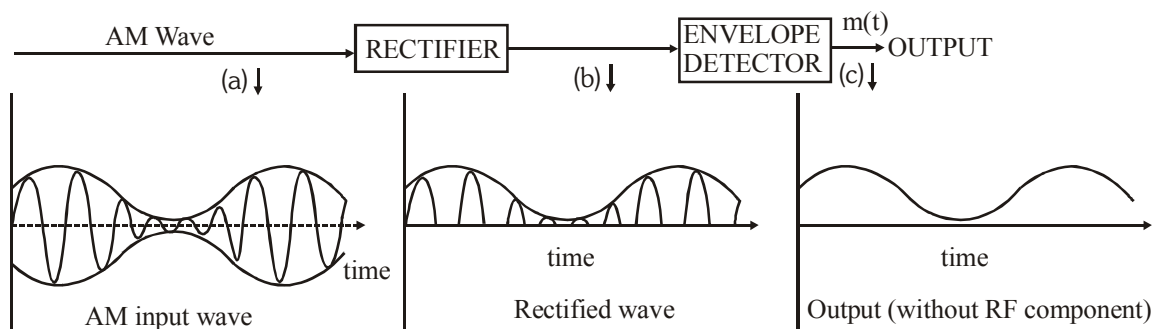


### Detection of Amplitude Modulated Wave :

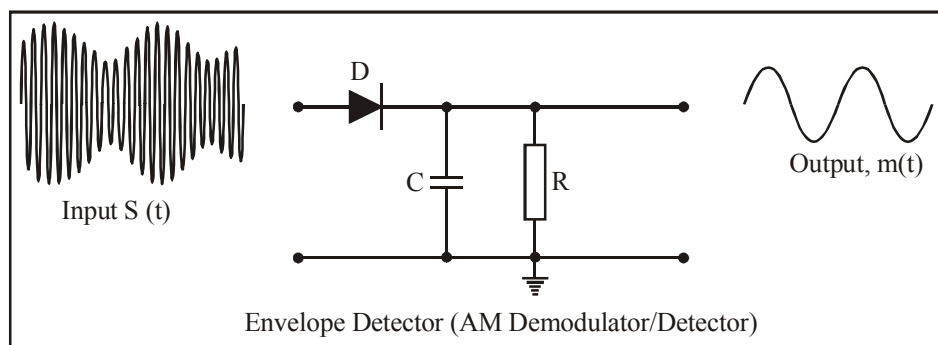
The transmitted message gets attenuated in propagating through the channel. The receiving antenna is therefore to be followed by an amplifier and a detector. In addition, to facilitate further processing, the carrier frequency is usually changed to a lower frequency by what is called an intermediate frequency (IF) stage preceding the detection. The detected signal may not be strong enough to be made use of and hence is required to be amplified. A block diagram of a typical receiver is shown in figure.



Detection is the process of recovering the modulating signal from the modulated carrier wave. We just saw that the modulated carrier wave contains the frequencies  $\omega_c$  and  $\omega_c \pm \omega_m$ . In order to obtain the original message signal  $m(t)$  of angular frequency  $\omega_m$ , a simple method is shown in the form of a block diagram in figure.



The modulated signal of the form given in (a) of above figure is passed through a rectifier to produce the output shown in (b). This envelope of signal (b) is the message signal. In order to retrieve  $m(t)$ , the signal is passed through an envelope detector (which may consist of a simple RC circuit).



This is essentially just a half wave rectifier which charges a capacitor to a voltage to the peak voltage of the incoming AM waveform,  $S(t)$ . When the input wave's amplitude increases, the capacitor voltage is increased via the rectifying diode. When the input's amplitude falls, the capacitor voltage is reduced by being discharged by a 'bleed' resistor,  $R$ . The main advantage of this form of AM Demodulator is that it is very simple and cheap!

The circuit relies upon the behaviour of the diode-allowing current through when the input is +ve with respect to the capacitor voltage, hence 'topping up' the capacitor voltage to the peak level, but blocking any current from flowing back out through the diode when the input voltage is below the capacitor voltage.

Consider what happens when we have a carrier frequency,  $f_c$ , and use an envelope detector whose

time constant,  $\tau = (RC)$ . The time between successive peaks of the carrier will be  $T = \frac{1}{f_c}$

Each peak will charge the capacitor to some voltage,  $V_{\text{peak}}$ , which is proportional to the modulated amplitude of the AM wave. Between each peak and the next the capacitor voltage will therefore be discharged to

$$V'_{\text{peak}} = V_{\text{peak}} \text{Exp} \{-T/\tau\}$$

which, provided that  $T \ll \tau$ , is approximately the same as

$$V'_{\text{peak}} \approx V_{\text{peak}} [1 - T/\tau]$$

The peak-to-peak size of the ripple,  $\Delta V$ , will therefore be

$$\Delta V \approx \frac{V_{\text{peak}} T}{\tau} = \frac{V_{\text{peak}}}{f_c \tau}$$

A sudden, large reduction in the amplitude of the input AM wave means that capacitor charge isn't being 'topped up' by each cycle peak. The capacitor voltage therefore falls exponentially until it reaches the new, smaller, peak value. In practice the modulating signal is normally restricted to a specific frequency range. This limits the maximum rate of fall of the AM wave's amplitude. We can therefore hope to avoid negative peak clipping by arranging that the detector's time constant  $\tau \ll t_m$  where

$$t_m = 1/f_m$$

and  $f_m$  is the highest modulation frequency used in a given situation.

The above implies that we can avoid negative peak clipping by choosing a small value of  $\tau$ . However, to minimise ripple we want to make  $\tau$  as large as possible. In practice we should therefore choose a value

$$1/f_m \gg \tau \gg 1/f_c$$

to minimise the signal distortions caused by these effects. This is clearly only possible if the modulation frequency  $f_m \ll f_c$ . Envelope detector only work satisfactorily when we ensure this inequality is true. With the modulation index and the resistor the capacitor can be computed by

$$C \leq \frac{\sqrt{\left(\frac{1}{m}\right)^2 - 1}}{2\pi R f_m (\text{max})}$$



## EXERCISE-S

1. Is it necessary for a transmitting antenna to be at the same height as that of the receiving antenna for 'line-of-sight communication? A TV transmitting antenna is 81m tall. How much service area can it cover if the receiving antenna is at the ground level?

**PC0001**

2. A carrier wave of peak voltage 12 V is used to transmit a message signal. What should be the peak 'voltage of the modulating signal in order to have a modulation index of 75%?

**PC0002**

3. For an amplitude modulated wave, the maximum amplitude is found to be 10V while the minimum amplitude is found to be 2V. Determine the modulation index  $\mu$ .

**PC0003**

4. If the sum of the heights of transmitting and receiving antennas in line of sight of communication is fixed at  $h$ , show that the range is maximum when the two antennas have a height  $h/2$  each.

**PC0004**

## EXERCISE-O

## SINGLE CORRECT TYPE QUESTIONS

1. If a carrier wave of 1000 kHz is used to carry the signal, the length of transmitting antenna will be equal to :-  
 (A) 3 m (B) 30m (C) 300 m (D) 3000 m  
 PC0005
2. The maximum range of ground or surface wave propagation depends on :-  
 (A) the frequency of the radio waves only (B) power of the transmitter only  
 (C) both of them (D) none of them  
 PC0006
3. For television broadcasting, the frequency employed is normally :-  
 (A) 30-300MHz (B) 30-300 GHz (C) 30-300KHz (D) 30-300Hz-  
 PC0007
4. The audio signal :-  
 (A) can be sent directly over the air for large distance  
 (B) cannot be sent directly over the air for large distance  
 (C) possess very high frequency  
 (D) none of the above  
 PC0008
5. For a carrier frequency of 100 kHz and a modulating frequency of 5 kHz what is the width of AM transmission-  
 (A) 5 kHz (B) 10kHz (C) 20 kHz (D) 200 KHz  
 PC0009
6. Three waves A, B and C of frequencies 1600 kHz, 5 MHz and 60 MHz, respectively are to be transmitted from one place to another. Which of the following is the most appropriate mode of communication :  
 (A) A is transmitted via space wave while B and C are transmitted via sky wave.  
 (B) A is transmitted via ground wave, B via sky wave and C via space wave.  
 (C) B and C are transmitted via ground wave while A is transmitted via sky wave.  
 (D) B is transmitted via ground wave while A and C are transmitted via space wave.  
 PC0010
7. A speech signal of 3kHz is used to modulate a carrier signal of frequency 1 MHz, using amplitude modulation. The frequencies of the side bands will be  
 (A) 1.003 MHz and 0.997 MHz. (B) 3001 kHz and 2997 kHz.  
 (C) 1003 kHz and 1000 kHz. (D) 1 MHz and 0.997 MHz.  
 PC0011
8. A message signal of frequency  $\omega_m$  is superposed on a carrier wave of frequency  $\omega_c$  to get an amplitude mediated wave (AM) The frequency of the AM wave will be  
 (A)  $\omega_m$  (B)  $\omega_c$  (C)  $\frac{\omega_c + \omega_m}{2}$  (D)  $\frac{\omega_c - \omega_m}{2}$   
 PC0012

9. A basic communication system consists of  
 (A) transmitter (B) information source. (C) user of information.  
 (D) channel (E) receiver.  
 Choose the correct sequence in which these are arranged in a basic communication system :-  
 (A) ABCDE (B) BADEC (C) BDACE (D) BEADC
- PC0013**
10. **Statement 1 :** Skywave can not be observed on moon.  
**Statement 2 :** Atmosphere of variable refractive index is require for propagation of skywave.  
 (A) Both Statement-1 and Statement-2 are true, and Statement-2 is the correct explanation of Statement-1  
 (B) Both Statement-1 and Statement-2 are true but Statement-2 is not the correct explanation of Statement- 1.  
 (C) Statement-1 is true but Statement-2 is false.  
 (D) Statement-1 is false but Statement-2 is true.
- PC0014**
11. **Statement 1 :** Ground wave communication is effective only allow frequencies in the range 500kHz to about 1500 kHz.  
**Statement 2 :** The decrease in the intensity of the signal due to absorption by the earth and its atmosphere is higher for higher frequencies.  
 (A) Both Statement-1 and Statement-2 are true, and Statement-2 is the correct explanation of Statement-1.  
 (B) Both Statement-1 and Statement-2 are true but Statement-2 is not the correct explanation of Statement-I .  
 (C) Statement-1 is true but Statement-2 is false.  
 (D) Statement-1 is false but Statement-2 is true.

**PC0015**

### MULTIPLE CORRECT TYPE QUESTIONS

12. An audio signal of 15kHz frequency cannot be transmitted over long distances without modulation because  
 (A) the size of the required antenna would be at least 5 km which is not convenient.  
 (B) the audio signal cannot be transmitted through skywaves.  
 (C) the size of the required antenna would be at least 20 km, which is not convenient.  
 (D) effective power transmitted would be very low, if the size of the antenna is less than 5 km.
- PC0016**
13. Audio sine waves of 3 kHz frequency are used to amplitude modulate a carrier signal of 1.5 MHz. Which of the following statements are true?  
 (A) The side band frequencies are 1506 kHz and 1494 kHz.  
 (B) The bandwidth required for amplitude modulation is 6kHz.  
 (C) The bandwidth required for amplitude modulation is 3 MHz.  
 (D) The side band frequencies are 1503 kHz and 1497 kHz.

**PC0017**

## EXERCISE-JM

1. This questions has Statement-1 and Statement-2. Of the four choice given after the statements, choose the one that best describes the two statements.

Statement-1 : Sky wave signals are used for long distance radio communication. These signals are in general, less stable than ground wave signals. [AIEEE-2011]

Statement-2 : The state of ionosphere varies from hour to hour, day to day and season to season.

- (1) Statement-1 is true, Statement-2 is true, Statement-2 is the correct explanation of statement-1  
 (2) Statement-1 is true, Statement-2 is true, Statrment-2 is not the correct explanation of Statement-1  
 (3) Statement-1 is false, Statement-2 is true  
 (4) Statement-1 is true, Statement-2 is false

PC0018

2. A radar has a power of 1 kW and is operating at a frequency of 10 GHz. It is located on a mountain top of height 500 m. The maximum distance up to which it can detect object located on the surface of the earth (Radius of earth =  $6.4 \times 10^6$  m) is : [AIEEE-2012]

- (1) 40 km (2) 64 (3) 80 km (4) 16km

PC0019

3. A diode detector is used to detect an amplitude modulated wave of 60% modulation by using a condenser of capacity 250 pico farad in parallel with a load resistance 100 kilo ohm. Find the maximum modulated frequency which could be detected by it. [JEE Main-2013]

- (1) 10.62kHz (2) 5.31 MHz (3) 5.31 kHz (4) 10.62MHz

PC0020

4. A single of 5 kHz frequency is amplitude modulated on a carrier wave of frequency 2 MHz. The frequencies of the resultant signal is/are - [JEE Main-2015]

- (1) 2005 kHz, 2000 kHz and 1995 kHz (2) 2000 kHz and 1995 kHz  
 (3) 2 MHz only (4) 2005 kHz and 1995 kHz

PC0021

5. Choose the correct statement : [JEE Main-2016]

- (1) In frequency modulation the amplitude of the high frequency carrier wave is made to vary in proportion to the frequency of the audio signal.  
 (2) In amplitude modulation the amplitude of the high frequency carrier wave is made to vary in proportion to the amplitude of the audio signal.  
 (3) In amplitude modulation the frequency of the high frequency carrier wave is made to vary in proportion to the amplitude of the audio signal.  
 (4) In frequency modulation the amplitude of the high frequency carrier wave is made to vary in proportion to the amplitude of the audio signal.

PC0022

6. In amplitude modulation, sinusoidal carrier frequency used is denoted by  $\omega_c$  and the signal frequency is denoted by  $\omega_m$ . The bandwidth ( $\Delta\omega_m$ ) of the signal is such that  $\Delta\omega_m \ll \omega_c$ . Which of the following frequencies is not contained in the modulated wave ? **[JEE Main-2017]**

(1)  $\omega_m + \omega_c$                       (2)  $\omega_c - \omega_m$                       (3)  $\omega_m$                       (4)  $\omega_c$

**PC0023**

7. A telephonic communication service is working at carrier frequency of 10 GHz. Only 10% of it is utilized for transmission. How many telephonic channels can be transmitted simultaneously if each channel requires a bandwidth of 5 kHz ? **[JEE Main-2018]**

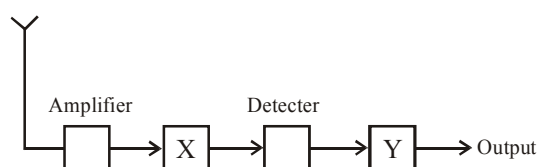
(1)  $2 \times 10^4$                       (2)  $2 \times 10^5$                       (3)  $2 \times 10^6$                       (4)  $2 \times 10^3$

**PC0024**

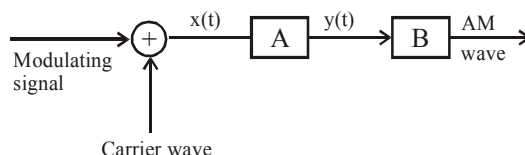
## CBSE PREVIOUS YEAR'S QUESTIONS

1. A.T.V. tower has a height of 400 m at a given place. Calculate as coverage range, if the radius of the earth is 6400 km. [2; CBSE-2004]
2. Why is shortwave band used for long distance radio broadcast? [1; CBSE-2004]
3. Distinguish between analog and digital communication. Write any two modulation techniques employed for the digital data. Describe briefly any one of the techniques used. [3; CBSE-2005]
4. A ground receiver stations receiving a signal at (a) 5 MHz and (b) 100 MHz, transmitted from a ground transmitter at a height of 300 m located at a distance of 100 km. Identify whether it is coming via space wave or sky wave propagation or satellite transponder. (Given the value of radius of the earth is 6400 km and maximum electron density,  $N_{\max} = 1012 \text{ m}^{-3}$ ). [3; CBSE-2005]
5. Consider an optical communication system operating at  $\lambda = 800 \text{ nm}$ . Suppose, only 1% of the optical source frequency is the available channel band-width for optical communication. How many channels can be accommodated for transmitting [3; CBSE-2006]
  - (a) audio-signals requiring a band-width of 8 kHz,
  - (b) video TV signals requiring an approximate band-width of 4.5 MHz? Support your answer with suitable calculations.
6. Distinguish between frequency modulation and amplitude modulation. Why is an FM signal less susceptible to noise than an AM signal? [3; CBSE-2006]
7. Give any one difference between FAX and e-mail systems of communication. [1; CBSE-2006]
8. What is modulation? Explain the need of modulating a low frequency information signal. With the help of diagrams, differentiate between PAM and PDM. [3; CBSE-2007]
9. What should be the length of dipole antenna for a carrier wave of frequency  $6 \times 10^8 \text{ Hz}$ ? [1; CBSE-2007]
10. A transmitting antenna at the top of a tower has a height of 36m and the height of the receiving antenna is 49m. What is the maximum distance between them for satisfactory communication in the LOS mode? (Radius of earth = 6400km). [3; CBSE-2008]
11. What is meant by term 'modulation'? Draw a block diagram of a simple modulator for obtaining an AM signal. [2; CBSE-2009]
12. Why is high frequency carrier waves used for transmission? [2; CBSE-2009]
13. By what percentage will the transmission range of a TV tower be affected when the height of tower is increased by 21%? [12; CBSE-2009]
14. Write two factors justifying the need of modulating a signal. A carrier wave of peak voltage 12 V is used to transmit a message signal. What should be the peak voltage of the modulating signal in order to have a modulation index of 75% ? [12; CBSE-2010]
15. Which mode of propagation is used by short wave broadcast services having frequency range from a few MHz upto 30 MHz ? Explain diagrammatically how long distance communication can be achieved by this mode. Why is there an upper limit to frequency of waves used in this mode ? [3; CBSE-2010]

16. What is sky wave communication ? Why is this mode of propagation restricted to the frequencies only upto few MHz? [2; CBSE-2011]
17. Write briefly any two factors which demonstrate the need for modulating a signal. Draw a suitable diagram to show amplitude modulation using a sinusoidal signal as the modulating signal. [3; CBSE-2011]
18. Mention three different modes of propagation used in communication system. Explain with the help of a diagram how long distance communication can be achieved by ionospheric reflection of radio waves. [3; CBSE-2012]
19. In the given block diagram of a receiver, identify the boxes labeled as X and Y and write their functions. [2; CBSE-2012]



20. In the block diagram of a simple modulator for obtaining an AM signal, shown in the figure, identify the boxes A and B. Write their functions. [12; CBSE-2013]



21. Name the type of waves which are used for line of sight (LOS) communication. What is the range of their frequencies?  
A transmitting antenna at the top of a tower has a height of 20 m and the height of the receiving antenna is 45 m. Calculate the maximum distance between them for satisfactory communication in LOS mode. (Radius of the Earth =  $6.4 \times 10^6$  m) [3; CBSE-2013]
22. Write the functions of the following in communication systems.  
(i) Transmitter (ii) Modulator [2; CBSE-2014]
23. Write two basic modes of communication Explain the process of amplitude modulation. Draw a schematic sketch showing how amplitude modulated signal is obtained by superposing a modulating signal over a sinusoidal carrier wave. [3; CBSE-2014]
24. What is the function of a band pass filter used in a modulator for obtaining AM signal ? [1; CBSE-2015]
25. Differentiate between amplitude modulated (AM) and frequency modulated (FM) waves by drawing suitable diagrams. Why is FM signal preferred over AM signal ? [2; CBSE-2015]
26. Name the three different modes of propagation in a communication system. [3; CBSE-2015]  
State briefly why do the electromagnetic waves with frequency range from a few MHz upto 30 MHz can reflect back to the earth. What happens when the frequency range exceeds this limit ?
27. A signal of 5 kHz frequency is amplitude modulated on a carrier wave of frequency 2 MHz. What are the frequencies of the side bands produced ? [CBSE-2016]

28. Why is base band signal not transmitted directly ? Give any two reasons. [CBSE-2016]
29. What is space wave propagation? State the factors which limit its range of propagation. Derive an expression for the maximum line of sight distance between two antennas for space wave propagation. [CBSE-2016]
30. (a) How is amplitude modulation achieved ?  
(b) The frequencies of two side bands in an AM wave are 640 kHz and 660 kHz respectively. Find the frequencies of carrier and modulating signal. What is the bandwidth required for amplitude modulation ? [CBSE-2017]
31. Draw a block diagram of a generalized communication system. Write the functions of each of the following: [CBSE-2017]  
(a) Transmitter  
(b) Channel  
(c) Receiver
32. Which modes of propagation is used by short wave broadcast services? [CBSE-2018]
33. A carrier wave of peak voltage 15 V is used to transmit a message signal. Find the peak voltage of the modulating signal in order to have a modulation index of 60%. [CBSE-2018]
34. (a) Give three reasons why modulation of a message signal is necessary for long distance transmission.  
(b) Show graphically an audio signal, a carrier wave and an amplitude modulated wave. [CBSE-2018]



## ANSWER KEY

### EXERCISE-S

- Ans.** No. Service area will be  $A = \pi d_7^2 = \frac{22}{7} \times 162 \times 6.4 \times 10^6 = 3258 \text{ km}^2$ .
- Ans.**  $\mu = 0.75 = \frac{A_m}{A_c}$ ;  $A_m = 0.75 \times 12 = 9\text{V}$
- Ans.** Since the AM wave is given by  $(A_c + A_m) \sin \omega_m t) \cos \omega_c t$ , the maximum amplitude is  $M_1 = A_c + A_m$ . Hence the modulation index is

$$m = \frac{A_m}{A_c} = \frac{8}{12} = \frac{2}{3}$$

with  $M_2 = 0$ , clearly,  $m = 1$  irrespective of  $M_1$

### EXERCISE-O

#### SINGLE CORRECT TYPE QUESTIONS

- |             |             |             |              |              |             |
|-------------|-------------|-------------|--------------|--------------|-------------|
| 1. Ans. (C) | 2. Ans. (C) | 3. Ans. (A) | 4. Ans. (B)  | 5. Ans. (B)  | 6. Ans. (B) |
| 7. Ans. (A) | 8. Ans. (B) | 9. Ans. (B) | 10. Ans. (A) | 11. Ans. (A) |             |

#### MULTIPLE CORRECT TYPE QUESTIONS

12. Ans. (A,B,D) 13. Ans. (B,D)

### EXERCISE-JM

- |             |             |             |             |             |             |
|-------------|-------------|-------------|-------------|-------------|-------------|
| 1. Ans. (1) | 2. Ans. (3) | 3. Ans. (3) | 4. Ans. (1) | 5. Ans. (2) | 6. Ans. (3) |
| 7. Ans. (2) |             |             |             |             |             |